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Tamás Rudas

Lectures on Advanced Topics in Categorical Data Analysis

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Preface

Writing a book makes only sense if reading it may help scholars to acquire and consolidate knowledge more easily than reading the relevant research publications. Published papers tend to emphasize novelty, and may not deal with the fundamentals that are assumed to be known by their readers. As opposed to research papers, a book creates opportunities to present a unifying view of its material and to work out the background details. This is the ambition of this book, too.

Words and expressions in the title try to explain what this book does.

Lectures: The topics that are dealt with are related, and together give a potentially useful selection of pieces of knowledge but this book is not an encyclopedia. No attempt is made to mention all possible methods, ideas, and contributions that have been published related to its topic. On the contrary, strong selection is applied as to what is relevant and what requires explanation.

Categorical Data: Categorical data are interpreted as observations of classifications, usually cross-classifications, into a finite number of possible categories. The view, represented by some authors, that categorical data arise as a result of poor observation of continuous variables, and the goal of the analysis would be to reconstruct those underlying variables, is not adopted here.

Data Analysis: Data analysis is used in the sense like data science defines itself relative to statistics. Methods of data collection that are certainly part of statistics are not dealt with. The book assumes that one has a data set, and considers ways to find a simple structure in it. The book contains no real data sets, and actual analyses are not performed. There are, however, several stylized research problems and types of data sets, and the reasons for their collection and analysis are mentioned. This is a matter of the choice of the coverage of the book and it is not to say that the way in which the data were collected would be irrelevant regarding the analysis. Most often, how the data were collected or came about is intrinsically connected to the relevant research question and also to certain quality aspects, but these are not discussed here.

Lectures on Categorical Data Analysis: The book builds heavily on my earlier volume with Springer with this title, Rudas [149]. Results in that book are often referred to by adding $LCDA$ to the reference.

Advanced Topics: The topics are advanced relative to much of *LCDA*, not in the sense that studying this book would require more background knowledge but rather in the sense that a lot from *LCDA* is used here.

The book gives a general purpose discussion of its material, not specified for a particular area of application. The presentation of the book is quite mathematical but equal importance is given to something that is part of statistics and of data science and is not part of mathematics: considerations as to how to decide what mathematical formulation is relevant for the real research problem at hand. Almost all claims are given rigorous proofs, but rigorous proofs are no substitutes for careful consideration of the applicability of the method to the given research question. The chapters in the book are somewhat different in their approaches, with varying depths of discussion, depending on whether the chapter presents an overview of widely known material, summarizes recent developments, or even has new results in it.

The readers of this book will, hopefully, learn about principles, structures, and connections more than simply the straightforward application of existing methods. The book prepares for reading research publications, perhaps also for contributing to research in theoretical statistics and designing algorithms, choosing modeling strategies, and giving more substantiated interpretations in data science.

In line with the current trends in statistics and data science, more emphasis is given to structures, simplification, and estimation of parameters than to formal testing of model fit. Traditional statistical analysis is aimed at finding some simple description of reality, and it is strongly interested in making sure (with hypothesis testing) that the simplification is not too far-reaching, but data science these days seems to be more interested in finding possibilities for interventions into processes in order to optimize them. Consequently, while the coverage envisaged for this book in the last chapter of *LCDA* is followed quite closely, ways of testing or measuring model fit were left out because of time and space limitations.

When closing the *Preface* to my book, I feel appropriate to mention other volumes on categorical data analysis that, although some of them may be seen as somewhat old, are worth studying because of their deep insight and lucid presentation: Agresti [1], Andersen [5], Bishop et al. [23], Clogg and Shihadeh [26], Fienberg [56], Goodman [69], Haberman [71], Kateri and Moustaki [78], Upton [165]. There are, of course, many more books related to categorical data analysis that may serve as useful sources of knowledge.

Budapest
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Tamás Rudas

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Chapter 1
Introduction



1.1 What Is in This Book?

The material in this book is organized into ten chapters. Chapters 2 and 3 cover graphical Markov models associated with undirected and directed graphs. The coverage in these chapters is fairly standard.

Chapters 4, 5, and 6 deal with various aspects of using Markov models associated with directed acyclic graphs (DAGs) to model causal structures. This theory, while a major intellectual breakthrough, is often used without criticism as if it could deal with any kind of causality, which does not appear to be the case. Chapter 4 postulates three simple properties that a real causal system has to have so that this theory applies to its analysis. Many of the fundamental results of the theory may be derived from these assumptions. Chapter 5 develops intuition around this type of causality by describing a simple and general model of it and also by discussing claims that may seem attractive intuitively but are not implied by the theory. Chapter 6 shows the applications and limitations of the theory. Among others, the propensity score method and the method of instrumental variables are embedded into this theory. As limitations, real causal systems where the theory does not seem to provide a relevant description are discussed.

Chapters 7, 8, 9, and 10 are primarily devoted to marginal log-linear parameters and models defined based on such parameterizations, but also to the smoothness of parameterizations of discrete distributions. One of the main motivations behind developing marginal log-linear parameterizations in Chap. 7 is that graphical Markov models associated with DAGs may be defined in a straightforward way using these parameterizations and, further, these parameters measure the magnitudes of the effects that are allowed in such a model. Chapter 8 deals with the smoothness of parameterizations of discrete distributions and the conditions which imply that models are smooth manifolds. These smoothness properties are needed for the standard asymptotic behavior of the maximum likelihood estimates and chi-squared

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related test statistics but are usually not discussed in statistics books. Here, a quite elementary introduction is given, and in Chap. 9 the smoothness of log-linear, mean-value, and mixed parameterizations, and also the smoothness of marginal log-linear parameterizations, is proved. Marginal models log-linear models are introduced in Chap. 10.

Chapter 11 deals with relational models that generalize log-linear models to sample spaces that do not have a Cartesian product structure and effects may be defined in ways not dependent on the variables, thus, these models are coordinate-free. In this generality, many of the properties seen so far do not hold. For instance, maximum likelihood estimates for Multinomial and multivariate Poisson sampling may not be proportional and some of the sufficient statistics may not be reproduced by the estimates.

The chapters are quite different as to the depth of the coverage, depending on whether standard material is presented or the material is new or, at least, presented in a somewhat novel way. Almost all results are given rigorous and mostly elementary proofs. Only minimal linear algebra and calculus are used.

The book may be used as a reference, as a guide to self-study, and also as a graduate-level textbook. The material presented in each chapter is more than what could be realistically covered in a 90- or 100-minute lecture. In addition to standard material, depending on the goal of the course, for example, the conceptual issues around causal analysis or the technical aspects of the models being smooth manifolds may be given a more detailed discussion in class. Alternatively, some classroom time may be devoted to exploring the relevant pieces of software that are only briefly referred to in the book.

1.2 Outline of the Contents of the Chapters

A bit more detailed outline of the contents of the chapters with comments regarding some of the novel aspects of the presentation is given next. References to the sources of the material, to possible alternative approaches, and to applications are deferred to the individual chapters.

Chapter 2. *Undirected Graphical Models*. Graphical log-linear models were introduced in *LCDA* and here a more detailed discussion is given. First, basic concepts related to undirected, directed, and mixed graphs are described. Then, the fundamental properties of conditional independence are proved. Conditional independence plays a central role in the entire book representing the simplification of the structure of the joint distribution of several variables. Next, the undirected Markov properties (pairwise, local, global) often associated with graphs are discussed. The implications among them and also their relationships with the Gibbs factorization, with or without the (strict) positivity assumption, are proved. Graphical Markov models defined by the assumption of one of the Markov properties associated with a graph are characterized as log-linear models, and then faithfulness is discussed. Faithfulness, intuitively, means that a distribution exhibits

those conditional independences only that are implied by a particular model. As a special feature of categorical data, the importance of using hypergraphs (and not only graphs) to describe the data structure is emphasized. Because graphical Markov models associated with undirected graphs are log-linear models, the maximum likelihood estimates have known properties, already presented in *LCDA*, and here algorithms are discussed to find the MLE. A type of faithfulness under which model search algorithms may be expected to work successfully is discussed briefly. Then, the case of mixed discrete and continuous variables is considered, and conditional Gaussian distributions are described that provide a context in which models specified by conditional independence assumptions may be defined. Finally, comments are given about the importance of interpreting graphical Markov models based on the conditional independences they stipulate, that is, based on the missing associations, rather than based on the associations they allow to exist.

Chapter 2 is quite standard. When discussing properties of conditional independence, attention is paid to a distinction between results that only hold for (strictly) positive distributions and those that do not require this assumption. The somewhat redundant language of (strict) positivity is used throughout the book to distinguish positivity from non-negativity of probability distributions.

Chapter 3. *Directed Graphical Models*. The chapter starts with introducing the so-called directed Markov properties (pairwise, local, global) that are often associated with directed acyclic graphs (DAGs). The relationships among them and also their relationships with a recursive factorization of the joint probability are proved. This recursive factorization contains terms, each of which depends only on a variable and its parents in the graph. Then, dependence separation is studied. Its equivalence with the directed global Markov property is proved. They are both related to the conditions under which the effect of a group of variables on another group of variables is blocked by keeping a third group fixed. Next, Bayesian networks are discussed. This is a name often used for directed Markov models in the context of expert systems. In expert systems, known facts are used to determine conditional probabilities of not directly or not easily observable characteristics of a complex system. This procedure is sometimes called knowledge propagation. The required calculations are largely simplified by the Markov property assumed. One of the most important algorithms of knowledge propagation, the junction tree algorithm, is given a detailed discussion. Finally, chain graph models, as a common generalization of undirected and directed graphs, are considered. Among other aspects, chain graphs are interesting because multiple, nonequivalent Markov properties were suggested to be associated with them in the literature, illustrating the fact that Markov models, in spite of their great and increasing importance, are not based on inherent properties of graphs; the graphs are only communication tools.

The material in Chap. 3 is also quite standard but the discussion of some aspects is more detailed than in most books. In particular, the concept of dependence separation and the proof of its relation with the directed global Markov property are given a rigorous treatment avoiding intuitively appealing but informal arguments. Also, the junction tree algorithm for knowledge propagation in Bayesian networks is given a precise presentation.

Chapter 4. *Causal Analysis Using DAGs: Theory*. First, a cornerstone of this theory is discussed: the fundamental difference between the simple observation of a system and the ability to manipulate or to apply intervention and then observe its response. Simple observation cannot reveal causal relationships, only the observation of the response to manipulation or intervention. Intervention may be setting variables to particular values or fixing them at particular values. Then, three assumptions are formulated that, if apply to the causal system, imply all the fundamental results of the theory. The assumptions, formulated somewhat informally, are that (1) the causal effects are described by a binary acyclic relation; (2) if the manipulation of a group of variables has an effect on a second group of variables, even if a third group of variables is kept fixed, then there must be a causal path from the first group to the second that avoids the third group; and (3) if the manipulation of a group of variables that are all nondescendant of a variable and contain all its parents has an effect on this variable, then the distribution of this variable that arises as the result of the manipulation is the same as its observable conditional distribution, when conditioned on the manipulated variables. The first assumption implies that a DAG may be used to describe the causal system; the second one says that if there is a causal effect, there is also a causal path in the DAG; and the third one is that the conditional distribution that is observable without manipulation reveals the effects of some types of manipulations. These results are then applied to the basic causal configurations of collider, chain, and fork already introduced in *LCDA*. Finally, causal discovery, that is, learning the causal structure through intervention and observation of the response to it, and also through observation without the possibility of intervention, is investigated.

The theory of causal analysis using DAGs, presented in Chap. 4, is also quite standard. However, the presentation does not rely on claims that may be seen as intuitively appealing to many, but rather explicit assumptions are made that need to hold for the real causal system for this theory to be applicable. The almost axiomatic approach is intended to encourage users to critically apply the theory, as opposed to believing that it may always be applied. Ways of learning aspects of the causal structure in cases when only observation and in cases when intervention and then observation are possible are clearly separated.

Chapter 5. *Causality and Intuition*. Depending on the kind of causality they encounter in their own fields, researchers may have very different intuitions regarding the behavior of a causal system, and this chapter aims to build intuition with respect to the kind of causality described previously. First, Berkson's paradox is discussed, which is closely connected to one of the not-so-obvious implications of the theory of causality described: conditioning on a common collider in a causal system makes the parents related if they are independent without the conditioning. The preferential selection that is assumed in Berkson's paradox is a kind of causal effect, and a general but very simple causal structure, called incremental probabilistic causation, is defined, using which the assumptions and many of their implications are illustrated. The second part of the chapter investigates the limitations of intuition. Several claims are discussed, which are often believed to be true but are all shown to be not implied by the theory of causality developed.

Berkson's paradox, a fact that is often regarded as an important building block of the causal theory presented here, is given a more precise discussion in Chap. 5 than it is usual. The very simple type of causal structure described here, incremental probabilistic causation, appears to be a good tool for associating the correct intuition with the assumptions made and properties like dependence separation. Also, by discussing claims that may be thought of as being implied by this theory of causality but, in fact, are not, some of the limitations of the theory are presented in a more straightforward way than in most publications on the topic.

Chapter 6. *Causal Analysis Using DAGs: Applications and Limitations*. Applications of the theory include an extended discussion of an example often considered in the literature. Further, the method of instrumental variables and the propensity score method to estimate causal effects from observational data already discussed in *LCDA* are derived from the theory developed. Then, Simpson's paradox, also considered in *LCDA*, is discussed from the perspective of causality. Among the limitations of the theory, real-world examples are presented, where the assumptions made do not hold, and the theory, as developed here, does not give a valid description. These examples include cases when the response to intervention also depends on the current status, when variables do not have individual causal effects but have a joint effect, cyclic causality, and causal systems with a memory, and others.

The applications of the causal theory shown in Chap. 6 to justify the method of instrumental variables, to embed the propensity score method, and to Simpson's paradox are not new and show the power of this theory. On the other hand, the list of examples given where the applicability of this theory is limited and further development would be needed, if at all possible, is usually not discussed in accounts of this theory of causality. Overall, the presentation of the DAG-based theory of causal analysis in Chaps. 4–6 is more critical than it is usual, and instead of suggesting the overall applicability of this theory, it rather encourages a careful consideration whether or not the theory is relevant for the given research problem.

Chapter 7. *Marginal Log-Linear Parameterization*. Graphical Markov models associated with undirected graphs impose conditional independences on the joint distribution and are log-linear models. Graphical Markov models associated with DAGs also impose conditional independences, but these do not necessarily involve all the variables, but rather apply to subsets of the variables, that is, to marginal distributions. Log-linear parameters cannot be used to impose such conditional independences, but log-linear parameters calculated in the relevant marginals can. The chapter develops such marginal log-linear parameters and also parameterizations using these. Marginal log-linear parameterizations are very flexible because of the possible choices of the marginals in which the parameters are determined and may be used in many statistical problems. Examples include, in addition to DAG models, repeated measurements and missing data problems.

The material in Chap. 7, although organized somewhat differently, closely follows published results in the area. It also gives a more detailed presentation than research papers introducing marginal log-linear parameters did.

Chapter 8. *Smoothness of Parameterizations of Discrete Distributions*. Smoothness is the central analytical property of statistical models, not only in the context of categorical data, that is needed for standard asymptotic behavior. In spite of this, statistics books usually only refer to general discussions of this property and do not present it in the context of their respective topics. In this chapter, a rigorous but elementary discussion of smoothness is given. Smoothness means that the distributions parameterized may be considered to be a continuous and differentiable image of an open set. In this case, the distributions together are a smooth manifold. A smooth manifold has a well-defined dimension and it determines the number of degrees of freedom associated with the model. It is proved that a model defined by considering a smooth parameterization and fixing some of the values of the parameters is smooth, and this result implies the smoothness of many important statistical models. A detailed discussion of log-linear parameters and of exponential families completes the material already presented in *LCDA*.

The material in Chap. 8 is standard but is usually not included in statistics texts. Instead, typically reference is made to the usual regularity conditions. The method of defining statistical models by considering a smooth parameterization of all distributions and then setting some of the parameters to zero is particularly relevant in the context of categorical data when the nonzero value of a parameter often allows the presence of an interaction in the joint distribution. In fact, all models in the book are defined like this, but the initial parameterizations and the resulting models are very different.

Chapter 9. *Smoothness of Marginal Log-Linear Parameterizations*. The chapter starts by discussing log-linear parameterizations and proves that the log-linear and the related mean-value and mixed parameterizations are smooth. While general results for exponential families imply these facts, in the case considered here, elementary and intuitive proofs may be given based on the structures of the partial derivative matrices associated with these parameterizations. Frequency and probability distributions are given separate but parallel considerations. Further, marginal log-linear parameterizations are also proved to be smooth, as long as the marginals in which the parameters are determined form a hierarchical and complete sequence.

None of the results in Chap. 9 are new, but they are usually derived from general considerations for parameterizations of exponential families. The proofs given here are direct and fairly elementary, and also shed light on the meaning of smoothness in the context log-linear and marginal log-linear parameterizations.

Chapter 10. *Marginal Log-Linear Models*. Marginal log-linear models are defined by selecting a marginal log-linear parameterization and setting some of the parameters equal to zero. Based on the results of the previous chapters, these are smooth models and the parameters not set to zero provide a smooth parameterization of them. Properties of the maximum likelihood estimates under marginal log-linear models for Multinomial and multivariate Poisson sampling are discussed. Standard algorithms and their modifications proposed in this context are considered to calculate the MLEs. The MLEs of the cell parameters under the two sampling

schemes are shown to be proportional. A brief overview of the software available to calculate the MLEs under marginal log-linear models is given. Finally, applications of marginal log-linear models, including models specified by arbitrary lists of conditional independences, are discussed. Specific applications include Markov models associated with DAGs, and with chain graphs, and also path models for categorical data.

Most of the existing knowledge about marginal log-linear parameterizations and the associated models is available in research-level papers, and very few publications exist with the goal of a detailed presentation and explanation. The material in Chap. 10 summarizes the main results and presents the details of many facts in their background.

Chapter 11. *Coordinate-Free Relational Models*. This chapter gives a concise account of yet another generalization of log-linear models that applies to cases when the sample space is not a full contingency table or does not have a product structure at all. Accordingly, the effects allowed by the models are not necessarily associated with groups of the variables, but rather with arbitrary subsets of the sample space. First, some of the situations when nonstandard sample spaces are relevant are reviewed. Then, relational models are defined as multiplicative representations of the cell parameters that may be probabilities when sampling is Multinomial or may be intensities when sampling is multivariate Poisson. A model is defined by selecting arbitrary subsets of the cells and associating a subset parameter with each of them. The subset parameters entering the multiplicative formulas for a cell parameter are those associated with the subsets containing this cell. The properties of these models depend heavily on whether or not such a reparameterization that has a common subset parameter in every cell is possible. If this is not possible, the MLEs have unexpected characteristics and are not proportional under the two sampling schemes. It is shown how relational models may also be defined using homogeneous or nonhomogeneous generalized odds ratios. Finally, an algorithm for the calculation of the MLEs is presented.

The material in Chap. 11 is not new but it appears to be the first attempt to summarize the relevant results that have been published in fairly recent research papers.

Chapter 2
Undirected Graphical Models



2.1 Introduction

Some graphical log-linear models were considered in *LCDA* 11.3. These models assumed that for the joint distribution of categorical variables, the local Markov property with respect to a given graph holds. It was proved that every decomposable log-linear model is graphical, and the main result was that a distribution is graphical with respect to a graph, if and only if it is Gibbs with respect to that graph. This chapter gives a more detailed discussion of graphical models. After summarizing in Sect. 2.2 the most important properties of the various types of graphs used in graphical modeling in the present and the next chapters, general properties of probabilistic conditional independence[1] are discussed in Sect. 2.3. Next, additional Markov properties are considered in Sect. 2.4 and their relationships are worked out in detail, with or without the assumption of (strict) positivity.[2]

For categorical data, all variants of graphical models imply a very strong simplification of the association structure, and although graphical models are widely studied and applied, the various types of graphical models considered do not necessarily contain a model that is an adequate description of the population distribution or of the data generating mechanism. More precisely, no graphical model may imply exactly the same conditional independences as those exhibited by the distribution to be analyzed or implied by the data-generating mechanism,

[1] There are many structures in mathematics that can be considered conditional independence, see, e.g., Bártfai and Rudas [14] and Dawid [40], and the language here emphasizes that conditional independence is considered a possible property of joint distributions of at least three variables.

[2] Although there is nothing strict or not strict about positivity (meaning that all probabilities are greater than zero), in the spoken language, "strict positivity" is often used to give emphasis to this property, as opposed to non-negativity. The language used retains this emphasis but also points to it being redundant.

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see Sect. 2.5.1. If there exists such a graphical model, the distribution is said to be faithful to it. A more general concept, faithfulness to a hypergraph is studied briefly to clarify the situation.

The iterative proportional fitting procedure (IPFP) to obtain maximum likelihood estimates under log-linear models was already considered in Section *LCDA* 12.2. Here, in addition, the Darroch-Ratcliff algorithm proposed in [38] and a heuristic algorithm dual to the IPFP, proposed by Rudas [146], are discussed in Sect. 2.5.2. For undirected graphical log-linear models that are decomposable, a closed-form MLE exists, see Theorem *LCDA* 12.4.

Conditional Gaussian (CG) distributions, as means of defining graphical models when some of the variables are discrete, and some are continuous, are introduced in Sect. 2.6, and aspects of the interpretation of undirected graphical models are discussed in Sect. 2.7.

There are many technical and substantive reasons why graphical models are used so widely. It has to be pointed out, however, that graphs do not have any inherent probabilistic or statistical meanings or implications. Rather, an important body of research associated graphs with probabilistic structures, and these are being applied in the statistical analysis of data. More precisely, graphical models associate various sets of (conditional or marginal) independence properties of multivariate distributions with graphs, and graphical modeling utilizes these structures as statistical models. The graphs do not only illustrate the assumed structure, but are also useful in communicating them and in developing intuition about their meanings in a theoretical or applied setting.

Important references for graphical modeling are Lauritzen [98] and Maathuis et al. [108] and for categorical data in particular Roverato [137].

2.2 Some Graph Theory

This section covers the basic concepts of graph theory for undirected and directed graphs. The material for directed graphs will only be used in the next chapter. The models to be based on graphs, discussed in this and the next chapter, intend to use undirected graphs to model associations (or, rather, lack of associations) and directed graphs to model effects (or the lack thereof). For a discussion of effects and associations, in particular for categorical data, see Chapters *LCDA* 6 and 8 and Rudas [148]. Mixed graphs are used in attempts to model more complex situations, either as to the kinds of associations or effects to be taken into account or as to the levels of measurements[3] of the variables to be analyzed.

[3] See Section *LCDA* 1.1.

2.2.1 Undirected Graphs

A graph G is a pair (V, E), where V is a finite set, the set of nodes, also called vertices,[4] and E is a set of (unordered) pairs of distinct nodes. Pairs of nodes listed in E are connected with an edge, and E is called the set of edges of the graph. When E is empty, the graph is called an empty graph, and when E contains all pairs of the vertices, the graph is called complete. The neighbors of a node V_i are the nodes, to which V_i is connected by an edge:

$$nb(V_i) = \{V_j \in V : \{V_i, V_j\} \in E\}.$$

Two nodes,[5] V_1 and V_j, are connected by a path, if there exist nodes $V_2, V_3, \ldots, V_{j-2}, V_{j-1}$, such that,

$$\{V_1, V_2\} \in E,\ \{V_2, V_3\} \in E,\ \ldots \{V_{j-2}, V_{j-1}\} \in E,\ \{V_{j-1}, V_j\} \in E.$$

If any two nodes in V are connected by a path, then G is said to be connected. It is easy to see that being connected by a path, as applied to two nodes, is an equivalence relation, and the implied partition defines the connected components of the graph.

If $U \subseteq V$ is a subset of the nodes, then $G_1 = (U, E \cap (U \times U))$ is a subgraph of G. If $G = (V, E)$ is a graph, then $\overline{G} = (V, \overline{E})$, where

$$\overline{E} = \{\{V_1, V_2\} : V_1, V_2 \in V, \{V_1, V_2\} \notin E\}$$

is called the complement of G.

If T, U, W are pairwise disjoint subsets of V and for every $V_1 \in T$ and $V_3 \in W$, any path connecting them contains a node in U, then U is said to separate T and W. Note that if T and W are (subsets of) different connected components of G, that is, there is no path connecting them, then the definition implies that the empty set separates them.

A subgraph of G, say G_1, that is complete and is also maximal complete, is called a clique of the graph G. Maximal here means that there does not exist another subgraph of G, say G_2, such that G_2 is complete and G_1 is a subgraph of G_2.

In addition to the simple concepts above, hypergraphs will also be needed to give a precise description of some of the limitations of graphical models in the case of categorical data and to account more precisely for relevant properties of the joint distribution. A hypergraph $\mathscr{G} = (V, \mathscr{E})$ consists of the finite set V of nodes and the set of hyperedges $\mathscr{E}$. Every hyperedge is a subset of V and these subsets are assumed to be incomparable with respect to inclusion. Here, the generalization of the concept of graph is that instead of edges containing two nodes, arbitrary subsets

[4] The vertices of the graphs considered will be identified with the variables of interest; therefore, the notation V is used.

[5] Perhaps with relabeling of the nodes.

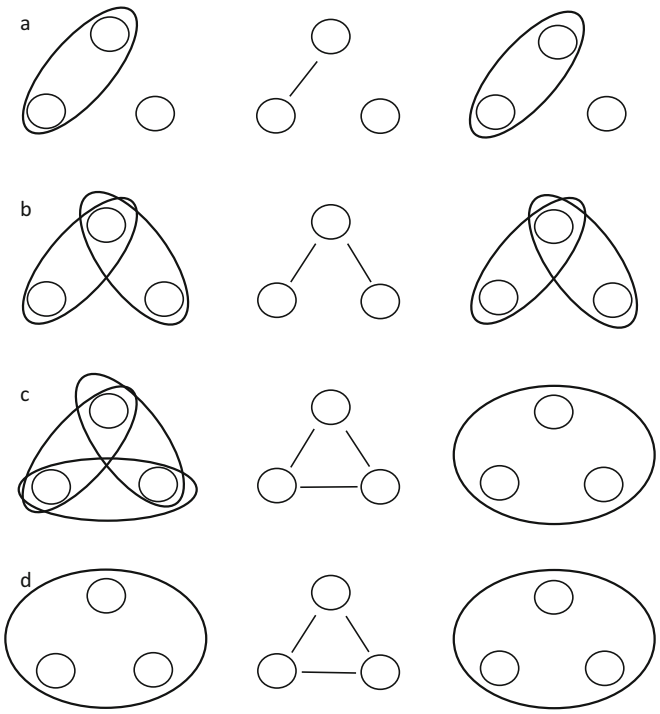


Fig. 2.1 Hypergraphs **a**, **b**, and **d**, are graphical, **c** is not. The figure shows hypergraphs with their respective 2-sections next to them and the clique hypergraphs obtained

of the nodes are considered as hyperedges. Hypergraphs are relevant, because the generating class of a log-linear model may be identified with a hypergraph: the nodes are the variables, and the hyperedges are the maximal interactions.

The hypergraph $\mathscr{G}$ generates a graph $\mathscr{G}_2 = (V, E)$, which is called its 2-section. In $\mathscr{G}_2$, two nodes are connected by an edge, if and only if there is a hyperedge in $\mathscr{E}$, which contains both nodes. There is also an operation that generates a hypergraph $G_c = (V, \mathscr{E})$ from a graph $G = (V, E)$. The nodes are the same, and the hyperedges in $\mathscr{E}$ are the maximal complete subgraphs of G. The hypergraph G_c is called the clique hypergraph of G.

For some hypergraphs, $(\mathscr{G}_2)_c$ is equal to $\mathscr{G}$, but for some others, the equality does not hold. If the equality holds, the hypergraph is called graphical. A few simple cases are illustrated in Fig. 2.1.

2.2.2 Directed Graphs

A directed graph is a pair (V, A), where V is a finite set and A is a set of ordered pairs of distinct elements from V. An ordered pair $(V_i, V_j) \in A, i \neq j$, is interpreted

as an arrow pointing from V_i to V_j. In this case, V_i is called a parent of V_j, and V_j is a child of V_i. The set of all parents of V_j is denoted as $pa(V_j)$.

A directed path starting at V_i and ending at V_j (or from V_i to V_j) is a sequence of nodes[6] $V_i, V_{i+1}, V_{i+2}, \ldots, V_{j-2}, V_{j-1}, V_j$, such that

$$(V_i, V_{i+1}) \in A,\ (V_{i+1}, V_{i+2}) \in A,\ \ldots (V_{j-2}, V_{j-1}) \in A,\ (V_{j-1}, V_j) \in A.$$

If a directed path leads from V_i to V_j, then V_j is called a descendant of V_i. The set of all descendants of V_i is denoted as $de(V_i)$. If there is no directed path from V_i to V_j, then V_j is a nondescendant of V_i. The set of all nondescendants of V_i is denoted as $nd(V_i)$. On the other hand, if a directed path leads from V_i to V_j, then V_i is called an ancestor of V_j. The set of all such ancestors of V_j is denoted as $an(V_j)$. A set of nodes is called ancestral, if it contains all ancestors of its elements.

The following is easy to see.

Proposition 2.1 *If two sets of nodes are ancestral, then so is their intersection.*

□

This then implies immediately the following result.

Proposition 2.2 *For a set of nodes, there is a smallest set that contains it and is ancestral.*

□

The smallest ancestral set containing a set of nodes U is denoted as $anc(U)$.

If G is a directed graph, its undirected moralization, G^{UM}, is an undirected graph obtained by first linking with and edge any pairs of nodes that have a common child and then replacing arrows with edges.

A directed cycle is a directed path ending in its starting point. From among the directed graphs, those that do not have any directed cycles play an important role in graphical modeling. Such graphs are called directed acyclic graphs (DAGs).

An important property of DAGs is that there always exists an ordering of the nodes, such that all arrows point in the same direction. More precisely, the following result (see Lauritzen et al. [99]), holds.

Theorem 2.1 *For every DAG $G = (V, A)$, there exists an ordering of the nodes V, such that for all $V_i \in V$,*

(i) all nodes in $pa(V_i)$ are before V_i and
(ii) all nodes in $de(V_i)$ are after V_i.

Proof A DAG always has a node that has no parents. To see this, choose an arbitrary node, then choose one of its parents, and then choose one of the parents of the last one. Continue this procedure. It either stops because one of the nodes selected does not have a parent, or because V is finite it arrives at a node already selected. But the

[6] Perhaps with some relabeling.

latter option is not possible, because in that case, the selected nodes would form a directed cycle (in the reverse order).

In the ordering to be defined, select a node in V that has no parent in G as the first node, say V_1. All other nodes will be after V_1 in the ordering, including its descendants. Then remove this node from V to obtain $V(1)$ and then consider the subgraph of G, say $G(1)$, defined by $V(1)$. This is obtained by omitting the selected node V_1 from V and by omitting all the arrows from A that contain V_1. Note that all such arrows start at V_1.

Every subgraph of a DAG is also a DAG. This is obvious, as creating a subgraph involves leaving out nodes and arrows and cannot create a directed cycle. Thus, $G(1)$ is also a DAG.

$G(1)$ also has a node with no parents in it, say $V_2 \in V(1)$. Choose this as the second one in the order to be created. All remaining nodes will be after V_2 in the order. The node V_2 does not have a parent among them, so if it has a parent in G, it is not after it. Further, V_1 is not a descendant of V_2, otherwise V_1 would have a parent in G, but V_1 does not have parents. Therefore, all descendants of V_2 are after it.

Remove V_2 from $V(1)$ to obtain $V(2)$ and consider the corresponding subgraph $G(2)$. Let V_3 be a node in $V(2)$ that has no parent in $G(2)$. The node V_2 is not a descendant of V_3, as V_2 does not have a parent in $G(1)$ and V_1 is also not a descendant of V_3 in G, as V_1 does not have parents. Thus, all the descendants of V_3 are among $V(2) \setminus \{V_3\}$, and thus they come later in the order, and as V_3 has no parent in $G(2)$, all its parents must be in $V \setminus V(2)$, that is, before it.

This procedure can be continued to include all nodes and provides the required ordering.

□

Such an ordering is called a well-numbering of the nodes.

In a graph, having an edge or not having an edge between two nodes is a symmetric relationship. In DAGs, having an arrow between two nodes is asymmetric. That is, if $(V_1, V_2) \in A$, then $(V_2, V_1) \notin A$, because otherwise V_1 and V_2 would be a directed cycle. In the theory of graphical models, further variants of (partially) directed graphs are also discussed. Some of these are presented next.

2.2.3 Mixed Graphs

One type of mixed graphs used in statistical modeling is called a chain graph. To define a chain graph $G = (V, C)$, consider a partitioning of the set of nodes into components $\mathscr{K}_1, \mathscr{K}_2, \ldots \mathscr{K}_k$. Within each component $\mathscr{K}_i$, an undirected and connected graph is defined with the edge set E_i and C contains these edge sets. Further, there are arrows in C, which run between nodes in different components, with the restriction that between components $\mathscr{K}_i$ and $\mathscr{K}_j$, all arrows point in the same direction: either all point from nodes in $\mathscr{K}_i$ to nodes in $\mathscr{K}_j$ or all point from nodes in $\mathscr{K}_j$ to nodes in $\mathscr{K}_i$. Let the set of these arrows be A, and then

$$C = \{E_1, E_2, \ldots, E_k, A\}.$$

A final condition is that the directed graph $G_d = (M, A_d)$ with nodes $M = \{\mathcal{K}_1, \mathcal{K}_2, \ldots \mathcal{K}_k\}$ and an arrow in A_d from $\mathcal{K}_i$ to $\mathcal{K}_j$ if and only if there was an arrow from any $V_i \in \mathcal{K}_i$ to any $V_j \in \mathcal{K}_j$ has to be acyclic.

As G_d is a DAG, the concepts of parent, descendant, and nondescendant can be extended to apply to the components, and Theorem 2.1 implies that there is a numbering of the components $\mathcal{K}_1, \mathcal{K}_2, \ldots, \mathcal{K}_k$, such that components in $pa(\mathcal{K}_i)$ precede $\mathcal{K}_i$ and the components in $de(\mathcal{K}_i)$ follow it.

Thus, chain graphs generalize undirected graphs and DAGs. If there is one component only, the chain graph is an undirected graph. If all components consist of a single node, the chain graph is a DAG. The intuition to be associated with chain graphs is that they represent situations, when there may be effects between certain groups of variables, and no direct effects between others, and within these groups, there may be associations or rather no direct associations present.

A more general type of mixed graph may contain directed arrows, bidirected arrows, and undirected edges between nodes. Between two nodes, there may be an arrow or a bidirected arrow, but not two arrows with the same or different directions, see Evans and Richardson [53] and Richardson [138]. Nodes connected by an edge are called neighbors and if they are connected by a double headed arrow, they are called spouses. The concepts of parent and child are used just like in the case of DAGs.

In this case, a directed cycle is defined as a cycle going through arrows in their direction (but not through bidirected arrows), and acyclic means that such directed cycles do not exist, and such graphs are called acyclic directed mixed graphs or ADMGs. If the ADMG has no bidirected edges, it is a DAG, so ADMGs can be seen as generalizing DAGs in a way different from how chain graphs do.

In a mixed graph of this type, a vertex V_1 is said to be anterior to a vertex V_2 if there is a path from V_1 to V_2, which is semi-directed in the sense that it contains (undirected) edges or arrows pointing away from V_1. For a subset of the nodes $W \subseteq V$, the set of its anteriors, $ant(W)$, contains the nodes that are anterior to any of the nodes in W. A mixed graph is called ancestral if for any node V, the following two properties hold. First, V is not anterior to any of its parents or spouses. Second, if V does have at least one neighbor, then it has neither parents nor spouses, see Richardson and Spirtes [139]. Note that the ancestral concept defined for mixed graphs is different from the ancestral concept for DAGs. It can be seen (see Richardson and Spirtes [139]) that undirected graphs and DAGs are ancestral graphs, thus, the class of ancestral graphs is a common generalization.

The intuition to be associated with ADMGs is that while arrows represent some kind of conditional effects, bidirected arrows represent marginal associations. The latter can be particularly useful when modeling latent variables but this is not going to be pursued in this book.

Another complex type of graph, called a summary graph, was considered in Cox and Wermuth [34] with solid edges, dashed edges, and arrows. These, just like the previous types of graphs, are relevant under conditioning or marginalization, see Wermuth [166].

A final type of more complex graphs does not have additional types of edges, but the nodes are marked, as some represent categorical, other continuous variables and will be used in Sect. 2.5.

To close the overview of the most important types of graphs used in graphical modeling, it has to be emphasized that graphs do not have inherent relevance for the statistical analysis, but only after appropriate distributional assumptions have been made. The graphs are, however, useful in certain manipulations of the associated probability distribution and also are useful in developing intuition. However, the intuitive language does not necessarily give a correct interpretation of the related models. See Sect. 2.7 for comments on how graphs are useful in interpreting graphical models.

2.3 Conditional Independence

To take the possibility of zero cell probabilities into account, the conditional independence of variables X and Y, given Z, under the distribution P is defined as

$$P(X = x, Y = y|Z = z) = P(X = x|Z = z)P(Y = y|Z = z), \text{ if } P(Z = z) > 0. \tag{2.1}$$

When $P(Z = z) = 0$, the probabilities in (2.1) are undefined but also not restricted by the definition. If any of the conditional probabilities on the right-hand side of (2.1) is zero, then so is the conditional probability on the left-hand side, and (2.1) holds, irrespective of the other conditional probability on the right-hand side.

An important equivalent formulation of (2.1) is that

$$P(X = x|Y = y, Z = z) = P(X = x|Z = z), \text{ if } P(Y = y, Z = z) > 0. \tag{2.2}$$

Note that the condition given in (2.2) implies that not only $P(Y = y, Z = z) > 0$ but also $P(Z = z) > 0$. In fact, one has

Proposition 2.3 *The definitions in (2.1) and in (2.2) are equivalent.*

Proof If (2.1) holds, then

$$\begin{aligned} P(X = x, Y = y, Z = z) &= P(X = x|Z = z)P(Y = y|Z = z)P(Z = z) \\ &= P(X = x|Z = z)P(Y = y, Z = z), \text{ if } P(Z = z) > 0. \end{aligned}$$

Suppose that $P(Y = y, Z = z) > 0$, otherwise (2.2) is nil. Then also

$$P(Z = z) = \sum_y P(Y = y, Z = z) > 0,$$

and from the previous formula, one obtains that

$$P(X = x|Y = y, Z = z) = \frac{P(X = x, Y = y, Z = z)}{P(Y = y, Z = z)} = P(X = x|Z = z).$$

Conversely, if (2.2) holds, one needs to see (2.1) for every z, for which $P(Z = z) > 0$ holds. Consider one such z. Then, there exist such categories y' of Y, for which $P(Y = y', Z = z) > 0$, and one has for all such y' that

$$P(X = x, Y = y', Z = z) = P(X = x|Z = z)P(Y = y', Z = z),$$

and thus

$$\begin{aligned} P(X = x, Y = y'|Z = z) &= \frac{P(X = x|Z = z)P(Y = y', Z = z)}{P(Z = z)} \\ &= P(X = x|Z = z)P(Y = y'|Z = z). \end{aligned}$$

If there are categories $y"$, such that $P(Y = y", Z = z) = 0$ (but $P(Z = z) > 0$), the second term on the right-hand side of (2.1) exists and is zero, and also the left-hand side is zero, as

$$0 \leq P(X = x, Y = y"|Z = z) \leq P(Y = y"|Z = z) = 0.$$

□

The above proposition is remarkable, as the definition (2.1) appears to be symmetric in X and Y, while (2.2) does not appear so. Further, the conditions of the two definitions are different. Definition (2.2) is often interpreted as saying that if Z is already known to be equal to z, learning that $Y = y$ in addition, does not provide further information with respect to the distribution of X. This interpretation is often used with graphical models that assume conditional independences. The concept of conditional independence can also be given other non-probabilistic interpretations, where this property is also relevant, see Dawid [40]. In the setting of this book, X, Y, and Z are categorical variables and they have a joint distribution; otherwise, the probabilities in the definitions (2.1) and (2.2) would not exist.

The following proposition is implied immediately by (2.1).

Proposition 2.4

$$X \perp\!\!\!\perp Y|Z \text{ if and only if } Y \perp\!\!\!\perp X|Z.$$

□

Let T be a categorical variable that is a function f of another categorical variable Y, in the sense that if y is known, so is t. Given the level of measurement of the variables involved here, assuming that the domain of T is the entire range of Y, T, if it is a function of Y, may only be a coarsening of Y. That is, T is obtained by considering a partitioning of the categories of Y and combining the categories in

subsets to form the categories of T. This can be relevant in data collection, when a less precise (but hopefully more reliable) measurement is needed. Note that a partition may contain all categories of Y separately, and in that case, $T = Y$ or T associates different labels with the categories of Y. But coarsening also occurs in a different context, which is more relevant for this book. Marginalization is also coarsening, when instead of the of the joint distribution of, say, AB, only the marginal distribution of B is considered. In general, for every category t, there is a subset of the categories y, such that $t = f(y)$ holds, and these subsets for different categories t of T are disjoint. Note that a category of T may have the same label as a category of Y.

Proposition 2.5 *If T is a function of Y, then*

$$X \perp\!\!\!\perp Y|Z \text{ implies } X \perp\!\!\!\perp T|Z.$$

Proof By Proposition 2.4, it is sufficient to show that if (2.2) holds, then also

$$P(X = x|T = t, Z = z) = P(X = x|Z = z), \text{ if } P(T = t, Z = z) > 0,$$

for any categorical variable $T = f(Y)$. Consider fixed t and z categories, for which the positivity condition above holds. Then, as

$$P(T = t, Z = z) = \sum_{y:f(y)=t} P(Y = y, Z = z),$$

there are categories y such that $f(y) = t$ and $P(Y = y, Z = z) > 0$. For all these y and z categories, (2.2) holds by assumption, and, therefore, the conditional probability $P(X = x|Y = y, Z = z)$ does not depend on y and is equal to the common value $P(X = x|Z = z)$. Then, the same holds for any collection of y categories combined, including those that make up t, as when the positivity condition for t, y, and z holds:

$$P(X = x|T = t, Z = z) = \frac{P(X = x, T = t, Z = z)}{P(T = t, Z = z)}$$

$$= \frac{\sum_{y:f(y)=t} P(X = x, Y = y, Z = z)}{\sum_{y:f(y)=t} P(Y = y, Z = z)}$$

$$= \frac{\sum_{y:f(y)=t} P(X = x|Y = y, Z = z)P(Y = y, Z = z)}{\sum_{y:f(y)=t} P(Y = y, Z = z)}$$

$$= \frac{\sum_{y:f(y)=t} P(X = x|Z = z)P(Y = y, Z = z)}{\sum_{y:f(y)=t} P(Y = y, Z = z)}$$

$$= \frac{P(X = x|Z = z) \sum_{y:f(y)=t} P(Y = y, Z = z)}{\sum_{y:f(y)=t} P(Y = y, Z = z)}$$

$$= P(X = x|Z = z),$$

as required.

□

The next property shows that including in the condition information about one of the variables preserves the conditional independence.

Proposition 2.6 *If T is a function of Y, then*

$$X \perp\!\!\!\perp Y|Z \text{ implies } X \perp\!\!\!\perp Y|(Z, T).$$

Proof Consider a pair (z, t), such that $P(Z = z, T = t) > 0$. In this case, $P(Z = z) > 0$, and by assumption, (2.1) holds. Then, (2.1) holds if Y is restricted to those categories, for which $f(y) = t$.

□

The previous proof argues that if there are more conditions, the conditional independence statement is only applied to a subset of the original condition, because conditioning is selection.

The last property to be discussed here, which holds for conditional independence without the assumption of (strict) positivity, is the following.

Proposition 2.7 *If*

$$X \perp\!\!\!\perp Y|Z \text{ and } X \perp\!\!\!\perp W|(Y, Z),$$

then

$$X \perp\!\!\!\perp (Y, W)|Z.$$

Proof Using the definition in (2.2), one has to see that for all (y, w, z), such that $P(Y = y, W = w, Z = z) > 0$, one has that

$$P(X = x|Y = y, W = w, Z = Z) = P(X = x|Z = z),$$

whenever the conditions of the proposition hold. The second condition implies that if $P(Y = y, W = w, Z = z) > 0$, then

$$P(X = x|Y = y, W = w, Z = Z) = P(X = x|Y = y, Z = Z).$$

In these cases, also $P(Y = y, Z = z) > 0$, and the first condition implies that

$$P(X = x|Y = y, Z = Z) = P(X = x|Z = z),$$

and these combined yield the required result.

□

The final general property of conditional independence to be considered here holds under (strict) positivity.

Proposition 2.8 *If the joint probability distribution of the variables X, Y, Z, W is (strictly) positive, then*

$$X \perp\!\!\!\perp Y|(Z, W) \text{ and } X \perp\!\!\!\perp Z|(Y, W),$$

imply

$$X \perp\!\!\!\perp (Y, Z)|W.$$

Proof Under positivity, for any category combination, definition (2.2) implies that if the assumptions hold, then

$$\begin{aligned} &P(X = x|Y = y, Z = z, W = w) \\ &= P(X = x|Z = z, W = w) = P(X = x|Y = y, W = w). \end{aligned} \tag{2.3}$$

This means that

$$P(X = x|Y = y_1, Z = z, W = w) = P(X = x|Y = y_2, Z = z, W = w),$$

because they are both equal to $P(X = x|Z = z, W = w)$, and, similarly, that

$$P(X = x|Y = y, Z = z_1, W = w) = P(X = x|Y = y, Z = z_2, W = w),$$

because they are both equal to $P(X = x|Y = y, W = w)$. Then,

$$\begin{aligned} P(X = x|Y = y_1, Z = z_1, W = w) &= P(X = x|Y = y_2, Z = z_1, W = w) \\ &= P(X = x|Y = y_2, Z = z_2, W = w), \end{aligned}$$

implying that $P(X = x|Y = y, Z = z, W = w)$ does not depend on neither y nor z. Let the common value be denoted as $u(x, z)$. With respect to definition (2.2), it is enough to show that

$$u(x, z) = P(X = x|W = w).$$

Indeed,

$$P(X = x|W = w) = \frac{P(X = x, W = w)}{P(W = w)}$$

$$= \frac{\sum_{y,z} P(X = x, Y = y, Z = z, W = W)}{\sum_{y,z} P(Y = y, Z = z, W = w)}$$

$$= \frac{\sum_{y,z} P(X = x|Y = y, Z = z, W = w)P(Y = y, Z = z, W = w)}{\sum_{y,z} P(Y = y, Z = z, W = w)}$$

$$= \frac{u(x, w) \sum_{y,z} P(Y = y, Z = z, W = w)}{\sum_{y,z} P(Y = y, Z = z, W = w)} = u(x.w).$$

□

Note that if positivity is not assumed, it may occur that the first expression in (2.3) does not exist, but one or two of the second and third ones do, and the equalities would not be implied for all category combinations by the conditions of the proposition.

Structures, including joint distributions of categorical variables but many other ones, with properties similar to the ones formulated in Propositions 2.4, 2.5, 2.6, and 2.7, are called semigraphoids. If also Proposition 2.8 holds, the structure is called a graphoid (Pearl and Paz [133]).

2.4 Undirected Markov Properties

There are three Markov properties that are usually considered in the theory of undirected graphical models: the pairwise, the local, and the global Markov properties. The second one of these was discussed in Section $LCDA$11.3, in the context of (strictly) positive probability distributions. All Markov properties refer to the joint distribution of variables $V = \{X_1, X_2, \ldots, X_v\}$, and they state conditional independences that are associated with a graph $G = (V, E)$, where the nodes are identified with the variables.

The pairwise Markov property is that

$$\text{If } \{X_i, X_j\} \notin E, \text{ then } X_i \perp\!\!\!\perp X_j | V \setminus \{X_i, X_j\}, \text{ for all } X_i, X_j \in V.$$

The local Markov property is that

$$X_i \perp\!\!\!\perp V \setminus nb(X_i) \setminus \{X_i\} | nb(X_i), \text{ for all } X_i \in V.$$

The global Markov property is that for pairwise disjoint subsets $\mathscr{A}, \mathscr{B}, \mathscr{C}$ of V,

$$\text{If } \mathscr{C} \text{ separates } \mathscr{A} \text{ and } \mathscr{B}, \text{ then } \{X_i : X_i \in \mathscr{A}\} \perp\!\!\!\perp \{X_i : X_i \in \mathscr{B}\} | \{X_i : X_i \in \mathscr{C}\},$$

or written in a more compact way,

$$\text{If } \mathscr{C} \text{ separates } \mathscr{A} \text{ and } \mathscr{B}, \text{ then } \mathscr{A} \perp\!\!\!\perp \mathscr{B} | \mathscr{C}.$$

The three Markov properties under the assumption of (strictly) positive probability distributions are equivalent, but if positivity is not assumed, the relationship is more complex. The following is seen easily.

Theorem 2.2 *For any probability distribution and undirected graph, if the global Markov property holds, then the local and the pairwise Markov properties hold as well.*

□

To see that the local Markov property implies the pairwise Markov property, some of the properties presented in the previous section are needed.

Theorem 2.3 *For any probability distribution and undirected graph, if the local Markov property holds, then the pairwise Markov property holds as well.*

Proof To simplify presentation, let $V = \{X_1, \ldots, X_7\}$, and let $nb(X_1) = \{X_2, X_3, X_4\}$, thus the non-neighbors of X_1 are X_5, X_6, and X_7. Then, the local Markov property for X_1 implies that

$$X_1 \perp\!\!\!\perp (X_5, X_6, X_7) | (X_2, X_3, X_4).$$

Then, let $Y = (X_5, X_6, X_7)$ and let $T(Y) = (X_6, X_7)$. Apply Proposition 2.6 to obtain that

$$X_1 \perp\!\!\!\perp (X_5, X_6, X_7) | (X_2, X_3, X_4, X_6, X_7).$$

Finally, apply Proposition 2.5 to obtain that

$$X_1 \perp\!\!\!\perp X_5 | (X_2, X_3, X_4, X_6, X_7),$$

which is the pairwise Markov property for X_1 and X_5. A similar argument yields the conditional independences involving X_6 and X_7.

□

Under (strict) positivity, the pairwise Markov property implies the local Markov property.

Theorem 2.4 *Assume that for a given graph $G = (V, E)$, the pairwise Markov property holds for a (strictly) positive probability distribution P on V. Then, the local Markov property also holds for G and P.*

Proof To simplify presentation, let $V = \{X_1, \ldots, X_7\}$, and let $nb(X_1) = \{X_2, X_3, X_4\}$. Then, by the pairwise Markov property

$$X_1 \perp\!\!\!\perp X_5 | (X_2, X_3, X_4, X_6, X_7), \text{ and } X_1 \perp\!\!\!\perp X_6 | (X_2, X_3, X_4, X_5, X_7).$$

Then, apply Proposition 2.8 with $X = X_1$, $Y = X_5$, $Z = X_6$, $W = (X_2, X_3, X_4, X_7)$ to obtain that

$$X_1 \perp\!\!\!\perp (X_5, X_6) | (X_2, X_3, X_4, X_7).$$

The pairwise Markov property also implies that

$$X_1 \perp\!\!\!\perp X_7 | (X_2, X_3, X_4, X_5, X_6).$$

Finally, Proposition 2.8 may be applied again, using the last two conditional independences, with $X = X_1$, $Y = (X_5, X_6)$, $Z = X_7$, and $W = (X_2, X_3, X_4)$ to obtain that

$$X_1 \perp\!\!\!\perp (X_5, X_6, X_7) | (X_2, X_3, X_4),$$

which is the local Markov property for X_1 in G.

□

The previous result is not necessarily true without the assumption of (strict) positivity. To illustrate this, let A be a binary variable with a uniform distribution, and let $B = A$ and $C = A$. This implies that the three variables have a joint distribution, in which only cells with the three indices being equal have positive probabilities. Consider the graph $G = (V, E) = (\{A, B, C\}, \{\{A, B\}\})$, shown in Fig. 2.2. The pairwise Markov property for this graph implies that

$$A \perp\!\!\!\perp C | B \text{ and } B \perp\!\!\!\perp C | A.$$

These hold true, as the only positive probabilities are those where all three indices are the same, thus, for the first conditional independence,

$$P(A = 0 | B = 0) = 1, \quad P(C = 0 | B = 0) = 1, \quad P(A = 0, C = 0 | B = 0) = 1,$$

$$P(A = 1 | B = 0) = 0, \quad P(C = 0 | B = 0) = 1, \quad P(A = 1, C = 0 | B = 0) = 0,$$

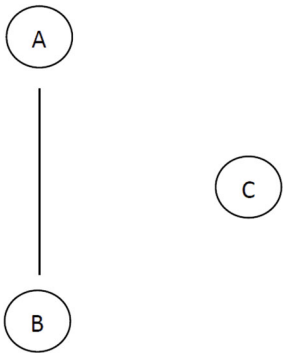


Fig. 2.2 If $A = B = C$, the pairwise Markov property for this graph holds, but the local does not

$$P(A=0|B=0)=1,\ P(C=1|B=0)=0,\ P(A=0,C=1|B=0)=0,$$

$$P(A=1|B=0)=0,\ P(C=1|B=0)=0,\ P(A=1,C=1|B=0)=0,$$

$$P(A=0|B=1)=0,\ P(C=0|B=1)=0,\ P(A=0,C=0|B=1)=0,$$

$$P(A=1|B=1)=1,\ P(C=0|B=1)=0,\ P(A=1,C=0|B=1)=0,$$

$$P(A=0|B=1)=0,\ P(C=1|B=0)=0,\ P(A=0,C=1|B=1)=0,$$

$$P(A=1|B=1)=1,\ P(C=1|B=1)=1,\ P(A=1,C=1|B=1)=1,$$

and similarly for the second one.

On the other hand, the local Markov property in this case would imply, as $nb(C)=\emptyset$, that

$$C \perp\!\!\!\perp (A,B),$$

which is clearly not true. For instance, $P(C=1)=0.5$, $P(A=0,B=0)=0.5$ but $P(A=0,B=0,C=1)=0$.

In the example above, the assumption that $A=B=C$ introduced second-order interaction, which a graph cannot capture; rather, a hypergraph would be needed for proper modeling of the distribution, see Sect. 2.5.

To complete the description of the relationships among the Markov properties, the last result shows that under the assumption of (strict) positivity, the other two Markov properties imply the global one.

Theorem 2.5 *Let $G=(V,E)$ be a graph and P a (strictly) positive probability distribution on the vertices. Then, if the pairwise or the local Markov property holds for P, then so does the global Markov property.*

Proof Under positivity, Theorem 2.4 says that the pairwise Markov property implies the local Markov property. Theorem *LCDA* 11.6 is that if the local Markov property holds, then P is also Gibbs with respect to G. This means that the maximal interactions in a log-linear representation of P are the cliques of G. More precisely, let $\mathscr{D}$ contain all subsets $V_i \subseteq V$, for which the subgraph $G_i=(V_i,E_i)$ is complete. Then $\mathscr{D}$ is a descending class of subsets of V. The distribution P is Gibbs with respect to the graph G if for any joint category v of the variables V,

$$P(v)=\prod_{W\in\mathscr{D}} \beta_v^W((v)_W). \tag{2.4}$$

The parameters β_w^W were defined in Section *LCDA* 10.3 and are the so-called balanced multiplicative parameters. The equality in (2.4) remains true, if the parameters are replaced by the corner multiplicative parameters γ_w^W defined in Section *LCDA* 11.3, see Theorem *LCDA* 11.5. In either case, the parameters

associated with a subset of the variables W for a cell (combination of indices of the variables in V) v depend only on the indices of the variables which are in W, denoted as $(v)_W$.

The condition that the global Markov property holds is that if the subsets of variables W_1, W_2, W_3 are such that every path from any variable in W_1 to any variable in W_3 contains a variable from W_2, then

$$W_1 \perp\!\!\!\perp W_3 | W_2,$$

which will now be shown to hold as a consequence of the Gibbs factorization.

If every path from a variable in W_1 to a variable in W_3 contains a variable from W_2, then no complete subgraph W contains nodes from both W_1 and W_3, because otherwise there would be a path (in fact an edge) connecting these variables in W_1 and in W_3, which does not contain a variable from W_2.

Set $W_4 = V \setminus W_1 \setminus W_2 \setminus W_3$. Out of the variables in W_4, let W_{4i} denote those that are connected by such a path to a variable in W_i, $i = 1, 3$ that does not go through W_2. Clearly, $W_{41} \cap W_{43} = \emptyset$, otherwise there would be a path between W_1 and W_3, which avoids W_2.

Set $W_{40} = W_4 \setminus W_{41} \setminus W_{43}$. Then $W_4 = W_{41} \cup W_{43} \cup W_{40}$ and this is a disjoint union.

Thus, for a complete subgraph W, exactly one of the following holds:

$$W \subseteq W_{40}$$

or

$$W \subseteq W_1 \cup W_2 \cup W_{41}$$

or

$$W \subseteq W_3 \cup W_2 \cup W_{43};$$

otherwise, the complete subgraph would yield a path which cannot exist.

Therefore, $P(v)$ in (2.4) may be rewritten as the product of the following terms:

$$\begin{aligned} P(v) = & \prod_{W \in \mathscr{D}, W \subseteq V_{40}} \beta_v^W((v)_W) \\ & \times \prod_{W \in \mathscr{D}, W \subseteq W_1 \cup W_2 \cup W_{41}} \beta_v^W((v)_W) \\ & \times \prod_{W \in \mathscr{D}, W \subseteq W_3 \cup W_2 \cup W_{43}} \beta_v^W((v)_W), \end{aligned} \tag{2.5}$$

If now i, j, k, l denote combined indices of the combined variables in W_{40}, W_2, $W_1 \cup W_{41}$, $W_3 \cup W_{43}$, respectively, then (2.5) may be rewritten as

$$P(v) = \gamma_i \delta_{jk} \kappa_{jl},$$

and the marginal distribution of $V \setminus W_{40}$ is

$$P((v)_{jkl}) = \delta_{jk} \kappa_{jl}. \tag{2.6}$$

Then, by Theorem *LCDA* 10.10, this means that

$$(W_1 \cup W_{41}) \perp\!\!\!\perp (W_3 \cup W_{43}) | W_2.$$

Finally, as the variables in W_1 are a function of the variables in $W_1 \cup W_{41}$, and the variables in W_3 are a function of the variables in $W_3 \cup W_{43}$, Proposition 2.5 implies that

$$W_1 \perp\!\!\!\perp W_3 | W_2,$$

as required.

□

Note that the proof that Gibbs factorization implies the global Markov property did not use the positivity. This means that one has the following result.

Theorem 2.6 *For a graph $G = (V, E)$ and joint distribution P of the variables V, Gibbs factorization implies the global Markov property.*

□

The global Markov property implies the local Markov property (see Theorem 2.2), and the latter, under positivity implies the Gibbs factorization (see Theorem *LCDA* 11.6). Therefore, under positivity, the global Markov property implies the factorization.

Theorem 2.7 *For a graph $G = (V, E)$ and (strictly) positive joint probability distribution P of the variables V, if the global Markov property holds for P, so does the Gibbs factorization.*

□

Theorem 2.7 is not true without the positivity assumption. An example given in Moussouris [117] illustrates this. Consider the graph $G = (V, E) = (\{A, B, C, D\}, \{\{A, B\}, \{B, C\}, \{C, D\}, \{D, A\}, \})$ for four binary variables with categories denoted as 1 and 2. This graph is called the cordless 4-cycle. The global Markov property for G is that

$$A \perp\!\!\!\perp C | (B, D), \text{ and } B \perp\!\!\!\perp D | (A, C). \tag{2.7}$$

A distribution P on V is Gibbs, that is, it factorizes according to G if

$$P(abcd) = \beta^{\emptyset}\beta^{A}(a)\beta^{B}(b)\beta^{C}(c)\beta^{D}(d)\beta^{AB}(ab)\beta^{BC}(bc)\beta^{CD}(cd)\beta^{DA}(da). \tag{2.8}$$

Let a distribution P be such that $P(abcd) = 0$, except for

$$P(1111),\ P(2111),\ P(2211),\ P(2221),$$
$$P(1112),\ P(1122),\ P(1222),\ P(2222),$$

which are all equal to $1/8$. The peculiar property of this distribution is in the structure of the conditional distributions

$$P(A, C|b, d) \text{ and } P(B, D|a, c),$$

for any choice of b, d and of a, c, respectively. These conditional distributions can be written in 2×2 contingency tables, and all these conditional distributions are such that either a row or a column of the conditional table contains zeros in every cell. For example, if $b = 1$ and $d = 1$, $P(1121) = P(2121) = 0$, thus, in the $P(A, C|B = 1, D = 1)$ conditional distribution, the second column, that is, $C = 2$, has zero probabilities.

As in a table with a zero row or zero column, independence holds, the global Markov property in (2.7) holds for P.

If P also had a Gibbs factorization as in (2.8), then, for instance, one would have

$$0 = P(1121) = \beta^{\emptyset}\beta^{A}(1)\beta^{B}(1)\beta^{C}(2)\beta^{D}(1)\beta^{AB}(11)\beta^{BC}(12)\beta^{CD}(21)\beta^{DA}(11),$$

and at least one of the multiplicative parameters would have to be equal to zero. But out of these nine parameters, the first, the second, the third, the fifth, the sixth, and the ninth would also appear in the factorization of $P(1111) = 1/8$, thus, none of these could be equal to zero. Further, the fourth and the seventh would appear in the factorization of $P(1122) = 1/8$, and the eighth would appear in the factorization of $P(2221) = 1/8$, thus, none of these could be zero either; consequently, the factorization does not exist.

The results obtained so far in this section are summarized in the next theorem, which is stronger than Theorem *LCDA* 11.6 and gives a complete characterization of undirected graphical models.

Theorem 2.8 *The following statements hold.*

(i) For distributions that are (strictly) positive, the three Markov properties are equivalent and they are also equivalent to the Gibbs factorization.

(ii) For all distributions, the Gibbs factorization implies the global Markov property, which implies the local Markov property, which implies the pairwise Markov property.

□

The three Markov properties are not equivalent in the case of all graphs, if the (strict) positivity assumption is not made. However, for many graphs, the equivalence still holds, even without this assumption, see Matus [113].

A useful aspect of graphical models is that graphs can be learned to be seen as intuitive representations of the association structure among variables. The proper interpretation that should be used to develop the intuition in interpreting graphical models will be discussed later in this chapter. But before doing so, this section is concluded with a brief discussion of the relationships between graphs and Markov properties. It is clear that given a graph $G = (V, E)$, any one of the three Markov properties generates a list of conditional independences. In a certain sense, the converse is also true.

Theorem 2.9 *Let V be a set of variables and $G = (V, E)$ be a graph. Consider one of the global, local, or pairwise Markov properties and the list of all conditional independence statements implied by it. Then, G can be generated from the list.*

Proof As either one of the lists is derived from a graph, for each list, there exists a graph which implies it. It is sufficient to show that for every possible pair of variables $\{V_1, V_2\}$, one can decide whether or not $\{V_1, V_2\} \in E$ holds.

All of the conditional independence statements are of the form

$$\mathscr{A} \perp\!\!\!\perp \mathscr{B} | \mathscr{C},$$

with $\mathscr{A}, \mathscr{B}, \mathscr{C} \subseteq V$, where these are disjoint subsets and, further, $\mathscr{A}$ consists of a single variable, $\mathscr{B}$ contains its non-neighbors, $\mathscr{C}$ contains its neighbors in the case of the local Markov property, and both $\mathscr{A}$ and $\mathscr{B}$ consist of a single variable, while $\mathscr{C}$ contains all other variables in the case of the pairwise Markov property.

Then, for any pair of variables $\{V_1, V_2\} \in V$, if $\{V_1, V_2\} \in E$, there can exist no triplet $\mathscr{A}, \mathscr{B}, \mathscr{C}$ in any of the lists, such that $V_1 \in \mathscr{A}$ and $V_2 \in \mathscr{B}$ or the other way around. Indeed, such a conditional independence could only exist in either one of the lists, if they were contained in two subsets that are connected through $\mathscr{C}$ only.

On the other hand, if $\{V_1, V_2\} \notin E$, then

$$V_1 \perp\!\!\!\perp V_2 | V \setminus \{V_1, V_2\}$$

is contained in all the lists because they contain all implied conditional independences.

Thus, for either one of the lists, $\{V_1, V_2\} \in E$, if and only if there exists no triplet $\mathscr{A}, \mathscr{B}, \mathscr{C}$, with any $\mathscr{C}$ such that $V_1 \in \mathscr{A}$ and $V_2 \in \mathscr{B}$ or the other way around.

□

Note that the above result does not claim the existence of probability distributions, thus, the question of (strict) positivity does not apply. On the other hand, listing all implied conditional independences can be quite an involved task. The result also implies that if two undirected graphs imply the same conditional independences according to any of the Markov properties, then they are identical.

As will be shown in the next chapter, this is not true for directed graphs, see also the discussion related to Figure *LCDA* 8.1.

If, instead of all the conditional independences implied by any of the Markov properties applied to a graph, an arbitrary list of conditional independences is considered, then even determining all the implied conditional independences for the variables involved may be a difficult task. For a discussion of this problem and references, see Studený [159].

The implications among the different Markov properties can be used to derive other relationships. For example, if for three variables A, B, C,

$$A \perp\!\!\!\perp B|C, \quad A \perp\!\!\!\perp C|B, \quad B \perp\!\!\!\perp C|A, \tag{2.9}$$

all hold, then in the case of (strictly) positive probability distributions P, the variables A, B, C are jointly independent. This is seen easily by noting that (2.9) means that for P, the pairwise Markov property holds with respect to the empty graph, and then P is also Gibbs, that is, it factorizes according to the empty graph, so $P(A = a, B = b, C = c) = P(A = a)P(B = b)P(C = c)$. As another proof, note that (2.9) is that all three conditional odds ratios of two of the variables, given any category of the third, are equal to 1, implying that the second-order odds ratio is also 1, and this means that the distribution belongs to the log-linear model with maximal interactions A, B, C.

2.5 Undirected Graphical Models and Log-Linear Models

Simple and intuitive graphical models that had been used in science before graphical modeling in the sense discussed here started include the concept of a Markov chain and the Ising model,[7] briefly discussed in Chapter *LCDA* 11. The graphs generating a Markov chain and an Ising model, respectively, are shown in Fig. 2.3. In these models, direct association (i.e., being connected by an edge) was implied by temporal or spatial adjacency. While applications along the same lines remain important, graphical models, in general, allow for the modeling of structures where such immediate considerations do not apply.

Graphical log-linear models in the undirected sense, originally proposed by Darroch et al. [39], were defined in Section *LCDA* 11.3 as those log-linear models where the maximal interactions are the cliques of a graph $G = (V, E)$, that is, which can be written as in (2.4). In that definition, (strict) positivity was assumed, but in view of part (ii) of Theorem 2.8, the existence of a Gibbs factorization based on the cliques of a graph may be used as the definition, without the positivity assumption.

[7] There is renewed interest in many of the properties of the Ising model, see, e.g., Marchetti and Wermuth [111].

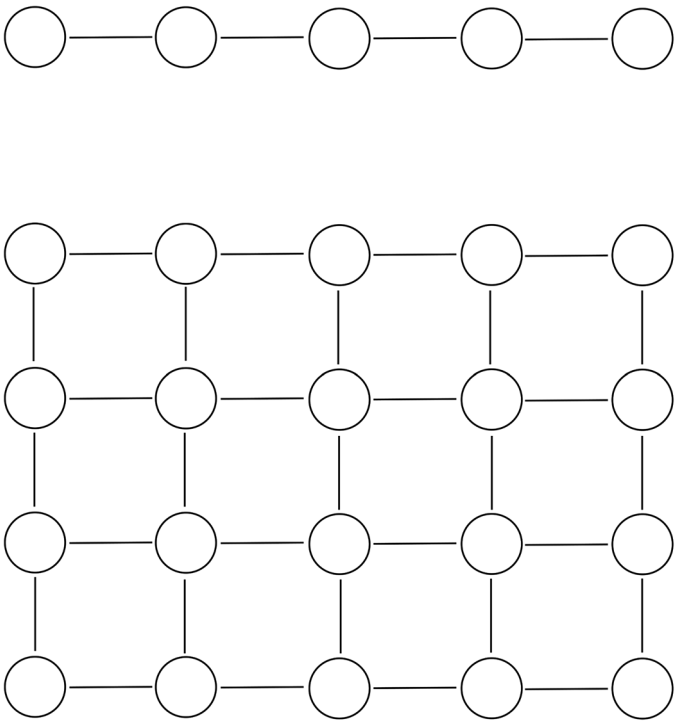


Fig. 2.3 The top graph is a Markov chain, and the bottom graph is an Ising model

As was mentioned above, a generating class on the variables V is a hypergraph $\mathscr{G} = (V, \mathscr{E})$, where $\mathscr{E}$ consists of the maximal interactions in the generating class. Theorem *LCDA* 11.3 stated that a generating class (i.e., a list of maximal interactions) is graphical (i.e., identical to the cliques of a graph) if and only if it does not have a forbidden configuration. A forbidden configuration is a subset $F \subseteq V$, such that every proper subset of F is an interaction but F itself is not an interaction.

In view of the definition of a graphical hypergraph given in Sect. 2.1, the following result is straightforward (see Proposition *LCDA* 11.3).

Theorem 2.10 *The following statements hold:*

(i) A log-linear model for the variables V with maximal interactions $\mathscr{E}$ is graphical, if and only if the hypergraph $\mathscr{G} = (V, \mathscr{E})$ is graphical.
(ii) Log-linear models based on the hypergraphs $\mathscr{G}_1 = (V, \mathscr{E}_1)$ and $\mathscr{G}_2 = (V, \mathscr{E}_2)$ are graphical equivalent if and only if the 2-sections of $\mathscr{G}_1$ and of $\mathscr{G}_2$ are identical.
(iii) The hypergraph associated with a graphical log-linear model is the clique hypergraph of its 2-section.

For example, hypergraphs c and d in Fig. 2.1 are graphical equivalent.

Undirected and directed graphical models for discrete and also for continuous data have found many applications. For overviews of these, see Dawid and Mortera [43], Li and Ma [104], Luger and Chakrabarti [106], Mortera and Dawid [116], and Mukherjee and Oates [118].

To further explore the specific nature of graphical models for categorical data, a brief discussion of the Gaussian case is needed. Details and proofs of these facts can be found in Lauritzen [98] and Whittaker [170], see also Sections *LCDA* 1.3 and 7.2. In a multivariate normal distribution, two components X_i and X_j are conditionally independent, given all the other components, if and only if the (i, j) entry in the inverse of the covariance matrix, that is, in the concentration matrix, is zero, see Dempster [45]. Therefore, if the graph G is empty, the pairwise Markov property implies that for a Gaussian distribution belonging to the associated graphical model, all off-diagonal elements of the concentration matrix are zero. Then, the inverse of this matrix, the covariance matrix, is also diagonal, and the components of the multivariate normal distribution are jointly independent.

Further, pairwise marginal independence in the multivariate normal case, if it holds for all pairs of variables, makes the covariance matrix diagonal, thus, the joint distribution has joint independence. This is in sharp contrast with the categorical case. This is well illustrated by the distribution in Table *LCDA* 1.2, where for three binary variables, all pairwise marginal independences hold, but this does not imply joint independence. On the contrary, the second-order odds ratio may take on arbitrary values. More generally, in the case of categorical data, marginal distributions on a descending class of subsets are variationally independent from the conditional odds ratios on the complement ascending class (see Theorem *LCDA* 10.2); therefore, association information taken from lower-dimensional marginals has no relevance with respect to the higher-order interactions. This implies that a graphical model may not describe the association structure for categorical data in an appropriate way. The next subsection explores the extent to which graphical models may be expected to provide adequate descriptions of multivariate categorical data.

2.5.1 Faithfulness

In the graphical modeling framework, a graph G (more precisely the implied graphical model) may be seen as an appropriate description of a distribution P,[8] if P is Markov according to G, and if also all the conditional independences present in P can be read off from (are implied by) G. In the latter case, P is called faithful to G.[9] One might think that if P is Markov according G, but it is not faithful to

[8] The distribution to be analyzed is usually the observed distribution, but it may be the result of some initial smoothing or modeling.

[9] See also the concept of parametric cancellation in Wermuth and Cox [167].

it, that is, it has additional conditional independences not implied by the Markov property associated with G, then some of the edges may be deleted from G, so that the resulting graph G' implies all the conditional independences present in P.

That this is not always possible, is illustrated now with an example. To be more precise, it has to be pointed out that the Markov properties occasionally imply conditional independences, with no variable in the condition.[10] For example, with $|V| = 2$, and E empty, the pairwise Markov property implies that the two variables are independent, with nothing in the condition. Similarly, if the graph consists of two connected components, $\mathscr{A}$ and $\mathscr{B}$, then by the global Markov property $\mathscr{A} \perp\!\!\!\perp \mathscr{B}$, without any variables in the condition, see also Fig. 2.2. Therefore, in the context of graphical models, marginal independence is considered a special type of conditional independence.

The example is adopted from Klimova et al. [88] and uses the distribution P in Table 2.1. As is seen easily, $C \perp\!\!\!\perp D|(A, B)$ in P, and thus P is Markov according to the graph G in Fig. 2.4. But also $A \perp\!\!\!\perp B$ in P, thus, P is not faithful to G. If one removes an edge from G, in the hope of finding a graph, according to which P

Table 2.1 A distribution that is Markov according to the graph in Fig. 2.4 but is not faithful to it

	A=1 B=1			A=1 B=2	
	D=1	D=2		D=1	D=2
C=1	0.02	0.08	C=1	0.12	0.03
C=2	0.03	0.12	C=2	0.08	0.02
	A=2 B=1			A=2 B=2	
	D=1	D=2		D=1	D=2
C=1	0.12	0.003	C=1	0.02	0.08
C=2	0.08	0.02	C=2	0.03	0.12

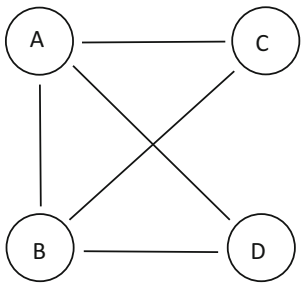


Fig. 2.4 A graph according to which the distribution in Table 2.1 is Markov but not faithful to it and is not Markov according to any of its subgraphs

[10] Such independences are called marginal.

is Markov and also faithful to it, the resulting graph, say G', will imply one of the following conditional independences:

$$A \perp\!\!\!\perp B|(C, D),\ \ A \perp\!\!\!\perp C|(B, D),\ \ A \perp\!\!\!\perp D|(B, C),\ \ B \perp\!\!\!\perp C|(A, D),\ \ B \perp\!\!\!\perp D|(A, C).$$

But none of these holds in P, as one obtains that

$$COR\,(A, B|1, 1) = 0.027778,\ \ COR\,(A, C|1, 1) = 0.44444,$$

$$COR\,(A, D|1, 1) = 0.0625,\ \ COR\,(B, C|1, 1) = 0.44444,$$

$$COR\,(B, D|1, 1) = 0.0625.$$

Therefore, P is not Markov according to G'. Whichever of the above conditional independences was implied by G', it would remain among the implications of any subgraph of G', thus, P is also not Markov to any subgraph of G'.

Obviously, there is no graph, at all, to which P would be faithful. As was seen above, P is not faithful to G, neither to any of its subgraphs, as it is not Markov according to any of the subgraphs. Any other graph on the same set of nodes contains the edge $\{C\,D\}$, and thus it does not imply $C \perp\!\!\!\perp D|(A, B)$, which holds for P.

Because the concept of faithfulness to a graph, in the case of categorical data, does not behave as one would expect intuitively, Klimova et al. [88] defined faithfulness to a hypergraph, as the property that the distribution belongs to the log-linear model defined by the hypergraph and the maximal nonvanishing conditional interactions of the distribution coincide with the hyperedges of the hypergraph.

Note that in the case of graphical log-linear models, at least in the (strictly) positive case, belonging to the model, which is the same as having a Gibbs factorization, implies the Markov properties, but for nongraphical log-linear models, there is no Markov interpretation. This is why the definition says belonging to the model, rather than being Markov according to a graph. Clearly, if a distribution is faithful to a hypergraph which is graphical, then it is also faithful to its 2-section and one has Markov properties implied by being in the model. The following result was given in Klimova et al. [88].

Theorem 2.11 *For any (strictly) positive joint distribution P of the variables V, there exists a hypergraph $\mathscr{H} = (V, \mathscr{E})$ to which P is faithful.*

Proof For a subset of variables $\mathscr{A}$, the conditional interaction

$$COR\,(\mathscr{A}|V \setminus \mathscr{A})$$

is said to vanish, if it is equal to 1 for all joint categories of the variables in $V \setminus \mathscr{A}$. In this case,

$$COR\,(\mathscr{A}'|V \setminus \mathscr{A}')$$

also vanishes for all $\mathscr{A}' \supset \mathscr{A}$ (see Proposition *LCDA* 6.4), thus, there are maximal nonvanishing interactions in P, in the sense that all bigger interactions vanish and

none of the smaller ones vanish. If a hypergraph $\mathscr{H}$ is defined with these as the hyperedges, then, by definition, P belongs to the log-linear model defined by $\mathscr{H}$.

The distribution P cannot have vanishing interactions other than those implied by $\mathscr{H}$, because any such interaction would pertain to a proper subset of one of the hyperedges, and then the interactions of this hyperedge would also vanish which would contradict its definition.

□

The above result does not say that there cannot be conditional odds ratios in P which are equal to 1 without this being implied by the model associated with $\mathscr{H}$. It says that there cannot be such conditional odds ratios that are equal to 1 for all possible category combinations of the conditioning variables without this being implied by $\mathscr{H}$.

The results of this subsection indicate that graphical models may be overly restrictive for categorical data, certainly more restrictive, than when graphical models are applied to data with multivariate normal distribution, of which a brief summary was given earlier. This means that when a graphical log-linear model gives a good fit, one has a substantial simplification of the structure, but it is by far not guaranteed that such a model could be found for a given set of data.

2.5.2 Estimation and Search

As the undirected graphical models discussed here are log-linear models, the results related to maximum likelihood estimation of log-linear models in Chapter *LCDA* 12, especially the iterative proportional fitting procedure (IPFP) in Section *LCDA* 12.2, apply. In that context, log-linear models were only applied to (strictly) positive probability distributions, and the convergence of the IPFP was proved assuming the observations were positive in every cell. Symbolically, if the hyperedges of the hypergraph defining the model are $\mathscr{C} = (\mathscr{C}_1, \ldots \mathscr{C}_l)$, the IPFP can be described as having the following steps.

Algorithm (2.10)

Step 0: Set $k = 0$, $t = 0$, and let P_k be the uniform distribution, and Q the observed distribution.
Step 1: For $j = 1, \ldots, l$, and $k = tl + j$,

$$P_k(V) = P_{k-1}(V \setminus \mathscr{C}_j)\frac{Q(\mathscr{C}_j)}{P_{k-1}(\mathscr{C}_j)}$$

Step 2: If no convergence is achieved, set $t = t + 1$ and go to Step 1.

The MLE in a log-linear model was characterized in Theorem *LCDA* 12.2. The note after this theorem illustrated that the condition of (strictly) positive observations cannot be weakened by requiring that only the sufficient statistics $(Q(\mathscr{C}_1), \ldots, Q(\mathscr{C}_l))$ are positive. An example was given illustrating that even if the

latter condition holds, there may exist no MLE. A necessary and sufficient condition for the existence of the MLE in terms of positivity will be given in Chap. 11, as part of the theory of a more general model class, called relational models.

A generalized version of the IPFP was proposed by Darroch and Ratcliff in [38]. When applied to log-linear models,[11] instead of the iterated updates based on marginal distributions, it collects all the adjustments that apply to a cell. The Darroch-Ratcliff algorithm can be described as follows.

Algorithm (2.11)

Step 0: Set $k = 0$, $t = 0$, and let P_k be the uniform distribution, and Q the observed distribution. Let $v_1, v_2, \ldots, v_c$ be the cells of the joint sample space of the variables V.

Step 1: For $i = 1, \ldots, c$, and $k = tc + i$,

$$P_k(v_i) = P_{k-1}(v_i) \prod_{\mathscr{C}_j \in \mathscr{C}} \frac{Q((v_i)_{\mathscr{C}_j})}{P_{k-1}((v_i)_{\mathscr{C}_j})}$$

Step 2: If no convergence is achieved, set $t = t + 1$ and go to Step 1.

In Step 1 of the algorithm, $(v_i)_{\mathscr{C}_j}$ means those indices out of v_i, which belong to the variables in $\mathscr{C}_j$.

It was shown in Csiszár [35] that the Darroch-Ratcliff algorithm can be seen as performing iterated I-projections onto linear families of probability distributions, each of which is defined by fixing one marginal distribution. Based on this, a proof similar to the one given for the convergence of the IPFP in Section *LCDA* 12.2 may be given.

Another algorithm, which may be seen as dual to IPFP, was proposed by Rudas [146] for the maximum likelihood estimation of graphical log-linear models. It was later pointed out by Matus [114] that the proof of convergence given was incomplete. In spite of this, the algorithm works well in many cases and may offer advantages over the IPFP.[12] The algorithm is based on Theorem *LCDA* 11.10, which stated that every graphical log-linear model is the intersection of conditional joint independence models. More precisely, if G is a graph, then let $\mathscr{C}_1, \ldots \mathscr{C}_l$ denote the cliques of the complement of G. These are the maximal empty subgraphs of G. Then, for each $\mathscr{C}_j$, $j = 1, \ldots, l$, define a generating class $\mathscr{D}_j$, in which the maximal elements are of the form

$$\{A\} \cup \overline{\mathscr{C}_j},$$

for all $A \in \mathscr{C}_j$, where $\overline{\mathscr{C}_j} = V \setminus \mathscr{C}_j$, so that

$$\mathscr{D}_j = \{\{\{A\} \cup \overline{\mathscr{C}_j}\}, A \in \mathscr{C}_j\}.$$

[11] The algorithm applies to models more general than the log-linear models considered here.

[12] See the discussion after the examples later.

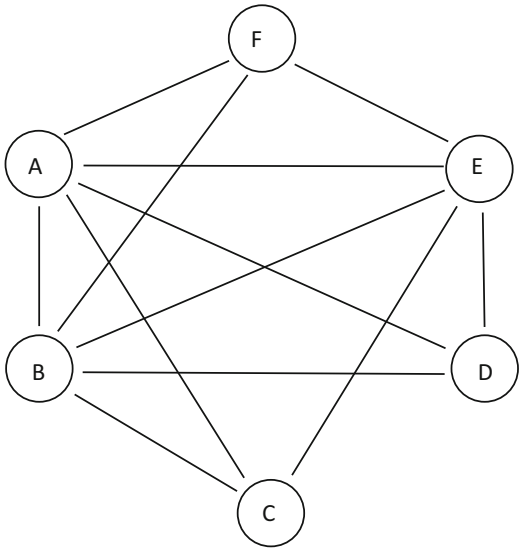


Fig. 2.5 The complement of this graph has one clique: CDF

Then, for the generated log-linear models,

$$LL(G) = \cap_{j=1}^{l} LL(\mathscr{D}_j). \tag{2.12}$$

Note that with the $\mathscr{D}_j$ above, $LL(\mathscr{D}_j)$ means the joint conditional independence of the variables in $\mathscr{C}_j$, given the variables in $\overline{\mathscr{C}_j}$ (see, e.g., (2.13)) and has a closed-form MLE.

For example, if G has the following edges: V_1V_2, V_1V_3, V_1V_4, V_1V_5,[13] then its complement graph has one clique, namely, $V_2V_3V_4V_5$, and the complement of the clique only contains V_1. Then $\mathscr{D}_1 = \{V_1V_2, V_1V_3, V_1V_4, V_1V_5\}$ which is the original model, of course.

As another example, let G be the graph in Fig. 2.5. Then, its complement graph has one clique, consisting of the nodes C, D, F, and by Theorem *LCDA* 11.10, the graphical log-linear model based on G is equivalent to the joint conditional distribution of C, D, F, given A, B, E being joint independence, or

$$P(a, b, c, d, e, f) = P(c|a, b, e)P(d|a, b, e)P(f|a, b, e)P(a, b, e). \tag{2.13}$$

Note that (2.13) is not a Markov property itself, but can be derived from the Markov properties. The local Markov property implies that $C \perp\!\!\!\perp DF|ABE$, thus,

[13] This is a star graph with variable V_1 in the center.

$$\begin{aligned} P(a,b,c,d,e,f) &= P(c|a,b,e)P(d,f|a,b,e)P(a,b,e) \\ &= P(c|a,b,e)P(a,b,d,e,f). \end{aligned} \tag{2.14}$$

The local Markov property also implies that $D \perp\!\!\!\perp CF|ABE$, from which Proposition 2.5 gives that $D \perp\!\!\!\perp F|ABE$, which means that

$$P(a,b,e,f,d) = P(d|a,b,e)P(f|a,b,e)P(a,b,e),$$

and combining this with (2.14) gives (2.13).

When the above example is formulated as a log-linear model, in addition to (2.13), one also has

$$COR(C,D|a,b,e,f)=1,\ COR(C,F|a,b,d,e)=1,\ COR(D,F|a,b,c,e)=1,$$

and the minimal elements of the ascending class are CD, CF, DF, and the maximal elements of the descending class are $ABEF$, $ABDE$, $ABCE$. Thus, the IPFP starts with a uniform distribution and cyclically enforces the observed marginal distributions of the three four-way maximal marginals in the descending class.

The dual algorithm, in the case of this example, starts with the observed distribution and enforces (2.13) on it, so it does not require iteration (see also Proposition *LCDA* 6.7) and is much faster than running the $IPFP$. In general, whether the $IPFP$ or the dual algorithm is faster depends on how dense is G, more precisely, whether G or $\overline{G}$ has more cliques. The sizes of the cliques are not important, because iterations occur over the cliques, while for every clique, the algorithms apply closed formulas.

The algorithm consists of the following steps:

Algorithm (2.15)

Step 0: Set $k = 0$, $t = 0$, and let P_k be the observed distribution.
Step 1: For $j = 1, \ldots, l$, and $k = tl + j$,

$$P_k(V) = P_{k-1}(\overline{\mathscr{C}_j}) \prod_{A \in \mathscr{C}_j} P_{k-1}(A|\overline{\mathscr{C}_j})$$

Step 2: If no convergence is achieved, set $t = t + 1$ and go to Step 1.

The IPFP starts with the uniform distribution and cyclically imposes the marginal distributions of the maximal interactions in the model. During this, the higher-order interactions, which were equal to 1 in the uniform distribution, do not change. The algorithm in (2.15) starts with the observed distribution and cyclically imposes the conditional independences defining the model, while the lower-order marginals do not change. The following is easy to see.

Proposition 2.9 *For data obtained through Multinomial or multivariate Poisson sampling, if the algorithm in (2.15) converges for a graphical model, then the limit is the maximum likelihood estimate under the model.*

□

When log-linear models are fitted to large amounts of data, the computational burden may become critical and non-iterative fitting is of great advantage. This is possible for a subclass of undirected graphical models. Decomposable models were introduced in Section *LCDA* 11.3. Theorem *LCDA* 12.4 gave a closed-form maximum likelihood estimate in the case of Multinomial or multivariate Poisson sampling and a decomposable log-linear model. For a model search algorithm among decomposable models, see Deshpande et al. [44].

Model search procedures are data-driven, in the sense that a model is sought for that provides an approximate description of the data but is simple enough to provide insight. For the illustration and limitations of standard model search procedures in the context of log-linear models, see Section *LCDA* 11.2.

A more general model search idea was considered by Edwards and Havranek in [49] and [50], see also Edwards [48]. It applies to any situation, when the model class in which the search is to be made has a lattice structure. A lattice is a partially ordered set, where any two elements have a largest lower bound (their meet) and a smallest upper bound (their join). The procedure is based on two principles. If a model is accepted (more precisely: is not rejected), then all larger models are considered as accepted, and if a model is rejected, all smaller models are considered as rejected. The procedure starts with testing a model and then proceeds with testing models that have not been deemed fitting or non-fitting thus far. The two principles, for reasons explained in Section *LCDA* 11.2, may fail in practice, in the sense that non-fitting models may be deemed fitting or fitting models may be deemed as non-fitting by this procedure. In many applications, however, the speed of the procedure makes up for the limited precision.

It has to be noted, however, that the practical properties of the procedure described above are different from its theoretical properties, that is, from its behavior if one has access, instead of a sample, to population data. In that case, whether or not a model fits can be decided exactly, instead of testing based on data, and one does not have to face the possibilities of committing errors associated with statistical tests. Indeed, if a distribution is contained in a model (a subset of distributions), it is also contained in any larger model, and if a distribution is not contained in a model, it is also not contained in any smaller model. Therefore, the decisions on which the algorithm is based are correct if it is applied to population data. This remains true, whether one conducts a search among all log-linear models or among all graphical log-linear models. If, however, the goal is to find a model that fits tightly, in the sense that the distribution belongs to the model, that is, the distribution is Markov according to the graph, but also the graph implies all conditional independences present in the distribution, that is, the distribution is faithful to the graph, then a search among all log-linear models will lead to a tightly fitting model (see

Theorem 2.11), but not necessarily so in the case when the search is restricted to the class of graphical log-linear models (see the example before Theorem 2.11).

General search algorithms to learn a model in the undirected and also in the directed case (see Spirtes et al. [157]) usually assume that the true distribution that produced the data upon which the search is based is faithful to the model, because otherwise the search may result in a false model. As was seen in the previous section, there are discrete distributions that are not faithful to any undirected graphical model, which limits the usefulness of search among graphical models in the case of discrete data, and thus search among all log-linear models may be more appropriate. For example, when applied to the distribution in Table *LCDA* 1.2, one algorithm, the PC algorithm (see Spirtes et al. [157]), that searches among graphical models would check first the pairwise conditional independences of the variables. These all hold, thus, the algorithm would delete the relevant edges and would conclude with an empty graph, implying joint independence, which does not hold.

To study the properties of such search procedures among hypergraphs, parametric strong faithfulness to a hypergraph was introduced in Klimova et al. [88] as the property that all interaction terms are bounded away from the no-interaction value by a common constant (see the condition of Theorem 2.12). The interaction terms may be derived from the logarithms of the conditional odds ratios, but other choices are also possible. The following result was obtained in Klimova et al. [88].

Theorem 2.12 *Let $\mathscr{H}$ be a hypergraph on the variables V, and let $\mathscr{M}(\mathscr{H}, \lambda, \delta)$ be the collection of probability distributions $P(V)$ such that for the logarithms of the conditional odds ratios,*

$$|\log COR(A|\overline{A} = \overline{a})| > \lambda, \text{ for all hyperedges } A \subseteq V \text{ of } \mathscr{H} \text{ and } \overline{A} = \overline{a}, \tag{2.16}$$

that is, parametric strong faithfulness holds, and

$$P(V = v) > \delta, \text{ for all } v.$$

Assume sample data from a distribution which belongs to the log-linear model defined by $\mathscr{H}$ with a Multinomial distribution are available and suppose the vector $\hat{\mathbf{e}}$ of maximum likelihood estimates of the vector $\mathbf{e}$ of all the log-interaction terms in (2.16) for the hyperedges of $\mathscr{H}$ exists and is unique. Then, over $\mathscr{M}(\mathscr{H}, \lambda, \delta)$, $\hat{\mathbf{e}}$ is a uniformly consistent estimator of $\mathbf{e}$.

□

The importance of this result is that under these conditions, with increasing sample sizes, one can have inference with increasing precision about the lack or existence of interaction terms, and the same precision applies uniformly, that is, to all distributions in $\mathscr{M}(\mathscr{H}, \lambda, \delta)$. The precision of the inference is measured by bounds on type I and type II error probabilities when testing the null hypothesis assuming that the maximal interactions in $\mathscr{H}$ vanish, see Klimova et al. [88].

2.6 Mixed Continuous and Discrete Variables

Undirected graphical interaction models have been developed by Lauritzen and Wermuth [102] for the case when some of the variables of interest are continuous, and some are discrete, under the assumption that the joint conditional distribution of the continuous variables, given the discrete ones, is multivariate normal, where both the conditional expectation vector and the conditional covariance matrix may depend on the categories in the condition. Such a distribution is called a conditional Gaussian (*CG*) distribution.

To be more specific, let D be the collection of the discrete variables and C be the collection of the continuous variables, and $V = \{D \cup C\}$. In this context, it is useful to treat the variables in a particular order, thus, $\mathbf{D}$ and $\mathbf{C}$ are the corresponding vectors and $\mathbf{V}$ is obtained from them via concatenation.

Following Wermuth and Lauritzen [169], to build the *CG* distribution, one determines first the (marginal) distribution of $\mathbf{D}$, by specifying the $P(\mathbf{d})$ probabilities that are assumed to be (strictly) positive. Next, the conditional distribution of the continuous variable is specified as

$$\mathbf{C}|\mathbf{D} = \mathbf{d} \sim N(\boldsymbol{\mu}(\mathbf{d}),\ \boldsymbol{\Sigma}(\mathbf{d})),$$

which is a multivariate normal distribution where the expectation vector and the variance-covariance matrix depend on $\mathbf{d}$. Thus, the joint probability density function of a *CG* distribution takes the form

$$f(\mathbf{v}) = f(\mathbf{d}, \mathbf{c}) = P(\mathbf{d})(2\pi)^{-q/2}(det(\Sigma(\mathbf{d})))^{-1/2}$$

$$\times \exp\left(-\frac{1}{2}(\mathbf{c} - \boldsymbol{\mu}(\mathbf{d}))'\boldsymbol{\Sigma}^{-1}(\mathbf{d})(\mathbf{c} - \boldsymbol{\mu}(\mathbf{d}))\right).$$

To define statistical models, it is useful to separate the components of the probability density function that do not depend on the continuous coordinates, that depend on them, and that depend on their products. This can be done in the following way:

$$f(\mathbf{v}) = f(\mathbf{d}, \mathbf{c}) = P(\mathbf{d})(2\pi)^{-q/2}(det(\Sigma(\mathbf{d})))^{-1/2}$$

$$\times \exp\left(-\frac{1}{2}\mathbf{c}'\boldsymbol{\Sigma}^{-1}(\mathbf{d})\mathbf{c} - \frac{1}{2}\boldsymbol{\mu}(\mathbf{d})'\boldsymbol{\Sigma}^{-1}(\mathbf{d})\boldsymbol{\mu}(\mathbf{d})\right.$$

$$\left.+\frac{1}{2}\mathbf{c}'\boldsymbol{\Sigma}^{-1}(\mathbf{d})\boldsymbol{\mu}(\mathbf{d}) + \frac{1}{2}\boldsymbol{\mu}(\mathbf{d})'\boldsymbol{\Sigma}^{-1}(\mathbf{d})\mathbf{c}\right)$$

$$= P(\mathbf{d})(2\pi)^{-q/2}(det(\Sigma(\mathbf{d})))^{-1/2}$$

$$\times \exp\left(-\frac{1}{2}\mathbf{c}'\boldsymbol{\Sigma}^{-1}(\mathbf{d})\mathbf{c} - \frac{1}{2}\boldsymbol{\mu}(\mathbf{d})'\boldsymbol{\Sigma}^{-1}(\mathbf{d})\boldsymbol{\mu}(\mathbf{d}) + \boldsymbol{\mu}(\mathbf{d})'\boldsymbol{\Sigma}^{-1}(\mathbf{d})\mathbf{c}\right).$$

The last term was obtained using the equality

$$\boldsymbol{\mu}(\mathbf{d})'\boldsymbol{\Sigma}^{-1}(\mathbf{d})\mathbf{c} = (\boldsymbol{\mu}(\mathbf{d})'\boldsymbol{\Sigma}^{-1}(\mathbf{d})\mathbf{c})' = \mathbf{c}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}(\mathbf{d}),$$

which holds, because the left-hand side is a number in $\mathbb{R}$, and $\boldsymbol{\Sigma}$ and thus also $\boldsymbol{\Sigma}'$ are symmetric.

Then, the density of a *CG* distribution may be written as

$$f(\mathbf{v}) = f(\mathbf{d}, \mathbf{c}) = \exp(g(\mathbf{d}) + \mathbf{h}(\mathbf{d})'\mathbf{c} - \frac{1}{2}\mathbf{c}'\boldsymbol{\Sigma}^{-1}(\mathbf{d})\mathbf{c}), \tag{2.17}$$

where

$$g(\mathbf{d}) = \log P(\mathbf{d}) - \frac{1}{2}\left(q\ \log(2\pi) + \log(det\,(\Sigma(\mathbf{d})) + \boldsymbol{\mu}(\mathbf{d})'\boldsymbol{\Sigma}^{-1}(\mathbf{d})\boldsymbol{\mu}(\mathbf{d})\right),$$

and

$$\mathbf{h}(\mathbf{d}) = \boldsymbol{\Sigma}^{-1}(\mathbf{d})\boldsymbol{\mu}(\mathbf{d}).$$

The following result is immediate.

Proposition 2.10 *Assume that the joint distribution of the variables* $\mathbf{V} = (\mathbf{D}, \mathbf{C})$ *is a CG distribution and* $\mathbf{G} \subseteq \mathbf{C}$. *Then, the marginal distribution of* $(\mathbf{D}, \mathbf{G})$ *is also a CG distribution.*

□

Marginalization over discrete variables generally leads to more complex distributions; however, one has the following result.

Proposition 2.11 *Assume that the joint distribution of the variables* $\mathbf{V} = (\mathbf{D}, \mathbf{C})$ *is a CG distribution and* $\mathbf{F} \subseteq \mathbf{D}$ *such that*

$$(\mathbf{D} \setminus \mathbf{F}) \perp\!\!\!\perp \mathbf{C}|\mathbf{F}. \tag{2.18}$$

Then, the marginal distribution $(\mathbf{F}, \mathbf{C})$ *is also a CG distribution.*

Note that in this chapter, conditional independence was defined for categorical variables only, and the result above uses conditional independence between categorical and continuous variables, given categorical variables. This can be defined similarly to the categorical variables only case. The condition in (2.18) means that if the categories of the variables $\mathbf{F}$ are kept fixed, then the conditional distribution of the variables $\mathbf{C}$, given all the variables $\mathbf{D}$, does not depend on the categories of the variables in $\mathbf{D} \setminus \mathbf{F}$. This, just like in the discrete case, implies that the conditional

distribution of $\mathbf{C}$ is the same, wether in the condition one has $\mathbf{D} = (\mathbf{D} \setminus \mathbf{F}) \cup \mathbf{F}$ or $\mathbf{F}$ only.

Proof Obviously, if the marginal distribution of $\mathbf{D}$ is (strictly) positive, so is the marginal distribution of $\mathbf{F}$. Further, in this case, the conditional distribution of $\mathbf{C}$ given $\mathbf{D}$ is the same as the conditional distribution of $\mathbf{C}$ given $\mathbf{F}$, namely, both are multivariate normal, which implies the claim of the proposition.

□

When marginalization happens for both discrete and continuous variables, say to $\mathbf{F} \subseteq \mathbf{D}$ and $\mathbf{G} \subseteq \mathbf{C}$, then the marginal distribution of $(\mathbf{F}, \mathbf{G})$ remains a *CG*, if

$$(\mathbf{D} \setminus \mathbf{F}) \perp\!\!\!\perp \mathbf{G} | \mathbf{F},$$

which can be seen by applying first Proposition 2.10 and then Proposition 2.11.

The family of *CG* distribution is closed under conditioning.

Proposition 2.12 *Assume that the joint distribution of the variables* $\mathbf{V} = (\mathbf{D}, \mathbf{C})$ *is a CG distribution; further let* $\mathbf{F} \subseteq \mathbf{D}$ *and* $\mathbf{G} \subseteq \mathbf{C}$*. Then, the conditional distribution of* $((\mathbf{D} \setminus \mathbf{F}), (\mathbf{C} \setminus \mathbf{G}))$*, given* $\mathbf{F} = \mathbf{f}$ *and* $\mathbf{G} = \mathbf{g}$ *is also CG.*

Proof Denoting the distribution of $\mathbf{X}$ by $\mathscr{L}(\mathbf{X})$, one has to see that

$$\mathscr{L}(((\mathbf{C} \setminus \mathbf{G}) | \mathbf{F} = \mathbf{d}_{\mathbf{F}}, \mathbf{G} = \mathbf{g}) | \mathbf{D} \setminus \mathbf{F} = \mathbf{d}_{(\mathbf{D} \setminus \mathbf{F})})$$

is normal. This is implied if

$$\mathscr{L}((\mathbf{C} | \mathbf{F} = \mathbf{d}_{\mathbf{F}}) | (\mathbf{D} \setminus \mathbf{F}) = \mathbf{d}_{(\mathbf{D} \setminus \mathbf{F})})$$

is normal, which is the same as

$$\mathscr{L}(\mathbf{C} | \mathbf{D} = \mathbf{d})$$

being normal, which is implied by $\mathscr{L}(\mathbf{D}, \mathbf{C})$ being *CG*.

□

To be able to model the effects associated with groups of variables on the joint distribution, and to eventually define Markov models, following Lauritzen and Wermuth [102], define interaction functions $\lambda^{\mathbf{F}}$ scalars, $\boldsymbol{\eta}^{\mathbf{F}}$ vectors, and $\boldsymbol{\Psi}^{\mathbf{F}}$ matrices for all $\mathbf{F} \subseteq \mathbf{D}$, such that when applied to a combination of indices $(\mathbf{d}, \mathbf{c})$, they depend on $(\mathbf{d}, \mathbf{c})_{\mathbf{F}} = (\mathbf{d})_{\mathbf{F}}$ only, and such that

$$g(\mathbf{d}) = \sum_{\mathbf{F} \subseteq \mathbf{D}} \lambda^{\mathbf{F}}_{(\mathbf{d})_{\mathbf{F}}}, \ \mathbf{h}(\mathbf{d}) = \sum_{\mathbf{F} \subseteq \mathbf{D}} \boldsymbol{\eta}^{\mathbf{F}}_{(\mathbf{d})_{\mathbf{F}}}, \ \boldsymbol{\Sigma}^{-1}(\mathbf{d}) = \sum_{\mathbf{F} \subseteq \mathbf{D}} \boldsymbol{\Psi}^{\mathbf{F}}_{(\mathbf{d})_{\mathbf{F}}}. \tag{2.19}$$

In general, there are many ways to define such functions. For the interaction function g, the parameters defined in Section *LCDA* 7.2, the λ log-linear parameters defined

in Section *LCDA* 10.3 or the corner parameters defined in Section *LCDA* 11.3 are examples of choices. In general, the Möbius inversion given in Proposition *LCDA* 11.2 may be applied to obtain the interaction functions. Of course, different choices for the interaction functions will have different meanings; more precisely, they represent different operationalizations of the concept of interaction or association. For the definition of the Markov property and the Gibbs factorization, it does not matter how the interaction functions are defined, only which ones are zero for which indices $(\mathbf{d})_{\mathbf{F}}$.

For *CG* distributions, Lauritzen and Wermuth [102] defined the pairwise Markov property, just like it was defined above. The graphs, according to which a *CG* distribution may be Markov, have V as their vertex set and some of the variables are discrete and some are continuous, but this does not affect the definition: the *CG* distribution is Markov with respect to the graph, if two variables are conditionally independent given all others, whenever they are not connected in the graph. A *CG* distribution is said to be Gibbs according to a graph G, if

$$\lambda^{\mathbf{F}}_{(\mathbf{d})_{\mathbf{F}}} = 0, \text{ unless } \mathbf{F} \text{ is complete in } G,$$

$$(\boldsymbol{\eta}^{\mathbf{F}}_{(\mathbf{d})_{\mathbf{A}}})_{C_i} = 0, \text{ unless } \mathbf{F} \cup \{C_i\} \text{ is complete in } G,$$

$$(\boldsymbol{\Psi}^{\mathbf{F}}_{(\mathbf{d})_{\mathbf{A}}})_{C_i C_j} = 0, \text{ unless } \mathbf{F} \cup \{C_i\} \cup \{C_j\} \text{ is complete in } G,$$

where C_i and C_j are arbitrary continuous components in $\mathbf{C}$ and $(.)_{C_i}$ is the component of the vector in the parentheses associated with C_i, and similarly, $(.)_{C_i C_j}$ is the component of the matrix in the parentheses associated with $\{C_i \cup C_j\}$. Then, Lauritzen and Wermuth [102] gave the following result.

Theorem 2.13 *A CG distribution is Markov according to a graph G if and only if it is Gibbs with respect to G.*

□

It may seem that the choice of the *CG* distributions as joint distributions for discrete and continuous data is somewhat arbitrary. *CG* distributions are characterized by two main features: first, the conditional distribution of the continuous variables, given the discrete ones is specified, and second, this conditional distribution is assumed to be multivariate normal. The advantages and disadvantages of the choice of the normal distribution as a model for continuous data were discussed in Chapter *LCDA* 1. Here we address the choice of conditioning on the discrete variables, as opposed to conditioning on the continuous ones. In theory, both choices seem reasonable. With *CG* distributions, one has expectations and variance-covariance matrices for each combination of the categorical variables. If conditioning occurred the other way around, one would have the probabilities of the joint categories of the discrete variables for all values of the continuous ones as parameters. In the first case, modeling (i.e., quest for simplicity) assumes a log-linear model for the categorical variables and a structured dependence of the expectations and variance-covariance matrices on the discrete categories. In the second case, the continuous

variables could have a graphical structure, and the discrete probabilities could be assumed to have a simple functional dependence on the continuous values. It may appear that both approaches could be appropriate to describe data, but the second one uses infinitely many parameters, while the first one uses finitely many parameters. Infinitely many parameters cannot be estimated from real data, for example, most of the possible values of the continuous variables to be conditioned upon will not be observed at all. Therefore, simplifying assumptions are needed, which cannot be tested or the tests will have little power. The simplifying assumptions, if approximately correct, can be very useful, but if wrong, are dangerous.

The hypothetical situation of conditioning on the continuous variables is similar to the different versions of logistic regression analysis where continuous explanatory variables are used and conditional odds are estimated. The inferiority of this modeling strategy to the approach of discrete explanatory variables and log-linear models was discussed in Chapter *LCDA* 11. These considerations support the choice of *CG* distributions as models for data with mixed levels of measurement.

2.7 On the Interpretation of Undirected Graphical Models

The great success of graphical models (in particular in the directed variant, to be discussed in the next chapter) is largely due to the intuition which statisticians and substantive scientists can build around using graphs, the ease with which graphs communicate the essence of the model and the apparently straightforward modifications that may be applied in model search. In spite of all this, it has to be emphasized that graphs have no intrinsic meanings as tools of statistical modeling. All the related statistical models, whatever intuitive they seem to be, are created and learned and are seen as intuitively clear and appealing only after this learning procedure.

The most obvious illustration of the fact that graphs have no intrinsic meanings for statistical modeling is that although the pairwise, the local, and the global Markov properties may all appear attractive, unless (strict) positivity is assumed, they do not coincide, see the examples in Sect. 2.4.

Further illustrations of the fact that graphs do not have intrinsic statistical meaning are provided by models associated with chain graphs. In this case, as will be discussed in the next chapter, for every chain graph, four different Markov properties have been suggested in the literature; see, e.g., Drton [47].

A crucial aspect of the interpretation of undirected graphical models is the duality of the interpretation of the existing edges or of the missing ones. Clearly, if an edge is present in the graph, it implies that the model also contains distributions, where the corresponding two variables are not conditionally independent, given all other variables, but it also may contain distributions, where there is a conditional independence, in spite of the existing edge, see the discussion of faithfulness in Sect. 2.5.1. Therefore, an existing edge allows for the presence of an interaction, but does not imply it. A graphical model based on a graph including an edge between

two variables does not say that there is (conditional) interaction between them, only that there may be.

A tighter interpretation of graphical models is obtained by interpreting the missing edges. These always imply (conditional) independences for the distributions in the model. In more detail, let two graphs G_1 and G_2 on the same set of nodes V be such that $E_1 \subsetneq E_2$, so that G_1 imposes the conditional independences of G_2, but also additional ones. Then, it is not correct to say that both models contain distributions with conditional associations in E_1 and, in addition, the model associated with G_2 contains distributions that also have the $E_2 \setminus E_1$ conditional associations. In reality, the pairs of variables in E_1 may or may not have conditional associations in distributions belonging to both models, and the distributions that are in G_2 but not in G_1 also may or may not have associations among the pairs in $E_2 \setminus E_1$. The only certain thing as far as the comparison of the two models is concerned is that the model associated with G_2 implies conditional independences of the pairs not in E_2, and the model associated with G_1 implies, in addition, the conditional independences for the pairs in $\overline{E_1} \setminus \overline{E_2}$, where the complements are taken with respect to the set of pairs of nodes. And this is the interpretation based on the missing edges.

For example, for three variables, let E_1 contain the AB edge, and E_2 also the AC edge. It is not true that the second model consists of distributions where $COR(A, C|B = b) \neq 1$, while in the first model, $COR(A, C|B = b) = 1$, because $COR(A, C|B = b)$ may be also equal to 1 in distributions in the second model. But it is correct to say that the second model contains distributions where the conditional associations vanish for the pairs of variables in $\overline{E_2}$, that is, for BC, and in the first model, in addition, the conditional associations vanish for the pairs in $\overline{E_1} \setminus \overline{E_2}$, that is, for AC also. Thus, interpretation in terms of the missing edges is preferable to the interpretation based on the existing edges.

There is also a deeper reason why it is better to think of a graphical model as stating that certain (conditional) interactions are not present rather than as stating that the other (conditional) interactions may be present. When the strength of association is measured with the odds ratio, the first formulation of the model says that some conditional odds ratios are equal to 1, and the second would formulate the model saying that the other conditional odds ratios may be different from 1. When these assumptions are taken as statistical hypotheses, their behaviors are very different. To simplify the discussion, a single odds ratios ratio is considered. When the null hypothesis is that a conditional odds ratio is equal to 1, then there is a neighborhood of the hypothesis, which is the region of acceptance, containing cases (i.e., observed data sets) with a weak association (odds ratio values close to 1), and it is reasonable not to reject the hypothesis based on such observations. The complement of the region of acceptance is the region of rejection, containing observed values far from the null hypothesis, and it is reasonable to reject the null when the observations fall in this region. Also, in this case, the region of acceptance contains the null (OR$=1$) and part of the alternative (OR$\neq 1$).

But if the null hypothesis is that the odds ratio may be different from 1, it is not a restriction of the parameter space, that is, the alternative is nonexistent and no

observed value would imply rejection. Even if one chooses as the null hypothesis that the odds ratio is different from 1, the null still contains all possible values of the odds ratio with the exception of OR=1, and it is not possible to identify observations far enough from the hypothesis (more precisely, unlikely enough under the null) which would warrant rejection. This is because any neighborhood of the null would contain the entire alternative. The power considerations in Chapter *LCDA* 5 are also relevant here.

Of course, in real data analytic situations, the question is not exactly whether or not a certain conditional independence holds, but rather whether the conditional association is small enough to be of no substantive significance. See the discussion of substantive significance in Sections *LCDA* 6.1 and 11.2 and also in Wermuth and Cox [167].

The intersection representation of undirected graphical models in (2.12) also provides a powerful interpretation. The model means that the joint conditional distribution of the variables in any maximal empty subgraph, given its complement, is jointly independent.

Which one is the most useful interpretation depends, of course, on the substantive problem at hand. See, for example, the regression-type and non-regression-type interpretations of log-linear models in Chapter *LCDA* 11.

2.8 Summary

This chapter summarized the most fundamental concepts and results of graphical models associated with undirected graphs. First, the basic properties of undirected and directed graphs were reviewed. In addition, hypergraphs and many variants of mixed graphs were considered.

In Sect. 2.3, a fairly detailed discussion of conditional independence was given for categorical variables, with proofs of many important properties, with or without the assumption of (strict) positivity of the joint distribution. In the next section, the discussion of Markov properties associated with undirected graphs that had already been started in *LCDA* was continued. The global, local, and pairwise Markov properties were introduced and their relationships with or without the (strict) positivity assumption were described.

Section 2.5, relying on results from *LCDA*, gave a discussion of undirected graphical models as special log-linear models. To better explore the relationship between distributions and models, the concept of faithfulness was introduced as the property that all conditional independences that hold in a distribution are implied by the graph, and it was shown that not every joint distribution is faithful to a graph. This implies a major limitation of algorithms that search for the best undirected graphical model to data. However, for every distribution, there exists a hypergraph, to which it is faithful. Three algorithms were described to find maximum likelihood estimated under undirected graphical models

Section 2.6 summarized the theory of conditional Gaussian (CG) distributions that may be used when the data contain both discrete and continuous variables. The

Gibbs factorization was formulated for CG distributions and the equivalence with a Markov property was formulated.

The last section discussed aspects of interpretation, and several arguments were given in favor of interpretation based on the missing edges of the graph, as opposed to interpretation based on the edges present in the graph.

The Things to Do section, in addition to questions related to the material presented, also covers material not discussed in the body of the chapter.

2.9 Things to Do

1. Prove that the relation defined for a pair of nodes of a graph as being connected by a path is an equivalence relation.
2. Create a graphical and a nongraphical hypergraph on five nodes.
3. Prove Proposition 2.1.
4. Prove Proposition 2.2.
5. Is there a way to define chain graphs without first identifying the components, rather by deriving them from the edges and arrows given? Is there a special condition for this?
6. Prove that if T and Y are categorical variables and T is a function of Y, such that the domain of T is the entire range of Y, then T is a coarsening of Y, that is, its categories are obtained by combining some of the categories of Y.
7. Give a numerical example that Proposition 2.8 does not hold without the assumption of (strict) positivity.
8. Do the conditions of Proposition 2.8 hold if $Y = Z$? Is the claim true in this case?
9. Consider the cells of a three-way contingency. Let $\mathscr{A}$ be the set of those cells, that can be defined using restrictions (including no restriction at all) on the first and the third variables. Define $\mathscr{B}$ similarly using the second and the third variables. Prove that $\mathscr{A}$ and $\mathscr{B}$ are algebras of sets. Prove that $\mathscr{A} \cap \mathscr{B}$ is the algebra of those sets which can be defined using restrictions on the third variable.
10. Prove that the algebras $\mathscr{A}$ and $\mathscr{B}$ obtained in the previous item are qualitatively conditionally independent. Qualitative conditional independence introduced by Bártfai and Rudas [14] applies, in general, to two algebras $\mathscr{A}$ and $\mathscr{B}$ of subsets of the same set, and it is defined with the property that for any $A \in \mathscr{A}$ and $B \in \mathscr{B}$, such that $A \cap B = \emptyset$, there is a set $C \in \mathscr{A} \cap \mathscr{B}$, such that $A \subseteq C$ and $B \subseteq C^c$, see also Arató et al. [8].
11. Prove that qualitative conditional independence of $\mathscr{A}$ and $\mathscr{B}$ in the previous item is equivalent to the following property: any (probability) measures P and Q given on them, with the property that $P(C) = Q(C)$ for all $C \in \mathscr{A} \cap \mathscr{B}$, may be extended to a (probability) measure R on the smallest algebra containing $\mathscr{A}$ and $\mathscr{B}$, so that

$$R(A \cap B) = P(A|C_A)Q(B|C_B)P(C_A) = P(A|C_A)Q(B|C_B)Q(C_B),$$

where C_A is the smallest set in $\mathscr{A} \cup \mathscr{B}$ that contains A and C_B is the smallest set in $\mathscr{A} \cup \mathscr{B}$ that contains B. Further, if $A \cap B \neq \emptyset$, then $C_A = C_B$.

12. Consider a finite set. Generate two qualitatively conditional independent algebras of their subsets.
13. Check whether properties parallel to those formulated in Propositions 2.4, 2.5, 2.6, 2.7, and 2.8 hold for qualitative conditional independence of algebras.
14. Prove that the neighbors of a node separate it from all the other nodes and thus the global Markov property implies the local Markov property.
15. Prove that if two nodes are not connected in a graph, then the rest of the nodes separate them and thus the global Markov property implies the pairwise Markov property.
16. Prove that if a variable is constant, then it is independent from any other variable.
17. Draw a graph on five nodes and list all conditional independences implied by any of the Markov properties. Do Propositions 2.4–2.7 imply additional conditional independences?
18. Show that the proofs given for Theorems 2.3 and 2.4 apply generally.
19. Give an alternative proof of the fact that under (strict) positivity, (2.9) implies joint independence.
20. Check whether (strict) positivity is needed for the result above to hold.
21. Give all conditional independences that apply to a variable in a Markov chain.
22. Give all conditional independences that apply to a variable in an Ising model.
23. Prove Theorem 2.10.
24. Create a numerical example illustrating the comment after Theorem 2.11.
25. Develop a proof of convergence of the Darroch-Ratcliff algorithm using the hint given after its description and the proof of the convergence of the IPFP in Section *LCDA* 12.2.
26. Prove Proposition 2.9.
27. Use descending and ascending classes to find the meet and the join of two log-linear models and thus prove that all the log-linear models on a set of variables are a lattice.
28. Study the proof of Theorem 2.12 in Klimova et al. [88].
29. Prove Proposition 2.10.
30. Give one possible definition of the interaction functions in (2.19) for a *CG* distribution on two categorical and two continuous variables.
31. Find a proof of Theorem 2.13.

Chapter 3
Directed Graphical Models



3.1 Introduction

Statistical or data analytic models associated with directed graphs have gained importance during the last decades not only in statistics but also in machine learning and artificial intelligence. In this chapter, only some of the basic aspects of such models are discussed. An early authoritative review of the field was given by Lauritzen [98], and a fairly recent summary that also includes applications is the edited volume by Maathuis et al. [108]. Roverato [141] gave an overview concentrating on categorical data.

Section 3.2 of this chapter introduces directed Markov properties of the pairwise, local, and global types and gives results about their equivalence. Two factorizations of the joint probability are also considered, one of them using the conditional probabilities of variables given their respective parents. The implications among these five properties and their equivalence under (strict) positivity are derived.

For directed graphs, the same Markov property appeared in the literature in two different-looking but equivalent forms. One definition, called the directed global Markov property (see, e.g., Kiiveri et al. [79] and Lauritzen [98]), discussed in this chapter, is closely related to the undirected global Markov property discussed in Sect. 2.4. The other, equivalent, variant is called dependence separation (or d-separation) (see Pearl [127]) and is used more often in the application of directed graphical models in causal analysis, to be discussed in Chaps. 4, 5, and 6.

Section 3.3 deals with the d-separation property. A detailed proof is given of the fact that d-separation of two subsets of the variables by a third one is equivalent to the latter one separating the former two in the undirected moralization of the ancestral graph of their union, which is the condition used in the directed global Markov property. This implies that conditional independence based on the d-separation is equivalent to the directed global Markov property.

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Markov models associated with DAGs are also called Bayesian networks. Their application in expert systems and, more generally, in artificial intelligence requires efficient algorithms to update the distributions of the variables of interest, when knowledge regarding other variables becomes available. Section 3.4 describes the Lauritzen-Spiegelhalter algorithm, see [101]. This algorithm operates in an (undirected) decomposable graphical model, which is called in this context a junction tree, that allows all the conditional independences defining the Bayesian network to be present. Knowledge concerning certain marginal distributions is propagated similarly to how the closed-form maximum likelihood estimate in a decomposable graphical model, which has marginal distributions equal to the observed distributions, is determined.

Section 3.5 deals with Markov models associated with chain graphs. Chain graphs have groups of variables (components) such that conditional independences may exist within them, described by a connected undirected graph within each component. There may also be arrows pointing from variables in a component to variables in other components, so that all arrows between two components point in the same direction and, further, such that the directed graph on the components with an arrow between two components if there is a variable in one, from which an arrow points to a variable in the other one, is acyclic. Remarkably, there are various sets of Markov properties associated with chain graphs proposed in the literature. This fact illustrates that the Markov properties are not intrinsic characteristics of the graphs but rather are associated with the graphs. We describe and illustrate these sets of Markov properties and quote some results about the corresponding factorizations.

The chapter concludes with a brief summary and a list of things to do.

The study of the statistical models discussed in this chapter will be continued in Chap. 10, where parameterization, estimation, and various properties of these models will be discussed in the context of a more general model class called marginal models. The reason for this is that while Markov models associated with undirected graphs, discussed in Chap. 2, assume conditional independences involving all variables, and thus are log-linear models, the models described in this chapter assume conditional independences involving only some of the variables. Thus, these models may be seen as being restrictions on certain marginal distributions of the joint distribution.

3.2 Directed Markov Properties

Similarly to the undirected case discussed in Sect. 2.4, the directed Markov property for DAGs also has pairwise, local, and global versions.

A distribution is said to obey the directed pairwise Markov property with respect to a DAG G if for every V_i and $V_j \in nd(V_i)$, $V_j \notin pa(V_i)$,

$$V_i \perp\!\!\!\perp V_j | nd(V_i) \setminus \{V_j\}.$$

The directed local Markov property is defined as

$$V_i \perp\!\!\!\perp nd(V_i) | pa(V_i). \tag{3.1}$$

Theorem 3.1 *The directed local Markov property implies the directed pairwise Markov property.*

Proof As $pa(V_i) \subseteq nd(V_i)$, by Proposition 2.6, the directed local Markov property implies that if V_j is a nondescendant but not a parent of V_i, then

$$V_i \perp\!\!\!\perp nd(V_i) | ndV_i \setminus \{V_j\}.$$

Then, Proposition 2.5 implies the directed pairwise Markov property.

□

The next theorem gives an alternative form of the directed local Markov property.

Theorem 3.2 *The directed local Markov property (3.1) is equivalent to the following:*

$$V_i \perp\!\!\!\perp nd(V_i) \setminus pa(V_i) | pa(V_i). \tag{3.2}$$

Proof Proposition 2.5 implies that (3.2) follows from (3.1). As the graph is a DAG, $pa(V_i) \subseteq nd(V_i)$, thus,

$$V_i \perp\!\!\!\perp pa(V_i) | nd(V_i),$$

always holds, and Proposition 2.7, with $X = V_i$, $Y = nd(V_i) \setminus pa(V_i)$, $Z = W = pa(V_i)$ gives the reverse implication.

□

Thus, (3.2) is equivalent to the directed local Markov property and this was the form used in Chapter *LCDA* 13.

The global Markov property (see Kiiveri et al. [79]), for DAGs, or the directed global Markov property, is that for the joint distribution of the nodes (variables), for any three pairwise disjoint subsets $\mathscr{A}, \mathscr{B}, \mathscr{C} \subseteq V$,

$$\mathscr{A} \perp\!\!\!\perp \mathscr{B} | \mathscr{C} \tag{3.3}$$

holds, whenever $\mathscr{C}$ separates $\mathscr{A}$ and $\mathscr{B}$ in the graph $G^{UM}_{anc(\mathscr{A} \cup \mathscr{B} \cup \mathscr{C})}$, which is the undirected moralization of the subgraph of G defined by the smallest ancestral set (subgraph) containing $\mathscr{A} \cup \mathscr{B} \cup \mathscr{C}$.

To illustrate the definition, consider the DAG in Fig. 3.1 and determine whether the directed global Markov property implies that

$$(A, B) \perp\!\!\!\perp (E, F) | C. \tag{3.4}$$

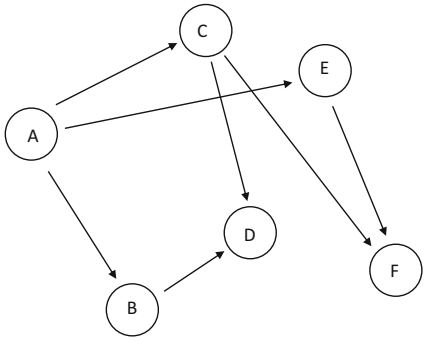


Fig. 3.1 Does the global Markov property imply that $(A, B) \perp\!\!\!\perp (E, F)|C$?

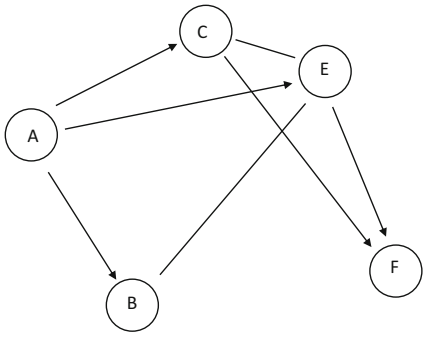


Fig. 3.2 The moralization (without the removal of direction) of the smallest ancestral subgraph containing $\{A, B, C, E, F\}$ from the DAG in Fig. 3.1

As the only node outside of $\{A, B, C, E, F\}$ is D, and it is not a parent of any other node, $anc(A, B, C, E, F)$ is obtained by deleting D and the moralization of the resulting graph is given in Fig. 3.2. Note that B and C would have to be connected, if moralization took place in the original graph, but as moralization takes place in the ancestral graph, they are not connected.

Now arrows in Fig. 3.2 need to be replaced by edges, and then one has to decide whether in the resulting graph, $\{A, B\}$ are separated from $\{E, F\}$ by $\{C\}$. The answer is negative, as there is an edge between A and E (and also between B and E). Therefore, the directed global Markov property does not imply (3.4).

Theorem 3.3 *If the directed global Markov property holds for the distribution P with respect to a DAG G, then the directed local Markov property also holds for P with respect to G.*

Proof It is enough to see that $pa(V_i)$ separates V_i and $nd(V_i) \setminus pa(V_i)$ in the graph $G^{UM}_{anc(\{V_i\}\cup nd(V_i))}$. In $\{V_i\} \cup nd(V_i)$, there are no children of V_i included, and this also holds for $anc(\{V_i\} \cup nd(V_i))$, because a child of V_i cannot be an ancestor of a nondescendant of V_i. Therefore, the moralization of $anc(\{V_i\}\cup nd(V_i))$ does not add any edges involving V_i. Thus, all the nodes connected to V_i in $G^{UM}_{anc(\{V_i\}\cup nd(V_i))}$ are

parents of it in the original graph. Consequently, $pa(V_i)$ separates V_i and $nd(V_i) \setminus pa(V_i)$ in $G^{UM}_{anc(\{V_i\} \cup nd(V_i))}$.

□

It will be shown that the directed global Markov property with respect to a DAG G is closely related to a recursive factorization of the joint probability distribution P of the variables in V. The components of this factorization are real functions f_i, for all $V_i \in V$, such that the domain of f_i is the marginal table of the variables $\{V_i\} \cup pa(V_i) \subseteq V$. Using the notation $(v)_{\{V_i\} \cup pa(V_i)}$ to select the indices of the variables in the subscript from v, the factorization is that

$$P(v) = \prod_{V_i \in V} f_i((v)_{\{V_i\} \cup pa(V_i)}). \tag{3.5}$$

When the variables in V are ordered according to a well-numbering (see Theorem 2.1), then f_i in (3.5) depends on the categories of V_i and some of the earlier variables, thus, the factorization is recursive. Of course, for any ordering of the variables, there is a factorization of any distribution P using conditional probabilities, conditioned on all earlier variables, if all these conditional probabilities exist:

$$P(v) = \prod_{i=1}^{|V|} P(v_i | v_1, \ldots, v_{i-1}), \tag{3.6}$$

where v_j is $(v)_{V_j}$ for $j = 1, \ldots, i$. The existence of the conditional probabilities in (3.6) is implied by the joint distribution being (strictly) positive, but this condition is only sufficient and is not necessary.

The important property of the factorization in (3.5) is that the functions f_i may be chosen to be the conditional probabilities; if they exist, as in (3.6) and instead of all the earlier variables, it is enough to condition only on the parents of the actual variable:

$$P(v) = \prod_{i=1}^{|V|} P(v_i | (v)_{pa(V_i)}). \tag{3.7}$$

More precisely, the relationship between the existence of the factorizations of the forms (3.5) and (3.7) is the following.

Theorem 3.4 *For the distribution P on the nodes of a DAG, the following statements are identical:*

(i) P has a factorization of the form (3.5) and for every $V_i \in V$ and every category combination v, $P(v_i|(v)_{pa(V_i)})$ exists.
(ii) P has a factorization of the form (3.7).

Proof First, we show that (3.5) implies (3.7) if the relevant conditional probabilities exist. The proof is by induction on $|V|$. For $|V| = 1$, there is nothing to prove. For $|V| \geq 2$, consider the variables in a well-numbering, see Theorem 2.1. For $|V| = 2$, because of the well-numbering, V_1 has no parent, and (3.7) holds, whether or not

V_1 is the parent of V_2. It is also clear that for $|V| \leq 2$,

$$f_i((v)_{\{V_i\}\cup pa(V_i)}) = P(v_i|(v)_{pa(V_i)}), \tag{3.8}$$

for $i = 1, 2$, is a possible choice.

Suppose the claim including (3.8) is proved for fewer variables than $|V|$. For a joint index $v = (v_1, \ldots, v_{|V|-1}, v_{|V|})$, consider the factorization in (3.5) and using the fact that $V_{|V|}$ is not a parent of any of the variables, write the $P(v_1, \ldots, v_{|V|-1})$ marginal probability as

$$P(v_1, \ldots, v_{|V|-1}) = \left(\sum_{v_{|V|}} f_{|V|}((v)_{\{V_{|V|}\}\cup pa(V_{|V|})})\right) \prod_{i=1}^{|V|-1} f_i((v)_{\{V_i\}\cup pa(V_i)})$$

$$= S \prod_{i=1}^{|V|-1} f_i((v)_{\{V_i\}\cup pa(V_i)}),$$

for simplicity of notation.

Then, because of the assumed existence of the relevant conditional probabilities, $P(v_1, \ldots, v_{|V|-1})$ is positive, and S is also positive, thus,

$$P(v_{|V|}|v_1, \ldots, v_{|V|-1}) = \frac{P(v_1, \ldots, v_{|V|})}{P(v_1, \ldots, v_{|V|-1})}$$

$$= \frac{\prod_{i=1}^{|V|} f_i((v)_{\{V_i\}\cup pa(V_i)})}{S\prod_{i=1}^{|V|-1} f_i((v)_{\{V_i\}\cup pa(V_i)})} = \frac{f_{|V|}((v)_{\{V_{|V|}\}\cup pa(V_{|V|})})}{S},$$

and using (3.5), the last two results and the induction assumption, one has that

$$P(v_1, \ldots, v_{|V|}) = \prod_{i=1}^{|V|} f_i((v)_{\{V_i\}\cup pa(V_i)})$$

$$= \left(\frac{f_{|V|}((v)_{\{V_{|V|}\}\cup pa(V_{|V|})})}{S}\right)\left(S \prod_{i=1}^{|V|-1} f_i((v)_{\{V_i\}\cup pa(V_i)})\right)$$

$$= P(v_{|V|}|v_1, \ldots, v_{|V|-1})P(v_1, \ldots, v_{|V|-1})$$

$$= P(v) = \prod_{i=1}^{|V|} P(v_i|(v)_{pa(V_i)}).$$

Obviously, the factorization in (3.7) is of the type (3.5), which proves the other direction and concludes the proof.

□

Note that in the case of (strictly) positive probability distributions, the condition that all the conditional probabilities exist holds.

The next result is parallel to Theorem *LCDA* 11.6.

Theorem 3.5 *For a joint distribution P of variables V and a DAG $G = (V, A)$, P factorizes as in (3.7) if and only if it obeys the directed global Markov property with respect to G and the conditional probabilities in (3.7) exist.*

Proof First, let P factorize as in (3.7). Then, the conditional probabilities $P(v_i|(v)_{pa(V_i)})$ for all $V_i \in V$ and all category combinations v exist. Let $\mathscr{A}$, $\mathscr{B}$, $\mathscr{C}$ be pairwise disjoint subsets of the variables. Determine the marginal distribution of P on $anc(\mathscr{A} \cup \mathscr{B} \cup \mathscr{C})$. This requires $P(v)$ to be summed for all variables $V_j \in V \setminus anc(\mathscr{A} \cup \mathscr{B} \cup \mathscr{C})$, for all possible categories v_j.

As G is a DAG, its subgraph induced by $V \setminus anc(\mathscr{A} \cup \mathscr{B} \cup \mathscr{C})$, say G', is also a DAG, thus, there is a variable $V_j \in V \setminus anc(\mathscr{A} \cup \mathscr{B} \cup \mathscr{C})$, which does not have a child in G'. As this V_j is not in the ancestral set $anc(\mathscr{A} \cup \mathscr{B} \cup \mathscr{C})$, it also does not have a child in $anc(\mathscr{A} \cup \mathscr{B} \cup \mathscr{C})$. Thus, V_j is not the parent of any node in G; therefore, in the factorization (3.7), it never appears in the condition. Consequently, marginalizing over the variable V_j leads to omitting the term $P(v_j|(v)_{pa(V_j)})$ from (3.7).

By repeated application of the argument above for $V \setminus \{V_j\}$ instead of V, and so on, one obtains for the marginal distribution that

$$P_{anc(\mathscr{A}\cup\mathscr{B}\cup\mathscr{C})}((v)_{anc(\mathscr{A}\cup\mathscr{B}\cup\mathscr{C})}) = \prod_{V_i \in anc(\mathscr{A}\cup\mathscr{B}\cup\mathscr{C})} P(v_i|(v)_{pa(V_i)}). \tag{3.9}$$

Note that $pa(V_i) \subseteq anc(\mathscr{A} \cup \mathscr{B} \cup \mathscr{C})$, if $V_i \in anc(\mathscr{A} \cup \mathscr{B} \cup \mathscr{C})$.

In the undirected moralization of the subgraph induced by $anc(\mathscr{A} \cup \mathscr{B} \cup \mathscr{C})$, the sets $\{V_i\} \cup pa(V_i)$ are complete, thus, with respect to that undirected graph (3.9) is a Gibbs factorization. Therefore, Theorem 2.6 applies and the directed global Markov property follows.

The converse is shown by induction on $|V|$. If $V = \{A, B\}$, then one node, say A, does not have a parent (such a node exists because G is a DAG) and

$$P(A, B) = P(A)P(B|A), \tag{3.10}$$

and the factorization holds for $|V| = 2$, whether or not A is the parent of B. If it is, then the conditional probability above exists by assumption. If not, then the global Markov property implies that $A \perp\!\!\!\perp B$, and then $P(B|A) = P(B)$.

Assume now that the result holds for all DAGs with fewer than $|V|$ nodes. Let G be a DAG with nodes V and let C be a node that has no child in G. Such a terminal node exists, because G is a DAG. Clearly, all directed paths from any node

in $nd(C) \setminus pa(C)$ to C go through $pa(C)$ and thus $nd(C) \setminus pa(C)$ and $\{C\}$ are separated by $pa(C)$ in G. Then, the separation also holds in any subgraph based on a subset of V, which contains $pa(C)$, including

$$anc(\{C\} \cup nd(C) \setminus pa(C) \cup pa(C)).$$

Thus, $nd(C) \setminus pa(C)$ and C are separated by $pa(C)$ in $G_{anc(\{C\} \cup nd(C) \setminus pa(C) \cup pa(C))}$. As C has no child in G, moralization does not add any edges containing C, thus, the separation also holds in $G^{UM}_{anc(\{C\} \cup nd(C) \setminus pa(C) \cup pa(C))}$.

Then, the directed global Markov property means that

$$C \perp\!\!\!\perp nd(C) \setminus pa(C) | pa(C).$$

As C has no children, $nd(C) = V \setminus \{C\}$, thus, one has

$$C \perp\!\!\!\perp V \setminus \{C\} \setminus pa(C) | pa(C).$$

As

$$\{C\} \cup (V \setminus \{C\} \setminus pa(C)) \cup pa(C) = V,$$

the conditional independence above implies that $P(V)$ has a factorization as

$$P(V) = P(V \setminus \{C\}) \frac{P(\{C\} \cup pa(C))}{P(pa(C))} = P(V \setminus \{C\}) P(C | pa(C)), \tag{3.11}$$

where the existence of the conditional distribution was assumed. In (3.11), the second term of the last formula depends on $\{C\} \cup pa(C)$ and the first term is the distribution of the variables $V \setminus \{C\}$. As $G_{V \setminus \{C\}}$ is a DAG with fewer than $|V|$ nodes, the induction assumption completes the proof.

□

For the relationship between the factorizations in (3.6) and (3.7), one has the following simple result.

Proposition 3.1 *If the directed global Markov property holds for the joint distribution P with respect to the DAG, and if $V_1, V_2, V_3, \ldots, V_k$ is a well-numbering of the variables V, then the factorizations in (3.6) and (3.7) coincide, in the sense that*

$$P(v_i | v_1, \ldots, v_{i-1}) = P(v_i | (v)_{pa(V_i)})$$

for all V_i, $i = 2, 3, \ldots, k$.

Proof As the directed global Markov property implies the directed local Markov property for P with respect to G (see Theorem 3.3),

$$V_i \perp\!\!\!\perp nd(V_i) \setminus pa(V_i) | pa(V_i).$$

Further, in the well-numbering, all descendants of V_i are after it, so the variables preceding V_i are all nondescendants of it. Then, by Proposition 2.5, one also has

$$V_i \perp\!\!\!\perp \{V_1, \ldots, V_{i-1}\} \setminus pa(V_i) | pa(V_i),$$

and then, the conditional distributions of V_i, when conditioned on $V_1, \ldots, V_{i-1}$ or only on $pa(V_i)$, are the same, see Proposition 2.3.

□

For example, if the directed global Markov property with respect to the DAG given in Fig. 3.1 holds for P, then

$$P(A, B, C, D, E, F) = P(A)P(B|A)P(C|A)P(D|B, C)P(E|A)P(F|C, E).$$

The results concerning the directed Markov properties given so far may be summarized by saying that for a DAG and a distribution, the directed global Markov property implies the local one and the local one implies the pairwise property. Further, the directed global Markov property is equivalent to the factorization (3.7), if the existence of the conditional probabilities of variables given their respective parents is also assumed. No (strict) positivity assumption was needed to prove these results, but if that condition holds, then, of course, the conditional probabilities exist. To complete the discussion of the implications among the directed Markov properties, we prove now that the directed local Markov property also implies the factorization in (3.7).

Theorem 3.6 *Suppose, for the joint distribution P of the variables V, the directed local Markov property with respect to a DAG $G = (V, A)$ holds and the $P(V_i|pa(V_i))$ conditional probabilities exist. Then, P has a factorization of the form (3.7).*

Proof Let $V = \{V_1, \ldots, V_k\}$. Because G is a DAG, it has a node with no child, and assume it is V_k. Then, V_k has no descendants and by the directed local Markov property,

$$V_k \perp\!\!\!\perp V \setminus \{V_k\} \setminus pa(V_k) | pa(V_k).$$

Thus,

$$P(V_k|V_1, \ldots, V_{k-1}) = P(V_k|pa(V_k)),$$

and

$$P(v) = P((v)_{(V_1,\ldots,V_{k-1})})P((v)_{(V_k)}|v_{pa(V_k)}).$$

Consider now the subgraph of G obtained by omitting V_k and the arrows which involve it, say G_k. Obviously, G_k is also a DAG. Further, the directed local

Markov property also holds for G_k. To see this, note that the directed local Markov property states conditional independences, where the parents of a variable are in the condition. As V_k has no child in G, if from all the conditional independences implied by the directed local Markov property for G, the variable V_k is omitted, no conditioning set is altered, thus, the reduced conditional independences remain valid, see Proposition 2.5. These are the conditional independences implied by the directed local Markov property for G_k.

Then, the starting argument of the proof may be applied to G_k, and recursively, one obtains the required factorization.

□

The equivalence of the directed pairwise and directed local Markov properties holds generally under the assumption of (strict) positivity only. Note that Proposition 3.1 gave the implication in one direction.

Theorem 3.7 *For a DAG $G = (V, A)$ and a (strictly) positive joint probability distribution P of the variables V, if the directed pairwise Markov property with respect to G holds for P, then so does the directed local Markov property.*

Proof For a variable $A \in V$, let $B, C \in nd(A)$, $B, C \notin pa(A)$. Then, the directed pairwise Markov property implies that

$$A \perp\!\!\!\perp B | \{C\} \cup nd(A) \setminus \{B\} \setminus \{C\}$$

and

$$A \perp\!\!\!\perp C | \{B\} \cup nd(A) \setminus \{B\} \setminus \{C\}.$$

In case of (strict) positivity, Proposition 2.8 implies that

$$A \perp\!\!\!\perp B, C | nd(A) \setminus \{B, C\}. \tag{3.12}$$

Then, let $D \in nd(A)$, $D \notin pa(A)$ different from B and C and apply, again, the directed pairwise Markov property yielding that

$$A \perp\!\!\!\perp D | \{B, C\} \cup nd(A) \setminus \{B, C\} \setminus \{D\}.$$

The conditional independence in (3.12) may be written as

$$A \perp\!\!\!\perp B, C | \{D\} \cup nd(A) \setminus \{B, C\} \setminus \{D\},$$

and the last two conditional independences, using Proposition 2.8 imply that

$$A \perp\!\!\!\perp B, C, D | nd(A) \setminus \{B, C, D\}.$$

By the same procedure, all variables in $nd(A) \setminus pa(A)$ may be moved to the right-hand side of the conditional independence, yielding that

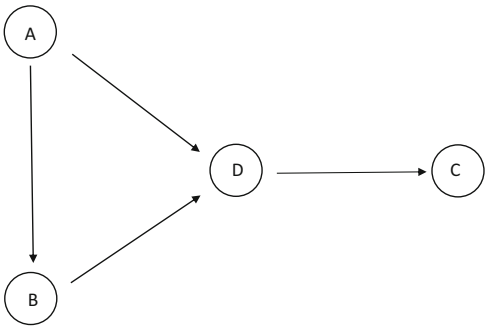


Fig. 3.3 DAG, according to which a distribution is directed pairwise Markov but not directed local Markov

$$A \perp\!\!\!\perp nd(A) \setminus pa(A) | pa(A),$$

which is the conditional independence required by the directed local Markov property for A.

□

The directed pairwise Markov property does not necessarily imply the directed local Markov property without the assumption of (strict) positivity. This is illustrated with an example taken from Lauritzen [98]. In this example, there are four variables, A, B, C, D, all binary with uniform distributions. As far as the joint distribution is concerned, $A = B = C$ and D is independent from the joint ABC distribution. Clearly, there are cells with zero probabilities in this distribution, namely, when A, B, C are not all equal to each other.

The distribution is directed pairwise Markov with respect to the DAG shown in Fig. 3.3. As seen easily, the directed pairwise Markov property means two conditional independences:

$$C \perp\!\!\!\perp A | B, D$$

and

$$C \perp\!\!\!\perp B | A, D.$$

These hold, because given the conditions, C is constant[1]. The directed local Markov property requires that

$$C \perp\!\!\!\perp A, B | D,$$

which does not hold, as the joint distribution of A, B, C does not depend on the condition, and in the joint distribution, $A = B = C$.

[1] See item #16 in Sect. 2.9.

The implications among the various directed Markov properties and the related factorizations are summarized in the next theorem.

Theorem 3.8 *For every probability distribution P of the variables in V and DAG $G = (V, A)$, the factorizations in (3.5) and (3.7) and the directed global Markov property and the directed local Markov property are equivalent, if the relevant conditional probabilities exist. Further, they imply the directed pairwise Markov property for P. If P is (strictly) positive, then the directed pairwise Markov property implies the other directed Markov properties and the factorizations, if the relevant conditional probabilities exist.*

□

In this section, Markov properties were given a strictly mathematical discussion, without reference to the reality one wants to model in a statistical analysis. One has to admit that, while many of the Markov properties defined so far had a clear intuitive meaning, the directed global Markov property does not easily lend itself to interpretation. In the next section, an equivalent but also very technical looking property is discussed. The intuitive meanings and potential uses of these properties will be discussed in the next chapter, where they will be linked to graphical models of certain causal structures.

3.3 Dependence Separation

Models associated with DAGs are used to model directed effects. The goal of the developments in this section is to build a formal treatment of the concept of effect or information originating in a subset of variables, traveling through the arrows of the DAG and reaching another subset of variables. A particularly relevant situation is when all the effect or information from, say, $\mathscr{A}$ to $\mathscr{B}$ is going through a third subset, say, $\mathscr{C}$. In this case, intuition suggests that if the variables in $\mathscr{C}$ are kept constant, that is, are not allowed to change, then they block the flow of information or effect from $\mathscr{A}$ to $\mathscr{B}$. If all the information from $\mathscr{A}$ to $\mathscr{B}$ goes through $\mathscr{C}$, then, when the flow is not blocked by $\mathscr{C}$ being kept constant, it is said that $\mathscr{C}$ induces a relationship.

Let $\mathscr{A}, \mathscr{B}, \mathscr{C}$ be disjoint subsets of the variables and let $G = (V, A)$ be a DAG.[2] Let $\mathscr{P}$ be a path, irrespective of the direction of the arrows,[3] connecting an $A \in \mathscr{A}$ and a $B \in \mathscr{B}$. From the perspective of blocking or inducing relationships (using other words: to have a closed or open path of the flow of information) between $\mathscr{A}$ and $\mathscr{B}$, obviously only the part of the path outside of these subsets is relevant. Therefore, related to the concept of dependence separation, only paths that contain only one $A \in \mathscr{A}$ and one $B \in \mathscr{B}$ are considered. Considering only paths that do not

[2] There is a bit of conflict of notation here but the precise meaning is always clear from the context: A denotes here the collection of the arrows of the graph, but A is also used for a variable, to keep the notation simple.

[3] A path that disregards the direction of the arrows is also called a trail, see, e.g., Geiger et al. [64].

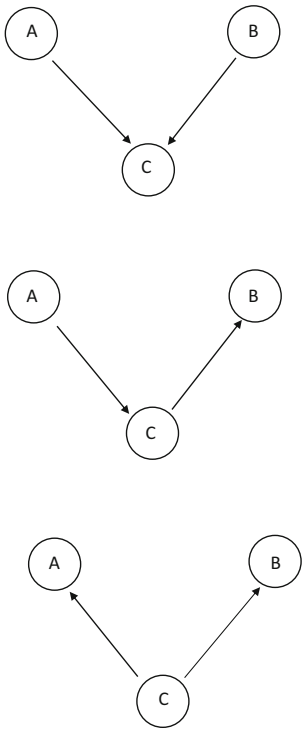


Fig. 3.4 Collider, intermediate (or chain), and fork configurations

visit more than one variable neither in $\mathscr{A}$ nor $\mathscr{B}$ is not a restriction of generality, as clearly every path that contains nodes from both $\mathscr{A}$ and $\mathscr{B}$ may be transformed into such a path by omitting nodes in the path from $\mathscr{A}$, except for the last such one, and omitting nodes from $\mathscr{B}$, except for the first one. Let the number of nodes along the path outside of $\mathscr{A}$ and of $\mathscr{B}$ be the length of the path. Every node in $\mathscr{P}$, except for A and B, has two arrows attached to it in $\mathscr{P}$, thus, $\mathscr{P}$ may contain intermediate nodes (incoming and outgoing arrows), forks (both arrows outgoing), or colliders (both arrows incoming). Intermediate nodes and fork nodes will be called noncolliders. These configurations are illustrated in Fig. 3.4, see also Figure *LCDA* 8.1.

The concepts developed here are used to model certain types of causal structures, to be discussed in the next chapter. In those types of causal structures, and possibly in other ones too, the three kinds of configurations have special roles in transmitting the information or inducing information. In particular, conditioning on a collider C or on a descendant of it implies association between the parents, say, A and B if without this conditioning, they are independent. On the other hand, conditioning on a noncollider C implies conditional independence between its neighbors.

Then, $\mathscr{A}$ and $\mathscr{B}$ are dependence connected (d-connected) (see Geiger et al. [64], Pearl [127], and Spirtes et al. [157]) by $\mathscr{C}$, if there exists a path $\mathscr{P}$ connecting an $A \in \mathscr{A}$ with a $B \in \mathscr{B}$ disregarding the direction of the arrows, such that

(i) if C is a collider in $\mathscr{P}$, then C or a descendant of it is in $\mathscr{C}$
and
(ii) no noncollider in $\mathscr{P}$ is in $\mathscr{C}$.

The intuitive meaning of being d-connected is that when conditioning on $\mathscr{C}$, then effect travels along the path $\mathscr{P}$. If conditioning happens on a collider or a descendant of it, then association is created[4] between the parents, and no association is blocked by conditioning on a noncollider, which would imply conditional independence, that is, would block the flow of information. It has to be emphasized, however, that this interpretation only applies to particular causal structures. In the current section, d-separation and the related d-connection are treated as abstract properties, without relying on any claim as to their deeper meanings or relevance.
Note that if an $A \in \mathscr{A}$ and a $B \in \mathscr{B}$ are connected by an arrow, then there is a path of length zero between $\mathscr{A}$ and $\mathscr{B}$, and (i) and (ii) will hold for any $\mathscr{C}$, including the empty set.
Dependence separation (d-separation) is the opposite of being d-connected. That is, $\mathscr{A}$ and $\mathscr{B}$ are d-separated by $\mathscr{C}$, if for every path $\mathscr{P}$ connecting an $A \in \mathscr{A}$ with a $B \in \mathscr{B}$ disregarding the direction of the arrows,

(iii) there is a collider in $\mathscr{P}$, such that neither this collider nor a descendant of it is in $\mathscr{C}$
or
(iv) there is a noncollider of $\mathscr{P}$ contained in $\mathscr{C}$.

For example, is $\{A, B, D\}$ d-separated from $\{F\}$ by $\{C, E\}$ in the graph in Fig. 3.1?

There are three paths connecting $\{A, B, D\}$ with $\{F\}$, namely, $A - C - F$, $A - E - F$, and $D - C - F$. In $A - C - F$, C is a noncollider; in $A - E - F$, E is a noncollider; and in $D - C - F$, C is a noncollider. Thus, (iv) holds for all paths connecting $\{A, B, D\}$ with $\{F\}$, thus, d-separation holds.

As another example, see whether in the graph in Fig. 3.1 $\{B\}$ is d-separated from $\{E, F\}$ by $\{C\}$. In the $B - D - C - F$ path, D is a collider without descendants that is not in $\{C\}$, so (iii) holds for this path. In the $B - A - E$ path, A is a noncollider and the length of this path is 1. As A is not in $\{C\}$, neither (iii) nor (iv) holds for this path. Finally, there is the $B - A - C - F$ path. It has two noncolliders, A and C, and as $C \in \{C\}$, (iv) holds. Thus, there is a path such that neither (iii) nor (iv) holds, and d-separation is not true. When conditioned on C, the $B - A - E$ path creates d-connection between $\{B\}$ and $\{E, F\}$. The intuitive meaning is that even if conditioned on C, A has influence on both B and E, creating a connection.

A joint distribution P of the variables in V is said to obey the d-separation rule according to a DAG $G = (V, A)$, if

$$\mathscr{A} \perp\!\!\!\perp \mathscr{B} | \mathscr{C}, \text{ whenever } \mathscr{C} \text{ d-separates } \mathscr{A} \text{ from } \mathscr{B}. \tag{3.13}$$

[4] This is strongly intuitive. For a precise result, see Theorem 4.18.

The concept of the d-separation rule has been introduced here with reference to a possible intuitive meaning with respect to causal analyses to be developed in the next chapter. As was mentioned earlier, so far no intuitive interpretation has been given for the directed global Markov property. The next result implies that these properties are equivalent.

Theorem 3.9 *For a set of variables V, let $G = (V, A)$ be a DAG. Then, for pairwise disjoint $\mathscr{A}, \mathscr{B}, \mathscr{C} \subseteq V$, the property that $\mathscr{C}$ separates $\mathscr{A}$ and $\mathscr{B}$ in the graph $G^{UM}_{anc(\mathscr{A}\cup\mathscr{B}\cup\mathscr{C})}$ is equivalent to d-separation.*

Proof Recall that $G^{UM}_{anc(\mathscr{A}\cup\mathscr{B}\cup\mathscr{C})}$ is the undirected moralization of the subgraph of G defined by the smallest ancestral set containing $\mathscr{A} \cup \mathscr{B} \cup \mathscr{C}$. So $G^{UM}_{anc(\mathscr{A}\cup\mathscr{B}\cup\mathscr{C})}$ is an undirected graph, which contains $\mathscr{A} \cup \mathscr{B} \cup \mathscr{C}$ and their ancestors, as nodes, and in addition to the undirected versions of the original arrows, also parents with a common child are connected.

The proof shows that the opposites of the two properties imply each other and are, thus, equivalent.

Let $\mathscr{P}_1$ be a path in $G^{UM}_{anc(\mathscr{A}\cup\mathscr{B}\cup\mathscr{C})}$ between $A \in \mathscr{A}$ and $B \in \mathscr{B}$, which does not go through $\mathscr{C}$, that is, $\mathscr{C}$ does not separate $\mathscr{A}$ and $\mathscr{B}$ in $G^{UM}_{anc(\mathscr{A}\cup\mathscr{B}\cup\mathscr{C})}$. A path $\mathscr{P}_2$ will be constructed by a series of modifications of $\mathscr{P}_1$, which d-connects $\mathscr{A}$ and $\mathscr{B}$ by $\mathscr{C}$.

Consider the path $\mathscr{P}_1$ in $G^{UM}_{anc(\mathscr{A}\cup\mathscr{B}\cup\mathscr{C})}$ semi-directed in the sense that equip it with the original arrows from $G_{anc(\mathscr{A}\cup\mathscr{B}\cup\mathscr{C})}$. In addition to arrows, $\mathscr{P}_1$ may also have undirected edges added during moralization. Recall that by assumption, with the exception of A and B, $\mathscr{P}_1$ runs entirely outside of $\mathscr{A} \cup \mathscr{B} \cup \mathscr{C}$.

If $\mathscr{P}_1$ does not contain undirected edges added because of the moralization, then set $\mathscr{P}_2 = \mathscr{P}_1$ and skip the next two paragraphs.

If $\mathscr{P}_1$ does contain undirected edges added because of the moralization, then the first step of the creation of $\mathscr{P}_2$ is to replace every undirected edge. If such an edge connects nodes E and F, then these were connected during moralization because they had a common child in $G_{anc(\mathscr{A}\cup\mathscr{B}\cup\mathscr{C})}$, say C. Then, replace the edge $E - F$ with $E - C - F$ with the original arrows, where now C is a collider. Assume, without restriction of generality, that E is closer to A than F is, and then also F is closer to B than E is. If the node C is in $\mathscr{A}$, then replace the part of the path connecting A to E with $C - E$. The new path connects $A = C \in \mathscr{A}$ and B; runs entirely outside of $\mathscr{A} \cup \mathscr{B}$, with the exception of A and B, but within $G_{anc(\mathscr{A}\cup\mathscr{B}\cup\mathscr{C})}$; and has one fewer undirected edges than the original path. If the node C is in $\mathscr{B}$, then replace the part connecting F to B with $F - C$.

Apply the procedure described in the previous paragraph for all undirected edges. $\mathscr{P}_2$ now connects A and B, with the exception of the endpoints is in $G_{anc(\mathscr{A}\cup\mathscr{B}\cup\mathscr{C})} \setminus (\mathscr{A} \cup \mathscr{B})$, and it only has arrows. $\mathscr{P}_2$ may have colliders (including the ones just added) in $\mathscr{C}$, but all of its noncolliders are outside of $\mathscr{C}$, because no noncollider was added.

Consider now the nodes in $\mathscr{P}_2$. If a node C is a collider, it is not one of the endpoints of the path. Then, it may be in $\mathscr{C}$ or may be an ancestor of one of the nodes in $\mathscr{A} \cup \mathscr{B} \cup \mathscr{C}$ but is not in $\mathscr{A} \cup \mathscr{B}$. If it is not in $\mathscr{C}$ and is not an ancestor of a

node in $\mathscr{C}$, but is an ancestor of an $A' \in \mathscr{A}$ (or is an ancestor of a $B' \in \mathscr{B}$), replace the part of the path from A to C (from C to B) by the path from A' to C (from C to B'), using A' (B') closer to C. Now one of the arrows points away from C, thus, C is not a collider. The new part of the path from A' to C (from C to B') does not go through $\mathscr{C}$, because in that case, C would be an ancestor of a node in $\mathscr{C}$. By repeated application of this, all colliders that are not in $\mathscr{C}$ and are not ancestors of nodes in $\mathscr{C}$ are removed from $\mathscr{P}_2$, which still connects $\mathscr{A}$ with $\mathscr{B}$. All the remaining colliders in $\mathscr{P}_2$ are either in $\mathscr{C}$ or are ancestors of nodes in $\mathscr{C}$. Thus, for $\mathscr{P}_2$, condition (i) of being d-connected holds.

As far as the noncolliders in the path $\mathscr{P}_2$ are concerned, they were either in the path before the transformations in the previous paragraphs were applied, and in that case outside of $\mathscr{C}$, or were added during those transformations and in that case are also outside of $\mathscr{C}$, which is condition (ii) of being d-connected.

To see the converse, suppose now that there is a path $\mathscr{P}_1$, connecting an $A \in \mathscr{A}$ with a $B \in \mathscr{B}$, with properties (i) and (ii). First, it will be shown that $\mathscr{P}_1$ is contained in $anc(\mathscr{A} \cup \mathscr{B} \cup \mathscr{C})$, and then another path, $\mathscr{P}_2$, will be constructed in $G^{UM}_{anc(\mathscr{A} \cup \mathscr{B} \cup \mathscr{C})}$ connecting $\mathscr{A}$ to $\mathscr{B}$, but not going through $\mathscr{C}$.

The first claim is true for the colliders in $\mathscr{P}_1$ because for every collider C in $\mathscr{P}_1$, (i) implies that $C \in \mathscr{C} \cup anc(\mathscr{C}) \subseteq anc(\mathscr{A} \cup \mathscr{B} \cup \mathscr{C})$. For every noncollider C, there is, at least, one arrow pointing away from it. When one follows the direction of this arrow as long as possible, then one either arrives at one of the endpoints A or B, and then C is in the ancestral set, or one arrives at a collider, which, as was seen above, is in the ancestral set, so also C is in the ancestral set.

The noncolliders of $\mathscr{P}_1$ are outside of $\mathscr{C}$ by (ii). Consider the colliders of $\mathscr{P}_1$. Let C be such a collider. Then, its parents in $\mathscr{P}_1$ cannot be colliders, as an arrow in the path points away from them toward C. Consequently, by (ii), the parents are not in $\mathscr{C}$. Create $\mathscr{P}_2$ from $\mathscr{P}_1$ by avoiding the colliders by going through the edges connecting their parents in the moral graph. Then, $\mathscr{P}_2$ is a path in $G^{UM}_{anc(\mathscr{A} \cup \mathscr{B} \cup \mathscr{C})}$ between $\mathscr{A}$ and $\mathscr{B}$, which avoids $\mathscr{C}$, thus, the proof is complete.

□

In the first example after the definition of d-separation, it was seen that $\{A, B, D\}$ and $\{F\}$ are d-separated by $\{C, E\}$ in the graph in Fig. 3.1. To check whether the separation, which is the condition of the directed global Markov property, holds, first note that as all nodes of G are involved, the ancestral graph is the same. As F has no children, the moralization is not going to add new edges involving it, so in the undirected moralization, $\{A, B, D\}$ and $\{F\}$ are separated by $\{C, E\}$.

While the above example illustrates that d-separation and the directed global Markov property are equivalent (both hold in the example), it cannot be seen as entirely satisfactory. It seems that the reasons why both properties hold are unrelated, and, if so, the example does not sufficiently illustrate that the two concepts are identical. Indeed, d-separation was implied by C and E not being colliders on any of the paths, and the directed global Markov property was implied by F being accessible from any of A, B, D only through C or E in the moralized graph, just like in the original one. In reality, these two facts are related. If C was a collider on the path $D - C - F$, then the directions of the CD and of the CF arrows would be

the opposites of what they really are. But in this case, moralization would connect D and F, creating a path between $\{A, B, D\}$ and $\{F\}$, which avoids $\{C, E\}$. Similarly, if E was a collider, moralization would add an edge between A and F.

To illustrate the argument of the proof of Theorem 3.9, the second example discussed after the definition of d-separation is used. It was shown that $\{B\}$ is not d-separated by $\{C\}$ from $\{E, F\}$. It was seen that in the $B - A - E$ path, A was a noncollider which is not in $\{C\}$, thus (ii) holds and there are no colliders in the path, so this path d-connects $\{B\}$ and $\{E, F\}$ by $\{C\}$. Let now $B - A - E$ be path $\mathscr{P}_1$ in the second part of the proof. There is no collider in the path, only a noncollider, A. An arrow points away from it, which arrives in E. Thus, A is in the ancestral set of $\{E, F\}$ and, consequently, of $\{B, C, E, F\}$. Therefore, the $\mathscr{P}_2$=$B - A - E$ path is in the undirected moralization of the ancestral set of $\{B, C, E, F\}$.

Conversely, to illustrate the first part of the proof, let now the $\mathscr{P}_1$=$B-A-E$ be a path in the undirected moralization of the ancestral set of $\{B, C, E, F\}$, which does not go through $\{C\}$. This path does not involve edges added during moralization, thus, $\mathscr{P}_1 = \mathscr{P}_2$. It only contains noncolliders, which are not in $\{C\}$, thus, (ii) holds for the path.

Now the equivalence of the d-separation rule and the global Markov property follows easily.

Theorem 3.10 *Let V be a set of variables, $G = (V, A)$ a DAG, and P the joint distribution of the variables. Then, the directed global Markov property holds for P with respect to G if and only if the distribution P obeys the d-separation rule with respect to G.*

□

The next section deals with applications of directed graphical models, which often appear under the name of Bayesian networks or probabilistic expert systems.

3.4 Bayesian Networks

Models defined by Markov properties associated with DAGs are also called Bayesian networks, see, e.g., Barber [9] and Geiger et al. [64]. This name refers to the fact that a Markov structure, that is, a list of conditional independences, largely simplifies the calculations of the probabilities including that of potentially underlying causes behind observed phenomena. For example, patients entering an emergency room have relatively easily observable symptoms, but different underlying medical conditions may cause the symptoms, although with varying probabilities. As treatment, ideally, targets the underlying condition (and not only the symptoms), probabilistic inference with respect to the underlying causes is needed. Software, incorporating relevant knowledge and providing estimates of the probabilities of the possible underlying causes, is often called a probabilistic expert system, see, e.g., Luger and Chakrabarti [106] and Mortera and Dawid [116]. Note that in the context of expert systems, it is usually existing substantive knowledge, including knowledge regarding cause and effect relationships, that is formulated

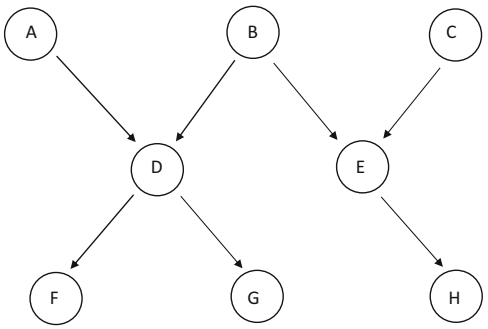


Fig. 3.5 A DAG model as an expert system. A, B, C are hard-to-test physiological functions of the human body; D, E are not easily observable medical conditions; and F, G, H are obvious symptoms

as a DAG model. In other applications of DAG models, to be addressed in the subsequent chapters, observations are used to discover and then describe possible causal relationships.

Expert systems are also being used, in addition to the backward reasoning described above, for forward reasoning, where probabilistic statements are made about possible outcomes of a process or provide labeling or classification of objects, often under the heading of artificial intelligence, e.g., spam detection in an email system.

Note that the name Bayesian network suggests that expert systems operate in the context of the Bayesian approach to statistics and data analysis, see, e.g., O'Hagan [126]. This is true in the sense that in the emergency room example above, not a single hypothesis is tested for the underlying true medical condition; rather, probabilities are associated with the different possible underlying conditions and these probabilities are updated during the analysis. On the other hand, the probabilities and conditional probabilities that enter the expert system are usually estimated values of frequentist probabilities characterizing the relevant populations and of the relative sizes of subpopulations, and not necessarily degrees of belief. (See the brief discussion of subjective and frequentist probabilities in Section *LCDA* 8.4.) Related to this, the interpretation of a Bayesian network may be that there is a known probability that a parent in the DAG causes a child, like lighting may cause thunder, but it also may mean that in a given fraction of the times when the parent is observed, also the child is observed, without the parent causing the child, as wind may or may not bring clouds.

Figure 3.5 is a very simple example. There are immediate symptoms, F, G, H, which may be observed easily when the patient enters the emergency room. These symptoms may be caused by medical conditions D, E, which may be determined by examinations by appropriately trained medical personnel and underlying, say, physiological causes, A, B, C, which may be determined by tests that may be expensive or may require a lot of time to yield results. So the three levels in Fig. 3.5 represent levels of knowledge, which require increasing amounts of resources to acquire.

The precise meaning associated with the model in Fig. 3.5 is that the joint distribution of the variables A, B, C, D, E, F, G, H factorizes as in (3.7) with respect to the graph:

$$P(a, b, c, d, e, f, g, h)$$

$$= P(a)P(b)P(c)P(d|a, b)P(e|b, c)P(f|d)P(g|d)P(h|e). \tag{3.14}$$

In general, the Bayesian network model associated with a DAG $G = (V, A)$ consists of those joint distributions of the variables in V, which factorize as in (3.7), according to G. The following result is readily implied by Theorems 3.8 and 3.10.

Corollary 3.1 *If a distribution belongs to the Bayesian network with respect to a DAG G, then the directed pairwise, local, and global Markov properties and also the d-separation rule apply to it with respect to G.*

□

If one has knowledge as to the conditional probabilities of facts at a level, given facts at a higher level, Bayes' theorem may be used to infer the probabilities of the underlying medical conditions or physiological processes behind the symptoms, see, e.g., Spiegelhalter et al. [156]. Such knowledge may be very useful in choosing a treatment. Similarly, knowing that some of the underlying conditions are present may be used to determine the probabilities of the symptoms to manifest themselves.

Such calculations, although theoretically straightforward, may be quite lengthy, in particular when they need to be performed in real time, that is, parallel to the data being observed, for example, in predicting component breakdowns in a telecommunications network.

For example, if all variables are binary and the probabilities

$$P(a)\ P(b),\ \ P(c)$$

and the conditional probabilities

$$P(d|a, b),\ \ P(e|b, c),$$

$$P(f|d),\ \ P(g|d),\ \ P(h|e)$$

for all categories a, b, c, d, e, f, g, h are given, and a patient arrives with symptoms f, g, h, then for the conditional probabilities of the possible medical conditions, one has that

$$P(d, e|f, g, h) = \frac{P(d, e)P(f, g, h|d, e)}{P(f, g, h)}. \tag{3.15}$$

There are four such probabilities for fixed f, g, h, and they sum to 1, and they also have the same denominator. Thus, the denominator is the sum of the numerators

and does not need to be determined directly. However, $P(d, e)$ is needed, which may be obtained as

$$P(d, e) = \sum_{a,b,c} P(d, e|a, b, c)P(a, b, c).$$

In determining the conditional probabilities above, one needs the joint distribution of the variables A, B, C, D, E and also of A, B, C. From (3.14) one gets that

$$P(a, b, c) = P(a)P(b)P(c)$$

and

$$P(d, e|a, b, c) = \frac{P(a, b, c, d, e)}{P(a, b, c)} = P(d|a, b)P(e|b, c).$$

also follows. To see this, note that the directed local Markov property implies that

$$D \perp\!\!\!\perp C, E, H|A, B$$

and

$$E \perp\!\!\!\perp A, D, F, G|B, C.$$

Then, by the properties of conditional independence (Propositions 2.5 and 2.6), one has that

$$D \perp\!\!\!\perp C, E|A, B$$

and

$$E \perp\!\!\!\perp A, D|B, C,$$

then,

$$D \perp\!\!\!\perp E|A, B, C$$

or that

$$P(d, e|a, b, c) = P(d|a, b, c)P(e|a, b, c)$$

and, applying the conditional independences again, finally that

$$P(d, e|a, b, c) = P(d|a, b)P(e|b, c).$$

Of course, this is also obtained from the factorization (3.14).

These are very strong simplification implied by the assumption of the DAG model. Thus,

$$P(d,e) = \sum_{a,b,c} P(d|a,b)P(e|b,c)P(a)P(b)P(c). \tag{3.16}$$

Similarly, one has that

$$F \perp\!\!\!\perp A, B, C, E, G, H|D,$$

$$G \perp\!\!\!\perp A, B, C, E, F, H|D,$$

$$H \perp\!\!\!\perp A, B, C, D, F, G|E,$$

thus,

$$H \perp\!\!\!\perp F, G|D, E,$$

$$H \perp\!\!\!\perp D|E,$$

$$F \perp\!\!\!\perp G|D, E,$$

so that

$$P(f,g,h|d,e) = P(f,g|d,e)P(h|d,e) = P(f|d,e)P(g|d,e)P(h|e).$$

Finally, as D separates F, G from E in the undirected moralization of their ancestral graph,

$$F, G \perp\!\!\!\perp E|D,$$

one has instead of the expression above that

$$P(f,g,h|d,e) = P(f|d)P(g|d)P(h|e). \tag{3.17}$$

Combining (3.16) and (3.17) and using them in (3.15), one has that

$$P(d,e|f,g,h) = \frac{\sum_{a,b,c} P(d|a,b)P(e|b,c)P(a)P(b)P(c)P(f|d)P(g|d)P(h|e)}{P(f,g,h)}$$

$$= \frac{\sum_{a,b,c} P(d|a,b)P(e|b,c)P(a)P(b)P(c)P(f|d)P(g|d)P(h|e)}{\sum_{d',e'}\left(\sum_{a,b,c} P(d'|a,b)P(e'|b,c)P(a)P(b)P(c)\right)P(f|d')P(g|d')P(h|e')}.$$

The complexity of the calculations and also the number of relevant conditional probabilities that may be needed to determine grow rapidly with the complexity of

the network. Therefore, more efficient ways to determine the effects of observations (also called evidence in this context) on the probabilities are of great interest. We emphasize that efficient does not mean ease of implementation, but rather the speed with which the calculations can be performed.

One such algorithm was given by Lauritzen and Spiegelhalter in [101] (see also Dawid [41]), and it will be described here informally, using the DAG in Fig. 3.5 as an example. In this section, we describe the intuitive meaning of the algorithm, link it to results in *LCDA*, and state some of the formal results that are needed for a precise proof of the properties of the algorithm.

To illustrate the fundamental idea of the algorithm to update probabilities, let there be three binary variables A, B, C and an undirected graph, with two edges AB and BC. Then, the joint distribution according to the graph is conditional independence, that is,

$$P(a,b,c) = \frac{P(a,b,+)P(+,b,c)}{P(+,b,+)}. \tag{3.18}$$

Because of the conditional independence of A and C, given B, assuming (strict) positivity,

$$P(C = c|B = b) = P(C = c|B = b, A = a),$$

see (2.2). This means that, under the conditional independence, learning that $A = a$ does not affect the conditional distribution of C given B.

For the $P(+, B, +)$ distribution, if $A = a$ becomes known, then one has, instead of the marginal distribution, the conditional distribution, given $A = a$. That is,

$$P_{A=a}(+,b,+) = \frac{P(a,b,+)}{P(a,+,+)},$$

where $A = a$ in the subscript denotes the probabilities after this fact was learned.

This gives the following simple result.

Proposition 3.2 *If the joint distribution of A, B, and C is (strictly) positive and*

$$A \perp\!\!\!\perp C|B,$$

then, having learned that

$$A = a,$$

the marginal probabilities of $(B = b, C = c)$ *become*

$$P_{A=a}(B = b, C = c) \propto \frac{P(a,b,+)}{P(+,b,+)} P(B = b, C = c), \tag{3.19}$$

where the multiplier

$$\frac{1}{P(a,+,+)}$$

applied to the right-hand side would imply equality.

Proof Note that

$$P_{A=a}(B=b,C=c)=P_{A=a}P(C=c|B=b)P_{A=a}(B=b)$$

$$=P(C=c|B=b,A=a)P_{A=a}(B=b)=P(C=c|B=b)P_{A=a}(B=b),$$

because of the assumed conditional independence. Then, one has

$$P_{A=a}(B=b,C=c)=\frac{P(B=b,C=c)}{P(B=b)}\frac{P(a,b,+)}{P(a,+,+)}$$

$$=\frac{P(B=b,C=c)}{P(+,b,+)}\frac{P(a,b,+)}{P(a,+,+)},$$

as claimed.

□

The proportionality in (3.19) means that to achieve equality instead of proportionality requires a multiplier that does not depend on the left-hand side.

The AB and BC marginals for binary variables after updating for $A=1$ as given in (3.19) are shown in Table 3.1. In the AB marginal, the $A=2$ row is empty, thus, the entries, although expressed in terms of the original probabilities, do not sum to 1. Of course, if normed, that is, are divided by their sum, which is $P(1,+,+)$, they give probabilities. In the BC marginal, the sum in Table 3.1 is also $P(1,+,+)$, and dividing with this would give a probability distribution. Quantities that after norming give a probability distribution are called potentials. For an example, see the comment after (3.15). The numerator of the right-hand side

Table 3.1 Potentials after the updating in (3.19)

	B=1	B=2
A=1	$P(1,1,+)$	$P(1,2,+)$
A=2	0	0

	C=1	C=2
B=1	$\frac{P(1,1,+)}{P(+,1,+)}P(+,1,1)$	$\frac{P(1,1,+)}{P(+,1,+)}P(+,1,2)$
B=2	$\frac{P(1,2,+)}{P(+,2,+)}P(+,2,1)$	$\frac{P(1,2,+)}{P(+,2,+)}P(+,2,2)$

of (3.15) is a (conditional) potential. Obviously, probabilities are also potentials. From now on, the notation Ψ will be used for potentials.

Note that the updating process modifies the AB and BC marginal distributions, but leaves the conditional independence (3.18) of the original distribution unchanged, so the resulting distribution remains Markov according to the graph.

From now on, the superscript o (for old) will denote the potentials before an updating step is performed, and the superscript n (for new) will denote the potentials after the current updating has been performed.

If further information that becomes available applies to the intersection of the cliques of the graph, e.g., in the current example, $B = 1$ is also given, then one has

$$\Psi^n(+, 1, +) = 1, \text{ and } \Psi^n(+, 2, +) = 0,$$

thus, the potentials for the BC marginal become, by retaining the conditional distribution of C, given B,

$$\Psi^n(B = 1, C = c|B = 1) = \frac{\Psi^n(+, 1, +)}{\Psi^o(+, 1, +)}\Psi^o(B = 1, C = c) \tag{3.20}$$

$$= \frac{1}{\Psi^o(+, 1, +)}\Psi^o(B = 1, C = c) = \Psi^o(C = c|B = 1),$$

and

$$\Psi^n(B = 2, C = c|B = 1) = \frac{\Psi^n(+, 2, +)}{\Psi^o(+, 2, +)}\Psi^n(B = 2, C = c) \tag{3.21}$$

$$= \frac{0}{\Psi^o(+, 1, +)}\Psi^n(B = 2, C = c) = 0,$$

and the updating procedure gave the correct values.

The potentials, when also the updating steps in (3.20) and (3.21) are applied, are given in Table 3.2. Note the simplification achieved by using potentials instead of probabilities. Also note that the notation for potentials only implicitly shows the subsets of variables to which they refer, namely, through their respective arguments. Therefore, the notation used in Table 3.2 includes all three arguments, to make it

Table 3.2 The structure of the potentials after the second updating for $B = 1$

	B=1	B=2
A=1	$\Psi(1, 1, +)$	0
A=2	0	0

	C=1	C=2
B=1	$\Psi(+, 1, 1)$	$\Psi(+, 1, 2)$
B=2	0	0

clear to which variables the actual numbers (categories) refer. Later on, a more precise notation will be used for the potentials.

Clearly, the updating above extends to the following slightly more general setup. Let the Markov model be based on two cliques, and the new potential of the intersection is not derived from a condition regarding some of the variables in their intersection but rather, from a modification of the potential of one of the cliques. This includes knowledge becoming available with respect to variables not in the intersection, in the intersection, or both.

Proposition 3.3 *Let an undirected graph for the variables V have two cliques, $\mathscr{C}_1$ and $\mathscr{C}_2$. Let the distribution P be Markov according to the graph and also (strictly) positive. Suppose the old potential of $\mathscr{C}_1$,*

$$\Psi^o(\mathscr{C}_1 = (v)_{\mathscr{C}_1})$$

changes to the new potential

$$\Psi^n(\mathscr{C}_1 = (v)_{\mathscr{C}_1}).$$

Then, the new potential for $\mathscr{C}_2$ is

$$\Psi^n(\mathscr{C}_2 = (v)_{\mathscr{C}_2}) =$$

$$= \frac{\Psi^n(\mathscr{C}_1 \cap \mathscr{C}_2 = (v)_{\mathscr{C}_1 \cap \mathscr{C}_2})}{\Psi^o(\mathscr{C}_1 \cap \mathscr{C}_2 = (v)_{\mathscr{C}_1 \cap \mathscr{C}_2})} \Psi^o(\mathscr{C}_2) = (v)_{\mathscr{C}_2}), \tag{3.22}$$

where

$$\Psi(\mathscr{C}_1 \cap \mathscr{C}_2) = (v)_{\mathscr{C}_1 \cap \mathscr{C}_2})$$

$$= \sum_{(v)_{\mathscr{C}_1 \setminus \mathscr{C}_2}} \Psi(\mathscr{C}_1 = (v)_{\mathscr{C}_1}), \tag{3.23}$$

with both the superscripts o and n.

□

The updating step (3.22) is the same as the updating in $IPFP$ (see (2.10)), although in a different context, namely, the distribution of $\mathscr{C}_1 \cap \mathscr{C}_2$ is set to what is implied by the knowledge regarding the distribution of $\mathscr{C}_1$, and the conditional distribution of $\mathscr{C}_2$, given $\mathscr{C}_1 \cap \mathscr{C}_2$ is retained.

Obviously, if the old potentials are derived from the distribution P, then

$$\Psi^o(\mathscr{C}_1 \cap \mathscr{C}_2) = (v)_{\mathscr{C}_1 \cap \mathscr{C}_2})$$

$$= \sum_{(v)_{\mathscr{C}_1 \setminus \mathscr{C}_2}} \Psi^o(\mathscr{C}_1 = (v)_{\mathscr{C}_1}) = \sum_{(v)_{\mathscr{C}_2 \setminus \mathscr{C}_1}} \Psi^o(\mathscr{C}_2 = (v)_{\mathscr{C}_2})$$

and also after the updating in (3.22)

$$\Psi^n(\mathscr{C}_1 \cap \mathscr{C}_2) = (v)_{\mathscr{C}_1 \cap \mathscr{C}_2})$$

$$= \sum_{(v)_{\mathscr{C}_1 \setminus \mathscr{C}_2}} \Psi^n(\mathscr{C}_1 = (v)_{\mathscr{C}_1}) = \sum_{(v)_{\mathscr{C}_2 \setminus \mathscr{C}_1}} \Psi^n(\mathscr{C}_2 = (v)_{\mathscr{C}_2}). \tag{3.24}$$

If there are two cliques in an undirected graphical model, then (3.22) shows how the marginal potential of one clique changes, if some of the variables in the other clique become fixed at certain categories. The change depends on how the potential of the intersection changes.

The updating procedure in Proposition (3.3) may be extended to several cliques, if the cliques are decomposable, see Section *LCDA* 11.3 and also Sect. 2.5.2. A particular representation of a decomposable class of cliques is helpful in doing so.

A graph is called a tree if there is exactly one path between any two nodes. A graph that is a tree on the set of the cliques as nodes is called a junction tree, if the following property holds: For any two nodes, $\mathscr{C}_i$ and $\mathscr{C}_j$, consider the unique path connecting them. Then, the intersection of any two nodes (cliques) that are on this path contains $\mathscr{C}_i \cap \mathscr{C}_j$. Every decomposable class of cliques has a junction tree graph on them. Indeed, one has the following result. For a proof, see, e. g., Lauritzen [98].

Proposition 3.4 *On every decomposable class of cliques, there exists a graph that is a junction tree.*

□

Of course, the result above applies to any class of the subsets of the variables. It is stated in this form because it will be used for cliques of a graph.

In the context of a junction tree, the intersection of two neighboring nodes is called a separator. Note that some subsets of variables may be the separator for distinct pairs of neighbors, in which case they appear repeatedly among the separators. The next result generalizes (3.18).

Proposition 3.5 *Consider a set of variables V and a graph G on them, such that the cliques $\mathscr{C}_1, \ldots \mathscr{C}_k$ of G are decomposable. Further, let T be a junction tree on the cliques $\mathscr{C}_1, \ldots \mathscr{C}_k$, with separators $\mathscr{S}_1, \ldots \mathscr{S}_l$. Finally, let the joint distribution of the variables V be Markov according to G. Then, for the respective potentials,*

$$\Psi_V(v) = \frac{\prod_{i=1}^k \Psi_{\mathscr{C}_i}((v)_{\mathscr{C}_i})}{\prod_{j=1}^l \Psi_{\mathscr{S}_j}((v)_{\mathscr{S}_j})}. \tag{3.25}$$

□

The formula above is closely related to the formula in Theorem *LCDA* 12.4, in particular to expression *LCDA* (12.3). The equalities in (3.24) imply that the marginal distributions of the intersection of two neighboring cliques are the same in the two cliques, a property called weak compatibility in Section *LCDA* 12, see also Sect. 7.2.4.

If there are two cliques only, and evidence arises in one of them, the other one is updated, as in Proposition 3.3. When there are more than two cliques in a junction tree, some order of the updating steps needs to be determined.

To determine the order of updating in a junction tree of cliques, choose any node (clique) and call it the root of the tree. Consider the paths connecting this root with all the other nodes. Because of the tree property, for every node, there is exactly one such path; however, the paths reaching from the root different nodes may partially coincide. Some nodes have two neighbors on these paths, and some have only one. The latter nodes are called the leaves of the tree. Start the updating procedure at each leaf, and move with the updating toward the root. If there is no information to be taken into account for the variables in a node (clique), no updating is needed. If, on such a path, two nodes have a non-empty intersection, then this intersection is part of all the intersections of neighbors on the path between them by the junction tree property, and the information is properly passed on. When all information was sent from the leaves and from the intermediate nodes to the root, then send out the information from the root to all leaves, including the nodes in between. Note that if two nodes have a non-empty intersection, but they are not on the same path connecting the root with a leaf, but rather on two such paths, then they will both receive the same information regarding their intersection, because the intersection is part of all nodes on the path between them, even if this path goes through the root.

Let $\mathscr{I}$ be a set of information, knowledge, or belief with respect to the statuses of some of the variables. It is assumed that $\mathscr{I}$ only contains information about individual variables, but not about joint distributions. For example, it may contain that $A = a$ and $B = b$ but may not contain $A = B$. To see why the latter information would be problematic, imagine that there are only two variables, A and B, and the graph is empty. This is a decomposable system, with two cliques, each containing a single variable. Then the separator is empty and the condition $A = B$ does not have an implication on either one of the cliques (only on both of them, jointly), thus, the information $A = B$ cannot be imposed by marginal adjustments.

When $\mathscr{I}$ is given and $\Psi^n(\mathscr{C}_1 \cap \mathscr{C}_2) = (v)_{\mathscr{C}_1 \cap \mathscr{C}_2}$ is determined in (3.23), then

$$\Psi^n(\mathscr{C}_1 = (v)_{\mathscr{C}_1}) \text{is obtained by applying } \mathscr{I} \text{ to } \Psi^o(\mathscr{C}_1 = (v)_{\mathscr{C}_1}). \tag{3.26}$$

Thus, one has the following algorithm to update (condition the original distribution on evidence becoming available) for clique distributions on a junction tree. The algorithm starts with all the potentials (cliques and separators) being equal to the corresponding probabilities.

Algorithm (3.27)

Step 0: Choose any node as the root.
Step 1: For all leaves, make the leaf the current node.
Step 2: For all current nodes $\mathscr{C}_1$, apply (3.26), (3.23), and (3.22), with $\mathscr{C}_2$ being its neighbor closer to the root.

Step 3: Repeat Step 2 till the root is updated from all the leaves.
Step 4: Make the root the current node.
Step 5: Apply Step 6 on all paths leading to the leaves.
Step 6: For all current nodes $\mathscr{C}_1$, apply (3.26), (3.23), and (3.22), with $\mathscr{C}_2$ being its neighbor farther from the root.
Step 7: Repeat Step 6 till all leaves are updated.

Note that this algorithm, once the junction tree structure is set up, may be performed parallel on every path connecting a leaf with the root. Clearly, different choices of the root may lead to very different running times of the updating procedure.

Then, one has the following result.

Proposition 3.6 *The potential obtained after the algorithm (3.27) was run for $\mathscr{I}$ is the conditional potential*

$$\Psi_V(v|\mathscr{I}).$$

□

Then, marginalization may be applied to this potential to obtain the potential of any subset of interest of the variables.

So far, we have presented an algorithm to incorporate evidence into the distribution, if the distribution is Markov according to a decomposable graphical model. The remaining question is whether this algorithm may be adapted for cases when the joint distribution P is not supposed to belong to a decomposable graphical model but rather is assumed to belong to a DAG model. The answer to this question is positive. The DAG model is replaced by an undirected graph such that its cliques are decomposable and is such that all conditional independences that hold in the latter are also conditional independences in the former. That is, the decomposable model assumes only some of the conditional independences assumed by G but, of course, allows all of them to hold.

To create this decomposable model for the DAG G, first replace G with G^{UM}, its undirected moralization. The cliques of the graph G^{UM} are not necessarily decomposable. That will be achieved in a second step. To see that all the conditional independences implied by the Markov property in G^{UM} are also conditional independences in G, use the global versions of the undirected and of the directed Markov properties.

Proposition 3.7 *For three pairwise disjoint subsets of the variables V, a joint distribution P, and a DAG G on V, whenever*

$$\mathscr{A} \perp\!\!\!\perp \mathscr{B}|\mathscr{C}$$

in the undirected graphical model associated with G^{UM}, the same conditional independence holds in the directed graphical model associated with G.

Proof The conditional independence

$$\mathscr{A} \perp\!\!\!\perp \mathscr{B} | \mathscr{C}$$

is implied by the undirected global Markov property in G^{UM}, if

$$\mathscr{C} \text{ separates } \mathscr{A} \text{ and } \mathscr{B} \text{ in } G^{UM}. \tag{3.28}$$

On the other hand, the same conditional independence is implied by the directed global Markov property in G if

$$\mathscr{C} \text{ separates } \mathscr{A} \text{ and } \mathscr{B} \text{ in } G^{UM}_{anc(\mathscr{A} \cup \mathscr{B} \cup \mathscr{C})}. \tag{3.29}$$

As $anc(\mathscr{A} \cup \mathscr{B} \cup \mathscr{C}) \subset V$, the nodes and the edges of $G^{UM}_{anc(\mathscr{A} \cup \mathscr{B} \cup \mathscr{C})}$ are, respectively, subsets of the nodes and edges of G^{UM}, and thus, (3.28) implies (3.29).

□

Thus, using instead of the DAG G, the undirected model G^{UM} does not impose new conditional independences relative to those implied by G.

The cliques of the graph G^{UM} are not necessarily decomposable. The following result offers a way to transform the graph further to achieve decomposability. For a proof, see, e. g., Lauritzen [98].

Proposition 3.8 *The cliques of a graph are decomposable if and only if the graph is triangulated*

□

A graph is triangulated (or chordal) if it does not have a cycle longer than three nodes, without two nonconsecutive nodes of it being connected by an edge. So if in a graph $G = (V, E)$, $(A, B) \in E$, $(B, C) \in E$, $(C, D) \in E$, and $(D, A) \in E$, then for it to be triangulated, either $(A, C) \in E$ or $(B, D) \in E$ (or both) must hold. Triangulation is the process of connecting nodes with additional edges, in order to make the resulting graph triangulated.

To obtain the decomposable graph needed to apply the algorithm, G^{UM} is triangulated. Denote the resulting graph by $T(G^{UM})$. This, in general, may be achieved in many ways and the choice may have a big effect on how fast the algorithm runs. As $T(G^{UM})$ is obtained from G^{UM} by adding edges, the undirected graphical model associated with it $T(G^{UM})$ implies fewer conditional independences than the one associated with G^{UM}. For $T(G^{UM})$, one has the following.

Proposition 3.9 *Let G be a DAG. The triangulation of the undirected moralization of G, $T(G^{UM})$, is such that its cliques are decomposable. Further, the graphical model associated with $T(G^{UM})$ does not impose additional conditional independences relative to those imposed by the directed graphical model associated with G.*

□

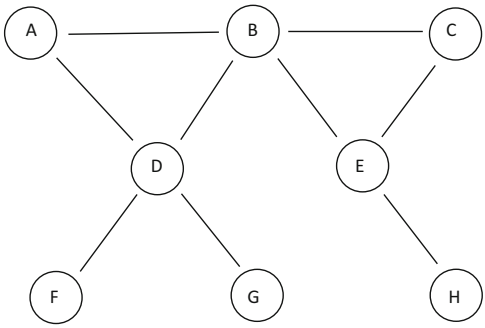


Fig. 3.6 Undirected moralization of the DAG in Fig. 3.5

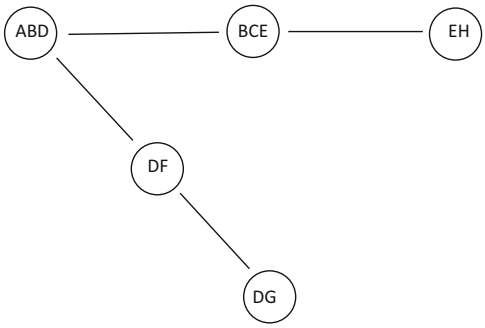


Fig. 3.7 A junction tree on the cliques of the undirected moralization of the DAG in Fig. 3.5

Then, algorithm (3.27) may be applied to the distribution P in the DAG model G, using the decomposable system of cliques from $T(G^{UM})$. It yields the conditional potentials derived from P, given $\mathscr{I}$. The conditional distribution may then be obtained and from it the marginal distributions of interest.

The algorithm is now illustrated for the Bayesian network in Fig. 3.5. The procedure starts with moralizing the DAG, and then replacing arrows with edges, to obtain the graph in Fig. 3.6

The undirected moral graph in Fig. 3.6 is triangulated (or chordal) in this case.

The cliques of the moralized graph are $\mathscr{C}_1 = ABD$, $\mathscr{C}_2 = BCE$, $\mathscr{C}_3 = DF$, $\mathscr{C}_4 = DG$, $\mathscr{C}_5 = EH$.

A junction tree on the cliques of Fig. 3.6 is shown in Fig. 3.7

Assuming (strict) positivity for simplicity, the potential representation according to (3.25) is

$$\Psi_{ABCDEFGH}(a,b,c,d,e,f,g,h)$$

$$= \frac{\mathscr{S}_{ABD}(a,b,d)\mathscr{S}_{BCE}(b,c,e)\mathscr{S}_{EH}(e,h)\mathscr{S}_{DF}(d,f)\mathscr{S}_{DG}(d,g)}{\Psi_B(b)\Psi_E(e)\Psi_D(d)\Psi_D(d)}. \tag{3.30}$$

Note that the separator D appears twice, as it is the separator of the cliques ABD and DF and also of DF and DG.

The starting values of the potentials are determined from the product form of the probability distribution in (3.7), based on the DAG in Fig. 3.5, as was given in (3.14):

$$\begin{aligned} &P(a,b,c,d,e,f,g,h) \\ &= P(a)P(b)P(c)P(d|a,b)P(e|b,c)P(f|d)P(g|d)P(h|e) \\ &= (P(a)P(b)P(d|a,b))\ P(c)(P(e|b,c))\ (P(h|e))\ (P(f|d))\ (P(g|d)) \\ &= P(a)P(b)\frac{P(a,b,d)}{P(a,b)}\ P(c)\frac{P(b,c,e)}{P(b,c)}\ \frac{P(e,h)}{P(e)})\ \frac{P(d,f)}{P(d)}\ \frac{P(d,g)}{P(d)}. \end{aligned}$$

Now, the (conditional) independences implied by the DAG, and not implied (but allowed) by the junction tree, are incorporated in the potentials. These are that A and B are independent and B and C are independent. This gives

$$\begin{aligned} &P(a,b,d))\ \frac{P(b,c,e)}{P(b)}\ \frac{P(e,h)}{P(e)}\ \frac{P(d,f)}{P(d)}\ \frac{P(d,g)}{P(d)} \\ &= \frac{P(a,b,d)P(b,c,e)P(e,h)P(d,f)P(d,g)}{P(b)P(e)P(d)P(d)}. \end{aligned}$$

Thus, the junction tree representation in (3.30) may be set up to use potentials equal to the corresponding marginal distributions of the joint distribution. The junction tree turned out to provide a way of writing the joint probability in terms of marginal probabilities of cliques and their intersections. At the beginning of the algorithm, the potentials determined have the superscript o.

In this example, the available knowledge is

$$\mathscr{I} = \{F = f^*,\ G = g^*,\ H = h^*\},$$

where the asterisks denote specific categories of the variables.

The junction tree in Fig. 3.7 has two leaves, EH and DG. Starting the updates (also called message passing) with the first one, $\mathscr{I}$ implies that for the corresponding potential

$$\Psi^n_{EH}(e,h^*) = \Psi^o_{EH}(e,h^*)$$

and

$$\Psi^n_{EH}(e,h) = 0,\ \text{if } h \neq h^*.$$

The separator of EH and its neighbor BCE is E, and its potential is obtained using (3.23):

$$\Psi_E^n(e) = \sum_h \Psi_{EH}^n(e,h) = \Psi_{EH}^o(e,h^*).$$

The clique potential of BCE is updated now using (3.22):

$$\Psi_{BCE}^n(b,c,e) = \frac{\Psi_E^n(e)}{\Psi_E^o(e)} \Psi_{BCE}^n(b,c,e).$$

In the next step, the root ABD is updated similarly.

Then, the other leaf DG is updated using the information in $\mathscr{I}$. The separator with DF is D, which is updated as before. There are two effects on the DF clique that need to be used. One is the information in $\mathscr{I}$, similarly to how it was done for H in the other leaf. And the other one is the effect from DG through the separator D, similarly to how it was done above with other separators. Next, the separator is D again, and its potential needs to be recalculated based on the clique DF. Then, the potential of the root ABD is updated.

Now, all updates from the leaves arrived at the root, and the second round of updates, from the root to the leaves, follows.

Finally, an updated potential representation of the form (3.25) is obtained, and by normalization, the joint probability distribution that is conditional on $\mathscr{I}$ is determined. The appropriate marginalization of this probability distribution answers the original question concerning the conditional distribution $P(d,e|f^*,g^*,h^*)$.

Recall that the example considered above is special in the sense that the undirected moral graph is triangular. An example, when this is not the case, is given in Fig. 3.8.

The undirected moralization of the graph in Fig. 3.8 is shown in Fig. 3.9. This graph is not triangulated, as $ABED$ is a chordless cycle. To triangularize the graph, add the edge AE, to obtain the graph in Fig. 3.10. Note that the other possible triangularization through including the edge BD could have been selected too.

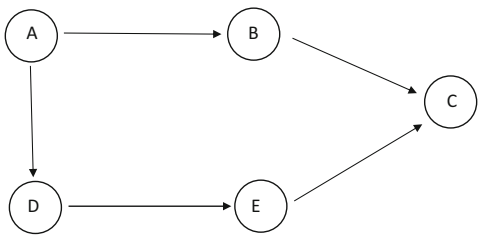


Fig. 3.8 Another DAG model of an expert system

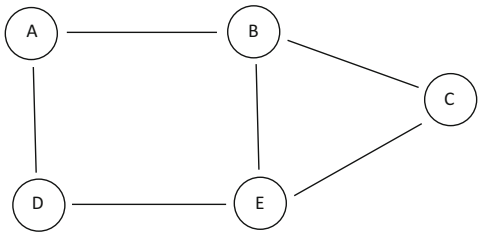


Fig. 3.9 The undirected moralization of the graph in Fig. 3.8

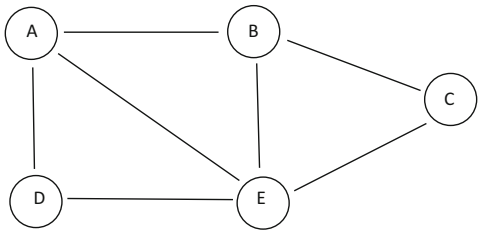


Fig. 3.10 The triangularization of the graph in Fig. 3.9

The cliques of the triangularized graph are ADE, ABE, BCE. This is a decomposable ordering and it also gives a junction tree. Indeed, the intersection of the first and last cliques is E, and it is also contained in the second clique. Based on this, the potential representation of the joint distribution is

$$\Psi_{ABCDE}(a,b,c,d,e) = \frac{\Psi_{ADE}(a,d,e)\Psi_{ABE}(a,b,e)\Psi_{BCE}(b,c,e)}{\Psi_{AE}(a,e)\Psi_{BE}(b,e)}.$$

On the other hand, the factorization implied by the original DAG is

$$\begin{aligned} P(a,b,c,d,e) &= P(a)P(b|a)P(d|a)P(e|d)P(c|b,e) \\ &= P(a)P(d|a)P(e|d)\ P(b|a)\ P(c|b,e) \\ &= P(a)P(d|a)P(e|d)\ \ P(b|a)\frac{P(a,e)}{P(a,e)}\ \ P(c|b,e)\frac{P(b,e)}{P(b,e)}. \end{aligned} \tag{3.31}$$

Thus, for the potentials of the cliques,

$$\Psi_{ADE}(a,d,e) = P(a)P(d|a)P(e|d) = P(a,d,e),$$

$$\Psi_{ABE}(a,b,e) = P(b|a)P(a,e) = P(b|a,e)P(a,e) = P(a,b,e),$$

$$\Psi_{BCE}(b,c,e) = P(c|b,e)P(b,e) = P(b,c,e),$$

and similarly for the potentials of the separators,

$$\Psi_{AE}(a, e) = P(a, e),$$

$$\Psi_{BE}(b, e) = P(b, e).$$

Note that in the case of $\Psi_{ABE}(a, b, e)$, the fact that $B \perp\!\!\!\perp E|A$ was used. The potential representation implied by the junction three allows A, B, E to have an arbitrary joint distribution, and it also allows the conditional independence implied by the original DAG to hold.

Thus, a representation of the joint probability as the product of the clique potentials, divided by the product of the separator potentials, is also possible in this case, even if an additional edge needed to be included to achieve decomposability and the existence of a junction tree representation.

In both examples, the junction tree or clique potential representation was possible, but because moralization added edges, some of the (conditional) independences of the original distributions ($P(a, b, c) = P(a)P(b)P(c)$ in the first example and $P(a, b, e) = P(a)P(b|a)P(e|a)$ in the second example) were not represented. However, these (conditional) independences could be imposed on the structure of the potentials.

In summary, the Lauritzen-Spiegelhalter algorithm, also called the junction tree algorithm, considers, instead of the original DAG model, a decomposable model and the equivalent junction tree. The latter model may impose fewer conditional independences than the original model, but, of course, allows the original conditional independences to be present. Using the junction tree, the propagation of the information among the potentials of the cliques is computationally easy, and the resulting marginal distributions make reading off the implications straightforward.

For a precise description, background, and applications of the algorithm, see, in addition to Lauritzen and Spiegelhalter [101], also Barber [9], Geiger, Verma, and Pearl [64], Luger and Chakrabarti [106], Mortera and Dawid [116], and Kiiveri, Speed, and Carlin [79].

3.5 Chain Graph Models

Chain graphs were introduced in Sect. 2.2.3, and the same notation is used here. They are generalizations of directed and undirected graphs, and the models associated with them are used to describe situations when there may be (undirected) associations among certain groups of variables, and there also may be (directed) effects present.

To define Markov properties associated with chain graphs, the concepts of parents, descendants, and nondescendants for components $\mathcal{K}$ of the chain graph are introduced. The set of parents of $\mathcal{K}$, $PA(\mathcal{K}) \subseteq V$ is the collection of those nodes, that are in such components from which an arrow points to $\mathcal{K}$. That is, the parents are a union of other components. The descendants of $\mathcal{K}$, $DE(\mathcal{K}) \subseteq V$ are the

nodes that are in such components that a semi-directed path leads to a node in this component from a node in $\mathcal{K}$. A semi-directed path may contain edges and arrows and the arrows all point away from the starting node in $\mathcal{K}$. As the nondirected parts are within components, the descendants may also be defined as the union of those components, into which a directed path leads from $\mathcal{K}$. The nodes that are not in $DE(\mathcal{K})$ are the nondescendants $ND(\mathcal{K})$ of the component.

There are various Markov properties that have been associated with chain graphs in the literature. It is customary to consider only (strictly) positive distributions in this context, see Frydenberg [63] and Drton [47]. These Markov properties are selected from the following five conditional independences, see, e.g., Drton [47].

M1: For all components $\mathcal{K}$,

$$\mathcal{K} \perp\!\!\!\perp ND(\mathcal{K}) \setminus PA(\mathcal{K}) | PA(\mathcal{K}).$$

Property M1 is the directed local Markov property for G_d, that is, the DAG of the chain components. In this graph, the nodes are identified by the components of the chain graph, and there is an arrow between two nodes if there was an arrow between the respective components.

M2a: For all components $\mathcal{K}$ and $W \subseteq \mathcal{K}$,

$$W \perp\!\!\!\perp \mathcal{K} \setminus W \setminus nb(W) | PA(\mathcal{K}) \cup nb(W),$$

where the neighbors of $W \subseteq \mathcal{K}$ are meant to be the neighbors within $\mathcal{K}$.

M2b: For all components $\mathcal{K}$ and $W \subseteq \mathcal{K}$,

$$W \perp\!\!\!\perp \mathcal{K} \setminus W \setminus nb(W) | PA(\mathcal{K}).$$

The properties M2a and M2b are similar to the local Markov property for the undirected graph on $\mathcal{K}$, but the independence in this context is conditional on the parents of the components. The two properties differ in also including or not including in the conditioning set the neighbors of W within $\mathcal{K}$.

M3a: For all components $\mathcal{K}$ and $W \subseteq \mathcal{K}$,

$$W \perp\!\!\!\perp PA(\mathcal{K}) \setminus pa(W) | pa(W) \cup nb(W),$$

M3b: For all components $\mathcal{K}$ and $W \subseteq \mathcal{K}$,

$$W \perp\!\!\!\perp PA(\mathcal{K}) \setminus pa(W) | pa(W),$$

These two properties give conditional independences linking parents of the chain graph component to the parents of the subset within the component and differ in whether or not the conditioning set also contains the neighbors.

Then, using these components, the Type I Markov property assumes M1, M2a, and M3a and is also called the Lauritzen-Wermuth-Frydenberg block recursive Markov property, see Frydenberg [63] and Lauritzen and Wermuth [102].

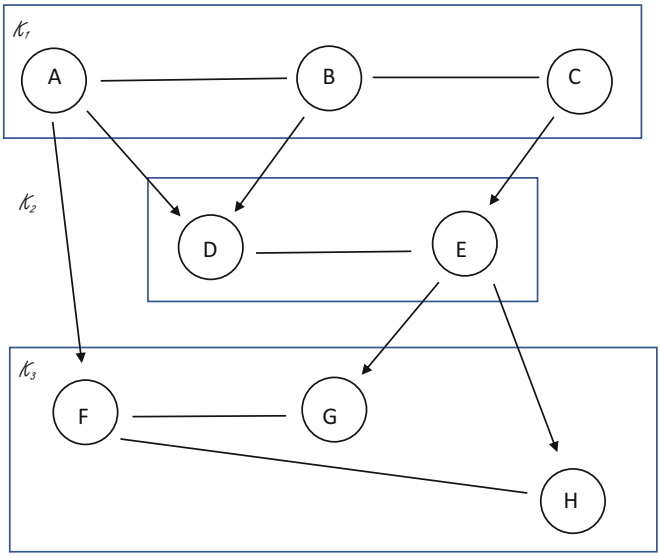


Fig. 3.11 A chain graph

The Type II Markov property for chain graphs assumes M1, M2a, and M3b and is also called the Andersson-Madigan-Perlman Markov property, see [6].

The Type III Markov property for chain graphs assumes M1, M2b, and M3a and is in a sense[5] dual to Type II, see Drton [47].

The Type IV Markov property is defined by M1, M2b, and M3b (see Cox and Wermuth [34] and Wermuth and Cox [168]) and is also called the multivariate regression-type Markov property.

To illustrate the definitions and their differences, consider the chain graph in Fig. 3.11. It has three chain components, $\mathcal{K}_1$, $\mathcal{K}_2$, and $\mathcal{K}_3$. M1, which is adopted by all four interpretations, implies no conditional independence, because the component $\mathcal{K}_1$ has no nondescendants, and for the components $\mathcal{K}_2$ and $\mathcal{K}_3$, all nondescendants are also parents.

M2a, when applied to G in the component $\mathcal{K}_3$, is that

$$G \perp\!\!\!\perp H | A, B, C, D, E, F,$$

while the competing M2b is that

$$G \perp\!\!\!\perp H | A, B, C, D, E.$$

M3a, when applied to F in the component $\mathcal{K}_3$, is that

[5] See the details later.

$$F \perp\!\!\!\perp B, C, D, E | A, G, H,$$

and M3b is that

$$F \perp\!\!\!\perp B, C, D, E | A.$$

When M3a is applied to $\{G, H\}$ in the last component, then one has

$$G, H \perp\!\!\!\perp A, B, C, D | E, F,$$

while M3b is that

$$G, H \perp\!\!\!\perp A, B, C, D | E.$$

All four of the interpretations assume M1, the directed local Markov property for the DAG of the components. Also, all interpretations assume M2a or M2b, a variant of the local Markov property within the components, of course after conditioning on the parents of the component. But the interpretations may differ, as to whether to have conditional independence between a set of variables W within a component, from its non-neighbors in the component, when conditioned on the parents of the component, also conditioning on the neighbors of W is needed. Similarly, all interpretations adopt M3a or M3b, that is, the conditional independence of a set of variables W within a component, from those variables that are parents of the component, but not parents of W, when conditioned on the parents of W. But these requirements differ in whether or not, in addition, conditioning on the neighbors of W is also needed.

Type I models assume that to have conditional independence of a subset W within a component from the other variables in the same component that are not neighbors of W, one has to condition, in addition to the parents of the component, also on the neighbors of W (M2a). Further, to have conditional independence of W from the variables that are parents of the component containing W except for the parents of W, one has to condition, in addition to the parents of W, also on its neighbors (M3a). Type II models do not require conditioning on the neighbors for the first conditional independence to hold (M2b), but do require it for the second one (M3a). Type III models do require conditioning on the neighbors for the first conditional independence (M2a), but do not require conditioning on the neighbors for the second one (M3b). Finally, Type IV models do not require conditioning on the neighbors for either of the conditional independences to hold (M2b, M3b). Type IV models only condition on parents, and this is why the Markov property consisting of M1, M2b, and M3b is called the multivariate regression-type Markov property.

There are factorization results for the different chain graph Markov models that are not discussed here in detail. Obviously, as all chain graph model types assume M1, the factorization equivalent to the directed Markov property associated with the directed graph on the components implied by the chain graph holds. As given in (3.7), see also Theorem 3.6,

$$P(v) = \prod_{\mathscr{K}} P(\mathscr{K}|PA(\mathscr{K})).$$

However, depending on the additional conditional independence assumptions made by the model, the conditional distributions above need to be restricted.

For Type I chain graph models, Frydenberg [63] showed that the equivalent factorization is

$$P(V) = \prod_{\mathscr{K}} \prod_{\mathscr{C}} g_{(\mathscr{K},\mathscr{C})}((v)_{\mathscr{C}}),$$

where the first product is over the components $\mathscr{K}$ of the chain graph, and the second product is over the cliques $\mathscr{C}$ of the moralization of the closure of $\mathscr{K}$, where the closure of $\mathscr{K}$ means

$$\mathscr{K} \cup \{\cup_{A\in\mathscr{K}} pa(A)\}.$$

Further, the functions $g_{(\mathscr{K},\mathscr{C})}$ may be different for different components $\mathscr{K}$ and cliques $\mathscr{C}$ and depend only on the indices of the variables in the clique $\mathscr{C}$.

Factorization results for Type II chain graph models were given by Peña [134], and Drton [47] studied the factorization of Type IV models.

Studený and Bouckaert [160] gave a separation rule for chain graphs similar to d-separation, which may be used to read off independences implied by chain graph Markov models of Type I. This rule is often called the c-separation rule and, just like in the directed case, is related to the moralization graphs of the components and their respective parents.

3.6 Summary

This chapter discussed several important topics in the theory of directed graphical models, but the coverage is far from being complete.

In Sect. 3.2, the pairwise, local, and global variants of the directed Markov property were introduced. It was proved that the last two are equivalent and imply the first one, but the first one implies the last two only for (strictly) positive distributions. The existence of a factorization consisting of the conditional distributions of the variables, given their respective parents, when these conditional distributions exist, is also equivalent to the directed global Markov property.

Another technically equivalent but very different-looking way of formulating the directed global Markov property was discussed in Sect. 3.3. This is related to d-separation, and a detailed proof was given of the equivalence of the directed global Markov property and of the d-separation rule. The d-separation rule was given an intuitive "explanation," and in the subsequent chapters of the book, this intuition

and possible uses of it for the analysis a certain types of causal structures will be studied further.

Section 3.4 considered an important field of application of models associated with DAGs, also called Bayesian networks, in the area of expert systems. In this application, a central task is to recalculate the conditional distributions of some of the variables, when knowledge regarding other variables becomes available. While, in theory, such calculations, often using Bayes' theorem, are straightforward, in practice, there is a need for fast algorithms to propagate knowledge or belief. A very heuristic but detailed description of one such algorithm was given, and finding the proofs of some of the underlying theoretical results, although some of the results were stated, was left to the reader. The algorithm uses a decomposable structure that imposes fewer conditional independences than the original one, although does not exclude the other conditional independences to be present. In such a decomposable structure, the effect of knowledge with respect to one clique propagates to the other cliques in a simple way, and a potential representation of the conditional distribution, given the available knowledge, is easily obtained.

Chain graph models that generalize both undirected and directed graphical models were discussed in Sect. 3.4. Quite remarkably, there are four related but different Markov interpretations of such graphs that have been published in the literature and were described in Sect. 3.5. This shows that the Markov properties are not inherent characteristics of graphs but are, rather, associated with them. The differences between these Markov properties were illustrated.

A common feature of Markov models associated with directed graphs (including chain graphs) is that the defining conditional independences do not necessarily involve all variables. For example, the local Markov property is a conditional independence on the marginal $\{A\} \cup nd(A)$, for any variable A. Therefore, these models fall into the more general class of marginal models to be discussed in Chap. 10. Properties of maximum likelihood estimates and algorithms to find them will be presented there, and that will complete the discussion of directed graphical models given in this chapter.

Many important concepts and results about various Markov models associated with some kinds of graphs were not discussed in this or the previous chapter. These include summary graphs (see Wermuth [166]), ancestral graphs (see Richardson and Spirtes [139]), and acyclic directed mixed graphs, in particular graphs with bidirected edges (see Evans and Richardson [53] and Richardson [138]).

3.7 Things to Do

1. The Markov chain was presented as an undirected graph in Chap. 2. Represent it now with a DAG, so that each variable has one parent, namely, the variable at the previous time point. Prove that the undirected and directed Markov properties coincide in this case.

2. Suppose the nodes of a DAG admit different well-numberings. Are the derived factorizations in (3.5) different?
3. Compare Theorem 3.4 to Proposition *LCDA* 6.2.
4. Is Theorem 3.5 true for (3.5) instead of (3.7), without the assumption of the existence of the conditional probabilities?
5. Prove that every DAG has a terminal node, which was used in the proof of Theorem 3.5.
6. Prove the claim after (3.10).
7. Consider an undirected graph where the nodes are categorical variables, the associated Gibbs factorization of the joint distribution of the variables, and the implied conditional independences. Now remove some of the functions from the factorization (or change their values to 1). Show that all the conditional independences remain implied after the change.
8. Apply the proof of Theorem 3.6 to the DAG in Fig. 3.1 by writing out the conditional independences implied by the directed local Markov property for each subgraph.
9. Use linear and dual multiplicative families of distributions like in Section *LCDA* 12.2 to explore how (3.25) and Theorem *LCDA* 12.4 are related.
10. Prove Proposition 3.3.
11. Find a proof of Proposition 3.4.
12. Prove Proposition 3.5.
13. Prove Proposition 3.6.
14. Find a proof of Proposition 3.8.
15. Show that the starting value of the separator potentials in the algorithm described in Sect. 3.4 is irrelevant.

Chapter 4
Causal Analysis Using DAGs: Theory



4.1 Introduction

The use of DAGs to model (certain types of) causal structures was briefly illustrated in Section *LCDA* 8.4. The discussion there was largely intuitive. The previous chapter of this book gave a fairly detailed description of DAGs and the related models, including the equivalent factorization of the joint probability, the Markov properties, and the d-separation rule. This theory is used in the present chapter to give a more disciplined description of the theory of causality that uses DAGs to model the causal structure. Much of this theory has been developed and is summarized in the books of Pearl [127, 129], and Spirtes et al. [157].

What distinguishes the approach of this chapter from most of the existing literature is that we do not take a position, whether the theory presented is a good or, perhaps, irrelevant description of the causal structure; rather, three explicit assumptions with respect to some of the characteristics of the real causal structure one wishes to describe will be made. These assumptions concern the structure of the causal effects, the observable behavior of the causal system, its reactions to interventions, and the relationships among these. It is then left to the researcher to decide, whether the reality to be modeled is similar enough to the behavior described in the assumptions for the application of this theory to be relevant. Chapter 6 of the book will, however, discuss the applicability of the assumptions to certain types of real causal problems, to illustrate how the three assumptions may be used as a checklist. It will be shown there that this theory applies quite well to some phenomena which one would call causal and does not to others. This theory provides a useful tool but not a universal one.

This chapter does not try to define what is causality, but only postulates some of the formal properties of the causal systems the behavior of which may be modeled using DAGs and the related Markov models. But the approach is not really axiomatic, as the three assumptions do not apply exclusively to some mathematical

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T. Rudas, *Lectures on Advanced Topics in Categorical Data Analysis*, Springer Texts in Statistics, https://doi.org/10.1007/978-3-031-55855-9_4

objects the causality is modeled with; rather, they assume implications of reality for some of these objects. Central to understanding of how a causal system works is observing its responses to manipulations and other types of interventions (roughly equivalent to the concept of experiment in more traditional statistical thinking, see, e.g., Section *LCDA* 8.3). The properties of such interventions are treated quite intuitively. That could be formalized to achieve an axiomatic presentation, but that would add little to precision or clarity but a lot to complexity.

There appear to be several ways to choose the assumptions from which the rest of the theory may be derived. The ones selected appeared the best choice, given the aspects of conceptual simplicity, ease of deciding the practical applicability, and intuitive value of the proofs of the other results.

Overall, the study of causality using DAGs is a rapidly developing area and this chapter can only cover its fundamentals. In particular, important issues such as confounding, direct versus indirect effects are not considered.

Section 4.2 discusses what manipulation of a causal system (an important variant of interventions) means and how it relates to simply observing the behavior of the causal system. The very simple example of rain and umbrellas is used. Observing that it is raining, that is, conditioning on it, goes with observing open umbrellas. Similarly, observing (conditioning on) open umbrellas goes with seeing falling rain. This symmetry tells the researcher nothing about the underlying causal structure (if there is one, at all). As statistics textbooks often say, association is not causation. But if the researcher is able to make rain (manipulation), then they will see people opening their umbrellas, and if the researcher is also able to make people open their umbrellas, they will see that rain is not implied. In general, the behavior of the causal system is described by intervention and observational distributions.

Section 4.3 contains the core material of the chapter. It provides a model for (certain types of) causality, a model in the sense of postulating the most important properties of the causal system. This is done by formulating Assumptions 1–3, which, informally, may be summarized as stating that the causal system is built up from individual effects that are acyclic, that there must be a causal path if there is an intervention effect, and that if all immediate causes (parents) are manipulated or conditioned upon, then the resulting intervention or observational distributions are the same. Many of the fundamental results of DAG-based causality are derived surprisingly easily from these assumptions. The section closes with revisiting the basic causal structures discussed in Section *LCDA* 8.1 and shows that the conditional independence statements associated with graphical representations on an intuitive basis there do follow mathematically in the current setup.

Section 4.4 deals with the most important characteristic of the theory, namely, that if it is valid, it makes it possible to discover or learn certain aspects of the causal structure not only when interventions to the causal system are possible, but more importantly, also when only observation is possible. This is true in particular when the intervention and observational distributions are faithful to the underlying causal structure, that is, when the existence of a causal path implies the existence of a causal effect.

The chapter closes with brief sections of conclusions and problems to work on.

For simplicity of presentation, except for the few places dealing with deterministic causation, all the probabilities are assumed to be (strictly) positive.

4.2 Manipulation and Observation

In statistical methodology, the gold standard of establishing causal relationships is to conduct a designed experiment, see Section *LCDA* 8.3.1. The central feature of an experiment is the ability to decide who gets which treatment, that is, to manipulate the treatments that individuals receive. This is in sharp contrast with observational studies where the researcher only observes the treatments the individuals who participate in the study received. In most experimental setups, it is not possible to administer first one treatment and then the other one and then to find out their respective effects by comparing the responses. The main reason for this is that the order of the treatments may influence the responses one obtains. Therefore, an attempt is made (through random allocation) to divide the participants into two identical groups and then to estimate the responses of the entire population one would obtain if one or the other treatment was applied, by administering each of them to the members of one of the groups only.

In the present context of causal analysis, it is assumed that the different treatments may be administered in an arbitrary order and even repeatedly, without the responses being influenced by earlier treatments and that the responses may be observed. As will be seen, this, in certain situations, may lead to exploring or learning the causal structure. On the other hand, the causal structure to be described may be much more complex than in a traditional experimental setup, where one is interested in the response to a treatment, typically not investigating the complexity of the factors that may influence the responses. It will be also illustrated that manipulating one variable and observing the response to it leads to information very different from that of simple observation. The example of rain and umbrellas, already discussed in Section *LCDA* 8.1, is used.

To study the information one can learn from manipulation, it is assumed that the variables $V \setminus \{A\}$ are set to certain values or categories, and A is observed. In general, it is not supposed that the other variables determine A exactly; rather, given the other variables, A has a conditional distribution. Based on the assumption that repeated observations are possible, the researcher may determine relative frequencies and the probability distributions in this and subsequent chapters are derived from these. When the variables $V \setminus \{A\}$ are set to categories $(c)_{V\setminus\{A\}}$, then let

$$Q(A = a | V \setminus \{A\} =_s (c)_{V\setminus\{A\}}) \tag{4.1}$$

denote the conditional distribution of A. The notation $=_s$ refers to the fact that this is not a conditional probability distribution in the sense that out of all the cases, those when $V \setminus \{A\} = (c)_{V\setminus\{A\}}$ happens to be true are selected, and the probability of $A = a$ is determined among these. Rather, the variables $V \setminus \{A\}$ are set to the

categories $(c)_{V\setminus\{A\}}$, and the probability of $A = a$ from among the responses is meant by (4.9). The distribution Q in (4.9) is called the manipulation distribution of A.

To illustrate the difference between the standard conditional distribution and the manipulation distribution, the example of rain and umbrellas already discussed in Chapter *LCDA* 8 is used. If rain and umbrellas are observed, then most, perhaps all, of the time either it is raining and the umbrellas are open or it is not raining and the umbrellas are closed. Thus, the observer will find that

$$\begin{aligned} &P(umbrellas\ are\ open \,|\, it\ is\ raining) \\ &= P(it\ is\ raining \,|\, umbrellas\ are\ open) = 1, \end{aligned}$$

and the observer can establish strong association and then needs to use subject matter knowledge, if available, to infer causation. The observer may also determine that

$$P(rain = yes \,|\, umbrellas = open) \neq P(rain = yes \,|\, umbrellas = closed), \tag{4.2}$$

and further that

$$P(umbrellas = open \,|\, rain = yes) \neq P(umbrellas = open \,|\, rain = no). \tag{4.3}$$

These relationships all follow from the fact that rain and open umbrellas are observed at the same time, just like no rain and closed umbrellas.

The situation is very different, when the observer is able to manipulate,[1] that is, to make people open or close their umbrellas and make or stop rain. The experimenter will find, using the $=_s$ notation to refer to a particular category is not simply being observed rather the variable is set to this category, that

$$Q(rain = yes \,|\, umbrellas =_s open) = Q(rain = yes \,|\, umbrellas =_s closed), \tag{4.4}$$

that is, whether people open or close their umbrellas will not alter the relative frequency and the manipulation probability of rain.

The situation is different when the experimenter makes or stops rain. People will open or close their umbrellas accordingly, thus,

$$\begin{aligned} &Q(umbrellas = open \,|\, rain =_s yes) = 1, \\ &Q(umbrellas = closed \,|\, rain =_s yes) = 0, \end{aligned}$$

[1] That is, the observer becomes an experimenter.

and

$$Q(umbrellas = closed \mid rain =_s no) = 1,$$
$$Q(umbrellas = open \mid rain =_s no) = 0.$$

That is, the probability[2] of open umbrellas changes, when rain is set to yes or to no:

$$Q(umbrellas = open \mid rain =_s yes) \neq Q(umbrellas = open \mid rain =_s no). \tag{4.5}$$

In other words, manipulating rain shows an effect on the umbrellas (the manipulation distribution does depend on the manipulation), but manipulating umbrellas does not have an effect on rain (the manipulation distribution does not depend on the manipulation). Of course, the anticipation of these findings is based on our subject matter knowledge that people open umbrellas if it rains and close them if it does not, while rain cannot be made or stopped by opening or closing umbrellas. If this was a real research problem, then this subject matter knowledge would not be available neither to the observer nor to the experimenter, but also would not be needed by the experimenter. If experiments may be made (manipulation is possible), the experimenter will find out that manipulating rain implies a change in the umbrellas, while manipulating umbrellas implies nothing for the rain.

Thus, an observer learns very different things from what an experimenter learns. The observer sees (4.2) and (4.3), while the experimenter sees (4.4) and (4.5). The conditional distribution of rain does depend on the status of the umbrellas (see (4.2)), while the manipulation distribution of rain does not depend on the status of umbrellas (see (4.4)). Further, the manipulation distribution of the umbrellas depends on how rain is manipulated (see (4.5)), while the manipulation distribution of rain does not depend on the manipulation of the umbrellas (see (4.4)).

While the observer can infer association only, the experimenter can infer causation.

The above result about the expected behavior of a causal system (rain and umbrellas) was based on our understanding of the causal mechanisms and is, therefore, intuitive. Applying similar ideas to the analysis of more complex causal systems requires an appropriate generalization of the approach presented here but also an explicit formulation of the conditions that imply this behavior. In the next section, such conditions will be formulated. No claim will be made, however, that all causal systems would behave like this.

4.3 A Model of Causality

In this section, the main definitions and assumptions underlying the approach to causal analysis developed mainly by Pearl [128–130] and Pearl and Mackenzie

[2] Of course, the probabilities may be slightly different from 1 and 0.

[132], using DAGs to describe the causal structure and to make inference with respect to it, are formulated. For related, fundamental work, see, among others, Glymour, Zhang, and Spirtes [66], Kiiveri, Speed, and Carlin [79], and Spirtes, Glymour, and Scheines [157] and for other approaches Dawid [42], Granger [70], Holland [74], and Imbens and Rubin [76]. This is a model of causality in the sense that it postulates some basic properties and analyzes real situations based on these. As opposed to the apparent belief and sometimes even claims made[3] by many of the authors working in this framework or in its applications, this theory (model) of causation and causal analysis cannot be considered universal. Before applying it, a careful consideration needs to be given to the substantive characteristics of the research problem to decide whether or not the following theory captures its essential features. The approach only applies to causal structures, to which the concepts and assumptions presented in this section with respect to their behavior are a reasonably good approximation.

4.3.1 Assumptions and Fundamental Results

In the approach presented in this chapter, a causal system consists of a finite set of variables V and a binary relation between them, say $\mathscr{C}$. In this book, the variables are categorical, but an extension to include other types of variables is possible.

The binary relation $\mathscr{C} \subseteq V \times V$ means that with $A, B \in V$, either $(A, B) \in \mathscr{C}$ or $(A, B) \notin \mathscr{C}$. In the former case, one says that A is an immediate cause of B or that B is an immediate effect of A. The relation $\mathscr{C}$ is assumed to have particular properties.

Note that we are not defining what is cause or effect; we only postulate some formal properties, which then may or may not give a good enough description of a real situation.

The relation $\mathscr{C}$ is irreflexive:

$$\text{For every } A \in V, \ (A, A) \notin \mathscr{C}. \tag{4.6}$$

and asymmetric:

$$\text{For every } A, B \in V, \ (A, B) \in \mathscr{C} \text{ implies } (B, A) \notin \mathscr{C}. \tag{4.7}$$

In fact, the theory to be developed requires a stronger property than (4.6) and (4.7). It is also assumed that the relation $\mathscr{C}$ is acyclic in the following sense. Call a sequence of variables $V_1, V_2 \ldots V_{k-1}, V_k$ a causal path from V_1 to V_k, if $(V_i, V_{i+1}) \in \mathscr{C}$ for all $i = 1, \ldots k-1$. The relation is acyclic, if no causal path exists from a variable A to itself. This will be assumed to hold for $\mathscr{C}$.

[3] For instance, Pearl [130] says the aim is "presenting a unifying theory, ... within which most (if not all) aspects of causation can be formulated, analyzed, and compared ...."

Assumption 1: (4.8)

The causal effects among variables V are described by a binary relation $\mathscr{C}$, which is acyclic.

Note that (4.8) also excludes paths with $k = 1$ starting and ending at the same variable and, thus, implies (4.6) and for $k = 2$ it implies (4.7).

A causal structure[4] is $(V, \mathscr{C})$, where V is a set of variables and $\mathscr{C}$ is an acyclic binary relation on it. The reason for excluding casual structures that are cyclic, that is, contain a directed cycle, is that in a directed cycle, cause and effect could not be distinguished.[5]

Such a causal structure may be represented by a directed graph, called the causal graph, where V is the set of nodes, and an arrow points from A to B if and only if $(A, B) \in \mathscr{C}$. The following is readily implied by (4.8).

Proposition 4.1 *The causal graph is acyclic.*

□

The graph-related language, including parent, descendant, and nondescendant, which applies to the causal graph, will be used for the causal structure as well.

The causal structure is assumed to be related to the real, observable properties of the joint behavior of the variables. This behavior does not depend on the observer and is described by a joint probability distribution $P(V)$ of the variables V, called the observational distribution. In some causal systems, this distribution may have zero probabilities for some combinations of the categories of the variables, including the possibility of deterministic causal systems but, in general, a probabilistic behavior is assumed. For example, if the real causal system consists of a switch and a light bulb controlled by it, then the system is deterministic: the light bulb will be on or off depending on the position of the switch. The probability of the switch being on but the light off is zero. This is a deterministic causal system. Of course, one may choose to describe the system with a stochastic model, allowing a small probability that the switch is on but the light is off, for example, to include in the model possible electricity outages or a blown fuse or a burnt light bulb.

In addition to observing the behavior of the causal system, interventions in the form of manipulating some of the variables or in the form of fixing some of the variables may also be possible in an experimental situation. Manipulation sets some of the variables to certain categories, and fixing prevents some of the variables from changing even if other variables would imply such a change.

In the rain and umbrellas example, making rain or stopping rain would be manipulations of the rain variable, and not allowing people to open their umbrellas, when the umbrellas are closed, even if rain would imply they do so, is fixing. It may happen that certain interventions need forcing the causal structure to behave in a way

[4] Sometimes called a causal system.

[5] However, see also the discussion in Chap. 6 regarding the cyclic nature of some causal structures.

it would not, if left alone. For instance, in the rain and umbrellas example, it may be the situation that both rain and umbrellas may be manipulated individually freely, but the manipulation of making rain to start and making umbrellas closed is different from how the system would behave under observation, that is, without intervention. For the deterministic switch and light bulb example, the situation where the switch is on but the light bulb is off is not possible, but it is assumed that the intervention may achieve this. In the physical reality, the intervention may be turning the switch on and unscrewing the light bulb. Of course, this example is not very useful, unless it is part of a larger causal system, because otherwise it would be an intervention affecting all variables and nothing would remain to be observed or learned about the behavior of the system.[6]

The intervention distribution Q describes the behavior of the system after particular interventions were applied to it. After intervention, some of the variables not directly affected by the intervention, that is, not manipulated or fixed, typically do change. They may take on particular categories in deterministic causation, but in more general stochastic causation, the category they take on may not be determined, but rather influenced. The behavior of the variables not manipulated or fixed during the intervention is described by their intervention distributions, which are probability distributions. In the rain and umbrellas example, if it is made to rain (manipulation), the intervention distribution of umbrellas will associate a probability equal or close to 1 with open umbrellas. On the other hand, if umbrellas are forced to close (intervention), the probability of rain is not affected: it remains the same whether umbrellas are manipulated to be closed or open.

When the intervention distribution $Q(V)$ is mentioned in the sequel without specifying the actual intervention, all the intervention distributions are meant, which may arise as a result of an intervention to the system $(V, \mathscr{C})$. This usage is different from the joint distribution $P(V)$ which, if known, implies all possible conditional distributions of subsets of the variables, conditioned on specific categories of some of the other variables. $Q(V)$ is not a distribution in this sense, but rather a collection of distributions. For a particular intervention, the intervention may be seen as an index of the intervention distribution of the variables not subject to the intervention directly.

The variables in a causal system are assumed to always have observational and intervention distributions, but the researcher may or may not be able to get to know (learn) them. Finding out the intervention distribution would require potentially extensive interventions and this is not always feasible. Very often, only the observational distribution is known[7]. But, as will be seen later, when the intervention distribution may be learned, then it may be used to reconstruct the causal structure, even if the latter is not known to the researcher.[8] Further, very

[6] The issue of causal discovery, that is, learning the causal system by observing it with or without intervention, is discussed in Sect. 4.4.

[7] More precisely: may be estimated from data

[8] This, of course, is meant in cases when the causal structure may be assumed to have the properties postulated here.

importantly, the observational distribution may also be used to find out certain characteristics of the causal structure, which may be seen as the main strength of this theory.

Of course, to have the intervention distribution linked to the causal structure and to be able to recover aspects of the causal structure from the observational distribution, particular assumptions are necessary, which link the causal structure to the intervention distribution and the observational distribution to the causal structure and the intervention distribution.

To give a more formal treatment to interventions, let $\mathscr{B}_1$ and $\mathscr{B}_2$ be disjoint subsets of V. Then, an intervention to the causal system, where the variables in $\mathscr{B}_1$ are set equal to the categories $(c)_{\mathscr{B}_1}$ and the variables in $\mathscr{B}_2$ are kept fixed at $(c)_{\mathscr{B}_2}$, is denoted as

$$\mathscr{I}(\mathscr{B}_1 =_s (c)_{\mathscr{B}_1}, \mathscr{B}_2 =_f (c)_{\mathscr{B}_2}).$$

This intervention may be applied when $\mathscr{B}_2 = (c)_{\mathscr{B}_2}$ and entails setting $\mathscr{B}_1$ to $(c)_{\mathscr{B}_1}$ and fixing $\mathscr{B}_2$ at $(c)_{\mathscr{B}_2}$. The variables in $\mathscr{B}_3 = V \setminus \mathscr{B}_1 \setminus \mathscr{B}_2$ are not fixed or directly manipulated by the intervention $\mathscr{I}(\mathscr{B}_1 =_s (c)_{\mathscr{B}_1}, \mathscr{B}_2 =_f (c)_{\mathscr{B}_2})$, but they are not prevented from changing as a consequence of manipulating the variables in $\mathscr{B}_1$. Any or even both of the sets $\mathscr{B}_1, \mathscr{B}_2$ may be empty. When both $\mathscr{B}_1$ and $\mathscr{B}_2$ are empty, there is no intervention.

It is possible to apply interventions one after the other, for example, first setting $\mathscr{B}_2$ to $(c)_{\mathscr{B}_2}$ and then setting $\mathscr{B}_1$ to $(c)_{\mathscr{B}_1}$ and fixing $\mathscr{B}_2$ at $(c)_{\mathscr{B}_2}$. For the effect of an intervention, only the actual categories of the variables into which they are being manipulated or at which they are fixed is what matters, and whether the variables are manipulated into certain categories or fixed at them does not matter. More precisely, the consequences of the interventions

$$\mathscr{I}(\mathscr{B}_1 =_s (c)_{\mathscr{B}_1}, \mathscr{B}_2 =_s (c)_{\mathscr{B}_2})$$

and

$$\mathscr{I}(\mathscr{B}_1 =_s (c)_{\mathscr{B}_1}, \mathscr{B}_2 =_f (c)_{\mathscr{B}_2}).$$

are the same, if the second one may also be applied. This is a very strong assumption, implying no memory of the effect of interventions. In many real situations, this may not be true. For example, how sweet a dessert feels may depend not only on its sugar content but also on what one has consumed before tasting it. In such a case, it may be useful to consider not only the dessert but also the previous dish being part of the treatment. See also Sect. 6.3.2 for related comments.

The intervention distribution of the variables in $\mathscr{A} \subseteq \mathscr{B}_3$ after the interventions above is denoted as

$$Q(\mathscr{A} | \mathscr{I}(\mathscr{B}_1 =_s (c)_{\mathscr{B}_1}, \mathscr{B}_2 =_f (c)_{\mathscr{B}_2})). \tag{4.9}$$

Note again that, as opposed to what the notation suggests, (4.9) is not a conditional distribution derived from $Q(\mathscr{A}, \mathscr{B}_1, \mathscr{B}_2)$; rather, it is a probability distribution on $\mathscr{A}$, which is associated with the intervention $\mathscr{I}(\mathscr{B}_1 =_s (c)_{\mathscr{B}_1}, \mathscr{B}_2 =_f (c)_{\mathscr{B}_2})$.

If there exist category combinations $(c)_{\mathscr{B}_1}$, $(d)_{\mathscr{B}_1}$ and $(e)_{\mathscr{B}_2}$ such that

$$Q(\mathscr{A}|\mathscr{I}(\mathscr{B}_1 =_s (c)_{\mathscr{B}_1}, \mathscr{B}_2 =_f (e)_{\mathscr{B}_2})) \neq Q(\mathscr{A}|\mathscr{I}(\mathscr{B}_1 =_s (d)_{\mathscr{B}_1}, \mathscr{B}_2 =_f (e)_{\mathscr{B}_2})), \tag{4.10}$$

then $(\mathscr{B}_1, \mathscr{B}_2)$ is said to have an intervention effect on $\mathscr{A}$. When $\mathscr{B}_2 = \emptyset$, then the intervention effect is called a manipulation effect. The situation described here is that even though $\mathscr{B}_2$ is fixed, there is manipulation to $\mathscr{B}_1$ that has an effect on $\mathscr{A}$.

To link the intervention effects to the causal structure, the latter is assumed to be complete in the following sense.

Assumption 2: (4.11)

> If $(\mathscr{B}_1, \mathscr{B}_2)$ have an intervention effect on $\mathscr{A}$, there there exists a causal path from a $B \in \mathscr{B}_1$ to an $A \in \mathscr{A}$ which does not go through $\mathscr{B}_2$.

This is a completeness property that has the intuitive meaning that in the reality, which governs the intervention behavior of the system, and thus the intervention distribution Q, there exists no intervention effect that is not accounted for by a causal path in the causal structure. If part of the intervention is fixing some of the variables, then a causal path exists which does not go through any of these variables. Note that it is possible that different intervention distributions belong to the same causal structure[9] and in a particular intervention distribution, there is no causal effect, in spite of a causal path being present. A related phenomenon was called parametric cancelation in Wermuth and Cox [167]. See also the discussion of faithfulness later in this chapter.

Completeness is a strong assumption, which may be very difficult to check in some of the real applications.[10] In particular, this assumption does not hold if existing causal paths, whether going through included or not included variables, are left out from the analysis. On the other hand, making Assumption 2 does not provide protection against the consequences of the eventual existence of variables or causal paths left out from the analysis.

It has to be emphasized that the intervention distribution may be very complex. In particular, the joint manipulation effect of two variables usually cannot be deducted from their individual manipulation effects.

[9] For example, in the rain and umbrellas example, in one country, rain may not be very common, people do not wear raincoats, and thus $Q(U = yes|R =_s yes) = 1$, while in another country, many people wear raincoats and only open their umbrellas if the rain is heavy, leading to $Q(U = yes|R =_s yes) < 1$.

[10] In engineered systems, this may be easier than in the social, behavioral, or life sciences.

Now a few fundamental implications are considered. Here, and in the sequel, Assumption 1 is supposed to hold for the causal system. The pair (P, Q) is called a realization of the causal structure $(V, \mathscr{C})$, if Assumption 2 holds. This will be completed, later on, with Assumption 3 also being true.

Theorem 4.1 *If (P, Q) is a realization of a causal structure $(V, \mathscr{C})$, then*

$$Q(A|\mathscr{I}(nd(A) \setminus pa(A) =_s (c)_{nd(A)\setminus pa(A)}, pa(A) =_f (e)_{pa(A)})) \tag{4.12}$$

$$= Q(A|\mathscr{I}(nd(A) \setminus pa(A) =_s (d)_{nd(A)\setminus pa(A)}, pa(A) =_f (e)_{pa(A)})),$$

for all joint indices c, d, and e.

Proof If (4.12) was not true, then the variables in $nd(A) \setminus pa(A)$ would have a manipulation effect on A, when the variables in $pa(A)$ are fixed[11]. In this case, there would be a causal path from a variable in $nd(A) \setminus pa(A)$ to A, which does not go through $pa(A)$, and this is not possible, because every path leading to A has to go through one of its parents.

□

The result formulated in Theorem 4.1 is that if its parents are kept fixed, then manipulation of the other nondescendants has no effect on the behavior of the variable. The language often used for the property above is that keeping the parents fixed blocks the manipulation effects of the other nondescendants.

The next result is about blocking effects in a sense more general than considered in Theorem 4.1.

There may be several causal paths between two variables. For two causal paths to be different, they do not need to be disjoint (except for the starting and ending variables); rather, it is sufficient that they are not identical. Then one has the following.

Theorem 4.2 *Let there be T causal paths between variables B and A in a causal system $(V, \mathscr{C})$, and let the t-th be $B = V_1^t, \ldots, V_{k_t}^t = A$, $t = 1, \ldots, T$. For every path, keep one variable, say $V_{i_t}^t$, $1 \le i_t \le k_t$, fixed and manipulate B. Then, the intervention distribution of A does not depend on the manipulation of B, that is,*

$$Q(A|\mathscr{I}(B =_s b, \{V_{i_1}^1, \ldots, V_{i_T}^T\} =_f (e)_{V_{i_1}^1,\ldots,V_{i_T}^T})) \tag{4.13}$$

does not depend on b, irrespective of the choice of $(e)_{V_{i_1}^1,\ldots,V_{i_T}^T}$.

Proof By Assumption 2, if the manipulation distribution (4.13) depended on b, then there would be causal path from B to A not going through any of $\{V_{i_1}^1, \ldots, V_{i_T}^T\}$. But any causal path between B and A contains one of these variables, by the condition of the theorem.

□

[11] That is, $(nd(A) \setminus pa(A), pa(A))$ would have an intervention effect on A.

The following result may be seen as a stronger variant of Assumption 2.

Theorem 4.3 *Let $\mathscr{A}$, $\mathscr{B}_1$, $\mathscr{B}_2$ be pairwise disjoint subsets of the variables such that $(\mathscr{B}_1, \mathscr{B}_2)$ have an intervention effect on $\mathscr{A}$, and let c, d, and e be the category combinations in (4.10). Further, let Assumption 2 hold, and let*

$$\mathscr{B}_1 = \mathscr{B}_1^{c=d} \cup \mathscr{B}_1^{c\neq d}$$

be such, that for every variable $B \in \mathscr{B}_1^{c=d}$, $(c)_B = (d)_B$, and for every variable $B \in \mathscr{B}_1^{c\neq d}$, $(c)_B \neq (d)_B$, and $\mathscr{B}_2^{c\neq d}$ is not empty. Then, there is a causal path from a variable in $\mathscr{B}_1^{c\neq d}$ to a variable in $\mathscr{A}$, which avoids $\mathscr{B}_1^{c=d} \cup \mathscr{B}_2$.

Proof Because (4.10) holds, the variables in the causal system may take such values that fixing $\mathscr{B}_2$ at $(e)_{\mathscr{B}_2}$ is possible.

First, apply the intervention

$$\mathscr{I}(\mathscr{B}_1 =_s (c)_{\mathscr{B}_1}, \mathscr{B}_2 =_f (e)_{\mathscr{B}_2}) = \mathscr{I}(\mathscr{B}_1^{c\neq d} = (c)_{\mathscr{B}_1^{c\neq d}}, \mathscr{B}_1^{c=d} =_s (c)_{\mathscr{B}_1^{c=d}}, \mathscr{B}_2 =_f (e)_{\mathscr{B}_2}),$$

and after it the intervention

$$\mathscr{I}(\mathscr{B}_1^{c\neq d} =_s (c)_{\mathscr{B}_1^{c\neq d}}, \mathscr{B}_1^{c=d} \cup \mathscr{B}_2 =_f (c)_{\mathscr{B}_1^{c=d}}, (e)_{\mathscr{B}_2}),$$

that is, first set $\mathscr{B}_1$ to $(c)_{\mathscr{B}_1}$ and keep $\mathscr{B}_2$ fixed, then set $\mathscr{B}_1^{c\neq d}$ to $(c)_{\mathscr{B}_1^{c\neq d}}$ and keep $\mathscr{B}_1^{c=d} \cup \mathscr{B}_2$ fixed. As none of the variables changes during the second intervention compared to the results of the first one, the intervention distribution remains the same:

$$Q(\mathscr{A}|\mathscr{I}(\mathscr{B}_1 =_s (c)_{\mathscr{B}_1}, \mathscr{B}_2 =_f (e)_{\mathscr{B}_2}))$$
$$= Q(\mathscr{A}|\mathscr{I}(\mathscr{B}_1^{c\neq d} =_s (c)_{\mathscr{B}_1^{c\neq d}}, \mathscr{B}_1^{c=d} \cup \mathscr{B}_2 =_f (c)_{\mathscr{B}_1^{c=d}}, (e)_{\mathscr{B}_2})).$$

Apply the same with d instead of c, noting that

$$(c)_{\mathscr{B}_1^{c=d}} = (d)_{\mathscr{B}_1^{c=d}},$$

to obtain that

$$Q\mathscr{A}|(\mathscr{I}(\mathscr{B}_1 =_s (d)_{\mathscr{B}_1}, \mathscr{B}_2 =_f (e)_{\mathscr{B}_2}))$$
$$= Q(\mathscr{A}|\mathscr{I}(\mathscr{B}_1^{c\neq d} =_s (d)_{\mathscr{B}_1^{c\neq d}}, \mathscr{B}_1^{c=d} \cup \mathscr{B}_2 =_f (c)_{\mathscr{B}_1^{c=d}}, (e)_{\mathscr{B}_2})).$$

As the left-hand sides of the equations above are different by the assumption that (4.10) holds, so are the right-hand sides, and thus, Assumption 2 implies that

there is a causal path from a variable in $\mathscr{B}_1^{c\neq d}$ to a variable in $\mathscr{A}$, which avoids $\mathscr{B}_1^{c=d} \cup \mathscr{B}_2$.

□

This means that in the case of an intervention effect, not only is there a causal path starting within the subset of manipulated variables, but there is also one that starts at a variable actually changed, that is, manipulated to be different.

The final property postulates the relationship between certain intervention and observational distributions.

Assumption 3: (4.14)

Let $\mathscr{I}(\mathscr{B}_1 =_s (c)_{\mathscr{B}_1}, \mathscr{B}_2 =_f (c)_{\mathscr{B}_2})$ be an intervention with $pa(A) \subseteq \mathscr{B}_1 \cup \mathscr{B}_2 \subseteq nd(A)$. Then, for all joint indices c, $Q(A|\mathscr{I}(\mathscr{B}_1 =_s (c)_{\mathscr{B}_1}, \mathscr{B}_2 =_f (c)_{\mathscr{B}_2}) = P(A|\mathscr{B}_1 = (c)_{\mathscr{B}_1}, \mathscr{B}_2 = (c)_{\mathscr{B}_2})$.

The assumption means that the intervention distribution of a variable, when the intervention applies to all of its parents and possibly to some of its other nondescendants, is the same as its observational distribution conditioned on the same categories of the variables to which the intervention applied. Assumption 3 also implies that for two interventions,

$$\mathscr{I}(\mathscr{B}_1 =_s (c)_{\mathscr{B}_1}, \mathscr{B}_2 =_f (c)_{\mathscr{B}_2})$$

and

$$\mathscr{I}(\mathscr{C}_1 =_s (d)_{\mathscr{C}_1}, \mathscr{C}_2 =_f (d)_{\mathscr{B}_2})$$

such that

$$\mathscr{B}_1 \cup \mathscr{B}_2 = \mathscr{C}_1 \cup \mathscr{C}_2,$$

$$pa(A) \subseteq \mathscr{B}_1 \cup \mathscr{B}_2 = \mathscr{C}_1 \cup \mathscr{C}_2 \subseteq nd(A),$$

and

$$(c)_{\mathscr{B}_1 \cup \mathscr{B}_2} = (d)_{\mathscr{C}_1 \cup \mathscr{C}_2},$$

one has that

$$Q(A|\mathscr{I}(\mathscr{B}_1 =_s (c)_{\mathscr{B}_1}, \mathscr{B}_2 =_f (c)_{\mathscr{B}_2})$$

$$= Q(A|\mathscr{I}(\mathscr{C}_1 =_s (d)_{\mathscr{C}_1}, \mathscr{C}_2 =_f (d)_{\mathscr{C}_2}).$$

That is, for the intervention distribution of a variable, as long as the intervention applies to its parents and possibly to some of its other nondescendants, it does matter to which variables the intervention applies and also the categories into which the variables are manipulated or at which they are fixed. But it does not matter, which out of these variables are manipulated and which are fixed.

Assumption 3 implies immediately the following.

Proposition 4.2 *For every variable* $A \in V$,

$$Q(A|\mathscr{I}(pa(A) =_s (c)_{pa(A)})) = P(A|pa(A) = (c)_{pa(A)}).$$

□

Note that one can observe (4.14) to be true in the rain and umbrellas example. Using our subject matter knowledge, set up the causal structure that R (rain) is the cause of U (status of the umbrellas). Then, the causal graph consists of an arrow pointing from R to U.

The variable R is the parent and the only nondescendant of the variable U. The intervention and observational distributions of U were the same whether rain was made (manipulation) or observed. However, the intervention and observational distributions of R were different when umbrellas were set to open or closed (in either one of these cases, one still had the probability of rain) and when umbrellas were observed to be open (leading to a conditional probability of 1) or observed to be closed (giving a conditional probability of 0), respectively.

From now on, (P, Q) is called a realization of the causal system $(V, \mathscr{C})$, if Assumptions 2 and 3 hold, and such a realization is shown in Tables 4.1, 4.2, 4.3.

Let $P(R = yes) = 0.3$, $P(U = open|R = yes) = 1$, and $P(U = open|R = no) = 0$. Then, the joint observational distribution is given in Table 4.1.

The conditional distributions derived from the observational distribution are given in Table 4.2.

As was mentioned above, the intervention distribution is not a single joint distribution, but rather the collection of all distributions that arise after different

Table 4.1 A realization of the rain and umbrella causal system: the observational distribution

	U=open	U=closed
R=yes	0.3	0
R=no	0	0.7

Table 4.2 A realization of the rain and umbrella causal system: the observed conditional distribution

Condition	Conditional probability	Conditional probability
$R = yes$	$P(U = open\|R = yes) = 1$	$P(U = closed\|R = yes) = 0$
$R = no$	$P(U = open\|R = no) = 0$	$P(U = closed\|R = no) = 1$
$U = open$	$P(R = yes\|U = open) = 1$	$P(R = no\|U = open) = 0$
$U = closed$	$P(R = yes\|U = closed) = 0$	$P(R = no\|U = closed) = 1$

Table 4.3 A realization of the rain and umbrella causal system: the intervention distribution

Manipulation	Manipulation probability	Manipulation probability
$\mathscr{I}(R =_s yes)$	$Q(U = open \mid \mathscr{I}(R =_s yes)) = 1$	$Q(U = closed \mid \mathscr{I}(R =_s yes)) = 0$
$\mathscr{I}(R =_s no)$	$Q(U = open \mid \mathscr{I}(R =_s no)) = 0$	$Q(U = closed \mid \mathscr{I}((R =_s no)) = 1$
$\mathscr{I}(U =_s open)$	$Q(R = yes \mid \mathscr{I}(U =_s open)) = 0.3$	$Q(R = no \mid V(U =_s open)) = 0.7$
$\mathscr{I}(U =_s closed)$	$Q(R = yes \mid VU =_s closed)) = 0.3$	$Q(R = no \mid \mathscr{I}(U =_s closed)) = 0.7$

interventions. Table 4.3 shows these for the example of rain and umbrellas, assuming that rain always causes umbrellas to be opened, and no rain always causes umbrellas to be closed.

One sees that in the example,

$$P(U = o|R = y) = Q(U = o|R =_s y),$$

but

$$P(R = y|U = o) \neq Q(R = y|U =_s o),$$

which illustrate Assumption 3. In the first case, the parent is manipulated, and in the second case, the child is manipulated.

With the previous results, Assumption 3 leads immediately to the following important property of the observational distribution:

Theorem 4.4 *Let* (P, Q) *be a realization of the causal system* $(V, \mathscr{C})$. *Then,*

$$P(A|nd(A) = (c)_{nd(A)})$$

depends on $(c)_{pa(A)}$ *only.*

Proof By Assumption 3,

$$P(A|nd(A) = (c)_{nd(A)}) = Q(A|\mathscr{I}(nd(A) =_s (c)_{nd(A)})),$$

and by Theorem 4.1, the right-hand side does not depend on $(c)_{nd(A)\setminus pa(A)}$.

□

Let now $A \in V$ and consider

$$\{Q(A|\mathscr{I}(nd(A) =_s (c)_{nd(A)})), \text{ for all possible joint indices } (c)_{nd(A)}\}.$$

This is a collection of probability distributions depending on $(c)_{nd(A)}$. Let R be a probability distribution on $nd(A)$, that is, on all the joint indices of these variables. Then,

$$Q_R((c)_A, (c)_{nd(A)}) = Q(A|\mathscr{I}(nd(A) =_s (c)_{nd(A)}))R((c)_{nd(A)}) \tag{4.15}$$

is a probability distribution on the marginal $\{A\} \cup nd(A)$ which combines the probabilistic response to a manipulation with the probability of the manipulation.

For this distribution, one has the following result.

Theorem 4.5 *If (P, Q) is a realization of the causal structure $(V, \mathscr{C})$, then for the probability distribution Q_R defined in (4.15), irrespective of the choice of R,*

$$A \perp\!\!\!\perp nd(A) \setminus pa(A) | pa(A),$$

for all $A \in V$.

Proof The conditional probability

$$Q_R(a' | nd(A) = (c)_{nd(A)})$$

from (4.15), when it exists, is equal to

$$\frac{Q(a' | \mathscr{I}(nd(A) =_s (c)_{nd(A)})) R((c)_{nd(A)})}{\sum_a Q(a | \mathscr{I}(nd(A) =_s (c)_{nd(A)})) R((c)_{nd(A)})}$$

$$\frac{Q(a' | \mathscr{I}(nd(A) =_s (c)_{nd(A)})) R((c)_{nd(A)})}{R((c)_{nd(A)})}$$

$$= Q(a' | \mathscr{I}(nd(A) =_s (c)_{nd(A)}))$$

$$= Q(a' | \mathscr{I}(nd(A) \setminus pa(A) =_s (c)_{nd(A) \setminus pa(A)}, pa(A) =_s (c)_{pa(A)})),$$

and the right-hand side, by Theorem 4.1, does not depend on $(c)_{nd(A) \setminus pa(A)}$. Then, this also holds for the conditional distribution $Q_R(A | nd(A) = (c)_{nd(A)})$. Thus, Proposition 2.3 applies and yields the claim of the theorem.

□

As the most important implication of the assumptions, one has the following result for the observational distribution.

Theorem 4.6 *If (P, Q) is a realization of the causal system $(V, \mathscr{C})$, then, for the observational distribution on the $\{A\} \cup nd(A)\}$ marginal,*

$$A \perp\!\!\!\perp nd(A) \setminus pa(A) | pa(A).$$

Proof Assumption 3 implies that on the $\{A\} \cup nd(A)\}$ marginal, $P = Q_P$ by definition (4.15), and that

$$Q_P(A | nd(A) = (c)_{nd(A)}) = Q(A | \mathscr{I}(nd(A) =_s (c)_{nd(A)})).$$

Thus, by Theorem 4.1,

$$P(A|nd(A) = (c)_{nd(A)})$$

does not depend on $(c)_{nd(A)\setminus pa(A)}$, and then Proposition 2.3 yields the result.

□

This theorem means that if (P, Q) is a realization of the causal system with Assumptions 1–3 being valid, then the directed local Markov property holds for the observational distribution P. Not only does this result justify the definition and study of the directed Markov properties in the previous chapter but also provides the researcher who only can observe (and cannot manipulate) the causal system, with tools to test causal assumptions. If the researcher has a causal hypothesis, that is, a hypothesis with respect to the relation $\mathscr{C}$, then the conditional independences implied by Theorem 4.6 would present themselves in the observational distribution, see Sect. 4.4.4.

In particular, in view of Theorem 3.8, one has the following.

Corollary 4.1 *If (P, Q) is a realization of the causal system, then the observational distribution P factorizes as in (3.7).*

□

Another implication of this result is that one does not need to require a property similar to (4.14) for subsets of variables, instead of a single variable A, because the joint observational distribution of the variables is subject to the factorization (3.7) which uses the conditional distributions of individual variables given their respective parents.

Because of the factorization, the properties proved for distributions with Markov properties hold for observational distributions that are realizations of a causal system, with the graph defined by the causal relation. Out of the general results, the one considered next with an elementary proof has a central role in causal theory. It is related to the situation considered in Theorem 4.2.

Theorem 4.7 *Let a causal structure $(V, \mathscr{C})$ consist of T causal paths between variables B and A. Let the t-th one be $B = V_1^t, \ldots, V_{k_t}^t = A$, $t = 1, \ldots, T$. Further, let (P, Q) be a realization of the causal structure. For every path, choose one variable, say $V_{i_t}^t$, $2 \leq i_t \leq k_t - 1$. Then, in the observational distribution,*

$$A \perp\!\!\!\perp B | V_{i_1}^1, \ldots, V_{i_T}^T.$$

Proof First, we show that every variable, except for B, A, and the conditioning variables $V_{i_t}^t$, $2 \leq i_t \leq k_t - 1$, is either between B and one of the conditioning variables or between one of the conditioning variables and A, but not both. For simplicity of presentation, the former situation will be referred to as being before, while the latter as after a conditioning variable.

We show that it is not possible that for a variable, say, C, other than B, A, and $V_{i_t}^t$, $2 \leq i_t \leq k_t - 1$, that it is both before one conditioning variable (on one path) and also after another conditioning variable (on another path). Suppose, to the contrary,

that C is such that $V^1_{i_1}$ is before it and $V^2_{i_2}$ is after it. (The indices 1 and 2 are just used to simplify the presentation.) This is only possible if C is contained in both the first and second causal paths between B and A. Let the index of the variable C be c_1 on the first path and c_2 on the second path. Then, by the assumption that contradicts the claim to be proved, $i_1 < c_1$ and $i_2 > c_2$. Then, combine the part of the first path before C with C and the part of the second path after C. That is, consider

$$B = V^1_1 - V^1_2 - \cdots V^1_{c_1-1} - V^1_{c_1} = C = V^2_{c_2} - V^2_{c_2+1} - \cdots - V^2_{k_1-1} - V^2 k_1 = A. \tag{4.16}$$

This is a causal path from B to A, and it has two conditioning variables, $V^1_{i_1}$ and $V^2_{i_2}$. Indeed, $i_1 < c_1$ and $i_2 > c_2$. This contradicts the assumption of one conditioning variable on every path.

Consequently, every variable (other than the conditioning variables) is before or after a conditioning variable. Then, all the variables appearing in the causal paths connecting B to A belong to exactly one of the following groups:

(a) Variables before conditioning
(b) Conditioning variables
(c) Variables after conditioning

Because of the factorization (3.7) (see Corollary 4.1), the observational distribution is the product of terms $P(D|pa(D))$, for $D \in V$. The parents of a variable D are the ones that are immediately before it in any of the paths that contain D. Thus, the terms $P(D|pa(D))$, when D is before conditioning, contain only variables that are before conditioning. Similarly, the terms $P(D|pa(D))$ for conditioning variables contain conditioning and before conditioning variables, and the terms $P(D|pa(D))$ for the after conditioning variables contain after conditioning and conditioning variables. Therefore,

$$P(V) = \prod_{D\in\text{before}} P(D|pa(D) \prod_{D\in\text{conditioning}} P(D|pa(D) \prod_{D\in\text{after}} P(D|pa(D)$$

$$= \prod_{D\in\text{before or conditioning}} P(D|pa(D) \prod_{D\in\text{after}} P(D|pa(D).$$

Then, the first term contains only variables that are either conditioned on or are before conditioning, and the second term contains only variables that are conditioned on or are after conditioning.

The required conditional independence is yielded by Theorem *LCDA* 10.10.

□

Note that Theorems 4.2 and 4.7 have different conditions. The first one shows the lack of manipulation effect through the blocked chains, irrespective of the other parts of the causal graph. This is a result about the intervention distribution. The second one, to prove conditional independence, has to exclude the existence of other variables that still may relate B to A in the observational distribution.

Of course, the most general results establishing conditional independence in the observational distribution are the directed global Markov property (3.3), its equivalence with the factorization (3.7), and the equivalent d-separation rule, see Sect. 3.3. The blocking property of the variables $V_{i_1}^1, \ldots, V_{i_T}^T$ in Theorem 4.7 may be seen as separation of variables B and A.

The developments in this section have given explicit forms of the assumptions which, if hold,[12] are sufficient for this model of causality to be applied and for the above results to be true. Under these assumptions, the traditional view that observational data carry no relevance for the underlying causal structure needs to be revised, as discussed in Sect. 4.4.4.

4.3.2 Basic Causal Configurations Revisited

The results thus far are sufficient to give a rigorous discussion of some of the claims presented on a somewhat intuitive basis in Section *LCDA* 8.4. Figure 3.4 showed the three fundamental causal configurations for three variables. These configurations also appeared in Figure *LCDA* 8.1 It was shown in Section *LCDA* 8.4 that if effect meant a linear effect like in *LCDA* (8.8), then graphs (configurations) [g],[n], [r] in Figure *LCDA* 8.1 implied conditional independence among X, Y, and Z. As will be seen now, Corollary 4.1 implies this property for all causal structures, where Assumptions 1–3 hold.

Theorem 4.8 *Let $V = \{A, B, C\}$ and let $\mathscr{C}$ be such that in the causal graph, C is the parent of both A and B and no other arrow is present (a fork configuration). Then, if (P, Q) is a realization of the causal structure, then in the observational distribution,*

$$A \perp\!\!\!\perp B|C.$$

Proof By Corollary 4.1, $P(V)$ factorizes as

$$P(A, B, C) = P(C)P(A|C)P(B|C).$$

Then, whenever $P(C = c) > 0$,

$$P(A = a, B = b|C = c) = P(A = a, B = b, C = c)/P(C = c)$$

$$= P(A = a|C = c)P(B = b|C = c).$$

□

[12] With reasonably good approximation of reality

Similarly, for the graphs [h], [i], [l], [m], [p], and [q] in Figure *LCDA* 8.1 (chain configurations), if (P, Q) is a realization of the causal structures depicted in the graphs, then one obtains the following.

Theorem 4.9 *Let $V = \{A, B, C\}$ and let $\mathscr{C}$ be such that in the causal graph A is the parent of C and C is the parent of B and no further arrow is present (a chain configuration). Then, if (P, Q) is a realization of the causal structure, then in the observational distribution,*

$$A \perp\!\!\!\perp B|C.$$

Proof By Corollary 4.1, $P(V)$ factorizes as

$$P(A, B, C) = P(A)P(C|A)P(B|C),$$

thus,

$$P(A = a, C = c) = P(A = a)P(C = c|A = a).$$

Then, whenever the conditional probabilities exist,

$$\begin{aligned} P(B = b|A = a, C = c) &= P(A = a, B = b, C = c)/P(A = a, C = c) \\ &= P(B = b|C = c), \end{aligned}$$

which implies the claim.

□

For longer chains, one obtains the following generalization of the previous theorem. Note that this result also follows directly from the directed global Markov property, but an elementary proof is given here.

Corollary 4.2 *Let the causal structure be a causal chain, such that for variables $V_1, \ldots, V_l$, V_i is the parent of V_{i+1}, $i = 1, \ldots, l-1$. Let $1 \leq i < j \leq l$. Then, in any realization of the causal structure, for the observational distribution*

$$V_1, \ldots V_{i-1} \perp\!\!\!\perp V_{j+1}, \ldots, V_l|.V_i, \ldots, V_j.$$

Proof Combine the three groups of variables into variables A, C, B, respectively, and apply the previous theorem.

□

Finally, for the DAGs [j], [k], and [o] in Figure *LCDA* 8.1 (collider configurations), one obtains marginal independence of two of the variables.

Theorem 4.10 *Let $V = \{A, B, C\}$ and let $\mathscr{C}$ be such that in the causal graph, A and B are the parents of C and no further arrow is present (collider configuration). Then, if (P, Q) is a realization of the causal structure, then in the observational distribution,*

$$A \perp\!\!\!\perp B.$$

Proof One has a factorization

$$P(A = a, B = b, C = c) = P(A = a)P(B = b)P(C = c|A = a, B = b),$$

and marginalization over C yields the result.

□

An important implication of the results of this section is that if Assumptions 1–3 hold for the realization of a causal system, then by Theorems 3.8 and 3.9, the d-separation rule applies to the observational distribution.

In the next section, we return to the issue of learning the causal structure when intervention or when observation only is possible with Assumptions 1–3 made for the causal system.

4.4 Causal Discovery

A main strength of the theory presented in this chapter is that it makes learning or discovering the causal structure possible. That this can be done if the researcher is able to manipulate some of the variables (Sect. 4.4.3) is not surprising, as manipulation is an extension of the classical experimentation. More surprising and useful is that important characteristics of the causal structure may also be discovered, when the researcher may only observe the behavior of the causal system, without being able to intervene (Sect. 4.4.4). A useful summary of the methods of causal discovery from an applied perspective is given in Glymour, Zhang, and Spirtes [66]. Of course, the observation of the behavior of a causal system (with or without intervention) may only reveal the underlying causal structure if the realization from which the observations come is faithful to the causal structure in the sense that, for instance, does not have conditional independences that are not implied by the causal structure. A formal discussion of this concept is given in Sect. 4.4.2. Before presenting some of these results on learning causal structures, a brief overview of the possible types of causal analyses is given in Sect. 4.4.1.

4.4.1 Types of Causal Analyses

There are many research scenarios that may be called causal analysis.[13] In some cases, the causal structure is known either from construction or previous experience

[13] The view of the different causal problems is different here from the distinction between the effects of causes and the causes of effects, see Holland [74] and Dawid [42] and also of what is called the ladder of causality in Pearl and Mackenzie [132].

in human created or engineered systems, and the goal is to estimate the magnitudes of causal effects. In other setups, the causal structure is not known but the researcher has a hypothesis with respect to it, and the task is to confront the hypothetical structure with the observed behavior. A third situation is when the causal structure is not known but the researcher has the opportunity to repeatedly observe its behavior, with or without the ability of intervention, and the task is to find out as much as possible about the causal structure. This latter activity is called causal learning or causal discovery.

Causal discovery based on observations, if Assumptions 1–3 are made, relies on the results of the previous section. First, we will deal with ways of exploring the causal structure through intervention with the aim of drawing a causal graph and thus reconstructing the causal relation $\mathscr{C}$. The possibilities of causal discovery, when only the observation of the behavior of the system is possible, will be discussed after that.

In the previous section, the causal system was assumed to be given and assumptions with respect to its behavior were made. In this section, the causal system is not assumed to be known, but it is assumed that it possesses the characteristics (4.8), (4.11), and (4.14) (Assumptions 1–3), and, further, it is assumed that an experimenter can manipulate or fix the variables, can do so repeatedly, and can also estimate the resulting intervention distributions.

In the approaches mentioned so far, most of the time, the set of variables V is given, and learning concentrates on the relation $\mathscr{C}$. One has to keep in mind that to realistically believe that Assumptions 1–3 hold for the causal system, sometimes additional variables would need to be taken into account and this is not always possible.

The findings of the experimenter, of course, refer to the empirical intervention distributions, but it is also assumed that a sufficiently large number of experiments may be made so that the empirical distributions approximate the true distributions reasonably well. To simplify the presentation, the results will be stated for the true intervention distributions, that is, as if they could be observed directly by the experimenter.

Before giving the relevant results, a further possible property of a realization is discussed, which may be very helpful in causal discovery.

4.4.2 Faithfulness

In the context of causal discovery, it is assumed that the experimenter cannot observe the causal system $(V, \mathscr{C})$ directly, but only a possible realization (P, Q) of it. This fact may limit the researcher's ability to learn the causal structure. For example, if $V = \{A, B\}$ and $\mathscr{C} = \{(B, A)\}$, then the structure allows a causal effect of B on A, but does not require it, thus, a (P, Q), where A and B are independent in P and have no manipulation effects on each other in Q, may be a realization. Indeed, Assumptions 1 and 3 are not affected and Assumption 2 is easily seen to hold. If the

experimenter has the ability to manipulate B, then they will find out that it has no manipulation effect on A and will not learn about the causal effect of B on A which is allowed by the causal structure. Indeed, the researcher will see that manipulating B does not have an effect on A, because

$$Q(A|\mathscr{I}(B =_s b))$$

does not depend on b and neither does manipulating A have an effect on B. Thus, the conclusion will be that the causal relation $\mathscr{C}$ is empty. In fact, this conclusion is correct for the actual realization, but not for the causal system.

One strategy to deal with this situation is to use the manipulations only to establish where a causal effect exists, but not to conclude based on the manipulation that it does not. The existence of a causal effect will be implied by the existence of a manipulation effect because of Assumption 2 and, where this cannot be established, leave the existence of a causal effect or the lack of it open. This approach renders the causal discovery partial.

When this is not satisfactory, one may assume that the intervention distribution the researcher can observe is faithful to the causal structure. The intervention distribution Q is said to be faithful to the causal structure $(V, \mathscr{C})$ if the existence of a causal path from a variable in $\mathscr{B}_1$ to a variable A implies that $\mathscr{B}_1$ has a manipulation effect on A in Q. More precisely, faithfulness is defined as the property that for every $\mathscr{B}_1 \neq \emptyset$ and $\mathscr{B}_2$,

$$Q(A|\mathscr{I}(\mathscr{B}_1 =_s (c)_{\mathscr{B}_1}, \mathscr{B}_2 =_f (c)_{\mathscr{B}_2}))$$

does depend on $(c)_{\mathscr{B}_1}$ if there is a causal path from a variable in $\mathscr{B}_1$ to variable A which does not go through $\mathscr{B}_2$. The latter part of the condition is necessary; otherwise, fixing the variables in $\mathscr{B}_2$ may block the manipulation effect. The following variant is readily implied.

Proposition 4.3 *If the intervention distribution Q is faithful to the causal structure and there is a causal path from a variable in $\mathscr{B}_1$ to a variable A, then*

$$Q(A|\mathscr{I}(\mathscr{B}_1 =_s (c)_{\mathscr{B}_1}))$$

does depend on $(c)_{\mathscr{B}_1}$.

Proof No path goes through $\mathscr{B}_2 = \emptyset$, and the definition of faithfulness implies the claim.

□

Note that faithfulness is defined as the converse of Assumption 2. Assumption 2 and faithfulness together imply that there is a causal effect of B on A if and only if there is a causal path from B to A in $\mathscr{C}$.

When, in a realization of the causal structure $(V, \mathscr{C})$, Q is faithful, then, by Assumption 3, P also has a faithfulness property, namely, that if $pa(A) \subseteq \mathscr{B}_1 \cup$

$\mathscr{B}_2 \subseteq nd(A)$, and if there is a causal path from a variable in $\mathscr{B}_1$ to A that avoids $\mathscr{B}_2$, then

$$P(A|\mathscr{B}_1 = (c)_{\mathscr{B}_1}, \mathscr{B}_2 = (c)_{\mathscr{B}_2}))$$

does depend on $(c)_{\mathscr{B}_1}$. Because of this, if Q is faithful, then (P, Q) is also called faithful.

The existence of such a faithful intervention distribution for every causal structure is stated in the next theorem. The proof relies on a construction to be given in the next chapter.

Theorem 4.11 *Let $(V', \mathscr{C})$ be a causal structure. Then, there exists a realization (P, Q), which is faithful to it.*

Proof For an example of a faithful intervention distribution that applies to all causal structures when the variables are binary, see the incremental probabilistic causation model ($IPCM$) constructed in Sect. 5.4, in particular (5.8) and (5.9). Theorem 5.3 states that the constructed distribution is faithful.

To extend the $IPCM$ construction to the not necessarily binary variables V', let V be another set of $0-1$ variables so that to every $A' \in V'$, there is exactly one binary variable $A \in V$, such that if A' has $c_{A'}+1$ categories, $1, 2, \ldots, c_{A'}, c_{A'}+1$, then A is equal to 0 if and only if $A' = 1, 2, \ldots, c_{A'}$ and A is equal to 1 if and only if $A' = c_{A'} + 1$.

Consider the causal structure $(V, \mathscr{C})$, and the $IPCM$ intervention distribution Q, which is faithful to $\mathscr{C}$.

For the intervention distribution Q on the binary variables V, define an intervention distribution Q' on the variables in V'. For a subset $\mathscr{B} \subseteq V$, let $\mathscr{B}' \subseteq V'$ be the subset consisting of the corresponding variables in V'. Every intervention probability

$$Q'(A' = (v')_{A'}|\mathscr{I}(\mathscr{B}'_1 =_s (v')_{\mathscr{B}'_1}, \mathscr{B}'_2 =_f (v')_{\mathscr{B}_2}))$$

is a function of

$$(v')_{\{A'\}\cup\mathscr{B}'_1\cup\mathscr{B}'_2} = w'.$$

The intervention probabilities Q' are defined by distributing equally the corresponding intervention probabilities in the binary case. That is, let

$$W' = \{A'\} \cup \mathscr{B}'_1 \cup \mathscr{B}'_2 = W'_1 \cup W'_2,$$

where W'_2 is the set of those variables that are in their last categories in w', and W'_1 is the set of the other variables in W'. Then, define

$$d_{w'} = \prod_{C' \in W'_1} \frac{1}{c_{C'}}.$$

The corresponding intervention probability in the binary case is

$$Q(A = (v)_A | \mathscr{I}(\mathscr{B}_1 =_s t, \mathscr{B}_2 =_f u)),$$

where the joint indices t and u are defined, to specify the corresponding intervention, as

$$(t)_{C'} = 0, \text{ if } C' \in \mathscr{B}'_1 \cap W'_1,$$

$$(t)_{C'} = 1, \text{ if } C' \in \mathscr{B}'_1 \cap W'_2,$$

$$(u)_{C'} = 0, \text{ if } C' \in \mathscr{B}'_2 \cap W'_1,$$

$$(u)_{C'} = 1, \text{ if } C' \in \mathscr{B}'_2 \cap W'_2.$$

Then, define

$$\begin{aligned} Q'(A' = (v')_{A'} | \mathscr{I}(\mathscr{B}'_1 =_s (v')_{\mathscr{B}'_1}, \mathscr{B}'_2 =_f (v')_{\mathscr{B}'_2})) \\ = d_{w'} Q(A = (v)_A | \mathscr{I}(\mathscr{B}_1 =_s t, \mathscr{B}_2 =_f u)). \end{aligned} \tag{4.17}$$

Note that other than being a probability distribution, there is no particular assumption as to what an intervention distribution may be, so the Q' defined above is possible.

To see that the defined intervention distribution Q' is faithful, assume, to the contrary, that there is a variable $A'(Q')$ and subsets of variables $\mathscr{B}'_1(Q')$ and $\mathscr{B}'_2(Q')$, such that there is a causal path from a variable in $\mathscr{B}'_1(Q')$ to $A'(Q')$ that does not go through $\mathscr{B}'_2(Q')$, yet

$$Q'(A'(Q') | \mathscr{I}(\mathscr{B}'_1(Q') =_s (v')_{\mathscr{B}'_1(Q')}, \mathscr{B}'_2(Q') =_f (v')_{\mathscr{B}'_2(Q')}))$$

does not depend on $(v')_{\mathscr{B}'_1(Q')}$. But if Q is the intervention distribution of the $IPCM$, then because of the definition (4.17), this would also be the case for Q, which contradicts Q being faithful.

□

When the manipulation distribution is not faithful, one may experience a phenomenon similar to blocking, as was described in Theorem 4.2. First, we need a preliminary result.

Theorem 4.12 *Suppose that the realization of a causal system $(V, \mathscr{C})$ is such that a variable B does not have a manipulation effect on any of the parents of one of its descendants A, that is,*

$$Q(pa(A) | \mathscr{I}(B =_s b))$$

does not depend on b. Then, B has no manipulation effect on A, that is,

$$Q(A|\mathscr{I}(B =_s b))$$

does not depend on b.

Proof If A is a descendant of B, then B is a nondescendant of A by Proposition 4.1. When an intervention is applied to (some of) the nondescendants of A, the intervention distribution of A depends only on the categories into which its parents are set or at which they are fixed by Theorem 4.1.

Applying first an intervention to the parents of A, and then fixing the parents as a manipulation of A, one obtains that for all $B \in nd(A)$, the manipulation probability may be written as

$$Q(A = a|\mathscr{I}(B =_s b))$$

$$= \sum_{(v)_{pa(A)}} Q(pa(A) = (v)_{pa(A)}|\mathscr{I}(B =_s b))Q(A = a|\mathscr{I}(pa(A) =_f (v)_{pa(A)}).$$

The first term in the product does not depend on b by assumption, and then $Q(A = a|\mathscr{I}(B =_s b))$ also does not depend on b, as required.

□

Then, one has the following result for a realization that is not faithful.

Theorem 4.13 *Let there be T causal paths between variables B and A in a causal system* $(V, \mathscr{C})$*, and let the t-th be* $B = V_1^t, \ldots, V_{k_t}^t = A$*,* $t = 1, \ldots, T$*. Assume that in a realization* (P, Q)*, for every path, there is one variable, say* $V_{i_t}^t$*,* $1 < i_t < k_t$*, on which B does not have a manipulation effect in the sense that*

$$Q(V_{i_t}^t|\mathscr{I}(B =_s b))$$

does not depend on b, for all $t = 1, \ldots, T$*. Then, B does not have a manipulation effect on A in the sense that*

$$Q(A|\mathscr{I}(B =_s b))$$

does not depend on b.

Proof First note that $V_{k_t-1}^t$, $t = 1, \ldots, T$, are parents of A. A may have further parents, but there is no causal path from B to any of these further parents, because then one would have additional causal paths between B and A. Therefore, B has no manipulation effect on any of the parents of A other than $V_{k_t-1}^t$, $t = 1, \ldots, T$ by Assumption 2.

The proof is by induction on the length of the longest path between B and A. The length of the t-th path is defined as k_t. It follows from the assumptions that $k_t \geq 3$, $t = 1, \ldots, T$.

If $k_t = 3$ for $t = 1, \ldots, T$, then, by assumption, B has no manipulation effect on V_2^t, for $t = 1, \ldots, T$, which are the parents of A on the causal paths. As B also does not have a manipulation effect on the other parents A, B has no manipulation effect on A by Theorem 4.12.

Suppose the claim of the theorem is true for all cases when all the causal paths between B and A are such that $k_t \leq j, t = 1, \ldots, T$, for some integer j.

Let $k_t \leq j + 1$ for $t = 1, \ldots, T$. For B to have a manipulation effect on A, by Theorem 4.12, B would need to have a manipulation effect on a parent of A. As B does not have a manipulation effect on any parent of A which is not on the paths considered, B would need to have a manipulation effect on $V_{k_u-1}^u$ for some u, $1 \leq u \leq T$. We show that this is not possible.

Consider the paths from B to $V_{k_u-1}^u$. All these paths are among the ones considered, because any path from B to $V_{k_u-1}^u$ continues to A, as $V_{k_u-1}^u$ is a parent of A. The lengths of all the paths from B to $V_{k_u-1}^u$ are less than or are equal to j. Further, on any path from B to A, there is one variable that is not affected by the manipulation of B. On all paths from B to $V_{k_u-1}^u$, this variable is either $V_{k_u-1}^u$, which completes the proof, or it is before $V_{k_u-1}^u$, which by the induction assumption implies that B does not have a manipulation effect on $V_{k_u-1}^u$, and in this case, the proof is also complete.

□

Note that in the proof above, it may be the case that $V_{k_u-1}^u$ is a parent of A, and, in addition, there is a causal path from $V_{k_u-1}^u$ to A, which is longer than the parent to child arrow. But this does not affect the validity of the proof, as also on that longer path, there is a parent of A, for which the argument applies.

By Assumption 3, the faithfulness of the intervention distribution has implications for the observational distribution too.

Theorem 4.14 *Let $(V, \mathscr{C})$ be a causal structure and (P, Q) a realization of it and $\mathscr{D} \subseteq nd(A)$ for a variable A. Then, if*

$$\mathscr{D} \supseteq pa(A)$$

holds, then for the observational distribution P,

$$A \perp\!\!\!\perp nd(A) \setminus \mathscr{D} | \mathscr{D}. \tag{4.18}$$

If Q is faithful, then the reverse implication also holds.

Proof Theorem 4.6 and Propositions 2.6 and 2.5 imply the first claim.

To see the converse statement, assume it is not true, so (4.18) holds, but $\mathscr{D} \subset pa(A)$. Then, there are variables in $pa(A) \setminus \mathscr{D} \subseteq nd(A) \setminus \mathscr{D}$, and there are causal paths (in fact, arrows) from the variables in $pa(A) \setminus \mathscr{D}$ to A. Then, by faithfulness, there is a manipulation effect of some of the variables in $pa(A) \setminus \mathscr{D}$ on A, even if the variables in $\mathscr{D}$ are fixed, as these paths (arrows) do not go through any variable in $\mathscr{D}$. That means that

$$Q(A|\mathscr{I}(nd(A) =_s (c)_{nd(A)},)) \tag{4.19}$$

$$= Q(A|\mathscr{I}(nd(A) \setminus \mathscr{D} =_s (c)_{nd(A)\setminus\mathscr{D}}, \mathscr{D} =_s (c)_{\mathscr{D}})$$

does depend on $(c)_{nd(A)\setminus\mathscr{D}}$. Then, by Assumption 3,

$$P(A|nd(A) \setminus \mathscr{D} = (c)_{nd(A)\setminus\mathscr{D}}, \mathscr{D} = (c)_{\mathscr{D}})$$

does depend on $(c)_{nd(A)\setminus\mathscr{D}}$ and by Proposition 2.3, (4.18) cannot hold.

□

This result means that if Q is faithful, then for $\mathscr{D} \subseteq nd(A)$,

$$A \perp\!\!\!\perp nd(A) \setminus \mathscr{D}|\mathscr{D}$$

holds if and only if

$$\mathscr{D} \supseteq pa(A).$$

4.4.3 Learning the Causal Structure Through Intervention

By observing the effects of manipulations, one may determine, for each variable $A \in V$, those variables the manipulation distributions of which never change upon manipulating A, irrespective of the values of the other variables.[14] Let $nc(A)$ denote these variables (nc for no change). Then one obtains easily the following.

Theorem 4.15 *Let (P, Q) be a realization of the causal structure $(V, \mathscr{C})$. Then,*

$$nc(A) \supseteq nd(A).$$

If the manipulation distribution is faithful, then

$$nc(A) = nd(A).$$

□

Based on the knowledge obtained through manipulating the system, for every variable $A \in V$, a further set of variables, denoted as $bs(A)$ (bs for blocking set), may be determined. The set $bs(A)$ is a subset of $nc(A)$ and is minimal with respect to the following property: if the variables in $bs(A)$ are fixed, then no manipulation of the remaining variables in $nc(A) \setminus bs(A)$ will change the distribution of A. That such a minimal set $bs(A)$ exists, if faithfulness is assumed, is proved next.

[14] Remember that the simplifying assumption that one can observe the theoretical distributions, not only empirical versions of them, was made.

Theorem 4.16 *Let (P, Q) be a realization of a causal system $(V, \mathscr{C})$, such that Q is faithful. Then, for every variable $A \in V$, there is a unique set $bs(A) \subseteq nc(A)$ which is minimal with the blocking property that if the variables in it are fixed, then the variables in $nc(A) \setminus bs(A)$ have no manipulation effect on A.*[15]

Proof If there is only one blocking set, then it is minimal.

Suppose there are sets of variables $S_i \subseteq nc(A)$, such that

$$Q(A|\mathscr{I}(nc(A) \setminus S_i =_s (c)_{nc(A)\setminus S_i}, S_i =_f (c)_{S_i})) \tag{4.20}$$

does not depend on $(c)_{nc(A)\setminus S_i}$.

This means that in any causal path starting in $nc(A) \setminus S_i$ and leading to A, after the last variable in $nc(A) \setminus S_i$, there has to be a variable from S_i. Indeed, consider the part of the causal path starting from the last variable in $nc(A) \setminus S_i$ and leading to A. Because of the faithfulness of Q, if this causal path avoided S_i, then (4.20) would depend on $(c)_{nc(A)\setminus S_i}$.

It is sufficient to show that if S_1 and S_2 are blocking sets, then

$$Q(A|\mathscr{I}(nc(A) \setminus S_1 \cap S_2 =_s (c)_{nc(A)\setminus S_i \cap S_2}, S_i \cap S_2 =_f (c)_{S_1 \cap S_2}))$$

also does not depend on $(c)_{nc(A)\setminus S_i \cap S_2}$ and thus, the set of variables $bs(A)$ is obtained as the intersections of all sets with the blocking property. Note that as V is finite, there can be only a finite number of sets with the blocking property.

Suppose this is not the case, and $nc(A) \setminus (S_1 \cap S_2)$ has a manipulation effect on A, which is not blocked by $(S_1 \cap S_2)$. Then, by Assumption 2, there is a causal path $\mathscr{P}$ starting from $nc(A) \setminus (S_1 \cap S_2)$ to A which does not go through $(S_1 \cap S_2)$, see also Theorem 4.2.

If $\mathscr{P}$ does not go through $(S_1 \cap S_2)$, then it may go through or start at a variable in $S_1 \setminus (S_1 \cap S_2) = S_1 \setminus S_2$, or it goes through or starts at a variable in $S_2 \setminus (S_1 \cap S_2) = S_2 \setminus S_1$, or it avoids $S_1 \cup S_2$ entirely.

In the first case consider only the part that starts in $S_1 \setminus S_2$, this, as $\mathscr{P}$ is blocked by S_2, also has to go through it.

In the second case consider only the part that starts in $S_2 \setminus S_1$, this is $\mathscr{P}$ is blocked by S_1, also has to go through it.

The third case is not possible: $\mathscr{P}$ has to go through both S_1 and S_2, as both block it, or it starts in one of them and goes through the other one.

In either case, $\mathscr{P}$ goes through both S_1 and S_2 or starts in one of them and goes through the other one. That is, $\mathscr{P}$ has variables both in S_1 and S_2, and as it does not go through $S_1 \cap S_2$, these are different variables. Let C be the last of these variables, and without restriction of generality, assume $C \in S_1$, and then $C \notin S_2$.

As $S_1, S_2 \subseteq nc(A)$, the part of $\mathscr{P}$ starting at C and leading to A is also a causal path in $nc(A) \setminus S_2$, and by assumption it is also blocked by S_2. But if there was no

[15] Minimal here does not mean of minimal cardinality, but rather being a subset of all sets with the blocking property

node from S_2 between C and A on the path $\mathscr{P}$, this blocking would not be possible. Thus, C cannot be the last variable in $\mathscr{P}$ from S_1 or S_2 if it does not belong to both S_1 and S_2.

But then $S_1 \cap S_2$ is not empty and has the claimed blocking property.

□

The next result links the blocking set of a variable to the set of its parents.

Theorem 4.17 *Let (P, Q) be a realization of a causal system $(V, \mathscr{C})$. Then*

$$bs(A) \subseteq pa(A).$$

If Q is faithful, then

$$bs(A) = pa(A).$$

Proof By Theorem 4.2, all parents are a blocking set, so the minimal blocking set $bs(A)$ is a subset of the set of parents.

If Q is faithful, then all parents have to be in all and thus also in the minimal blocking set. If there was a parent not in the blocking set, then the causal effect of this parent, which by faithfulness exists even if eventual alternative paths from this parent to the child A are blocked, would not be blocked.

□

This means that if the intervention distribution is faithful, the observer who can manipulate may determine, for every variable A, all parents of it, and then the causal relation $\mathscr{C}$ may be recovered. If faithfulness does not hold, then only some of the parents may be revealed.

4.4.4 Learning the Causal Structure Through Observation Only

A particular strength of the theory developed so far is that causal discovery is also possible, if one cannot intervene, only observe the behavior of the system. The main tool of learning the causal system when only observation is possible is the d-separation rule developed in Sect. 3.2. In fact, the real importance of the d-separation rule lies in this application.

The concept of d-connectedness of the subsets of variables $\mathscr{A}$ and $\mathscr{B}$ by $\mathscr{D}$ was defined as the existence of a path $\mathscr{P}$ (disregarding the direction of the arrows) between an $A \in \mathscr{A}$ and a $B \in \mathscr{B}$ such that

(i) if C is a collider on $\mathscr{P}$, then C or a descendant of it is in $\mathscr{D}$
and
(ii) no noncollider in $\mathscr{P}$ is in in $\mathscr{D}$.

Then, the subsets of variables $\mathscr{A}$ and $\mathscr{B}$ are d-separated by the subset of variables $\mathscr{D}$ if they are not d-connected. That is, if for every path $\mathscr{P}$ connecting an $A \in \mathscr{A}$ with a $B \in \mathscr{B}$, disregarding the direction of the arrows,

(iii) there is a collider in $\mathscr{P}$, such that neither this collider nor a descendant of it is in $\mathscr{D}$
or
(iv) there is a noncollider of $\mathscr{P}$ contained in $\mathscr{D}$.

The d-separation rule is that if $\mathscr{A}$ and $\mathscr{B}$ are d-separated by $\mathscr{D}$, then

$$\mathscr{A} \perp\!\!\!\perp \mathscr{B} | \mathscr{D}.$$

As implied by Theorem 3.10, the d-separation rule is equivalent to the directed global Markov property. Corollary 4.1 is that if (P, Q) is a realization of a causal system (Assumptions 1–3 hold), then the observational distribution P factorizes as in (3.7). Finally, as the factorization (3.7) is equivalent to the directed global Markov property (see Theorem 3.8), one has the following result.

Theorem 4.18 *If P is the observational distribution from the causal structure $(V, \mathscr{C})$, then P obeys the d-separation rule with respect to the DAG $(V, \mathscr{C})$.*

□

The first related results were already given in Sect. 4.3.2 for the basic configurations depicted in Fig. 3.4, using the factorization property.

If for three variables, in a fork configuration, C is the parent of both A ad B, then on the $A - C - B$ path, C is a noncollider and d-separation holds, implying

$$A \perp\!\!\!\perp B | C,$$

as was shown in Theorem 4.8 using the factorization.

For a chain $A - C - B$, the path $A - C - B$ connects A with B and C is a noncollider on this path, thus, d-separation of A and B by C holds and implies

$$A \perp\!\!\!\perp B | C,$$

as was shown in Theorem 4.9 using the factorization.

If both A and B are parents of C, then on the $A - C - B$ path, C is a collider, thus, d-separation of A and B by C does not hold.

When $\mathscr{D}$ is selected to be the empty set, one has the following. If there is a collider on the path $A - C - B$, it is not contained in $\emptyset$, so d-separation by $\emptyset$ holds, and if there is a noncollider on the path, then it is not contained in $\emptyset$, and d-separation does not hold. This means that

$$A \perp\!\!\!\perp B$$

holds when C is a collider, but does not hold when there is a chain or fork configuration. Note that these facts follow when there are only three variables in V.

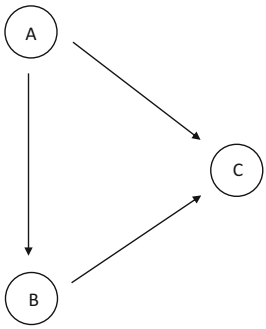


Fig. 4.1 A DAG, where conditional independence does not necessarily imply d-separation

Table 4.4 The manipulation distribution of B, when A is manipulated

Manipulation	Manipulation probability	Manipulation probability
$\mathscr{I}(A =_s 1)$	$Q(B = 1 \mid \mathscr{I}(A =_s 1)) = 7/11$	$Q(B = 2 \mid \mathscr{I}(A =_s 1)) = 4/11$
$\mathscr{I}(A =_s 2)$	$Q(B = 1 \mid \mathscr{I}(A =_s 2)) = 8/19$	$Q(B = 2 \mid \mathscr{I}(A =_s 2)) = 11/19$

One might expect that when the intervention distribution is faithful, then the converse of the d-separation rule also holds, that is, conditional independence of two subsets of variables given a third implies that they are d-separated by it. This, however, is not true, as is illustrated now by an example.

Let the causal graph for three binary variables A, B, C with categories 1 and 2 be as given in Fig. 4.1. Then, for every P, including those in a realization of the causal structure,

$$P(A = a, B = b, C = c) = P(A = a)P(B = b|A = a)P(C = c|A = a, B = b).$$

Variable A has no nondescendants, and assume its (manipulation) distribution is $(11/30, 19/30)$. Variable B has one nondescendant, and its manipulation distribution is given in Table 4.4.

Finally, Table 4.5 gives the intervention probabilities of C, when both A and B are manipulated.

The intervention probabilities show that Q is faithful.

The probabilities of the joint observational distribution, using the factorization (3.7) and Assumption 3, are obtained[16] as follows:

$$P(1, 1, 1) = 11/30 \times 7/11 \times 6/7 = 6/30,$$

$$P(1, 1, 2) = 11/30 \times 7/11 \times 1/7 = 1/30,$$

$$P(1, 2, 1) = 11/30 \times 4/11 \times 2/4 = 2/30,$$

[16] It does not matter which panel of Table 4.5 one uses, see the comment after Assumption 3.

Table 4.5 The intervention distribution of the collider C

Intervention	Intervention probability	Intervention probability
$\mathscr{I}(A =_s 1,\ B =_s 1)$	$Q(C = 1 \mid \mathscr{I}(A =_s 1,\ B =_s 1)) = 6/7$	$Q(C = 2 \mid \mathscr{I}(A =_s 1,\ B =_s 1)) = 1/7$
$\mathscr{I}(A =_s 1,\ B =_s 2)$	$Q(C = 1 \mid \mathscr{I}(A =_s 1,\ B =_s 2)) = 2/4$	$Q(C = 2 \mid \mathscr{I}(A =_s 1,\ B =_s 2)) = 2/4$
$\mathscr{I}(A =_s 2,\ B =_s 1)$	$Q(C = 1 \mid \mathscr{I}(A =_s 2,\ B =_s 1)) = 3/8$	$Q(C = 2 \mid \mathscr{I}(A =_s 2,\ B =_s 1)) = 5/8$
$\mathscr{I}(A =_s 2,\ B =_s 2)$	$Q(C = 1 \mid \mathscr{I}(A =_s 2,\ B =_s 2)) = 1/11$	$Q(C = 2 \mid \mathscr{I}(A =_s 2,\ B =_s 2)) = 10/11$
$\mathscr{I}(A =_s 1,\ B =_f 1)$	$Q(C = 1 \mid \mathscr{I}(A =_s 1,\ B =_f 1)) = 6/7$	$Q(C = 2 \mid \mathscr{I}(A =_s 1,\ B =_f 1)) = 1/7$
$\mathscr{I}(A =_s 1,\ B =_f 2)$	$Q(C = 1 \mid \mathscr{I}(A =_s 1,\ B =_f 2)) = 2/4$	$Q(C = 2 \mid \mathscr{I}(A =_s 1,\ B =_f 2)) = 2/4$
$\mathscr{I}(A =_s 2,\ B =_f 1)$	$Q(C = 1 \mid \mathscr{I}(A =_s 2,\ B =_f 1)) = 3/8$	$Q(C = 2 \mid \mathscr{I}(A =_s 2,\ B =_f 1)) = 5/8$
$\mathscr{I}(A =_s 2,\ B =_f 2)$	$Q(C = 1 \mid \mathscr{I}(A =_s 2,\ B =_f 2)) = 1/11$	$Q(C = 2 \mid \mathscr{I}(A =_s 2,\ B =_f 2)) = 10/11$
$\mathscr{I}(A =_f 1,\ B =_s 1)$	$Q(C = 1 \mid \mathscr{I}(A =_f 1,\ B =_s 1)) = 6/7$	$Q(C = 2 \mid \mathscr{I}(A =_f 1,\ B =_s 1)) = 1/7$
$\mathscr{I}(A =_f 1,\ B =_s 2)$	$Q(C = 1 \mid \mathscr{I}(A =_f 1,\ B =_s 2)) = 2/4$	$Q(C = 2 \mid \mathscr{I}(A =_f 1,\ B =_s 2)) = 2/4$
$\mathscr{I}(A =_f 2,\ B =_s 1)$	$Q(C = 1 \mid \mathscr{I}(A =_f 2,\ B =_s 1)) = 3/8$	$Q(C = 2 \mid \mathscr{I}(A =_f 2,\ B =_s 1)) = 5/8$
$\mathscr{I}(A =_f 2,\ B =_s 2)$	$Q(C = 1 \mid \mathscr{I}(A =_f 2,\ B =_s 2)) = 1/11$	$Q(C = 2 \mid \mathscr{I}(A =_f 2,\ B =_s 2)) = 10/11$
$\mathscr{I}(A =_f 1,\ B =_f 1)$	$Q(C = 1 \mid \mathscr{I}(A =_f 1,\ B =_f 1)) = 6/7$	$Q(C = 2 \mid \mathscr{I}(A =_f 1,\ B =_f 1)) = 1/7$
$\mathscr{I}(A =_f 1,\ B =_f 2)$	$Q(C = 1 \mid \mathscr{I}(A =_f 1,\ B =_f 2)) = 2/4$	$Q(C = 2 \mid \mathscr{I}(A =_f 1,\ B =_f 2)) = 2/4$
$\mathscr{I}(A =_f 2,\ B =_f 1)$	$Q(C = 1 \mid \mathscr{I}(A =_f 2,\ B =_f 1)) = 3/8$	$Q(C = 2 \mid \mathscr{I}(A =_f 2,\ B =_f 1)) = 5/8$
$\mathscr{I}(A =_f 2,\ B =_f 2)$	$Q(C = 1 \mid \mathscr{I}(A =_f 2,\ B =_f 2)) = 1/11$	$Q(C = 2 \mid \mathscr{I}(A =_f 2,\ B =_f 2)) = 10/11$

$$P(1, 2, 2) = 11/30 \times 4/11 \times 2/4 = 2/30,$$

$$P(2, 1, 1) = 19/30 \times 8/19 \times 3/8 = 3/30,$$

$$P(2, 1, 2) = 19/30 \times 8/19 \times 5/8 = 5/30,$$

$$P(2, 2, 1) = 19/30 \times 11/19 \times 1/11 = 1/30,$$

$$P(2, 2, 2) = 19/30 \times 11/19 \times 10/11 = 10/30,$$

showing that A and B are conditionally independent, given C.

If the converse of the d-separation rule was true, this conditional independence would imply that C d-separates A and B. In fact, C d-connects A and B: for the path $A - C - B$, which connects A and B, both (i) and (ii) hold. Indeed, C is a collider on the path and it is in the conditioning set and there is no noncollider on the path.

In light of this negative result, one cannot expect that observation (without intervention) of the causal system could entirely reveal the causal structure, even if the intervention distribution is faithful. However, a hypothetical causal structure has implications for the observational distribution which may be tested using the d-separation rule. If, as for simplicity of presentation is assumed here, the exact (theoretical) observational distribution may be observed, then conditional independences implied by the hypothetical causal structure may be checked via the d-separation rule directly. In the more realistic scenario when only the empirical version of the observational distribution may be observed, the tests of conditional independence models may be used. For the related multiple testing problem, see some of the comments in Section *LCDA* 11.2.

Also in this context, a stronger faithfulness property, similar to the one discussed for undirected graphical models (see Sect. 2.5.1), could be considered. This faithfulness concept would assume that the observational distribution exhibits such conditional independences only, which are implied by the d-separation rule applied to the DAG of the causal structure, see Glymour et al. [157] and Zhang and Spirtes [175]. While such an assumption is convenient in that it makes causal discovery more effective, in view of the previous example, it seems very strong and is impossible to check in a real data analytic situation.

On the other hand, if the researcher has a hypothetical causal structure, its implications via the d-separation rule may be tested independently of any faithfulness assumptions, even if observation only is possible.

4.5 Summary

This chapter presented the fundamentals of the theory that uses DAGs to model causality. A major contribution of this theory is a clear distinction between merely observing the behavior of the causal system or observing after some intervention to it was made. In the traditional approach of statistics, the former has been known to say little or nothing about the underlying causal structure, and the latter has been the basic component of experimenting and then obtaining causal conclusions. This approach to causal analysis shows that, under particular assumptions, certain types of observations may reveal characteristics of the causal structure, but also that manipulations of or more generally interventions to the system are useful beyond experimenting based on random allocation.

The theory, as presented here, relies on three assumptions, and before one would apply it to a real research problem, the relevance of these (or of other, equivalent) assumptions needs to be checked carefully. When this theory applies, it gives a rigorous basis for the somewhat intuitive discussion given to the applications of

graphs to model causal structures presented in *LCDA* and, more generally, implies rules as to how and to what extent the causal system may be learned from the results of interventions or of observations only.

As with most theories, some of the implications are intuitively attractive and make some of the users think "yes, this is the right theory to deal with causality," and others are less obvious, making other users think "I have to be careful using this theory to my research problem." The main methodological implication is that each researcher has to choose one of the positions above with respect to each and every research problem they work on. Applying this theory and the powerful methods it leads to routinely is not advised. But, on the other hand, abstaining from the application of this theory without a careful evaluation of its relevance to the actual research problem is also not advised.

The three main assumptions were that the causal relation and the causal graph have an acyclic structure, that if there is an intervention effect, then there is also a causal path, and that the intervention distribution, when the intervention applies to the parents and possibly some of the other nondescendants, is the same as the corresponding (i.e., appropriately conditioned) observational distribution. In addition, a few more, and not less important, assumptions were made, including that effects are described by a binary relation, that is, effects are individual, or that interventions may be applied one after the other, and if the final result is the same, it does not matter in which order the interventions were applied. The most important result was that these assumptions imply the Markov properties associated with DAGs, which were studied in the previous chapter.

The next two chapters of the book investigate further the intuitive appeal of this theory and also some of its application and limitations.

4.6 Things to Do

1. Find examples of causal systems where Assumption 1 does not hold.
2. Find examples of causal systems where Assumption 2 does not hold.
3. As mentioned in the text, it may happen that the causal path, the existence of which is postulated in Assumption 2, goes through included but also not included variables. This is the view if the variables that may be included in the analysis are known. But if the researcher does not know of variables not yet included, either one of two choices needs to be made. Either looking for additional variables through which the causal path goes or just including a causal effect in the DAG without including new variables. Discuss the possible reasons for either one of these choices.
4. Show that Theorem 4.2 implies Theorem 4.1.
5. Find examples of causal systems where Assumption 3 does not hold.
6. Use the fact that the relative frequency of an event converges in the probability to the probability of this event (see after Proposition *LCDA* 3.4) to develop a concept of an observed (discrete) distribution converging to the underlying (discrete) probability distribution.

7. Using the concept developed in the previous item, describe how the empirical intervention distribution relates to the theoretical intervention distribution.
8. Apply the description of the previous item to observing no intervention effect.
9. Prove Theorem 4.15.
10. The d-separation rule is defined for pairwise disjoint subsets of the variables. Why is this? What happens if this assumption is not made?
11. Tables 4.4 and 4.5 provide intervention distributions in an example. Is there any compatibility condition these distributions need to meet to be the intervention distributions associated with a causal structure?
12. Show that the intervention distribution given in Tables 4.4 and 4.5, with the manipulation distribution of variable A given in the text, is faithful.
13. Using the intervention distributions given, determine $bs(B)$ in the example with Fig. 4.1, without relying on the graph.

Chapter 5
Causality and Intuition



5.1 Introduction

This chapter aims at building intuition around the model of causality developed in Chap. 4 and also at showing some of the limitations of the intuition, including expectations one may have with respect to such a theory. As will be seen, some of the concepts and results may be given intuitive underpinnings, but once one works within a well-defined model of causality, not all intuitively "clear" claims prove to be true.

Berkson's paradox, discussed in Sect. 5.2, which is often mentioned in connection with theories of causality, may be seen to illustrate that conditioning on a collider may introduce dependence. The paradoxical fact is that when two features are independent, and the selection of individuals depends on these two features, then among those selected, the features may be dependent. More precisely, if the presence of one or both features increases the probability of selection, then, if originally independent, they will be dependent either among those who were or among those who were not selected. The section describes this more complete version of the paradox. The increased probability of selection may be seen as the (causal) effect of the presence of the features on selection. Thus, in the implied casual graph, looking at only those who were or only those who were not selected is conditioning on the collider.

The increased probability of selection, when the desired features are present in Berkson's paradox, may be seen as a simple case of preferential selection. By applying preferential selection for all causal effects in a causal system, Sect. 5.3 builds, for every causal system where Assumption 1 of Chap. 4 holds, an intervention distribution, for which Assumption 2 is true and then uses Assumption 3 to define the corresponding observational distribution. This shows that the model of causality developed in Chap. 4 is not empty, that is, there are realizations of it. Further, the realization developed in Sect. 5.3 is faithful. The intervention

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distribution, for binary variables, is such that the probability that a feature is present is a simple monotone increasing function of the expectations of its parents under that intervention. The higher are the expectations of the parents, the larger is the probability that the feature is present. The effects are individual and additive. This is called the incremental probabilistic causation model (*IPCM*). Many components of the theory are illustrated by direct proofs for the *IPCM*, including the fact that independent variables, when conditioned on a common child or a descendant of it, become associated which is a generalization of Berkson's paradox and was a basic component of the d-separation theory.

While the *IPCM* gives an intuitively clear and sufficiently concrete example of a realization of a causal structure that may be well described with the theory in Chap. 4, this does not mean that the theory would account for all possible expectations with respect to the behavior of the causal system. Section 5.4 considers five claims sometimes made by authors applying variants of this theory and shows that every one of them is wrong in the sense that they are not implied by the theory—although may be occasionally true in reality. These wrong claims are related to certain causal conclusions made based entirely on the observational distribution. Most of the arguments are based on counterexamples.

At the end of the chapter, a brief summary is given and a list of things to do is included.

5.2 Berkson's Paradox

In this section, a phenomenon that is important on its own right, but is also relevant for understanding the background and implications of conditioning on a collider and, more generally, also of d-separation (see Sect. 3.3) is discussed.

Berkson's paradox (see Berkson [22]) is a particular variant of the phenomenon that if unequal selection probabilities are used to collect the data but are not taken into account when estimating a quantity, the estimates may be biased. Although this is a simple fact and may not seem paradoxical at all, Berkson's paradox is widely discussed in the biomedical literature (see, e.g., Woodfine and Redelmeier [172]), in the form that unequal selection probabilities may introduce correlation among variables that are independent of the population. Some authors call it Berkson bias (see, e.g., Westreich [171]), which appears to be a more appropriate name of what it is. Thus, the phenomenon is relevant for understanding the association structure after conditioning on the outcome of the selection, whether or not the data arise as a result of proper sampling. It is also relevant from the more general perspective that these days data scientists often analyze data and draw conclusions, without a proper understanding of how the data came about. More than that, studying this phenomenon will lead to insights that are highly relevant for causality and directed graphical models.

To illustrate Berkson's paradox, let two binary features or variables, say A and B, be independent and distributed jointly with population probabilities p_{ij}, $i, j = 0, 1$. Independence of A and B means that

$$OR_p = \frac{p_{00}p_{11}}{p_{01}p_{10}} = 1.$$

Let the available data be the result of a selection procedure where in category (i, j), the selection probability is s_{ij}. Note that the numbers s_{ij} are not a probability distribution; they simply mean that in cell (i, j), the probability of selection into the data is s_{ij}, while the probability of not selecting an individual in that joint category is $1 - s_{ij}$, independently of the selection in other category combinations. Thus, the expectation of the observed frequency in cell (i, j) is proportional to $p_{ij}s_{ij}$, and the value of the odds ratio at the expectation is

$$OR_s = \frac{p_{00}p_{11}}{p_{01}p_{10}} \frac{s_{11}s_{22}}{s_{12}s_{21}} = \frac{s_{11}s_{22}}{s_{12}s_{21}}$$

and as long as $OR_s \neq 1$, the observed odds ratio will vary around a value different from 1, although the true value is 1 because of the assumed independence of A and B.

Were the selection probabilities s_{ij} all equal, then one would have $OR_s = 1$, and the estimate of the odds ratio based on the data would be unbiased. Even if the selection probabilities are unequal, as was discussed in Section *LCDA* 4.1.4, weights inversely proportional to the selection probabilities may be used to obtain unbiased estimates.

In the situations that are seen as paradoxical by many, the selection process that yields the data does not have the aim of representing some underlying population, as is the case with proper sampling; rather, some preferential selection is applied. The selection is called preferential, say, for category 1 over category 0, if

$$s_{1j} \geq s_{0j}, \text{ for all } j, \text{ and there exists a } j^* \text{for which } s_{1j^*} > s_{0j^*}, \tag{5.1}$$

and, similarly

$$s_{i1} \geq s_{i0}, \text{ for all } i, \text{ and there exists an } i^* \text{for which } s_{i^*1} > s_{i^*0}. \tag{5.2}$$

That is, for any fixed category of one of the variables, the probability of selection when the other variable is 1 is greater than or equal to the selection probability when the latter variable is 0 and there exists a category of the first variable, in the case of which one has strict inequality.

Standard examples include admitting students to a university, choosing a restaurant to have dinner, or becoming a Hollywood star. In all these examples, there is a pair of characteristics which, if present, individually increase the chances of being selected. Universities are happy to admit students who are good academically or who are good athletes (or both) and are less interested in admitting students who are weak on both accounts. Also, from among two applicants, who appear equally good academically, the one who is a better athlete has a higher chance of being admitted, in particular if they are both weak academically. Similarly, from among two equally

Table 5.1 Preferential selection probabilities s_{ij} that preserve the odds ratio

	B=0	B=1
A=0	0.1	0.3
A=1	0.3	0.9

good athletes, the one academically better has a higher chance of being admitted, in particular if they are not very good in sports. When choosing a restaurant, one may go for good food or pleasant ambiance (or both) and would avoid restaurants with neither of these. Again, out of two restaurants equally positioned in one aspect, one will choose based on the other one. One either has to be a talented actor or have very good looks (or both) to become a Hollywood star; others have little chance.

In the preferential selection examples above, as the selection probability of individuals who do not possess any of the desired characteristics is lower than of those who have at least one of these features, the odds ratio of the two features will tend to be less than 1 in the data, even if the features are originally independent. Of course, this only happens if the selection probability of those possessing both characteristics is not increased further to compensate for the lower selection probability of those not possessing any of the features.

For example, if the selection probability of those not possessing any of the features is 0.1, and is 0.2 in all of the other three categories, then the observed odds ratio will vary around $OR_s = 1/2$. Even if the selection probability of those possessing both characteristics is increased over the selection probability of those having only one feature, for example, keeping 0.1, for no, increasing to 0.2 for 1, and to 0.3 for two features present, the odds ratio, in expectation, will be still less than 1. Very often, this phenomenon is referred to by saying that selection introduces negative association. The intuitive explanation is that in the university student population, the odds of finding an academically good student among those who are not good athletes is higher than among those who are good athletes. This is because students in the first group (not so good athletes) need to be academically good to be admitted, but students in the second group (good athletes) need not. Of course, there are choices of the selection probabilities that lead to an odds ratio equal to or more than 1,[1] see Table 5.1 for an example.

In some other situations, the selection process is based on convenience. For example, patients already hospitalized are selected to participate in an experiment; see, e.g., Schwartzbaum, Ahlbom, and Feychting [161]. Berkson's original example was in the context of hospitalized patients and two diseases that appeared to be positively associated among hospital patients, although they were independent of the population. The positive association could be explained by the simple assumption that people having more diseases were seeking hospital care more often than people

[1] In the previous numerical example, this will happen if the selection probability for those possessing both characteristics is increased to 0.4 or more.

with fewer ones. Note that this setup is related to but also somewhat different from the structure discussed so far. But also in this situation, unequal selection probabilities imply that OR_s is not equal to OR_p and the observed odds ratio is a biased estimate of the population odds ratio. More complex examples for Berkson's paradox or bias include case control studies using among hospital patient (see, e.g., Snoep et al. [155]) but that variant is not needed for the topic of this chapter.

It is clear from the foregoing discussion that it is not true that selection or preferential selection in the sense defined above always leads to a reduced OR. For example, the selection probabilities in Table 5.1 preserve the OR, as $OR_s = 1$, and selection is still preferential. Here, the increased preference for both features being present compensates the value of the odds ratio for the reduced preference for the case when both are missing.

Berkson's paradox may be seen as relevant for probabilistic causality, because the preferred characteristics, if present, increase the probability of being selected. The analysis of those selected is conditional based on having been selected. Conditional analysis, however, has a more general meaning too: Conditioning on the outcome of the selection, that is, separate analyses for those who were and for those who were not selected. This may be seen as irrelevant for many of the examples discussed so far, but there are contexts in which this is highly relevant. If in a study, the individuals available are distributed into two treatment groups by preferential selection, it is relevant to know what happens with the association among two variables that were independent originally, within the two groups. There are many such important designs. For instance, a sample of those who applied but were not selected to university admission may be offered a university course, and aspects of their performances may be compared to those who were admitted, to see how successful was the admission procedure. Similar designs could be used in the reject inference in credit scoring, see, e.g., Feelders [54]. In market segmentation (see, e.g., Tynan and Drayton [164]), individuals with the highest incomes or highest levels of education are more likely to be placed into the top segment of customers, but of interest is not only this segment, rather all of them.

The considerations so far illustrate that if one analyzes data for those who were selected, the fact of having been selected may have implications on the association among the desired features, upon which the selection was based. This may be avoided by an appropriate choice of the selection probabilities. Table 5.2 considers both those selected and those not selected and shows the conditional probabilities of selection and of no selection in the numerical example in Table 5.1. In this case, the odds ratio of the two, originally independent, features among those selected remains equal to 1. However, if one looks at the odds ratio among those not selected, one finds an OR equal to $9/49$, indicating a negative association. Thus, preserving independence among those selected preferentially happened at the cost of inducing (negative) association among those not selected. This may be seen as a more comprehensive version of Berkson's paradox. If preferential selection based on two independent variables is used to divide the population into two groups (those selected and those not selected), the two variables will not be independent of one of the groups, in this numerical example.

Table 5.2 The complete structure of preferential selection—conditional probabilities of selection and of no selection

Selected			Not selected		
	B=0	B=1		B=0	B=1
A=0	0.1	0.3	A=0	0.9	0.7
A=1	0.3	0.9	A=1	0.7	0.1

The next result shows that this cannot be avoided. Preferential selection into two groups based on two variables cannot preserve the original odds ratio among them within both groups.

Theorem 5.1 *If the joint probability distribution of two binary variables is (strictly) positive, then in the case of selection based on these two variables, there exist no preferential selection probabilities such that the (expectation of the) odds ratio among those selected and also among those not selected would be both equal to the odds ratio of the two features, that is,*[2]

$$\frac{p_{yy}p_{nn}}{p_{yn}p_{ny}} = \frac{p_{yy}p_{nn}}{p_{yn}p_{ny}}\frac{s_{yy}s_{nn}}{s_{yn}s_{ny}} = \frac{p_{yy}p_{nn}}{p_{yn}p_{ny}}\frac{(1-s_{yy})(1-s_{nn})}{(1-s_{yn})(1-s_{ny})}$$

is not possible.

Note that when the selection probabilities are given in advance to govern the selection process, that is, the selection probabilities are the expectations of the observed selection fractions, then the claim is about the expectation of the resulting odds ratios. When the selection probabilities are determined after the selection as ratios of numbers of selected to the relevant group sizes, then the claim is about the odds ratios.

Proof Suppose, to the contrary, that a choice of the selection probabilities, such that

$$\frac{s_{yy}s_{nn}}{s_{yn}s_{ny}} = \frac{(1-s_{yy})(1-s_{nn})}{(1-s_{yn})(1-s_{ny})} = 1 \tag{5.3}$$

is possible. Then,

$$(1-s_{yy})(1-s_{nn}) = (1-s_{yn})(1-s_{ny})$$

or

$$1 - s_{yy} - s_{nn} + s_{yy}s_{nn} = 1 - s_{yn} - s_{ny} + s_{yn}s_{ny}. \tag{5.4}$$

Also from the indirect assumption (5.3), one obtains that

[2] Here the more intuitive notation of y (yes) and n (no) is used instead of 1 and 0, and yes is preferred to no.

$$s_{yy}s_{nn} = s_{yn}s_{ny}, \tag{5.5}$$

and subtracting this from (5.4) yields that

$$1 - s_{yy} - s_{nn} = 1 - s_{yn} - s_{ny}$$

or that

$$s_{yy} + s_{nn} = s_{yn} + s_{ny}.$$

Thus,

$$s_{yy} = s_{yn} + s_{ny} - s_{nn}, \tag{5.6}$$

and plugging this into (5.5) yields that

$$s_{yn}s_{nn} + s_{ny}s_{nn} - s_{nn}^2 = s_{yn}s_{ny},$$

or that

$$s_{yn}(s_{nn} - s_{ny}) = s_{nn}(s_{nn} - s_{ny}),$$

which may be written as

$$(s_{nn} - s_{yn})(s_{nn} - s_{ny}) = 0.$$

The above equation is only possible if

$$s_{nn} = s_{yn} \text{ or } s_{nn} = s_{ny}. \tag{5.7}$$

If $s_{nn} = s_{yn}$ holds, then by preferential selection, $s_{ny} < s_{yy}$, and these contradict (5.6). Similarly, if $s_{nn} = s_{ny}$ holds, then by preferential selection $s_{yn} < s_{yy}$, and these also contradict (5.6).

□

This result is going to be fundamental in understanding some of the characteristics of the causal theory developed in the previous chapter, in particular in the following form.

Corollary 5.1 *Let the binary variables A and B be independent. Let C be the binary variable that is the indicator of preferential selection with (5.1) and (5.2), based on A and B. Then,*

$$A \perp\!\!\!\perp B|C$$

does not hold.

□

5.3 Incremental Probabilistic Causation

In this section, a simple variant of causality, called incremental probabilistic causation, is introduced, which is helpful in understanding the essence of the assumptions made in Sect. 4.3.1 and is also an important example of the existence of observational distributions and, in particular faithful ones, for every causal structure.

Let for a set of binary variables V, a DAG G, or the equivalent causal structure $(V, \mathscr{C})$ be given. Let the categories of the variables be 0 and 1, indicating the lack or presence of a feature associated with the variable. Let $\mathscr{C}$ be as required by Assumption 1, that is, acyclic.

Now, an intervention distribution Q is determined as part of a realization of the causal structure $(V, \mathscr{C})$, which will be proved to possess the property in Assumption 2, and the relevant observational distribution P will be defined by the property in Assumption 3 and the factorization (3.7).

Consider the variables in V in a well-numbering, see Theorem 2.1. The first variable has no parents. The variables that have no parents will be uniformly distributed. For the other variables, the definition is recursive. Assume for a variable A that the intervention distributions of all previous variables in the well-numbering have already been determined. This includes the intervention distributions of all nondescendants of A, in particular of all its parents, as these precede it in the well-numbering.

For arbitrary $A \notin \mathscr{D}_1, \mathscr{D}_2 \subseteq V$ disjoint subsets and category combination c, set

$$Q(A = 1|\mathscr{I}(\mathscr{D}_1 =_s (c)_{\mathscr{D}_1}, \mathscr{D}_2 =_f (c)_{\mathscr{D}_2}))$$

$$= Q(A = 1|\mathscr{I}(\mathscr{D}_1 \cap nd(A) =_s (c)_{\mathscr{D}_1 \cap nd(A)}, \mathscr{D}_2 \cap nd(A) =_f (c)_{\mathscr{D}_2 \cap nd(A)}))$$

that is, the intervention distribution depends only on the part of the intervention which applies to nondescendants. Further, the intervention distribution, when the intervention is applied to nondescendants of A, will only depend on the intervention distributions of the parents of A. Note that nondescendants of A are also nondescendants of its parents, by Assumption 1, and, thus, when only the part of the intervention which applies to the nondescendants of A is taken into account, this is also an intervention that applies to nondescendants of the parents.

Let now the intervention $\mathscr{I}(\mathscr{D})$ apply to a subset of the nondescendants of A. For every parent B of A, determine its expectation under the intervention, $E_{\mathscr{I}(\mathscr{D})}B$. Of course, when $\mathscr{I}(\mathscr{D})$ happens to set a parent to a particular value or to fix it at that value, then the expectation is that value. In other cases, the intervention distribution and the expectation have already been determined by the induction assumption.

Then, set

$$Q(A = 1|\mathscr{I}(\mathscr{D})) = 0.05 + 0.90\frac{\sum_{B \in pa(A)} E_{\mathscr{I}(\mathscr{D})}B}{u}, \tag{5.8}$$

where u is the number of parents of A. In this intervention distribution, the more features associated with the parents of A are set to be present or kept being present, the more likely is the feature associated with A to be also present. The constants 0.05 and 0.90 imply that all the probabilities generated by (5.8) are between 0.05 and 0.95.

If $B \in pa(A)$ is set to 1, then the value of (5.8) exceeds its value in the case when B is set to 0, by $0.9/u$, if the other parents are kept fixed.

If $\mathscr{B}_1$ and $\mathscr{B}_2$ are such that $pa(A) \subseteq \mathscr{B}_1 \cup \mathscr{B}_2 \subseteq nd(A)$, then by Assumption 3,

$$Q(A = 1|\mathscr{I}(\mathscr{B}_1 =_s (c)_{\mathscr{B}_1}, \mathscr{B}_2 =_f (c)_{\mathscr{B}_2}))$$

$$Q(A = 1|\mathscr{I}(\mathscr{B}_1 \cap pa(A) =_s (c)_{\mathscr{B}_1 \cap pa(A)}, \mathscr{B}_2 \cap pa(A) =_f (c)_{\mathscr{B}_2 \cap pa(A)}))$$

$$P(A = 1|\mathscr{B}_1 \cap pa(A) = (c)_{\mathscr{B}_1 \cap pa(A)}, \mathscr{B}_2 \cap pa(A) = (c)_{\mathscr{B}_2 \cap pa(A)}),$$

where the first equality comes from the definition of the intervention distribution. These conditional probabilities yield the joint observational distribution of all variables because of the factorization property (3.7).

The definition given so far may be extended to sets of variables. For $\mathscr{A} = \{A_1, A_2, \ldots, A_k\}$ (in the order of the well-numbering), and $\mathscr{B} = \mathscr{B}_1 \cup \mathscr{B}_2 \subseteq V \setminus \mathscr{A}$, and intervention $\mathscr{I}(\mathscr{B}_1 =_s (c)_{\mathscr{B}_1}, \mathscr{B}_2 =_f (c)_{\mathscr{B}_2})$,

$$Q(\mathscr{A}|\mathscr{I}(\mathscr{B}_1 =_s (c)_{\mathscr{B}_1}, \mathscr{B}_2 =_f (c)_{\mathscr{B}_2})) \tag{5.9}$$

$$= \prod_{A_i \in \mathscr{A}} Q(A_i|\mathscr{I}(\mathscr{B}_1 \cap nd(A_i) =_s (c)_{\mathscr{B}_1 \cap nd(A)}, \mathscr{B}_2 \cap nd(A_i) =_f (c)_{\mathscr{B}_2 \cap nd(A)})).$$

The terms in the product on the right-hand side of (5.9) need to be determined in the order of the well-numbering so that the intervention distributions and then the intervention expectations of the parents of each variable A are determined before the intervention distribution of A is determined, and, thus, (5.8) may be applied.

The causal structure and the realization of it defined in (5.8) and (5.9) is called an incremental probabilistic causation model (*IPCM*). This type of causality is incremental, because increasing the probability that one parent of a variable is equal to 1 increases the probability that the child is also equal to 1. It is probabilistic, because making the feature represented by a parent present will not necessarily make the effect present in the child, but it will only increase the probability that it is present.

To illustrate the construction of an *IPCM* given in (5.8) and (5.9), consider four binary variables A, B, C, D with categories 0 and 1. Let the causal structure be as represented by the DAG in Fig. 5.1. In this DAG, A, B, C, D is a well-numbering. As an example, determine

$$Q(C|\mathscr{I}(B =_s 1)).$$

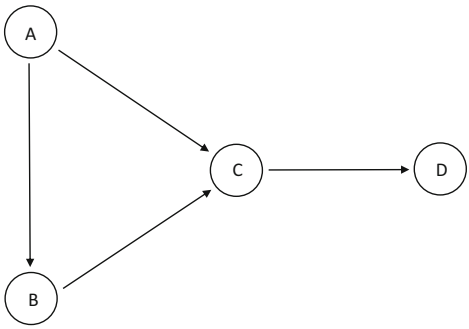


Fig. 5.1 The DAG for an illustration of *IPCM*

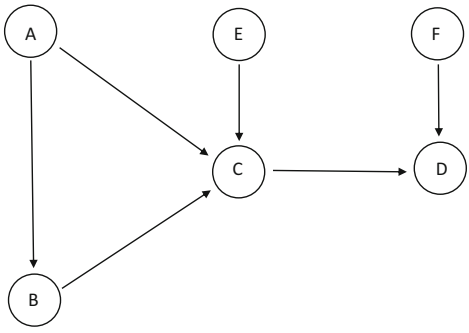


Fig. 5.2 The DAG of another illustration of *IPCM*

Here, variable C has two parents, A and B. A has no parents, thus, it has a uniform distribution and then $E_{\mathscr{I}}(A) = 0.5$. The next variable in the well-numbering is B. As it is set to 1, its expectation is not affected by its parent: $E_{\mathscr{I}}(B) = 1$. Thus,

$$Q(C = 1|\mathscr{I}(B =_s 1)) = 0.05 + 0.90\frac{0.5 + 1}{2} = 0.725.$$

With no intervention, in this example, $Q(C|\mathscr{I}(\emptyset))$ is obtained as follows. The distribution of A is uniform, thus, $E_{\mathscr{I}}(A) = 0.5$. The variable B has one parent, A, thus, $Q(B = 1|\mathscr{I}(\emptyset))$ is $0.05 + 0.90(0.5/1) = 0.5$, and $E_{\mathscr{I}}(B) = 0.5$. Finally, C has two parents, thus, $Q(C|\mathscr{I}(\emptyset)) = 0.05 + 0.90(0.5 + 0.5)/2$, implying that $Q(C|\mathscr{I}(\emptyset))$ is also uniform.

For a slightly more complex example, consider the DAG in Fig. 5.2. Let the variables without parents be distributed uniformly. Assume that $B = 0$, and apply the intervention

$$\mathscr{I}(A =_s 1, E =_s 0, B =_f 0)$$

and determine the joint intervention distribution of C, D. One needs to determine the intervention distributions of C and D separately first. To find the intervention distribution of C, one needs for each of its parents their expectations under the intervention. These are the following: $E_{\mathscr{I}}(A) = 1$ (set by the intervention), $E_{\mathscr{I}}(B) = 0$ (fixed by the intervention), and $E_{\mathscr{I}}(E) = 0$ (set by the intervention). Thus,

$$Q(C = 1|\mathscr{I}(A =_s 1, E =_s 0, B =_f 0)) = 0.05 + 0.90\frac{1+0+0}{3} = 0.35.$$

This also implies that $E_{\mathscr{I}}(C) = 0.35$. The variable D, in addition to C, has another parent, F, which is not affected by the intervention, and as it has no parents, thus $E_{\mathscr{I}}(F) = 0.5$. This gives that

$$Q(D = 1|\mathscr{I}(A =_s 1, E =_s 0, B =_f 0)) = 0.05 + 0.90\frac{0.35+0.5}{2} = 0.4325.$$

For the joint intervention distribution of C, D, one obtains that

$$Q(C = 1, D = 1|\mathscr{I}(A =_s 1, E =_s 0, B =_f 0)) = 0.35 * 0.4325 = 0.1512$$

or

$$Q(C = 0, D = 1|\mathscr{I}(A =_s 1, E =_s 0, B =_f 0)) = 0.65 * 0.4325 = 0.2811.$$

Now a proof of the existence, for every causal structure, of realizations that are *IPCM*s is given.

Theorem 5.2 *Let $(V, \mathscr{C})$ be a causal structure for $0-1$ variables. Then, there exists a realization (P, Q), where Q is defined in (5.8) and in (5.9).*

Proof Out of the three assumptions made for causal structures and their representations, only Assumption 2 contains a restriction for the intervention distribution, and it requires, for pairwise disjoint subsets of variables $\mathscr{A}$, $\mathscr{B}_1$, and $\mathscr{B}_2$, that if there is an effect of the manipulation of the variables in $\mathscr{B}_1$ on the variables in $\mathscr{A}$ even if the variables in $\mathscr{B}_2$ are kept fixed, then there exists a causal path from a variable in $B \in \mathscr{B}_1$ to a variable $A \in \mathscr{A}$, which does not go through $\mathscr{B}_2$.

To see that Assumption 2 holds, consider a variable $A \in \mathscr{A}$ that is affected by the manipulation of a variable in $\mathscr{B}_1$, even if the variables in $\mathscr{B}_2$ are kept fixed. For this to be the case, one of the parents of A needs to be affected or manipulated too, as seen in (5.8). Let one such variable be A_1. If A_1 itself is manipulated, then it is in B_1. Otherwise, for A_1 to be affected by the manipulation, one of its parents also needs to be affected or manipulated. In the latter case, the parent is in $\mathscr{B}_1$, or else A_1 has an affected parent. Let A_2 be such a variable. Continue this procedure. It cannot visit a variable repeatedly because of Assumption 1; therefore, it terminates after a finite number of steps. Either it arrives at a variable A_k in $\mathscr{B}_1$ defining a causal path from $A_k \in \mathscr{B}_1$ to A_1 or else it arrives at a variable with no affected parent (including

no parent at all). But the latter is not possible, as such a variable is not affected by the manipulation, thus, the process terminates in $\mathscr{B}_1$. As the variables $A_1, \ldots A_k$ are all affected by the manipulation, none of them can be in $\mathscr{B}_2$, as those variables are kept fixed by the intervention.

Finally, the observational distribution is obtained from the intervention distribution using the factorization (3.7) and Assumption 3.

□

The intervention distribution constructed is also faithful. To see this, another result about the effect of changing a variable from 0 to 1 is needed.

Proposition 5.1 *For the intervention distribution constructed in (5.8) and (5.9), and every $A, B \in V$,*

$$Q(A = 1|\mathscr{I}(B =_s 1)) \geq Q(A = 1|\mathscr{I}(B =_s 0)).$$

Proof Suppose, to the contrary, that there are variables A and B, such that

$$Q(A = 1|\mathscr{I}(B =_s 1)) < Q(A = 1|\mathscr{I}(B =_s 0)).$$

Then, A would also have a parent, which had a lower expectation when $B =_s 1$ than when $B =_s 0$. This variable also would have to have such a parent and so on. By continuing this procedure, one cannot arrive at a variable already visited because of Assumption 1. Therefore, the procedure will arrive at a variable where it cannot be continued. This could happen if the last variable had no parents at all, but then its expectation would not be affected by setting B to a particular value, thus, this is not possible. Or, the procedure could stop if the last variable had parents but all had higher (or equal) expectations when $B =_s 1$ than when $B =_s 0$, but this is not possible, either, because in that case the last variable could not have a lower expectation when $B =_s 1$ than when $B =_s 0$. This means that A cannot have a parent that has a lower expectation when $B =_s 1$ than when $B =_s 0$. But then this applies to A too.

□

Note that Proposition 5.1 says that no variable has smaller expectation when B is set to 1 than when B is set to 0, irrespective of the causal relationship between B and A in an *IPCM*.

Theorem 5.3 *The intervention distribution constructed in (5.8) and (5.9) is faithful. In particular, if there is a causal path from B to A, then, irrespective of the rest of the causal structure,*

$$Q(A = 1|\mathscr{I}(B =_s 1)) > Q(A = 1|\mathscr{I}(B =_s 0)). \tag{5.10}$$

Proof The existence of a causal path from B to A means that there is a variable B_k, which is a parent of A; a variable B_{k-1}, which is a parent of B_k; and so on, and finally B is a parent of a B_1.

Note that in addition to this causal path, there may be other causal paths leading from B to A. Other than through the causal paths from B to A, B cannot have an intervention effect on A, because to have such an effect, B would have to have an intervention effect on one of the parents of A. For this, B would have to an intervention effect on a parent of this parent and so on, defining an additional causal path from B to A.

If B is set to 1, its expectation is 1 and when it is set to 0, its expectation is 0. Out of the parents of B_1, by Proposition 5.1, none has a smaller expectation in the first than in the second case, and one parent, namely, B, has a larger expectation in the first than in the second case. Thus, the expectation of B_1 is greater when $B =_s 1$ than when $B =_s 0$:

$$E_{\mathscr{I}(B=_s 1)} B_1 > E_{\mathscr{I}(B=_s 0)} B_1.$$

The next variable on the causal path is B_2. By Proposition 5.1, none of its parents has smaller expectation when $B =_s 1$ than when $B =_s 0$, and according to the inequality above, one of its parents, B_1, has a larger expectation in the first than in the second case. Thus,

$$E_{\mathscr{I}(B=_s 1)} B_2 > E_{\mathscr{I}(B=_s 0)} B_2.$$

The same argument can now be applied to B_3 instead of B_2 and so on, yielding that A has a greater expectation when $B =_s 1$ than when $B =_s 0$.

□

Observational distributions from an *IPCM* possess the following property.

Theorem 5.4 *Under an IPCM, for every* $B \in pa(A)$,

$$\begin{aligned} 1 > P(A = 1 | B = 1, pa(A) \setminus \{B\} = (c)_{pa(A)\setminus\{B\}}) \\ > P(A = 1 | B = 0, pa(A) \setminus \{B\} = (c)_{pa(A)\setminus\{B\}}) > 0, \end{aligned} \tag{5.11}$$

holds for every joint index $(c)_{pa(A)\setminus\{B\}}$ *of the variables in* $pa(A) \setminus \{B\}$.

Proof The claim is true for the intervention distributions instead of the observational distribution:

$$\begin{aligned} 1 > Q(A = 1 | \mathscr{I}(B =_s 1, pa(A) \setminus \{B\} =_s (c)_{pa(A)\setminus\{B\}})) \\ > Q(A = 1 | \mathscr{I}(B = 0, pc(A) \setminus \{B\} =_s (c)_{pa(A)\setminus\{B\}})) > 0. \end{aligned}$$

Indeed, irrespective of $(c)_{pa(A)\setminus\{B\}}$, setting B to 1 as opposed to setting it to 0 increases the probability of A being 1 directly by definition. The manipulation may change the expectations of other parents of A, but it will not reduce them by Proposition 5.1, thus, the effects of these further parents do not reduce the expectation of A.

As the manipulations apply to some of the nondescendants of A, including all its parents, the inequality for the intervention distributions implies the same inequality for the observational distribution by Assumption 3.

□

The structure of an *IPCM* is a generalization of preferential selection discussed in Sect. 5.2. Indeed, for three variables, one has the following.

Proposition 5.2 *For three binary variables A, B, C, and a causal structure where A and B are the parents of C that is, C is a collider, in the case of an IPCM, the observational distribution has the preferential selection properties (5.1) and (5.2).*

Proof The claim follows from (5.11).

□

This results creates a simple opportunity to illustrate the implications of conditioning on a collider, as discussed in Sect. 4.3.2. Of course, the general result applies in the special case of an *IPCM*. Corollary 5.1 and Proposition 5.2 immediately yield the following.

Corollary 5.2 *For binary variables A, B, C, let A and B be the parents of C. If A and B are independent in the observational distribution that is an IPCM, then conditioned on C, A and B are not conditionally independent.*

□

In fact, one may see similar behavior, when conditioned on a descendant of the collider.

Theorem 5.5 *Consider the binary variables A, B, C, $D_1, \ldots, D_d$ and a causal structure in which A and B are the parents of C. Further, let C be the parent of D_1 and D_i the parent of D_{i+1}, $i = 1, \ldots d-1$. Let (P, Q) be a realization of the causal structure, which is an IPCM, and where A and B are independent. Then, in the observational distribution, conditioned on any D_i, $i = 1, \ldots, d$, A and B are not independent.*

Proof In this causal structure, for any $i = 1, \ldots, d$, there is a causal path from A to D_i and also one from B to D_i.

Consider the intervention $\mathscr{I}(A =_s a, B =_s b)$, where $a, b, = 0, 1$. Then, by the construction of Theorem 5.3,

$$Q(D_i = 1|\mathscr{I}(A =_s 1, B =_s b)) > Q(D_i = 1|\mathscr{I}(A =_s 0, B =_s b)),$$

for all b, and similarly,

$$Q(D_i = 1|\mathscr{I}(A =_s a, B =_s 1)) > Q(D_i = 1|\mathscr{I}(A =_s a, B =_s 1)),$$

for all a.

Note that Assumption 3 cannot be applied directly to the inequalities above to obtain similar relationships for the observational distribution, as the intervention does not apply to the parent of D_i. Instead, an induction argument is used.

Denote $D_0 = C$ and consider D_1. It has one parent, D_0, thus,

$$P(D_1 = 1|A = a, B = b) = \sum_{c=0,1} P(D_1 = 1|D_0 = c)P(D_0 = c|A = a, B = b),$$

and thus

$$\begin{aligned}
&P(D_1 = 1|A = 1, B = 0)\\
&= P(D_1 = 1|D_0 = 0)P(D_0 = 0|A = 1, B = 0)\\
&\quad + P(D_1 = 1|D_0 = 1)P(D_0 = 1|A = 1, B = 0)\\
&= P(D_1 = 1|D_0 = 0)(1 - P(D_0 = 1|A = 1, B = 0))\\
&\quad + P(D_1 = 1|D_0 = 1)P(D_0 = 1|A = 1, B = 0)\\
&= P(D_1 = 1|D_0 = 0) + P(D_0 = 1|A = 1, B = 0)\\
&\quad \times (P(D_1 = 1|D_0 = 1) - P(D_1 = 1|D_0 = 0)).
\end{aligned}$$

Similarly,

$$\begin{aligned}
&P(D_1 = 1|A = 0, B = 0)\\
&= P(D_1 = 1|D_0 = 0)P(D_0 = 0|A = 0, B = 0)\\
&\quad + P(D_1 = 1|D_0 = 1)P(D_0 = 1|A = 0, B = 0)\\
&= P(D_1 = 1|D_0 = 0)(1 - P(D_0 = 1|A = 0, B = 0))\\
&\quad + P(D_1 = 1|D_0 = 1)P(D_0 = 1|A = 0, B = 0)\\
&= P(D_1 = 1|D_0 = 0) + P(D_0 = 1|A = 0, B = 0)\\
&\quad \times (P(D_1 = 1|D_0 = 1) - P(D_1 = 1|D_0 = 0)).
\end{aligned}$$

This gives that

$$\begin{aligned}
&P(D_1 = 1|A = 1, B = 0) - P(D_1 = 1|A = 0, B = 0)\\
&= (P(D_0 = 1|A = 1, B = 0) - P(D_0 = 1|A = 0, B = 0))\\
&\quad \times (P(D_1 = 1|D_0 = 1) - P(D_1 = 1|D_0 = 0))
\end{aligned}$$

and both terms of the product are positive because of the *IPCM*. Thus,

$$P(D_1 = 1|A = 1, B = 0) - P(D_1 = 1|A = 0, B = 0) > 0.$$

Similar arguments show that

$$P(D_1 = 1|A = 0, B = 1) - P(D_1 = 1|A = 0, B = 0) > 0,$$

$$P(D_1 = 1|A = 1, B = 1) - P(D_1 = 1|A = 1, B = 0) > 0,$$

$$P(D_1 = 1|A = 1, B = 1) - P(D_1 = 1|A = 0, B = 1) > 0.$$

Therefore, the joint distribution of the variables A, B, D_1 possesses the preferential selection properties (5.1) and (5.2).

Suppose now that for some $i \leq d-1$, it has been proved that the joint distribution of A, B, D_i has the preferential selection property. Then, apply the argument above with D_i instead of D_0 and with D_{i+1} instead of D_1 to finally obtain the claim because of Corollary 5.1.

□

Note that for the independence condition in the previous theorem, the factorization (3.7) implies the following.

Proposition 5.3 *If neither of the variables A and B have parents in the causal structure, then in any realization of the causal structure, A and B are independent of the observational distribution.*

□

The structure of an *IPCM* is a simple and intuitive illustration of how the type of causality discussed here works. However, one has to be careful not to assume that all intuitively clear and attractive properties are necessarily true. This is investigated in the next section.

5.4 Intuition Has Its Limitations

The theory of causality presented in the previous chapter is seen by many researchers as a major breakthrough and is considered with criticism by others. Many publications, in addition to describing and developing the approach, also try to propagate it, occasionally making claims which are intuitively appealing but would require proofs, including a clear formulation of the conditions under which they hold. Applied papers, on the other hand, sometimes use these methods because of their intuitive appeal, instead of an explicit consideration of the actual arguments for and against its application. As is so often the case, intuition needs to be educated before can be relied on.

In this section, we consider several claims, which may be appealing intuitively but are not true or not true without further assumptions.

Claim 1. For $0-1$ variables,

$$P(A = 1|B = 1) > P(A = 1|B = 0)$$

may be interpreted as suggesting that variable B has a causal effect on variable A.

Claim 1 is wrong.

In fact, one has the following.

Proposition 5.4 *In the case of (strictly) positive probabilities,*

$$P(A = 1|B = 1) > P(A = 1|B = 0)$$

implies

$$P(B = 1|A = 1) > P(B = 1|A = 0).$$

Proof By assumption,

$$P(A = 1|B = 1) > P(A = 1|B = 0),$$

which gives that

$$P(A = 0|B = 1) < P(A = 0|B = 0).$$

These two inequalities imply that

$$\begin{aligned} 1 &< \frac{P(A = 1|B = 1)P(A = 0|B = 0)}{P(A = 1|B = 0)P(A = 0|B = 1)} \\ &= \frac{\frac{P(A=1,B=1)}{P(B=1)}\frac{P(A=0,B=0)}{P(B=0)}}{\frac{P(A=1,B=0)}{P(B=0)}\frac{P(A=0,B=1)}{P(B=1)}} \\ &= \frac{\frac{P(A=1,B=1)}{P(A=1)}\frac{P(A=0,B=0)}{P(A=0)}}{\frac{P(A=1,B=0)}{P(A=1)}\frac{P(A=0,B=1)}{P(A=0)}} \\ &\frac{P(B = 1|A = 1)P(B = 0|A = 0)}{P(B = 1|A = 0)P(B = 0|A = 1)}, \end{aligned}$$

using that all probabilities are positive by assumption. This inequality may be written as

$$P(B = 1|A = 1)P(B = 0|A = 0) > P(B = 1|A = 0)P(B = 0|A = 1)$$

or

$$P(B = 1|A = 1)(1-P(B = 1|A = 0)) > P(B = 1|A = 0)(1-P(B = 1|A = 1))$$

or

$$\begin{aligned} &P(B = 1|A = 1) - P(B = 1|A = 1)P(B = 1|A = 0)) \\ &> P(B = 1|A = 0) - P(B = 1|A = 0)P(B = 1|A = 1)) \end{aligned}$$

or

$$P(B = 1|A = 1) > P(B = 1|A = 0). \tag{5.12}$$

□

If Claim 1 was true, then whenever one would see evidence for B having a causal effect on A, at the same time, one would also see evidence for the causal effect going in the opposite direction, and most analyst appear to agree that effect cannot go in both ways. This was formulated in Assumption 1. Here, one has association in the (observational) distribution but not causation.

Also, this situation shows clearly the fundamental difference between the intervention and observational distributions. The observational distribution is a joint distribution of all variables in V, and the simple calculation in the proof was possible because both conditional distributions could be derived from the joint distribution. The intervention distribution of A, when B is manipulated, has nothing to do with the intervention distribution of B, when A is manipulated. Therefore, no similar calculation is possible for intervention distributions, which would lead to the reversal seen in the proof of Proposition 5.4.

Note that the situation discussed in Sect. 4.4.4 was different. There, it was assumed that one had a hypothesis and that hypothesis was confronted with reality, through testing one of its implications. Such a hypothesis should rely on substantive knowledge and/or previous experience. Here, Claim 1 tries to create a hypothesis without using substantive knowledge or previous experience, based only on the available data, and this is not justified in the current case, as one could infer the opposite just as well.

Claim 2. If variable A is independent from B in the observational distribution,[3] then this is evidence for neither one of them having a causal effect on the other one.

Claim 2 is wrong.

Two reasons need to be mentioned why Claim 2 is wrong.

First, if the realization from which the observations come is not faithful, then it may happen that there is a causal effect of B on A, but they are independent of the actual realization of the causal structure. Faithfulness may, of course be assumed, under which this may not happen, but faithfulness is usually hard, often impossible, to test in practice.

Second, $A \perp\!\!\!\perp B$ in the observational distribution does not necessarily imply that they are also independent in all intervention distributions. Assumption 3 only implies the equality of the observational and intervention distributions under special conditions.

Both Claim 1 and Claim 2 are related to the idea of exploring the underlying causal structure by studying the properties of the observational distribution. While in

[3] In reality, of course, one only can observe an empirical version of the observational distribution. In that case, independence refers to the hypothesis of independence not being rejected based on the data.

Table 5.3 The joint observational distribution of A and B

	B=yes	B=no
A=yes	70/300	40/300
A=no	80/300	110/300

Table 5.4 Selection probabilities in the joint categories of A and B

Selected			Not selected		
	B=yes	B=no		B=yes	B=no
A=yes	60/70	20/40	A=yes	10/70	20/40
A=no	30/80	10/110	A=no	50/80	100/110

the present framework, this is sometimes possible (see, for instance, the discussion of the instrumental variable method in Sect. 6.2.2), it does not work in all settings.

There is a big difference in a confirmatory[4] application of testing the relationships among variables in the observational distribution and a purely exploratory approach with the intention to derive the causal structure from the study of the associations among pairs of variables in the observational distribution. See also the related comments in Sect. 4.4.4. Note also that the joint distribution of categorical variables can never be fully explored by studying their pairwise relationships; see also Sect. 2.5.1 and the many relevant parts of *LCDA*.

Claim 3. Conditioning on a collider or on a descendant of a collider makes the parents dependent in the observational distribution.

Claim 3 is wrong.

Theorem 5.5 was that in the case of an *IPCM*, if the parents are independent, then conditioning on the collider or a descendant of it makes them dependent. In the general case, when only the three variables of a basic collider configuration were present, Theorem 4.10 was that if the two parents are marginally independent and Corollary 5.1 was that in the case of preferential selection with independent causes, when conditioned on the common effect, the causes became dependent. One has to be careful not to read any of these as if they said that conditioning on a collider or on a descendant of it introduces dependence. This is not true, which is now illustrated with a numerical example.

The example is essentially the one given in Fig. 4.1 and Tables 4.4 and 4.5, with categories of A and B called yes and no, instead of 1 and 2. Under the conditions described there, the observational distribution of the AB marginal is as given in Table 5.3. Variable C, the collider, is interpreted as a selection indicator, with categories selected (1) and not selected (0) instead of 1 and 2. The selection probabilities, based on the intervention probabilities in Table 4.5, are shown in Table 5.4. Here, $s_{yy} = 60/70$, $s_{yn} = 20/40$, $s_{ny} = 30/80$, and $s_{nn} = 10/110$, thus, (5.1) and (5.2) hold indeed.

[4] That is, when the goal is to test a scientifically grounded hypothesis

Table 5.5 The expected result of the selection process, leading to conditional independence

Selected			Not selected		
	B=yes	B=no		B=yes	B=no
A=yes	60/300	20/300	A=yes	10/300	20/300
A=no	30/300	10/300	A=no	50/300	100/300

The expected distribution with respect to selected and not selected is shown in Table 5.5. Clearly,

$$A \perp\!\!\!\perp B|C$$

holds. This example is not isolated in the sense that there would be only a single choice of the selection probabilities which leads to conditional independence of the originally not independent features A and B.

Corollary 5.2 says that in the case of an *IPCM*, conditioning on a collider makes independent causes dependent; see also Theorem 4.10. This is an important characteristic of causal models discussed here and is usually justified intuitively, with reference to Berkson's paradox (although we have seen that this may not happen with those selected, rather this happens when both those selected and not selected are taken into account)[5]. Our approach in the previous section yields this property as a consequence of a particular, but quite general, model of how causality may work. In situations when the independence of A and B is not known, it is useful to remember Corollary 5.2 as saying that conditionally on C, A and B may be dependent.

It is, of course, possible that for three binary variables, A and B are both marginally and conditionally (on C) independent, but then preferential selection of C based on A and B is not possible.

The next two claims are related to the concept of d-connectedness and to the d-separation rule. These are often explained by arguments like the following.

D-connectedness of variables B and A by a (not directed, i.e., not causal) path $\mathscr{P}$, when conditioned on a set of variables $\mathscr{D}$, means that for any three consecutive variables in $\mathscr{P}$, say B_{i-1}, B_i, B_{i+1}, in the observational distribution

(a) if they are in a chain configuration, conditioning on $\mathscr{D}$ does not involve conditioning on the noncollider B_i, which would make in the conditional distribution B_{i-1} and B_{i+1} independent; therefore, the effect may propagate from B_{i-1} to B_{i+1},
(b) if they are in a fork configuration, conditioning on $\mathscr{D}$ does not involve conditioning on the noncollider B_i, which would make in the conditional distribution B_{i-1} and B_{i+1} independent, therefore effect may propagate from B_{i-1} to B_{i+1},

[5] See also the car example in Sect. 6.2.1.

(c) if they are in a collider configuration, conditioning on $\mathscr{D}$ does include conditioning on the noncollider B_i or on one of its descendants, which may make in the conditional distribution B_{i-1} and B_{i+1} dependent; therefore, the effect may propagate from B_{i-1} to B_{i+1}.

And then, the argument continues, the effect may propagate step by step from B to A. Finally, the d-separation rule is often explained as being implied by the fact that if there is no path through which effect could propagate, one has to have independence of B and A conditioned on $\mathscr{D}$.

While the argument above is useful in remembering d-connectedness and the d-separation rule, one should not believe that this is a proof or that it would be correct without strong assumptions being made. This is illustrated now.

Claim 4. Effect in the observational distribution is transitive (propagates) in the sense that if B has an effect on C and C has an effect on A, then these imply that B has an effect on A.

This claim has the intuitive meaning that the effect of A would propagate to C through B. If it was true, it would be helpful in understanding the d-separation rule. For example, if a path is $B - D - C - E - A$ and D and E are colliders, and $D, E \in \mathscr{D}$, the conditioning on D may imply that B is not independent of C and conditioning on E may imply that C is not independent of A, so if transitivity would hold, then B would not be independent of A. But transitivity does not hold, in general.

Claim 4 is wrong.

First, as discussed in the case of Claim 1, the existence of the effect of one variable on another one cannot be deducted from the observational distribution, so even the conditions are problematic. But even if effect is replaced by association, Claim 4 remains wrong. To find a language that applies to the observational distribution strictly, and allows the existence of a causal structure in the background, say that variable B stimulates variable A, for $0 - 1$ variables, if

$$P(A = 1|B = 1) > P(A = 1|B = 0).$$

This was the case with a realization of an *IPCM*, when there was a causal path from B to A. But Claim 4 remains wrong even in the following form:

Claim 4a. Stimulation is transitive in the sense that if A stimulates B and B stimulates C, then A stimulates C.

Claim 4a is also wrong.

This is now illustrated with two examples. The first one shows a situation where stimulations are transitive, and the second one a situation when it is not.

The first example shows a probability distribution where

$$P(B = 1|A = 1) > P(B = 1|A = 0)$$

and

$$P(C = 1|B = 1) > P(C = 1|B = 0)$$

Table 5.6 Observational distribution where A stimulates B, B stimulates C, and also A stimulates C

C=0			C=1		
	B=0	B=1		B=0	B=1
A=0	0.1	0.05	A=0	0.05	0.09
A=1	0.05	0.09	A=1	0.3	0.27

Table 5.7 Bivariate marginals of the distribution in Table 5.6

	B=0	B=1		C=0	C=1		C=0	C=1
A=0	0.15	0.14	B =0	0.15	0.35	A =0	0.15	0.14
A=1	0.35	0.36	B=1	0.14	0.36	A =1	0.14	0.57

and also

$$P(C = 1|A = 1) > P(C = 1|A = 0),$$

that is, the effect of A propagates to C: A stimulates B and B stimulates C and these together imply that A also stimulates C.

Table 5.6 shows the joint distribution, and the bivariate marginal distributions are shown in Table 5.7.

In the distributions in Table 5.7,

$$P(B = 1|A = 1) = 0.507 > P(B = 1|A = 0) = 0.483,$$

$$P(C = 1|B = 1) = 0.72 > P(C = 1|B = 0) = 0.7,$$

$$P(C = 1|A = 1) = 0.803 > P(C = 1|A = 0) = 0.483,$$

thus, A stimulates C.

The next example is a joint distribution, where

$$P(B = 1|A = 1) > P(B = 1|A = 0)$$

and

$$P(C = 1|B = 1) > P(C = 1|B = 0)$$

hold, but

$$P(C = 1|A = 1) < P(C = 1|A = 0),$$

thus, the stimulation is not transitive here. The joint distribution is given in Table 5.8, and the marginal distributions are given in Table 5.9.

Table 5.8 Distribution where A stimulates B and B stimulates C, but A does not stimulate C

C=0			C=1		
	B=0	B=1		B=0	B=1
A=0	0.05	0.05	A=0	0.22	0.15
A=1	0.01	0.063	A=1	0.02	0.347

Table 5.9 Bivariate marginals of the distribution in Table 5.8

	B=0	B=1		C=0	C=1		C=0	C=1
A=0	0.27	0.2	B =0	0.15	0.24	A =0	0.1	0.37
A=1	0.12	0.41	B=1	0.113	0.497	A =1	0.163	0.367

In the distributions in Table 5.9,

$$P(B = 1|A = 1) = 0.774 > P(B = 1|A = 0) = 0.426,$$

$$P(C = 1|B = 1) = 0.815 > P(C = 1|B = 0) = 0.615,$$

$$P(C = 1|A = 1) = 0.692 < P(C = 1|A = 0) = 0.787,$$

thus, the effect A does not stimulate C, although it does stimulate B and B also stimulates C.

Whether or not stimulation is transitive depends on a number of factors. If the realization is an *IPCM*, and there is a causal path between B and A, then stimulations along this path will be transitive, irrespective of the other parts of the causal graph, see Proposition 5.1, Theorem 5.3, and Assumption 3. This is not necessarily true for realizations that are not *IPCM*s. If there are additional causal paths between B and A, they may alter the effect of B on A, relative to what the path considered first suggests.

Even in the simple case of three variables and a single causal path $A - B - C$, the marginal distributions of AB and of BC do not determine the AC marginal, not even the marginal odds ratios. In the example in Table 5.7, the three marginal odds ratios are 1.10, 1.10, and 4.36, all greater than 1. In the example in Table 5.9, the marginal odds ratios are 4.61, 2.75, and 0.61, indicating the same change in the direction of the association as the change in stimulation status seen. The issue may also be described using the second-order odds ratios, see *LCDA* (1.4) and Section *LCDA* 6.2. The second-order odds ratio is 7.2 for the first set of data, while it is 0.02 for the second set of data. The existence of higher-order associations, an important characteristic of categorical data, poses special problems when this causal theory, based on the individual effects of variables, is applied to it. See Sect. 6.3.3 for further comments.

All this shows that it would be a very difficult task to create a general proof of the d-separation rule based on transitive effects and underlines the importance

of the proof via the directed global Markov property given in Sect. 3.3 but also of Assumptions 1–3 that imply the Markov properties.

Claim 5. The distributions used as counterexamples in this section are very special and cannot be part of a realization of a causal structure with Assumptions 1–3.

Claim 5 is wrong.

In fact, any joint distribution of the variables V may be obtained as part of a realization of a causal structure.

Theorem 5.6 *For a set V of variables, let $P(V)$ be a probability distribution. Then, there exists a causal relation $\mathscr{C}$ and an intervention distribution Q, such that Assumptions 1–3 hold and (P, Q) is a realization of $(V, \mathscr{C})$.*

Proof Consider the variables in an arbitrary order, say $V_1, \ldots, V_k$. Define $\mathscr{C}$ as

$$(V_i, V_j) \in \mathscr{C} \text{ if and only if } i < j,$$

that is, for every variable, its parents are the earlier variables. Then $\mathscr{C}$ is acyclic; otherwise, an index smaller than j, for some j, would be also greater than it. Thus, Assumption 1 holds.

Let $V_j \notin \mathscr{B} \subset V$ and let $\mathscr{B}_1, \mathscr{B}_2$ be a disjoint partition of $\mathscr{B}$. Define the intervention distribution as follows:

$$Q(A = 1|\mathscr{I}(\mathscr{B}_1 =_s (c)_{\mathscr{B}_1}, \mathscr{B}_2 =_f (c)_{\mathscr{B}_2}))$$

$$= Q(A = 1|\mathscr{I}(\mathscr{B}_1 \cap nd(A) =_s (c)_{\mathscr{B}_1 \cap nd(A)}, \mathscr{B}_2 \cap nd(A) =_f (c)_{\mathscr{B}_2 \cap nd(A)}))$$

$$= Q(A = 1|\mathscr{I}(\mathscr{B} \cap nd(A) =_s (c)_{\mathscr{B} \cap nd(A)}))$$

$$= P(A = 1|\mathscr{B} \cap nd(A) = (c)_{\mathscr{B} \cap nd(A)}).$$

As $nd(A) = pa(A)$, this definition implies that Assumption 3 holds.

Finally, $(\mathscr{B}_1, \mathscr{B}_2)$ has an intervention effect on A in the sense used in Assumption 2 if and only if $pa(A) \cap \mathscr{B}_1 \neq \emptyset$. But in this case, there is a causal path, in fact an arrow, from the parent in $\mathscr{B}_1$ to A, and it avoids any other variable, thus, Assumption 2 holds.

□

Note that Theorem 5.6 says that for all distributions, there is a causal structure of which it is a realization. This does not contradict Theorem 4.18, which says that realizations of a particular causal structure have to obey the d-separation rule.

5.5 Summary

This chapter dealt with the intuition—correct or not—around the theory of causality developed in Chap. 4.

The phenomenon called Berkson's paradox is often referred to as one of the justifications for building a theory of causality in which conditioning on a collider may induce association, as was seen in d-separation. Berkson's paradox is the simple fact that if in a 2×2 contingency table the cell probabilities are multiplied by unequal selection probabilities, then the resulting odds ratio may be different from the original one. If one thinks of this as a selection process, then when the selection is preferential, that is, the presence of any of the features implies higher selection probabilities than the lack of that feature, and if the odds ratio is equal to 1 originally, then after selection, it may become less than 1. A precise formulation of this phenomenon was given in Sect. 5.2, showing that if the odds ratio is equal to 1, then it will be always different from one in at least one of the groups of those selected and those not selected. If, however, a causal interpretation is given, namely, that the presence of either one of the features has a causal effect on the selection, through increasing its probability, then the selection is a collider, and conditioning on it introduces association among the two causes if they were independent originally.

Incremental probabilistic causation is a generalization of this, where the intervention probability that a feature is present depends through a simple additive formula on the expectations of the parents under the given intervention. The properties of such realizations were studied in Sect. 5.3, and many of the general results obtained in Chap. 4 were shown to hold in the case of *IPCM*s, using simple arguments. Among others, it was proved that realizations that are *IPCM*s are faithful.

Section 5.3 showed that not all realizations of causal systems behaved as intuitively as *IPCM*s did. A number of attractive-sounding claims were shown not to be implications of the theory developed. These included showing that causal effects are not necessarily transitive. This fact was related to the higher-order associations among variables, a possibility that requires lots of attention when this theory of causality is applied to categorical variables.

The next chapter discusses important applications of the causal theory developed, including its relationship with the propensity score method and instrumental variables, presented in *LCDA*, but also many scenarios where there is a clear causal relationship, but this theory does not apply or where higher-order associations are present and restrict the usefulness of this theory.

5.6 Things to Do

1. Find situations which may serve as additional examples of Berkson's paradox. Think about applying for a developmental grant or winning a sports competition involving subjective assessment criteria, such as ice skating or selecting a breakfast cereal.
2. Study the original context of the paradox as described in Berkson [22].
3. Consider a definition of an *IPCM*, instead of (5.8), with

$$Q(A = 1|\mathscr{I}(nd(A))) = 0.05 + 0.9\frac{\sum_{D\in pa(A)} a(D)E_{\mathscr{I}(nd(A))}(D)}{v}, \qquad (5.13)$$

where $a(D)$ are nonnegative weights. How v needs to be defined to obtain an intervention probability with the formula above? What is the meaning of $a(D)$?

4. Which of the results given for *IPCM*s remain valid if the more general definition in (5.13) is used instead of (5.8) ?
5. May the weights $a(D)$ be allowed to be negative in (5.13)?
6. Why is incremental probabilistic causation defined in (5.8) taking into account interventions to the parents of A?
7. Generalize the results given for a fork configuration in the context of an *IPCM*. Assume variables A and B have a common ancestor. Does it follow that they are not independent?
8. In the setup of the previous item, is it true that conditioning on all common ancestors of A and B makes them conditionally independent of the observational distribution?
9. Consider Claim 2 from the perspective of the relationship between the intervention and observational distributions. What condition would have to be true for Assumption 3 to imply that if two variables are independent of the observational distribution, they are also independent of all the intervention distributions that arise after all possible interventions?
10. Show that stimulation is equivalent to the odds ratio being greater than 1.
11. Prove that if in the observational distribution

$$A \perp\!\!\!\perp C|B,$$

then

$$P(B = 1|A = 1) > P(B = 1|A = 0)$$

and

$$P(C = 1|B = 1) > P(C = 1|B = 0)$$

imply

$$P(C = 1|A = 1) > P(C = 1|A = 0),$$

that is, if A stimulates B and B stimulates C, then also A stimulates C.

12. For four variables A, B, C, D, draw all the causal graphs that imply the conditional independence in the previous item.
13. Show that if in the causal graph no path (disregarding direction) connects A and B, then they are independent of the observational distribution.

Chapter 6
Causal Analysis Using DAGs: Applications and Limitations



6.1 Introduction

In Chap. 4, the fundamentals of a theory of causality originally developed by Pearl and his coworkers were presented. This approach to modeling and analyzing causality has become very influential during the past two decades. In this chapter, we present some of its important applications and also discuss real situations, when this theory fails to give a sufficiently good description. With this, both the power and the limitations of the approach will be illustrated.

The first part of the chapter presents applications of the theory. Section 6.2.1 looks at a simple causal system, introduced by Pearl [127], and used by many authors to explain the fundamentals of the theory, and shows how added knowledge may be used to update and expand the causal model.

The method of instrumental variables, which under certain conditions makes it possible to infer the direction of an effect from observational data, was discussed in Section *LCDA* 8.4 on an intuitive basis. Section 6.2.2 shows that the method is implied directly by the assumptions made in Chap. 4.

Propensity score-based matching to estimate causal effects was discussed in Section *LCDA* 8.4, as an alternative and also a very popular approach to causality. Section 6.2.3 fits that theory into the framework discussed here. Using d-separation and the implied conditional independence, we show that given the propensity score and the treatment received, the response variable is conditionally independent of a design variable indicating whether having received treatment or control was observed in an observational study or imposed in an experiment. Thus, response to an observed treatment category (treatment or control), given the propensity score, has the same distribution as if the treatment received was imposed, and the former may be used to estimate the latter. This is the essence of propensity score-based matching.

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As the last topic to show applications of the theory, Simpson's paradox (see Chapter *LCDA* 9) is revisited. The argument presented here (see Sect. 6.2.4) is different from the discussion earlier and may be used to decide whether the marginal or the conditional conclusion to rely on, when these are different and the causal structure is known.

In the second part of the chapter, limitations of the theory are discussed. This is done in terms of presenting real-life examples, where the general setup of the theory does not appear to be relevant or Assumptions 1–3 are unrealistic. These limitations show that the theory of causality based on DAGs is not as general, as some of its proponents appear to believe.[1]

In Sect. 6.3.1, an example of a causal system is presented, where the effect of the intervention also depends on the current status. Thus, the intervention distributions, as used in Chap. 4, do not properly describe the behavior of the causal structure. Section 6.3.2 considers a possible extension, where the causal system also has memory, as was seen with Markov chains.

Another aspect where the current theory seems to need an extension is the issue of individual or joint effects. As a binary causal relation is used and is depicted in a DAG, the effects are individual in the current theory. Section 6.3.3 presents an example, where the individual effects do depend on the status of the system, and are poorly described by the intervention distributions, yet the two causes together determine the response fully.

In Sect. 6.3.4 an example is discussed where causality is cyclic, in sharp contradiction with Assumption 1 made here. The key to this example is a causal structure, where effects take place after finite amounts of time. Thus, in spite of cyclic effects, the system is stable and well-defined. If one assumes that intervention effects take place immediately, cyclic causality is, indeed, hard to think of. This, again, underlies the need to model the temporal aspect more explicitly.

Section 6.3.5 illustrates that depending on the substantive research context, including a causal path in the model, when there is a causal effect, as required in that context by Assumption 2, may be intellectually or scientifically challenging. This, of course, is not a limitation of the theory, but shows that the assumptions made cannot be seen as being true automatically.

The final example in Sect. 6.3.6 is related to Assumption 3 and illustrates that, occasionally, making relevant interventions to a causal system may be problematic. If a feature of the causal system remains unknown to the experimenter, then naive interventions may not reveal its behavior, and, among other implications, intervention distributions and observational distributions may not coincide, even if parents are manipulated.

[1] For example, Pearl writes in [130] "This survey aims at making these advances more accessible to the general research community by, first, contrasting causal analysis with standard statistical analysis, second, presenting a unifying theory, called "structural," within which most (if not all) aspects of causation can be formulated, analyzed, and compared, ..." Based on the examples presented, this does not seem to be the case.

The material in this chapter shows that the theory of causality based on DAGs is very powerful in dealing with a wide variety of causal problems but cannot be considered appropriate in all possible situations when a causal analysis is needed.

6.2 Applications

To take a closer look at the scope and limitations of the approach to use directed acyclic graphs to describe known causal structures, we discuss first a famous example due to Pearl [127]. In this example, the everyday experience that if there is no gas in your car or the battery is dead, the car won't start is formalized.

Later parts of this section show how the methods of instrumental variables and propensity score-based matching, discussed in *LCDA*, can fit into and be justified by DAG-based models of causality. Finally, Simpson's paradox is revisited and is shown that when the marginal and conditional analyses yield contradicting conclusions, studying the causal structure between treatment, response, and the conditioning variable, whenever this is possible, provides useful insight into which conclusion to rely on.

6.2.1 The Car Will Not Start If the Battery Is Discharged or There Is No Gas

Both no gas and dead battery lead to the car not starting, and this relationship may be described by the graph in Fig. 6.1. This graph has the same structure as graphs [j], [k], and [o] in Figure *LCDA* 8.1 and also the first graph in Fig. 3.4.

The nodes of the DAG in Fig. 6.1 need to be interpreted as discrete variables in order to be the DAG of a causal structure as introduced in Chap. 4. For example, instead of dead battery, one has to consider a binary variable indicating whether the

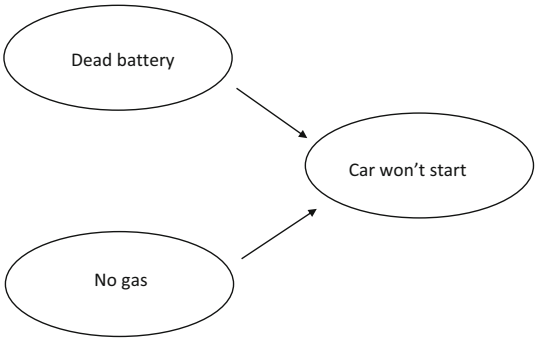


Fig. 6.1 A simplified causal structure described with a DAG—roadside assistance

battery is dead, instead of no gas, a binary variable indicating whether the tank is empty, and instead of the car won't start, an indicator whether or not the car will start.

The causal structure in Fig. 6.1 can have many realizations. As in reality, where the car won't start if there is no gas or the battery is discharged, in every possible realization, the probability of the car starting with no gas or no electricity is zero. Note that this holds true, whether the car is observed to have no gas or no electricity (observational distribution) or the car is made to have no gas or no electricity (intervention distribution).

Useful realizations may differ in terms of what is the probability that the car starts if both gas and electricity are present. It is obviously not true that in this case, the car will surely start. A realization in which the probability of starting is 1 in this case (i.e., a deterministic causal model) may be a useful description of reality if one has reason to exclude any other possible cause why the car may not start, either by general considerations or based on actual knowledge in the situation to be analyzed. This would require one to know all possible causes why the car won't start and either to establish that there cannot be any cause present other than the ones considered or to have checked all possible other causes and establish that they are not present actually.

As discussed in Section *LCDA* 8.1, such a clear-cut dealing with causes is usually not possible in the social and behavioral sciences or in the medical field, and in such cases the assumption of knowing all possible causes can only be seen as an approximation—good or bad—of reality. But this view of causality is very relevant in strongly engineered contexts, like a computer code or man-made machines or systems; however, later examples in this chapter also illustrate counterexamples to this.

After the simplifying assumption that the car will surely start if there is gas and electricity, one has a clear deterministic causal structure in this example, which is illustrated in Table 6.1. A + sign indicates a possible scenario, while a − indicates an impossible scenario. After the simplifying assumptions were made, the status of gas and of the battery will imply whether the car will start. If there is gas, and the battery works, the car will start. If there is no gas and/or the battery is dead, the car won't start.

Table 6.1 also illustrates that in this deterministic model, if a dead battery and no gas are assumed to be independent then, conditioned on the collider whether or not the engine started, they become dependent both when the engine started and when it did not.

Table 6.1 The structure of the battery—gas—car example

Car will start			Car won't start		
	Gas	No gas		Gas	No gas
Battery works	+	−	Battery works	−	+
Battery dead	−	−	Battery dead	+	+

Obviously, a model, like the one in Fig. 6.1, is an oversimplification of reality. This, however, not to say that such a model is not useful. In fact, it is very similar to what the roadside assistance technician would use, if you need to call help because the engine did not start. The goal of the technician is not to understand what is wrong with the car, rather to find out if there is anything that can be done on the spot to make the engine start. And with most modern cars, one cannot do much more on the spot than adding gas or giving it a jump start. This is characteristic for many data analysis problems: the kind of analysis which is needed is determined by the policy decision or intervention that is to be decided upon.

In reality, there are several other factors that influence whether the car will or will not start. Because of this, most realizations of the causal structure will be stochastic and associate a probability with the car starting, even if gas and electricity are present.

If the car does not start and the battery is charged and there is gas, it is usually towed to a garage. The car mechanics there may be able to do much more than their roadside assistance colleagues and, therefore, use a more sophisticated causal structure. One such causal structure is shown in Fig. 6.2.

The causal structure in Fig. 6.2 says that for the engine to start, the starter motor, ignition, and fuel injection need to work properly, and these depend on battery and gas. It is, of course, still a simplification, because all these systems have many individual components, which may be the cause of failure to work properly.

In this more complex model, Battery and Gas are still d-connected by engine. The undirected paths between battery and gas are the following:

$$B \rightarrow S \rightarrow E \leftarrow In \leftarrow G,$$

$$B \rightarrow Ig \rightarrow E \leftarrow In \leftarrow G,$$

$$B \rightarrow In \leftarrow G.$$

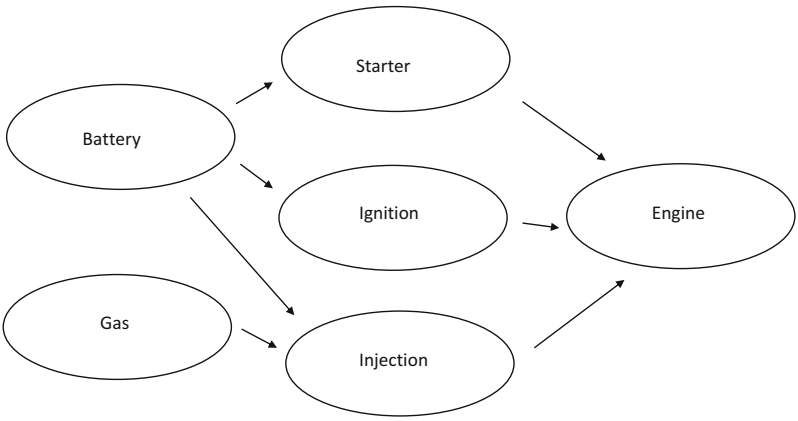


Fig. 6.2 A more complete causal structure—mechanic in the garage

The first two paths between battery and gas go through engine, and in both of them, engine is a collider. The third path goes through injection, which is a collider and engine is a descendant of it. Thus, conditioned on engine, battery, and gas may be dependent.

To illustrate the complexity of causal modeling even in this simple case, and the decisions the analyst has to make before the formal analysis could be applied, assume now that the starter and ignition are known to work fine (perhaps because they were recently fixed in the garage or because they could be tested independently from the other components). Then, one may condition on starter and ignition, but not on engine. The reason for this decision may be that the kind of certainty with respect to starter and ignition is different from the certainty concerning engine. While the former two were fixed or tested to work independently of the other components, the latter one was observed not to work, and it may or may not be defective.

In the first path, starter is a noncollider and in the second path, ignition is a noncollider, and these are in the conditioning set. In the third path, injection is a collider, and it is not in the conditioning set and neither is its descendant engine. Thus, battery and gas are d-separated by starter and ignition, and if the model is correct, they are conditionally independent.

Compared to the simpler causal structure in Fig. 6.1, working with the more complex causal structure in Fig. 6.2 provides more certainty that if all conditions are met, and the engine is not defective, it will start. That is, if the starter, the ignition, and the injection all work properly (which assume that a (charged) battery and gas are also present), then one can be more certain that the engine will start (if not defective), than when only gas and electricity are known to be present but one does not know about the other components. In many cases, scientific research proceeds similarly. Accumulating knowledge makes it possible to build more detailed models which lead to a better understanding and more precise prediction of the phenomena studied.

The next three sections revisit topics discussed in *LCDA*, where the theory of causality based on DAGs provides relevant insight.

6.2.2 *Instrumental Variables Revisited*

The instrumental variable method was described in Section *LCDA* 8.4. It offers an answer to the question whether X has an effect on Z or vice versa, that is, whether $(X, Z) \in \mathscr{C}$ or rather $(Z, X) \in \mathscr{C}$. The framework developed in Sect. 4.3 makes it possible to give a more rigorous discussion of the instrumental variable method. The variable Y is an instrumental variable, if the following holds:

$$(Y, Z) \in \mathscr{C}, \ (Y, X) \notin \mathscr{C}, \ (X, Y) \notin \mathscr{C},$$

that is, if Y is a direct cause of Z but not a direct cause of X, neither is X a direct cause of Y. In other words, in the causal DAG, Y and X are not linked by an arrow and there is an arrow from Y to Z.

The question concerning the direction of the effect between X and Z is then equivalent to choosing between graphs [k] and [m] in Figure *LCDA* 8.1. In the former case, $(X, Z) \in \mathscr{C}$, and in the latter case $(Z, X) \in \mathscr{C}$.

If Assumptions 1–3 are made, then in the observational distribution, if (P, Q) is a realization of the causal system, the two possibilities imply different factorizations of the joint distribution. If $(X, Z) \in \mathscr{C}$, that is, the DAG is [k], then the implied factorization is

$$P(X, Y, Z) = P(X)P(Y)P(Z|X, Y),$$

leading to

$$X \perp\!\!\!\perp Y.$$

In case when $(Z, X) \in \mathscr{C}$, that is, the DAG is [m], the factorization is

$$P(X, Y, Z) = P(Y)P(Z|Y)P(X|Z),$$

leading to

$$X \perp\!\!\!\perp Y|Z,$$

see Sect. 4.3.2.

Then, data may be used to test which of these seems to be a more reasonable assumption regarding the joint distribution. For a straightforward application of the instrumental variable method, see, e.g., Németh and Rudas [121].

It has to be emphasized that it may happen that neither $X \perp\!\!\!\perp Y$ nor $X \perp\!\!\!\perp Y|Z$ has to be rejected based the data or that both are to be rejected.

The instrumental variable method is applied in a situation when one does not know the direction of the arrow between Z and X, but is convinced of the existence of a direct causal effect between them. The latter piece of knowledge, in this situation, usually cannot come from knowledge of the data generating (causal) mechanism, because if that was known, not only the existence but also the direction of the arrow between Z and X would be known.

When the existence of an arrow that is, a direct causal effect between X and Z is not certain but the direct effect of Y on Z is known, also the factorization

$$P(X, Y, Z) = P(X)P(Y)P(Z|Y)$$

is possible. In this case,

$$X \perp\!\!\!\perp Y$$

and

$$X \perp\!\!\!\perp (Y, Z)$$

also hold. Thus, if marginal independence of X and Y does not need to be rejected, this may refer to X having a causal effect on Z, as was seen above, but also to no direct causal effect in any direction between X and Z, as in the latter case. This underlies the importance of the assumption of an existing direct causal effect of unknown direction between X and Z in the instrumental variable method.

6.2.3 Propensity Score Revisited

In this section, the propensity score-based estimation of causal effects using observational data, presented in Section *LCDA* 8.4, is discussed from the perspective of the DAG-based approach to causal analysis.

The general context of the problem of estimating causal effects based on observational data is that the causal effect to be estimated has to describe what would happen if one or the other of the treatments was applied to the entire population. The observations, on which such estimates should be based, however, show what the responses to the treatments were of those who received the treatments, and, because of the observational nature of the data, those receiving a treatment may be different from those receiving another treatment. Thus, what the response to a treatment in the entire population would be is hard or impossible to estimate based on those who actually received it.

As pointed out in Section *LCDA* 8.4, the fundamental idea on which the propensity score method relies is that allocation into the treatment categories behaves as if it was a random process within different subsets of the population[2] but allocation into treatment categories happened with different probabilities within the different subsets.

In order to give a more precise description, let the variable X comprise all relevant characteristics[3] of the individuals in the population to which the estimate of the causal effect should refer, and let T be the treatment received by an individual. Then, in a designed experiment with random allocation

$$X \perp\!\!\!\perp T.$$

Independence is a property of random variables. X is a random variable because of random sampling into the study from among individuals in the population of interest,[4] and the usual random allocation process implies that T is also a random

[2] Of course, if allocation was random among the individuals available for the study, then one would have a designed experiment.

[3] See the comments later in this section concerning the meaning of this assumption.

[4] Unfortunately, in many experiments involving human beings, the selection of the participants is not random. This is the issue of external validity, see Section *LCDA* 8.3.1.

variable and that they are independent.[5] The allocation procedure may be thought of as first randomly selecting an individual from the population to participate in the experiment, which makes the characteristics X a random variable, and then, without knowing the actual value x, selecting randomly a treatment for this individual. The independence brought about by this procedure implies that the conditional distributions of X in the various treatment groups are identical:

$$P(X|T = tr_i) = P(X). \tag{6.1}$$

In turn, this implies that for the response variable R, the expectation of $R|T = tr_i$ is the same if determined for those actually treated with treatment tr_i or for the entire population. For details, see Theorem *LCDA* 8.1.

For observational data, as the allocation process is not controlled by the experimenter, the independence $X \perp\!\!\!\perp T$ cannot be assumed, and the previous argument fails. In fact, T should be considered as being potentially dependent on X. The propensity score theory relies on the fact that if conditioned further on an appropriate variable, called a balancing score, one obtains the lack of dependence of the conditional distribution of X from the actual treatment received. It turns out (see Proposition *LCDA* 8.1) that[6] the propensity score, defined as

$$S(x) = P(T(x) = tr_1),$$

is such a balancing score, so that

$$X \perp\!\!\!\perp T|S, \tag{6.2}$$

and then

$$P(X|T = tr_1, S = s) = P(X|S = s)$$

and the magnitude of the causal effect may be estimated within the categories of the propensity score as if one had experimental data, because then (6.1) holds within each propensity score category.

To further clarify the relationship between X and S, let the population consist of individuals i, each with their relevant characteristics $X(i) = x(i)$. Out of all the individuals, for whom $X(i) = x$, some received treatment tr_1, and their theoretical fraction is $S(x)$. Theoretical fraction means here the probability of receiving the

[5] More precisely, the independence considered here is a property of the joint distribution of X and T. In the case of a designed experiment, that joint distribution exists by construction, that is, by the sampling and allocation processes and their independence.

[6] Usually, tr_1 is identified with the treatment the effect of which is to be estimated, and tr_2 is control to which the effect of the treatment is to be compared.

treatment that is supposed to govern the observed fractions (relative frequencies) in actual studies.[7]

The value of $S(x)$ is assigned to every individual i, for whom $X(i) = x$. This is the theoretical fraction, may be estimated from a sample of individuals. Of course, it may occur that $S(x)$ happens to be the same for different values of x.

Usually, X is a combination of several, perhaps very many, variables. In practical applications of this theory, knowledge of X is only needed to estimate the propensity score. One cannot seriously believe to have access to all relevant variables, thus, an estimate of the propensity score may only be an approximation, even if one had population data, instead of sample data. It is assumed here that the propensity score may be estimated with sufficient precision for the analysis based on it to be relevant. In practice, observations with the same or similar value(s) of X are grouped together and s is estimated for them. If the estimates for two groups turn out the same or similar, the groups are combined.

As the calculation of the causal effect (or attributable effect, see Section *LCDA* 8.4) happens within subsets of individuals characterized by (approximately) the same propensity score, S is assumed to be categorical.

The property which is relevant for the present argument is the following.

Proposition 6.1 *With the notation above, for a binary treatment variable (e.g., treatment and control)*

$$P(T = tr_j | S = s, X = x) = P(T = tr_j | S = s)$$

for all s and x, when the condition on the left-hand side is possible.[8]

□

This is, of course, equivalent to (6.2). A heuristic explanation of why this is true may be given like this: The second condition of the left-hand side selects individuals i, for whom $X(i) = x$ and, by definition, $S(i) = S(x(i))$. As S defines a partition of the population, both conditions on the left-hand side are possible at the same time only if $S(x) = s$. The partition of the population generated by X is finer than the partition generated by S, that is, individuals with the same x have the same s. Whatever is their x value, as long as they have the same s, the probability of receiving tr_j is the same.

[7] Note however that, in general, there may be a particular fraction of a group receiving a treatment, without this treatment being randomly allocated. This may happen, for instance deterministically, if certain individuals from the group receive one treatment, the others the other one. Thus, $S = 1$ or $S = 0$, and when conditioned on S, T is constant and (6.2) holds.

[8] Again, if X contains "all" characteristics, then P is a probability, because in that case individuals who are identical in every aspect are given different treatments, thus, allocation has an unexplainable, say random, component in it. If not "all" characteristics are known, then P is a descriptive fraction, and assuming that allocation is random is an approximation.

To set up a $(V, \mathscr{C})$ causal structure, let X be all[9] the characteristics of individuals in the population of interest. In order to model the relationship between what one observes in an observational study and what the responses would be, if one of the treatments was applied to the entire sample, let D be a design variable with $three$ categories. $D = o$ (observation) is the observational study, and individuals undergo treatments that are selected by a mechanism, which is unspecified other than the role of the propensity score discussed above. $D = tr_1$ is the decision to apply tr_1 (often treatment) to everybody in the sample, and $D = c$ is the decision to apply tr_2 (often control) to everybody in the sample. Let T be the variable showing the treatment an individual received, with categories tr_1 and tr_2. Further, one needs to include the propensity score S and the response to the treatment received, R.

To better understand the context in which the causal structure will be built up, note that, in theory, there may be several studies available concerning the effect of T on R. These studies may be different as to D and as to the actual sample of individuals on which they were carried out. The analyst chooses one of these randomly that is with equal probability of selection. Further, R may also have uncertainty, that is, even the same individual may give different responses if the same treatment is applied repeatedly to them. This uncertainty is assumed to be not related to a memory effect, rather to measurement error, and is also considered to be probabilistic. Then, the variables are random variables, and they have a joint distribution and expectations are meaningful, namely, expectations are over all possible choices of D, of the sample of individuals, and of the uncertainty of responses.

Of course, in reality, the analyst will apply intervention to the system, namely, will set $D =_s o$. Clearly, neither one of the interventions $D =_s tr_1$ and $D =_s tr_2$ is a good design, because it would not tell which treatment is better. If the analyst could apply both of the latter interventions, then they could perform a designed experiment, too. The goal here is to estimate from the observations obtained from $\mathscr{I}(D =_s o)$ what would happen if tr_1 or tr_2 was applied to the entire population.

We define a causal structure that yields the same analysis as the propensity score method. The design variable has a direct causal effect on the treatment variable T. The propensity score variable S also has a direct effect T. Of course, X has a direct effect on S but also on the response variable R. Finally, the treatment variable T has a direct effect on response R. The resulting causal relationship is the following:

$$\mathscr{C} = \{(D, T), (T, R), (X, S), (S, T), (X, R), (X, R)\}. \tag{6.3}$$

The causal DAG is shown in Fig. 6.3.

[9] As mentioned repeatedly, this assumption is problematic. When the population consists of human beings, the list of all characteristics is not finite and, considering only those characteristics that are relevant, is not a task that may be easily carried out in practice. Unfortunately, quite often in applications, the available variables are assumed to be the only relevant ones. That assumption cannot be tested within the context of the analysis, and may lead to false results.

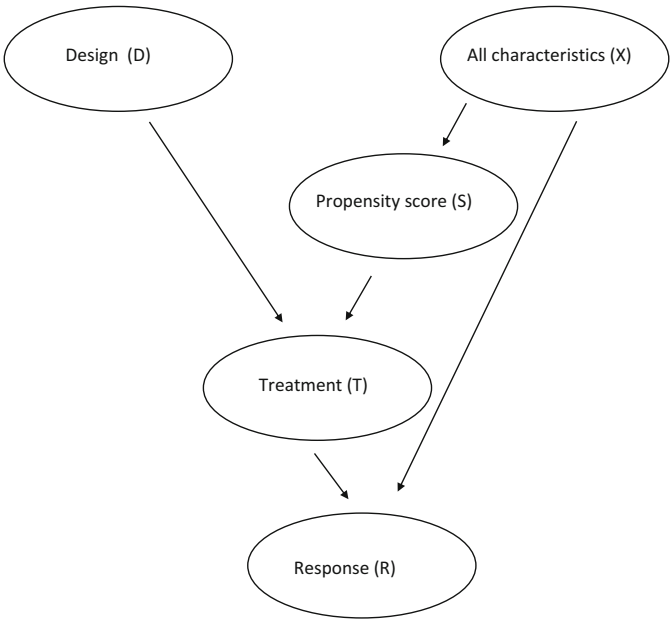


Fig. 6.3 DAG representation of the propensity score method

There is no argument presented here whether or not the causal structure (6.3) is correct, because such a decision should depend on the actual circumstances of the data collection procedure, yet comments are needed concerning some of the arrows that are not included.

The variables D and X are considered to be directly not related, that is, in their marginal distribution, they are independent. This is true in reality, if proper sampling from among the members of the relevant population is possible. However, very often in practice, there is a stronger or weaker element of self-selection present. Those who are ready to participate when $\mathscr{I}(D =_s o)$, that is, in an observational study, may not be ready to participate, if the design imposes one of the treatments on them.

Another arrow that is assumed to be not present may connect D to R. That is, D has no direct effect on R (of course, it has an indirect effect through T). This assumption may or may not be true for the actual research problem. If the potential response to the treatment is "objective," for example, a pill may reduce blood sugar level, then it is reasonable to assume that the effect will or will not happen, independently whether subjects chose the pill they took based on, say, its color (D=o) or the pill was given to them by their physicians ($D = tr_1$ or $D = tr_2$). If, however, the response is a matter of judgment, for example, a movie may or may not be liked by those who watch it, it is quite possible that some of those who chose this movie and liked it would not like it if, instead of their own choice, someone forced them to watch it. Call the assumption that the response does not depend on the design, objective response. If this assumption was not made, one would have to

Table 6.2 The intervention distribution of treatment

Intervention	Intervention probability of tr_1
$\mathscr{I}(D =_s o, \quad X =_s x, \ S =_s s)$	$Q(T = tr_1 \mid \mathscr{I}(D =_s o, \quad X =_s x, \ S =_s s)) = s$
$\mathscr{I}(D =_s tr_1, \ X =_s x, \ S =_s s)$	$Q(T = tr_1 \mid \mathscr{I}(D =_s tr_1, \ X =_s x, \ S =_s s)) = 1$
$\mathscr{I}(D =_s tr_2, \ X =_s x, \ S =_s s)$	$Q(T = tr_1 \mid \mathscr{I}(D =_s tr_2, \ X =_s x, \ S =_s s)) = 0$

model the direct effect of design on response, e.g., how reported liking of a movie depends on whether the movie was selected by the individual or they were forced to watch it. Without the objective response assumption, it appears to be impossible to estimate the responses in case the treatment was imposed, based on data when the treatment was selected. The assumption of objective response seems implicit also in the original theory of propensity score matching. Also note that the assumption of objective response is related to but is different from the stable unit treatment value assumption (SUTVA) applied in Rubin's theory, see, e.g., Imbens and Rubin [76]. SUTVA requires that the response of an individual to a treatment does not depend on what treatments other received.

The intervention distributions of T are given in Table 6.2. The nondescendants of T are D, X, S, and D and S are its parents. By Theorem 4.1, the intervention distribution depends only on the parents. When the $D = tr1$, the probability that $T = tr_1$ is 1, and when $D = tr_2$, then the probability that $T = tr_1$ is 0. When $D = o$, the probability $T = tr_1$ is the value of the propensity score, s. Of course, interventions when $X =_s x$ and $S =_s s$, but $s \neq S(x)$, are not possible. Here we assume that treatment allocation is probabilistic. As was mentioned in Section *LCDA* 8.4, this is a very strong assumption, and is much stronger than observing that allocation (by some unknown mechanism) resulted in receiving one of the treatments in a given fraction of cases. The assumption means that for those with the same propensity score, allocation is the same random mechanism. This assumption is related to the assumption that X contains all (relevant) information and is not discussed here further.

The equivalent of Theorem *LCDA* 8.1 is formulated as follows.

Theorem 6.1 *If (P, Q) is a realization of the causal structure $(V, \mathscr{C})$ defined above, then for the observational distribution*

$$D \perp\!\!\!\perp R | (S, T).$$

Proof The proof is based on the d-separation rule.

Indeed, there are two undirected paths between D and R:

$$\mathscr{P}_1 = D \to T \to R$$

and

$$\mathscr{P}_2 = D \to T \leftarrow S \leftarrow X \to R.$$

In $\mathscr{P}_1$, T is a noncollider, which is contained in the conditioning set $\{S, T\}$. In $\mathscr{P}_2$, S is a noncollider, which is contained in the conditioning set $\{S, T\}$.

□

This result implies that

$$P(R|S = s, T = tr_i, D = d) = P(R|S = s, T = tr_i),$$

and, in particular, that

$$P(R|S = s, T = tr_1, D = o) = P(R|S = s, T = tr_1, D = tr_1)$$

and

$$P(R|S = s, T = tr_2, D = o) = P(R|S = s, T = tr_2, D = tr_2),$$

that is, given a fixed value s of the propensity score S, the distribution of responses to treatment (to control) is the same whether the treatment (control) is the observed treatment in an observational study, or the treatment (control) is imposed on everybody by design. Then, the same equalities apply to the respective expectations, thus,

$$E(R|S = s, T = tr_1, D = tr_1) - E(R|S = s, T = tr_2, D = tr_2)$$

$$= E(R|S = s, T = tr_1, D = o) - E(R|S = s, T = tr_2, D = o),$$

which is the estimate of the causal/attributable effect among those whose propensity score is s, as suggested by the propensity score method. Note that the different conditional probabilities and the different expectations refer to different subsets of the individuals participating in the study. This is the point of the propensity score method: within a propensity score category, use estimates based only on those who chose a treatment as an estimate of what the response would be if everybody was given that treatment.

Note that the same argument is not true without conditioning on the propensity score. If one conditioned on T only, then D and R would be d-connected by $\mathscr{P}_2$.

In this section we have seen that the approaches to causal analysis associated with Rubin and with Pearl are closely related. When the assumptions behind both hold, then the method of propensity scores is obtained from a DAG-based analysis, where the causal structure also includes a design variable. It has to be pointed out that this is not a general proof of the "correctness" of the propensity score method, just like no claim was made that the DAG-based methods would be "correct." The results here concern their relationship, namely, the propensity score method was embedded into the DAG method.

For further approaches to identify the causal effect of a treatment, see Pearl [130].

6.2.4 Simpson's Paradox and DAG Models of Causality

Simpson's paradox was discussed in detail in Chapter *LCDA* 9. It is the phenomenon that the direction of the effect of a treatment T on a response R is positive (negative), but when conditioned on a variable C, the direction of the effect reverses in all categories of C from positive to negative (from negative to positive).

The essence of the analysis of the situation given there, following Rudas [147], was that the usual way of reading off the direction of the effect based on the odds ratio disregards relevant features of the data collection procedure. When the data come from a designed experiment, then the number of those choosing one treatment or another is determined by the experimenter and the outcome of the randomization procedure and is, therefore, not informative with respect to the effect which could be expected if the treatment was applied to everyone in the population. On the other hand, the relative popularities of the treatment options in an observational study are potentially informative with respect to the effect which may be expected if one of the treatments is applied to the entire population. The popularity of a particular treatment is relevant when that treatment is made available to the entire population and also when it is imposed on the entire population. In the former case, it has an effect on the fraction of the population accepting the treatment;[10] in the latter case it may have an effect on the reaction to the policy decision, see Rudas [148].

Consequently, the odds ratio, which is variation independent of the marginal distributions (see Section *LCDA* 6.1.2), should not be used when the T marginal is informative. It was shown (see Theorems *LCDA* 9.2 and 9.3) that subject to mild assumptions, every measure which is variation independent from the marginals commits Simpson's paradox for some set of data.

Instead of the odds ratio (or cross product ratio), the cross sum ratio was proposed as a measure of treatment success in case of data from observational studies, which is sensitive to the marginal distributions and never commits Simpson's paradox. It was also shown, that subject to mild assumptions, all measures of treatment success which never commit the paradox reach the same conclusion as to which treatment is better, as the cross product ratio does, see Theorem *LCDA* 9.1.

In view of the developments in Chap. 4, an alternative analysis of Simpson's paradox is possible, which was first proposed by Pearl in [131]. In fact, the argument in Pearl [131] used the so-called backdoor rule (see Pearl [128]), not discussed here, but the argument can be presented based on the tools developed so far.

The argument concentrates on the causal structure among the variables T, R, C. It is obvious that in the situations when Simpson's paradox may occur, one is interested in the effect of T on R, more precisely in the distribution of R, when $T = tr_1$ or $T = tr_2$ is administered. In other words, one is interested in the intervention distribution of R, when T is manipulated. It was seen in Sect. 4.2 that

[10] This also implies that if the selected treatment is to be made available, then an observational study is a more appropriate data collection procedure to select the better treatment. When the selected treatment is to be made mandatory, a designed experiment is more useful.

conditional distributions do not always give relevant information with respect to the manipulation distributions. In the simplest example, the conditional distribution of umbrellas given rain revealed the intervention distribution of umbrellas when rain was manipulated, but the conditional distribution of rain given umbrellas said nothing about the intervention distribution of rain when umbrellas were manipulated.

The situation here is a bit more complex, as in addition to T and R, there is also a conditioning variable C. In most applications, the categories of C select meaningful subgroups of individuals. The goal of conditioning on C is to see whether the same treatment is preferable in all groups, or perhaps in one group one treatment is preferable, and in another one the other treatment. For example, if treatment is administering drug D_1 or drug D_2 to patients with headache, it may be the case that D_1 is preferable over D_2 for all groups of patients, but it may just as well be the case that for younger patients, D_1 is preferable while for older patients D_2 is preferable. Such a result could not be found without conditioning on age groups.

When the better treatment suggested by the marginal odds ratio (or any other measure of effect or association) of T and R is different from the one suggested by the conditional odds ratio (or any other measure of effect or association) of T and R conditioned on C, then the researcher has to choose which conclusion to accept. The answer proposed here depends on the causal structure among the variables.

As one is interested in the effect of a treatment on response, all causal structures considered allow T to be a parent of R, and are different with respect to the other possible arrows, between T and C and between C and R. The latter arrows, if both are present, create a path between T and R via C. This path may or may not be a causal path depending on the direction of the arrows.

If any of the arrows connecting T with C and C with R is not present in the causal structure, then there is no path between T and R other than the arrow pointing from T to R, and the causal effect of T on R goes through this arrow (see Assumption 2). Consequently, the marginal effect is relevant. The effect seen when conditioned on C has no meaning and should not be considered.

The arrows between T and C and C and R, when both are present, may be in three configurations, as the fourth one would create a directed cycle (see Assumption 1) with the arrow pointing from T to R. The possible DAGs are shown in Figs. 6.4, 6.5, and 6.6.

The analysis presented here is based on the principle that the correct choice (marginal or conditional analysis) depends on the causal structure and not on the particular realization one has. Therefore, it is sufficient to find the correct choice for a particular realization in the case of each causal structure. For each of the causal structures, this will be a realization which is not faithful in the sense that T has no causal effect on R. It is easy to see that such a realization does exist for each of the causal structures in Figs. 6.4, 6.5, and 6.6. If in the intervention distribution there is no effect of T on R, then these realizations are, at the same time, also realizations of causal structures which are obtained from the DAGs in Figs. 6.4, 6.5, and 6.6 by omitting the arrow from T to R. For each of these reduced structures, consider the $IPCM$ defined in Sect. 5.3.

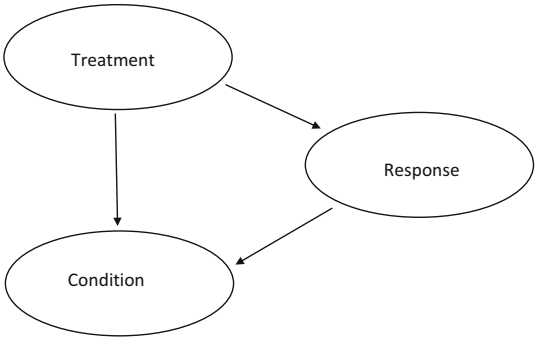


Fig. 6.4 Causal structure where a marginal analysis reveals the correct effect

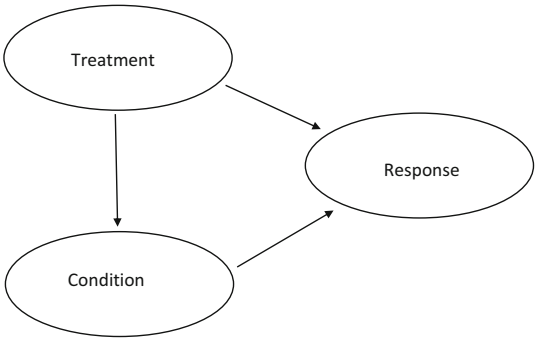


Fig. 6.5 Another causal structure where a marginal analysis reveals the correct effect

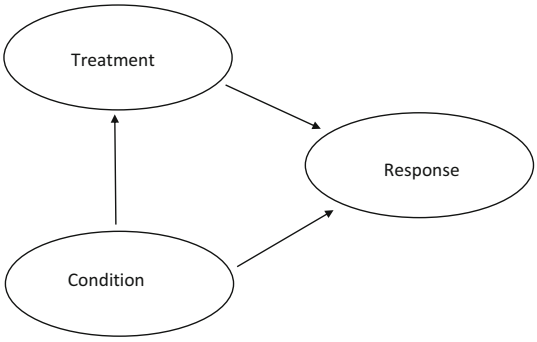


Fig. 6.6 Causal structure where a conditional analysis reveals the correct effect

To find out whether one should rely on the marginal or on the conditional effect, investigate what these choices would imply in the cases of these realizations.

When the arrow from T to R is omitted from Fig. 6.4, T and R are (marginally) independent in the observational distribution, and C is a collider. The path between T and R is not a causal path and conditioning on C makes, when the realization

is an $IPCM$, the marginally independent T and R dependent, see Corollary 5.2.[11] Thus, conditioning on C suggests that T and R are not independent, which is the wrong conclusion. Therefore, one has to rely on the marginal effect in the case of this causal structure.

When the arrow from T to R is omitted in Fig. 6.5, the causal path from T to R through C remains there, and in any faithful realization, including the $IPCM$, there is a causal effect of T on R. If one conditions on C, T and R become conditionally independent, see Theorem 4.9. This is misleading, as there is a causal path from T to R and in any faithful realization there is a causal effect of T on R through C. Thus, conditioning on C suggests a wrong conclusion, and one has to rely on the marginal effect in the case of this causal structure.

Finally, if the arrow from T to R is removed from the DAG in Fig. 6.6, then there is no causal path between T and R. There is a fork configuration, and conditioning on C would make the T and R conditionally independent, see Theorem 4.8. Thus, conditioning on C suggests the correct conclusion, and one has to rely on the conditional effect in the case of this causal structure.

Note that the analyses above relied on the assumption that no variable other than T, C, and R is part of the causal structure, that is, no other variable has a causal effect on these and these have no causal effect on any other variable.

The analysis of the observational distributions (conditional or marginal) may or may not reveal the relevant intervention distributions. For a correct estimation of the causal effect of T on R, one has to estimate:

$$Q(R|\mathscr{I}(T =_s t)),$$

when the causal structure is as in Figs. 6.4 and 6.5 (marginal analysis) or[12]

$$Q(R|\mathscr{I}(T =_s t, C =_f c))$$

when the causal structure is as in Fig. 6.6 (conditional analysis).

In general, these are not equal to

$$P(R|T = t)$$

and to

$$P(R|T = t, C = c).$$

Assumption 3 is directly relevant here, and it says that if

$$pa(R) \subseteq \{T\} \subseteq nd(R), \tag{6.4}$$

[11] But also see Claim 3 in Sect. 5.4.

[12] For the intervention distribution, it is irrelevant whether T and C are set or fixed. Fixing C and setting T is one possibility which expresses the different roles these variables play in the analysis.

then

$$Q(R|\mathscr{I}(T =_s t)) = P(R|T = t),$$

and if

$$pa(R) \subseteq \{T, C\} \subseteq nd(R), \tag{6.5}$$

then

$$Q(R|\mathscr{I}(T =_s t, C =_f c)) = P(R|T = t, C = c).$$

The left-hand sides are the quantities of interest which may not be directly observable, but the right-hand sides are directly observable.

In the causal structure in Fig. 6.4, T is a nondescendant, and the only parent of R, thus, condition (6.4) holds, and the marginal observational distribution is equal to the marginal intervention distribution which is relevant when determining the causal effect of T on R.

In the causal structure in Fig. 6.5, T and C are nondescendants and also parents of R, thus, condition (6.5) holds. As was seen above, to find out the causal effect of T on R, one has to use the marginal effect:

$$P(R|T = t) \tag{6.6}$$

but it is the conditional observational distribution and, consequently, the conditional effect:

$$P(R|T = t, C = c)$$

which is equal to the conditional intervention distribution

$$Q(R|\mathscr{I}(T =_s t, C =_f c))$$

by Assumption 3.

In fact, also the marginal observational distribution (6.6) may be shown to be equal to the marginal intervention distribution:

$$Q(R|\mathscr{I}(T =_s t)).$$

To see this, note that

$$pa(C) \subseteq \{T\} \subseteq nd(C), \tag{6.7}$$

and thus

$$Q(C|\mathscr{I}(T =_s t)) = P(C|T = t)$$

by Assumption 3. This implies that

$$P(R=r|T=t)=\sum_c P(R=r,C=c|T=t)=\sum_c \frac{P(R=r,C=c,T=t)}{P(T=t)}$$

$$=\sum_c \frac{P(T=t)P(C=c|T=t)P(R=r|T=t,C=c)}{P(T=t)}$$

$$=\sum_c P(C=c|T=t)P(R=r|T=t,C=c)$$

$$=\sum_c Q(C=c|\mathscr{I}(T=_s t))Q(R=r|\mathscr{I}(T=_f t,C=_s c))$$

$$=Q(R=r|\mathscr{I}(T=_f t))=Q(R=r|\mathscr{I}(T=_s t)),$$

where the intervention $\mathscr{I}(T=_f t)$ was built up in three steps, first setting T to t, then fixing T and setting C to c, and then summing the intervention probabilities for c.

In the causal structure in Fig. 6.6, T and C are nondescendants and also parents of R, thus, condition (6.5) holds. In this case, as shown above, the conditional observational distribution is to be used to determine the direction of the effect, and it also reveals the relevant intervention distribution.

In summary, this argument suggests that if the treatment may have a causal effect on the conditioning variable, then conditioning may be misleading and the marginal conclusion is relevant. If the conditioning variable may have a causal effect on treatment, then the conditional conclusion is relevant. Note, however, that this analysis was based on the principle that the same marginal or conditional conclusion has to be applied irrespective of the realization, only depending on the causal structure, and this principle may also require justification.

With this discussion of Simpson's paradox, we have completed the review of the applications of the DAG-based causal theory in various research problems, and turn to discussing some of its limitations. It will be illustrated that there are research problems where the approach to causality based on DAGs does not appear to provide a good description.

6.3 Limitations

In this section, a few real or realistic situations are reviewed, where causality, in the everyday sense of the word, is present, yet the theory of causality discussed so far cannot give a relevant description. In some of these cases, further developments appear possible, not so much in others.

6.3.1 When the Effect of Manipulation Depends on Current Status, the Intervention Distribution Is Not Useful

Simple electric devices, like a reading lamp, often have push-button switches. When the button is pushed, the light comes on; if it was off, and goes off, it was on. Thus, the push-button switch, when operated, changes the status of the light but does not set it to on or off.

There can be no question that the switch affects the light; this is a causal system, but the theory developed in Chap. 4 does have serious difficulties describing it. The switch does not have distinct positions, into which it may be set or at which it may be fixed during an intervention. It may be questioned whether the switch may be described by the concept of a variable. That would mean the switch has specific positions and it may be manipulated into one of those. A possibility is to consider two statuses of the switch: one when it is not pushed and another when it is pushed.

Just like the result of pushing the button depends on the status of the light when the button is pushed, the behavior which may be learned from observing responses to intervention depends on the situation, when the intervention is applied. Imagine the experimenter applies an intervention at such time points that in a fraction p of them the light is on, and in fraction $1-p$, the light is off. The experimenter, trained in the theory of Chap. 4, applies interventions independently of the current status of the light, because nothing in our theory suggests (or makes it possible for that matter) that the effect of an intervention to the parent also depends on the status of the child when the intervention is applied. To come to that idea, the experimenter would have to have a more general theory, when the system remembers its own status and joint effects of the switch and of the light itself, as opposed to individual effects only, are also possible.

Based on the current theory, the manipulation of "not pushing the switch" will lead to the light being on in a fraction of p of the times applied. The manipulation "pushing the switch" will lead to the light being on in a fraction of $1-p$ of the times applied. What the experimenter sees depends on p. If p is 0.5, the experimenter will find that the manipulation distributions are the same and will conclude that pushing the button has no effect. If p is small, the experimenter will conclude that pushing the switch increases the probability that the light is on. If p is large, the experimenter will conclude that pushing the switch increases the probability that the light is off. All these conclusions are wrong. Pushing the switch changes the status of the light deterministically.

The effect of manipulating the switch cannot be described by the conditional probabilities of the light being on or off, that is, with the intervention distribution. The issue is discussed further in the next subsection.

6.3.2 Causal Systems with Memory and Time Series

One possibility to extend the theory developed so far to make it applicable to the push-button switch example of the previous subsection is to allow the status of the child to also influence the effect of the intervention, when the parent is manipulated. Thus, the response to intervention would depend not only on the intervention applied but also on the before-intervention status of the child. Then, the result of the intervention would depend not only on the manipulation and fixing of variables but also on the status of the system, when the intervention is applied.

This requires allowing an effect of A on itself, by including for $A \in V$, $(A, A) \in \mathscr{C}$. In the DAG representation, this would mean an arrow pointing from variable A to itself.

Apparently, if an arrow goes from A to itself, that could be seen as a violation of the assumption of not directed cycle being present. More importantly, A would be a descendant of A, and, at the same time A would be a parent of A. Therefore, a parent would not anymore be a nondescendant necessarily.

A better way to implement this situation is to assume that interventions are possible at set time points t and for every variable in V, define one copy for every t, which is its version before[13] the intervention in time point t is implemented. This gives a simple way to deal with the push-button example.

At time point t, the effect of pushing or not pushing the button is

$$Q(Light_t = On|\mathscr{I}(Switch_t =_s Pushed, Light_{t-1} =_f On)) = 0,$$

$$Q(Light_t = On|\mathscr{I}(Switch_t =_s Pushed, Light_{t-1} =_f Off)) = 1,$$

$$Q(Light_t = Off|\mathscr{I}(Switch_t =_s Pushed, Light_{t-1} =_f On)) = 1,$$

$$Q(Light_t = Off|\mathscr{I}(Switch_t =_s Pushed, Light_{t-1} =_f Off)) = 0,$$

$$Q(Light_t = On|\mathscr{I}(Switch_t =_s NotPushed, Light_{t-1} =_f On)) = 1,$$

$$Q(Light_t = On|\mathscr{I}(Switch_t =_s NotPushed, Light_{t-1} =_f Off)) = 0,$$

$$Q(Light_t = Off|\mathscr{I}(Switch_t =_s NotPushed, Light_{t-1} =_f On)) = 0,$$

$$Q(Light_t = Off|\mathscr{I}(Switch_t =_s NotPushed, Light_{t-1} =_f Off)) = 1.$$

The reason why this does not really fit into the theory built here is that Light in the previous time point is not fixed at a position and, also, it cannot be manipulated

[13] A similar theory is possible with copies describing the after-intervention positions.

into a position directly (only via operating the switch, that is, by intervention of its parent); it is, rather, observed to be in a particular position.

The simplest example of a causal structure with time-dependent copies of the same variable is the Markov chain, see Sections 2.5 and *LCDA* 11.3. One has to be careful though to clearly distinguish two types of effects (or their mixtures) the previous copy (like yesterday's weather) has on the current copy (like today's weather). One is the causal effect, which, although not defined, is investigated in the current and the previous chapters, and the other is the predictive effect, which means that the previous copy is relevant to predicting the current copy. To illustrate this, one can be fairly certain that after spring, summer will come (predictive effect), but one may be reluctant to say that summer is caused by spring. Some of the related topics include Granger causality (see [70]) and the Kalman filter (see [77]), which are not discussed here.

6.3.3 No Individual Effects, Only Joint Effect

Many houses and apartments have a hall. This is the first room where you enter and other rooms are accessible through the hall. These halls tend to have light switches next to each door, so that when one walks through the hall, the lights can be turned on when entering (e.g., through the entrance), and can be turned off, when leaving the hall (e.g., entering the living room), even if one uses different doors. Such switches and the circuits behind them are called alternating. For simplicity, we assume here that the hall has only two doors and two switches, $S1$ and $S2$. Each of those switches has two positions, say U and D for up and down, and whenever any of the switches is flipped, the light changes its status. If it was off, it goes on; if it was on, it goes off. The individual effect of any of the switches is similar to the push-button switch in Sect. 6.3.1, but they have two distinct positions into which they can be manipulated.

If one assumes normal operating conditions (electricity supply, fuse, wiring, light bulb all functioning properly), then the positions of the two switches determine the status of the light entirely: if both are in the same position (both in U or both in D), then the light is on; otherwise, it is off. This is a fully engineered context, and the simplifying assumption that everything else is in good working order was made, so we have a deterministic causal structure which is described in Table 6.3, with $+$ denoting a possible scenario, and $-$ denoting an impossible one.

Table 6.3 The structure of the alternating switches example

Light is on			Light is off		
	S1=U	S1=D		S1=U	S1=D
S2=U	$+$	$-$	S2=U	$-$	$+$
S2=D	$-$	$+$	S2=D	$+$	$-$

Here, unlike in the push-button switch example, $S1$ and $S2$ are variables, with *two* positions each. But when one of them, say $S1$, is manipulated, while $S2$ is fixed at U, manipulating $S1$ to U turns on the light, and manipulating it to D switches off the light, giving a deterministic intervention distribution. When $S2$ is fixed at D, the effect of manipulating $S1$ is exactly the opposite:

$$Q(Light = On|\mathscr{I}(S_1 =_s U, S_2 =_f U)) = 1,$$

$$Q(Light = On|\mathscr{I}(S_1 =_s U, S_2 =_f D)) = 0,$$

$$Q(Light = On|\mathscr{I}(S_2 =_s U, S_1 =_f U)) = 1,$$

$$Q(Light = On|\mathscr{I}(S_2 =_s U, S_1 =_f D)) = 0.$$

That is, the effect of a switch may be identified, when the other switch is fixed in a particular position. While manipulating any of the switches has a deterministic effect, this effect depends on the position in which the other switch is fixed.

Therefore, one cannot tell what is the effect of setting $S1$ to U in terms of turning the light on or off: either can happen. Exactly the same can be said about the effect of setting $S1$ to D, and the same applies to $S2$. In the push-button example, the effect of pushing the button depended on the actual status of the light; in this example, the effect of operating a switch depends on the position of the other switch. As opposed to $IPCM$ realizations discussed earlier, where setting a parent to be present increased the probability that the feature represented by the child was also present, irrespective of the other parents, here the effect of manipulating one parent fully depends on the other parent. However, it is also true that operating the switch changes the status of the light.

The relationship between the two switches and the light is closely related to the distribution given in Table *LCDA* 1.2. With the choice of $a = 1/8$, that distribution is a possible joint distribution of the three variables considered here, that one would observe if the switches were set to their possible positions with probabilities equal, and independently from each other. Then, the experimenter would see no individual effects.

While it is very hard to associate individual effects with $S1$ and $S2$, their joint effect completely determines whether the light is on or off. The two causes ($S1$ and $S2$) have no individual effects, but they have a deterministic joint effect.

The graphs, like the one in Fig. 6.1, do not properly depict this situation, because they show two individual effects, instead of a joint effect. This is the case, in spite of the context being strongly engineered and the assumption of no other causes being present has been made. Perhaps, hypergraphs could be used to give a proper description, but at this point there seems to be no widely accepted theory of how hypergraphs could be used to model causality.

6.3.4 Cyclic Causality

The examples presented so far in this section have illustrated that causal systems may involve effects of a variable on itself or the system may have memory and also the DAGs may not give a proper description of the causal structure, all being limitations as to the applicability the theory described in the previous chapters.

Nevertheless, no scholar seems to question Assumption 1, that is, that the causal relation $\mathscr{C}$ is acyclic. As the next example illustrates, this may very well be the case, and in such a situation, the machinery developed thus far does not apply.

Imagine a very delicate production process happens in room P mostly automatically. When adjustment or maintenance is required, workers enter room P through a room C, where cleaning of their clothes and hands happens. When they finished the work in room P, they leave and enter room W where they wait till the next adjustment or maintenance becomes necessary, when they will enter room C first, and then room P, again.

Each room has a light L, which is controlled by a switch S. When the workers leave room W, they switch off LW using SW. For their convenience, the switches are connected, and when they switch off SW, the switch in room C, SC, gets into the on position and the light in room C, LC, comes on. When they finished cleaning, they switch off SC, LC goes off and automatically, the switch in room P, SP, goes into the on position, and also the light LP comes on. Similarly, when they switch off the light in room P, the switch and the light in room W come on. This is a (highly) engineered environment and a fully understood clear causal structure. SW controls LW and SC, SC controls LC and SP, and SP controls LP and SW. The structure is illustrated in Fig. 6.7.

In this example, there is a directed cycle containing the switches, thus, Assumption 1 does not hold, and the machinery developed cannot be used to describe this simple causal structure.

The cyclic effects in this example also underline the issue of dealing with the role of time in the theory developed. If one takes into account that the effects in this example do not take place instantaneously, rather after some time, then one may wish to use different copies of the variables in the different time points. Then, say, SW_1 has effect on SC_1 and the latter has effect on SP_1. Then, SP_1 has an effect on SW_2, which has an effect on SC_2, and so on. This approach has two limitations. First, it fails to take into account that SW_1 is the same switch as SW_2, just in different time points (and similarly for the other switches). The model is the same as the one that would apply if there were several waiting—cleaning—production room triplets. Second, as long as V is assumed to be finite, this model assumes that the effect may go around a finite number of times, but in reality, the number of times workers may go around in the rooms is not bounded. Ultimately, the real data analytic problem determines whether these limitations are important or not.

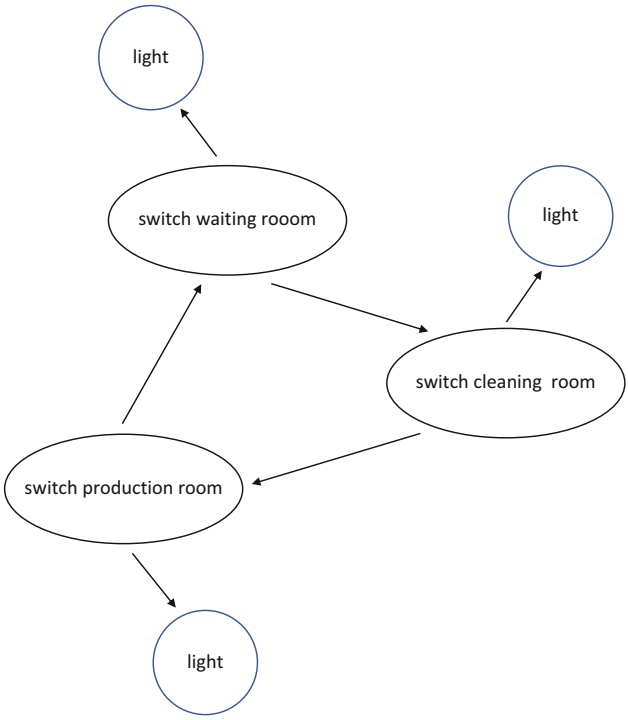


Fig. 6.7 An example of cyclic causality

6.3.5 Causal Effects and Causal Paths

Assumption 2 plays different roles in different types of causal analyses (see Sect. 4.4.1). When the causal structure is known, e.g., in an engineered environment, then Assumption 2 means that if there is no causal path leading from variable B to variable A, then the former cannot have an intervention effect on the latter. In this case, the assumption seems quite agreeable.

When the causal structure is not known, and is rather to be explored based on data arising from interventions and/or observations, Assumption 2 may pose certain difficulties (the universality of the other assumptions is discussed in other sections). In these situations, if there is an intervention effect, it is deemed to be a causal effect, and there has to be a causal path from the cause to the effect. This is a completeness assumption, meaning that no causal effect present among the variables is disregarded in the analysis.

Sometimes, a closer look at a causal effect, and the effort to incorporate a causal path in the model accordingly, suggest including new variables through which the causal effect takes place. This is closely related to the tradition of not believing in the existence of a causal effect until the causal mechanism behind it is understood. Causal effects are usually established by conducting experiments,

and in the theory of causality presented in the previous chapters by observing the results of interventions with the goal of estimating the intervention distribution. Experimental evidence should, however, be handled carefully, keeping in mind the limitations of experiments. These include the issue of external validity, that of sampling variations depending on the actual participants, possible measurement errors, and that randomization creates identical treatment groups in expectation but not necessarily in every actual experimental situation, see Section *LCDA* 8.3.1. This issue is discussed further after presenting two examples which span a very wide range with respect to how easily one incorporates a reported causal effect in the casual structure.

There are situations when an explanation of why and how a causal effect takes place is readily available and cases when it is not.

For example, a car driver may be told by the roadside assistance mechanic that they could not start the engine because the battery was down. A very naive driver[14] may ask the mechanic: "Why do I need electricity? This is not an electric car; it runs on gas." And then the mechanic may explain what is depicted in Fig. 6.2. Even though the car runs on gasoline, it requires electricity for the starter motor, for fuel injection, and for ignition. From the perspective of the very naive car driver, the mechanic is in possession of all necessary and possible knowledge, and their word is taken for fact. The driver redraws their causal graph, and includes the effect of electricity on whether or not the car will start, through the variables starter, injection, and ignition. The latter variables are included to be able to properly account for the causal effect of electricity on whether or not the engine starts.

In many research situations however, there is no unquestionable guidance available to the scientist. Occasionally, the final decision whether or not to admit the existence of a causal path and to include it in the model does depend not so much on scientific evidence like the lack or existence of an intervention effect, rather on *Weltanschauung*. Research on the healing power of intercessory prayer illustrates this situation. There have been many such studies conducted (for a review, see, e.g., Andrade and Radhrakrishnan [7]), and according to the published descriptions, they were proper designed experiments with randomization and double-blind participation. The general scheme of such a study is that a group of medical patients is selected and is randomized into a treatment and a control group. Patients in the treatment group are then being prayed for by a group of individuals who, independently of the experiment, regularly participate in religious practices. Of course, the patients do not know whether they are or are not being prayed for, neither do the medical personnel. The response observed is recovery time or time to release from a hospital ward or some score of medical status. Many such experiments reported a positive effect of prayer: the treatment group had better scores than the control group, see, e.g., Harris et al. [72]. The difference in many cases was "statistically significant." According to current practice, authors mean

[14] As scientists, we are often very naive because we may lack the relevant background knowledge to properly interpret our observations.

by this that after (often unspecified) distributional assumptions were made, the hypothesis that the population averages for those who were being prayed for and those who were not were equal had to be rejected at some level.[15]

Designed experiments are the gold standard of establishing causal effects. Some published experiments appeared to prove the causal effect of intercessory prayer on healing. Is it that straightforward to include a causal path between prayer and healing? Of course, individuals who believe in the existence of metaphysical forces and in their effects, with or without the intervention of a deity, see no need to conduct such experiments. Others are not likely to be convinced based on a single experiment. Other than a metaphysical effect, one does not understand the causal mechanism and may be reluctant to assume the existence of a causal path.

An important question when a new, hitherto unknown causal effect is to be included in the model is whether the causal effect (e.g., of intercessory prayer on healing) takes place directly or through other variables. In the former case, there is an arrow to be included creating a parent-child relationship; in the latter case, the variables need to be identified through which the effect takes place. These variables may already be present in the analysis, but there may be a need to include new variables. This is important not only from the perspective of whether the effect may be blocked but also from the point of view of the tradition mentioned above which suggests not considering the existence of an effect proven until the actual mechanism through which it takes place is understood.

This view also suggests caution with taking a few published experimental results as proof of the existence of effect of prayer even without the individuals who are being prayed for being aware of this. Even if in reality there is no effect of intercessory prayer on healing, 5% (or 2.5%) of the experiments are expected to provide results showing the effect, which are significant at the 5% level if a one-sided (or two-sided) alternative is used. Further, some journal editors may be inclined to publishing "surprising" findings, leading to publication or reporting bias.

In fact, later on researchers also published results of similar experiments when prayer did not appear to have an effect on complication-free recovery, see, e.g., Benson et al. [15]. Note, however, that these results may not be seen interesting or worth publishing in themselves, only after publications reporting the existence of an effect made them relevant.

We are not trying to find out what is the truth of the matter; simply note that sometimes the existence of an effect may be controversial and, in such cases, one may find including a causal path in the causal structure problematic, while in other cases this is quite straightforward.

[15] An apparent problem with this usage of tests of hypotheses is that these populations do not exist; consequently they have no population averages. One has to make a counterfactual assumption to associate a precise meaning with the test: there are two copies of the population of potential patients and members of one copy are being prayed for and members of the other copy are not being prayed for. Then one has—counterfactual—recovery times, which may or may not have the same expectations. Further—as always with experiments—one has to believe that the actual participants are proper samples from the respective populations.

Although Assumption 2 is a completeness assumption in the sense that it postulates that no existing causal effect among the variables considered remains unaccounted for, making this assumption does not guarantee that all variables that have a causal effect on a specific variable are included in the analysis. If such variables are left out, the results may be quite misleading.

6.3.6 Limitations of Interventions

Many cellular phones take screenshots if two designated buttons are pressed at the same time. This means that it is not enough to have both buttons in the pushed-down position at the same time; rather, they have to go from the not pushed-down position to the pushed-down position at the same time (say, within a half-second interval).

This is a clear and simple causal system, from the perspective of the user: the *two* buttons, say A and B, are the parents and the screenshot, say S, is the response. However, in this case, the status of the parents, being pushed down or not being pushed down, does not imply response. Response only occurs if A and B are pushed down at the same instance. This is another example of a case when there is joint effect but no individual effects.

The intervention distribution of A is deterministic, whether B is fixed in the down or not down position: if A is pressed, no screenshot is taken, and if A is not pressed, also no screenshot is taken, and similarly for B. If both A and B are manipulated, the experimenter—not knowing of the time sensitivity—may push them down one after the other, and no screenshot will be taken, either.

However, when the researcher observes the system, the owner of the phone—knowing of time sensitivity—will not only push down both A and B occasionally, but will push them down in all cases at the same time and a screenshot will be taken. Thus, in the observational distribution, pushing down both A and B leads to a screenshot being taken, but not necessarily in the intervention distribution, if the experimenter does not have the relevant information.

The example illustrates that Assumption 3 about the identical manipulation and observational distributions, when nondescendants, including all the parents, are manipulated or conditioned upon, is not necessarily true for all causal systems. A particular knowledge (time sensitivity, in this example) is needed to apply manipulations in such a way that Assumption 3 holds. Such knowledge may not always be available to the experimenter.

This also has relevance for the possibility of learning the causal system through intervention. The experimenter may not find out that the two buttons need to be pushed at the same time, and even if they do, there seems to be no way to describe this knowledge within the current theory.

Finally, this may also be seen as an example when the joint effect is different from the individual effects (when the two buttons are pushed down at the same time, screenshot is taken; when they are pushed down individually, no screenshot is taken).

6.4 Summary

This chapter has shown many useful applications of the graphical theory of causality and also illustrated several real situations, when causality is clearly present, but it is more or less outside of what DAGs and Assumptions 1–3 can describe.

Among the successful applications, Pearl's car example was presented and expanded, and three topics discussed in *LCDA* were revisited. In the case of the instrumental variable method (inferring the direction of the effect based on observational data), Assumptions 1–3 provided justification instead of the heuristic argument used before. The method of propensity score-based matching, which can be used to estimate the expected effect if a treatment is applied to the entire population, using only observational data, was also proved, with a few further assumptions, to be implied by Assumptions 1–3. In the case of Simpson's paradox, the causal structure among treatment, response, and a potential conditioning variable was used to decide, whether the marginal or the conditional direction of the effect was to be relied on, if these effects had different directions.

Among the limitations, examples were presented, where the concept of variable was not very useful to describe the real entities in the causal relationship and when the individual effects modeled by DAGs were of little relevance while when evaluated jointly, there was a deterministic causal effect. Further, all three assumptions made were shown to be not universally applicable. An example of cyclic causality was shown, and also one where the researcher was very unlikely to make relevant interventions because of a hidden feature, and thus observational and intervention distribution could not be expected to coincide. Perhaps the most disturbing example illustrated that even if a causal effect appears to be present, establishing a causal chain through which it takes place may be problematic.

In order to be able to better describe such situations, the theory would have to be able to account for the temporal aspect of causality and incorporate joint effects which are not implied by the component individual effects. Also, restricting the analysis of a causal problem to considering relatively few variables may be relevant in engineered environments, but may often only be seen as an approximation in the social, behavioral, and medical sciences.

6.5 Things to Do

1. Show that in Table 6.1, battery and gas are not conditionally independent given whether or not the car starts.
2. Find an example of a design that is neither experimental nor observational. Hint: a group of friends goes on a hike and they decide by majority vote which trail to take.
3. In Sect. 6.2.4, whether the effect of T on R should be assessed marginally or conditioned on C in case of the causal structures in Figs. 6.4, 6.5, and 6.6

was decided using particular realizations of these casual structures. In these realizations T had no causal effect on R. Generate all three $IPCM$s based on the DAGs obtained by omitting the arrow from T to R.

4. Can a causal structure describing the behavior of a self-regulatory system like the heating in a house be given? Let the heating system consist of the water boiler, the radiator in a room, and a thermostat measuring room temperature and regulating the operation of the boiler. How does the causal graph look like?
5. Draw a hypergraph expressing that in the alternating switches example the two switches together cause the light coming on or going off.
6. Give a full causal description of the alternating switches example completed with indicators for someone entering the room through one door or the other, and turning on the light.
7. Explain why both switches belong to the blocking set of the light bulb in the sense used in Sect. 4.4.3.
8. Modify the example in Sect. 6.3.4 so that each room has two switches, both of the push-button type. The first one turns on the light in that room and is operated automatically by one of the switches in the previous room, and the second one that switches off the light in that room and switches on the first switch in the next room, which then turns on the light in that room. Can now the system be described with a DAG?
9. May the first switches in the previous example be left out from the causal structure, by assuming that the second switches, when operated, turn on the light in the next room?
10. Find further examples when the researcher, although in the position of applying interventions, may not be able to find out what interventions may be relevant.
11. Find examples of causal systems where Assumption 1 does not hold.
12. Find examples of causal systems where Assumption 2 does not hold.
13. Find examples of causal systems where Assumption 3 does not hold.

Chapter 7
Marginal Log-Linear Parameterization



7.1 Introduction

This chapter defines marginal log-linear parameterizations of joint distributions of several categorical variables. These parameters are log-linear parameters calculated in various marginals of the contingency table. There is a great deal of flexibility as to what marginals to choose and what log-linear parameters to calculate, and the approach is appropriate to formulate parameterizations that are relevant in a large variety of statistical problems. Some of these examples are discussed in Sect. 7.2, including the definition of DAG models, models for repeated measurements, including panel studies, incomplete data (observed or by design), and distributions with specified marginal distributions; or data fusion.

Section 7.3 discusses marginal log-linear parameters and parameterizations. The models mentioned apply restrictions to marginal distributions of the joint distribution. Some of these are that certain marginal associations do not exist, thus the relevant marginal log-linear parameters are zero, while others assume equalities among certain parameters. The marginals, in which the marginal log-linear parameters are calculated, may, subject to a few simple rules, be selected freely and are determined based on the actual model to be defined. There is also flexibility as to what kind of log-linear parameters are used. In addition to the balanced and corner (effect) log-linear parameters related to the local and spanning cell odds ratios in non-binary tables, the use of global odds ratios is also discussed. Marginal log-linear parameters are determined not only for (strictly) positive probability distributions on a contingency table, but for any (strictly) positive distribution, including frequency distributions consisting of integers, but also general positive distributions, where positive real numbers are associated with the cells of the table. This generalization will become very important in the next two chapters, where smoothness properties of marginal log-linear parameters are investigated.

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It is proved that these marginal log-linear parameters provide a parameterization of the joint distribution, as long as the marginals in which they are calculated form a hierarchical and complete sequence and a variation independence property among some of the parameters is derived. Later chapters of the book will discuss the smoothness of these parameterizations and the properties of the models obtained by restricting them. It will also be shown later that when a statistical model is defined by restricting the values of some of the marginal log-linear parameters, then the distributions in the model are parameterized by the unrestricted marginal log-linear parameters.

Marginal log-linear parameterizations in this generality were introduced by Bergsma and Rudas [20], and Rudas and Bergsma [151] give an overview of many recent developments.

7.2 Motivation

In this section, several common data analytic situations are considered, which all point to the necessity of modeling the marginals of a joint distribution. The most important examples are DAG models. These models were introduced in Chap. 3 and used extensively in Chaps. 4, 5, and 6 to model causal structures. These models impose conditional independences on the marginals of a joint distribution. Other examples include the analysis of repeated measurements on the same subjects and the very common task of estimating the joint distribution of several variables based on incomplete observations.

7.2.1 DAG Models

As was discussed in Sect. 2.2.1, log-linear models may be identified with hypergraphs on the variables involved, in the sense that the generating class of the log-linear model (the maximal interactions allowed to be present among the variables) are the hyperedges of the hypergraph. In case when the generating class of the log-linear model does not have forbidden configurations (subsets of variables which are not interactions but all their subsets are interactions), then Theorem *LCDA* 11.3 stated that the corresponding hypergraph consists of the cliques (maximal complete subgraphs) of a graph. Such log-linear models are called graphical, and for every graph on the variables in the model, there is exactly one graphical log-linear model, namely, the one associated with its clique hypergraph, see Theorem 2.10. Further, belonging to the graphical log-linear model is equivalent to having a Gibbs factorization, and Theorem *LCDA* 11.6 was that this is equivalent to possessing a Markov property. The equivalence of the local, pairwise, and global Markov properties with each other and the Gibbs factorization was given in Theorem 2.8.

Therefore, graphical Markov models, in the undirected case, are log-linear models, and, in addition to the Markov properties, also the values of the log-linear parameters may be used to interpret distributions belonging to such a model, see Chapter *LCDA* 11. In fact, the following is true.

Theorem 7.1 *Let V be a set of variables, and*

$$V = \mathscr{A}_i \cup \mathscr{B}_i \cup \mathscr{C}_i$$

disjoint partitions for $i = 1, \ldots, k$. Suppose that there exists a (strictly) positive probability distribution P on V, such that under P

$$\mathscr{A}_i \perp\!\!\!\perp \mathscr{B}_i | \mathscr{C}_i, \tag{7.1}$$

for all $i = 1, \ldots, k$. Then, P belongs to the log-linear model, where the generating class consists of all subsets $W \subseteq V$, such that

$$W \neq \mathscr{A}_i' \cup \mathscr{B}_i' \cup \mathscr{C}_i,$$

for all $i = 1, \ldots, k$, where

$$\emptyset \neq \mathscr{A}_i' \subseteq \mathscr{A}_i$$

and

$$\emptyset \neq \mathscr{B}_i' \subseteq \mathscr{B}_i.$$

Proof The claim follows directly from Proposition *LCDA* 6.5 and Theorem *LCDA* 10.10.

□

Theorem 7.1 means that if one has a model defined through the conditional independences in (7.1), then this model is a log-linear model with the generating class given. Here, all conditional independences involve all the variables in V.

When a list of conditional independences is such that some of them involve only some of the variables, then Theorem 7.1 does not hold. To illustrate this, consider the DAG in Fig. 3.1. Using the directed local Markov property, the DAG implies the following conditional independences:

$$B \perp\!\!\!\perp CEF | A,$$

$$C \perp\!\!\!\perp BE | A,$$

$$D \perp\!\!\!\perp AEF | BC,$$

$$E \perp\!\!\!\perp BCD | A,$$

$$F \perp\!\!\!\perp ABD | CE.$$

Table 7.1 Conditional independences implied by the DAG in Fig. 3.1

Marginal	Conditional independence
$ABCE$	$C \perp\!\!\!\perp BE \mid A$
$ABCEF$	$B \perp\!\!\!\perp CEF \mid A$
$ABCDE$	$E \perp\!\!\!\perp BCD \mid A$
$ABCDEF$	$D \perp\!\!\!\perp AEF \mid BC,\ F \perp\!\!\!\perp ABD \mid CE$

The conditional independences implied by the directed graphical Markov model are summarized in Table 7.1.

The last two conditional independences are easily imposed by setting the log-linear parameters of the following effects equal to zero

$$ABCD,\ BCDE,\ BCDF,\ ABCDE,$$

$$ABCDF,\ BCDEF,\ ABCDEF$$

and

$$ACEF,\ BCEF,\ CDEF,\ ABCEF,$$

$$ACDEF,\ BCDEF,\ ABCDEF.$$

These restrictions define a log-linear model. But the other conditional independences, which apply to various (proper) subsets of the variables, cannot be imposed by setting log-linear parameters to zero and, thus, defining a log-linear model.

To see the simplest possible example, if one wished to impose

$$A \perp\!\!\!\perp B$$

in a $2 \times 2 \times 2$, $A \times B \times C$ table, then one would need to have:

$$\log(P(1,1,1) + P(1,1,2)) + \log(P(2,2,1) + P(2,2,2))$$

$$= \log(P(1,2,1) + P(1,2,2)) + \log(P(2,1,1) + P(2,1,2)).$$

Here

$$\log(P(i,j,1) + P(i,j,2))$$

$$= \log(\beta^{\emptyset}\beta_i^A\beta_j^B\beta_1^C\beta_{ij}^{AB}\beta_{i1}^{AC}\beta_{j1}^{BC}\beta_{ij1}^{ABC} + \beta^{\emptyset}\beta_i^A\beta_j^B\beta_2^C\beta_{ij}^{AB}\beta_{i2}^{AC}\beta_{j2}^{BC}\beta_{ij2}^{ABC})$$

$$= \log(\beta^{\emptyset}\beta_i^A\beta_j^B\beta_{ij}^{AB}(\beta_1^C\beta_{i1}^{AC}\beta_{j1}^{BC}\beta_{ij1}^{ABC} + \beta_2^C\beta_{i2}^{AC}\beta_{j2}^{BC}\beta_{ij2}^{ABC}))$$

$$= \lambda^{\emptyset} + \lambda_i^A + \lambda_j^B + \lambda_{ij}^{AB} + log(\beta_1^C \beta_{i1}^{AC} \beta_{j1}^{BC} \beta_{ij1}^{ABC} + \beta_2^C \beta_{i2}^{AC} \beta_{j2}^{BC} \beta_{ij2}^{ABC}))$$

$$= \lambda^{\emptyset} + \lambda_i^A + \lambda_j^B + \lambda_{ij}^{AB} + log(\beta_1^C \beta_{i1}^{AC} \beta_{j1}^{BC} \beta_{ij1}^{ABC} + \frac{1}{\beta_1^C \beta_{i1}^{AC} \beta_{j1}^{BC} \beta_{ij1}^{ABC}})),$$

and then the marginal independence would require

$$\lambda_{11}^{AB} + log(\beta_1^C \beta_{11}^{AC} \beta_{11}^{BC} \beta_{111}^{ABC} + \frac{1}{\beta_1^C \beta_{11}^{AC} \beta_{11}^{BC} \beta_{111}^{ABC}}))$$

$$+ \lambda_{22}^{AB} + \log(\beta_1^C \beta_{21}^{AC} \beta_{21}^{BC} \beta_{221}^{ABC} + \frac{1}{\beta_1^C \beta_{21}^{AC} \beta_{21}^{BC} \beta_{221}^{ABC}}))$$

$$- \lambda_{12}^{AB} + \log(\beta_1^C \beta_{11}^{AC} \beta_{21}^{BC} \beta_{121}^{ABC} + \frac{1}{\beta_1^C \beta_{11}^{AC} \beta_{21}^{BC} \beta_{121}^{ABC}}))$$

$$- \lambda_{21}^{AB} + \log(\beta_1^C \beta_{21}^{AC} \beta_{11}^{BC} \beta_{211}^{ABC} + \frac{1}{\beta_1^C \beta_{21}^{AC} \beta_{11}^{BC} \beta_{211}^{ABC}})) = 0.$$

This definition of the two-way marginal independence in a three-way table is far from the simplicity of the definition of log-linear models, where the models were obtained by first considering a parameterization of all distributions on the contingency table, and then by setting some of the parameters to zero. The remaining parameters provided a parameterization of all distributions in the model, and degrees of freedom were easily determined. For details see Chapters *LCDA* 10–12. If, however, one had a parameterization, which contained the log-linear parameters calculated in the AB marginal of the table, then the independence of A and B could be easily imposed. This is the idea to be developed in the sequel.

The marginal log-linear models, that are based on marginal log-linear parameterizations, provide a comparably simple way of defining, estimating, and testing models which impose conditional independences on marginal distributions, like the graphical models associated with directed acyclic graphs do.

But there are many other problems in data analysis that lead to similar structures.

7.2.2 Repeated Measurements and Panel Studies

Another group of statistical problems where marginal models provide a useful approach is that of repeated measurements. Such setups occur frequently in the context of measuring or testing the existence of the effect of some treatment. For example, before a particular language teaching method is applied to a class of students, their language proficiency is measured, and after the application of the teaching method, the language proficiency is tested again. Of course, ideally, one

would also have to have a control group, where the standard language teaching method is applied, and the members of that group are also tested before and after the language course.

Such procedures may measure several variables both before and after treatment. For example, the richness of the vocabulary (V) and the grammatical correctness (G) may be measured both before and after the treatment. A simple structure in this case is independence of V and G both before and after treatment.[1] It is straightforward to test independence both before and after treatment, but these tests are not independent, as the variables before and after the treatment are measured on the same individuals. Taken this into account, the testing problem may be seen as two hypotheses being tested on the same data and methods of multiple testing (see Section *LCDA* 11.2) are relevant.

If an index 1 denotes the variables before treatment, and an index 2 after treatment, then the hypothesis to be tested is

$$V_1 \perp\!\!\!\perp G_1 \text{ and } V_2 \perp\!\!\!\perp G_2. \tag{7.2}$$

Here, the two independences are defined on two-way marginals of the four-way table, so this may be seen as some kind of a marginal model.

Note that the model in (7.2) is not an undirected graphical model, as the independences do not pertain to all the variables and is also not a DAG model. If (7.2) was a DAG model, then none of the variables had parents, implying that the DAG is empty. The associated Markov property would be joint independence of all the variables (which is both a directed and an undirected graphical model) which is a much stronger assumption. That model would imply that V_1 and V_2 are independent, similarly G_1 and G_2, which is hardly true in reality. The model in (7.2) allows association or effect between V_1 and V_2 and also between G_1 and G_2.

Note that the model in (7.2) is an example of models of marginal independence briefly discussed in Sect. 7.2.1. They require certain log-linear parameters defined in marginals to be zero.

Marginal log-linear parameterizations and the resulting models, as defined here, provide a convenient tool for formulating such models.

An important variant of repeated measurements are panel studies, see, e.g., Frees and Kim [62]. In a panel study, the same sample of individuals is interviewed repeatedly over a period of time. For example, a sample of 1000 individuals is selected and then interviewed every month about their party preferences. When a new sample is used every month to estimate the distribution of party preferences, the design is called a cross-sectional study. When comparing these two designs, both have advantages and disadvantages.

The main advantage of the panel design is that transition probabilities may be directly observed. In the example above, the transition probabilities are the

[1] In a real research situation, it is usually more relevant to investigate the departure from this simple structure.

probabilities with which someone supporting party A in one month chooses party B the next month. Transition probabilities are important in modern politics, as they show where new supporters of a party are coming from and for which party do former supporters leave. In a panel study, the earlier party preferences of the respondents have been observed, while in a cross-sectional study, respondents are asked about previous and current preferences, which leads to additional error. The process of the changing party preferences is often a Markov chain, see Section *LCDA* 11.3.

Panel designs also have disadvantages when compared to cross-sectional studies. As time passes by, the individuals in the sample will tend to become too old, compared to the population, because after one year, the sample will be older by one year, but the average age of the population typically changes much less or remains the same. Also, members of the panel are usually more and more difficult to find as time passes by, and members drop out (called panel attrition), introducing a self-selection aspect, even if the original sample was randomly selected. Finally, the interest for and knowledge of current events and also the related opinions may be influenced if someone knows they would be interviewed about them.

In practice, various mixed designs are used to have the advantages and to avoid the disadvantages of panel designs. For example, for every new wave of interviewing, part of the current panel sample is retained and part is replaced by a new sample. This is called a rolling design.

An important model to analyze panel data is the assumption that it is a Markov chain. For example, if observations are available in time points $t = 1, \ldots, T$, and P_t denotes the distribution of the variable of interest (e.g., the preferred political party) at time point t, then the model of a two-step Markov chain is that

$$P_t | P_{t-1}, P_{t-2}, Pt - 3, \ldots, P_1 = P_t | P_{t-1}, P_{t-2}, \text{ for } t = 4, \ldots, T,$$

which is equivalent to

$$P_t \perp\!\!\!\perp P_1, \ldots, P_{t-3} | P_{t-1}, P_{t-2} \text{ for } t = 4, \ldots, T.$$

This means in the political party preference example that this month's preferences may depend on the last 2 months' preferences, but if these are given, the preferences in the earlier months have no additional effect.

This is a marginal model, as the conditional independence assumptions apply to the marginals of the first t observations, $t = 4, \ldots, T$.

7.2.3 Incomplete Data

Incomplete data occur in many statistical problems in various contexts. A complete data set is a matrix, where the rows are usually associated with the units observed (e.g., respondents in a survey, individuals in a registry containing those suffering

from a rare disease, countries which are members of an international organization) and the columns represent the variables on which data are collected. When an entry is missing from this matrix, the data are called incomplete. Incomplete data may arise as an observed phenomenon during the data collection procedure but may also be implied by design.

When the data matrix is incomplete because not all responses from all individuals targeted (those in the sample) were collected, one usually speaks of missing or incomplete data,[2] see Little and Rubin [105]. Missingness may occur for a number of reasons, including that some of the individuals who were planned to be interviewed were not found (no contact) or found but not ready to participate (unit nonresponse) or were found, participated but failed to answer some of the questions (item nonresponse). In the first two situations, the corresponding row in the data matrix is completely empty, and in the last case, only some of the entries are missing. Or, in a registry of traffic accidents, fewer circumstances may be recorded in some cases than in others.

Sometimes, such an incomplete data structure is implied by the design of the data collection procedure. When the questionnaire is deemed too long, different subsets of the respondents may be asked only (overlapping) parts of the questionnaire. The main reason for splitting up the questionnaire is the experience that respondents are usually less willing to participate in long surveys, and if they do, the precision of the responses may deteriorate with time. For example, the questions are divided into the following groups: demographic questions (A), job satisfaction questions (B), questions about the personal network (C), and questions about health (D). Then, the sample is divided into the disjoint parts of X, Y, Z, and individuals in X are asked questions A and B, those in part Y questions A and C, and those in part Z questions A and D. Such a design is called a split questionnaire (see, e.g., Rhemtulla and Little [136]), and it also leads to an incomplete data matrix.

If the data matrix is incomplete, whether because of nonresponse or by design, one may identify the subsets of respondents who answered the same questions, and the joint distribution of every such subset of the questions may be estimated. In the split questionnaire example, the A marginal of the $ABCD$ distribution may be estimated from all observations, the AB marginal from the observations X, the AC marginal from the observations Y, and the AD marginal from the observations Z. Then, one may wish to estimate the joint distribution of all variables using the marginal estimates.[3] In the previous problems, conditional independences were prescribed on various marginal distributions; in this one, various marginal distributions are estimated, and one may want to find a distribution with these marginals, or one that is optimal in some sense if there are many such distributions.

[2] Apparently, missing refers to the part of the data not observed, incomplete refers to the part observed, but the two names are used almost interchangeably.

[3] Whether such a procedure is possible indeed will be discussed in the next subsection.

When the marginal information stems from different data sources, the problem above is called data fusion (see, e.g., Cocchi [27]) or statistical matching[4] (see, e.g., D'Orazio et al. [46]).

The next subsection gives results for the existence of distributions with prescribed marginals.

7.2.4 Existence of Distributions with Specified Marginal Distributions

In the previous subsections, two types of problems were illustrated. In one, marginal distributions of a joint distribution were given, and the task was to generate a joint distribution with these marginal distributions. In the other one, conditional independences on marginals were prescribed and the task was similar: to generate and perhaps parameterize joint distributions with the given independences on the marginals.

With respect to the problem of given marginal distributions, an example given in Section *LCDA* 12.2 is directly relevant. Table *LCDA* 12.1 gave three 2×2 probability distributions. When interpreted as the AB, the AC, and the BC marginal distributions of an ABC joint distributions, these were weakly compatible. Weak compatibility here means that the marginal distributions of any variable, as implied by the two-way marginals containing it, are the same. In the example given, all variables had a uniform distribution. Clearly, weak compatibility is a necessary condition for the existence of a joint distribution with the given marginal distributions. If, from the AB marginal, A is uniform but from the AC marginal A is not uniformly distributed, then there cannot be a three-way table with the given two-way marginal distributions.

More formally, for a set of variables V, and for marginals $\mathscr{M}_i \subseteq V$, let the marginal distributions $P_{\mathscr{M}_i}$, $i = 1, \ldots, k$ be given. Let $\mathscr{A} \subseteq V$ be arbitrary and let for all $i = 1, \ldots, k$, for which $\mathscr{A} \subseteq \mathscr{M}_i$, $P_{\mathscr{M}_i}|\mathscr{A}$ denote the implied marginal distribution on $\mathscr{A}$. Then, the marginal distributions $P_{\mathscr{M}_i}$, $i = 1, \ldots, k$ are said to be weakly compatible, if for all $\mathscr{A} \subseteq V$,

$$P_{\mathscr{M}_i}|\mathscr{A} = P_{\mathscr{M}_j}|\mathscr{A}, \text{ if } \mathscr{A} \subseteq \mathscr{M}_i \cap \mathscr{M}_j, \ 1 \leq i, j \leq k.$$

The following is easy to see.

[4] The word matching is used here in a sense different from how it was used in the context of propensity score matching, and how it is sometimes used in the context of experiments. In those contexts, matching means finding a pair to a subject, with which comparison is made. For instance, treated and not-treated individuals, who are similar enough, e.g., according to their propensity scores, are matched and their responses are compared to find out the effect of treatment. In the context of incomplete data, matching means combining pieces that fit together.

Proposition 7.1 *Let P be a joint distribution of the variables V and let $\mathcal{M}_i \subseteq V$, $i = 1, \ldots, k$ be arbitrary subsets. Set*

$$P_{\mathcal{M}_i} = P|\mathcal{M}_i, \ i = 1, \ldots, k,$$

that is, the restrictions of P on the marginals. Then $P_{\mathcal{M}_i}$, $i = 1, \ldots, k$ are weakly compatible.

□

The marginal distributions $P_{\mathcal{M}_i}$, $i = 1, \ldots, k$ are called strongly compatible, if there exists a joint distribution P of the variables V, of which they are the marginal distributions, that is, such that

$$P_{\mathcal{M}_i} = P|\mathcal{M}_i, \ i = 1, \ldots, k.$$

Proposition 7.1 says that strong compatibility implies weak compatibility. The reverse implication does not hold, in general, as illustrated by the example given in Table *LCDA* 12.1.

If one tries to determine, under what conditions does the converse hold, then it is enough to consider classes of marginals which are incomparable, as weak compatibility implies that the distribution given on a smaller (contained) marginal is the one implied by the bigger (containing) marginal and thus an extension of the distribution on the bigger marginal is also an extension of the distribution on the smaller marginal.

In fact, weak compatibility implies strong compatibility for all possible marginal distributions, if the set of marginals is decomposable.

Decomposability in the context of a class of subsets of variables that constituted the generating class, that is, the maximal interactions allowed in a log-linear model, was defined in Section *LCDA* 11.3. As in hierarchical log-linear models, which were discussed in *LCDA*, a subset of an interaction is also an interaction, a generating class was assumed to contain only maximal elements. Accordingly, decomposability was defined in a way that it applied only to classes where all subsets were maximal with respect to inclusion. The definition said that the class needs to have fewer than three elements, or the following running intersection property has to hold for the elements of the class: there is an order, in which the maximal subsets are

$$\mathcal{M}_{m_1}, \ldots, \mathcal{M}_{m_k},$$

such that for every $m_3 \leq m_j \leq m_k$, there exists an $m_3 \leq m_i \leq m_j - 1$, such that

$$\left(\mathcal{M}_{m_1} \cup \ldots \cup \mathcal{M}_{m_j-1}\right) \cap \mathcal{M}_{m_j} = \mathcal{M}_{m_i} \cap \mathcal{M}_{m_j}. \tag{7.3}$$

Then, Theorem *LCDA* 12.4 directly implies the following.

Theorem 7.2 *Let the incomparable subsets $\mathscr{M}_1, \ldots \mathscr{M}_k$ be decomposable. Further, let the probability distributions $P_{\mathscr{M}_i}$, $i = 1, \ldots, k$ be weakly compatible. Then, $P_{\mathscr{M}_i}$, $i = 1, \ldots, k$ are also strongly compatible.*

□

Decomposability is not only sufficient for weak compatibility to imply strong compatibility for any distribution but is also necessary.

Theorem 7.3 *Suppose, the incomparable subsets of the variables $\mathscr{M}_1, \ldots \mathscr{M}_k$ are not decomposable. Then, there exist probability distributions $P_{\mathscr{M}_i}$, $i = 1, \ldots, k$ on them, which are weakly compatible but not strongly compatible.*

Proof The proof is a construction of weakly compatible distributions on a non-decomposable class of marginals, which are not strongly compatible, by embedding the example in Table *LCDA* 12.1 into an arbitrary non-decomposable class. The following construction applies to any ordering of the subsets, and yields a counterexample. Consider the subsets in a fixed order and the subset which is the first such that the running intersection property is violated. For simplicity of presentation, let this subset be $\mathscr{M}_o$, and denote the two earlier subsets with which it has nonempty intersections, which are together not parts of the intersection of $\mathscr{M}_o$ with any of the earlier subsets as $\mathscr{M}_m$ and $\mathscr{M}_n$.

The example to be constructed shows that there are weakly compatible distributions on the marginals $\mathscr{M}_m$, $\mathscr{M}_n$, and $\mathscr{M}_o$, which are not strongly compatible. Thus, these distributions cannot be extended to a joint distribution on the entire table, because that would imply an extension to $\mathscr{M}_m \cup \mathscr{M}_n \cup \mathscr{M}_o$, too.

Using the fact that categorical variables may be combined to yield another categorical variable, define the variable $\mathscr{A}$ as the combination of the variables in

$$(\mathscr{M}_m \cap \mathscr{M}_n) \setminus \mathscr{M}_o),$$

and, similarly, $\mathscr{B}$ as the combination of variables in

$$(\mathscr{M}_m \cap \mathscr{M}_o) \setminus \mathscr{M}_n),$$

and $\mathscr{C}$ as the combination of variables in

$$(\mathscr{M}_n \cap \mathscr{M}_o) \setminus \mathscr{M}_m).$$

Let $\mathscr{D}$ be the combination of the variables in the intersection of all three subsets. The sets $\mathscr{A}, \mathscr{B}, \mathscr{C}$ are not empty, but $\mathscr{D}$ may be empty. Then,

$$\mathscr{A}, \mathscr{B} \in \mathscr{M}_m, \ \mathscr{A}, \mathscr{C} \in \mathscr{M}_n, \ \mathscr{B}, \mathscr{C} \in \mathscr{M}_o.$$

Number the joint categories of $\mathscr{A}, \mathscr{B}$, and $\mathscr{C}$ as $1, 2, \ldots$.

Define now the joint distributions of $\mathscr{A} \times \mathscr{B}$, $\mathscr{A} \times \mathscr{C}$, and $\mathscr{B} \times \mathscr{C}$ so that all cells, where any of the indices is different from 1 or 2, have zero probability and the

cells (1, 1) and (2, 2) have the probability 0.05 and the cells (1, 2) and (2, 1) have the probability of 0.45. That is, the distributions given on the $\mathscr{A} \times \mathscr{B}$, $\mathscr{A} \times \mathscr{C}$, and $\mathscr{B} \times \mathscr{C}$ marginals are all concentrated to a 2×2 subtable, and there coincide with the distributions given in Table *LCDA* 12.1.

After the $\mathscr{A} \times \mathscr{B}$, $\mathscr{A} \times \mathscr{C}$, and $\mathscr{B} \times \mathscr{C}$ distributions were defined, the joint distributions of the variables in $\mathscr{M}_m$, $\mathscr{M}_n$, and $\mathscr{M}_o$, respectively, are specified. For $\mathscr{M}_m$, the $\mathscr{A} \times \mathscr{B}$ marginal is already given and the conditional distributions of the other variables, given $\mathscr{A} \times \mathscr{B}$, are uniform. The joint distributions of the other marginals are specified similarly, using $\mathscr{A} \times \mathscr{C}$ and $\mathscr{B} \times \mathscr{C}$, respectively.

This definition gives the same distribution for $\mathscr{D}$ in all three subsets. Indeed, if the variables in

$$\mathscr{M}_m \setminus (\mathscr{M}_n \cup \mathscr{M}_o)$$

are combined into $\mathscr{X}$, so that $\mathscr{M}_m = \mathscr{A} \cup \mathscr{B} \cup \mathscr{D} \cup \mathscr{X}$ is a disjoint union, then the conditional distribution

$$\mathscr{X} \times \mathscr{D} | \mathscr{A} \times \mathscr{B}$$

was set to uniform, and so is its marginal distribution

$$\mathscr{D} | \mathscr{A} \times \mathscr{B}$$

and similarly for the other two marginals. The distributions of $\mathscr{A} \cup \mathscr{B}$, $\mathscr{A} \cup \mathscr{C}$, $\mathscr{B} \cup \mathscr{C}$ are identical on their respective supports, which are the 2×2 subtables with indices 1 and 2. Thus, the $\mathscr{A} \times \mathscr{B} \times \mathscr{D}$, $\mathscr{A} \times \mathscr{C} \times \mathscr{D}$, $\mathscr{B} \times \mathscr{C} \times \mathscr{D}$ distributions are also identical, and weak compatibility of the distributions on $\mathscr{M}_m$, $\mathscr{M}_n$, $\mathscr{M}_o$ marginals holds.

Suppose now that there is a distribution, of which the $\mathscr{M}_m$, $\mathscr{M}_n$, and $\mathscr{M}_o$ marginal distributions are as prescribed and consider its $\mathscr{A} \times \mathscr{B} \times \mathscr{C}$ marginal distribution. Of this, the $\mathscr{A} \times \mathscr{B}$, $\mathscr{A} \times \mathscr{C}$, and $\mathscr{B} \times \mathscr{C}$ marginal distributions are as specified. As these distributions are concentrated to the $\{1, 2\}^2$ subtables, respectively, the distribution on $\mathscr{A} \times \mathscr{B} \times \mathscr{C}$ would be concentrated on the $\{1, 2\}^3$ subtable. But, as shown with the example in Table *LCDA* 12.1, all 8 cells of the $\{1, 2\}^3$ subtable would have probabilities not exceeding 0.05, which is not possible.

□

Note that if the class of marginals on which the distributions are given does not only contain incomparable subsets, rather some of the subsets are contained in others, then weak compatibility implies that if, for instance, $\mathscr{M}_i \subset \mathscr{M}_j$, then the distribution $P_{\mathscr{M}_j}$, if restricted to $\mathscr{M}_i$, yields the distribution $P_{\mathscr{M}_i}$. Consequently, if a distribution is an extension of $P_{\mathscr{M}_j}$, then it is also an extension of $P_{\mathscr{M}_i}$. This implies the following.

Corollary 7.1 *Any weakly compatible measures given on a class of marginals are strongly compatible if and only if the maximal ones from among the marginals are decomposable.*

□

With respect to the data fusion (or statistical matching) problem mentioned in Sect. 7.2.3, the previous result implies that if the maximal marginals of the class are not decomposable, then weak compatibility does not necessarily imply strong compatibility.

In such cases, one may estimate the joint distribution which maximizes some function of the likelihoods of the observed marginal distributions.

7.3 Marginal Log-Linear Parameterization

In this section, a flexible way of parameterizing a joint distribution is discussed. With these parameterizations, the various models discussed in Sect. 7.2 may be defined by setting some of the parameters to zero, and then, the remaining parameters provide a parameterization of all the distributions in the model. The parameters used are some kind of log-linear parameters, and the (strict) positivity of the probability distributions considered is assumed in the sequel.

To illustrate the fundamental idea, consider a $2 \times 2 \times 2$ contingency table, with variables A, B, C, and suppose one is interested in the model assuming (marginal) independence B and C. This is a very simple model, but it is not straightforward to say what restriction it implies for the log-linear parameters (see Chapter *LCDA* 10) or for their multiplicative form. This is because the independence assumption applies to a marginal. Indeed, with the notation of Section *LCDA* 10.3

$$P(i,j,k) = \beta^{\emptyset}\beta_i^A\beta_j^B\beta_k^C\beta_{ij}^{AB}\beta_{ik}^{AC}\beta_{jk}^{BC}\beta_{ijk}^{ABC},$$

and

$$P(+,j,k) = \sum_{i=1,2} \beta^{\emptyset}\beta_i^A\beta_j^B\beta_k^C\beta_{ij}^{AB}\beta_{ik}^{AC}\beta_{jk}^{BC}\beta_{ijk}^{ABC},$$

and

$$P(+,j,+) = \sum_{i,k=1,2} \beta^{\emptyset}\beta_i^A\beta_j^B\beta_k^C\beta_{ij}^{AB}\beta_{ik}^{AC}\beta_{jk}^{BC}\beta_{ijk}^{ABC},$$

and

$$P(+,+,k) = \sum_{i,j=1,2} \beta^{\emptyset}\beta_i^A\beta_j^B\beta_k^C\beta_{ij}^{AB}\beta_{ik}^{AC}\beta_{jk}^{BC}\beta_{ijk}^{ABC}.$$

Thus, using these, the model may be written as

$$\sum_{i=1,2} \beta^{\emptyset}\beta_i^A\beta_j^B\beta_k^C\beta_{ij}^{AB}\beta_{ik}^{AC}\beta_{jk}^{BC}\beta_{ijk}^{ABC} =$$

$$\left(\sum_{i,k=1,2} \beta^{\emptyset}\beta_i^A\beta_j^B\beta_k^C\beta_{ij}^{AB}\beta_{ik}^{AC}\beta_{jk}^{BC}\beta_{ijk}^{ABC}\right)$$

$$\times\left(\sum_{i,j=1,2} \beta^{\emptyset}\beta_i^A\beta_j^B\beta_k^C\beta_{ij}^{AB}\beta_{ik}^{AC}\beta_{jk}^{BC}\beta_{ijk}^{ABC}\right),$$

and some simplification is obtained as

$$\beta^{\emptyset}\beta_j^B\beta_k^C\beta_{jk}^{BC}\sum_{i=1,2} \beta_i^A\beta_{ij}^{AB}\beta_{ik}^{AC}\beta_{ijk}^{ABC} =$$

$$\left(\beta^{\emptyset}\beta_j^B\sum_{i,k=1,2} \beta_i^A\beta_k^C\beta_{ij}^{AB}\beta_{ik}^{AC}\beta_{jk}^{BC}\beta_{ijk}^{ABC}\right)$$

$$\times\left(\beta^{\emptyset}\beta_k^C\sum_{i,j=1,2} \beta_i^A\beta_j^B\beta_{ij}^{AB}\beta_{ik}^{AC}\beta_{jk}^{BC}\beta_{ijk}^{ABC}\right),$$

or

$$\beta_{jk}^{BC}\sum_{i=1,2} \beta_i^A\beta_{ij}^{AB}\beta_{ik}^{AC}\beta_{ijk}^{ABC} =$$

$$\left(\sum_{i,k=1,2} \beta_i^A\beta_k^C\beta_{ij}^{AB}\beta_{ik}^{AC}\beta_{jk}^{BC}\beta_{ijk}^{ABC}\right)$$

$$\times\left(\beta^{\emptyset}\sum_{i,j=1,2} \beta_i^A\beta_j^B\beta_{ij}^{AB}\beta_{ik}^{AC}\beta_{jk}^{BC}\beta_{ijk}^{ABC}\right),$$

but there does not seem to be a very simple way of writing the assumption[5].

[5] See also the related discussion at the end of Sect. 7.2.1. The two attempts to define marginal independence of two variables in a three-way table are related to the constraint and freedom specifications of the model, see Sect. 10.3.

If the model applies to the entire table, instead of a marginal of it, e.g., it is conditional independence A and B, given C, then it can be easily defined by assuming that

$$\beta_{ijk}^{ABC} = 1 \text{ and } \beta_{jk}^{AB} = 1.$$

It is not the conditional independence that leads to this easy definition, but rather the fact that it applies to the entire table. If the model is the independence of A and BC, then the model assumes that

$$\beta^{\emptyset}\beta_i^A\beta_j^B\beta_k^C\beta_{ij}^{AB}\beta_{ik}^{AC}\beta_{jk}^{BC}\beta_{ijk}^{ABC}$$

$$= \left(\sum_{i=1,2} \beta^{\emptyset}\beta_i^A\beta_j^B\beta_k^C\beta_{ij}^{AB}\beta_{ik}^{AC}\beta_{jk}^{BC}\beta_{ijk}^{ABC}\right)$$

$$\times \left(\sum_{j,k=1,2} \beta^{\emptyset}\beta_i^A\beta_j^B\beta_k^C\beta_{ij}^{AB}\beta_{ik}^{AC}\beta_{jk}^{BC}\beta_{ijk}^{ABC}\right).$$

This may be imposed by assuming that

$$\beta_{ij}^{AB} = \beta_{ik}^{AC} = \beta_{ijk}^{ABC} = 1.$$

Indeed, in this case the model assumption is that

$$\beta^{\emptyset}\beta_i^A\beta_j^B\beta_k^C\beta_{jk}^{BC}$$

$$= \left(\sum_{i=1,2} \beta^{\emptyset}\beta_i^A\beta_j^B\beta_k^C\beta_{jk}^{BC}\right)$$

$$\times \left(\sum_{j,k=1,2} \beta^{\emptyset}\beta_i^A\beta_j^B\beta_k^C\beta_{jk}^{BC}\right).$$

or that

$$\beta^{\emptyset}\beta_i^A\beta_j^B\beta_k^C\beta_{jk}^{BC}$$

$$= \left(\beta_j^B\beta_k^C\beta_{jk}^{BC}\sum_{i=1,2} \beta^{\emptyset}\beta_i^A\right)$$

$$\times \left(\beta^{\emptyset}\beta_i^A\sum_{j,k=1,2} \beta_j^B\beta_k^C\beta_{jk}^{BC}\right).$$

The right-hand side of this equation may be written as

$$\left(\beta^{\emptyset}\beta_i^A\beta_j^B\beta_k^C\beta_{jk}^{BC}\sum_{i=1,2}\beta^{\emptyset}\beta_i^A\right)\left(\sum_{j,k=1,2}\beta_j^B\beta_k^C\beta_{jk}^{BC}\right)$$

$$=\beta^{\emptyset}\beta_i^A\beta_j^B\beta_k^C\beta_{jk}^{BC}\left(\sum_{i=1,2}\beta^{\emptyset}\beta_i^A\right)\left(\sum_{j,k=1,2}\beta_j^B\beta_k^C\beta_{jk}^{BC}\right)$$

$$=\beta^{\emptyset}\beta_i^A\beta_j^B\beta_k^C\beta_{jk}^{BC}\left(\sum_{i,j,k=1,2}\beta^{\emptyset}\beta_i^A\beta_j^B\beta_k^C\beta_{jk}^{BC}\right)$$

$$=\beta^{\emptyset}\beta_i^A\beta_j^B\beta_k^C\beta_{jk}^{BC}\left(\sum_{i,j,k=1,2}P(i,j,k)\right),$$

and as the term in the parenthesis is equal to 1, the required independence is implied, indeed.

In general, one has the following result about the definition of independence using log-linear parameters. Recall that the λ log-linear parameters are the logarithms of the β multiplicative parameters, see Section *LCDA* 10.3.

Corollary 7.2 *Let V be the set of variables and $\mathscr{A}$, $\mathscr{B}$, $\mathscr{C}$ a disjoint partition of it. Then, for a (strictly) positive probability distribution on V,*

$$\mathscr{A} \perp\!\!\!\perp \mathscr{B}|\mathscr{C}$$

holds, if and only if for $\emptyset \neq W \subseteq V$,

$$\lambda_w^W = 0, \text{ for all category combinations } w \text{ of } W,$$
$$\text{whenever } W \cap \mathscr{A} \neq \emptyset \text{ and } W \cap \mathscr{B} \neq \emptyset.$$

Proof This follows immediately from Theorem 7.1.

□

Note that the above result also holds if any of the sets in the partition is empty.

As the examples in Sect. 7.2 illustrate, a simple way of defining and parameterizing models with conditional independence or other properties on the marginals would be useful. A general and flexible class of parameterizations was suggested in Bergsma and Rudas [20], which may be used for this task. These parameterizations are called marginal log-linear parameterizations and essentially consist of log-linear parameters calculated not on the entire table but rather on a collection of lower-dimensional marginals.

The general definition given applies not only to probability distributions on a contingency table but also to frequency distributions of integers associated with the cells of the table (like, for instance, observed frequencies) and also to arbitrary distributions, where real numbers are associated with the cells of the table (like, for instance, estimated frequencies). Thus, distribution, in the sequel, means any of the above, and when a result only applies to one type of them, it will be mentioned explicitly. This generality will be useful in the next two chapters, where smoothness properties of marginal log-linear and of other parameters will be discussed, and are possible, because the log-linear parameters may be defined for any (strictly) positive distribution, not only for probability distributions.

To define a marginal log-linear parameterization, consider a sequence of different marginals[6] $\mathscr{M}_1, \mathscr{M}_2, \ldots \mathscr{M}_k$ of the contingency table. A sequence of marginals is called hierarchical if

$$i < j \text{ implies } \mathscr{M}_i \not\supseteq \mathscr{M}_j, \ \ 1 \leq, i, j \leq k,$$

and is complete, if

$$\mathscr{M}_k = V.$$

Every hierarchical and complete sequence of marginals defines a marginal log-linear parameterization of the distributions on V. The flexibility of the approach comes from every hierarchical and complete sequence defining a different parameterization, and in different parameterizations, different models have straightforward definitions and parameterizations.

The marginal log-linear parameterization defined by the hierarchical and complete sequence of marginals $\mathscr{M}_1, \mathscr{M}_2, \ldots \mathscr{M}_k$ consists of the log-linear parameters of every subset of variables $W \subseteq V$. The log-linear parameters of the subset of variables W are calculated in the first marginal in the sequence $\mathscr{M}_1, \mathscr{M}_2, \ldots \mathscr{M}_k$, which contains W as a subset. For a subset W, let $i(W)$ be the index of the marginal which contains it first, that is,

$$W \subseteq \mathscr{M}_{i(W)} \text{ and } W \nsubseteq \mathscr{M}_j, \text{ for } j < i(W).$$

When the λ log-linear parameters (see Section *LCDA* 10.3) are used, as originally introduced in Bergsma and Rudas [20], then the notation is

$$\lambda_W^{\mathscr{M}_{i(W)}}$$

that is the marginal (subset of variables) in which the parameter is calculated, and the subset of variables (also called the effect), for which the parameter is calculated, are shown. Of course, this parameter has values for all category combinations of

[6] Note that the marginals are also considered as subsets of the variables.

the variables in W, and the notation above refers to all of these. When the marginal log-linear parameter for a particular category combination $(c)_W$ of the variables in W is meant, the notation

$$\lambda_W^{\mathscr{M}_{i(W)}}((c)_W)$$

is used, and these, for all possible choices of $(c)_W$, are called the components of the parameter.

To illustrate, for three variables A, B, and C, the following are examples of hierarchical and complete sequences of marginals:

$$ABC,$$

or

$$BC,\ ABC,$$

or

$$C,\ A,\ BC,\ ABC,$$

or

$$A,\ AB,\ BC,\ ABC.$$

If the sequence considered is BC, ABC, then the log-linear parameters for the subsets

$$\emptyset,\ B,\ C,\ BC$$

are calculated in the BC marginal, and the log-linear parameters for the subsets

$$A,\ AB\ AC,\ ABC$$

are calculated in the marginal ABC, that is, in the whole table.

Note that as V is a hierarchical and complete sequence of marginals, one can consider the implied marginal log-linear parameters, which are the ordinary log-linear parameters. Therefore, all results for marginal log-linear parameters, parameterizations, and models obtained by restrictions on these also apply to log-linear parameters, parameterizations, and models.

Marginal log-linear parameters generalize not only ordinary log-linear parameters but also the multivariate logistic parameters of Glonek and McCullagh [65] which are marginal log-linear parameters defined when the hierarchical and complete sequence of marginals consists of all subsets of V. In this case, every log-

linear parameter (except for the one belonging to the empty set) is defined in the marginal equal to it.

To motivate using marginal log-linear parameters to parameterize distributions, we note, before any general result is given, that it was illustrated at the beginning of this chapter, how difficult it is to impose the marginal independence of two variables say B and C in a three-way table, when log-linear parameters are used to describe the joint distribution. When the above marginal log-linear parameters are used, then the simple condition

$$\lambda_{BC}^{BC} = 0,$$

where the equality is understood to apply to all components of the parameter, imposes

$$B \perp\!\!\!\perp C$$

in a straightforward way.

To prove that marginal log-linear parameters constitute a parameterization of the joint distribution, a preliminary result about the standard log-linear parameters is needed.

Theorem 7.4 *Consider a set of variables V, and let P be their joint distribution.*[7] *Suppose that the marginal distributions of the subsets of variables $W_1, \ldots, W_l$ are given. If, in addition, the log-linear parameters of all subsets W not contained in any of these*

$$\lambda_W^V, \ W \subseteq V, \ W \nsubseteq W_i, \ \textit{for all } i = 1, \ldots, l$$

are also given, then P may be determined.

Proof Consider the class of subsets of V:

$$\mathscr{D} = \{W : W \subseteq W_i, \ \text{for some } i = 1, \ldots, l\}.$$

Then, $\mathscr{D}$ is a descending class of subsets of V. By assumption, all log-linear parameters of the subsets of the complement ascending class, say $\mathscr{A}$, are given.

Using the given log-linear parameters, the conditional odds ratios of the sets in $\mathscr{A}$ may be determined, when conditioning is always on a category combination of all variables not contained in the set. Indeed, for arbitrary $T \in \mathscr{A}$, when all the variables in T are binary with indices 1 and 2

$$COR(T|V \setminus T = (c)_{V\setminus T}) = \prod_{(c)_T} P((c)_T, (c)_{V\setminus T})^{i((c)_T)},$$

[7] Recall that distribution is used in the general sense, not only for probability distributions.

where $i((c)_T)$ is 1 or -1, depending on whether the number of 1's in $(c)_T$ has or does not have the same parity as the number of variables in T. Then write, using the multiplicative parameters

$$\beta = \exp(\lambda),$$

$$COR(T|V \setminus T = (c)_{V\setminus T}) = \prod_{(c)_T} \left(\prod_{U\subseteq V} \beta^U_{(c)_U} \right)^{i((c)_T)} . \tag{7.4}$$

$$= \prod_{(c)_T} \left(\left(\prod_{U\in\mathscr{D}} \beta^U_{(c)_U} \right)^{i((c)_T)} \left(\prod_{U\in\mathscr{A}} \beta^U_{(c)_U} \right)^{i((c)_T)} \right)$$

$$= \left(\prod_{U\in\mathscr{D}} \prod_{(c)_T} (\beta^U_{(c)_U})^{i((c)_T)} \right) \left(\prod_{U\in\mathscr{A}} \prod_{(c)_T} (\beta^U_{(c)_U})^{i((c)_T)} \right).$$

The first product in the expression above is equal to 1. To see this note that $T \in \mathscr{A}$ and $U \in \mathscr{D}$, thus, $T \setminus U \neq \emptyset$. For given T and U and fixed $(c)_U$, there are as many indices $(c)_{T\setminus U}$ which make $i((c)_T = 1$, as such that make $i((c)_T = -1$. Thus, $\beta^U_{(c)_U}$ appears in the numerator as many times as it appears in the denominator.

This implies that

$$COR(T|V \setminus T = (c)_{V\setminus T}) = \prod_{U\in\mathscr{A}} \prod_{(c)_T} (\beta^U_{(c)_U})^{i((c)_T)},$$

that is, the conditional odds ratios of the subsets of the variables in the ascending class may be determined using their log-linear parameters.

When not all the variables in T are binary, then the argument above needs to be applied to all (local or spanning cell) odds ratios, which are defined in binary subtables.

Finally, the claim of the theorem follows from the marginal distributions on a descending class and the conditional odds ratios on the complement ascending class constituting a mixed parameterization of all distributions, see Sections *LCDA* 7.2 and 10.1.

□

Then, one has the following result for marginal log-linear parameters.

Theorem 7.5 *Let $\mathscr{M}_1, \ldots, \mathscr{M}_k$ be a hierarchical and complete sequence of marginals, that is, subsets of the variables V. Then, the marginal log-linear parameters based on these marginals are a parameterization of all distributions on V.*

Proof The marginal log-linear parameters may be determined if the distribution P is given. One needs to see that the distribution may be reconstructed from the marginal log-linear parameters.

For the subsets of variables in marginal $\mathscr{M}_1$, all log-linear parameters are given, thus, the marginal distribution $P|\mathscr{M}_1$ may be determined.

The rest of the proof is by induction. Suppose that for a $2 \leq j \leq k$, the marginal distributions

$$P|\mathscr{M}_1, \ldots, P|\mathscr{M}_{j-1}$$

have been determined. Then, consider the descending class $\mathscr{D}$ with maximal elements equal to the maximal ones from:

$$\mathscr{M}_j \cap \mathscr{M}_1, \ldots, \mathscr{M}_j \cap \mathscr{M}_{j-1}.$$

The distributions of these subsets of variables are all given by the induction assumption, and also the log-linear parameters of the subsets which are in the ascending class which is the complement of $\mathscr{D}$ with respect to all subsets of $\mathscr{M}_j$.

Then, Theorem 7.4 implies that the distribution of $\mathscr{M}_j$ can be determined, which completes the induction argument and the proof.

□

Obviously, the above results hold not only for the λ balanced log-linear parameters but also for the effect (or corner) log-linear parameters, see Sections *LCDA* 10.2 and 11.3.

As the proof of Theorem 7.4 illustrated (see also the discussion in Section *LCDA* 10.3), the log-linear parameters are closely related to the conditional odds ratios and to their local or spanning cell generalizations in the case of non-binary variables. This is true for any kind of distribution. Marginal parameters and parameterizations may also be defined using other ways of describing the association structure of the contingency table, see the end of this section.

As the marginal log-linear parameters are a parameterization of the joint distribution, the following is obvious.

Proposition 7.2 *For a hierarchical sequence of marginals $\mathscr{M}_1, \ldots, \mathscr{M}_k$, suppose that for some $3 \leq j \leq k$*

$$\mathscr{M}_j \setminus \cup_{i<j} \mathscr{M}_i \neq \emptyset.$$

Then, the marginal log-linear parameters that are determined in the marginal $\mathscr{M}_j$ determine the conditional distribution of the variables in

$$\mathscr{M}_j \setminus \cup_{i<j} \mathscr{M}_i,$$

given the variables in

$$\cup_{i<j} (\mathscr{M}_j \cap \mathscr{M}_i).$$

□

In addition to the conditional distribution described in Proposition 7.2, the marginal log-linear parameters on the hierarchical sequence of marginals also yield what is needed to define the joint distribution of the variables:

$$\mathscr{M}_j \cap \left(\cup_{i<j} \mathscr{M}_i\right),$$

in addition to the distributions of the variables in

$$\mathscr{M}_j \cap \mathscr{M}_i, \ i < j, \tag{7.5}$$

that is, the (marginal) log-linear parameters on the ascending class which is complement to the descending class defined by the subsets (7.5), see Rudas and Bergsma [151]. This means, for example, that if $\mathscr{M}_1 = \{A\}$, $\mathscr{M}_2 = \{B\}$, $\mathscr{M}_3 = \{ABC\}$, then the marginal log-linear parameters yield not only the A and B marginal distributions but also the information needed to define the AB joint distribution.

To further study the properties of marginal log-linear parameters, a more general concept of decomposability is needed than the one considered so far, which also applies to marginals, some of which may be comparable with respect to inclusion, like the marginals in a hierarchical and complete sequence.

The more general concept is called ordered decomposability (see Bergsma and Rudas [20]), which is relevant with respect to the sequence of marginals in a marginal log-linear parameterization. The class of marginals (i.e., subsets of variables) $\mathscr{M}_1, \ldots, \mathscr{M}_k$ is ordered decomposable, if $k \leq 2$, or if there is an order of them

$$\mathscr{M}_{m_1}, \ldots, \mathscr{M}_{m_k},$$

which is hierarchical and such that for all $3 \leq l \leq k$, the maximal ones from among

$$\mathscr{M}_{m_1}, \ldots, \mathscr{M}_{m_l}$$

are decomposable.

The difference between the two definitions is that decomposability applies to classses of incomparable subsets ad requires the running intersection property to hold for an ordering, and ordered decomposability applies to classes of arbitrary subsets of variables and requires the running intersection property to hold for those which are maximal in each slice $\mathscr{M}_{m_1}, \ldots, \mathscr{M}_{m_k}$, $k = 3, \ldots, l$, for a hierarchical order.

Thus, for example, in the case of the class consisting of AB, AC, BC, ABC is not decomposable, and neither is ordered decomposable. It is not decomposable, because that definition does not apply to it, as its elements are not incomparable, and is not ordered decomposable, because in the order ABC, AB, AC, BC the running

intersection property holds, but this order is not hierarchical. In any hierarchical order, ABC is the last subset, and then the running intersection property does not hold for the first three subsets.

To illustrate further, to the class $\{ABD,\ ACE,\ BCF\}$, both definitions apply, but it is neither decomposable nor ordered decomposable because although any ordering is hierarchical, the running intersection property fails to hold. The class $\{ABCE,\ ABD,\ ACE,\ BCF\}$ is not decomposable but is ordered decomposable with the order $\{ACE,\ ABCE,\ ABD,\ BCF\}$.

The following is easy to see.

Proposition 7.3 *A class consisting of incomparable subsets of variables is ordered decomposable if and only if it is decomposable.*

□

The next results completes Proposition 7.2.

Theorem 7.6 *Let the sequence of marginals $\mathscr{M}_1, \ldots, \mathscr{M}_k$ be hierarchical and ordered decomposable in this order. Then, the marginal log-linear parameters*

$$\lambda_{\mathscr{E}}^{\mathscr{M}_j},\ \mathscr{E} \subseteq \mathscr{M}_j,\ \mathscr{E} \nsubseteq \mathscr{M}_i,\ i < j,\ \ j = 1, \ldots, k \tag{7.6}$$

are a one-to-one function of the conditional distributions of the variables in

$$\mathscr{M}_j \setminus \cup_{i<j} \mathscr{M}_i,$$

given the variables in

$$\cup_{i<j} (\mathscr{M}_j \cap \mathscr{M}_i)$$

Further, the marginal log-linear parameters in (7.6) are variation independent from all marginal log-linear parameters determined in earlier marginals.

Proof Ordered decomposability means that the intersection of $\mathscr{M}_j$ with all previous marginals is equal to its intersection with one of the earlier marginals, say $\mathscr{M}_{j'}$, with $j' < j$, for all $j = 1, \ldots, k$. By hierarchy, $\mathscr{M}_j \setminus \mathscr{M}_{j'}$ is not empty. Thus, in this case, Proposition 7.2 means that the marginal log-linear parameters determine the conditional distribution of the variables in

$$\mathscr{M}_j \setminus \mathscr{M}_{j'}, \tag{7.7}$$

given the distribution of the variables in

$$\mathscr{M}_{j'}. \tag{7.8}$$

To see the converse, that is, that the conditional distribution of (7.7), given the distribution of (7.8), determines the marginal log-linear parameters defined on $\mathscr{M}_j$,

use the argument in the proof of Theorem 7.4 in the reverse direction to see that the marginal log-linear parameters determined in $\mathscr{M}_j$ depend on the conditional odds ratios of the variables in $\mathscr{M}_j \setminus \mathscr{M}_{j'}$. The latter depend on the conditional distribution of the variables in $\mathscr{M}_j \setminus \mathscr{M}_{j'}$, given the variables in $\mathscr{M}_{j'}$.

The second claim of the theorem is implied by the first one, as the conditional distribution of the variables in $\mathscr{M}_j \setminus \mathscr{M}_{j'}$, given the variables in $\mathscr{M}_{j'}$, is variation independent of the latter.

□

The previous theorem is not true without the assumption of ordered decomposability. If the sequence of marginals is $AB,\ AC,\ BC$, which is a hierarchical order but not decomposable, then the λ_{BC}^{BC} parameter is not variation independent from the λ_{AB}^{AB} and λ_{AC}^{AC} parameters. This, again, is seen in the example given in Table *LCDA* 12.1. These parameters may be set (together with the λ_A^{AB}, λ_B^{AB} and λ_C^{AC} parameters to imply the marginal distributions given in Table *LCDA* 12.1, and then they cannot be the marginal log-linear parameters of a three-way distribution.

For the AB marginal, marginal log-linear parameters are standard log-linear parameters in the marginal table, and, using the formulas given in Section *LCDA* 10.3, one obtains that

$$\lambda_{\emptyset}^{AB} = \frac{2}{4}(\log(0.05) + \log(0.45)) = -1.90,$$

$$\lambda_A^{AB} = \left(\frac{1}{2}(\log(0.05) + \log(0.45)) - \lambda_{\emptyset}^{AB},\ \frac{1}{2}(\log(0.45) + \log(0.05)) - \lambda_{\emptyset}^{AB}\right)'$$

$$= (0,\ 0)',$$

$$\lambda_B^{AB} = \left(\frac{1}{2}(\log(0.05) + \log(0.45)) - \lambda_{\emptyset}^{AB},\ \frac{1}{2}(\log(0.45) + \log(0.05)) - \lambda_{\emptyset}^{AB}\right)'$$

$$= (0,\ 0)',$$

$$\lambda_{AB}^{AB} = (\log(0.05) - 0 - 0 + 1.90,\ \log(0.45) - 0 - 0 + 1.90,$$

$$\log(0.45) - 0 - 0 + 1.90,\ \log(0.05) - 0 - 0 + 1.90)'$$

$$= (-1.10,\ 1.10,\ 1.10,\ -1.10)'.$$

The marginal log-linear parameters in the AC and BC marginals are determined similarly. Although these marginal log-linear parameters imply the same one-way marginal distributions, they cannot be the marginal log-linear parameters determined from a joint distribution of the three variables with the given two-way marginals, as such does not exist, illustrating the lack of variation independence.

In addition to the local and spanning cell odds ratios, there are further variants of odds ratios, and their generalizations may be useful in understanding and modeling

the association structure of the data. In the case of ordinal variables, one of these is the global odds ratio. The basic idea is that in a two-way table, the column and the row of a cell divide the table into four rectangles, and the global odds ratio is calculated from the sums of probabilities or frequencies in these rectangles as if they were probabilities or frequencies in a 2×2 table.

As an example, suppose that a large corporation has salary bands A, B, C, D, E, F for those who are in managerial positions, with A being the lowest and F the highest salary band, and they also label the different units that the managers oversee as Team, Group, Section, DEpartment, and DIvision. The managers may be classified into a 6×5 table, and one is interested in measuring the strength of association between salary band and unit size. In this case, it is informative, to pick one category of each variable, say C and Se, and combine the observations in salary bands A, B, and C, and also those in salary bands D, E, and F, and also combine the observations in units T, G, and S, and also in DE and DI. The cell (C, S) is used as a cut point. The combination of the cells, also called lumping, associates a 2×2 table with the cell (C, S), and one may compute the odds ratio in this 2×2 lumped table, the global odds ratio.

For an $I \times J$ table, any cell (i, j) with $1 \leq i < I$ and $1 \leq j < J$ may be used as a cut point, thus, there are $(I-1)(J-1)$ global odds ratios.

As shown by Bartolucci et al. [11], these global and several related variants of odds ratios may be generalized to multidimensional tables and may also be used to build marginal log-linear parameterizations.

To define a global odds ratio in a k-way table, let the indices of the jth variable, $j = 1, \ldots, k$, be $1, \ldots, I_j$. For each variable, specify an index $1 \leq i_j^* < I_j$. Then, the joint index $(i_1^*, \ldots, i_k^*)$, the cut point, defines a k-way binary table derived from the original table.

Let, for every j, l_j be 1 or 2. Then, define:

$$S_j(1) = \{1, 2, \ldots, i_j^*\}$$

and

$$S_j(2) = \{i_j^* + 1, j_j^* + 2, \ldots, I_j\}.$$

The probabilities (and similarly the frequencies) in the lumped 2^k table are

$$P(l_1, l_2, \ldots, l_k) = \sum_{((i_1, i_2, \ldots, i_k) \in S_1(l_1) \times S_2(l_2) \times \cdots \times S_k(l_k)} P(i_1, i_2, \ldots, i_k)$$

The higher-order and conditional odds ratios calculated in the binary table are called global odds ratios of the original table.

The logarithms of these global odds ratios for all possible choices of $(i_1^*, \ldots, i_k^*)$ were used in Bartolucci et al. [11] as a description of the association structure of the joint distribution and marginal parameterizations using these were developed,

adding further flexibility to the marginal approach to analyzing contingency tables. The models obtained by restricting these and other related parameters are called hierarchical marginal models.

7.4 Summary

Many statistical problems require the analysis of models imposing restrictions on lower-dimensional marginals of a joint distribution. The examples discussed in this chapter included Markov models associated with directed acyclic graphs, repeated measurements, and missing data. In the case of DAG models, the conditional independence statements of the Markov property apply to marginals consisting of variables and their respective parents in the graph. As was shown, while conditional independences applying to the entire table are straightforward to describe with log-linear parameters, there appears to be no easy way to define conditional independences applying to marginals. Similar problems arise in the definition of models that are relevant with repeated measurements or panel studies.

Dealing with missing data and data fusion often requires building up a joint distribution from a class of marginal distributions. It was shown that whether or not this is possible depends on the structure of marginal distributions given. More precisely, weak compatibility of the marginal distributions implies strong compatibility if and and only if the marginals are decomposable.

To be able to formulate marginal restrictions on joint distributions, marginal log-linear parameters were introduced. These are log-linear parameters calculated from marginal distributions. If the marginals used are in a hierarchical order, and the last marginal is the entire table, further, the log-linear parameter of every effect is calculated in the first marginal which contains it, then the marginal log-linear parameters are a parameterization of the joint distribution. When the class of marginals used is ordered decomposable, then there is a variation independence between the parameters calculated in different marginals.

In Chap. 9, smoothness properties of marginal log-linear parameterizations are proved, and in Chap. 10, models defined by restricting the values of some of the marginal log-linear parameters are investigated, which also include Markov models associated with DAGs. Chapter 8 deals with the concept of smoothness of parameterizations and of models defined by restricting values of some of the parameters.

7.5 Things to Do

1. Show that a graphical model associated with an empty graph, both in the undirected and directed sense, is equivalent to joint independence.
2. Why is self-selection of members of a sample problematic?

3. How does nonresponse relate to self-selection?
4. Prove Proposition 7.1.
5. Prove Theorem 7.2.
6. Prove that the sets of variables $\mathscr{A}$, $\mathscr{B}$, $\mathscr{C}$ in the proof of Theorem 7.3 are not empty.
7. Prove Corollary 7.1.
8. Show that for any three variables, the conditional independence of B and C, given A, is equivalent to $\beta_{ijk}^{ABC} = 1$ and $\beta_{jk}^{BC} = 1$.
9. Prove that Corollary 7.2 holds if any of $\mathscr{A}$, $\mathscr{B}$, or $\mathscr{C}$ is empty.
10. In a two-way table, determine local odds ratios from the global odds ratios and vice versa.
11. Prove Proposition 7.2.
12. Prove Proposition 7.3.
13. Work out the details of the example given after Proposition 7.3.
14. Work out the details of the proof of Theorem 7.6.
15. Determine the marginal log-linear parameters as described in the example after Theorem 7.6.
16. For an $I \times J$ table, consider all the lumped odds ratios for cut points (i, j), with $1 \leq i < I$ and $1 \leq j < J$. Are these variation independent from the one-way marginal distributions?
17. Do the lumped odds ratios considered above, together with the one-way marginal distributions, provide a parameterization of the joint distribution on the $I \times J$ table?

Chapter 8
Smoothness of Parameterizations of Discrete Distributions



8.1 Introduction

A parameterization of a family of distributions is smooth, if it is one-to-one between open sets, the parameterization and its inverse (the parameter) are twice continuously differentiable, and the Jacobian matrices consisting of the partial derivatives are of full rank over their entire domains. In a more intuitive form, this means that the parameterization may be well approximated locally by a linear transformation. If a family has a smooth parameterization, it is called a smooth manifold or, simply, smooth. Smooth manifolds have their own dimensions, depending on the characteristics of their parameterizations, and are often subsets of higher-dimensional spaces. The smoothness of a statistical model is a highly desirable property for two main reasons. First, it implies the applicability of standard asymptotic theory and, second, when the parameterization is smooth, the parameter values provide a proper description of the distributions in the model: the parameter space and the distributions in the model are isomorphic, at least locally.

This chapter starts by introducing the concepts of smoothness and that of smooth manifolds. In Sect. 8.2, linear parameterizations and their ranks are discussed in detail, where smoothness is easily established and is used later for the logarithms of exponential families. It is also shown that under variation independence of two groups of parameters of a smooth family, when the values of one group are fixed, the resulting family is also smooth, which is an important way of defining statistical models with desirable properties. For example, this implies that every log-linear model is a smooth manifold.

Section 8.3 discusses the linear parameterization of the logarithms of all general or probability distributions with log-linear parameter, including the balanced and the corner (or effect) log-linear parameters. For the non-redundant log-linear parameters, the respective full-rank design matrices are given.

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The linear parameterization of the logarithms of all distributions is essentially an exponential family representation, and Sect. 8.4 provides variants of the parameterization that are of full rank and take into account that in the case of probability distributions, the probabilities have to sum to 1, and thus the dimension of the manifold is one less than the number of the cells of the sample space. Reparameterizations connecting the overall effect and normalization are presented, and, finally, full-rank exponential family representations are developed for both general and probability distributions

The chapter closes with a brief summary and list of problems in Sect. 8.5 and Sect. 8.6.

8.2 Smooth Manifolds

Smoothness of a parameterization, in particular in the context of the statistical models obtained by restricting the values of some of the parameters, implies many desirable characteristics. Smoothness is a necessary condition for the applicability of standard large sample theory, including the asymptotic chi-squared distribution of certain test statistics and also for the relevance of the concept of degree of freedom associated with them. For simple cases, see Chapters *LCDA* 5, 11, 12, and 13.

Heuristically, smoothness means that small changes within the parameter space are possible in any direction, these changes may be made irrespective of each other and the number of directions in which the small changes are possible is the same in the entire sample space,[1] and these small changes imply small changes in the resulting distributions. When smoothness holds, the dimension of the parameter space is constant and is equal to the dimension of the distributions that are parameterized. Further, smooth parameters may be interpreted instead of the distributions themselves. Two distributions are close, if their parameters are close. The existence of smooth parameterizations plays a central role in understanding the important characteristics of marginal parameterizations defined in Chap. 7 and of marginal models to be defined in the sequel.

Recall that any function defined on the distributions (all that are relevant for a sample space or possibly those constituting a model) is a parameter and a parameter is called a parameterization, if it is invertible on its range. The concepts of parameter and parameterization were extended to any distribution in Sect. 7.3 from the original concept that applied to probability distributions only. The term distribution is used for any collection of real numbers associated with the elements of a sample space (most often cells of a contingency table). The observed frequency distribution consists of integers, but an estimated frequency distribution, e.g., according to the principle of maximum likelihood, typically consists of non-integer values.

[1] In order to make the number of these directions finite, one may assume that the directions are perpendicular to each other.

Probability distributions are also distributions, obtained if the sum of the entries associated with the cells of the sample space is made or happens to be equal to 1.

Often in practice, it is not parameters that are defined first; rather functions of parameters are used to define the distributions in the statistical model of interest. For example, in the case of a 2×2 contingency table, one may assume—based on substantive knowledge—that the joint probability distribution of the variables is such that the two marginal distributions are independent. In that case, two parameters, say α and β, both between 0 and 1 may be selected to parameterize the joint distribution, so that $p_{11} = \alpha\beta$, $p_{12} = \alpha(1-\beta)$, $p_{21} = (1-\alpha)\beta$, and $p_{22} = (1-\alpha)(1-\beta)$. This usage is in slight contrast with the definition above, which starts with a parameter and calls it a parameterization, if it is invertible. Here, first a parameterization is given, that is, the inverse of a parameter in the sense above. The parameters are, of course, obtained as $\alpha = p_{1+}$ and $\beta = p_{+1}$, in this case.

Parameters are useful if they capture relevant properties of the distributions in the model. If the model consists, e.g., of the univariate normal distributions, then the parameter with the mean and variance as components captures properties that are relevant. There are several possible parameters that may be useful or relevant in different applications. In the context of categorical data, often used parameters include various conditional odds ratios, conditional logits, marginal distributions, or log-linear parameters (see Section *LCDA* 10.1).

Parameters do not necessarily give a parameterization that are not necessarily invertible. If there are three categories in the population, with respective probabilities p_1, p_2, and p_3, then $p_1 + p_2$ is a parameter but is not a parameterization. Also, $(p_1+p_2, |p_1-p_2|)'$ is a parameter, but not a parameterization. An interesting feature of this example is that given $p_1 + p_2$, the parameter $|p_1 - p_2|$ is not redundant in the sense that it would be implied by the first parameter, yet it does not lead to a parameterization. It provides additional information, but not enough to fully determine the distribution. Of course, $(p_1 + p_2, p_1 - p_2)'$ is a parameterization.

When a parameter is a parameterization, that is, if a possible value of it is attained by only one distribution in the model, then it characterizes that distribution. For instance, in the case of 2×2 contingency tables, the two marginal distributions and the odds ratio characterize the distribution, meaning that they contain all the information in it. Parameterizations not only contain all the information in the distributions but also structure it in a useful or more easily understandable way. Of course, it depends on the actual application, what kind of structuring that is, the use of what parameters gives useful insight.

Parameterizations are also useful in associating a dimension with a statistical model, which is of central importance, when, as mentioned above, the deviation between the observed and the estimated distributions is assessed. In the context of categorical data in an asymptotic setting, this is usually done using the likelihood ratio or the Pearson chi-squared or other statistics, with an asymptotic chi-squared distribution (see Section *LCDA* 5.2 and for a more detailed exposition Bishop et al. [23] or Read and Cressie [140]). In this case, the dimension of the model for a contingency table of a particular size is characterized by the number of degrees of freedom. More precisely, the number of (residual) degrees of freedom is the

difference between the dimension of the unrestricted distributions on the table and the dimension of the model of interest. For example, in a 2×2 table, as will be seen later, the dimension of all unrestricted probability distributions is 3, and the dimension of the ones where the two marginal distributions are independent is 2, thus, the (residual) degree of freedom associated with the model of independence is 1.

A first notion of dimension may be obtained by comparing two subsets of $\mathbb{R}^2$. The first one is $\{(x, y)\}$, with $0 < x, y < 1$, the open unit square, and the other one is $\{(x, y)\}$ with $0 < x < 1$ and $y = x^2$. Although both of these are subsets of $\mathbb{R}^2$, and neither is a subset of $\mathbb{R}^1$, it is only the first one which is two-dimensional and the second one is one-dimensional in a certain sense. The latter one is a line, more precisely a one-dimensional curve in $\mathbb{R}^2$. As $y = x^2$ is a function that is differentiable, it can be well approximated locally by a (one-dimensional) straight line. Differentiability of $y = x^2$ means, as described in many books on calculus (see, e.g., Scherbert and Bartle [160]), that $|x^2 - (x_0^2 + 2x_0(x - x_0)|$ may be made arbitrarily small, if x is close enough to x_0, where $2x_0$ is the derivative of x^2 evaluated as x_0. The curve $\{(x, x^2) : x \in (0, 1)\}$ is a very simple example of a smooth one-dimensional manifold in the two-dimensional space.

In general, a subset $\mathcal{V} \subseteq \mathbb{R}^k$ is an l-dimensional smooth manifold ($l \leq k$), if (see, e.g., Oprea [125]):

(i) there exist a $\mathcal{U} \subseteq \mathbb{R}^l$ open set, and open sets $\mathcal{U}_u \subseteq \mathbb{R}^l$ for every $u \in \mathcal{U}$, such that,
(ii) there exist functions $f_u : \mathcal{U}_u \to \mathbb{R}^k$ for all $u \in \mathcal{U}$, such that f_u is one-to-one, f_u and its inverse (on $f_u(\mathcal{U}_u)$) are infinitely many times differentiable, further
(iii) such that

$$\cup_{u \in \mathcal{U}} f_u(\mathcal{U}_u) \supseteq \mathcal{V}$$

and
(iv) the vectors consisting of the partial derivatives

$$\left(\frac{\partial f_u(u)_1}{\partial u_i}, \ldots, \frac{\partial f_u(u)_k}{\partial u_i} \right)'$$

for $i = 1, \ldots, l$ are linearly independent,[2] and finally
(v) if for $u, t \in \mathcal{U}$, $f_u(\mathcal{U}_u) \cap f_t(\mathcal{U}_t) \neq \emptyset$, then the function

$$f_t^{-1} \circ f_u : f_u^{-1}(f_u(\mathcal{U}_u) \cap f_t(\mathcal{U}_t)) \to f_t^{-1}(f_u(\mathcal{U}_u) \cap f_t(\mathcal{U}_t)),$$

where $\circ$ denotes the composite function, has differentiable coordinate functions.

[2] The somewhat compact notation means the partial derivatives of the k coordinate functions of f_u according to the different components of the vector $u \in \mathbb{R}^l$, evaluated at u.

The definition may be interpreted as saying that $\mathscr{V}$ is an l-dimensional smooth manifold in $\mathbb{R}^k$, if it is covered by the $f_u(\mathscr{U}_u)$ images of open sets $\mathscr{U}_u$ through very smooth functions f_u (very smooth meaning that both f_u and its inverse are infinitely many times differentiable) with a full rank partial derivative matrix,[3] and with the property that if there is an overlap between $f_u(\mathscr{U}_u)$ and $f_t(\mathscr{U}_t)$, then $f_u(\mathscr{U}_u)$ and $f_t(\mathscr{U}_t)$ connect smoothly.

If the assumptions above hold, then l is the dimension of the smooth manifold. Manifolds that are one-dimensional are called curves, and those that are two-dimensional are called surfaces in a higher-dimensional Euclidean space.

For example, in the case of a contingency table with $I = 2$ rows and $J = 3$ columns, the model of independence of the two marginal distributions applied to arbitrary (strictly) positive distributions[4] on the table is an $I+J-1 = 4$ dimensional smooth manifold in the $IJ = 6-$dimensional Euclidean space. In the notation of the definition, $\mathscr{V} \subseteq \mathbb{R}^6$ is the collection of those points in the six-dimensional Euclidean space, where the six components are the frequencies[5] of a (strictly) positive distributions. This $\mathscr{V}$ is a smooth manifold. For (i), $\mathscr{U}$ and $\mathscr{U}_u$ for $u \in \mathscr{U}$ may be all selected to be the subset of $\mathbb{R}^4$ defined as

$$\mathscr{U} = \{u \in \mathbb{R}^4 : u_l > 0,\ 1 \le l \le 4,\ \sum_{l=3}^{4} u_l < \sum_{l=1}^{2} u_l\},$$

which is an open set. To make the notation more intuitive, let us denote u_1 as r_1, u_2 as r_2, u_3 as c_1, and u_4 as c_2, for rows and columns. The inequality in the definition of $\mathscr{U}$ above assures that the sums of the row and of the column marginals are the same, with an implied and positive third column marginal equal to $r_1+r_2-(c_1+c_2)$.

For (ii), let there be a single $f_u = f$ with components:

$$f(i, j) = \frac{r_i c_j}{r_1 + r_2},\ \text{if } 1 \le i \le 2,\ 1 \le j \le 2,$$

and

$$f(i, 3) = \frac{r_i(r_1 + r_2 - (c_1 + c_2))}{r_1 + r_2},\ \text{for } i = 1, 2,$$

where $r_1 + r_2$ is the total of the frequencies. Then, (iii) holds with equality. The partial derivatives in (iv), using a lexicographic order of the cells to arrange the coordinates in the six-dimensional space as

[3] This condition will be of central importance in the applications considered and will be discussed in detail.

[4] Out of the definitions of independence, the one requiring the local or the spanning cell odds ratios being equal to 1, applies to general distributions, as well.

[5] For simplification of the presentation, the entries of a general distribution associated with the cells of contingency table will sometimes be called frequencies.

$$(1,1),\ (1,2),\ (1,3),\ (2,1),\ (2,2),\ (2,3),$$

and order of the parameters r_1, r_2, c_1, c_2, written as a 6×4 matrix, omitting the division of each derivative by $(r_1 + r_2)^2$ is the following:

$$\begin{pmatrix} r_2c_1 & -r_1c_1 & r_1(r_1+r_2) & 0 \\ r_2c_2 & -r_1c_2 & 0 & r_1(r_1+r_2) \\ (r_1+r_2)^2 - r_2(c_1+c_2) & r_1(c_1+c_2) & -r_1(r_1+r_2) & -r_1(r_1+r_2) \\ -r_2c_1 & r_1c_1 & r_2(r_1+r_2) & 0 \\ -r_2c_2 & r_1c_2 & 0 & r_2(r_1+r_2) \\ r_2(c_1+c_2) & (r_1+r_2)^2 - r_1(c_1+c_2) & -r_2(r_1+r_2) & -r_2(r_1+r_2) \end{pmatrix}.$$

This matrix, containing the partial derivatives, is called the Jacobian[6] and is easily seen to be of full (column) rank. Indeed, suppose there are coefficients α, β, γ, δ, not all of them equal to zero, so that the linear combination of the column vectors with these coefficients is a vector of zeros. If this was true, then this would apply to the sum of the first three rows, as well. This sum vector is

$$((r_1 + r_2)^2, 0, 0, 0),$$

thus $\alpha = 0$. A similar argument for the last three rows shows that $\beta = 0$. Then, because of the zeros in the last two columns, to have a zero linear combination, one needs to have $\gamma = 0$ and $\delta = 0$, so all coefficients would have to be equal to zero.

To see that (v) holds, too, note that $\mathscr{U}_t = \mathscr{U}_u$ and $f_t = f_u$.

If one considers (strictly) positive probability, rather than general, distributions on the 2×3 contingency table, then one needs three parameters to define independence. The parameter u_1 may be identified with the first row probability, r_1, and u_2 and u_3 with c_1 and c_2, the first and second column probabilities. The ranges are $0 < r_1 < 1$, $0 < c_1, c_2$ and $c_1 + c_2 < 1$. With these parameters, the parameterization is

$$P(1,1) = r_1c_1,\ \ P(1,2) = r_1c_2,\ \ P(1,3) = r_1(1-c_1-c_2),$$

$$P(2,1) = (1-r_1)c_1,\ \ P(2,2) = (1-r_1)c_2,\ \ P(2,3) = (1-r_1)(1-c_1-c_2).$$

In this case, the partial derivatives arranged in a 6×3 matrix, with the columns containing the partial derivatives according to r_1, c_1 and c_2, respectively, are

[6] Some authors call the determinant of this matrix the Jacobian.

$$\begin{pmatrix} c_1 & r_1 & 0 \\ c_2 & 0 & r_1 \\ 1-c_1-c_2 & -r_1 & -r_1 \\ -c_1 & 1-r_1 & 0 \\ -c_2 & 0 & 1-r_1 \\ c_1+c_2-1 & r_1-1 & r_1-1 \end{pmatrix}.$$

To see that no nonzero coefficients α, β, γ can make a linear combination of the column vectors equal to a vector of zeros, consider the sum of the first three rows, which is

$$(1, 0, 0),$$

and if the linear combination gave a vector of zeros, then $\alpha 1 + \beta 0 + \gamma 0$ would give 0, as well, thus, $\alpha = 0$ would follow. Using $\alpha = 0$, the sum of the 2nd and 3rd rows would be

$$-\beta r_1 + \gamma 0,$$

which, if equal to zero, implies that $\beta = 0$. Finally, the 2nd row implies that $\gamma = 0$, too.

The other requirements hold similarly to the case of general frequencies.

Thus, the collection of all independent distributions on a 2×3 contingency table is a smooth manifold of dimension 4, and the collection of all independent probability distributions on a 2×3 table is a smooth manifold of dimension 3. Both of these lower-dimensional smooth manifolds are subsets of the six-dimensional Euclidean space.

In fact, all the probability distributions on the 2×3 table are also a smooth manifold. To understand the dimension of this manifold, first the simplest possible cases are considered that give further insight into the concept of a smooth manifold. If there are only two categories, then a probability distribution (p_1, p_2) may be identified with a point in the two-dimensional Euclidean space, with coordinates p_1 and p_2. Do all these points yield a two-dimensional set? As $p_1 + p_2 = 1$, the collection of all the points with coordinates (p_1, p_2) gives a line segment connecting $(0, 1)$ with $(1, 0)$, not containing these points if the probabilities are assumed to be (strictly) positive. In this case, the line segment is a one-dimensional object, a one-dimensional smooth manifold. If there are three categories, the three probabilities may be illustrated in the three-dimensional Euclidean space, but all these points lie in a triangle connecting the points $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$. And this triangle is a two-dimensional object, a smooth manifold. Similarly, for four categories, if represented in the four-dimensional space, the probability distributions give a tetrahedron, spanned by the four points which have one 1 and three 0s among their coordinates, and this is a three-dimensional smooth manifold. These objects are called the probability simplexes of the given dimensions.

When there are c categories, as the simple examples above indicate, all c probabilities do not give a c-dimensional smooth manifold. Indeed, assume that $P = P(\phi)$ is a parameterization. As

$$p_c = 1 - \sum_{i=1}^{c-1} p_i,$$

one also has, for any component ϕ_i of ϕ that

$$\frac{\partial p_c}{\partial \phi_i} = -\sum_{i=1}^{c-1} \frac{\partial p_i}{\partial \phi_i},$$

thus the vectors of partial derivatives in condition (iv) above are not linearly independent. Thus, the c probabilities are not a c-dimensional smooth manifold. Rather, they are a $c-1$-dimensional smooth manifold in the c-dimensional space. To see this, choose $p_1, \ldots, p_{c-1}$ as the parameters, and obtain p_c as $1-(p_1+\ldots+p_{c-1})$. Then the Jacobian, written as a $c \times (c-1)$ matrix, has the structure of a $(c-1) \times (c-1)$ identity matrix in the first $c-1$ rows and its last row has -1 in every position.

The c rows of this Jacobian sum to zero, of course, but the $c-1$ columns are linearly independent. Indeed, if one considers the first $c-1$ rows only, then, for every column, a 1 is found only in a single row, thus, no column can be a linear combination of other columns. This gives the following.[7]

Proposition 8.1 *The collection of (strictly) positive probability distributions with c categories is a smooth manifold of dimension* $c-1$.

□

Thus, in the case of a 2×3 table, the unrestricted (strictly) positive probability distributions have dimension 5, and when independence is assumed, the dimension is 3. The difference is the number of parameters fixed by the model, the two local odds ratios. See the related discussion in Section *LCDA* 5.4.

The (strictly) positive general distributions on the 2×3 table are also a smooth manifold, namely, the positive orthant of $\mathbb{R}^6$. This has dimension 6 (a formal argument would use the identity mapping on $\mathbb{R}^6$) and that the positive orthant is an open set. The model of independence is a smooth manifold of dimension 4, thus, the number of (residual) degrees of freedom is 2, also in this case, equal to the number of restrictions defining the model of independence.

In statistical analysis, the models of interest are often smooth manifolds. In all the applications considered in this book, manifolds will be defined, just like in the examples discussed, in such a way that in the definition above, $\mathscr{U}_u$ may be selected

[7] Here, and often in the sequel, for smoothness results only the Jacobian is proved to be of full rank, as the other conditions are straightforward.

to not depend on u, thus, there is a single function $f = f_u$, and condition (iii) implies condition (v). This will lead to a simplified definition, see below.

To give further heuristic illustrations of how properties of parameterizations relate to the dimension of a model, let a parameterization of a class of probability distributions be $P = P(\phi)$, with $\phi \in \Phi$, where Φ is an open set in the l-dimensional Euclidean space. This means that for every $\phi \in \Phi$, there is an open ball around it, such that it is contained in Φ. This implies that one may move in the direction of any of the l coordinates (dimensions) away from ϕ, at least for a small distance, so that every value of the parameter still corresponds to a distribution in the model. Does this possibility of l-dimensional free movement of the parameters values around ϕ imply that the set of distributions also contains an open l-dimensional ball around $P(\phi)$? In other words, does it imply that one may move away from $P(\phi)$ in any of the directions of the l-dimensions and still find a probability distribution in the model there? Perhaps somewhat surprisingly, the answer to this question is negative.

To develop a very simple example[8] that probability distributions do not necessarily inherit the dimensions of their parameterizations, let the population be divided into four categories, with probabilities p_1, p_2, p_3, and $1 - p_1 - p_2 - p_3$, and assume that these are (strictly) positive. Parameterize the first three probabilities as

$$p_1 = 2\alpha + 2\gamma, \tag{8.1}$$

$$p_2 = \beta,$$

$$p_3 = 3\alpha + 3\beta + 3\gamma,$$

and set $p_4 = 1 - p_1 - p_2 - p_3$. Here, α, β, and γ are real parameters, and they are unrestricted in the sense that if a particular choice of them, say, α_0, β_0, γ_0 yields a probability distribution, say P_0, then there is a three-dimensional neighborhood, say $\mathscr{B}$, of this parameter, such that every point in $\mathscr{B}$ yields a probability distribution, too. The set $\mathscr{B}$ is three-dimensional in this sense. Of course, only such values of the parameters are allowed that yield (strictly) positive probabilities when the formulas above are applied. Consider now the image of $\mathscr{B}$, $P(\mathscr{B})$, that is, all the probability distributions that are yielded by the parameters in $\mathscr{B}$.

The probability distributions contain four probabilities; they may be three-dimensional, at most, as was shown above. But the structure imposed by the parameterization implies further a reduction of dimensionality.

To make the last claim a bit clearer, let $Q, R \in P(\mathscr{B})$ be probability distributions in the image of $\mathscr{B}$. Consider these as two points in $\mathbb{R}^4$. Then, the difference of the two vectors Q and R may be written as a linear transform of the differences of their three parameters:

[8] The example is discussed very heuristically and serves only building the right intuition. In particular, the dimension of a set is used in a heuristic manner.

$$\begin{pmatrix} Q_1 - R_1 \\ Q_2 - R_2 \\ Q_3 - R_3 \\ Q_4 - R_4 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 2 \\ 0 & 1 & 0 \\ 3 & 3 & 3 \\ -5 & -4 & -5 \end{pmatrix} \begin{pmatrix} \alpha_Q - \alpha_R \\ \beta_Q - \beta_R \\ \gamma_Q - \gamma_R \end{pmatrix}.$$

This may be seen as suggesting that the image of $\mathscr{B}$ is three-dimensional, which is the same as the dimension of $\mathscr{B}$. But this is not true. In fact, one also has the following representation of the difference:

$$\begin{pmatrix} Q_1 - R_1 \\ Q_2 - R_2 \\ Q_3 - R_3 \\ Q_4 - R_4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1.5 & 3 \\ -2.5 & -4 \end{pmatrix} \begin{pmatrix} \delta_Q - \delta_R \\ \epsilon_Q - \epsilon_R \end{pmatrix},$$

with the new parameters

$$\delta = 2\alpha + 2\gamma \text{ and } \epsilon = \beta.$$

This representation shows that the image of $\mathscr{B}$ is two-dimensional, at most. Using the parameters δ and ϵ, one has

$$p_1 = \delta, \tag{8.2}$$

$$p_2 = \epsilon,$$

$$p_3 = 1.5\delta + 3\epsilon.$$

Thus, the transformation matrix for the first three probabilities is

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1.5 & 3 \end{pmatrix}.$$

As the parameterization used in (8.2) is linear, the transformation matrix is also the Jacobian, and it has two independent columns. On the other hand, the Jacobian of the original parameterization in (8.1) is

$$\begin{pmatrix} 2 & 0 & 2 \\ 0 & 1 & 0 \\ 3 & 3 & 3 \end{pmatrix}.$$

This matrix also has two linearly independent columns.

Irrespective of the parameterization used, one sees that the model for the first three probabilities is a smooth manifold[9] of dimension 2.

If one wishes to parameterize all four probabilities, then one has

$$p_1 = 2\alpha + 2\gamma,$$

$$p_2 = \beta,$$

$$p_3 = 3\alpha + 3\beta + 3\gamma,$$

$$p_4 = 1 - 5\alpha - 4\beta - 5\gamma,$$

and the Jacobian is

$$\begin{pmatrix} 2 & 0 & 2 \\ 0 & 1 & 0 \\ 3 & 3 & 3 \\ -5 & -4 & -5 \end{pmatrix}$$

which also has two linearly independent columns only. The parameterization in (8.2), if extended to include p_4 too, has the Jacobian

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1.5 & 3 \\ -2.5 & -4 \end{pmatrix}$$

and the two columns are linearly independent.

The possibility of reparameterization in the case when the columns of the transformation matrix are not linearly independent just seen is present in the case of all linear parameterizations, that is, when the probabilities or frequencies are linear functions of the parameters.

Theorem 8.1 *For a distribution f with c categories, let*

$$f = \mathbf{A}\alpha,$$

where $\mathbf{A}$ is a $c \times e$ matrix and α is a parameterization of dimension e. Let $\mathbf{A}$ have d linearly independent columns. Then, there is a matrix $\mathbf{B}$ of size $c \times d$ and a vector β of dimension d, such that

$$f = \mathbf{B}\beta.$$

[9] The definition of a smooth manifold has further requirements, but those are easy to check.

Proof For simplicity of presentation, assume that the first d columns of $\mathbf{A}$ are linearly independent. Then, denoting the lth column of $\mathbf{A}$ with a^l, one has that

$$a^l = \sum_{k=1}^{d} \delta_{lk} a^k,$$

for $l = d+1, \ldots, e$, for some coefficients δ_{lk}. Then, denoting the (i, l) entry in $\mathbf{A}$ by a_i^l, one has

$$a_i^l = \sum_{k=1}^{d} \delta_{lk} a_i^k,$$

for $l = d+1, \ldots, e$.

Thus, the original parameterization may be written as

$$f_i = \sum_{j=1}^{e} a_i^j \alpha_j = \sum_{j=1}^{d} a_i^j \alpha_j + \sum_{j=d+1}^{e} a_i^j \alpha_j$$

$$= \sum_{j=1}^{d} a_i^j \alpha_j + \sum_{l=d+1}^{e} \sum_{k=1}^{d} \delta_{lk} a_i^k \alpha_l = \sum_{j=1}^{d} a_i^j \alpha_j + \sum_{k=1}^{d} a_i^k \sum_{l=d+1}^{e} \delta_{lk} \alpha_l$$

$$= \sum_{j=1}^{d} a_i^j (\alpha_j + \sum_{l=d+1}^{e} \delta_{lj} \alpha_l).$$

This means that the d linearly independent column vectors of $\mathbf{A}$ may be retained to obtain $\mathbf{B}$, but the first d components of the original parameterization need to be replaced by

$$\alpha_j + \sum_{l=d+1}^{e} \delta_{lj} \alpha_l,$$

$j = 1, \ldots d$, to obtain the same distribution f. The new parameterization is a linear transformation of the original parameterization.

□

This theorem means that if its conditions hold, then for arbitrary $A \subseteq \mathbb{R}^e$, there exists a $B \subseteq \mathbb{R}^d$, such that

$$\{f : \ f = \mathbf{A}\alpha, \ \alpha \in A\} = \{f : \ f = B\beta, \ \beta \in B\}.$$

Of course, the same applies to probability distributions, too.

Corollary 8.1 *For a probability distribution P with c categories, let*

$$P = \mathbf{A}\alpha,$$

where $\mathbf{A}$ *is a* $c \times e$ *matrix and* α *is a parameterization of dimension* e*. Let* $\mathbf{A}$ *have* d *linearly independent columns. Then, there is a matrix* $\mathbf{B}$ *of size* $c \times d$ *and a parameter* β *of dimension* d*, such that*

$$P = \mathbf{B}\beta.$$

Proof If the sum of the frequencies is 1 then, this does not change when using the new parameterization developed in the proof of Theorem 8.1.

□

The previous theorem showed that in a linear parameterization of a distribution, the linearly independent columns of the transformation matrix are sufficient for the parameterization. The next theorem is about linearly independent rows of the transformation or design matrix.

Theorem 8.2 *For a class of distributions f with c categories, let α be a linear parameterization with*

$$f = \mathbf{A}\alpha,$$

where $\mathbf{A}$ *is a* $c \times e$ *matrix. Let* $\mathbf{A}$ *have* d *linearly independent rows. Then, there is a matrix* $\mathbf{C}$ *of size* $c \times d$ *and a parameterization* γ *of dimension* d*, such that*

$$f = \mathbf{C}\gamma.$$

Proof For simplicity of presentation, assume that the first d rows of $\mathbf{A}$ are linearly independent and, with a_i denoting the ith row of $\mathbf{A}$

$$a_i = \sum_{l=1}^{d} \epsilon_{il} a_l,$$

for $i = d + 1, \ldots, c$. For $i = 1, \ldots, d$, define ϵ_{il} as 1, if $i = l$, and zero if $i \neq l$.

Then, for all $i = 1, \ldots, c$,

$$a_i = \sum_{l=1}^{d} \epsilon_{il} a_l.$$

Thus, with a_i^j denoting the (i, j) entry in $\mathbf{A}$, for all $i = 1, \ldots, c$ and $j = 1, \ldots, e$

$$a_i^j = \sum_{l=1}^{d} \epsilon_{il} a_l^j,$$

and

$$f_i = \sum_{j=1}^{e} a_i^j \alpha_j = \sum_{j=1}^{e} \alpha_j \sum_{l=1}^{d} \epsilon_{il} a_l^j = \sum_{l=1}^{d} \epsilon_{il} \sum_{j=1}^{e} a_l^j \alpha_j = \sum_{l=1}^{d} \epsilon_{il} f_l.$$

Thus, the choice

$$\gamma = (f_1, \ldots, f_d)'$$

and for the design matrix

$$\mathbf{C} = \begin{pmatrix} 1 & 0 & \ldots & 0 \\ 0 & 1 & 0 \ldots & 0 \\ \vdots & & & \\ 0 & . & \ldots 0 & 1 \\ \epsilon_{d+1,1} & \epsilon_{d+1,2} & \ldots & \epsilon_{d+1,d} \\ \epsilon_{d+2,1} & \epsilon_{d+2,2} & \ldots & \epsilon_{d+2,d} \\ \vdots & & & \\ \epsilon_{c,1} & \epsilon_{c,2} & \ldots & \epsilon_{c,d} \end{pmatrix},$$

is as required.

□

Note that the parameterization γ given in the proof may be replaced with other parameterizations, so that the top part of the matrix constructed in the proof is not necessarily diagonal; this was just the simplest example of the existence of a parameterization which used as many components as the number of linearly independent rows of the design matrix. However, the particular design matrix given shows that the following is also true.

Corollary 8.2 *For a frequency distribution f with c categories, let α be a parameterization with*

$$f = \mathbf{A}\alpha,$$

where $\mathbf{A}$ is a $c \times e$ matrix. Let $\mathbf{A}$ have d linearly independent rows. Then, there are $c - d$ of the c frequencies in f that may be expressed as linear combinations of the other d frequencies.

□

Both Theorem 8.2 and Corollary 8.2 are true for probability distributions, too. Thus, the last two theorems and their corollaries may be summarized as follows.

Theorem 8.3 *Let a collection of (general or probability) distributions with c categories have a linear parameterization using a design matrix* $\mathbf{A}$ *of size $c \times e$. Then, it also has a linear parameterization with a design matrix of size $c \times d$, where d is the smaller of the number of linearly independent rows and the number of linearly independent columns of* $\mathbf{A}$.

□

The quantity d in this theorem is called the rank of $\mathbf{A}$.

In the example with the parameterization (8.1), the first two rows of both design matrices are linearly independent, and the last row (when three probabilities are modeled) may be obtained as linear combinations of the first two. This illustrates the situation described in Corollary 8.2. In the example, both parameterizations imply that

$$p_3 = 1.5p_1 + 3p_2.$$

The requirement that the four probabilities sum to 1 implies that the sum of the first three probabilities is less than 1 or that

$$5\alpha + 4\beta + 5\gamma < 1,$$

which determines the domain of the parameterization (and the range of the parameters).

This example gave a simple illustration of parameterizations where the Jacobian is not of full rank. In this case, the number of parameters cannot be assumed to determine the dimension of the manifold. The model is, rather, a smooth manifold of dimension equal to the number of linearly independent columns or rows of the Jacobian.

When the parameterization is not linear, the rank of the Jacobian still determines its dimension in a local sense.

As a simple example, let there be three categories, and assume that $p_1 = p_2$. Choose as parameters $0 < \alpha,\ \beta < 1$, and let

$$p_1 = \alpha\beta,\ \ p_2 = \alpha\beta.$$

Of course, one sees that the function above is not the inverse of a parameter, as knowing the probabilities does not imply particular values of α and β, only of their product. Some authors would say that α and β are unidentified parameters, but they are not parameters in the sense parameters and parameterizations are used here. In more complex setups, however, this may not be obvious, see e.g., Section *LCDA* 10.3. This is, again, a situation, where the Jacobian is helpful. In this case, the Jacobian is 3×2, with the rows identified with the probabilities and the columns with the components of the parameterization. As

$$p_3 = 1 - p_1 - p_2 = 1 - 2\alpha\beta,$$

the Jacobian is

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ -2\,\beta & -2\,\alpha \end{pmatrix}.$$

This matrix is not of full (column) rank everywhere, because if $\alpha = \beta$, the two columns are identical. In fact, the model is a smooth manifold of dimension 1 in $\mathbb{R}^3$, which is often referred to by saying that the model has 1 free parameter. Indeed, with the obvious parameterization

$$p_1 = \gamma,\ \ p_2 = \gamma,\ \ p_3 = 1 - 2\gamma,\ \ 0 < \gamma < 0.5,$$

one has a smooth manifold of dimension 1.

The relevance of the Jacobian regarding the local dimension of a model comes from the fact that—as discussed above in the example of $y = x^2$—the existence of partial derivatives means that the parameterization may be well approximated by a linear transformation locally. The actual linear transformation may be different in different points. The Jacobian evaluated at a point (parameter value) is the local slope of the (multivariate) linear transformation which gives the best linear approximation, when it exists. When the Jacobian is of full rank, the slopes in different directions are not linear combinations of each other, thus, in the linear approximation, change in one dimension is not implied by the changes in other directions. Thus, changes of the probabilities in as many dimensions as the number of columns of the Jacobian are, so to say, independent[10] Further, when the best linear approximation is with a Jacobian which is of size $c \times e$, but only $d < e$ columns of the Jacobian are linearly independent, then, as the proof of Theorem 8.1 implies, the best linear approximation that uses e parameters may be replaced with one using d parameters, thus, the manifold is locally d-dimensional only.

A family of distributions is said to have a smooth parameterization if it is a smooth manifold. Similarly to Lauritzen [98] and Bergsma and Rudas [20], the following definition of a smooth parameterization is used in the sequel. This is a simplification of the definition of a smooth manifold given above which has the same properties in the cases considered here.

Let a population or sample space be divided into c categories, and consider all probability or general distributions f on it.

Let $f = f(\alpha)$ be a parameterization.[11] This parameterization is called smooth, if:

[10] This independence is not the same as the one used elsewhere in the book rather means the changes do not depend on each other.

[11] This is a slight abuse of notation, where f denotes both the frequencies and the function yielding the frequencies.

(i) it is one-to-one between open sets
(ii) it and its inverse are twice continuously differentiable and
(iii) it has a Jacobian of full rank everywhere on the domain of the parameterization.

Note that (ii) refers to the partial derivatives in the multidimensional case.

A parameterization is a function defined on the space of the parameters with its range being a class of distributions, including the case of all distributions, general or probability. Smoothness is a property of this function, thus, the same concept will sometimes be applied to arbitrary functions or transformations from a Euclidean space to another Euclidean space, even if the vectors in the range are not interpreted as distributions. This language will be used, in particular, when the result of a smooth transformation of parameters is not a distribution, rather another set of parameters.

A straightforward consequence of the definition is stated next.

Proposition 8.2 *If a parameterization $f = g(\alpha)$ of a class of (probability or general) distributions is smooth, then the parameter*

$$\alpha = g^{-1}(f),$$

exists and the following hold:

(a) it is one-to-one between open sets
(b) it and its inverse are twice continuously differentiable and
(c) it has a Jacobian of full rank on its entire domain.

Proof This is an immediate implication of the definition of a parameterization and of the so-called inverse function theorem, see, e.g., Fleming [57] or Lax [103].

□

The next result refers to the situation, when one has a (not necessarily proper) subset of the distributions on the sample space, which is a smooth manifold, and then a model is defined by fixing the values of some of the parameters in a smooth parameterization. Such a way of model definition is especially useful, when there is variation and/or likelihood independence among the fixed and the not fixed components of the parameter, see the discussion in *LCDA*, in particular in Chapter *LCDA* 6.

As a simple example, consider all probability distributions on a 2×2 contingency table. By Proposition 8.1, this is a smooth three-dimensional manifold. The distributions may be parameterized by the following parameters:

$$\alpha = P(1, 1),$$

$$\beta = P(1, +),$$

$$\gamma = P(+, 1) - P(1, +).$$

Indeed,

$$P(1,1) = \alpha,$$

$$P(1,2) = P(1,+) - P(1,1) = \beta - \alpha,$$

$$P(2,1) = P(+,1) - P(1,1) = \gamma + \beta - \alpha,$$

$$P(2,2) = 1 - P(1,1) - (P(1,+) - P(1,1)) - (P(+,1) - P(1,1))$$

$$= 1 - \alpha - (\beta - \alpha) - (\gamma + \beta - \alpha) = 1 + \alpha - 2\beta - \gamma.$$

This is a smooth parameterization, with the following Jacobian:

$$\begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 1 \\ 1 & -2 & -1 \end{pmatrix}$$

which is of full rank, that is, the columns are linearly independent. To see this, note that the 1st column cannot be obtained as a linear combination of the last two columns because of the first row. To obtain the 2nd column would require the 1st column to have a coefficient of 0 because of the first row, which would then imply that the entry in the second row cannot be obtained. To reproduce the third column would imply a coefficient of zero for the first column because of the first row and then also to the second column because of the second row.

The model of marginal homogeneity is defined by setting

$$P(+,1) - P(1,+) = \gamma = 0, \tag{8.3}$$

or

$$P(+,1) = P(1,+),$$

and then also

$$\beta = P(+,1).$$

The distributions in the model are parameterized by the remaining parameters. Indeed, under marginal homogeneity

$$P(1,1) = \alpha,$$

$$P(1,2) = P(1,+) - P(1,1) = \beta - \alpha,$$

$$P(2,1) = P(+,1) - P(1,1) = \beta - \alpha,$$

$$P(2,2) = 1 + P(1,1) - 2P(1,+) = 1 + \alpha - 2\beta.$$

This gives the following Jacobian:

$$\begin{pmatrix} 1 & 0 \\ -1 & 1 \\ -1 & 1 \\ 1 & -2 \end{pmatrix}$$

that is clearly of full rank and consists of the first two columns of the Jacobian of the original parameterization.

It will be shown now, in general, that fixing some of the parameters of a smooth manifold yields a smooth manifold. In particular, the Jacobian of the restricted model (manifold) is obtained from the original Jacobian by omitting the columns corresponding to the parameters that are fixed.

Theorem 8.4 *Let the family of distributions*

$$\mathscr{F} = \{f = f(\alpha), \alpha \in A\}$$

be a smooth manifold of dimension $l \leq k$ *in* $\mathbb{R}^k$*, with a smooth parameterization* $f = f(\alpha)$*, where* $\alpha \in A \subseteq \mathbb{R}^l$ *is a parameter, and* A *is an open set.*

Let $\alpha^0 \in A$ *be a fixed vector, and consider the subfamily defined as*

$$\mathscr{F}_{\alpha^0,m} = \{f = f(\alpha) \in \mathscr{F} : \ \alpha_1 = \alpha_1^0, \ldots, \alpha_m = \alpha_m^0\},$$

with an $m < l$*. Then,* $\mathscr{F}_{\alpha^0,m}$ *is a smooth manifold of dimension* $l - m$ *in* $\mathbb{R}^k$*, and the unrestricted components of the parameters constitute a smooth parameterization of the subfamily, with the Jacobian consisting of those columns of the original Jacobian that are associated with the unrestricted parameters* $(\alpha_{m+1}, \ldots, \alpha_l)$*.*

Proof The subfamily $\mathscr{F}_{\alpha^0,m}$ is not empty, because $\alpha^0 \in A$. Define

$$B_{\alpha^0,m} = \{(\alpha_{m+1}, \ldots, \alpha_l)' : (\alpha_1^0, \ldots, \alpha_m^0, \alpha_{m+1}, \ldots, \alpha_l)' \in A\}.$$

The set $B_{\alpha^0,m}$ is not empty, as it contains $(\alpha_{m+1}^0, \ldots, \alpha_l^0)'$. Further, it is an open set in $\mathbb{R}^{l-m}$. To see this, note that A is open in $\mathbb{R}^l$, so every $\alpha \in A$ has an open neighborhood containing it, say $N(\alpha)$ that is entirely in A. This means that for every $\alpha \in A$ and $i = 1, \ldots, l$, there exists an open interval $I_i(\alpha)$, such that $\alpha_i \in I_i$, and

$$I_1(\alpha) \times I_2(\alpha) \times \ldots \times I_l(\alpha) \subseteq N(\alpha) \subseteq A.$$

Because

$$(\alpha_1^0, \alpha_2^0, \ldots, \alpha_m^0)' \in I_1(\alpha^0) \times I_2(\alpha^0) \times \ldots \times I_m(\alpha^0),$$

one has that

$$\{\alpha_1^0\} \times \{\alpha_2^0\} \times \ldots \times \{\alpha_m^0\} \times I_{m+1}(\alpha^0) \times I_{m+2}(\alpha^0) \times \ldots \times I_l(\alpha^0) \subseteq A,$$

implying that

$$I_{m+1}(\alpha^0) \times I_{m+2}(\alpha^0) \times \ldots \times I_l(\alpha^0) \subseteq B_{\alpha^0,m}.$$

and it follows that $B_{\alpha^0,m}$ is an open set in $\mathbb{R}^{l-m}$.

Define

$$g : B_{\alpha^0,m} \to \mathbb{R}^k,$$

so that for $\beta \in B_{\alpha^0,m}$

$$g(\beta) = f(\alpha_1^0, \ldots, \alpha_m^0, \beta_1, \ldots, \beta_{l-m}),$$

that is, g gives for the last $l-m$ coordinates of the parameter the same distribution as f does for the fixed first m components and these $l-m$ components concatenated.

Then, g is a one-to-one function between $B_{\alpha^0,m}$ and $\mathscr{F}_{\alpha^0,m}$, and g, and its inverse are twice continuously differentiable.

The Jacobian of g with domain $B_{\alpha^0,m}$ is obtained from the Jacobian of f with domain A by retaining the last $l-m$ columns and deleting the first m columns that are fixed, and they are not in the argument of g. As the columns of the Jacobian of f are linearly independent so are its last $l-m$ columns.

Finally, the vectors in $B_{\alpha^0,m}$, namely, $(\beta_1, \ldots, \beta_{l-m})'$, parameterize $\mathscr{F}_{\alpha^0,m}$ because f is one-to-one on A, thus, if two distributions, say, f_1 and f_2 are different, also their parameters, say, a^1 and a^2, are different. If both f_1 and f_2 are contained in $\mathscr{F}_{\alpha^0,m}$, then their first m parameters are the same:

$$(\alpha_1^1, \ldots, \alpha_m^1)' = (\alpha_1^2, \ldots, \alpha_m^2)' = (\alpha_1^0, \ldots, \alpha_m^0)',$$

thus, their last $l-m$ parameters in the parameterization α have to be different, and these are their β parameters.

□

Note that openness of the domain implies a local version of variation independence.

A very important application of this result to all log-linear models is given in Corollary 9.4, see also the example using (8.11).

The final results of this section will be powerful tools of investigating the smoothness properties of log-linear and related parameterizations in the next chapter.

Let a manifold in $\mathbb{R}^k$ have a linear parameterization

$$f = \mathbf{A}\alpha,$$

where $\alpha \in \mathbb{R}^l$ and $\mathbf{A}$ is of size $k \times l$. Such a parameterization is called minimal linear, if there exists no matrix $\mathbf{B}$ of size $k \times m$, with $m < l$, such that a parameterization of the form

$$f = \mathbf{B}\beta$$

with parameters β would be possible.

Theorem 8.5 *For a manifold $\mathscr{F}$ in $\mathbb{R}^k$, let*

$$\mathscr{F} = \{f = \mathbf{A}\alpha,\ \alpha \in A\}$$

for a subset $A \subseteq \mathbb{R}^l$ be a minimal linear parameterization. Then, $\mathbf{A}$ is of full (column) rank.

Proof As the minimal parameterization has l parameters, $\mathbf{A}$ is of size $k \times l$. Suppose, to the contrary of the claim, that $\mathbf{A}$ has only $m < l$ linearly independent columns. Then, Theorem 8.2 implies that the parameterization is not minimal linear. □

The results for smooth linear parameterizations have relevance for log-linear parameterizations considered in the next chapter through the following result.

Theorem 8.6 *Let $\mathscr{F} = \{f(\alpha) :\ \alpha \in A\}$ be a smooth l-dimensional manifold in $\mathbb{R}^k$, $(l \leq k)$ with a smooth parameterization*

$$f = \mathbf{A}\alpha,\ \textit{for}\ \alpha \in A.$$

Further, let $h : \mathbb{R}^k \to \mathbb{R}^k$ be a one-to-one, twice continuously differentiable function with a full rank Jacobian. Then,

$$\{h(f(\alpha)) :\ \alpha \in A\}$$

is also a smooth l-dimensional manifold in $\mathbb{R}^k$.

Proof The so-called chain rule (see, e.g., Lax [103]) implies that the Jacobian of the composite function $h(f)$ is the product of the individual Jacobians. The Jacobian of f, $\mathbf{J}_f$, is of size $k \times l$ and of rank l, and the Jacobian of h, $\mathbf{J}_h$, is $k \times k$ and of rank k, and the ranks are constant over the respective domains because of the assumed

smoothness. Then, the product $\mathbf{J}_h\mathbf{J}_f$ is of size $k \times l$. To see that this is of full rank, that is, of rank l, let β be an $l \times 1$ vector of coefficients and consider the vector that is the linear combination of the columns of $\mathbf{J}_h\mathbf{J}_k$ with the coefficients β:

$$(\mathbf{J}_h\mathbf{J}_f)\beta = \mathbf{J}_h(\mathbf{J}_f\beta), \tag{8.4}$$

where the equality follows from the associative property of matrix multiplication. If the product $(\mathbf{J}_h\mathbf{J}_f)$ is not of full rank, then there exists a vector β different from the zero vector, such that the linear combination of the columns of $(\mathbf{J}_h\mathbf{J}_f)$ with these coefficients, that is, on the left-hand side of (8.4), is a zero vector. But as the equation shows, in this case, the combination of the columns of $\mathbf{J}_h$ with the coefficient vector $\mathbf{J}_f\beta$ is zero. As $\mathbf{J}_h$ is of full rank, this is only possible if $\mathbf{J}_f\beta$ itself is zero, too. But as $\mathbf{J}_f$ is of full rank, this is only possible if β is a zero vector, which is a contradiction. □

Note that the theorem above remains true, even if the original parameterization is not assumed to be linear.

In the next section, properties of log-linear parameterizations are discussed, which will be needed to prove their smoothness in the next chapter.

8.3 Log-Linear Parameters Revisited

Although log-linear parameterizations, using the balanced log-linear parameters developed in Section *LCDA* 10.3 or the corner (or effect) log-linear parameters developed in Section *LCDA* 11.3, are widely considered as smooth, one has to be careful in formulating this precisely, taking into account two facts.

The first one is that the balanced log-linear parameters, when summed in any of their indices, sum to 0. For example, if variable A has 2 categories, say 1 and 2, then

$$\lambda_1^A + \lambda_2^A = 0,$$

or

$$\lambda_{12}^{AB} + \lambda_{22}^{AB} = 0,$$

and similarly for the other parameters. Thus, if λ_1^A is given some value, then λ_2^A cannot take on an arbitrary value, only $-\lambda_1^A$; therefore, the set in which the balanced log-linear parameters are defined is not open in the Euclidean space with as many dimensions as the number of balanced log-linear parameters. If it was, then from a given pair of λ_1^A and λ_2^A values, both could be moved away a little in every possible

direction, independently[12] of each other. Rather, in reality, if one is increased, the other one decreases, thus, both cannot be made smaller or larger at the same time. However, if from among all balanced log-linear parameters the ones in which there is an index equal to the last[13] index of any of the variables included in the effect are omitted, then the domain of the remaining parameters is an open set in the Euclidean space of dimension equal to the number of remaining parameters. Let the collection of the remaining balanced log-linear parameters be denoted as $\Lambda^{\sim}$.

For example, in a 2×2 table, using the parameters in $\Lambda^{\sim}$ gives the following linear parameterization of the log probabilities:

$$\begin{aligned}
\log P(1,1) &= \lambda^{\emptyset} + \lambda_1^A + \lambda_1^B + \lambda_{11}^{AB}, \\
\log P(1,2) &= \lambda^{\emptyset} + \lambda_1^A - \lambda_1^B - \lambda_{11}^{AB}, \\
\log P(2,1) &= \lambda^{\emptyset} - \lambda_1^A + \lambda_1^B - \lambda_{11}^{AB}, \\
\log P(2,2) &= \lambda^{\emptyset} - \lambda_1^A - \lambda_1^B + \lambda_{11}^{AB}.
\end{aligned} \tag{8.5}$$

If A has categories 1, 2, and B has categories 1, 2, 3, then

$$\begin{aligned}
\lambda_{11}^{AB} + \lambda_{21}^{AB} &= 0, \\
\lambda_{12}^{AB} + \lambda_{22}^{AB} &= 0, \\
\lambda_{13}^{AB} + \lambda_{23}^{AB} &= 0,
\end{aligned}$$

and thus

$$\lambda_{23}^{AB} = -\lambda_{21}^{AB} - \lambda_{22}^{AB} = \lambda_{11}^{AB} + \lambda_{12}^{AB}, \tag{8.6}$$

so the redundant log-linear parameter λ_{23}^{AB} is the sum of all the non-redundant ones in the $A \times B$ table.

The precise meaning of the term non-redundant in the current context is the following. Let α denote the elements of a vector space.[14] Then, out of the vectors in the current context

$$\alpha_1, \ldots, \alpha_k,$$

[12] Independence here means variation independence, of course.

[13] An arbitrary ordering of the categories may be used to select the last index.

[14] That is, let summation and multiplication by real numbers be meaningful for the αs.

the non-redundant ones are an arbitrary maximal linearly independent subset. The remaining, redundant, ones may be expressed as linear combinations of the non-redundant ones. In particular, if

$$\alpha_1 = 0, \ldots, \alpha_k = 0,$$

is required, it is sufficient to require this for the non-redundant ones.

The following result gives the number of log-linear parameters in $\Lambda^{\sim}$.

Theorem 8.7 *The number of log-linear parameters in $\Lambda^{\sim}$ and also of the corner log-linear parameters is the product of the numbers of the categories of the variables.*

Proof The proof is by induction on the number of variables. If there is one variable only with c categories, then omitting the parameter associated with the last category yields $c - 1$ parameters, and there is one more parameter for the empty set.

If there are 2 variables, A and B, with respective numbers of categories c_A and c_B, then $\Lambda^{\sim}$ contains 1 parameter for $\emptyset$, $c_A - 1$ parameters for A, $c_B - 1$ parameters for B, and $(c_A - 1)(c_B - 1)$ parameters for AB, in both parameterizations. This is

$$1 + c_A - 1 + c_B - 1 + (c_A - 1)(c_B - 1) = c_A c_B$$

parameters, which is equal to the number of cells in the two-way table formed by the ranges of the variables..

Assume now the claim is proved for contingency tables formed by v variables. Consider a table with $v + 1$ variables, say $V_1, \ldots, V_v, V_{v+1}$. Every subset of the variables either contains V_{v+1} or does not. The subsets not containing V_{v+1} are exactly the subsets of $V_1, \ldots, V_v$, with

$$\prod_{i=1}^{v} c_i,$$

parameters in $\Lambda^{\sim}$ by the induction assumption, where c_i is the number of categories of variable $V_i, i = 1, \ldots, v, v+1$. Each subset, which also contains V_{v+1}, has in $\Lambda^{\sim}$ as many parameters as for the variables other than V_{v+1}, multiplied by $(c_{v+1} - 1)$. Thus, the total number of parameters that also contain in their effects V_{v+1} is

$$(c_{v+1} - 1) \prod_{i=1}^{v} c_i.$$

Therefore, the total number of log-linear parameters in $\Lambda^{\sim}$ for $V_1, \ldots, V_v, V_{v+1}$ is

$$\left(\prod_{i=1}^{v} c_i \right) + \left((c_{v+1} - 1) \prod_{i=1}^{v} c_i \right) = c_{v+1} \prod_{i=1}^{v} c_i = \prod_{i=1}^{v+1} c_i.$$

The number of the corner parameters is the same as the number of non-redundant balanced parameters.

□

The term non-redundant log-linear parameters will be used to refer to the balanced log-linear parameters in $\Lambda^{\sim}$ or to the corner log-linear parameters[15].

When one uses the non-redundant log-linear parameters, the design matrix needs to be modified. Variants and special cases of the construction to follow were presented in Chapters *LCDA* 10, 11, and 12.

Let the variables and also the categories of the variables be ordered. The rows of the design matrix are identified with the cells of the table in an arbitrary, say lexicographic, order. The columns of the design matrix are identified with the effects of the log-linear parameters. To specify an order, let an effect consisting of fewer variables precede one with more variables; let the effects containing the same number of variables be ordered lexicographically, and among effects of the same subset of variables, let there be a lexicographic order according to the indices of the variables in the effect. Thus, λ_2^A precedes λ_{13}^{BC} and the latter one precedes λ_{21}^{BC}, which precedes λ_{11}^{BD}.

For example, in the case of a 2×3, $A \times B$ table, with categories 1, 2 and 1, 2, 3, respectively, the rows of the design matrix are associated with

$$((1,1),\ (1,2),\ (1,3),\ (2,1),\ (2,2),\ (2,3))',$$

and the columns in the case of balanced non-redundant parameters are associated with

$$(\lambda^{\emptyset},\ \lambda_1^A,\ \lambda_1^B,\ \lambda_2^B,\ \lambda_{11}^{AB},\ \lambda_{12}^{AB}).$$

Then, for the non-redundant balanced log-linear parameters, one has the following structure of the design matrix $\mathbf{T}$.

Let the i-th row of $\mathbf{T}$ be associated with cell v_i of the table and let i_v the row, which is associated with cell v.

The columns of $\mathbf{T}$ are associated with the non-redundant log-linear parameters. Thus, by Theorem 8.7, $\mathbf{T}$ is a square matrix. If λ_w^W is a non-redundant log-linear parameter, where $W \subseteq V$ is a subset of the variables and w is an index combination of it, then let W_j denote the subset associated with column j, and let w_j denote the index combination associated with the column j. Further, let j_w be the column associated with the indices w, so that $j_{w_j} = j$. The column j implies the combination of indices w_j and that implies W_j, that is, the subset of variables with

[15] Note, however, that in the case of probability distributions, one way to make the sum of probabilities equal to 1 is to choose an appropriate value of the log-linear parameter associated with the empty set. This also may be seen as making that parameter redundant, in a sense different from the one used here. This issue will be discussed later.

which the jth column is associated. Note that w_j, as a function of j, is one-to-one[16] but W_j is not. All of the combinations of indices w_j are such that no variable A in W_j has its last index in w_j. Let J_W denote the collection of all indices of columns of the design matrix associated with W.

In the design matrix $\mathbf{T}$

$$t_{i,1} = 1$$

for all i, as this entry is associated the log-linear parameter of subsets of variables $\emptyset \subseteq V$, and this is the overall effect in the sense that it appears in every cell of the table.

To define the $t_{i,j}$ entry in $\mathbf{T}$ for the other columns, let the row i be associated with the index combination v_i of the variables V and let W_j be the subset of variables associated with the column j and let w_j be their particular index combination associated with column j. Note that the index combination w_j is not redundant, that is, no variable is in its last category.

If $(v_i)_{W_j} = w_j$, that is, the selection of indices of the variables in W_j from among v_i, yields w_j and thus it is associated with a non-redundant combination of the indices, then

$$t_{i,j} = 1,$$

and the other entries $t(i, j)$ in row i that are associated with the same W_j are zero.

To determine the entry $t_{i,j}$ for those (i, j) combinations, when $(v_i)_{W_j}$ yields a redundant combination, express the log-linear parameters for every $W \subseteq V$ and category combination w as a linear combination of the non-redundant ones as

$$\lambda_w^W = \sum_{j \in J_W} \epsilon(w, w_j) \lambda_{w_j}^W.$$

Note that on the right-hand side only the non-redundant balanced log-linear parameters appear, while on the left-hand side all balanced log-linear parameters do.

Of course, for particular values of the log-linear parameters, several choices of the coefficients $\epsilon(w, w_j)$ may be possible. This implicit definition of the coefficients is meant to be the one that works irrespective of the actual values of the log-linear parameters. If there is such a $j \in J_W$ that $w = w_j$, that is, the log-linear parameter on the left-hand side is non-redundant, then $\epsilon(w, w_j) = 1$ for this j, and is zero for the other js in J_W. If the log-linear parameter on the left-hand side, λ_w^W is redundant, then let for every variable $A \in V$, c_A denote its last category and write:

$$w = ((w)_A : A \in W) = (((w)_A : A \in W, (w)_A = c_A), ((w)_A : A \in W, (w)_A \neq c_A)),$$

[16] That is, a collection of indices, say (2, 1, 3), is considered different, when these are the indices of variables, say, A, B, C, or when these are the indices of the variables, say, B, D, E.

that is, separate those variables that are and those that are not equal to their respective last categories in w. Then, define:

$$I_w = \{w' : (w')_A = (w)_A, \text{ if } (w)_A \neq c_A, \text{ and } (w')_A \neq (w)_A, \text{ if } (w)_A = c_A\},$$

as the collection of those index combinations w' of the indices of the variables $A \in W$ that, on the one hand, are equal to w on the variables the indices of which are different from their respective last ones in w, and, on the other hand, retain the indices of the variables that had an index in w, other than their last one.

The indices $w' \in I_w$ are all the non-redundant indices such that

$$(w')_{A \in W, (w)_A \neq c_A} = (w)_{A \in W, (w)_A \neq c_A}.$$

Thus,

$$\lambda_w^W = (-1)^{l(w)} \sum_{w' \in I_w} \lambda_{w'}^W,$$

where $l(w)$ is the number of variables $A \in W$ that have their last index in w, see (8.5) and (8.6).

This determines $\epsilon(w, w_j)$ as

$$\epsilon(w, w_j) = (-1)^{l(w)} \text{ if } w \in I_{w_j}$$

$$\epsilon(w, w_j) = 0 \text{ otherwise.}$$

Then, for every row and column combination (i, j), when $(v_i)_{w_j}$ is redundant, set:

$$t_{i,j} = \epsilon(w, w_j).$$

Note that if λ_w^W is not redundant, then $I_w = \{w\}$ and $|l(w)| = 0$, thus, the definition of $t_{i,j}$ given in the case when $(v_i)_{W_j}$ is redundant yields the definition given above for the case when $(v_i)_{W_j}$ is not redundant.

For example, in the case of the 2×3 table, the first non-empty subset is $\{A\}$ and $J_{\{A\}} = \{2\}$. Here, $w_2 = (1)$. Further, the possible indices are (1) and (2), and

$$\lambda_1^A = \lambda_1^A,$$

and

$$\lambda_2^A = -\lambda_1^A.$$

Thus, when $(v)_A = 1$, that is, for $i = 1, 2, 3$, one has

$$t_{i,2} = 1,$$

and when $(v)_A = 2$, that is, for $i = 4, 5, 6$, one has

$$t_{i,2} = -1.$$

The next non-empty subset is B, with three categories. Here, $J_B = \{3, 4\}$ and $w_3 = (1)$, $w_4 = (2)$. Further,

$$\lambda_1^B = \lambda_1^B,$$

$$\lambda_2^B = \lambda_2^B,$$

$$\lambda_3^B = -\lambda_1^B - \lambda_2^B.$$

Then,

$$t_{1,3} = 1,\ t_{1,4} = 0,$$

$$t_{2,3} = 0,\ t_{2,4} = 1,$$

$$t_{3,3} = -1,\ t_{3,4} = -1,$$

$$t_{4,3} = 1,\ t_{4,4} = 0,$$

$$t_{5,3} = 0,\ t_{5,4} = 1,$$

$$t_{6,3} = -1,\ t_{6,4} = -1.$$

The final subset is $\{A, B\}$ and $J_{\{A,B\}} = \{5, 6\}$, with $w_5 = (1, 1)$ and $w_6 = (1, 2)$. Then, for all pairs of indices, one has

$$\lambda_{1,1}^{AB} = \lambda_{1,1}^{AB},$$

$$\lambda_{1,2}^{AB} = \lambda_{1,2}^{AB},$$

$$\lambda_{1,3}^{AB} = -\lambda_{1,1}^{AB} - \lambda_{1,2}^{AB},$$

$$\lambda_{2,1}^{AB} = -\lambda_{1,1}^{AB},$$

$$\lambda_{2,2}^{AB} = -\lambda_{1,2}^{AB},$$

$$\lambda_{2,3}^{AB} = \lambda_{1,1}^{AB} + \lambda_{1,2}^{AB},$$

where the last equality is obtained either from $-\lambda_{13}^{AB}$ or from $-\lambda_{2,1}^{AB} - \lambda_{2,2}^{AB}$. Thus,

$$t_{1,5} = 1,\ t_{1,5} = 0,$$

$$t_{2,5} = 0,\ t_{2,6} = 1,$$

$$t_{3,5} = -1,\ t_{3,6} = -1,$$

$$t_{4,5} = -1,\ t_{4,6} = 0,$$

$$t_{5,5} = 0,\ t_{5,6} = -1,$$

$$t_{6,5} = 1,\ t_{6,6} = 1.$$

The design matrix $\mathbf{T}$ is the following 6×6 matrix:

$$\begin{pmatrix} 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & -1 & 1 & 0 & -1 & 0 \\ 1 & -1 & 0 & 1 & 0 & -1 \\ 1 & -1 & -1 & -1 & 1 & 1 \end{pmatrix}. \tag{8.7}$$

In the case of the corner log-linear parameters in the 2×3 example, the columns are associated with, using μ for the corner parameters:

$$(\log p_{11},\ \mu_2^A,\ \mu_2^B,\ \mu_3^B,\ \mu_{22}^{AB},\ \mu_{23}^{AB}),$$

with the following definition of the parameters:

$$\log p_{11} = \log p_{11},$$

$$\mu_2^A = \log p_{21} - \log p_{11},$$

$$\mu_2^B = \log p_{12} - \log p_{11},$$

$$\mu_3^B = \log p_{13} - \log p_{11},$$

$$\mu_{22}^{AB} = \log p_{22} - \log p_{12} - \log p_{21} + \log p_{11},$$

$$\mu_{23}^{AB} = \log p_{23} - \log p_{13} - \log p_{21} + \log p_{11}.$$

For a general definition of the corner parameters, see Proposition *LCDA* 11.2, the so-called Möbius inversion.

In general, the columns of the design matrix other than the first one are associated with the corner log-linear parameters $\mu_{(u)_W}^W$, such that none of the variables in W is

in its first category in u. The parameters are ordered according to W, as before, and within that in the lexicographic order of the indices $(u)_W$. The rows are ordered as before. Then, if row i is associated with cell v and column j is associated with the parameter $\mu^W_{(u)_W}$, then the design matrix is as follows:

$$t_{i,1} = 1,$$

$$t_{ij} = 1 \text{ if } j > 1 \text{ and } (v)_W = (u)_W,$$

and

$$t_{ij} = 0 \text{ otherwise.}$$

In the 2×3 example, the design matrix is the following 6×6 matrix:

$$\begin{pmatrix} 1&0&0&0&0&0 \\ 1&0&1&0&0&0 \\ 1&0&0&1&0&0 \\ 1&1&0&0&0&0 \\ 1&1&1&0&1&0 \\ 1&1&0&1&0&1 \end{pmatrix}.$$

In the sequel, when the non-redundant balanced log-linear parameters in $\Lambda^\sim$ or the corner parameters are used, the design matrices $\mathbf{T}^\sim$ defined above for the non-redundant balanced and the corner parameters will be used. For either type of parameters, the notation $\nu^\sim$ will be used. When needed, $\nu^\sim$ will be considered a vector with the order of the components used in the definitions of the design matrices.

Log-linear parameters may be determined for any distribution[17] exactly the same way as for probability distributions.

Theorem 8.8 *The design matrix $\mathbf{T}^\sim$ is of full rank, and the $\nu^\sim$ log-linear parameters constitute a smooth parameterization of the (strictly) positive distributions on a contingency table. The range of these parameters is the Euclidean space of as many dimensions as the number of cells of the table.*

Proof As the distributions considered may have any positive number in any cell of the table, their logarithms[18] may take on any real value. Thus, if there are c cells in

[17] Consisting of real, but not necessarily integer, numbers associated with the cells of the table. As mentioned before, this definition makes it possible to consider estimated frequency distributions, too, which do not necessarily consist of integers.

[18] The base of the logarithm is not important here but is traditionally e-based, that is, natural logarithms are used.

the contingency table, every point in $\mathbb{R}^c$ corresponds to the logarithms of a (strictly) positive distribution. For such a distribution

$$\log f = \mathbf{I}\alpha,$$

where $\alpha \in \mathbb{R}^c$ and $\mathbf{I}$ is the c-dimensional identity matrix, is a linear parameterization. Because the components of f are variation independent, so are the components of $\log f$, and this is a minimal linear representation.

Thus, the collection of all $\log f$ vectors is a smooth manifold of dimension c by Theorem 8.5.

Another linear representation is obtained if the log-linear parameters are calculated for the distribution (see, e.g., *LCDA* (10.21))

$$\log f = \mathbf{T}^{\sim}\nu^{\sim}. \tag{8.8}$$

This a minimal linear parameterization of $\log f$, because the number of non-redundant parameters is c, see Theorem 8.7.

Then, by Theorem 8.5, $\mathbf{T}^{\sim}$ is of full rank and (8.8) is a smooth parameterization.

The $\nu^{\sim}$ parameters are obtained from $\log f$ as

$$\nu^{\sim} = (\mathbf{T}^{\sim})^{-1} \log f.$$

To see that the range of the $\nu^{\sim}$ parameters is $\mathbb{R}^c$, let $\nu^{\sim}$ be an arbitrary vector in $\mathbb{R}^c$. This is obtained as the parameter belonging to the distribution

$$f = \exp(\mathbf{T}^{\sim}\nu^{\sim}). \tag{8.9}$$

Finally, apply Theorem 8.6 with the transformation of exponentiating every component of a vector, to the smooth parameterization (8.8), to obtain that (8.9) is also a smooth parameterization of all frequency distributions.

□

Note that the result above does not rely on any particular type of log-linear parameters being used. It does apply to any c parameters, of which the log frequencies are full rank linear functions.

The second issue to consider, referred to at the beginning of this section, in the case of parameterizations of probability distributions is that the number of non-redundant log-linear parameters is equal to the number of cells of the contingency table, which is also the number of probabilities. As the latter sum to 1, one cannot expect the non-redundant parameters to be unrelated, in the case of probability distributions; however, the exact form of this relationship depends on how the parameters are related to the probabilities. In the case of the balanced parameters, the effect associated with the empty set appears in every cell. In the case of the corner parameters, the effect associated with the cell with all indices equal to 1 appears in every cell. These are called overall effects. A straightforward way to ensure that the probabilities sum to 1 is to select the other log-linear parameters

arbitrarily, and then the value of the overall effect is set to make the sum of the probabilities equal to 1. Note, however, that although the value of the overall effect is implied by the values of the other log-linear parameters, this is not a linear relationship.

Implications of the linear relationship among the probabilities for smooth parameterizations are discussed further in the next section.

8.4 Exponential Families

In this section, all or classes of general and probability distributions are parameterized as exponential families. In the next chapter, smoothness of log-linear, mean value, and mixed parameterizations will be proved using this context.

Note that the parameterization (8.9) in the proof of Theorem 8.8 used the exponential family structure of all (strictly) positive distributions. Exponential families were introduced in Chapter *LCDA* 7. Our discussion, of course, is restricted to the categorical case. Let $\mathcal{S}$ be a contingency table of arbitrary dimension, and let s denote a cell of it. Let $\{q(s),\ s \in \mathcal{S}\}$ be a (strictly) positive distribution on $\mathcal{S}$, and $u : \mathcal{S} \to \mathbb{R}^l$ a function. Let $\tau \in \mathbb{R}^l$ and consider:[19]

$$f_\tau(s) = q(s) \exp(u(s), \tau). \tag{8.10}$$

Then, $\{f_\tau(s),\ s \in \mathcal{S}\}$ is a family of (strictly) positive distribution.

The functions u are called canonical statistics, and the parameters τ are called the canonical parameters. The components of the canonical statistic may be arranged into the design matrix, denoted as $\mathbf{U}$. Like in Chapter *LCDA* 7, the columns of the design matrix contain the components of the canonical statistic. Each component is a vector of length equal to the number of cells in the table. The cells of the table are arranged in a vector with an arbitrary but fixed, say lexicographic, order, and this defines the order in which the values of the canonical statistics are arranged within the columns of $\mathbf{U}$. That is, the number of rows of the design matrix is equal to the number of cells c in the table (the sample space), and the number of columns is equal to the number of components of the canonical statistic, say, l. The entry of the design matrix in the row that corresponds to the cell s and in the j-th column is $u_j(s)$. A special case of this setup was used earlier with the log-linear parameters.

If $l = c$ and the design matrix $\mathbf{U}$ contains c linearly independent columns, and thus it spans $\mathbb{R}^c$, a reparameterization in such a way that one of the columns is a vector of 1's is possible. The parameter associated with this column of the design matrix was called the overall effect above. Recall that both the balanced and the corner log-linear parameters have an overall effect.

[19] The notation here is unrelated to the one used earlier in the definitions of design matrices.

As noted in the proof of Theorem 8.8, as long as $\mathbf{U}$ has c columns and is of full (column) rank, (8.10) yields all (strictly) positive distributions on $\mathscr{S}$.

The distribution $q(s)$ in (8.10) is called the dominating measure. In more general approaches, $q(s) = 0$ for a particular $s \in \mathscr{S}$ implies that $f(s) = 0$, too. In the context of log-linear parameterizations, $f(s) = 0$ is excluded and so is $q(s) = 0$. With that assumption, the role of q is to allow models in which the conditional odds ratios are not set to be 1, as in log-linear models, rather they are set to values different from 1.

For example, in a 2×2 table, the non-redundant balanced log-linear parameters yield all (strictly) positive distributions through the following design matrix, see (8.5)[20]

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}. \tag{8.11}$$

The columns in this design matrix are the components of the canonical statistic, and the parameters are

$$(\lambda^{\emptyset},\ \lambda_1^A,\ \lambda_1^B,\ \lambda_{11}^{AB}).'$$

Then, the model of independence is obtained by setting, $\lambda_{11}^{AB} = 0$. This is a model definition as described in Theorem 8.4, and all the frequency distribution in this model may be obtained if the following design matrix is used in (8.10):

$$\mathbf{T}_{ind} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \\ 1 & -1 & -1 \end{pmatrix}. \tag{8.12}$$

Then, using

$$f = \mathbf{T}_{ind}(\lambda^{\emptyset},\ \lambda_1^A,\ \lambda_1^B)',$$

the odds ratio for all the distributions in the model is

$$\frac{f(1,1)f(2,2)}{f(1,2)f(2,1)} = \frac{q(1,1)q(2,2)}{q(1,2)q(2,1)},$$

[20] The design matrix given after *LCDA* (10.11) is obtained from that formula when the categories are denoted by 1 and 2, and yields this design matrix when the categories are denoted by 0 and 1.

that is, the odds ratio of q. For details and applications, see Rudas [144, 145], Rudas and Leimer [154], and for further background, the discussion of log-affine marginal models in Sect. 10.2 and the references given there.

In the sequel, unless stated explicitly otherwise, the distribution q will be assumed to be uniform, which implies that the odds ratio above is equal to 1. In this case, the constant value of $q(s)$ may be absorbed in the overall effect, and q may be omitted from (8.10), yielding the formula (8.8) used in Theorem 8.8.

To account for the fact that the sum of the cell probabilities is 1, a related parameterization of all (strictly) positive probability distributions on $\mathscr{S}$, with the following form of the definition of the exponential family, is used. Let

$$p_\theta(s) = q(s)a(\theta)\exp(t(s), \theta), \tag{8.13}$$

with $c = |\mathscr{S}|$ linearly independent and c-dimensional statistics $t(s)$ and parameter $\theta \in \mathbb{R}^c$, where

$$a(\theta) = \frac{1}{\sum_{s\in\mathscr{S}} q(s)\exp(t(s), \theta)}.$$

Then, $\{p_\theta(s),\ s \in \mathscr{S}\}$ is a probability distribution, indeed. As all probability distributions may be obtained by the normalization of a (general) distribution, (8.13) yields all probability distributions on $\mathscr{S}$.

Although θ in (8.13) is often called a parameter, in fact, it is not a parameter in the sense that it could be determined if p_θ on the left-hand side of (8.13) was given. To illustrate this, let $|\mathscr{S}| = 2$ and let the design matrix be the 2×2 unit matrix. Then, whenever $\theta_1 = \theta_2$, one gets the same uniform distribution $p_1 = p_2 = 0.5$. One may say that θ in (8.13) is a parameterization but not a parameter. Note that this is the case, even though the design matrix is of full rank.

In addition to applying normalization, as in (8.13), there is another way of obtaining probability distributions from the definition (8.10). This is to restrict the parameter space in such a way that, when all other non-redundant log-linear (or other) parameters are given, the only value of the overall effect would be the one that makes the sum of the frequencies equal to 1, that is, which gives a probability distribution. In this case, one still does not have parameters and the parameter space would have a complex structure. In fact, it would not be an open set in $\mathbb{R}^c$.

Another alternative is to use (8.10) to parameterize the probability distributions with the log-linear (or other) parameters derived from probability distributions. If $\mathscr{P}$ is the collection of all (strictly) positive probability distributions on $\mathscr{S}$, then consider

$$T = \{\tau(p) :\ p \in \mathscr{P}\},$$

that is, parameters, that can be obtained from a $p \in \mathscr{P}$ and use (8.10) with the restriction that $\tau \in T$. In this case, the τ's are parameters, but in this generality, T is not an open set in the c-dimensional space.[21]

In other words, (8.13) gives a probability distribution for every $\theta \in \mathbb{R}^c$, but (8.10) gives a probability distribution if and only if $\tau \in T$. The advantage of the second definition is that in this case τ is a parameter, and this is true more generally than only in the case of log-linear parameters. Indeed, when the design matrix is of full rank in a minimal parameterization, one has the following.

Theorem 8.9 *For the exponential family (8.10), $\{p_\tau, \tau \in T\}$, with*

$$p_\tau(s) = q(s) \exp(u(s), \tau), \tag{8.14}$$

if the design matrix $\mathbf{U}$ *of size $c \times c$ and is minimal and of full rank, then τ is a parameter and a parameterization.*

Proof When $\tau \in T$ is given, the probability distribution p_τ may be determined using (8.14). One needs to see that this function is invertible on T, that is, it is also a parameter.

Assume, to the contrary, that for two different τ values in T, say τ_1 and τ_2, the related probability distributions are the same:

$$p_{\tau_1}(s) = p_{\tau_2}(s) \text{ for all } s \in \mathscr{S}.$$

If this was true, then one would obtain that

$$\exp(u(s), \tau_1) = \exp(u(s), \tau_2),$$

for every s by positivity of $q(s)$, or that

$$u(s)'\tau_1 = u(s)'\tau_2$$

for every s, where $u(s)'$ denotes the row[22] of the design matrix associated with cell s. Thus, one would have

$$u(s)'(\tau_1 - \tau_2) = 0,$$

for every s, or

$$\mathbf{U}(\tau_1 - \tau_2) = 0, \tag{8.15}$$

where the zero denotes a vector consisting of zeros.

[21] See also the discussion around (9.14).

[22] The notation does not refer to any transpose; it is used to emphasize that $u(s)'$ is a row, as opposed to other vectors that are considered to be columns.

As the design matrix (that consists of the components of the canonical statistic) is of full column rank, and it is a square matrix, it has to have full row rank. This follows from the construction in the proof Theorem 8.2, because if $\mathbf{U}$ did not have full row rank, then the family consisting of the distributions[23]

$$\log \frac{p_\tau(s)}{q(s)}$$

could be parameterized with fewer than c parameters, but then $\mathbf{U}$ would not be minimal. Thus, (8.15) is only possible if

$$\tau_1 = \tau_2.$$

This shows that the inverse of the function that determines the probabilities, given the canonical statistics, exists. This inverse is a parameter of the distributions in the exponential family.

To obtain an expression for τ, when p_τ is given, consider:

$$\log \frac{p_\tau(s)}{q(s)} = u(s)'\tau,$$

for every $s \in \mathscr{S}$ where, again, $u(s)'$ is the row of $\mathbf{U}$ associated with s. This is a system of linear equations for the components of τ, where the matrix of coefficients is $\mathbf{U}$:

$$\log(p_\tau/q) = \mathbf{U}\tau,$$

where the logarithm is applied componentwise and p_τ/q is a vector with components $p_\tau(s)/q(s)$. As $\mathbf{U}$ is of full rank, the system of linear equations has a unique solution

$$\tau = \mathbf{U}^{-1} \log(p_\tau/q)$$

which yields τ.

□

For a particular type of design matrices, the reparameterization connecting the parameterization in (8.13) to the one in (8.14) is obtained easily.

Theorem 8.10 *Let $c = |\mathscr{S}|$, and the $c \times c$ matrix $\mathbf{T}$ be such that the c-dimensional vector of 1s is a linear combination of its columns. Then, the parameterization (8.14) is obtained from the parameterization (8.13) with $\mathbf{U} = \mathbf{T}$ and*

$$T = \{\tau : \tau_j = \alpha_j \log(a(\theta)) + \theta_j,\ \ j = 1, \ldots, k,\ \ \theta \in \mathbb{R}^c\},$$

[23] The function log is the inverse of the exp function, that is, the natural logarithm function.

where the coefficients α_j are such that for the vector of 1*s*

$$1 = \sum_{j=1}^{k} \alpha_j t_j,$$

where t_j is the jth column of $\mathbf{T}$.

Proof One needs to see that every distribution, which may be written as

$$p_\theta(s) = q(s)a(\theta)\exp(t(s), \theta)$$

for some $\theta \in \mathbb{R}^c$, may also be written as

$$r_\tau(s) = q(s)\exp(t(s), \tau(\theta)),$$

for some parameter $\tau(\theta)$ depending on θ, and the collection of these parameters for $\theta \in \mathbb{R}^c$ is T.

As the column vector 1 (having a 1 in every position) is a linear combination of the column vectors t_j of $\mathbf{T}$, there are coefficients such that

$$1 = \sum_{j=1}^{k} \alpha_j t_j,$$

or that for every $s \in \mathscr{S}$,

$$1 = \sum_{j=1}^{k} \alpha_j t_j(s),$$

where 1 stands for the number. Then,

$$p_\theta(s) = q(s)a(\theta)\exp(t(s), \theta) = q(s)\exp(\log(a(\theta)) + (t(s), \theta))$$

$$= q(s)\exp\left(\log(a(\theta))\sum_{j=1}^{k}\alpha_j t_j(s) + \sum_{j=1}^{k} t_j(s)\theta_j\right)$$

$$= q(s)\exp\left(\sum_{j=1}^{k}(\alpha_j \log(a(\theta)) + \theta_j)t_j(s)\right)$$

$$= q(s)\exp\left(\tau(\theta), (t(s))\right),$$

with

$$\tau_j(\theta_j) = \alpha_j \log(a(\theta)) + \theta_j.$$

Thus,

$$T = \{\tau : \tau_j = \alpha_j \log(a(\theta)) + \theta_j,\ \ j = 1, \ldots, k,\ \ \theta \in \mathbb{R}^c\}.$$

□

Note that in Theorem 8.10, no assumption was made for the rank of $\mathbf{T}$. However, if $\mathbf{T}$ is of full rank, then the condition of the theorem holds. This includes the case when the design matrix $\mathbf{T}$ has an overall effect, that is, contains a column of 1s, just like the design matrix $\mathbf{T}^\sim$ for the non-redundant log-linear parameters $\nu^\sim$. The reparameterization given in the proof above is sometimes referred to as absorbing the normalization in the parameterization.

Let now $\mathbf{T}^*$ denote the $c \times (c-1)$ matrix obtained from $\mathbf{T}^\sim$ by omitting the column of 1s. As the $c \times c$ matrix $\mathbf{T}^\sim$ is of full rank (see Theorem 8.8), there is no linear combination of its c columns, which would give the zero vector, except if all coefficients are equal to zero. This, of course, also applies to its $c-1$ columns retained by $\mathbf{T}^*$. Thus, one has the following.

Corollary 8.3 *The $c \times (c-1)$ matrix $\mathbf{T}^*$ is of full rank, that is, its rank is $c-1$.*

□

Let also ν^* denote the parameters obtained from $\nu^\sim$ by omitting the overall effect, that is, the parameter belonging to the empty cell in the case of the balanced parameters and the effect belonging to the cell with all indices equal to 1 in the case of the corner parameters. With this design matrix, one still has a parameterization of all probability distribution on $\mathscr{S}$ when the formula in (8.13), that is, the one with normalization, is used. Indeed, the overall effect may be absorbed in the normalization.

Theorem 8.11 *The distributions P in the family of (strictly) positive probability distributions $\mathscr{P}$ on $\mathscr{S}$ may be written with the log-linear parameters as*

$$P(s) = q(s)b(\nu^*)\exp(t^*(s), \nu^*),\ \ \nu^* \in \mathbb{R}^{|\mathscr{S}|-1}.$$

Proof Theorem 8.8 and, specifically, (8.9) say that

$$f = \exp(\mathbf{T}^\sim \nu^\sim)$$

is a smooth parameterization of all distributions on $\mathscr{S}$, and then the normalized version as in (8.13)

$$P(s) = q(s)a(\nu^\sim)\exp(t^\sim(s), \nu^\sim),$$

where $t^{\sim}(s)$ is the row[24] of $\mathbf{T}^{\sim}$ associated with cell s, is a parameterization of all probability distributions on $\mathscr{S}$, using the normalization $a(\nu^{\sim})$ defined after (8.13).

One needs to see that the overall effect in $\nu^{\sim}$ may be absorbed in the normalization and then, instead of $\mathbf{T}^{\sim}$, $\mathbf{T}^{*}$ may be used. Indeed, $\mathbf{T}^{*}$ is obtained from $\mathbf{T}^{\sim}$ by omitting its first column that consists of 1s in every row, and ν^{*} is obtained from $\nu^{\sim}$ by omitting its first component, say $\nu_1^{\sim}$. Thus,

$$\mathbf{T}^{\sim}\nu^{\sim} = (\nu_1^{\sim}) + \mathbf{T}^{*}\nu^{*},$$

where $(\nu_1^{\sim})$ is a vector containing of $\nu_1^{\sim}$ that is the first component of $\nu^{\sim}$ in every position.

Then, (8.13) may be written as

$$\begin{aligned} P(s) &= q(s)a(\nu^{\sim})\exp(t^{\sim}(s), \nu^{\sim}) \\ &= q(s)a(\nu^{\sim})\exp(\nu_1^{\sim} + (t^{*}(s), \nu^{*})) \\ &= q(s)a(\nu^{\sim})\exp(\nu_1^{\sim})\exp(t^{*}(s), \nu^{*}) \\ &= q(s)b(\nu^{*})\exp(t^{*}(s), \nu^{*}). \end{aligned}$$

Here, $a(\nu^{\sim})\exp(\nu_1^{\sim})$ depends on $\nu^{\sim}$, but instead of this expression, the term $b(\nu^{*})$ appears, which depends on the last $c-1$ components of $\nu^{\sim}$ that is on ν^{*}. This is correct, as

$$\begin{aligned} 1 = \sum_{s\in\mathscr{S}} P(s) &= \sum_{s\in\mathscr{S}} q(s)a(\nu^{\sim})\exp(\nu_1^{\sim})\exp(t^{*}(s), \nu^{*}) \\ &= a(\nu^{\sim})\exp(\nu_1^{\sim}) \sum_{s\in\mathscr{S}} q(s)\exp(t^{*}(s), \nu^{*}) = 1, \end{aligned}$$

thus

$$a(\nu^{\sim})\exp(\nu_1^{\sim}) = \frac{1}{\sum_{s\in\mathscr{S}} q(s)\exp(t^{*}(s), \nu^{*})},$$

therefore, $a(\nu^{\sim})\exp(\nu_1^{\sim})$ depends on ν^{*}, and may be written as $b(\nu^{*})$.

[24] Earlier in this section, the row of the matrix $\mathbf{U}$ associated with cell $s \in \mathscr{S}$ was denoted as $u(s)'$, to emphasize that it was a row vector although it was not the transpose of a vector $u(s)$. Such a notation is not used any more.

To complete the proof, it needs to be seen that in the parameterization obtained, the range of ν^* is $\mathbb{R}^{c-1}$, where $c = |\mathscr{S}|$. To see this, note that the domain of $\nu^{\sim}$ in (8.8) is $\mathbb{R}^c$, as the range of $\log f$ is $\mathbb{R}^c$, too. Further, all the non-redundant log-linear parameters in $\nu^{\sim}$, with the exception of the overall effect, are such that they are invariant against the division of all the frequencies by the same value, including the normalization that transforms the (general) distribution on $\mathscr{S}$ into a probability distribution. This follows immediately from the definition of the log-linear parameters, whether balanced or corner parameters are used, because they are functions of ratios of (strictly) positive frequencies[25] or probabilities. Thus, when the ν^* parameters are calculated from

$$f = \exp(\mathbf{T}^{\sim}\nu^{\sim}),$$

that is, from

$$f(s) = \exp(t^{\sim}(s), \nu^{\sim})$$

or from

$$P(s) = q(s)a(\nu^{\sim})\exp(\nu_1^{\sim})\exp(t^*(s), \nu^*),$$

one obtains the same ν^* parameters and their range is $\mathbb{R}^{c-1}$, and all such parameters yield a probability distribution using

$$P(s) = q(s)b(\nu^*)\exp(t^*(s), \nu^*).$$

□

Note that the domains of the parameters ν^* and of $\nu^{\sim}$ contain all parameters, for which $\{P(s),\ s \in \mathscr{S}\}$ is a probability distribution. However, the two domains are different, of course.

The previous results mean that whether one uses a design matrix with an overall effect but no normalization factor as in (8.10) or a design matrix with the overall effect omitted but with a normalizing factor as in (8.13), with an appropriate choice of the parameters, one obtains the same family. The advantage of the second approach is that also for probability distributions, the $\mathbf{T}^*$ $c \times (c-1)$ design matrix obtained after the overall effect was omitted is of full rank, and the domain of the parameterization is $\mathbb{R}^{c-1}$, that is, unrestricted. For more results relating to the role of the overall effect in exponential families, see Klimova and Rudas [84].

[25] Recall that in the case of a general distribution, the frequencies may be equal to any real number.

8.5 Summary

This chapter dealt with the smoothness of parameterizations, which in turn implies the smoothness of the resulting manifolds. Most of the statistical models used in practice, at least in the parametric approach to statistical analysis, are smooth manifolds. Smoothness not only implies the applicability of standard asymptotic theory[26] but also justifies the description of the observed and estimated distributions through the interpretation of their parameters. This makes the choice and precise understanding of the parameters and parameterization of central importance.

A further reason why smooth parameterizations are so important is that they offer a straightforward way of defining smooth statistical models. As was proved, if a family of distributions (a model or all the distributions) has a smooth parameterization, and the values of some of the parameters that are variation independent from the other parameters are fixed at one of their values, then the resulting model is also smooth. This is exactly the case with all log-linear models.[27]

Log-linear parameterizations were discussed in detail. In the case of the balanced log-linear parameters, special attention was paid to the redundancy among the parameters, and, in the case of probability distributions, to the complications resulting from the fact that they sum to 1.

It was proved that when there are c categories in the sample space, all (strictly) positive distributions are a smooth manifold of dimension c, and all (strictly) positive probability distributions are a smooth manifold of dimension $c - 1$. When using the non-redundant log-linear parameters $\Lambda^{\sim}$, for (strictly) positive general distributions, the $c \times c$ full rank design matrix $\mathbf{T}^{\sim}$ was developed for both the balanced and the corner log-linear parameters, and for (strictly) positive probability distributions, the $c \times (c-1)$ $\mathbf{T}^{*}$ design matrix was developed, when the overall effect is replaced with normalization. For the (strictly) positive probability distributions, the columns of $\mathbf{T}^{*}$ may be used, with normalization, as canonical statistics in an exponential family representation.

Although the contents of this chapter may not appear to have direct applications in the practice of data analysis, they will be used in the next chapter to prove the smoothness of log-linear, mean value, mixed, and also marginal log-linear parameterizations.

8.6 Things to Do

1. Prove that $\{(x, x^2) : \ 0 < x < 1\}$ is a 1-dimensional smooth manifold.
2. What is the range of (p_1, p_2) in a probability distribution for three categories? Prove that this is a smooth manifold.

[26] This aspect was not proved in this chapter.

[27] That log-linear parameterizations are smooth will only be proved in the next chapter.

3. What is the range of (p_1, p_2, p_3) in a probability distribution for four categories? Prove that this is a smooth manifold.
4. Why was in Proposition 8.1 (strict) positivity assumed?
5. In the example with the parameterization (8.2), not all the conditions of being a smooth manifold were checked. Prove that the remaining conditions hold.
6. Let $f = \mathbf{A}\alpha$ and let the first d rows of $\mathbf{A}$ be linearly independent, and let the remaining rows be linear combinations of them. Prove that, in this case, f cannot be parameterized with fewer than d parameters.
7. Is the previous claim true for probability distributions p instead of general distributions f?
8. Is Theorem 8.1 true for probability distributions, instead of general distributions?
9. Prove Proposition 8.2.
10. Show that if a linear parameterization of a manifold $f = \mathbf{A}\alpha$ is not minimal, then α is not a parameter.
11. In the proof of Theorem 8.8, the transformation of exponentiating all components of a vector was used. Prove that this is a transformation to which Theorem 8.6 applies.
12. Prove that the expression (8.13) gives a probability distribution for all $\theta \in \mathbb{R}^k$.
13. The proof of Theorem 8.11 used the fact that all log-linear parameters with the exception of the overall effect are invariant against the division of all frequencies by the same number. Why is this so?

Chapter 9
Smoothness of Marginal Log-Linear Parameterizations



9.1 Introduction

This chapter considers the issue of smoothness of log-linear, mean value, mixed, and marginal log-linear parameters and relies heavily on the concepts and results presented in the previous chapter.

A detailed discussion of the smoothness of log-linear, marginal, and mixed parameterizations is given in Sect. 9.2. General and probability distributions are parameterized as members of exponential families, already introduced in Chapter *LCDA* 7, using the design matrices developed in Sect. 8.4. These results are special cases of the corresponding results for exponential families (see, e.g., Barndorff-Nielsen [10]), but in the setup considered here, relatively simple and intuitive proofs may be given. The results are discussed first in the case of general distributions (Sect. 9.2.1), and then they are shown to apply to probability distributions, as well (Sect. 9.2.2).

The main result of the chapter is the smoothness of marginal log-linear parameterizations, presented in Sect. 9.3. The proof follows the one given in Bergsma and Rudas [20], but is presented in a detailed and rigorous way. The argument is based on a mixed parameterization and then a series of reparameterizations applied to it, which bring in the marginal log-linear parameters step by step, and retain smoothness.

This smoothness will be used in the next chapter to imply the smoothness of marginal log-linear models, defined using the method described in Theorem 8.4.

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9.2 Smoothness of Log-Linear, Mean Value, and Mixed Parameterizations

Creating full rank parameterizations and achieving smoothness results are a bit more difficult for probability than for general distributions because the former have to sum to 1. This also applies to the exponential family framework discussed in Sect. 8.4; therefore, the case of general distributions is considered first in Sect. 9.2.1 and then the case of probability distributions in Sect. 9.2.2. In both cases, results about the smoothness of log-linear, of mean value, and of mixed parameterizations are proved. These are true for general exponential families (see Barndorff-Nielsen [10]), but in the categorical case considered here, simple and intuitive proofs may be given.

9.2.1 General Distributions

In order to obtain simple results for the smoothness of log-linear and related parameterizations, first parameterizations of distributions are investigated. Recall that general distributions are allowed to consist of any positive real values associated with the cells of the sample space. The exponential family representation in (8.10) is used with the parameters and parameterizations $\nu^{\sim} \in \mathbb{R}^c$ and design matrix $\mathbf{T}^{\sim}$.

The next result gives the Jacobian of this parameterization.

Theorem 9.1 *For the family in (8.10), with the design matrix* $\mathbf{T}^{\sim}$ *and parameters* $\nu^{\sim}$*, the Jacobian is*

$$\mathbf{D}_f \mathbf{T}^{\sim},$$

where $\mathbf{D}_f$ *is a* $c \times c$ *diagonal matrix with the components of* $f_{\nu^{\sim}}$ *in the main diagonal and zeros elsewhere. Further, the Jacobian is of full rank.*

Proof The partial derivative of the component of $f_{\nu^{\sim}}$ associated with $s \in \mathscr{S}$, $f_{\nu^{\sim}}(s)$ according to $\nu_j^{\sim}$, $j = 1, \ldots, c$, is

$$\frac{\partial f_{\nu^{\sim}}(s)}{\partial \nu_j^{\sim}} = q(s) exp(t^{\sim}(s), \nu^{\sim}) t_j^{\sim}(s) = f_{\nu^{\sim}}(s) t_j^{\sim}(s).$$

In matrix notation, the Jacobian is $\mathbf{T}^{\sim}$ with every row multiplied by $f_{\sim}(s)$, or

$$\mathbf{D}_f \mathbf{T}^{\sim}.$$

To see that the Jacobian is full rank, suppose, to the contrary, that there exists a linear combination of its column vectors that gives the zero vector but not all the coefficients are zero, that is,

$$\sum_{j=1}^{c} \alpha_j f(s)_{\nu^\sim} t_j^\sim(s) = 0,$$

for all s. But then one would have, as $f(s)_{\nu^\sim} \neq 0$ that

$$\sum_{j=1}^{c} \alpha_j t_j^\sim(s) = 0,$$

for all s, but that contradicts the linear independence of the columns of the design matrix, see Theorem 8.8.

□

As the other conditions of smoothness are obvious, the previous theorem gives an alternative proof of the fact that the log-linear parameters $\nu^\sim$ provide a smooth parameterization of the (strictly) positive frequency distributions, as was already seen in Theorem 8.8.

The Jacobian of the inverse transformation is given next.

Theorem 9.2 *For the family in (8.10), with the design matrix* $\mathbf{T}^\sim$ *and parameters* $\nu^\sim$*, the Jacobian of the parameter as a function of the frequencies is*

$$(\mathbf{T}^\sim)^{-1}\mathbf{D}_{1/f},$$

where $\mathbf{D}_{1/f}$ *is a* $c \times c$ *diagonal matrix with the entries*

$$1/f_{\nu^\sim}(1), 1/f_{\nu^\sim}(2), \ldots, 1/f_{\nu^\sim}(c)$$

in the main diagonal and zeros elsewhere.

Further, the Jacobian is of full rank.

Proof To obtain the inverse of the transformation in (8.10) with $\mathbf{T}^\sim$ and $\nu^\sim$, write:

$$\log \frac{f}{q} = \mathbf{T}^\sim \nu^\sim,$$

where log is the inverse of the exponential function and is applied componentwise, f/q is a vector with components $f(s)/q(s)$ and $\nu^\sim$ is interpreted as a vector.

As $\mathbf{T}^\sim$ is of full rank, it has an inverse, thus,

$$(\mathbf{T}^\sim)^{-1} \log \frac{f}{q} = \nu^\sim$$

and, consequently, for every component of the parameter vector (indexed with j) and for components of the frequency vector (indexed with s), one has

$$\nu_j^{\sim} = \sum_s ((\mathbf{T}^{\sim})^{-1})_{(j,s)} \log \frac{f(s)}{q(s)},$$

where the term after the summation sign denotes the entry of the inverse of the design matrix in the position (j, s).

Therefore, the partial derivative matrix has the following entries:

$$\frac{\partial \nu_j^{\sim}}{\partial f_{\nu^{\sim}}(s)} = (\mathbf{T}^{\sim})^{-1})_{(j,s)} \frac{q(s)}{f(s)} \frac{1}{q(s)}.$$

Written in matrix form, one has

$$(\mathbf{T}^{\sim})^{-1} \mathbf{D}_{1/f}.$$

As the matrix $(\mathbf{T}^{\sim})^{-1}$ has an inverse, namely, $\mathbf{T}^{\sim}$, it is of full rank. The rest of the proof of the Jacobian being of full rank is like in Theorem 9.1.

□

Note that the two Jacobians obtained in Theorems 9.1 and 9.2 are inverses of each other, just like as is implied by the general result referred to in the proof of Proposition 8.2, namely, that the Jacobian if the inverse function is the inverse of the Jacobian, see, e.g., Lax [103]. Indeed,

$$(\mathbf{T}^{\sim})^{-1} \mathbf{D}_{1/f} \mathbf{D}_f \mathbf{T}^{\sim} = (\mathbf{T}^{\sim})^{-1} \mathbf{I} \mathbf{T}^{\sim}$$

$$= (\mathbf{T}^{\sim})^{-1} \mathbf{T}^{\sim} = \mathbf{I},$$

where $\mathbf{I}$ is the $c \times c$ identity matrix.

Now we turn to the discussion of the mean value parameters in the exponential family representation of the distributions in (8.10). The mean value parameters are defined as in the case of probability distributions, namely, the weighted averages of the canonical statistics:[1]

$$E_f(t_j^{\sim}) = \sum_s f(s) t_j^{\sim}(s),$$

or in matrix notation

$$E_f(t^{\sim}) = (\mathbf{T}^{\sim})' f, \tag{9.1}$$

[1] The same notation is used as in the case of probability distributions.

where $t^{\sim}$ is the vector with components identified with the components of the canonical statistic[2] and f is the vector of frequencies.

Clearly, $E_f(t^{\sim})$ are parameters of f. They are also a parameterization, which is seen directly from

$$f = ((\mathbf{T}^{\sim})')^{-1} E_f(t^{\sim}); \tag{9.2}$$

see also the discussion before and after Proposition *LCDA* 7.1. Note that as $\mathbf{T}^{\sim}$ is of full rank, so is its transpose, thus, the inverse above exists.

The mean value parameters defined in (9.1) are non-redundant in the sense that, in general, none of them is obtained as a linear combination of the others.

Theorem 9.3 *The mean value parameters are linearly independent in the sense that no vector of constants α exists, such that not all of its components are zero but*

$$\alpha' E_f(t^{\sim}) = 0,$$

for all distributions f.

Proof Indeed, if such coefficients existed, then one would have

$$\alpha' E_f(t^{\sim}) = \alpha'(\mathbf{T}^{\sim})' f = (\alpha'(\mathbf{T}^{\sim})') f = 0,$$

for all distributions f. This could only happen if one had

$$\alpha'(\mathbf{T}^{\sim})' = 0,$$

but that is not possible, as $(\mathbf{T}^{\sim})'$ is of full rank.

□

Of course, the result above (and also its proof) is equivalent to the mean value parameters being a parameterization, see above.

As the mean value parameters and the frequencies are linked to each other via linear transformations, the Jacobians are obtained easily.

Theorem 9.4 *The Jacobian of the frequencies, when the mean value parameters are used to parameterize them as in (9.2), is*

$$((\mathbf{T}^{\sim})')^{-1},$$

[2] Recall that the components of the canonical statistics are functions defined on the table $\mathscr{S}$ and the vector $t^{\sim}$ has one component for each of these functions.

and the Jacobian of the inverse transformation, when the frequencies are used to determine the mean value parameters as in (9.1), is

$$(\mathbf{T}^{\sim})'.$$

Further, both Jacobians are of full rank.

□

Mixed parameterizations may be defined for general distributions similarly to the definition for probability distributions given in Sections $LCDA$ 7.2 and 10.1. To define a mixed parameterization, recall that the parameters (log-linear or mean value) are all related to subsets of variables. In particular, for every subset W of the variables, there is a column j_w of the design matrix $\mathbf{T}^{\sim}$ associated with all such combinations w of the indices of the variables that either no variable is in its last category (in the case of the balanced non-redundant parameters) or no variable is in its first category (in the case of the corner parameters).

In a mixed parameterization, the mean value parameters are used for a descending class of subsets of the variables, and the canonical parameters are used for the complement ascending class of subsets. Similarly to the case of probability distributions, the convergence of the $IPFP$ (see Section $LCDA$ 12.2), with a proof identical to the one given there, implies that these parameters constitute a parameterization. It also implies that the mean value parameters and the canonical parameters used in a mixed parameterization are variation independent. The next result is about the Jacobian of the mixed parameterization.

Theorem 9.5 *The Jacobian of a mixed parameterization of the (strictly) positive distributions is of full rank.*

Proof Let $\mathscr{D}$ be a descending class of subsets of the variables, and let $\mathscr{A}$ be the complement ascending class. In the mixed parameterization, the mean value parameters are used for subsets $W \in \mathscr{D}$, and the canonical parameters are used for subsets $W \in \mathscr{A}$. Thus, the mixed parameterization is

$$\{E_f(t_j^{\sim}),\ \text{if } W_j \in \mathscr{D},\ \text{and } v_j^{\sim},\ \text{if } W_j \in \mathscr{A},\ \ j = 1, \ldots, c\}, \tag{9.3}$$

where, as before, W_j denotes the subset of variables with which the j-th column of the design matrix is associated.

The partial derivatives of $f(s)$, according to the canonical parameters and according to the mean value parameters, were given in Theorems 9.1 and 9.4. In the case of a mixed parameterization, the column j of the Jacobian is column j from the respective Jacobians, depending on whether the canonical or the mean value parameters are used for the set of variables W_j.

Thus, the column j of the Jacobian of the mixed parameterization is equal to

column j of $((\mathbf{T}^{\sim})')^{-1}$ if $W_j \in \mathscr{D}$,

or to

column j of $\mathbf{D}_f \mathbf{T}^{\sim}$ if $W_j \in \mathscr{A}$.

As for any square matrix $\mathbf{A}$

$$(\mathbf{A}')^{-1} = (\mathbf{A}^{-1})',$$

the composition of the Jacobian may be written as

column j of $((\mathbf{T}^{\sim})^{-1})'$ if $W_j \in \mathscr{D}$,

or

column j of $\mathbf{D}_f \mathbf{T}^{\sim}$ if $W_j \in \mathscr{A}$.

Further, as the columns of a matrix are the rows of its transpose, one also has for the composition of the columns of the Jacobian

transpose of the row j of $(\mathbf{T}^{\sim})^{-1}$ if $W_j \in \mathscr{D}$,

or

column j of $\mathbf{D}_f \mathbf{T}^{\sim}$ if $W_j \in \mathscr{A}$.

Let $\mathbf{M}$ denote the Jacobian of the mixed parameterization (9.3) and let m_j its j-th column. One needs to see that this matrix is of full rank. That is, no column m_j may be written as a linear combination of the other columns. Suppose, to the contrary, that there is a column, say column j', $1 \leq j' \leq c$, which is a linear combination of the other columns:

$$m_{j'} = \sum_{j \neq j'} \alpha_j m_j,$$

for suitable coefficients α_j which cannot be all equal to zero.

This can be written as

$$m_{j'} = \left(\sum_{j \neq j', W_j \in \mathscr{D}} \alpha_j m_j + \sum_{j \neq j', W_j \in \mathscr{A}} \alpha_j m_j \right)$$

$$= \left(\sum_{j \neq j', W_j \in \mathscr{D}} \alpha_j (r^j)' + \sum_{j \neq j' W_j \in \mathscr{A}} \alpha_j \mathbf{D}_f c_j \right),$$

where r^j is the j-th row[3] of $(\mathbf{T}^{\sim})^{-1}$ and c_j is the j-th column of $\mathbf{T}^{\sim}$.

Here, $W_{j'}$ is either in $\mathscr{D}$ or in $\mathscr{A}$. In the former case, rewrite the equation above as

$$m_{j'} - \sum_{j \neq j', W_j \in \mathscr{D}} \alpha_j (r^j)' = \sum_{j: W_j \in \mathscr{A}} \alpha_j \mathbf{D}_f c_j, \tag{9.4}$$

[3] An upper index is used to emphasize the fact that r is a row vector, as opposed to the column vectors used otherwise.

where the expression on the left-hand side is a linear combination of column vectors that are taken from the rows of the matrix $(\mathbf{T}^{\sim})^{-1}$ and the right-hand side is a linear combination of column vectors taken from the matrix $\mathbf{D}_f\mathbf{T}^{\sim}$. Also, the indices of the rows which appear as columns on the left-hand side are different from the indices of the vectors on the right-hand side, because for every j, W_j is either in $\mathscr{D}$ or in $\mathscr{A}$.

Then, consider the vectors:

$$x = m_{j'} - \sum_{j \neq j', W_j \in \mathscr{D}} \alpha_j (r^j)' \text{ and } y = \sum_{j: W_j \in \mathscr{A}} \alpha_j c_j.$$

We show that x and y are orthogonal to each other and thus are not equal.[4] First note that any of the vectors $m_{j'}$ and those denoted as $(r^j)'$ in x are orthogonal to any of the vectors denoted as c_j in y, because the latter are columns of a matrix and the former are rows of the inverse of this matrix, in positions all different from the positions of the columns. Then, the inner product (x, y) is zero. To see this, define $\alpha_{j'} = -1$ and write:

$$(x, y) = \left(\left(m_{j'} - \sum_{j \neq j', W_j \in \mathscr{D}} \alpha_j (r^j)' \right), \left(\sum_{k: W_k \in \mathscr{A}} \alpha_k c_k \right) \right) \tag{9.5}$$

$$= \left(\left(- \sum_{j: W_j \in \mathscr{D}} \alpha_j (r^j)' \right), \left(\sum_{k: W_k \in \mathscr{A}} \alpha_k c_k \right) \right)$$

$$= - \sum_s \sum_{j: W_j \in \mathscr{D}} \sum_{k: W_k \in \mathscr{A}} \alpha_j r^j(s) \alpha_k c_k(s)$$

$$= - \sum_{j: W_j \in \mathscr{D}} \sum_{k: W_k \in \mathscr{A}} \alpha_j \alpha_k \sum_s r^j(s) c^k(s) = 0,$$

as the last sum is the inner product of r^j and of c^k, which is zero.

Return now to the indirect assumption written in the form of Eq. (9.4). Suppose one has α_j for $j = 1, \ldots, c$, and let us see with what choice of the distribution f may the equation hold. To simplify notation, write (9.4) as

$$x = \mathbf{D}_f y.$$

[4] If x and y were equal, then their inner product would be positive and not zero. Note that neither x nor y may be a zero vector, because the Jacobians of the mean value and of the canonical parameterizations are of full rank.

Note that because $\mathbf{D}_f$ is a diagonal matrix, the right-hand side of (9.4) may, indeed, be written as

$$\sum_{j:W_j\in\mathscr{A}} \alpha_j \mathbf{D}_f c_j = \mathbf{D}_f \sum_{j:W_j\in\mathscr{A}} \alpha_j c_j = \mathbf{D}_f y.$$

According to (9.5), the vectors x and y are orthogonal, and neither of them can be zero vectors, because both $\mathbf{T}^{\sim}$ and its inverse are of full rank, thus, a linear combination of their rows or columns cannot give a zero vector, unless all coefficients are zero which is excluded here. Thus, there is an s, such that

$$x(s)y(s) < 0;$$

otherwise (x, y) would be positive and not zero. For this s, the signs of $x(s)$ and $y(s)$ are different, and for any choice of the (strictly) positive $f(s)$, the signs of $x(s)$ and of $f(s)y(s)$ remain different, thus, the equality (9.4), which is, an equality of vectors and has to be true for every component s, cannot hold, as was to be seen.

In the case when $W_{j'} \in \mathscr{A}$, one has, instead of (9.4),

$$m_{j'} - \sum_{j\neq j', W_j\in\mathscr{A}} \alpha_j \mathbf{D}_f c_j = \sum_{j:W_j\in\mathscr{D}} \alpha_j r^j,$$

but otherwise, the proof goes the same way as in the case of $W_{j'} \in \mathscr{D}$: one shows that the two vectors on the left-hand side and on the right-hand side of this equation with $\mathbf{D}_f$ omitted are orthogonal and then that no choice of f may lead to the equality above, because the opposing signs are not altered by multiplication with the positive $f(s)$.

□

As the other conditions hold obviously, one has the following.

Corollary 9.1 *Every mixed parameterization of the (strictly) positive distributions on a contingency table using mean value parameters on a descending class of subsets of the variables and using log-linear parameters on the complement ascending class of subsets is smooth.*

□

This fact plays an important role in the proof result that marginal log-linear parameterizations are smooth, which is presented in the Sect. 9.3.

9.2.2 Probability Distributions

In this section, the smoothness results obtained in the previous subsection for general distributions in the case of log-linear, mean value, and mixed parameterizations are extended to probability distributions.

First, a general way of deriving the smoothness of a parameterization of probability distributions when a smooth parameterization of general distributions is given. The link between these smooth parameterizations is, however, not through normalization, as in (8.13), rather by taking certain ratios.

Theorem 9.6 *Let the (strictly) positive distributions $f \in \mathbb{R}^{c-1}$ with $c-1$ categories be smoothly parameterized as*

$$f = g(\theta), \ \theta \in A \subseteq \mathbb{R}^{c-1} \tag{9.6}$$

with a set A. Then, there exists a function $h : \mathbb{R}^{c-1} \to \mathbb{R}^c$, such that for the c-category (strictly) positive probability distributions

$$P = h(g(\theta)), \ \theta \in A \subseteq \mathbb{R}^{c-1} \tag{9.7}$$

is a smooth parameterization.

Proof For a contingency table with c cells or joint categories, and any (strictly) positive probability distribution p on it, define a $(c-1)$-dimensional vector f as

$$f_i = p_{i+1}/p_1, \ i = 1, \ldots, c-1.$$

Then, the vectors f are (strictly) positive and their range is unrestricted, that is, they are (strictly) positive distributions in $\mathbb{R}^{c-1}$. Further, the probability distribution p may be obtained from f as $p = h(f)$, defined as

$$p_1 = \frac{1}{1'f+1}, \ p_i = \frac{f_{i-1}}{1'f+1}, \ i = 2, 3, \ldots c, \tag{9.8}$$

where 1 is a $(c-1)$-dimensional vector of 1s.

Thus, the function yielding f as a function of p is one-to-one, and it and its inverse h given in (9.8) are obviously twice continuously differentiable. The domains of these functions are the (strictly) positive probability distributions on the table and the positive orthant in $\mathbb{R}^{c-1}$, respectively, both open sets.

We show next that the Jacobian of h

$$\frac{\partial P}{\partial f} \tag{9.9}$$

is of full rank. The Jacobian is a $c \times (c-1)$ matrix.

When $i = 1$, then for all j

$$\frac{\partial p_1}{\partial f_j} = \frac{-1}{(1'f+1)^2}.$$

For $i = 2, \ldots, c$, when $j = i - 1$, then

$$\frac{\partial p_i}{\partial f_{i-1}} = \frac{-f_{i-1} + 1'f + 1}{(1'f + 1)^2},$$

and when $j \neq i - 1$, then

$$\frac{\partial p_i}{\partial f_j} = \frac{-f_{i-1}}{(1'f + 1)^2}.$$

Thus, the Jacobian (9.9) is the following matrix:

$$C \begin{pmatrix} -1 & -1 & -1 & \ldots & -1 \\ -f_1 + 1'f + 1 & -f_1 & -f_1 & \ldots & -f_1 \\ -f_2 & -f_2 + 1'f + 1 & -f_2 & \ldots & -f_2 \\ -f_3 & -f_3 & -f_3 + 1'f + 1 & \ldots & -f_3 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ -f_{c-1} & -f_{c-1} & -f_{c-1} & \ldots & -f_{c-1} + 1'f + 1 \end{pmatrix},$$

with

$$C = \frac{1}{(1'f + 1)^2}.$$

To see that this matrix is of full column rank, let its j-th column be denoted as u^j, $j = 1, \ldots, c - 1$. If it was not of full column rank, then there would exist constants α_j, such that

$$\sum_{j=1}^{c-1} \alpha_j u^j = 0,$$

where 0 is a vector of 0-s, but not all α_j is equal to zero. From the first row of the Jacobian, it would then follow that

$$\sum_{j=1}^{c-1} \alpha_j = 0. \tag{9.10}$$

For the other rows $i = 2, \ldots, c$, one would have

$$\sum_{j=1}^{c-1} \alpha_j f_{i-1} - \alpha_{i-1}(1'f + 1) = 0,$$

or

$$f_{i-1} \sum_{j=1}^{c-1} \alpha_j - \alpha_{i-1}(1'f + 1) = 0.$$

Here, because of (9.10), the first term is zero, and $1'f + 1$ is positive. As at least one of the $\alpha_j = \alpha_{i-1}$ is not zero, the second term cannot be zero for all $i = 2, \ldots, c$.

Thus, the parameterization of the probability distributions using the general distributions (with one fewer categories) is smooth. Further, the parameterization of the $c-1$ dimensional distributions with $c-1$ dimensional unrestricted parameters θ using (9.6) is also smooth by assumption. The Jacobian of the composite function $h \circ g$ is the product of the Jacobians of h and of g by the chain rule, see, e.g., Lax [103]. The argument in the proof of Theorem 8.6 applies here, too, and implies that the Jacobian of $h \circ g$ is of rank $c-1$, that is, it is of full rank. As the other conditions are obviously fulfilled, the parameterization of the probability distributions with $\theta \in A \subseteq \mathbb{R}^{c-1}$ in (9.7) is also smooth.

□

The strength of this result is that it does not depend on a particular parameterization of the distributions f; it only requires the parameterization to be smooth. On the other hand, when applied to log-linear or mean value parameterizations of general distributions, it does not yield the same parameterizations for probability distributions.

The next result shows that log-linear parameterizations of (strictly) positive probability distributions are also smooth. For this, a variant of the parameterization (8.13) is used, such that the design matrix is $\mathbf{T}^*$, that is, the column of $\mathbf{T}^{\sim}$ containing 1's in every position is omitted. The remaining $|\mathscr{S}| - 1$ columns are denoted as t_j^* and a particular entry in this column is $t_j^*(s)$. Accordingly, the overall effect is omitted from the parameter, and ν^* is used. Then, (8.13) becomes:

$$P_{\nu^*}(s) = q(s)a(\nu^*) \exp\Big(\sum_{j=1}^{|\mathscr{S}|-1} t_j^*(s)\nu_j^* \Big), \tag{9.11}$$

with

$$a(\nu^*) = \frac{1}{\sum_{s \in \mathscr{S}} q(s) \exp\left(\sum_{j=1}^{|\mathscr{S}|-1} t_j^*(s)\nu_j^*\right)}.$$

Theorem 9.7 *In the parameterization of all (strictly) positive probability distributions on the contingency table $\mathscr{S}$ as given in (9.11), the (s, j) entry of the Jacobian is*

$$P_{\nu^*}(s) \left(t_j^*(s) - E_{P_{\nu^*}} t_j^* \right),$$

where $E_{P_{\nu^*}}$ denotes expectation according to P_{ν^*}. Further, the Jacobian is of full rank.

Proof For a fixed $\overline{s} \in \mathscr{S}$ and $\overline{j} \in \{1, \ldots, |\mathscr{S}| - 1\}$, the partial derivative of (9.11) is

$$\frac{\partial P_{\nu^*}(\overline{s})}{\partial \nu^*_{\overline{j}}}$$

$$= \frac{q(\overline{s}) \exp(\sum_{j=1}^{|\mathscr{S}|-1} t^*_j(\overline{s})\nu^*_j)\, t^*_{\overline{j}}(\overline{s}) \left(\sum_{s\in\mathscr{S}} q(s) \exp(\sum_{j=1}^{|\mathscr{S}|-1} t^*_j(s)\nu^*_j)\right)}{\left(\sum_{s\in\mathscr{S}} q(s) \exp(\sum_{j=1}^{|\mathscr{S}|-1} t^*_j(s)\nu^*_j)\right)^2}$$

$$- \frac{\left(q(\overline{s}) \exp(\sum_{j=1}^{|\mathscr{S}|-1} t^*_j(\overline{s})\nu^*_j)\right) \left(\sum_{s\in\mathscr{S}} q(s) \exp(\sum_{j=1}^{|\mathscr{S}|-1} t^*_j(s)\nu^*_j) t^*_{\overline{j}}(s)\right)}{\left(\sum_{s\in\mathscr{S}} q(s) \exp(\sum_{j=1}^{|\mathscr{S}|-1} t^*_j(s)\nu^*_j)\right)^2}$$

$$= P_{\nu^*}(\overline{s}) t^*_{\overline{j}}(\overline{s}) \sum_{s\in\mathscr{S}} P_{\nu^*}(s)$$

$$- P_{\nu^*}(\overline{s}) \sum_{s\in\mathscr{S}} P_{\nu^*}(s) t^*_{\overline{j}}(s)$$

$$= P_{\nu^*}(\overline{s}) \left(t^*_{\overline{j}}(\overline{s}) - E_{P_{\nu^*}} t^*_{\overline{j}}\right).$$

The Jacobian is a matrix of size $|\mathscr{S}| \times (|\mathscr{S}|-1)$. One needs to see that its columns are linearly independent. Suppose, to the contrary, that coefficients $\alpha_1, \ldots, \alpha_{|\mathscr{S}|-1}$ exist, such that not all of them are equal to 0 and

$$\sum_{j=1}^{|\mathscr{S}|-1} \alpha_j P_{\nu^*}(s) \left(t^*_j(s) - E_{P_{\nu^*}} t^*_j\right) = 0 \tag{9.12}$$

for all s. Then, as the probabilities are (strictly) positive, also

$$\sum_{j=1}^{|\mathscr{S}|-1} \alpha_j \left(t^*_j(s) - E_{P_{\nu^*}} t^*_j\right) = 0$$

for all s, so that

$$\sum_{j=1}^{|\mathscr{S}|-1} \alpha_j t^*_j(s) = \sum_{j=1}^{|\mathscr{S}|-1} \alpha_j \, E_{P_{\nu^*}} t^*_j$$

for all s. Note that the right-hand side does not depend neither on s nor on j, and by denoting it by c_P, one has that

$$\sum_{j=1}^{|\mathscr{S}|-1} \frac{\alpha_j}{c_P} t_j^*(s) = 1,$$

for all s. This means that a linear combination of the columns of the matrix $\mathbf{T}^*$ yields the column vector of 1's. But $\mathbf{T}^*$ was obtained from $\mathbf{T}^{\sim}$ by omitting the columns of 1's. Thus, if (9.12) was true, $\mathbf{T}^{\sim}$ would not be of full rank, which contradicts Theorem 8.8.

□

As the other conditions are clearly true, one has the following result.

Corollary 9.2 *The parameterization of all (strictly) positive probability distributions on $\mathscr{S}$ given in (9.11) is smooth.*

□

Instead of referring to the general result in Proposition 8.2, it is instructive to see why the inverse transformation of (9.11), yielding the log-linear parameters from the probabilities, is also smooth. The argument is closely related to that of the proof of Theorem 8.10 and essentially uses the normalizing constant as the overall effect.

Rewrite (9.11) as

$$\log \frac{P_{\nu^*}(s)}{q(s)} = -\log \sum_{r \in \mathscr{S}} q(r) \exp\Big(\sum_{j=1}^{|\mathscr{S}|-1} t_j^*(r)\nu_j^* \Big) + \sum_{j=1}^{|\mathscr{S}|-1} t_j^*(s)\nu_j^*$$

$$= b(\nu^*) + \sum_{j=1}^{|\mathscr{S}|-1} t_j^*(s)\nu_j^*,$$

or in matrix notation

$$\log \frac{P_{\nu^*}}{q} = 1 b(\nu^*) + \mathbf{T}^*\nu^* = (1, \mathbf{T}^*) \begin{pmatrix} b(\nu^*) \\ \nu^* \end{pmatrix},$$

where 1 is a vector of 1s. But

$$(1, \mathbf{T}^*) = \mathbf{T}^{\sim},$$

and this was shown to be of full rank in Theorem 8.8. Further, $b(\nu^*)$ is the value of the overall effect, as it appears in every cell of the table, thus,

$$\begin{pmatrix} b(\nu^*) \\ \nu^* \end{pmatrix} = \nu^{\sim}, \tag{9.13}$$

and

$$\nu^{\sim} = (\mathbf{T}^{\sim})^{-1} \log \frac{P_{\nu^*}}{q}.$$

This is the situation described in Theorem 8.6. The full rank transformation of $\mathbb{R}^{|\mathscr{S}|}$ defined as

$$\nu^{\sim}(P) = (\mathbf{T}^{\sim})^{-1} \log \frac{P}{q} : \mathbb{R}^{|\mathscr{S}|} \to \mathbb{R}^{|\mathscr{S}|} \tag{9.14}$$

is applied to the $\mathbb{R}^{|\mathscr{S}|-1}$-dimensional smooth manifold of the (strictly) positive probability distributions (see Proposition 8.1). Consequently, the collection of the non-redundant log-linear parameters $\nu^{\sim}$ for (strictly) positive probability distributions is a smooth manifold of dimension $|\mathscr{S}| - 1$ in $\mathbb{R}^{|\mathscr{S}|}$.

The ν^* parameters, as was seen, are the last $c - 1$ components of the $\nu^{\sim}$ parameters, and the latter are a $|\mathscr{S}| - 1$-dimensional smooth manifold in $\mathbb{R}^{|\mathscr{S}|}$. Further, the remaining parameter, $b(\nu^*)$, is a function of ν^*. The situation here is similar to the (x, x^2) example discussed in Sect. 8.2. The similarity is more apparent if (x, x^2) is written as a vector $(x, x^2)'$ and is compared to (9.13). As long as the domain of x is a 1-dimensional open set, for instance, an open interval or the union of open intervals, (x, x^2) is a smooth one-dimensional manifold in the two-dimensional Euclidean space. Even if the domain of x is a one-dimensional open set, the collection of the points (x, x^2) is not an open set in the two-dimensional space, thus, the dimension of the smooth manifold is 1. If the domain of x was not a one-dimensional smooth manifold, for example, it was a single point,[5] say x_0, then $\{(x_0, x_0^2)\}$ would not be a one-dimensional smooth manifold, although it would be still two-dimensional.

Somewhat similarly, the c probabilities yield a $c - 1$-dimensional smooth manifold, as one of them is a function of the others, but the "other" $c - 1$ probabilities constitute a $c - 1$-dimensional smooth manifold in the $c - 1$-dimensional space.

When the $c-1$ parameters ν^* are calculated for all (strictly) positive distributions on $\mathscr{S}$, by their definition, they yield an open $(c-1)$-dimensional set that is a smooth manifold[6] in the $c-1$-dimensional space. As $b(\nu^*)$ is a function of theirs, the points with the coordinates $(b(\nu^*), \nu^*) = \nu^{\sim}$ do not yield an open set in the c-dimensional space. Rather, this set is a smooth $(c-1)$-dimensional manifold, as was seen above. And as only one component is a function of the others, the "other" components, namely, those in ν^*, have to be a smooth manifold of dimension $c-1$. Note that the latter argument, although correct, is heuristic and requires proof.

[5] A single point is not an open set, thus, it cannot be a smooth manifold.

[6] For instance, using the identity mapping.

Turning to the mean value parameterization of probability distributions, the definition in (9.1) remains valid:

$$E_P(t^{\sim}) = (\mathbf{T}^{\sim})' P, \tag{9.15}$$

but now the multiplication by the full rank $(\mathbf{T}^{\sim})'$ matrix is applied to the elements of a $|\mathscr{S} - 1|$-dimensional smooth manifold, the (strictly) positive probability distributions on $\mathscr{S}$, instead of a $|\mathscr{S}|$-dimensional smooth manifold, the (strictly) positive distributions on $\mathscr{S}$ however, both of these manifolds are in the $|\mathscr{S}|$-dimensional space. The transformation in (9.15) is smooth, thus, by Theorem 8.6, the mean value parameters are a $|\mathscr{S} - 1|$-dimensional smooth manifold. Again, the mean value parameters

$$E_P(t^*)$$

are of full rank, that is, are a $|\mathscr{S} - 1|$-dimensional smooth manifold in $\mathbb{R}^{|\mathscr{S}-1|}$.

The smoothness of the mixed parameterization for (strictly) positive probability distributions may be proved just like in the case of (strictly) positive distributions. In the proof of Theorem 9.5, the Jacobian of a mixed parameterization was shown to have been taken from various matrices derived from $\mathbf{T}^{\sim}$. In the case of probabilities, matrices derived similarly from $\mathbf{T}^*$ have to be used.

Corollary 9.3 *The (strictly) positive probability distributions may be smoothly parameterized with*

(i) the non-redundant log-linear parameters without the overall effect, or with
(ii) the mean value parameters without the sum of the probabilities or with
(iii) a mixed parameterization, where the log-linear parameters in (i) associated with an ascending class of subsets of the variables and the mean value parameters in (ii) associated with the complement descending class are used.

□

Log-linear models were defined in Chapter *LCDA* 10 by selecting an ascending class of subsets of the variables, then considering a mixed parameterization with log-linear parameters on the ascending class and mean value parameters on the descending class, and then assuming that the log-linear parameters on the ascending class were all equal to 1. The smoothness of this parameterization established in Corollary 9.3 implies that Theorem 8.4 applies to log-linear models, and gives the following.

Corollary 9.4 *Let $\mathscr{A}$ be an ascending class of subsets of V and let $\mathscr{D}$ be its complement descending class. Let $LL(\mathscr{A})$ be the log-linear model defined for the (strictly) positive probability distributions on the contingency table $\mathscr{S}$ formed by the ranges of the variables V, by*

$$COR(W|V \setminus W = (s)_{V\setminus W}) = 1 \text{ for all joint indices } (s)_{V\setminus W} \text{ and for all } W \in \mathscr{A}.$$

Then, $LL(\mathscr{A})$ is a smooth manifold of dimension e in $\mathbb{R}^{|\mathscr{S}|}$, where

$$e = |\mathscr{S}| - 1 - f,$$

with

$$f = |\{j : W_j \in \mathscr{A}\}|,$$

where j stands for the columns of the design matrix $\mathbf{T}^*$. Further, the mean value parameters associated with the columns of the design matrix $\{j : W_j \in \mathscr{D}\}$ parameterize the model.

□

Note that the quantity f above is called the number of degrees of freedom of the model.[7]

For example, in a 2×3 two-way table, for probability distributions, the $\mathbf{T}^*$ design matrix is 6×5, and the last two columns are associated with the subset of variables containing both. Thus, in the model of independence of the two variables, the last two log-linear parameters are set to zero, the number of degrees of freedom is 2, and the remaining 3 mean value parameters parameterize all 2×3 independent probability distributions, which is a smooth manifold of dimension 3 in the six-dimensional space. When the same model is applied to general distributions, the design matrix $\mathbf{T}^{\sim}$ is 6×6, the number of degrees of freedom is 2, but the manifold is four-dimensional.[8]

The results presented here are used in the next section to prove the smoothness of marginal log-linear parameterizations.

9.3 Smoothness of Marginal Log-Linear Parameterizations

Marginal log-linear parameterizations of (strictly) positive general or probability distributions were defined in Sect. 7.3. In this section, their smoothness is proved. The argument follows the one given in Bergsma and Rudas [20], but many of the details not discussed in that paper are worked out here.

First, the following simple result establishes the number of non-redundant marginal log-linear parameters for general distributions.

Theorem 9.8 *The number of the non-redundant marginal log-linear parameters, whether balanced or corner parameters are considered, is equal to the product of the numbers of the categories of the variables, irrespective of the hierarchical and complete sequence of marginals used to define the parameters.*

[7] For the meaning of this expression, see the comments after Theorem $LCDA$ 5.9.

[8] The definitions of the design matrices for 2×3 tables were given before Theorem 8.8.

Proof The claim follows from Theorem 8.7, because the argument presented there does not depend on the marginals in which the parameters are determined as long as the marginals contain the effects as subsets.

□

Next, a general result is given about replacing some of the coordinates of a smooth parameterization with a smooth parameterization of theirs. The result states that the new parameterization remains smooth. The theorem is formulated for smooth manifolds, and also applies to classes of distributions.

Theorem 9.9 *Let $T \subseteq \mathbb{R}^l$ be an open set, and let $G = \{g(\tau),\ \tau \in T\}$ be a smooth l-dimensional manifold in $\mathbb{R}^k$, with g being a smooth parameterization, so that g has an inverse, and both g and g^{-1} are twice continuously differentiable; further, $g(\tau)$, for $\tau \in T$, has a full rank Jacobian.*

Let τ have two components, $\tau = (\tau_1, \tau_2)$, with τ_1 having m coordinates, and define $T_1 \subseteq \mathbb{R}^m$ as $T_1 = \{\tau_1 : \exists \tau_2, (\tau_1, \tau_2) \in T\}$. Then, T_1 is an open set in $\mathbb{R}^m$.

Further, let $U_1 \subseteq \mathbb{R}^m$ be an open set, and $h_1 : U_1 \to T_1$ be a smooth transformation so that h_1 has an inverse, both h_1 and h_1^{-1} are twice continuously differentiable, and $h(u_1)$ has a full rank Jacobian for $u_1 \in U_1$.

Let $U \subseteq \mathbb{R}^l$ be defined as $U = \{(u_1, u_2) : (h_1(u_1), u_2) \in T\}$, that is, $U = \{(h_1^{-1}(\tau_1), \tau_2) : (\tau_1, \tau_2) \in T\}$. Finally, let $h = (h_1, id) : U \to T$, where id is the identity function.

Then, $G = \{g(h(u)) : u \in U\}$ is a smooth parameterization, using the parameters $u \in U$.

Proof The first claim is that T_1 is an open set in $\mathbb{R}^m$. If this was not true, then there would exist a sequence $(e_i) \subseteq \mathbb{R}^m$, such that $e_i \to e$ for some $e \in T_1$, such that $e_i \notin T_1$, $i = 1, 2, \ldots$. As $e \in T_1$, there is a τ_2, such that $(e, \tau_2) \in T$. Then, the sequence $(e_i, \tau_2) \notin T$, as $e_i \notin T_1$, but $(e_i, \tau_2) \to (e, \tau_2)$, and this contradicts to T being open.

To see the second claim, one needs to see that U is an open set in $\mathbb{R}^l$ and $g \circ h$ is one-to-one, that both $g \circ h$ and its inverse are twice continuously differentiable, and that $g \circ h$ has a full rank Jacobian on U.

To see that U is an open set in $\mathbb{R}^l$, choose an arbitrary $u = (u_1, u_2) \in U \subseteq \mathbb{R}^l$, and let (e_i, f_i) be a sequence converging to it. It is enough to see that for indices $i > i_0$, $(e_i, f_i) \in U$. Consider $(h_1(u_1), u_2) \in T$ and the sequence $(h_1(e_i), f_i)$. As h_1 is continuous, $(h_1(e_i), f_i) \to (h_1(u_1), u_2)$, and as T is open, for indices larger than some threshold i_0, $(h_1(e_i), f_i) \in T$. Then, for the same indices, $(e_i, f_i) = (h_1^{-1}(h_1(e_i)), f_i) \in U)$, proving that U is open in $\mathbb{R}^l$.

The composite function $g \circ h$ is one-to-one because both g and h are one-to-one, the latter because both h_1 and id are one-to-one.

The Jacobian of the composite function $g \circ h$ is the product of the Jacobians of the two functions, each evaluated at the appropriate arguments, see, e.g., Lax [103]. It was assumed that the Jacobian of g, say $\mathbf{J}_g$, is of full rank over its entire domain. It will be shown that the Jacobian of h, say $\mathbf{J}_h$, is also of full rank over its entire domain Thus, the Jacobian of $g \circ h$, $\mathbf{J}_g\mathbf{J}_h$ is a product of a $k \times l$ matrix with linearly independent columns and of an $l \times l$ matrix with both rows and columns being

linearly independent. The matrix $\mathbf{J}_g\mathbf{J}_h$ is of size $k \times l$ and it needs to be seen that its columns are linearly independent. An identical fact was proved in the proof of Theorem 8.6, see (8.4).

To determine the Jacobian of h, note that its first m coordinates do not depend on the last $l - m$ components of the argument, and its last $l - m$ coordinates do not depend on the first m arguments. Therefore, the Jacobian has a block structure. The upper left $m \times m$ block is the Jacobian of h_1, say, $\mathbf{J}_1$, and the lower right $(l - m) \times (l - m)$ block is the identity matrix:

$$\mathbf{J}_h = \begin{pmatrix} \mathbf{J}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}.$$

This matrix is of full rank. No one of the first m columns may be obtained as a linear combination consisting of the last $l - m$ columns only, because of the first m rows. If any linear combination of other columns gave any of the first m columns, then the first m rows of the linear combination would not change if the coefficients of the last $l - m$ columns were changed to zero. But then a linear combination of the first m columns gave the first m rows of one of the first m columns, which is not possible as $\mathbf{J}_1$ is of full rank. A similar argument applies to the last $l - m$ columns, and this completes the proof.

□

Corollary 9.3 stated that, when applied to an ascending and the complement descending classes, respectively, the non-redundant log-linear parameters without the overall effect and the mean value parameters yield a smooth parameterization of the (strictly) positive probability distributions on a contingency table. Theorem 9.9 goes further and, when applied to this situation, says, for instance, that if the mean value parameters are replaced by a smooth transformation yielding them, then smoothness is still retained.

Before the smoothness of marginal log-linear parameterization may be proved, one needs to see that the mean value parameters, that is, the expectations of the columns of the design matrix, depend only on the marginal distribution of that subset with which they are associated. More precisely, one has the following.

Theorem 9.10 *The $E_f(t_j^{\sim})$ mean value parameters depend on the W_j marginal distributions only in the sense that if f and g are two distributions, such that for the marginal distributions*

$$f|_W = g|_W,$$

for some $W \subseteq V$, then

$$E_f(t_j^{\sim}) = E_g(t_j^{\sim}),$$

whenever the column j of the design matrix is associated with the subset of the variables W.

Proof Because the definitions of the design matrices for balanced or corner log-linear parameters given in Sect. 8.3 depend on which ones of the log-linear parameters enter in a log-linear representation of the value of the distribution in a cell, if two cells of the table v_1 and v_2 are such that

$$(v_1)_{W_j} = (v_2)_{W_j},$$

where W_j is the subset of variables associated with column j of the design matrix, then

$$t_j^{\sim}(v_1) = t_j^{\sim}(v_2). \tag{9.16}$$

Thus, the

$$E_f(t_j^{\sim}) = \sum_v f(v) t_j^{\sim}(v)$$

mean value parameter may be written as

$$E_f(t_j^{\sim}) = \sum_{w_j} \left(t_j^{\sim}(w_j) \sum_{v:(v)_{W_j}=w_j} f(v) \right) = \sum_{w_j} t_j^{\sim}(w_j) f(w_j), \tag{9.17}$$

where the last term $f(w_j)$ is the marginal value of the distribution f in the marginal cell w_j.

□

Equation (9.17) immediately implies the following.

Corollary 9.5 *If for a $W \subseteq V$, the $f|_W$ marginal distribution is given, that is, for all category combination w of the variables W, $f(w)$ is given, then $E_f(t_{j_W}^{\sim})$ may be determined for all components of the canonical statistic $j_W \in J_W$ which are associated with W and they are the same as if they were determined from f. Further, the*

$$E_f(t_{j_{W'}}^{\sim}), \quad j_{W'} \in J_{W'}, \quad W' \subseteq W$$

values are the mean value parameters of the distribution $f|_W$.

□

Theorem *LCDA* 10.4 gave a related result, but for binary tables only.

Note that Theorem 9.10 and Corollary 9.5 apply to probability distributions, with the modification that instead of $E_f(t_{j_W}^{\sim})$, one has to use $E_f(t_{j_W}^{*})$.

Now, the smoothness of marginal log-linear parameterizations may be obtained, through a series of reparameterizations, based on Theorem 9.9, using mean value

and log-linear parameters, see Bergsma and Rudas [20]. The result is formulated for (strictly) positive distributions. Recall that in this chapter, a positive distribution means arbitrary positive numbers associated with the cells of a table.

To illustrate the argument of the proof, first an example is given. Consider a $2 \times 3 \times 2$, $A \times B \times C$ table and the (strictly) positive distributions f_{ABC} on it. The non-redundant log-linear parameters for the table, calculated from f_{ABC}, are denoted as $\Lambda^{\sim}_{ABC}$. The $W \subseteq \{A, B, C\}$ marginal distribution derived from f_{ABC} are denoted as f_W.

The columns j of the design matrix $\mathbf{T}^{\sim}$ and the mean value parameters that are associated with them belong to subsets W_j and, in particular, category combinations w_j of W_j, as described in the definitions of the design matrices for the non-redundant balanced and also for the corner log-linear parameters. In this example, the design matrix is of size 12×12, and, in the case of the balanced log-linear parameters, the category combinations for the columns are as follows:

Column 1: $\emptyset$
Column 2: $A = (1)$
Column 3: $B = (1)$
Column 4: $B = (2)$
Column 5: $C = (1)$
Column 6: $AB = (1, 1)$
Column 7: $AB = (1, 2)$
Column 8: $AC = (1, 1)$
Column 9: $BC = (1, 1)$
Column 10: $BC = (2, 1)$
Column 11: $ABC = (1, 1, 1)$
Column 12: $ABC = (1, 2, 1)$.

Let the hierarchical and complete sequence of marginals for the parameterization be

$$A, AB, AC, ABC. \tag{9.18}$$

The procedure showing that the marginal log-linear parameterization based on this sequence of marginals is smooth starts with the log-linear parameterization of f_{ABC}:

$$\lambda_{\emptyset}^{ABC}, \lambda_{A}^{ABC^{\sim}}, \lambda_{B}^{ABC^{\sim}}, \lambda_{C}^{ABC^{\sim}}, \lambda_{AB}^{ABC^{\sim}}, \lambda_{AC}^{ABC^{\sim}}, \lambda_{BC}^{ABC^{\sim}}, \lambda_{ABC}^{ABC^{\sim}},$$

where the $^{\sim}$ notation to denote the non-redundant parameters is applied to the groups of parameters. Accordingly, for instance, $\lambda_{AB}^{ABC^{\sim}}$ consists of the parameters λ_{11}^{ABC} and λ_{12}^{ABC}, because A has two categories and B has three categories.

Next, consider the marginals prior to ABC in (9.18):

$$A, AB, AC, \tag{9.19}$$

and then the descending class generated by them, that is,

$$\emptyset, A, B, C, AB, AC,$$

and the mixed parameterization, where the log-linear parameters associated with these subsets of the variables are replaced by the corresponding mean value parameters:[9]

$$E(t_1^{\sim}), E(t_2^{\sim}), E(t_3^{\sim}), E(t_4^{\sim}), E(t_5^{\sim}), E(t_6^{\sim}), E(t_7^{\sim}), E(t_8^{\sim}), \lambda_{BC}^{ABC^{\sim}}, \lambda_{ABC}^{ABC^{\sim}}. \tag{9.20}$$

Note that the last two groups of log-linear parameters each contain 2 parameters, associated with the 9th and 10th, and the 11th and 12th columns of the design matrix, respectively.

This parameterization is smooth, as it is a mixed parameterization using log-linear parameters on the ascending class $\{BC, ABC\}$, and mean value parameters on the complement descending class.

The last subset of the variables in (9.19) is AC, and the mean value parameters associated with columns $j = 1, 2, 5, 8$ of the design matrix constitute a mean value parameterization of f_{AC}. Indeed, $W_1 = \emptyset$, $W_2 = \{A\}$, $W_5 = \{C\}$, $W_8 = \{AC\}$. The parameterization is

$$E(t_1^{\sim}), E(t_2^{\sim}), E(t_5^{\sim}), E(t_8^{\sim}), \tag{9.21}$$

and it is smooth, see Corollary 9.5. This mean value parameterization will be replaced with a mixed parameterization of f_{AC}, using

$$E(t_1^{\sim}), E(t_2^{\sim}), \lambda_{C}^{AC^{\sim}}, \lambda_{AC}^{AC^{\sim}}. \tag{9.22}$$

On the AC marginal of the table, this is a smooth parameterization of f_{AC} using the log-linear parameters on the ascending class C, AC, and the mean value parameter on the complement descending class. The choice of the parameters that are replaced comes from selecting the subsets that do have parameters associated with them, and are such that AC is the first marginal that contains them. Such subsets are an ascending class within AC, namely, (C, AC) in the present case. The log-linear parameters are calculated on the AC marginal, and not on the entire table, so these are marginal log-linear parameters. Further, the retained mean value parameters, which were originally calculated from the entire table ABC remain the same, if calculated from the marginal table AC, see Corollary 9.5. Thus, they may be considered mean value parameters calculated from f_{AC}. After this reparameterization, one has the following parameterization of f_{ABC}:

[9] The $^{\sim}$ notation in $E(t_j^{\sim})$ refers to the fact that these are columns of the design matrix $\mathbf{T}^{\sim}$, and, at the same time, they are not redundant; otherwise, $\mathbf{T}^{\sim}$ would not be of full rank.

$$E(t_1^{\sim}), E(t_2^{\sim}), E(t_3^{\sim}), E(t_4^{\sim}), E(t_6^{\sim}), E(t_7^{\sim}), \lambda_C^{AC^{\sim}} \lambda_{AC}^{AC^{\sim}}, \lambda_{BC}^{ABC^{\sim}}, \lambda_{ABC}^{ABC^{\sim}}. \tag{9.23}$$

In fact, the last reparameterization replaced $E(t_8^{\sim})$ with $\lambda_{AC}^{AC^{\sim}}$ and $E(t_5^{\sim})$ with $\lambda_C^{AC^{\sim}}$, that is, introduced marginal log-linear parameters. Application of Theorem 9.9 shows that the parameterization (9.23) is smooth.

Indeed, (9.20) is a smooth parameterization of the distributions f_{ABC}. To obtain (9.23), the components of (9.20) given in (9.21) are replaced with parameters of which they are a smooth function, namely, with the parameters in (9.22). The smoothness of the latter transformation comes from first generating f_{AC} from (9.22) and then (9.21) from f_{AC}. These are both smooth transformations by Corollary 9.3, and by the fact that the inverse of a smooth transformation is also smooth, see Proposition 8.2, further, their combination is also smooth, see the comment after the proof of Theorem 8.6. Then, Theorem 9.9 applies and yields that (9.23) is a smooth parameterization of the distributions f_{ABC}.

The penultimate subset in (9.19) is AB, and f_{AB} is smoothly parameterized by the first six parameters in (9.23), because these are the mean value parameters associated with the subsets of AB. These parameters are replaced with a group of parameters that also yield a smooth parameterization of f_{AB}, namely, the mean value parameters on $\emptyset$ and A, and the log-linear parameters of B and AB, of course, the latter calculated from the AB marginal. This yields the following parameterization:

$$E(t_1^{\sim}), E(t_2^{\sim}), \lambda_B^{AB^{\sim}}, \lambda_{AB}^{AB^{\sim}}, \lambda_C^{AC^{\sim}}, \lambda_{AC}^{AC^{\sim}}, \lambda_{BC}^{ABC^{\sim}}, \lambda_{ABC}^{ABC^{\sim}}, \tag{9.24}$$

the smoothness of which may be seen similarly to the smoothness of (9.23). Note that $\lambda_B^{AB^{\sim}}$ and $\lambda_{AB}^{AB^{\sim}}$ each contain two non-redundant marginal log-linear parameters, as B has three categories.

Finally, $E(t_1^{\sim})$ and $E(t_2^{\sim})$, which yield a smooth parameterization of f_A, are replaced by $\lambda_{\emptyset}^{A^{\sim}}$ and $\lambda_A^{A^{\sim}}$, respectively, with smoothness retained, yielding

$$\lambda_{\emptyset}^{A^{\sim}}, \lambda_A^{A^{\sim}}, \lambda_B^{AB^{\sim}}, \lambda_{AB}^{AB^{\sim}}, \lambda_C^{AC^{\sim}}, \lambda_{AC}^{AC^{\sim}}, \lambda_{BC}^{ABC^{\sim}}, \lambda_{ABC}^{ABC^{\sim}}. \tag{9.25}$$

This is the marginal log-linear parameterization for the hierarchical and complete sequence of marginals (9.18).

The next theorem shows that such a series of reparameterizations is always possible and retains smoothness.

Theorem 9.11 *Let $\mathscr{M}_1, \mathscr{M}_2, \ldots, \mathscr{M}_k = V$ be a hierarchical and complete sequence of marginals of a contingency table formed by the ranges of the variables V. Then, the non-redundant marginal log-linear parameters determined in these marginals are a smooth parameterization of all the (strictly) positive distributions on the contingency table.*

Proof If $k = 1$, then the marginal log-linear parameterization is the log-linear parameterization and is smooth, see Theorem 8.8.

If $k = 2$, then the let $\Lambda_2^{\sim}$ denote the non-redundant log-linear parameters (balanced or corner) associated with the subset

$$\mathscr{M}_2 \setminus \mathscr{M}_1, \tag{9.26}$$

and let Δ_2 denote[10] the mean value parameters on $\mathscr{M}_1$. Then, $(\Delta_2, \Lambda_2^{\sim})$ is a mixed parameterization of the distributions $f_{\mathscr{M}_2}$, and is smooth, see Corollary 9.1.

The parameters Δ_2 are mean value parameters determined from $f_{\mathscr{M}_2}$, for the set of variables $\mathscr{M}_1$, but are the same as the corresponding mean value parameters determined from $f_{\mathscr{M}_1}$, see Corollary 9.5. The transformation from the non-redundant log-linear parameterization of $f_{\mathscr{M}_1}$, $\Lambda_1^{\sim}$, to $f_{\mathscr{M}_1}$ is smooth, see Theorem 8.8. Also, the transformation from $f_{\mathscr{M}_1}$ to its mean value parameterization Δ_2 is smooth, see Theorem 9.4. The combination of these two smooth parameterizations is a smooth transformation from $\Lambda_1^{\sim}$ to Δ_2 (see the comment after Theorem 8.6), thus, Theorem 9.9 implies that $(\Lambda_1^{\sim}, \Lambda_2^{\sim})$ is a smooth parameterization of all distributions $f_{\mathscr{M}_2}$.

The subset of variables in (9.26) is exactly the one that is contained in $\mathscr{M}_2$ but not in $\mathscr{M}_1$, thus, $\Lambda_2^{\sim}$ contains non-redundant log-linear parameters calculated from $f_{\mathscr{M}_2}$ for the sets of variables not contained in $\mathscr{M}_1$, and $\Lambda_1^{\sim}$ contains non-redundant marginal log-linear parameters calculated from $f_{\mathscr{M}_1}$, associated with the subsets of $\mathscr{M}_1$. Thus, the smooth parameterization consists of marginal log-linear parameters for the hierarchical sequence of marginals $\mathscr{M}_1, \mathscr{M}_2$.

If $k \geq 3$, then, set $\mathscr{D}_0$ be the empty set and for $l = 2, \ldots, k$, let

$$\mathscr{D}_l = \mathscr{D}(\mathscr{M}_1, \ldots, \mathscr{M}_{l-1})$$

be the descending class generated by the marginals prior to $\mathscr{M}_l$ in the hierarchical sequence of marginals, and let

$$\mathscr{A}_l = \mathbb{P}(\mathscr{M}_l) \setminus \mathscr{D}_l$$

be the complement ascending class in $\mathscr{M}_l$.

The rest of the proof is by induction. For $l = k$, consider a mixed parameterization of the distributions f_k on $\mathscr{M}_k$, consisting of the non-redundant log-linear parameters associated with the sets in $\mathscr{A}_k$, and determined in the marginal $\mathscr{M}_k$, say $\Lambda_{\mathscr{A}_k}^{\mathscr{M}_k\sim}$, and of the mean value parameters associated with the sets[11] in $\mathscr{D}_k$ and determined in the marginal $\mathscr{M}_k$, say Δ_k. Thus, the parameterization

$$\Pi_k = (\Delta_k, \Lambda_{\mathscr{A}_k}^{\mathscr{M}_k\sim}), \tag{9.27}$$

[10] The choice of the index 2 is in line with the notation used in more general versions of the argument presented later in the proof.

[11] More precisely, the mean value parameters associated with the columns j of the design matrix, for which $W_j \in \mathscr{D}_k$.

with[12]

$$\Delta_k = (E(t_j^{\sim}) : W_j \in \mathscr{D}_k),$$

$$\Lambda_{\mathscr{A}_k}^{\mathscr{M}_k\sim} = (\lambda_W^{\mathscr{M}_k\sim} : W \in \mathscr{A}_k)$$

is a smooth parameterization of $f_{\mathscr{M}_k}$, see Corollary 9.1.

Although Δ_k consists of mean value parameters for sets in $\mathscr{D}_k$, calculated from $f_{\mathscr{M}_k}$, they are the same as if were calculated from the respective marginal distributions of the subsets in $\mathscr{D}_k$, see Corollary 9.5.

The Δ_k part of (9.27) is replaced with the parameters Φ_{k-1}, such that the mean value parameters associated with subsets in $\mathscr{D}_k \setminus \mathscr{D}_{k-1}$ are replaced by the non-redundant log-linear parameters (balanced or corner) calculated for[13] $f_{\mathscr{M}_{k-1}}$ for the subsets in $\mathscr{D}_k \setminus \mathscr{D}_{k-1}$, and the other parameters in Δ_k are retained:

$$\Phi_{k-1} = (E(t_j^{\sim}) : W_j \in \mathscr{D}_{k-1};\ \lambda_W^{\mathscr{M}_{k-1}\sim} : W \in \mathscr{D}_k \setminus \mathscr{D}_{k-1}). \tag{9.28}$$

The transformation[14]

$$\Phi_{k-1} \to \Delta_k \tag{9.29}$$

will be shown to be smooth, and, thus, Theorem 9.9 implies that the parameterization

$$\Pi_{k-1} = (\Phi_{k-1}, \Lambda_{\mathscr{A}_k}^{\mathscr{M}_k\sim}) \tag{9.30}$$

$$= (E(t_j^{\sim}) : W_j \in \mathscr{D}_{k-1};\ \lambda_W^{\mathscr{M}_{k-1}\sim} : W \in \mathscr{D}_k \setminus \mathscr{D}_{k-1};\ \lambda_W^{\mathscr{M}_k\sim} : W \in \mathscr{A}_k)$$

of $f_{\mathscr{M}_k}$ is also smooth.

To see that (9.29) is smooth, observe that

$$\mathscr{D}_k \setminus \mathscr{D}_{k-1} = \mathscr{A}_{k-1},$$

as both contain those subsets of $\mathscr{M}_{k-1}$ that are not subsets of any of the earlier marginals in the given hierarchical sequence of marginals. Using this, write (9.28) in the form:

$$\Phi_{k-1} = (E(t_j^{\sim}) : W_j \in \mathscr{D}_{k-1} \setminus (\mathbb{P}(\mathscr{M}_{k-1}) \setminus \mathscr{A}_{k-1}); \tag{9.31}$$

[12] Not using a set notation refers to the sequential order of reparameterizations in the proof.

[13] Note that $\mathscr{M}_{k-1}$ is the last marginal in $\mathscr{D}_k$.

[14] Here $\to$ denotes the function that is defined on the vectors on the left-hand side and yields the vectors on the right-hand side.

$$E(t_j^{\sim}) : W_j \in \mathbb{P}(\mathscr{M}_{k-1}) \setminus \mathscr{A}_{k-1};$$

$$\lambda_W^{\mathscr{M}_{k-1}\sim} : W \in \mathscr{A}_{k-1}).$$

Here, $\mathscr{A}_{k-1}$ is an ascending class of subsets of $\mathscr{M}_{k-1}$, and $\mathbb{P}(\mathscr{M}_{k-1}) \setminus \mathscr{A}_{k-1}$ is its complement with respect to the class all subsets of $\mathscr{M}_{k-1}$, a descending class. Therefore, the last two parts of (9.31) constitute a mixed parameterization of $f_{\mathscr{M}_{k-1}}$. Further, the first and second parts of (9.31) are parameters that appear in (9.28).

Then, treating the first part of (9.31) as if it was retained in (9.29), as implied by Theorem 9.9, (9.29) is smooth, if the following transformation is smooth:

$$E(t_j^{\sim}) : W_j \in \mathbb{P}(\mathscr{M}_{k-1}) \setminus \mathscr{A}_{k-1};\ \lambda_W^{\mathscr{M}_{k-1}\sim} : W \in \mathscr{A}_{k-1}) \tag{9.32}$$

$$\rightarrow (E(t_j^{\sim}) : W_j \in \mathbb{P}(\mathscr{M}_{k-1})). \tag{9.33}$$

The smoothness of this transformation is easily seen by first transforming (9.32) into $f_{\mathscr{M}_{k-1}}$ and then $f_{\mathscr{M}_{k-1}}$ into (9.33), as the first one of these is a distribution determined from a mixed parameterization of it, and the second is the mean value parameters determined from the same distribution, and both of these transformations are smooth, and so is their combination.

This means that

$$\Pi_{k-1} = (\Phi_{k-1}, \Lambda_{\mathscr{A}_k}^{\mathscr{M}_k\sim}) \tag{9.34}$$

$$= (E(t_j^{\sim}) : W_j \in \mathscr{D}_{k-1};\ \lambda_W^{\mathscr{M}_{k-1}\sim} : W \in \mathscr{A}_{k-1};\ \lambda_W^{\mathscr{M}_k\sim} : W \in \mathscr{A}_k)$$

is smooth.

The argument from (9.27) to (9.34) may now be repeated, instead of $l = k$, for every $l = k-1, k-2, \ldots, 1$, with $\mathscr{D}_0 = \emptyset$, to obtain that the smoothness of the parameterization of $f_{\mathscr{M}_k}$ with

$$(E(t_j^{\sim}) : W_j \in \mathscr{D}_{l-1};\ \lambda_W^{\mathscr{M}_{l-1}\sim} : W \in \mathscr{A}_{l-1};\ \lambda_W^{\mathscr{M}_l\sim} : W \in \mathscr{A}_l;\ \ldots ;\lambda_W^{\mathscr{M}_k\sim} : W \in \mathscr{A}_k)$$

implies the smoothness of the parameterization as

$$(E(t_j^{\sim}) : W_j \in \mathscr{D}_{l-2};\ \lambda_W^{\mathscr{M}_{l-2}\sim} : W \in \mathscr{A}_{l-2};\ \ldots ;\lambda_W^{\mathscr{M}_k\sim} : W \in \mathscr{A}_k),$$

yielding that the parameterization

$$(\lambda_W^{\mathscr{M}_1\sim} : W \in \mathscr{A}_1;\ \lambda_W^{\mathscr{M}_2\sim} : W \in \mathscr{A}_2;\ \ldots ;\lambda_W^{\mathscr{M}_k\sim} : W \in \mathscr{A}_k) \tag{9.35}$$

is smooth. Given that $\mathscr{A}_l$ is the collection of subsets that are contained in $\mathscr{M}_l$ but not in any of the earlier marginals in the hierarchical sequence of marginals, (9.35) is the marginal log-linear parameterization of $f_{\mathscr{M}_k}$.

□

For the case of probability distributions, essentially the same argument applies, except that the non-redundant log-linear parameters do not contain the overall effect in that case rather, normalization applies, and the mean value parameters are the expectations of the columns of the $\mathbf{T}^*$ rather than of the $\mathbf{T}^{\sim}$ matrix. In notation, one would have to write ν^* instead of $\lambda^{\sim}$ and t^* instead of $t^{\sim}$. Also, the references to smoothness results established earlier would have to be changed from results applying to general distributions to the same results applying to probability distributions.

Theorem 9.11 says that the hierarchical nature of the sequence of marginals on which the marginal log-linear parameterization is based and considering parameters for every effect in the first marginal that contains it are sufficient for the parameterization to be smooth. These rules are violated if, for example, the marginal log-linear parameterization contains marginal log-linear parameters for the same effect in two different marginals. For such cases, Bergsma and Rudas [20] gave a formula as to how the marginal log-linear parameters are related, and based on this showed that in a three-way table, if a parameterization contains the log-linear parameter for the effect AB in both the AB and the ABC marginals, then the parameterization cannot be smooth as the Jacobian is not of constant rank.

The condition of Theorem 9.11, however, is not necessary. For further results concerning the smoothness of marginal log-linear parameterizations, including results about how parameters determined in different marginals relate, algorithmic approaches to decide smoothness when the conditions presented here do not hold and classes of smooth models, see Bergsma and Rudas [20], Colombi and Forcina [30], Drton [47], Evans [51], Forcina [59], Forcina et al. [61], and Rudas and Bergsma [151].

The smoothness of marginal log-linear parameterization, together with Theorem 8.4, will be used in the next chapter to define marginal log-linear models, and to establish some of their properties.

9.4 Summary

In the first part of this chapter, we gave rigorous proofs of the smoothness properties of various parameterizations of discrete distributions. These are applied to the balanced and the corner log-linear parameters, using the respective design matrices developed in the previous chapter; to the mean value parameters defined as the integrals of the canonical statistics that are in the columns of the design matrix; and to mixed parameterizations, which consist of the log-linear parameters on an ascending class of subsets and the mean value parameters on the complement descending class. Both general and probability distributions were considered, and the design matrices differed in the presence or lack of the overall effect. However, in the case of probabilities, normalization played the role of the overall effect. In these cases, relatively simple proofs were sufficient to see the smoothness.

The second part of the chapter relied on the above results in proving the smoothness of marginal log-linear parameterizations, as defined in Chap. 7. The proof of smoothness given used a general result about replacing some of the components of a smooth parameterization with other parameters, from which the former ones are obtained through a smooth transformation. The resulting parameterization retains smoothness. The proof of the smoothness of marginal log-linear parameterizations was based on a series of such transformations, starting with a mixed parameterization of the entire table, and gradually replacing some of the mean value parameters with mixed parameterizations of marginal distributions, and thus introducing the marginal log-linear parameters.

9.5 Things to Do

1. Prove Theorem 9.4.
2. Prove that the inverse of a full rank matrix is also of full rank.
3. Prove that the probabilities as a function of the mean value parameters are continuous.
4. Prove Corollary 9.1.
5. Apply the method of Theorem 9.6 to all general distributions on a 2×2 table with log-linear parameters and show that the resulting parameterization of the probability distributions on the table is not log-linear.
6. Prove Corollary 9.2.
7. Work out the details of the proof sketched before Corollary 9.3.
8. Show that Theorem 9.9 implies Corollary 9.1.
9. Check that (9.16) with appropriately selected g and A holds for both the balanced and the corner log-linear parameters.
10. Prove Corollary 9.5.
11. Where did the proof of Theorem 9.11 use that the sequence of marginals is hierarchical and complete?
12. Work out the details of the proof of Theorem 9.11 for probability distributions.
13. Many results in Sects. 9.3 and 9.4 claim smoothness of various parameterizations, but oftentimes only the relevant Jacobian was proved to be of full rank. Complete the proofs.

Chapter 10
Marginal Log-Linear Models



10.1 Introduction

In Chap. 7, a number of statistical problems related to conditional independence restrictions that do not apply to the entire joint distribution, rather to some of the marginals of it, were considered. The most important ones from among these were Markov models associated with DAGs, where each defining conditional independence applied to a marginal consisting of a variable and its nondescendants. In this chapter, marginal log-linear models, which not only formulate these properties but also generalize them widely, are defined. The model definition is based on a marginal log-linear parameterization, using a hierarchical and complete sequence of marginals. The models are defined by setting some of the marginal log-linear parameters to zero, and the remaining parameters parameterize the distributions in the models. Smoothness and standard asymptotic behavior follow immediately. Different possibilities for the choice of the hierarchical and complete sequence of marginals provide flexibility in terms of the choice of the parameters that are used to describe the distributions in the model.

Marginal log-linear models are defined in Sect. 10.2, and a general result about smoothness and the number of degrees of freedom associated with such models is given.

Section 10.3 deals with maximum likelihood estimation. The Multinomial and multivariate Poisson likelihoods are shown to provide the same MLEs. The standard Newton-Raphson and Fisher scoring algorithms are described, and an adaptation is used to obtain MLEs for marginal log-linear models. A brief overview of the available software is also included.

Among applications in Sect. 10.5, first a general sufficient condition for a list of conditional independences to define a smooth marginal log-linear model is given. Then, DAG models, chain graph models, and path models are considered as marginal log-linear models.

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10.2 Model Definition

Following Bergsma and Rudas [20], marginal log-linear models are defined by choosing a hierarchical and complete sequence of marginals, defining the marginal log-linear parameterization implied by it (see Chap. 7), and assuming that some of the marginal log-linear parameters are equal to zero.

This definition is a broad generalization of that of log-linear models, where in the log-linear parameterization, some of the log-linear parameters were assumed to be equal to zero. Such log-linear models may be interpreted as assuming that certain higher-order conditional associations vanish, and marginal log-linear models have the same meaning; however, the vanishing higher-order conditional associations apply, instead of the whole table, to certain marginals of it.

The great flexibility of marginal log-linear models comes from the several possibilities of choosing the hierarchical sequence of marginals, in addition to the choices of the log-linear parameters to be set to zero. The models considered are hierarchical in the sense that if for a certain marginal $\mathscr{M}$, the log-linear parameters $\lambda_W^{\mathscr{M}}$ is assumed to be zero, then this implies that $\lambda_{W'}^{\mathscr{M}}$ is also zero for all $W \subseteq W' \subseteq \mathscr{M}$. However, such implications do not exist in different marginals, that is, $\lambda_W^{\mathscr{M}} = 0$ does not imply that $\lambda_{W'}^{\mathscr{M}'} = 0$, if $\mathscr{M} \neq \mathscr{M}'$, even in $W \subseteq W'$.

To illustrate the flexibility of marginal log-linear models, consider an $A \times B \times C$ table. The following hierarchical sequences of marginals are possible:

$$(ABC). \tag{10.1}$$

$$(A, ABC),\ (B, ABC),\ (C, ABC),$$

$$(A, B, ABC),\ (A, C, ABC),\ (B, C, ABC).$$

$$(A, B, C, ABC).$$

$$(AB, ABC),\ (AC, ABC),\ (BC, ABC).$$

$$(A, AB, ABC),\ (B, AB, ABC),\ (C, AB, ABC),$$

$$(A, BC, ABC),\ (B, BC, ABC),\ (C, BC, ABC).$$

$$(A, AC, ABC),\ (B, AC, ABC),\ (C, AC, ABC).$$

$$(A, B, AB, ABC),\ (A, C, AB, ABC),\ (B, C, AB, ABC).$$

$$(A, B, AC, ABC),\ (A, C, AC, ABC),\ (B, C, AC, ABC).$$

$$(A, B, BC, ABC),\ (A, C, BC, ABC),\ (B, C, BC, ABC).$$

$$(A, B, C, AB, ABC),\ (A, B, C, AC, ABC),\ (A, B, C, BC, ABC).$$

$$(AB, AC, ABC),\ (AB, BC, ABC),\ (AC, BC, ABC).$$

$$(A, AB, AC, ABC),\ (A, AB, BC, ABC),\ (A, AC, BC, ABC).$$

$$(B, AB, AC, ABC),\ (B, AB, BC, ABC),\ (B, AC, BC, ABC).$$

$$(C, AB, AC, ABC),\ (C, AB, BC, ABC),\ (C, AC, BC, ABC).$$

$$(A, B, AB, AC, ABC),\ (A, B, AB, BC, ABC),\ (A, B, AC, BC, ABC).$$

$$(A, C, AB, AC, ABC),\ (A, C, AB, BC, ABC),\ (A, C, AC, BC, ABC).$$

$$(B, C, AB, AC, ABC),\ (B, C, AB, BC, ABC),\ (B, C, AC, BC, ABC).$$

$$(A, B, C, AB, AC, ABC),\ (A, B, C, AB, BC, ABC),\ (A, B, C, AC, BC, ABC),$$

$$(AB, AC, BC, ABC).$$

$$(A, AB, AC, BC, ABC),\ (B, AB, AC, BC, ABC),\ (C, AB, AC, BC, ABC),$$

$$(A, B, AB, AC, BC, ABC),\ (A, C, AB, AC, BC, ABC),$$

$$(B, C, AB, AC, BC, ABC),\ (A, B, C, AB, AC, BC, ABC).$$

Each of these hierarchical sequences of marginals leads to a marginal log-linear parameterization, and each parameterization facilitates a number of different models. Many of these hierarchical sequences are structurally identical, but very different in a real data analytical setup. For example, if variable C is potentially a response to the explanatory variables A and B, then the parameterization implied by the sequence (AB, AC, ABC) contains a parameter for the effect of A on C, disregarding (i.e., marginalized over) B, while the parameterization based on (A, B, C, ABC) contains only parameters for the effect of A on C, when conditioned on the different categories of B. That these two pieces of information are distinct and, in general, there is no way to infer one from the other, was illustrated with the discussion of Simpson's paradox in Chapter *LCDA* 9. Models that would contain a parameter for the marginal effect of A on C, and at the same time, a parameter on the conditional effect of A on C, given B, cannot be formulated as part of marginal log-linear parameterizations, as described here. See the discussion at the end of Sect. 9.3 concerning how such a parameterization would fail to be smooth.

For a contingency table formed by the ranges of variables V, let $\mathscr{F}$ denote the (strictly) positive frequency distributions and $\mathscr{P} \subseteq \mathscr{F}$ the (strictly) positive probability distributions. Note that if k is the number of cells of the contingency

table, then $\mathscr{F}$ is the positive orthant of $\mathbb{R}^k$. A subset of $\mathscr{F}$ or of $\mathscr{P}$ is a model. As introduced by Bergsma and Rudas [20], marginal log-linear models are defined by considering a hierarchical and complete sequence of marginals of V, $\mathscr{M}_1, \mathscr{M}_2, \ldots, \mathscr{M}_k$, and then considering the marginal log- linear parameterization based on this sequence of marginals and assuming that some of these marginal log-linear parameters, except for the overall effect, are equal to zero.

In the model definition, the marginal log-linear parameter $\lambda_W^{\mathscr{M}}$ for $W \subseteq \mathscr{M} \subseteq V$ may only be assumed to be zero if $\mathscr{M}$ is the first marginal in the hierarchical sequence that contains W, and in that case, the parameter is assumed to be zero for all category combinations[1] of the variables in W. Of course, when $\lambda_W^{\mathscr{M}} = 0$ for all combination of the indices of the variables in W, then

$$\lambda_{W'}^{\mathscr{M}} = 0 \tag{10.2}$$

for all category combinations of the variables W' such that $W \subseteq W' \subseteq \mathscr{M}$. Note that all these marginal log-linear parameters appear in the parameterization, because these W' sets of the variables are contained in $\mathscr{M}$, and $\mathscr{M}$ is the first marginal in the hierarchical sequence that contains them, because it is also the first marginal to contain their subset W.

For example, in a four-way table with variables A, B, C, D, and hierarchical and complete sequence of marginals A, BD, ABD, $ABCD$, if $\lambda_D^{BD} = 0$ is assumed, then $\lambda_{BD}^{BD} = 0$ is implied, but for instance, $\lambda_{BCD}^{ABCD} = 0$ is not implied. Although $D \subseteq BCD$, the marginal log-linear parameter of the effect BCD is calculated in a marginal different from the one where the parameter of the effect D is calculated.

When the hierarchical and complete sequence of marginals consists of a single marginal, the entire contingency table, then marginal log-linear models are log-linear models. This fact also illustrates how much wider and how much more flexible is the class of marginal log-linear models than the class of log-linear models. In the case of a three-way table, the log-linear models are based on the marginal ABC, while marginal log-linear models may be based on any of the sequences of marginals in the list (10.1). All results for marginal log-linear models apply to log-linear models, of course.

In spite of this flexibility, no marginal log-linear model is empty.

Theorem 10.1 *A marginal log-linear model is not empty.*

Proof Consider the uniform distribution on the contingency table. All implied marginal distributions are also uniform, thus, all conditional odds ratios are equal to 1 and then all log-linear parameters except for the overall effect, irrespective of the marginal where they are calculated, are equal to zero.

□

[1] However, see at the end of this section the references to work on context-specific independences about the possibility of relaxing this assumption.

Note that setting the overall effect to zero was excluded in the model definition and the above argument cannot be extended to apply to the overall effect. In the case of a uniform distribution, all cell frequencies in the marginal $\mathscr{M}$, where the $\lambda_{\emptyset}^{\mathscr{M}}$ overall effect is determined, are equal to, say, a. In the case of the balanced log-linear parameters, the overall effect is the average of their logarithms, and cannot be zero.

A more general class of models, where the marginal log-linear parameters are assumed to belong to an affine subspace, was also considered by Bergsma and Rudas [20]. A subset A of $\mathbb{R}^k$ is called an affine subspace, if there is a vector $b \in \mathbb{R}^k$, such that $\{a - b : a \in A\}$ is a linear subspace. These models were called marginal log-affine models and generalize the log-affine models considered in Haberman [71], Lauritzen [98], Rudas and Leimer [154], and the multivariate logistic models in Glonek and McCullagh [65], McCullagh, and Nelder [115]. See also the discussion of the dominating measure q in (8.13) in Sect. 8.4 and the references given there.

Marginal log-linear models are obtained from marginal log-affine models when $b = 0$.

Theorem 9.11 implies that Theorem 8.4 applies to marginal log-linear models and yields the following.

Theorem 10.2 *Let V be a set of variables and $\mathscr{S}$ the contingency table formed by their ranges. Consider for the (strictly) positive general or probability distributions on the contingency table a marginal log-linear model assuming that m of the non-redundant marginal log-linear parameters are zero. Then, the model is a smooth manifold in the $\mathbb{R}^{|\mathscr{S}|}$ dimensional Euclidean space of dimension $|\mathscr{S}| - m$, in the case of the general distributions, and of dimension $|\mathscr{S}| - m - 1$ in the case of probabilities. The unrestricted non-redundant marginal log-linear parameters constitute a smooth parameterization of the model.*

□

Smoothness is a very desirable property of statistical models. Among others, it implies asymptotic normality of the maximum likelihood estimates (see the next section for methods of determining the MLEs) and the asymptotic chi-squared distributions of the Pearson chi-squared and of the likelihood ratio statistics (see Section *LCDA* 5.2) with m degrees of freedom in the notation of Theorem 10.2. These results are described in many places. In particular, Lauritzen [98] gives a concise summary, using exactly the same form of the definition of smoothness, as it was defined in Chap. 8. For rigorous presentations of this theory in the discrete exponential family context used here, see Andersen [4] and Berk [21].

For extensions and applications of the theory of marginal log-linear models and for related results, see, among others, Qaqish and Ivanova [135] on multivariate logistic models; Shpitser et al. [158] on nested Markov models; Roverato et al. [142] and Lupparelli and Roverato [107] on mean linear parameterizations; Colombi and Giordano [32] on hidden Markov models; Colombi and Forcina [31] on testing order restrictions; Colombi and Forcina [30] and Nicolussi and Cazzaro [123] on context-specific independences, where the marginal log-linear parameters are not supposed

to be equal to zero for all categories of the conditioning variables, only for some; Bergsma et al. [18] on extensions of the theory to clustered and longitudinal data, and ways to incorporate the restrictions implied by complex sampling designs; and Rudas and Bergsma [151] for references to many further results.

10.3 Maximum Likelihood Estimation

In this section, various aspects of the maximum likelihood estimation under marginal log-linear models are presented. First, an algorithm is discussed that may seem intuitively appealing but is shown not to work in all cases. Also, the role of smoothness in the existence and properties of maximum likelihood estimates is discussed. Then, as a preparation for the study of the likelihood equations that are appropriate when estimating marginal log-linear models, the Multinomial and the multivariate Poisson likelihoods are considered, and, in the case of marginal log-linear models, are shown to be maximized by closely related distributions. This is a consequence of the homogeneity of marginal log-linear models. Then, the likelihood equations are developed using the Lagrange multiplier method, originally suggested by Aitchison and Silvey [2]. The Newton-Raphson and Fisher scoring algorithms are described briefly, as general methods of solving such equations, and, finally, algorithms proposed to estimate the parameters under marginal log-linear models are presented. It is shown that the algorithms rely on the model and its parameterization being smooth.

10.3.1 A Naive Algorithm

In view of Theorem 7.6 stating that the marginal log-linear parameters determined in marginal $\mathcal{M}_j$ are a one-to-one function of the conditional distribution of the variables in

$$\mathcal{M}_j \setminus \cup_{i<j} \mathcal{M}_i,$$

given the variables in

$$\cup_{i<j} \mathcal{M}_i,$$

and that the marginal log-linear parameters calculated in $\mathcal{M}_j$ are variation independent of the marginal log-linear parameters determined in earlier marginals, one may think that the following recursive algorithm may be used to obtain maximum likelihood estimates in marginal log-linear models. For $\mathcal{M}_1$, the marginal log-linear model implies a log-linear model, with known methods to calculate the MLEs, see Chapter *LCDA* 12. If the MLEs for the marginal distributions in $\mathcal{M}_1, \ldots, \mathcal{M}_{j-1}$

have been determined, then consider the distributions on $\mathscr{M}_i \cap \mathscr{M}_j, i = 1, \ldots, j-1$ as if they were observed lower-dimensional marginals of the distribution on $\mathscr{M}_j$. Then, estimate the higher-order odds ratios in $\mathscr{M}_j$ from the observed data, and then generate a synthetic set of data for $\mathscr{M}_j$ using these as a mixed parameterization. Finally, consider the synthetic data as if they were observed, and fit the log-linear model on $\mathscr{M}_j$ implied by the marginal log-linear model.

To illustrate this algorithm in the case of an $A \times B \times C$ table, let the hierarchical sequence of marginals be A, AB, ABC and let the model assume that $\lambda_{AB}^{AB} = 0$ and $\lambda_{ABC}^{ABC} = 0$. Then, there is no restriction on the A marginal, so the MLE is the observed distribution. The model assumes independence for the AB marginal, so the MLE uses the A marginal distribution from the previous marginal, the observed B marginal distribution and assumes they are independent to estimate the AB marginal distribution. Finally, for the ABC marginal, the assumption is no-second-order interaction, thus, the already estimated AB marginal is used, the AC and BC marginals are estimated to be the same as observed (because the already estimated A and B are also as observed), thus, weak compatibility holds and then these are combined with a second-order odds ratio equal to 1.

Unfortunately, the iterative algorithm described above does not work, in general. The main reason for this is that after the marginal distributions estimated on $\mathscr{M}_1, \ldots, \mathscr{M}_{j-1}$ have been determined, then the distributions on $\mathscr{M}_i \cap \mathscr{M}_j$, $i = 1, \ldots, j-1$, may not be strongly compatible.[2] To illustrate this, for an $A \times B \times C$ table, let the hierarchical and complete sequence of marginals be

$$AB, \quad AC, \quad BC, \quad ABC,$$

and let the model assume that

$$\lambda_{BC}^{BC} = 0, \text{ and } \lambda_{ABC}^{ABC} = 0.$$

Further, let the observed data in relative frequencies be as shown in Table 10.1.

The model implies no restrictions for the AB and AC marginals, thus, the MLEs in these two marginal tables are the observed distributions, as given in Table 10.2.

For the BC marginal, the model prescribes independence, and the two one-way distributions of B and of C have already been estimated: both are uniform, as can

Table 10.1 Observed data for the example where the stepwise estimation procedure fails

$C = 1$			$C = 2$		
	$B = 1$	$B = 2$		$B = 1$	$B = 2$
$A = 1$	0.47	0.01	$A = 1$	0.01	0.01
$A = 2$	0.01	0.01	$A = 2$	0.01	0.47

[2] See Sect. 7.2.4.

Table 10.2 MLEs for the AB and AC marginals

	$B=1$	$B=2$
$A=1$	0.48	0.02
$A=2$	0.02	0.48

	$C=1$	$C=2$
$A=1$	0.48	0.02
$A=2$	0.02	0.48

Table 10.3 MLEs for the BC marginal

	$C=1$	$C=2$
$B=1$	0.25	0.25
$B=2$	0.25	0.25

be seen from Table 10.2. Thus, the MLE in the BC marginal is also uniform, and is given in Table 10.3.

Now the algorithm turns to estimating the ABC distribution, by using the already estimated two-way marginal distributions and combining them with a vanishing second-order association. But, in fact, the three two-way marginal distributions, given in Tables 10.2 and 10.3, are weakly compatible but not strongly compatible. Indeed, if one had a common extension P, then one would have

$$P(1,1,2) + P(2,1,2) = 0.25,$$

but

$$P(1,1,2) + P(1,2,2) = 0.02,$$

thus

$$P(1,1,2) < 0.02,$$

and

$$P(2,1,1) + P(2,1,2) = 0.02,$$

thus

$$P(2,1,2) < 0.02,$$

and so

$$P(1,1,2) + P(2,1,2) < 0.04,$$

which is a contradiction.

Thus, the recursive algorithm does not work for all marginal log-linear models.

In general, there are two types of desirable results about maximum likelihood estimates for a model class. One that gives sufficient (perhaps also necessary)

conditions for the existence of a (unique) MLE and another one about the asymptotic behavior of the MLE, stating that it is unique[3] with probability converging to 1 and that it has an asymptotic normal distribution. However, these results, if hold, are stated under fundamentally different conditions. Conditions for existence results for the MLE usually formulate properties of data sets but make no assumptions about the true distribution that generated the data. For example, Theorem *LCDA* 12.2 gives a necessary and sufficient condition for the existence of the MLE under a log-linear model, in terms of properties of the observed distribution. On the other hand, the asymptotic results assume that the true distribution belongs to the model of interest.

It has to be pointed out that smoothness, in itself, does not imply the existence of a unique MLE; more precisely, if the model is a smooth manifold, it may not yield a unique MLE for all observed data.

As a very simple example, consider a population divided into two categories, and let the respective fractions be P_1 and $P_2 = 1 - P_1$. The (strictly) positive probability distributions (P_1, P_2) are on the line segment connecting $(1, 0)$ with $(0, 1)$, not including the endpoints. This is a smooth manifold of dimension 1 in $\mathbb{R}^2$. To obtain a minimal linear parameterization, identify each point (P_1, P_2) with its distance α from $(0, 1)$. That is

$$\alpha(P_1, P_2) = \sqrt{P_1^2 + (1 - P_2)^2} = \sqrt{2(P_1)^2} = \sqrt{2}P_1. \tag{10.3}$$

Then, α is a parameter and its range is $(0, \sqrt{2})$. The parameter α is also a parameterization and its smoothness is obvious.

Define now a model assuming that $0 < \alpha < 1$. This model is a smooth manifold, with the same parameterization α and Jacobian as all distributions, the only difference is that now the domain of α is $(0, 1)$, while it was $(0, \sqrt{2})$ before.

If for the observed distribution (Q_1, Q_2), $\alpha(Q_1, Q_2) \geq 1$, then there exists no MLE, because on the line segment connecting $(1, 0)$ with $(0, 1)$, the closer is a distribution (P_1, P_2) to (Q_1, Q_2), the larger is the likelihood of the latter, and there is no closest (P_1, P_2) to (Q_1, Q_2) in the model.

As far as the existence of unique MLEs is concerned for marginal log-linear models, there appears to be no general result stating this. Although a sufficient condition was proposed by Bergsma [16], it does not appear to have an intuitive meaning for applications. In fact, Bergsma [16] described the following example for a simple marginal log-affine model with no unique MLE. The example is for an $A \times B \times C$ table, and the model is that

$$OR(A, B) = OR(A, C) = 4, \;\; OR(B, C) = \frac{1}{4}. \tag{10.4}$$

Then, for the data with observed frequency equal to 1 in every cell of the table, there are two distributions, each yielding the same value of the likelihood, which is

[3] More precisely, that the probability that the likelihood function has a global maximum goes to 1.

Table 10.4 Distribution (probabilities multiplied by 8) yielding the highest possible value of the likelihood for the model (10.4) with 1 observation in every cell

$C = 1$			$C = 2$		
	$B = 1$	$B = 2$		$B = 1$	$B = 2$
$A = 1$	0.52	1.98	$A = 1$	1.98	0.37
$A = 2$	0.14	0.52	$A = 2$	0.52	1.98

Table 10.5 Another distribution (probabilities multiplied by 8) yielding the highest possible value of the likelihood for the model (10.4) with 1 observation in every cell

$C = 1$			$C = 2$		
	$B = 1$	$B = 2$		$B = 1$	$B = 2$
$A = 1$	1.98	0.52	$A = 1$	0.52	0.14
$A = 2$	0.37	1.98	$A = 2$	1.98	0.52

the maximum of the likelihood values. These distributions are given in Tables 10.4 and 10.5

Several algorithms have been published in the literature for the maximum likelihood estimation of marginal log-linear models. These algorithms, if converge, may only yield a local maximum of the likelihood function, and, in practice, one needs to run them several times with different starting values, because other local maxima may be higher than the one found.

It has to be emphasized, however, that there is a growing number of applications, mostly in the area of machine learning, where the goal is not to find the parameters that maximize the likelihood but rather to see if the model is able to provide a description of the data that is good enough according to some criterion. The criteria in this context are often related to an expected improvement of the outcome of a procedure (e.g., a 10% expected increase in the sales volume). These applications require finding the parameters associated with a good enough local maximum where one may expect to achieve the goal, and there is not necessarily a need to find the global maximum, see, e.g., Murphy [119].

10.3.2 Multinomial and Multivariate Poisson Sampling

For the marginal log-linear models discussed here,[4] it is possible to derive the MLEs under a Multinomial distribution from the MLEs determined under a Poisson distribution, and the latter task is usually simpler, as one does not have to deal with the requirement that the cell probabilities sum to 1.

[4] The next chapter will discuss models when this is not necessarily true, and will show that many properties of the MLEs are fundamentally different.

The marginal log-linear parameters, with the exception of the overall effect, are such that for two distributions g and h that are proportional in the sense that there exists a number a, so that $h = ag$, they have the same values. This is true, as the assumed proportionality applies to all the marginal distributions, too, and then one obtains the same marginal odds ratios for the two distributions, and thus also the same marginal log-linear parameters, as these are functions of the relevant marginal odds ratios, see Theorem *LCDA* 10.6.

Thus, if a model is defined by restrictions on the marginal log-linear parameters, except for the overall effect, like marginal log-linear models are, then if g is in the model, so is ag. This is called the homogeneity of the model.

Homogeneity of the model implies that if the general distribution g is in the model, then the probability distribution

$$P(g) = \frac{1}{1'g} g$$

is also in the model.

The Multinomial and multivariate Poisson sampling procedures were introduced in Chapter *LCDA* 4. In the multivariate Poisson case, in each cell of the table, one has an intensity, and these, together, are a general distribution on the sample space. Let this be denoted by g. If s is a cell of the sample space, then as shown in Section *LCDA* 4.1.2, the kernel of the log-likelihood function, with observations f, is

$$\sum_{s \in S} f(s) \log g(s) - g(s),$$

or in vector form

$$f' \log g - 1'g, \tag{10.5}$$

where 1 is a vector of 1s.

In the Multinomial case, with cell probabilities P and sample size N, the kernel of the Multinomial log-likelihood (see Section *LCDA* 4.1.1) is

$$f' \log P. \tag{10.6}$$

The domains of the g's and of the P's are, of course, different.

For the relationship between the unrestricted MLEs when Poisson or Multinomial sampling is assumed, Theorem *LCDA* 4.9 stated that if the intensities g of the Poisson distribution are proportional to the probabilities P of the Multinomial distribution with a known constant of proportionality, then the kernels of the log-likelihoods for an observed distribution are the same, and, consequently, do not depend on the known constant of proportionality.

The next result compares the likelihoods of two Poisson distributions for the same set of data.

Theorem 10.3 *Let the population be divided into k categories, and let the observed data be the frequencies* $(f_1, f_2, \ldots, f_k)$. *Suppose the sampling procedure was multivariate Poisson, and the intensities of the categories are to be estimated in a homogeneous model. Let* $\kappa_1, \kappa_2, \ldots, \kappa_k$ *be a set of intensities in the model of interest, and let*

$$\mu_i = \frac{\sum_{j=1}^{k} f_j}{\sum_{j=1}^{k} \kappa_j} \kappa_i, \ \ i = 1, 2, \ldots, k$$

be another set of intensities. Then, the likelihood of the data is higher with the μ *intensities than with the* κ *intensities, except when* $\mu_i = \kappa_i$ *for all i.*

Proof As the model is homogeneous, if the κ parameters are in it, so are the μ parameters.

The ratio of the two likelihoods is

$$\frac{\prod_{i=1}^{k} \frac{\mu_i^{f_i}}{f_i!} \exp(-\mu_i)}{\prod_{i=1}^{k} \frac{\kappa_i^{f_i}}{f_i!} \exp(-\kappa_i)} = \prod_{i=1}^{k} \left(\frac{\mu_i}{\kappa_i} \right)^{f_i} \exp(\kappa_i - \mu_i),$$

and its logarithm, with

$$t = \frac{\sum_{j=1}^{k} f_j}{\sum_{j=1}^{k} \kappa_j} \text{ and } N = \sum_{i=1}^{k} f_i$$

is

$$\sum_{i=1}^{k} f_i \log \left(\frac{\mu_i}{\kappa_i} \right) + \kappa_i - \mu_i = \left(\sum_{i=1}^{k} f_i \log(t) \right) + \frac{N}{t} - N = N \log(t) + \frac{N}{t} - N.$$

To prove the claim, it is enough to see that the last expression is positive for all t, with the exception of the case when $\mu_i = \kappa_i$ for all i, that is, when $t = 1$.

Write the expression as

$$N(\log(t) + \frac{1}{t} - 1),$$

so it is enough to see that

$$\log(t) + \frac{1}{t} - 1 > 0 \text{ if } t > 0 \text{ and } t \neq 1.$$

For $t = 1$, the function is zero. It is then sufficient to show that the function is monotone increasing for $t > 1$ and is monotone decreasing for $0 < t \leq 1$. Its

derivative is

$$\frac{1}{t} - \frac{1}{t^2},$$

and for $t > 1$,

$$\frac{1}{t} > \frac{1}{t^2},$$

so the derivative is positive. Finally, for $0 < t < 1$

$$t^2 < t,$$

thus

$$\frac{1}{t} < \frac{1}{t^2},$$

thus the derivative is negative.

□

The theorem above says that if two sets of intensities are proportional, but one is such that the sum of the intensities is the observed sample size, then this set of intensities yields a higher likelihood than the other one. This immediately implies the following result for the MLE under Poisson sampling for a homogeneous model.

Corollary 10.1 *If the MLEs for a homogeneous model under multivariate Poisson sampling exist, then are such that the sum of the intensities is equal to the total number of observations.*

□

From this, one gets the relationship between MLEs under Multinomial and multivariate Poisson sampling for a homogeneous model.

Theorem 10.4 *For every set of data f, if the MLEs for a homogeneous model exist, then they are proportional under Multinomial and multivariate Poisson sampling, in the sense that for the MLEs of the cell intensities and cell probabilities, one has*

$$\hat{\lambda}_i = N\hat{P}_i,$$

where N is the sample size (the total number of observations), for every cell of the table.

Proof Note that the sample size N is determined in advance under Multinomial sampling but is observed under multivariate Poisson sampling.

Assuming the existence of the MLEs, Corollary 10.1 says that the MLEs under Poisson sampling sum to the sample size N. Let $\overline{\lambda}$ denote such collections of intensities, and then it is sufficient to look for the MLE among such parameters.

If the kernel of the Poisson log likelihood given in (10.5) is greater for such an $\overline{\lambda}^*$ than for another such parameter, that is,

$$f' \log \overline{\lambda}^* - 1'\overline{\lambda}^* > f' \log \overline{\lambda} - 1'\overline{\lambda},$$

then, using that both parameters sum to N, that is, $1'\overline{\lambda}^* = \mathrm{N}$ and $1'\overline{\lambda} = \mathrm{N}$, one has that

$$f' \log \overline{\lambda}^* > f' \log \overline{\lambda},$$

or that

$$f'(\log \overline{\lambda}^* - \log N) > f'(\log \overline{\lambda} - \log N).$$

That is,

$$f' \log \frac{\overline{\lambda}^*}{N} > f' \log \frac{\overline{\lambda}}{N}.$$

This means that the Multinomial likelihood (10.6) is greater for the probability distribution $\overline{\lambda}^*/N$ than for the probability distribution $\overline{\lambda}/N$, and this completes the proof.

□

Consequently, for marginal log-linear models, it is sufficient to maximize the expression (10.5) yielding the MLEs under multivariate Poisson sampling, and the MLEs for Multinomial sampling may be obtained by dividing the estimated intensities by the sample size. The advantage of this approach is that the MLE in the Poisson case may be any (strictly) positive general distribution, as opposed to estimation of the probabilities where maximization is restricted to probability distributions.

10.3.3 The Likelihood Equations

To give an overview of the algorithms that have been proposed to maximize the likelihood function in the case of marginal log-linear models, recall that the log-linear parameters are linear combinations of the logarithms of cell probabilities or frequencies, and in the case of marginal log-linear parameters, one uses marginal cell probabilities or frequencies. So, if the marginal log-linear parameters are calculated from the frequencies f, then they may be written as

$$\mathbf{L} \log \mathbf{M} f,$$

where $\mathbf{M}$ is the matrix generating the marginal frequencies, the logarithm applies componentwise, and $\mathbf{L}$ is the matrix generating the marginal log-linear parameters from the logarithms of the marginal frequencies or probabilities. If the sample space has $|\mathscr{S}|$ cells, then f is a column vector of size $|\mathscr{S}|$; if the parameterization uses m marginal frequencies, then $\mathbf{M}$ is of size $m \times |\mathscr{S}|$ and $\mathbf{L}$ is $|\mathscr{S}| \times m$, because of Theorem 9.8.

For example, in the case of a 2×3 table, the matrix generating all marginal distributions has $1 + 2 + 3 + 6 = 12$ rows, for the entries in the $\emptyset$, A, B, AB marginals, respectively. Obviously, there is a lot of redundancy here, but that will be removed later. In this case, the matrix $\mathbf{M}$ with the lexicographic ordering of the cells is

$$\begin{pmatrix} 1&1&1&1&1&1 \\ 1&1&1&0&0&0 \\ 0&0&0&1&1&1 \\ 1&0&0&1&0&0 \\ 0&1&0&0&1&0 \\ 0&0&1&0&0&1 \\ 1&0&0&0&0&0 \\ 0&1&0&0&0&0 \\ 0&0&1&0&0&0 \\ 0&0&0&1&0&0 \\ 0&0&0&0&1&0 \\ 0&0&0&0&0&1 \end{pmatrix}. \tag{10.7}$$

Which ones of the rows of the matrix above are needed depends on the marginals used in the parameterization. If, for instance, the A and the AB marginals are used in a hierarchical and complete sequence of marginals, then one needs in $\mathbf{M}$ rows 2, 3, 7, 8, 9, 10, 11, 12 of (10.7). The non-redundant balanced marginal log-linear parameters in the parameterization may be chosen as

$$\lambda^A_\emptyset,\ \lambda^A_A(1),\ \lambda^{AB}_B(1),\ \lambda^{AB}_B(2),\ \lambda^{AB}_{AB}(11),\ \lambda^{AB}_{AB}(12).$$

Note that to impose a marginal log-linear model, it is enough to make sure that the non-redundant marginal log-linear parameters are equal to zero; this will imply that the corresponding redundant ones are also equal to zero. For example, if $\lambda^A_A(1) = 0$, then $\lambda^A_A(2) = 0$ follows.

The non-redundant marginal log-linear parameters listed above are obtained from the marginal frequencies using the following transformation:

$$\begin{pmatrix} 1/2 & 1/2 & 0 & 0 & 0 & 0 \\ 1/2 & -1/2 & 0 & 0 & 0 & 0 \\ 1/3 & -1/6 & -1/6 & 1/3 & -1/6 & -1/6 \\ -1/6 & 1/3 & -1/6 & -1/6 & 1/3 & -1/6 \\ 1/3 & -1/6 & -1/6 & -1/3 & 1/6 & 1/6 \\ -1/6 & 1/3 & -1/6 & 1/6 & -1/3 & 1/6 \end{pmatrix} \log \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} f_{11} \\ f_{12} \\ f_{13} \\ f_{21} \\ f_{22} \\ f_{23} \end{pmatrix}. \tag{10.8}$$

Observe that in the example above, the first two lines of the matrix $\mathbf{M}$ give the $f(1,+)$ and $f(+,2)$ values, that is, the A marginal distribution, and the last four lines give the AB marginal distribution. The coefficients in the matrix $\mathbf{L}$ are derived from the definitions of the log-linear parameters. In particular, the last four lines of $\mathbf{L}$ are the last four lines of the inverse of the design matrix (8.7) developed in Sect. 8.3.

Thus, λ_j, the jth non-redundant marginal log-linear parameter in the marginal log-linear parameterization on which the model is based, when applied to a (strictly) positive distribution g, is

$$\lambda_j = \sum_i \mathbf{L}_{ji} \left(\log \left(\sum_l \mathbf{M}_{il} g_l \right) \right), \tag{10.9}$$

and thus the partial derivative of λ_j according to g_k for fixed j and k is

$$\frac{\partial \lambda_j}{\partial g_k} = \sum_i \mathbf{L}_{ji} (\sum_l \mathbf{M}_{il} g_l)^{-1} \mathbf{M}_{ik}, \tag{10.10}$$

or, if the partial derivatives are arranged in a matrix with rows indexed by j and columns indexed by k, then in matrix notation, denoting the vector of the non-redundant marginal log-linear parameters in the marginal log-linear parameterization, on which the marginal log-linear model is based as Λ, one has

$$\frac{\partial \Lambda}{\partial g} = \mathbf{L} \mathbf{D}_{\mathbf{M}g}^{-1} \mathbf{M}, \tag{10.11}$$

where $\mathbf{D}_{\mathbf{M}g}$ is a diagonal matrix with $\mathbf{M}g$ in its diagonal, and thus $\mathbf{D}_{\mathbf{M}g}^{-1}$ is a diagonal matrix with $(\sum_l \mathbf{M}_{il} g_l)^{-1}$ in its position (i,i). This derivative is going to be used in the determination of the MLEs under marginal log-linear models. It is the Jacobian of Λ with the parameters[5] g, and Theorem 9.11 is that its inverse, $\partial g / \partial \Lambda$ is of full

[5] Parameters in the sense of a smooth manifold.

rank. Thus, (10.11) is also of full rank because of the inverse function theorem, see Lax [103].

In the case of log-linear models, one uses the log-linear parameters, thus, there is no marginalization and the matrix $\mathbf{M}$ in (10.9) may be taken to be the identity matrix. In that case, (10.11) yields the Jacobian given in Theorem 9.2.

The marginal log-linear model may be written, with an appropriate matrix $\mathbf{C}$ of size $r \times |\mathscr{S}|$, if there are r non-redundant marginal log-linear parameters set to zero by the model, as

$$\mathbf{C}\Lambda = 0, \tag{10.12}$$

or if Λ is written out in detail based on (10.9)

$$\mathbf{CL} \log \mathbf{M} g = 0,$$

where 0 is a vector of 0s of length r. It will be assumed without restriction of generality that the rows of $\mathbf{C}$ are linearly independent and that $r \leq |\mathscr{S}|$.

For example, if in the 2×3 table considered above, if the model assumes that $\lambda_B^{AB} = 0$, then it can be written as

$$\begin{pmatrix} 0\,0\,1\,0\,0\,0 \\ 0\,0\,0\,1\,0\,0 \end{pmatrix} \begin{pmatrix} \lambda_{\emptyset}^{A} \\ \lambda_{A}^{A}(1) \\ \lambda_{B}^{AB}(1) \\ \lambda_{B}^{AB}(2) \\ \lambda_{AB}^{AB}(11) \\ \lambda_{AB}^{AB}(12) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \tag{10.13}$$

In addition to the model definition in (10.12), there is another way to define the same model. In the example, as $\lambda_B^{AB} = 0$, there are, in fact, only four marginal log-linear parameters to be estimated from the data. In other words, these are free, that is, unspecified by the model and are to be estimated, and while the model definition (10.12) is called the constraint specification, this definition is called the freedom specification. In the example, the model is obviously equivalent to the following freedom specification:

$$\begin{pmatrix} \lambda_{\emptyset}^{A} \\ \lambda_{A}^{A}(1) \\ \lambda_{B}^{AB}(1) \\ \lambda_{B}^{AB}(2) \\ \lambda_{AB}^{AB}(11) \\ \lambda_{AB}^{AB}(12) \end{pmatrix} = \begin{pmatrix} 1\,0\,0\,0 \\ 0\,1\,0\,0 \\ 0\,0\,0\,0 \\ 0\,0\,0\,0 \\ 0\,0\,1\,0 \\ 0\,0\,0\,1 \end{pmatrix} \begin{pmatrix} \lambda_{\emptyset}^{A} \\ \lambda_{A}^{A}(1) \\ \lambda_{AB}^{AB}(11) \\ \lambda_{AB}^{AB}(12) \end{pmatrix}. \tag{10.14}$$

Note that the proper interpretation of the model specification in (10.13 is that the frequencies (or probabilities) are such that when all[6] the non-redundant marginal log-linear parameters are calculated, then the product with the specified matrix gives zeros. On the other hand, the proper interpretation of (10.14) is that the probabilities or frequencies behind the marginal log-linear parameters listed on the right-hand side are such that, when the marginal log-linear parameters on the right-hand side are multiplied by the given matrix, they yield all the non-redundant marginal log-linear parameters of the distributions in the model.

If the matrix in (10.14) is denoted by $\mathbf{K}$ and $\underline{\Lambda}$ is a proper subset of Λ, then the freedom specification is

$$\Lambda = \mathbf{K}\underline{\Lambda}. \tag{10.15}$$

It is easy to see for the matrix $\mathbf{C}$ on the left-hand side of (10.13), that $\mathbf{CK} = \mathbf{0}$, and the matrix $[\mathbf{C}', \mathbf{K}]$ is of full rank because its columns are the columns of the 6×6 identity matrix, and thus are linearly independent.

In more general cases, when the model is more complex than setting some of the parameters to zero, the actual matrices $\mathbf{C}$ and $\mathbf{K}$ are different, and, importantly, the right-hand side of the freedom specification is not a subset of the left-hand side; rather it contains unspecified parameters, but the two properties in the paragraph above may be assumed to hold. In such cases, the model assumes that parameters, say α, of a specified dimension but otherwise unrestricted exist, such that for a matrix $\mathbf{K}$, fully specified by the model:

$$\Lambda = \mathbf{K}\alpha. \tag{10.16}$$

For a general discussion of the constraint and freedom specifications of models for categorical data, see Aitchison and Silvey [2] and Lang [93].

In most of this chapter, the constraint model specification (10.12) is used, and Lagrange multipliers will be applied to impose this on the kernel of the Poisson log likelihood (10.5), see Section *LCDA* 4.1. This approach was first suggested by Aitchison and Silvey [2] and then adapted to marginal categorical data setups by Lang [92] and Lang and Agresti [97].

The Lagrange multiplier μ is a vector of length r, if there are r restrictions in the model. Then, the kernel of the augmented log-likelihood is

$$L(g, \mu) = f' \log g - 1'g + \mu'\mathbf{C}\Lambda = f' \log g - 1'g + \mu'\mathbf{CL} \log \mathbf{M}g. \tag{10.17}$$

The partial derivative of (10.17) with respect to g is written as a row vector of size $|\mathscr{S}|$, as (10.17) is a scalar function of $|\mathscr{S}|$ variables. Using (10.11), one has that

[6] All non-redundant marginal log-linear parameters implied by the given hierarchical and complete sequence of marginals.

$$\frac{\partial L(g,\mu)}{\partial g} = \frac{f'}{g'} - 1' + \frac{\partial \mu' \mathbf{C}\Lambda}{\partial g} \tag{10.18}$$

$$= \frac{f'}{g'} - 1' + \mu' \mathbf{C} \frac{\partial \Lambda}{\partial g} = \frac{f'}{g'} - 1' + \mu' \mathbf{CL} \mathbf{D}_{\mathbf{M}g}^{-1} \mathbf{M}.$$

Note that the derivative above depends on $\mathbf{CL}$, where the matrix $\mathbf{L}$ created the non-redundant marginal log-linear parameters from the logarithms of the marginal cell frequencies or probabilities, and $\mathbf{C}$ picked those parameters that were set to zero by the model assumption. The model (and its likelihood) depends on which of the parameters are set to zero, and it does not depend on the choice of the other parameters.

The derivative of the augmented kernel of the log-likelihood according to g above is written as a row vector,[7] where the division is meant componentwise. The derivative is written as a row vector to have the same structure as in the derivatives of the non-redundant marginal log-linear parameters, where the derivatives of the components of the parameters were in the rows of the Jacobian, while the columns were identified with the components of g. Indeed, the sizes of the matrices in the last expression in (10.18) are $1 \times r$, $r \times |\mathscr{S}|$, $|\mathscr{S}| \times m$, $m \times m$, $m \times |\mathscr{S}|$, respectively, so one gets a row vector of size $1 \times |\mathscr{S}|$.

There is also a partial derivative according to the Lagrange multiplier μ, namely,

$$\frac{\partial L(g,\mu)}{\partial \mu} = \mathbf{C}\Lambda = \mathbf{CL} \log \mathbf{M}g, \tag{10.19}$$

written as a column vector of r components, of course. To have comparable structures of the components of the likelihood equations, (10.18) is rewritten now and will be used in the sequel in a column[8] form:

$$\frac{\partial L(g,\mu)}{\partial g} = \frac{f}{g} - 1 + \mathbf{M}' \mathbf{D}_{\mathbf{M}g}^{-1} \mathbf{L}' \mathbf{C}' \mu. \tag{10.20}$$

In general, there is no closed-form solution to the system of equations:

$$\frac{f}{g} - 1 + \mathbf{M}' \mathbf{D}_{\mathbf{M}g}^{-1} \mathbf{L}' \mathbf{C}' \mu = 0, \tag{10.21}$$

$$\mathbf{CL} \log \mathbf{M}g = 0, \tag{10.22}$$

and various iterative procedures may be used to find a root.

[7] That is, the expression $\frac{\partial a}{\partial g}$, where a is scalar, means a row vector consisiting of the partial derivatives of a according to the components of g.

[8] The column form will be useful when the second derivative is computed in Sect. 10.3.5.

10.3.4 The Newton-Raphson and Fisher Scoring Algorithms

A very general procedure to find the solution of

$$T(t) = 0,$$

if it exists, is the following. Select a starting point t_0 of the iteration. If $T(t_0) = 0$, then one is ready; otherwise, determine the derivative of T, say dT/dt and evaluate it at t_0. If T was a linear function, then dT/dt would be its slope, irrespective of t_0, and the derivative could not be zero; otherwise, T would be constant and there would be no solution, except for $T = 0$. If t^* was such that

$$T(t^*) = 0,$$

then one would have

$$\frac{dT}{dt}[t_0] = \frac{T(t_0)}{t_0 - t^*},$$

so one could write

$$t^* = t_0 - \frac{T(t_0)}{\frac{dT}{dt}[t_0]}.$$

Of course, one does not need iterations, if T is linear. Otherwise, one still may use the above formula with the current value of the derivative, to obtain a next approximation to the solution

$$t_{k+1} = t_k - \frac{T(t_k)}{\frac{dT}{dt}[t_k]}.$$

This iterative method is called the Newton-Raphson method, see, e.g., Dahlquist and Björck [36].

If one is interested in finding an extremal value of a function and, because of this, the root of its derivative is of interest, then the iterations use the first and also the second derivative of the function. So, if $L(f, g)$ is the kernel of the log-likelihood of data f under a particular model and particular sampling scheme, and g is the parameter to be estimated, by maximizing the kernel of the log-likelihood, then the iterations take the form

$$g_{k+1} = g_k - \frac{L'[g_k]}{L''[g_k]},$$

where L' and L'' denote the first and the second derivatives of L in g, for the fixed f, respectively. Less heuristic descriptions of the method would use the Taylor expansion to derive the formulas above, see, e.g., Dahlquist and Björck [36].

This problem is important, as for a wide class of setups, the root of the derivative of $L(f, g)$ maximizes the likelihood, and, in theory, the Newton-Raphson method may be used to find the root.[9]

In the case, when g is a vector of c components, one uses the partial derivative matrix (or gradient):

$$\frac{\partial L}{\partial g},$$

which is a row vector of size $1 \times c$, and the derivative of this, that is, the second derivative of L, which is often called the Hessian or Hessian matrix,

$$\frac{\partial^2 L}{\partial g \partial g'},$$

which is a $c \times c$ matrix, and has in its (i, j) position

$$\frac{\partial \frac{\partial L}{\partial g_i}}{\partial g_j},$$

that is, the first derivative vector is considered in the calculation of the Hessian as a column,[10] and its derivatives are in the rows. Then, the Newton-Raphson method consists of the following iteration:

$$g_{k+1} = g_k - \left(\frac{\partial^2 L}{\partial g \partial g'}[g_k] \right)^{-1} \left(\frac{\partial L}{\partial g}[g_k] \right)'. \tag{10.23}$$

The derivative of the kernel of the log-likelihood is often called the score function, and the equation

$$\frac{\partial L(f, g)}{\partial g} = 0,$$

is the score equation.

Of course, in general, the partial derivative and the Hessian may or may not exist, and the Hessian may not be invertible, and even if these problems are not present, these need to be evaluated at every step of the iteration, which may be a serious sometimes even prohibitive computational burden. A variant of the algorithm is the following.

Note that $L(f, g)$, as it depends on the observed data f, is a random variable, and so are its derivatives, including the Hessian. The expectation of the negative Hessian

[9] See the details of the maximization in a simple setup in the argument leading to Theorem *LCDA* 4.1.

[10] This arrangement is different from the one used so far. The notation $\partial g \partial g'$ is a reminder of the first column, then row arrangement.

is called the Fisher information. In a variant of the Newton-Raphson method called Fisher scoring, instead of (10.23), one has

$$g_{k+1} = g_k + E\left(-\left(\frac{\partial^2 L}{\partial g \partial g'}[g_k]\right)^{-1}\right)\left(\frac{\partial L}{\partial g}[g_k]\right)'.$$

The Fisher information has many deep and interesting properties. Of prime importance for the above iterative process is that it is positive definite, and thus has an inverse, when the likelihood is to be maximized under a particular model, if this model is smoothly parameterized by the parameters g, which is the case with marginal log-linear models, as seen in Theorem 10.2. For other properties of the Fisher information in the context of exponential families, see, e.g., Barndorff-Nielsen [10].

10.3.5 Algorithms for Estimating Marginal Log-Linear Models

The iterative procedure suggested by Aitchison and Silvey [2] and adapted to categorical data by Lang [92] and Lang and Agresti [97], when applied to the current situation, considers the Eqs. (10.21) and (10.22) which need to hold for the MLE $\hat{g}$. Assume now that instead of the MLE, the iteration is at a g that is not too far from the MLE so that first-order approximations may be used. Symbolically, write the two equations as

$$\frac{\partial A}{\partial g}(\hat{g}) + \frac{\partial B}{\partial g}(\hat{g})\mu = 0 \tag{10.24}$$

and

$$B(\hat{g}) = 0,$$

where $A(.)$ is the kernel of the log-likelihood under Poisson sampling and $B(.)$ is the expression used to define the model that is,

$$A(\hat{g}) = f' \log \hat{g} - 1'\hat{g},$$

and

$$B(\hat{g}) = \mathbf{C}\mathbf{L} \log \mathbf{M}\hat{g}.$$

Note that A is a scalar and B is an $r \times 1$ vector. Recall that the partial derivatives appearing in (10.24) are arranged in such a way that the first term is of size $|\mathscr{S}| \times 1$

and in the second term the partial derivative is of size $|\mathscr{S}| \times r$ and the second term is, thus, of size $|\mathscr{S}| \times 1$.

Assume first, as an approximation, that

$$\frac{\partial B}{\partial g}(\hat{g}) = \frac{\partial B}{\partial g}(g),$$

to obtain, instead of the previous equations, that

$$\frac{\partial A}{\partial g}(\hat{g}) + \frac{\partial B}{\partial g}(g)\mu = 0$$

and

$$B(\hat{g}) = 0,$$

Then, apply first-order Taylor approximations for the terms depending on $\hat{g}$ in both equations, which gives that

$$\frac{\partial A}{\partial g}(g) + \frac{\partial A^2}{\partial g \partial g'}(g)(\hat{g} - g) + \frac{\partial B}{\partial g}(g)\mu = 0$$

and

$$B(g) + \left(\frac{\partial B}{\partial g}(g)\right)'(\hat{g} - g) = 0.$$

Note that in the second equation, the transpose is needed, because with the current convention of writing the partial derivative matrix, $\partial B/\partial g$ is a matrix of size $|\mathscr{S}| \times r$.

The two equations may be rearranged as follows:

$$\begin{pmatrix} -\frac{\partial A^2}{\partial g \partial g'}(g) & -\frac{\partial B}{\partial g}(g) \\ -\left(\frac{\partial B}{\partial g}(g)\right)' & 0 \end{pmatrix} \begin{pmatrix} \hat{g} - g \\ \mu \end{pmatrix} = \begin{pmatrix} \frac{\partial A}{\partial g}(g) \\ B(g) \end{pmatrix}. \tag{10.25}$$

If the matrix on the left-hand side of (10.25) had an inverse, then one would have

$$\begin{pmatrix} \hat{g} - g \\ \mu \end{pmatrix} = \begin{pmatrix} -\frac{\partial A^2}{\partial g \partial g'}(g) & -\frac{\partial B}{\partial g}(g) \\ -\left(\frac{\partial B}{\partial g}(g)\right)' & 0 \end{pmatrix}^{-1} \begin{pmatrix} \frac{\partial A}{\partial g}(g) \\ B(g) \end{pmatrix}, \tag{10.26}$$

and from this, $\hat{g} - g$ could be determined, and as g is the actual distribution, an approximation of $\hat{g}$ could be obtained. This process could then be iterated yielding the following algorithm.

Algorithm

Step 0: Set g_0, $i = 0$.
Step 1: Determine $\hat{g}_i$ from g_i using (10.26).
Step 2: If no convergence, set $g_{i+1} = \hat{g}_i$ and $i = i + 1$ and go to Step 1.

In this algorithm, convergence is said to have been achieved if $\hat{g}_i$ and g_i are closer than a predefined threshold or the derivative of the kernel of the augmented log-likelihood is sufficiently close to zero.

This algorithm was suggested by Aitchison and Silvey [2], with the simplification that instead of

$$-\frac{\partial A^2}{\partial g \partial g'}(g),$$

its expectation, that is, the Fisher information for the multivariate Poisson distribution may be used. The advantage of this is that one does not need to compute the inverse[11] of a new matrix in every step. Also, existence and asymptotic distribution results are given in Aitchison and Silvey [2]. Another simplification, suggested by Bergsma [16], is to express μ as a function of g. This reduces the dimension of the parameter space $|\mathscr{S}| + r$ it to $|\mathscr{S}|$, and in a smaller dimensional parameter space, faster convergence may be expected. It was also noted by Bergsma [16] that a stepsize may be needed to be introduced in Step 1 of the algorithm to improve convergence properties. For further details also see Bergsma et al. [18] and Rudas and Bergsma [151].

The key requirement for the algorithm to work is that the matrix

$$\begin{pmatrix} -E\left(\frac{\partial A^2}{\partial g \partial g'}(g)\right) & -\frac{\partial B}{\partial g}(g) \\ -\left(\frac{\partial B}{\partial g}(g)\right)' & 0 \end{pmatrix} \tag{10.27}$$

is invertible. We show now that this is the case.

Theorem 10.5 *The matrix (10.27) is invertible.*

Proof To simplify notation, write this matrix as

$$\begin{pmatrix} \mathbf{T} & \mathbf{U} \\ \mathbf{U}' & \mathbf{0} \end{pmatrix}.$$

Then, it can be verified directly that the following matrix, if exists, is its inverse:

[11] Why this inverse is needed will be seen shortly.

$$\begin{pmatrix} \mathbf{T}^{-1} - \mathbf{T}^{-1}\mathbf{U}(\mathbf{U}'\mathbf{T}^{-1}\mathbf{U})^{-1}\mathbf{U}'\mathbf{T}^{-1} & \mathbf{T}^{-1}\mathbf{U}(\mathbf{U}'\mathbf{T}^{-1}\mathbf{U})^{-1} \\ (\mathbf{U}'\mathbf{T}^{-1}\mathbf{U})^{-1}\mathbf{U}'\mathbf{T}^{-1} & -(\mathbf{U}'\mathbf{T}^{-1}\mathbf{U})^{-1} \end{pmatrix}.$$

Thus, one needs the existence of the inverses of

$$\mathbf{T} = -E\left(\frac{\partial A^2}{\partial g \partial g'}(g)\right) \tag{10.28}$$

and of

$$\mathbf{U}'\mathbf{T}^{-1}\mathbf{U} = -\left(\frac{\partial B}{\partial g}(g)\right)'\left(E\left(\frac{\partial A^2}{\partial g \partial g'}(g)\right)\right)^{-1}\left(\frac{\partial B}{\partial g}(g)\right). \tag{10.29}$$

The matrix (10.28) is the Fisher information for the multivariate Poisson distribution and is obtained as follows:

$$\begin{aligned} \frac{\partial A}{\partial g} &= \frac{\partial f' \log g - 1'g}{\partial g} \\ &= \frac{f}{g} - 1, \end{aligned}$$

as a column, and

$$\frac{\partial A^2}{\partial g \partial g'} = \mathbf{D}_{-f/g^2},$$

where $-f/g^2$ is a vector with $-f_i/g_i^2$ in its ith position. This matrix was also obtained in *LCDA* (4.8). Taking the expectation gives $\mathbf{D}_{-1/g}$, and the Fisher information is

$$\mathbf{D}_{1/g}.$$

The matrix has an inverse, namely,

$$\left(\mathbf{D}_{1/g}\right)^{-1} = \mathbf{D}_g.$$

To prove that the matrix (10.29) has an inverse, we use the well-known fact that this is equivalent to the matrix being positive definite, which means that

$$z'\left(\frac{\partial B}{\partial g}(g)\right)'\left(E\left(\frac{\partial A^2}{\partial g \partial g'}(g)\right)\right)^{-1}\left(\frac{\partial B}{\partial g}(g)\right)z = 0 \tag{10.30}$$

if and only if

$$z = 0,$$

for any vector of length r.

As

$$\left(E\left(\frac{\partial A^2}{\partial g \partial g'}(g) \right) \right)^{-1}$$

is positive definite, (10.30) can only hold, if

$$z' \left(\frac{\partial B}{\partial g}(g) \right)' = 0. \tag{10.31}$$

Recall that

$$B = \mathbf{C}\Lambda,$$

where $\mathbf{C}$ is a matrix of size $r \times |\mathscr{S}|$, with r linearly independent rows. Rewrite (10.31) as

$$0 = z' \left(\frac{\partial \mathbf{C}\Lambda}{\partial g}(g) \right)' = z'\mathbf{C} \left(\frac{\partial \Lambda}{\partial g}(g) \right). \tag{10.32}$$

Indeed, in the (i, j) position of the partial derivative matrix of $\mathbf{C}\Lambda$, one has the partial derivative of the ith component of $\mathbf{C}\Lambda$, according to the jth component of g that is

$$\frac{\partial \sum_{k=1}^{|\mathscr{S}|} \mathbf{C}_{(i,k)} \Lambda_k}{\partial g_j} = \sum_{k=1}^{|\mathscr{S}|} \mathbf{C}_{(i,k)} \frac{\partial \Lambda_k}{\partial g_j}.$$

In matrix notation this is

$$\mathbf{C} \frac{\partial \Lambda}{\partial g}.$$

As the last expression in (10.32) contains the Jacobian of the marginal log-linear parameters, which is of full rank (see Theorem 9.11), one can only have a zero product, if

$$z'\mathbf{C} = 0;$$

otherwise, a linear combination of the rows of the full rank Jacobian, with not all components equal to zero, would give the zero vector. But $\mathbf{C}$ is a matrix of linearly

independent rows, thus, a similar argument implies that $z = 0$, and this completes the proof.

□

Using the freedom specification of marginal log-linear models in (10.14) – (10.16), Colombi and Forcina [29] suggested another algorithm, but Evans and Forcina [52] proved the two algorithms to be identical.

For further details regarding these algorithms, and also for a brief description of the application of the generalized estimating equations (GEE) approach to estimating marginal log-linear models, see Rudas and Bergsma [151].

Software in R for fitting marginal log-linear models include the `cmm` by Wicher Bergsma and Andries van Ark [17] and for many details and examples of its application Németh [120]. This program uses concepts and notations very similar to those presented here. An R package with rich functionality is `hmm`, written by Roberto Colombi, Sabrina Giordano, and Manuela Cazzaro, with contributions by Joseph Lang, see [28] and [33]. This package can fit, in addition to marginal log-linear models, as presented here, also the marginal log-linear models based on lumped odds ratios (see Sect. 7.3), developed by Bartolucci et al. [11], and also various hidden Markov models, see Colombi and Giordano [32]. Also, `hmm` can test inequality constraints imposed on the parameters. Certain marginal models described by Lang [94] and [95] may also be fitted with the R program `mph.fit` by Joseph Lang [96].

10.4 Applications

There are a large number of statistical problems and methods, where the application of marginal models or the marginal view of models already discussed in the literature is useful. The most important examples are various graphical models, also mentioned among the motivating examples to consider marginal log-linear parameterizations and marginal log-linear models. These will be discussed here as special cases of a more general result about collections of conditional independences that may be formulated as marginal models.

10.4.1 Collections of Conditional Independences as Marginal Log-Linear Models

Let $\mathscr{A}_i \neq \emptyset$, $\mathscr{B}_i \neq \emptyset$, $\mathscr{C}_i \subseteq V$ for $i = 1, \ldots, k$ be pairwise disjoint subsets of variables and consider those joint distributions of the variables that are (strictly) positive, and for which the following conditional independences hold:

$$\mathscr{A}_i \perp\!\!\!\perp \mathscr{B}_i | \mathscr{C}_i, \quad i = 1, \ldots, k. \tag{10.33}$$

Many statistical models may be defined in the form (10.33). For example, undirected or directed graphical log-linear models may be written in this form (see Sect. 7.2.1) and also the models associated with repeated measurements and panel studies, discussed in Sect. 7.2.2. Rudas et al. [153] gave a number of results about models of the form (10.33). First, define:

$$\mathbb{D}_i = \mathbb{P}(\mathscr{A}_i \cup \mathscr{B}_i \cup \mathscr{C}_i) \setminus (\mathbb{P}(\mathscr{A}_i \cup \mathscr{C}_i) \setminus \mathbb{P}(\mathscr{B}_i \cup \mathscr{C}_i), \quad i = 1, \ldots, k. \tag{10.34}$$

That is, for every i, $\mathbb{D}_i$ is the class of those subsets of $\mathscr{A}_i \cup \mathscr{B}_i \cup \mathscr{C}_i$ that contain subsets from both $\mathscr{A}_i$ and $\mathscr{B}_i$. Then, one has the following.

Theorem 10.6 *Let V be a set of variables, and let*

$$\mathscr{M}_1, \mathscr{M}_2, \ldots, \mathscr{M}_m$$

be a hierarchical and complete sequence of their marginals with the property that for every $i = 1, \ldots, k$, and every $\mathscr{E} \in \mathbb{D}_i$,

$$\mathscr{C}_i \subseteq \mathscr{M}(\mathscr{E}) \subseteq (\mathscr{A}_i \cup \mathscr{B}_i \cup \mathscr{C}_i), \tag{10.35}$$

where $\mathscr{M}(\mathscr{E})$ is the first marginal in the sequence that contains $\mathscr{E}$. Then, a (strictly) positive distribution P is in the model defined by (10.33) if and only if

$$\lambda_{\mathscr{E}}^{\mathscr{M}(\mathscr{E})}(P) = 0, \text{ for all } \mathscr{E} \in \cup_{i=1}^{k} \mathbb{D}_i. \tag{10.36}$$

Proof Because of (10.35), for every $i = 1, \ldots, k$, and every $\mathscr{E} \subseteq \mathbb{D}_i$,

$$(\mathscr{A}_i \cap \mathscr{M}(\mathscr{E})) \cup (\mathscr{B}_i \cap \mathscr{M}(\mathscr{E})) \cup \mathscr{C}_i = (\mathscr{A}_i \cap \mathscr{M}(\mathscr{E})) \cup (\mathscr{B}_i \cap \mathscr{M}(\mathscr{E})) \cup (\mathscr{C}_i \cap \mathscr{M}(\mathscr{E}))$$

$$= (\mathscr{A}_i \cup \mathscr{B}_i \cup \mathscr{C}_i) \cap \mathscr{M}(\mathscr{E}) = \mathscr{M}(\mathscr{E}).$$

Further, (10.33) implies that

$$(\mathscr{A}_i \cap \mathscr{M}(\mathscr{E})) \perp\!\!\!\perp (\mathscr{B}_i \cap \mathscr{M}(\mathscr{E})) | \mathscr{C}_i$$

thus, for the marginal log-linear parameter that is the log-linear parameter in the marginal $\mathscr{M}(\mathscr{E})$ for the effect $\mathscr{E}$, one has

$$\lambda_{\mathscr{E}}^{\mathscr{M}(\mathscr{E})}(P) = 0,$$

as $\mathscr{E}$ contains variables from both $\mathscr{A}$ and $\mathscr{B}$, see Corollary 7.2.

To see the converse, for every $j = 1, \ldots, m$ define

$$I_j = \{i \leq k : \ \mathscr{C}_i \subseteq \mathscr{M}_j \subseteq \mathscr{A}_i \cup \mathscr{B}_i \cup \mathscr{C}_i\}.$$

Then, (10.33) implies that

$$(\mathscr{A}_i \cap \mathscr{M}_j) \perp\!\!\!\perp (\mathscr{B}_i \cap \mathscr{M}_j) | \mathscr{C}_i, \text{ for } i \in I_j. \tag{10.37}$$

Note that (10.37), for every j, is a conditional independence within $\mathscr{M}_j$.

Not only (10.33) implies (10.37), but also (10.37) implies (10.33) if (10.35) holds. To see this, let

$$\mathscr{E}_i = \mathscr{A}_i \cup \mathscr{B}_i \cup \mathscr{C}_i \in \mathbb{D}_i,$$

and then (10.35) implies that for all $i \leq k$, there is a $j_i \leq m$, such that

$$\mathscr{A}_i \cup \mathscr{B}_i \cup \mathscr{C}_i = \mathscr{M}_{j_i}.$$

This means that $i \in I_{j_i}$, that is, for every $i \leq k$, there is a j, such that $i \in I_j$. Then, for $j = j_i$, (10.37) is the conditional independence required by (10.33) for the particular i in $\mathscr{M}_j$. Therefore, as $\mathscr{M}_m = V$, (10.37) implies (10.33).

Thus, it is sufficient to show that (10.36) implies (10.37) for every $j \leq m$. As $\mathscr{M}_m = V$, it is also sufficient to show this implication for the restriction of P to $\mathscr{M}_j$, say P_j, for $j \leq m$. The rest of the proof is based on two embedded induction arguments. The outer one is in j for $j = 1, \ldots, m$, and the inner one is an iteration to produce P_j showing that it has the required property, if $P_1, \ldots, P_{j-1}$ had it.

First, let $j = 1$. In this case, either I_j is empty, in which case (10.37) is nil, or the required implication is for standard log-linear parameters and for conditional independence, and does hold for P_1. Let now the (outer) induction assumption be that the required implication holds for P_j for all $j < l$ for some $l \leq m$, and it is going to be proved now for $j = l$.

To show that (10.37) holds for P_l, one needs to show that P_l has a factorization using functions, each depending on a subset of the variables, but no subset has a non-empty intersection with both $\mathscr{A}_i$ and $\mathscr{B}_i$, for any $i \in I_l$, see Theorem *LCDA* 10.10. This property, for brevity, will be called factorization according to (10.37). To prove that P_l factorizes according to (10.37), a sequence of distributions on $\mathscr{M}_l$ is defined that converges to it, and then it is showed that each member of the sequence factorizes according to (10.37), and so does the limit P_l.

The sequence of distributions on $\mathscr{M}_l$ converging to P_l starts from a distribution Q^0. Q^0 is any (strictly) positive distribution on $\mathscr{M}_l$ with the following two properties. On the one hand, for every effect $\mathscr{E} \subseteq \mathscr{M}_l \setminus (\cup_{j<l} \mathscr{M}_j)$, the marginal log-linear parameter calculated in $\mathscr{M}_l$ of Q^0 is the same as that of P_l. On the other hand, (10.37) holds for Q^0.

To see that these two properties do not exclude each other, note that the latter applies to $i \in I_l$, and requires the log-linear parameters of the effects that have a non-empty intersection with both $\mathscr{A}_i$ and $\mathscr{B}_i$ to be zero. But this is what (10.36) implies for the marginal log-linear parameters of P_l, and so also for that of Q^0.

This means (see Theorem *LCDA* 10.10) that Q^0 factorizes according to (10.37). Of course, the $\mathscr{M}_1 \cap \mathscr{M}_l, \ldots, \mathscr{M}_{l-1} \cap \mathscr{M}_l$ marginal distributions of Q^0 are not necessarily equal to that of P_l.

The latter distributions $Q^1, Q^2, \ldots$ in the sequence converging to P_l are obtained by applying the *IPFP* (see Section *LCDA* 12.2), to sequentially update the marginals:

$$\mathscr{M}_1 \cap \mathscr{M}_l, \mathscr{M}_2 \cap \mathscr{M}_l, \ldots \mathscr{M}_{l-1} \cap \mathscr{M}_l$$

of the current distributions to obtain the marginals of P_l. Each step updates the $\mathscr{M}_j \cap \mathscr{M}_l$ marginal for $j = 1, \ldots, l-1$ cyclically, so that

$$Q^h(\mathscr{M}_l) = Q^{h-1}(\mathscr{M}_l) \frac{P_j(\mathscr{M}_j \cap \mathscr{M}_l)}{Q^{h-1}(\mathscr{M}_j \cap \mathscr{M}_l)}, \tag{10.38}$$

where $j = h(mod(l-1))$.

The (inner) induction argument will show now that Q^h, $h = 1, 2, \ldots$ factorizes according to (10.37). Indeed, Q^0 was already seen to factorize according to (10.37) on $\mathscr{M}_l$, and suppose now that Q^{h-1} factorizes according to (10.37) on $\mathscr{M}_l$. By the outer induction assumption, $P_j(\mathscr{M}_j \cap \mathscr{M}_l)$, for $j = 1, \ldots, l-1$, also factorizes according to (10.37) for $i \in I_j$. That $P_j(\mathscr{M}_j \cap \mathscr{M}_l)$ also factorizes according to (10.37) for the indices $i \in I_l \setminus I_j$ is shown by the following argument.

If $i \in I_l \setminus I_j$ and $\mathscr{M}_j \cap \mathscr{M}_l \notin \mathbb{D}_i$, then the factorization holds because then $\mathscr{M}_j \cap \mathscr{M}_l$ does not contain subsets from both $\mathscr{A}_i$ and $\mathscr{B}_i$.

The other possibility is that $i \in I_l \setminus I_j$ and $\mathscr{M}_j \cap \mathscr{M}_l \in \mathbb{D}_i$. Then, because of (10.35)

$$\mathscr{C}_i \subseteq \mathscr{M}(\mathscr{M}_j \cap \mathscr{M}_l) \subseteq \mathscr{A}_i \cup \mathscr{B}_i \cup \mathscr{C}_i. \tag{10.39}$$

If $\mathscr{M}(\mathscr{M}_j \cap \mathscr{M}_l)$ was $\mathscr{M}_j$, then (10.39) would imply that $i \in I_j$, which is not true. Let

$$\mathscr{M}(\mathscr{M}_j \cap \mathscr{M}_l) = \mathscr{M}_{j'},$$

for some $j' \neq j$. As $\mathscr{M}_j \supseteq \mathscr{M}_j \cap \mathscr{M}_l$, one must have $j' < j$. Then, (10.39) implies that

$$\mathscr{C}_i \subseteq \mathscr{M}_{j'} \subseteq \mathscr{A}_i \cup \mathscr{B}_i \cup \mathscr{C}_i,$$

meaning that $i \in I_{j'}$. That is, the factorization of $P_{j'}$ according to (10.37), which is part of the outer induction assumption, implies the desired factorization for i.

Then, as $\mathscr{M}_j \cap \mathscr{M}_l \subseteq \mathscr{M}_{j'}$, $P_j(\mathscr{M}_j \cap \mathscr{M}_l) = P_{j'}(\mathscr{M}_j \cap \mathscr{M}_l)$, also factorizes according to (10.37), even if $i \in I_l \setminus I_j$.

Thus, all components on the right-hand side of (10.38) factorize as required and so does $Q^h(\mathscr{M}_i)$. The iteration converges by Theorem *LCDA* 12.6. The limit has the mixed parameterization of P_l, thus, it is P_l. By the assumed positivity, the limit P_l also factorizes, and this completes the inner induction.

Then, the outer induction completes the proof.

□

Theorem 10.6 gave a sufficient condition (the condition given was necessary and sufficient, if (10.35) was true) for a list of conditional independences to be a marginal log-linear model, which was defined by (10.36). The list of conditional independences in (10.33) may also refer to different marginals in the table in the sense that

$$\mathscr{A}_i \perp\!\!\!\perp \mathscr{B}_i | \mathscr{C}_i \tag{10.40}$$

is a conditional independence in the marginal

$$\mathscr{A}_i \cup \mathscr{B}_i \cup \mathscr{C}_i$$

and these may be different for different is, as they are in the case of Markov models associated with directed acyclic graphs.

In the case of Markov models associated with undirected graphs, the defining list of conditional independences refers to the entire table in the sense that

$$\mathscr{A}_i \cup \mathscr{B}_i \cup \mathscr{C}_i = V, \tag{10.41}$$

for all i. In that case, Theorem 10.6 simplifies to the following

Corollary 10.2 *Let V be a set of variables, and let conditional independences be given as in (10.40) and (10.41). Further, let*

$$\mathscr{M}_1, \mathscr{M}_2, \ldots, \mathscr{M}_m$$

be a hierarchical and complete sequence of marginals with the property that for every $i = 1, \ldots, k$, and every $\mathscr{E} \in \mathbb{D}_i$,

$$\mathscr{C}_i \subseteq \mathscr{M}(\mathscr{E}), \tag{10.42}$$

where $\mathscr{M}(\mathscr{E})$ is the first marginal in the sequence that contains $\mathscr{E}$. Then, a (strictly) positive distribution P is in the model defined by (10.40) and (10.41) if and only if

$$\lambda_{\mathscr{E}}^{\mathscr{M}(\mathscr{E})}(P) = 0, \text{ for all } \mathscr{E} \in \cup_{i=1}^{k} \mathbb{D}_i. \tag{10.43}$$

□

If $m = 1$, one obtains the log-linear definition of the undirected graphical models.

To illustrate Corollary 10.2, let there be three variables, and let the model assume the conditional independence $A \perp\!\!\!\perp B|C$. Further, let the sequence of marginals be AB, ABC, which is hierarchical and complete. The conditional independence of A and B does not imply their marginal independence, and Corollary 10.2 does not say that λ_{AB}^{AB} would be zero, because condition (10.42) does not hold.

Theorems 10.2 and 10.6 immediately imply the following.

Corollary 10.3 *The model described in Theorem 10.6 is a smooth manifold in the $|\mathscr{S}|$-dimensional Euclidean space of dimension $|\mathscr{S}| - m$ for general distributions and of dimension $|\mathscr{S}| - m - 1$ for probability distributions, where m is the number of non-redundant marginal log-linear parameters set to zero in (10.36).*

□

The most important types of motivating examples to study marginal models mentioned in Chap. 7 were graphical models. Next, Theorem 10.6 is used to provide smooth marginal log-linear definitions and parameterizations of them.

10.4.2 Directed Graphical and Chain Graph Models

Theorem 10.6 provides ways to define graphical models associated with DAGs as marginal log-linear models, and also gives various smooth parameterizations of them. This is illustrated now for the DAG in Fig. 10.1. Using the local Markov property, it is equivalent to the following list of conditional independences:

$$A \perp\!\!\!\perp B,$$

$$D \perp\!\!\!\perp A|(B, C),$$

$$E \perp\!\!\!\perp (A, B)|(C, D).$$

The sequence of marginals will be chosen according to (10.35). The marginals, in this example, where the conditional independences are defined are AB, $ABCD$, $ABCDE$. Note that these are, in the notation of (10.33)

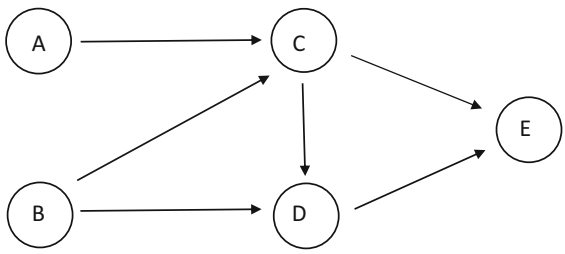


Fig. 10.1 A DAG, for which the marginal log-linear parameterizations are worked out

$$\mathscr{A}_i \cup \mathscr{B}_i \cup \mathscr{C}_i, \quad i = 1, \ldots, k, \tag{10.44}$$

in a hierarchical order. To be able to apply Theorem 10.6, one needs to have marginals with (10.35).

In the example, the model could be defined using the marginals in (10.44), by setting the appropriate marginal log-linear parameters equal to zero, and the smoothness result of Theorem 10.6 would hold. But many other sequences of marginals may be used, and these differ in the types of parameters that describe the model.

As far as $A \perp\!\!\!\perp B$ is concerned, the only set in $\mathbb{D}_1$ is $\mathscr{E} = \{A, B\}$, thus, each of the following is possible as the beginning of the sequence of marginals:

$$AB,$$

$$A, AB,$$

$$B, AB,$$

$$A, B, AB.$$

For $D \perp\!\!\!\perp A|B, C$, $\mathbb{D}_2$ only contains $\mathscr{E} = \{A, D\}$, and then

$$\{B, C\} \subseteq \mathscr{M}(A, D) \subseteq \{A, B, C, D\}$$

is required by the condition (10.35). This excludes, for example, ABD from among the marginals as the next one, and leaves only one possibility, namely, $\mathscr{M}(AD) = \{A, B, C, D\}$. Yet, there remain many other possibilities for marginals to be included prior to it, for example,

$$ABC, ABCD,$$

$$ABD, ABCD,$$

$$AB, AC, ABCD,$$

and several others. For the third conditional independence defining the model, $\mathbb{D}_3 = \{AE, BE\}$ and the condition requires

$$\{CD\} \subseteq \mathscr{M}(AE) \subseteq \{A, B, C, D, E\}.$$

and

$$\{CD\} \subseteq \mathscr{M}(BE) \subseteq \{A, B, C, D, E\}.$$

There are, again, several possible choices for the marginals, including

$$ABCD, ABCDE,$$

$$CDE, ABCDE,$$

$$CDA, CDE, ABCDE.$$

In this case, marginals from the three lists above may be combined freely. For example, one possible choice of the hierarchical and complete system of marginals is

$$A, AB, AC, ABCD, CDE, ABCDE.$$

In this case, the $A \perp\!\!\!\perp B$ independence is imposed via the marginal log-linear parameters in AB, but the A effect is calculated in the marginal A, while the B effect is calculated in the AB marginal that is conditioned on A. If, instead, one used

$$A, B, AB, AC, ABCD, CDE, ABCDE,$$

then also the B effect would be calculated marginally. Thus, the values of different parameters would be used to parameterize the distributions in the model.

General marginal log-linear definition, parameterization, and smoothness results for Markov models associated with DAGs may be obtained by using the so-called well-numbering Markov property, see Lauritzen et al. [99] and also Sect. 2.2. This also applies to certain chain graph models, as common generalizations of undirected and directed graphical models, defined in Sect. 3.5. Let the vertices of a DAG be well-numbered, and denote the variables before V_i in the well-numbering by $pre(V_i)$. Then, the local well-numbering Markov property is that

$$V_i \perp\!\!\!\perp pre(V_i) | pa(V_i).$$

Then, Lauritzen et al. [99] gave the following.

Theorem 10.7 *The directed local Markov property is equivalent to the local well-numbering Markov property.*

Proof In a well-numbering, all descendants of a variable are after it, thus, all the nodes before it are nondescendant. Thus, by Proposition 2.5, the directed local Markov property implies the local well-numbering Markov property.

The converse is shown by induction on the number of variables. If there is one variable, there is nothing to prove. For two variables, if variable A has a descendant, then it is after it and there is nothing to prove. If A has a parent, then $nd(A) \setminus pa(A) = \emptyset$, and there is nothing to prove. If A has a nondescendant that is

not a parent of it, then A and the other variable are not connected by an arrow, and irrespective of their order in the well-numbering, the local well-numbering Markov property implies that they are independent, just like the directed local Markov property.

Let there be k variables, and let V_k be the last one in the well-numbering. The variables before it are all nondescendants of it because of the well-numbering, and no variables are after it, because it is the last one. Therefore, the local well-numbering Markov property for V_k in G_k, implies the directed local Markov property for V_k in G_k.

To see the implication for the rest of the variables, assume the claim is true for directed acyclic graphs (DAGs) with fewer than k nodes and omit V_k from G_k. The original well-numbering remains a well-numbering of the first $k-1$ variables in the induced subgraph G_{k-1}. As V_k is not a parent of any of the earlier variables, $pa_k(V_i) = pa_{k-1}(V_i)$, $1 \leq i \leq k-1$ (and the common notation $pa(V_i)$ will be used). Consequently, the local well-numbering Markov property for G_k implies the same for G_{k-1}.

Then, by the induction assumption, the local well-numbering Markov property in G_{k-1} for a variable V_i, $1 \leq i \leq k-1$ implies the directed local Markov property, which is

$$V_i \perp\!\!\!\perp (nd_k(V_i) \setminus \{V_k\}) | pa(V_i), \tag{10.45}$$

where

$$nd_k(V_i) \setminus \{V_k\} = nd_{k-1}(V_i). \tag{10.46}$$

If $V_k \notin nd_k(V_i)$, then $nd_{k-1}(V_i) = nd_k(V_i)$ and (10.45) is also equivalent to the directed local Markov property in G_k for variable V_i, $1 \leq i \leq k-1$.

If $V_k \in nd_k(V_i)$, then also

$$pa(V_k) \subseteq nd_k(V_i). \tag{10.47}$$

The local well-numbering Markov property for V_k in G_k is that

$$V_k \perp\!\!\!\perp (V_1, \ldots, V_{k-1}) | pa(V_k).$$

Then, as $nd_k(V_i) \setminus \{V_k\} \subseteq \{V_1, \ldots, V_{k-1}\}$, Proposition 2.5 implies that

$$V_k \perp\!\!\!\perp (V_1, \ldots, V_{k-1}) | pa(V_k) \cup nd_k(V_i) \setminus \{V_k\}.$$

Further, as $pa(V_i) \subseteq \{V_1, \ldots, V_{k-1}\}$, because V_k is the last variable in the well-numbering, one has that

$$V_k \perp\!\!\!\perp (V_1, \ldots, V_{k-1}) | pa(V_k) \cup (nd_k(V_i) \setminus \{V_k\}) \cup pa(V_i)$$

or, by (10.47),

$$V_k \perp\!\!\!\perp (V_1, \ldots, V_{k-1}) | (nd_k(V_i) \setminus \{V_k\}) \cup pa(V_i).$$

Then, by Proposition 2.5,

$$V_k \perp\!\!\!\perp V_i | (nd_k(V_i) \setminus \{V_k\}) \cup pa(V_i),$$

or, by 10.46,

$$V_k \perp\!\!\!\perp V_i | nd_{k-1}(V_i) \cup pa(V_i).$$

Finally, the last conditional independence together with (10.45), by Proposition 2.7 imply that

$$V_i \perp\!\!\!\perp (nd_{k-1}(V_i) \cup \{V_k\}) | pa(V_i),$$

which is the directed local Markov property in G_k for V_i, because of (10.46).

□

This result implies the marginal log-linear definition, smoothness, and number of degrees of freedom of Markov models associated with DAGs.

Theorem 10.8 *Let G be a DAG, and let the vertices $V = (V_1, \ldots, V_k)$ be well-numbered. Then,*

$$\{V_i\} \cup pre(V_i), \quad i = 1, \ldots, k$$

is a hierarchical and complete sequence of marginals, with property (10.35), and the Markov model associated with G may be defined as in Theorem 10.6.

Proof By Theorem 10.7, the Markov model associated with G is obtained if the conditional independences implied by the local well-numbering Markov property are imposed. These conditional independences are, for every i,

$$V_i \perp\!\!\!\perp pre(V_i) | pa(V_i).$$

For every i and $\mathscr{E} \in \mathbb{D}_i$,

$$\mathscr{E} = \{V_i\} \cup \mathscr{F}, \text{ where } \mathscr{F} \subseteq pre(V_i).$$

Clearly,

$$\mathscr{E} \subseteq \{V_i\} \cup pre(V_i).$$

As

$$V_i \notin \{V_j\} \cup pre(V_j) \text{ for any } j < i,$$

it follows that

$$\mathscr{M}(\mathscr{E}) = \{V_i\} \cup pre(V_i).$$

Finally, as $pa(V_i) \subseteq \{V_i\} \cup pre(V_i)$, (10.35) follows.

□

Using Theorem 10.7, Rudas et al. [153] gave the following.

Theorem 10.9 *For a chain graph consisting of the well-numbered components $\mathscr{K}_1, \ldots, \mathscr{K}_T$ and for Type IV Markov properties, the following hierarchical and complete sequence of marginals has property (10.35):*

$$\left([PA(\mathscr{K}_t) \cup W : W \subseteq \mathscr{K}_t]^*, \ \cup_{i=1}^{t}(\mathscr{K}_i), \ \ t = 1, \ldots, T\right),$$

where $[.]^*$ *denotes a hierarchical ordering of the sets listed in* $[.]$.

Proof Type IV chain graph models were defined in Sect. 3.5 for each component by the conditional independences M1:

$$\mathscr{K} \perp\!\!\!\perp ND(\mathscr{K}) \setminus PA(\mathscr{K}) | PA(\mathscr{K}),$$

and M2b:

$$W \perp\!\!\!\perp \mathscr{K} \setminus W \setminus nb(W) | PA(\mathscr{K}),$$

and M3b:

$$W \perp\!\!\!\perp PA(\mathscr{K}) \setminus pa(W) | pa(W).$$

When written in the form of (10.33), the conditioning sets $\mathscr{C}_i$ are $PA(\mathscr{K}_t)$ or a subset of theirs, $pa(W) \subseteq PA(\mathscr{K}_t)$, for $W \subseteq \mathscr{K}_t$. The unions $\mathscr{A}_i \cup \mathscr{B}_i \cup \mathscr{C}_i$ for M2b and M3b are subsets of the form $PA(\mathscr{K}_t) \cup W$, for some $W \subseteq \mathscr{K}_t$, though this W may be different from the ones in M2b and M3b.

That is, using indices i for the conditional independences implied for the component $\mathscr{K}_t$ by M2b and M3b, for all of these,

$$\mathscr{C}_i \subseteq PA(\mathscr{K}_t) \cup W,$$

for all $W \subseteq \mathscr{K}_t$ and

$$PA(\mathscr{K}_t) \cup W \subseteq \mathscr{A}_i \cup \mathscr{B}_i \cup \mathscr{C}_i,$$

for some $W \subseteq \mathscr{K}_t$. Further, for any $\mathscr{E} \in \mathbb{D}_i$ (the latter was defined in (10.34)), $\mathscr{M}(\mathscr{E})$ is one of the marginals in the hierarchical and complete sequence given in the claim. As all $\mathscr{E} \in \mathbb{D}_i$ contains a variable from $W \subseteq \mathscr{K}_t$, $\mathscr{M}(\mathscr{E})$ cannot be equal

to a marginal associated with an earlier component, as variables in $W \subseteq \mathcal{K}_t$ cannot be included neither in earlier components nor in parents of them, as the graph of the components is acyclic.

Thus, for the conditional independences implied by M2b and M3b

$$[PA(\mathcal{K}_t) \cup W : W \subseteq \mathcal{K}_t]^*, \quad t = 1, \ldots T,$$

have property (10.35).

For the conditional independences implied my M1, note that these are the ones meaning the directed local Markov property for the DAG of the components. Theorem 10.7 was that this is equivalent to the local well-numbering Markov property, which requires that

$$\mathcal{K}_t \perp\!\!\!\perp (\mathcal{K}_1, \ldots, \mathcal{K}_{t-1}) | PA(\mathcal{K}_t) \text{ for } t = 2, \ldots, T. \tag{10.48}$$

For any $t = 2, \ldots, T$, these are conditional independences in the marginal:

$$\mathcal{K}_1 \cup \mathcal{K}_2 \cup \ldots \cup \mathcal{K}_t. \tag{10.49}$$

Any set $\mathcal{E} \in \mathbb{D}_i$, where the index i refers now to the conditional independence in (10.48) for this t, contains a variable from $\mathcal{K}_t$, and thus cannot be in

$$\mathcal{K}_1 \cup \mathcal{K}_2 \cup \ldots \cup \mathcal{K}_u$$

for any $u < t$, thus, the sets (10.49) need to be included in the hierarchical and complete sequence of marginals, for every t.

□

Then, the previous results of this section and the smoothness of marginal log-linear models imply the following.

Corollary 10.4 *For (strictly) positive probability distributions, Type IV chain graph models are marginal log-linear models and are, thus, smooth. The number of degrees of freedom associated with the model is equal to the number of linearly independent marginal log-linear parameters set to zero in (10.36).*

□

Drton [47] also obtained the smoothness of Type IV chain graph models, but the approach here provides interpretable parameterizations and also the number of degrees of freedom. Marchetti and Lupparelli [109] illustrated with examples that Type IV chain graph models are marginal log-linear models.

For Type I chain graph models, the results of Frydenberg [63] imply smoothness.

Type II chain graph models are not smooth in general (see Drton [47]), but using a marginal log-linear parameterization, Marchetti and Lupparelli [110] identified a smooth subclass.

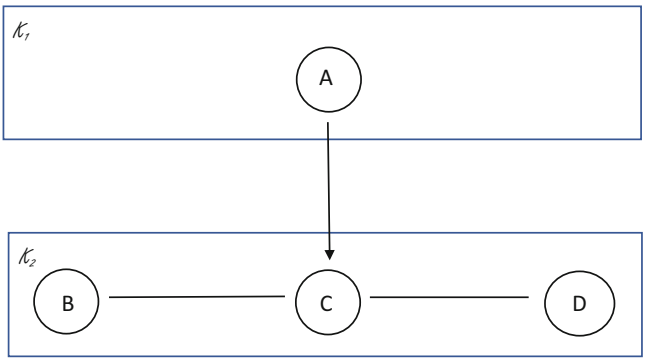


Fig. 10.2 A chain graph where the Type II model is not smooth

To illustrate the scope and limitations of Theorem 10.6 to prove smoothness of chain graph models, we consider an example given in Drton [47]. For the chain graph in Fig. 10.2, one easily sees that the different chain graph Markov interpretations give the following conditional independences:

$$\text{Type I: } B \perp\!\!\!\perp D|(A,C), \quad A \perp\!\!\!\perp (B,D)|C,$$

$$\text{Type II: } B \perp\!\!\!\perp D|(A,C), \quad A \perp\!\!\!\perp (B,D),$$

$$\text{Type III: } B \perp\!\!\!\perp D|A, \quad A \perp\!\!\!\perp (B,D)|C,$$

$$\text{Type IV: } B \perp\!\!\!\perp D|A, \quad A \perp\!\!\!\perp (B,D).$$

For the Type I model, one has

$$\mathscr{C}_1 = AC, \quad \mathscr{A}_1 \cup \mathscr{B}_1 \cup \mathscr{C}_1 = ABCD, \quad \mathbb{D}_1 = BD,$$

$$\mathscr{C}_2 = C, \quad \mathscr{A}_2 \cup \mathscr{B}_2 \cup \mathscr{C}_2 = ABCD, \quad \mathbb{D}_1 = AB, AD, ABD.$$

The first one of these implies $\mathscr{M}(BD) = ABCD$, while the second one requires $C \subseteq \mathscr{M}(AB) \subseteq ABCD$, $C \subseteq \mathscr{M}(AD) \subseteq ABCD$, and $C \subseteq \mathscr{M}(ABD) \subseteq ABCD$. For example, the sequence of marginals ABC, ACD, $ABCD$ allows the formulation of the conditional independences, thus, Theorem 10.6 can be used to prove smoothness.

This is an example where the Type II interpretation is not smooth, see Drton [47]. For the Type II model, one has

$$\mathscr{C}_1 = AC, \quad \mathscr{A}_1 \cup \mathscr{B}_1 \cup \mathscr{C}_1 = ABCD, \quad \mathbb{D}_1 = BD,$$

$$\mathscr{C}_2 = \emptyset, \quad \mathscr{A}_2 \cup \mathscr{B}_2 \cup \mathscr{C}_2 = ABD, \quad \mathbb{D}_1 = AB, AD, ABD.$$

The first one of these implies $\mathscr{M}(BD) = ABCD$, while the second one requires $\mathscr{M}(AB) \subseteq ABD$, $\mathscr{M}(AD) \subseteq ABD$, and $\mathscr{M}(ABD) \subseteq ABD$. The last requirement would imply that ABD is a marginal in the sequence used in the parameterization, but that already contains BD, and this contradicts the first requirement. Thus, Theorem 10.6 cannot be used to prove smoothness.

On the other hand, for the Type IV model, one has

$$\mathscr{C}_1 = A, \quad \mathscr{A}_1 \cup \mathscr{B}_1 \cup \mathscr{C}_1 = ABD, \quad \mathbb{D}_1 = BD,$$

$$\mathscr{C}_2 = \emptyset, \quad \mathscr{A}_2 \cup \mathscr{B}_2 \cup \mathscr{C}_2 = ABD, \quad \mathbb{D}_1 = AB, AD, ABD.$$

The first one of these implies $\mathscr{M}(BD) = ABD$, while the second one requires $\mathscr{M}(AB) \subseteq ABD$, $\mathscr{M}(AD) \subseteq ABD$, and $\mathscr{M}(ABD) \subseteq ABD$. There are no conflicting requirements, thus, Theorem 10.6 can be used to prove smoothness, for example, with the sequence ABD, $ABCD$.

Recall that in Sect. 9.3, references to other works related to smoothness were given.

10.4.3 Path Models and Other Applications

Path models were important motivations for the development of the theory of graphical modeling. Their history goes back to Wright [173], at least. Path models aim at decomposing the effect one variable has on another one, into components through different paths[12] between the two variables. However, as discussed repeatedly, in the case of categorical data, effects are not always individual effects of variables on other ones; rather, joint effects may exist, a characteristic that is in sharp contrast with Gaussian data, see, among others, Section *LCDA* 1.3 and Sect. 6.3.3. Here, we describe a method of applying marginal log-linear models to define path models for categorical data, inspired by Goodman [68], as suggested in Rudas et al. [152]. The essence of the approach is to consider a DAG model in a marginal log-linear parameterization, and restrict it further by assuming that no higher than first-order interactions exist. Then, all the existing effects are associated with either individual variables or with the effect of one variable on another one, represented by an arrow in the DAG. The marginal log-linear parameters not set to zero parameterize the distributions in the model, and quantify the effects that are allowed to exist by the model. Also, the number of degrees of freedom is implied.

To illustrate the procedure, consider the DAG in Fig. 10.1 in Sect. 10.4.2. It was shown there that there are several marginal log-linear parameterizations of the

[12] Recall that in the discussion of causality in Chap. 4, Assumption 2 postulated that if there is an intervention effect of one variable on another one, then there also must be a causal path from the former one to the latter one.

Table 10.6 Marginal log-linear parameters for the DAG in Fig. 10.1, using (10.50)

Marginal	Effect
A	$\emptyset, A$
AB	B, AB
AC	C, AC
$ABCD$	$D, AD, BC, BD, CD, ABC, ABD, ACD, BCD, ABCD$
CDE	E, CE, DE, CDE
$ABCDE$	$AE, BE, ABE, ACE, ADE, BCE, CDE, ABCE, ABDE, ACDE, BCDE, ABCDE$

Table 10.7 Marginal log-linear parameters for the DAG in Fig. 10.1, using (10.50), after the Markov model is imposed

Marginal	Effect
A	$\emptyset, A$
AB	B
AC	C, AC
$ABCD$	D, BC, BD, CD, ACD, BCD
CDE	E, CE, DE, CDE
$ABCDE$	

associated Markov model, as a marginal log-linear model, possible. These included using the hierarchical and complete sequence of marginals:

$$A, AB, AC, ABCD, CDE, ABCDE. \tag{10.50}$$

Using these marginals, one has the marginal log-linear parameters given in Table 10.6.

The Markov properties associated with the DAG are imposed by setting the parameters of such effects to zero, which contain variables from two subsets of variables that are conditionally independent. This is according to Theorem 10.6, and the condition there requires that such parameters are determined in marginals that also contain the respective conditioning sets which is true in the case of the sequence in 10.50.

The conditional independences are

$$A \perp\!\!\!\perp B,$$

$$D \perp\!\!\!\perp A|(B, C),$$

$$E \perp\!\!\!\perp (A, B)|(C, D),$$

thus, the marginal log-linear parameters containing the effects AB, AD, AE, BE, ABE are set to zero. The remaining marginal log-linear parameters are listed in Table 10.7. They parameterize the distributions in the DAG model.

Table 10.8 Marginal log-linear parameters for the path model associated with the DAG in Fig. 10.1, using (10.50)

Marginal	Effect
A	$\emptyset, A$
AB	B
AC	C, AC
$ABCD$	D, BC, BD, CD
CDE	E, CE, DE
$ABCDE$	

The comparison of Tables 10.6 and 10.7 shows how strong a simplification of the structure is the assumption of the Markov model. Nevertheless, because of the categorical data we are modeling here, there are joint effects that are allowed by the model. For instance, λ_{CDE}^{CDE} is a measure of the strength of the joint effect of C and D on E. This is parameterized now marginally, that is, in the CDE marginal. On the other hand, the possible joint effect of B and C on D is parameterized in the $ABCD$ marginal, that is, conditionally on A.

While the joint effect of C and D may exist, it is not an effect that the path model would allow. Only effects associated with nodes (variables) and with arrows (direct effects) in the graph are allowed. Note that already in the Markov model, no effects associated with the arrows not in the DAG are present. To define a path model, all effects including more than two variables are assumed to be zero. This leads to the model definition and parameterization given in Table 10.8.

In the path model, there is a parameter for the main effects of all of the variables, and there is a parameter for every effect that is allowed in the model based on the arrows. These effects are calculated in different marginals. For example, the effect of A on C in the AC marginal, that is, marginally; the separate effects of B on C and on D in the $ABCD$ marginal, that is, conditionally; the effect of C on D is also determined in this marginal, thus conditioned on A and B; the effects of C and D on E are calculated in the CDE are marginal, thus, each is conditioned on the other one. There are no further effects, according to the model definition, which, of course, should be tested using data.

If one chose a different hierarchical and complete sequence of marginals, the effects would be the same, but the conditioning structure would be different. For example, using

$$AB, ABCD, ABCDE$$

would lead to a different parameterization of the path model given in Table 10.8. Here, for instance, the effect of A on C would be measured not marginally, but rather conditioned on B and D, which illustrates the flexibility of the approach (see Table 10.9).

For applications of this type of path modeling to real data, see Németh and Rudas [121, 122], and Rudas et al. [152]. The estimated values of the marginal log-linear

Table 10.9 Marginal log-linear parameters for the path model associated with the DAG in Fig. 10.1, using different marginals

Marginal	Effect
AB	$\emptyset, A, B$
$ABCD$	C, D, AC, BC, BD, CD
$ABCDE$	E, CE, DE

parameters retained in the path model make it possible to reconstruct the MLEs for the data, and if the of testing model fit does not lead to rejection, then they may be seen as containing all relevant information in the original data set.

Topics of further published applications and extensions of marginal log-linear models include latent class analysis (see Forcina [58], Bergsma et al. [18], Rudas and Bergsma [151]); capture-recapture problems (see Bartolucci and Forcina [12], Turner [163]); testing equality, inequality, and monotonicity assumptions (see Bartolucci et al. [11], Dardadoni et al. [37]); rating and test score analysis (see Kuijpers et al. [90, 91]); disclosure limitation (see Bon et al. [24]); incorporating complex sampling designs into the analysis (see Bergsma et al. [19]); and Bayesian approaches (see Bartolucci et al. [13], Ntzoufras et al. [124]). Also, Bergsma et al. [18] and Rudas and Bergsma [150, 151] discuss general aspects of applications of marginal log-linear models.

10.5 Summary

This chapter defined marginal log-linear models by choosing a hierarchical and complete sequence of marginals, and assuming that some of the marginal log-linear parameters determined in these marginals are equal to zero. Earlier results implied that such models are smooth manifolds of dimension equal to the dimension of the unrestricted distribution minus the number of non-redundant zero assumptions. By general theory, smoothness implies the asymptotic normality of the MLE and the asymptotic chi-squared distributions of several test statistics, with the number of degrees of freedom derived from the dimension of the manifold.

Maximum likelihood estimation of marginal log-linear models is a complex problem. It was shown that the MLEs are identical,[13] whether the data set was obtained via Multinomial or multivariate Poisson sampling. The model definitions were given both in the constraint and in the freedom specification forms, and the likelihood equations were worked out. To find a root of the derivative of the kernel of the augmented log-likelihood, the Newton-Raphson algorithm was introduced. It is a generalization of the simple idea that if a function f is not zero at a particular x, then a straight line going through $(x, f(x))$, with the derivative as

[13] In the next chapter, models will be discussed, where this is not necessarily true.

its slope, may intersect the x-axis closer to the root. In fact, for linear functions, it will give the root exactly, and for other functions iteration may work. Fisher's scoring is a simplification that uses, instead of the actual derivative at each iteration, the expectation of the derivative. There have been further simplifications of this procedure suggested in the literature to find the MLE under marginal log-linear models. These modifications and the available software were also described.

In terms of applications, first a sufficient condition was given that a model defined by a list of conditional independences may be formulated as a marginal log-linear model and is, therefore, smooth. This condition was used then to provide marginal log-linear definitions and parameterizations of Markov models associated with DAGs and also of Type IV chain graph models. Using the marginal modeling approach, path models were defined for categorical data. These start with DAG models and then assume that all the marginal log-linear parameters with an effect of more than two variables are zero, that is, effects are limited to individual variables and the arrows present in the DAG.

The chapter concluded with brief hints to further results in the literature.

10.6 Things to Do

1. Prove (10.2).
2. Check that marginal log-linear models may be used to deal with all the statistical tasks mentioned in Sect. 7.2.
3. Prove Theorem 10.2.
4. How does one know in the application of Theorem 8.4 in the proof of Theorem 10.2 that the zeros are possible values of the marginal log-linear parameters?
5. Prove that in the example with (10.3) the likelihood of the observed distribution (Q_1, Q_2) for the hypothetical true distribution (P_1, P_2) is a monotone decreasing function of
$$|\alpha(Q_1, Q_2) - \alpha(P_1, P_2)|\,.$$
6. Check that (10.8) yields the non-redundant marginal log-linear parameters.
7. Consider in an $A \times B \times C$ table and the model of marginal independence of A and B. Work out the likelihood equations assuming Multinomial sampling.
8. In the previous problem, do the likelihood equations depend on the marginals selected to parameterize the model?
9. Prove the comment before Corollary 10.3 in Sect. 10.4.1.
10. Prove that the well-numbering in Theorem 10.9 is possible.
11. Prove Corollary 10.4.
12. Investigate the smoothness of Type III models for the chain graph in Fig. 10.2.
13. What is the precise interpretation of the values of a marginal log-linear parameter pertaining to two variables in a path model?

Chapter 11
Coordinate-Free Relational Models



11.1 Introduction

This chapter introduces models that apply to sample spaces less structured than considered so far. Contingency tables are Cartesian products of the ranges of the variables of interest, and the analysis of data takes place in this multidimensional discrete space. Accordingly, the effects considered in such an analysis are associated with variables or with groups of variables, called interactions. There are situations when the sample space is an incomplete Cartesian product or does not have a product structure. Relational models generalize log-linear models to such sample spaces, by allowing multiplicative effects on the cell parameters that are associated with arbitrary subsets of the cells.

Section 11.2 discusses broad statistical problems where sample spaces, other than Cartesian products, are relevant. Registers that are becoming increasingly important data sources contain records associated with certain events, for example, traffic violations, and describe their features. They do not contain records with no feature present. Another incomplete Cartesian product structure arises when an initial position is cross-classified with change in that position, as a response to some intervention. Those in the highest position cannot go up and those in the lowest position cannot go down. For such incomplete Cartesian product sample spaces, quasi log-linear models are described. Further examples include analyzing the relationship between the categories of a single variable and modeling the effect of features that are not coordinates of a product.

Relational models are defined in Sect. 11.3. Each cell of the sample space is characterized by a cell parameter. This may be a probability, when sampling is Multinomial, or an intensity, when sampling is multivariate Poisson. The separate treatments of these two cases, which are not necessary in the case of log-linear or marginal log-linear models, are necessary here, as will be seen later. A relational model is defined by specifying subsets of the cells of the sample space, and

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T. Rudas, *Lectures on Advanced Topics in Categorical Data Analysis*, Springer Texts in Statistics, https://doi.org/10.1007/978-3-031-55855-9_11

associating a subset parameter[1] with each. Then, the cell parameter of any cell is assumed to be the product of the subset parameters of all subsets that contain this cell. The indicators of the cells in the subsets defining the model may be arranged into the columns of a design matrix. This may be used to have a linear form of the model for the logarithms of the cell parameters. Log-linear models are the special case, when the sample space is a Cartesian product and the subsets are cylinder sets. As a generalization of log-linear models, relational models may be defined by setting generalized odds ratios equal to one. These are ratios of products of cell parameters. When the number of cells in the numerator is the same as that of in the denominator, then the generalized odds ratio is homogeneous; otherwise, it is non-homogeneous.[2]

Whether the columns of the design matrix may be combined into the vector of 1s has a fundamental effect on the properties of relational models. If yes, there is a reparameterization that contains an overall effect present in every cell. In that case, the relational model may be defined with homogeneous generalized odds ratios. Otherwise, one generalized odds ratio in the definition is non-homogeneous.

Maximum likelihood estimation under a relational model is discussed in Sect. 11.4. When the overall effect is present in the model, then the usual proportionality of the MLEs of the cell probabilities and of the cell intensities holds. When the relational model does not have an overall effect, this is not true. The MLEs of cell probabilities reproduce the sums of the observed relative frequencies in the defining subsets only up to a constant of proportionality, and the MLEs for the cell intensities do not reproduce the total sample size.

Relational models are shown to be smooth manifolds, which implies standard asymptotic behavior of the MLEs and of certain test statistics. A necessary and sufficient condition for the existence of the MLE is given. Finally, an algorithm is described to find the MLEs which is also implemented in an R package.

A brief summary and a list of things to do conclude the chapter.

11.2 Nonstandard Sample Spaces

While log-linear models are classical tools of the analysis of multivariate categorical data, graphical models, in particular in the directed variant, have gained popularity during the last decades, and marginal log-linear models provide a very flexible generalization of these, all these models have limitations. The limitations that are in the focus of interest of this chapter are partly related to the structure of the sample space and partly to the types of effects and interactions modeled. In

[1] In the sequel, subset parameters will often be called parameters, but cell parameters will always be referred to like that.

[2] More precisely, the sums of the exponents of the cells in the numerator and in the denominator have to be equal for the generalized odds ratio to be homogeneous.

the approaches discussed so far, the sample space is a contingency table, more technically a Cartesian product of the ranges of the variables involved. Accordingly, the effects that one tries to model (identify, test, interpret) in all these approaches are associated with groups of variables. The effects may be marginal or conditional but are always associated with subsets of the variables.

There are, however, important analytical problems where this general framework does not appear to provide a satisfactory description of the structure. These include situations when the sample space has an incomplete Cartesian product structure but also cases when there seems to be no product structure at all and the sample space can be any finite set. Consequently, also the concept of an effect needs to be defined in terms more general than necessarily being associated with subset of the variables. This is the second aspect in which the models to be discussed in this chapter generalize earlier approaches. An effect may be present in any subset of the cells of the table, not only in subsets of cells associated with subsets of variables.

The most important examples of sample spaces that are incomplete Cartesian products are registers. Registers contain records of particular events, and each record lists some of the features characterizing the event. For example, health authorities of a country may operate a register of congenital malformations. The events registered are the births of babies with congenital malformations, and the features registered are the types of malformation present in the newborn baby, and often also demographic characteristics and comorbidities. If there are k malformations monitored, then the sample space is not a 2^k contingency table, because babies with no malformation do not enter the register. It will be seen later that the lack of that particular cell from the Cartesian product structure of the sample space leads to important difficulties. In this example, the data may be completed by the number of healthy newborns, which is certainly available, and then one has a Cartesian product as the sample space. The drawback of this completion approach is that the frequency in the no-malformation cell will be huge relative to the frequency of any other cell, and whatever way of measuring model fit is applied, fit will depend mostly on correctly reproducing the number of healthy babies, and will do little in terms analyzing the associations among the malformations.

But there are situations when such an imputation is not possible. Cash register data from supermarkets contain the goods purchased, their prices, and some additional information. There is no cash register data about those customers who leave the supermarket without buying anything, and their number is not known. In this example, an empty record (that is one with no feature present) is meaningful, but the number of such records is not known.

In a register of traffic violations or of visits of a hospital emergency ward, features of the event are marked. Not only do the registers contain no record when no rule was violated or no visit happened because there was no reason for it, but such things do not exist, and one cannot impute their numbers. It makes no sense to ask in how many instances was no traffic rule violated or in how many instances had no one a reason to see the doctor.

The foregoing considerations suggest that the standard theory of random or sampling and structural zeros in a contingency table (see, e.g., Bishop et al. [23])

Table 11.1 An incomplete 2×2 contingency table

		F_2	
		No	Yes
F_1	No	–	(0, 1)
	Yes	(1, 0)	(1, 1)

Table 11.2 The marginal probabilities cannot be extended to the table

		F_2	
		0.8	0.2
F_1	0.7	–	0.7
	0.3	0.8	???

needs to be revised to include the possibility of nonexistent cells. There are cells that may be empty in the data but not in the underlying population. There are cells that may be empty in a particular population (and, consequently, in any sample from that population), but it is not impossible that they are not empty in other populations. And there may be cells that are logically[3] impossible and are, consequently, empty in any population and any data set.

These differences need to be reflected upon by the methods and models used in data analysis. If a cell is not possible, it does not seem to be a good analysis that estimates or assumes a positive probability of frequency there. Even independence in a 2×2 table associates positive probability with all four cells, if the marginal probabilities are positive, and it does not appear to be an easy task to modify the model in case some of the cells are nonexistent and, consequently, need to remain empty. Table 11.1 shows the structure of an incomplete 2×2 table with two features F_1 and F_2, and 1 indicating the presence and 0 the absence of the feature.

Compatibility issues of marginal distributions become more complicated than on complete Cartesian products. When one considers the two one-way marginals, they are decomposable, as they are disjoint, they are always weakly compatible, yet not always strongly compatible, thus, Theorem 7.2 is not true in this case. This is illustrated in Table 11.2. If

$$P(F_1 = 0) = 0.7,\ P(F_1 = 1) = 0.3 \text{ and } P(F_2 = 0) = 0.8,\ P(F_2 = 1) = 0.2, \tag{11.1}$$

then there is no extension of these measures to the incomplete Cartesian product.

Formulated slightly more generally, the problem is that the algebras (of events) generated by the individual variables are not qualitatively independent,[4] see Marczewski [112], as

[3] Logically here means according to the standard rules of thinking.

[4] Qualitative independence requires that any two non-empty events, each from one of the algebras, have a non-empty intersection.

$$\{F_1 = 0\} \cap \{F_2 = 0\} = \emptyset.$$

Therefore, if an extension P existed, then one would have:

$$P(F_1 = 0, F_2 = 1) = P(F_1 = 0)$$

and

$$P(F_1 = 1, F_2 = 0) = P(F_2 = 0),$$

thus these probabilities from the unrelated one-way distributions would appear in the two-way table, which is not possible because their sum is more than 1.

The independence of two variables has been characterized by two properties. The joint distribution preserves the marginal distributions, that is, it is an extension of them, and the odds ratio (or an appropriate generalization of it) is equal to 1. As extensions do not always exist in the cases of sample spaces that are incomplete Cartesian products, independence in the classical sense does not exist, either. Also, there is no odds ratio if the sample space has the structure of Table 11.1.

The situation is similar in the case of the model of conditional independence. If there are three binary variables F_1, F_2, F_3, and the sample space is a complete Cartesian product, then if probability distributions (or, more generally, measures), say P and Q, are given on the (F_1, F_2) and on the (F_2, F_3) marginals, respectively, and these distributions are weakly compatible that is

$$P(+, F_2) = Q(F_2, +),$$

then there always exists a common extension of P and of Q to the three-dimensional space, including the one where

$$F_1 \perp\!\!\!\perp F_3 | F_2.$$

If the cell $(0, 0, 0)$ does not exist, that is, the Cartesian product is incomplete, then this is not the case. As an example, Table 11.3 gives the distribution P on the F_1, F_2 marginal, and Table 11.4 gives the distribution Q on the (F_2, F_3) marginal. These are weakly compatible because they coincide on the F_2 marginal. In spite of this, there exists no common extension to the three-way table, as is illustrated in Table 11.5.

Table 11.3 An (F_1, F_2) marginal distribution

		F_2	
		No	Yes
F_1	No	0.6	0.1
	Yes	0.1	0.2

Table 11.4 An (F_2, F_3) marginal distribution

$F_3 = \text{No}$		$F_3 = \text{Yes}$	
$F_2 = \text{No}$	$F_2 = \text{Yes}$	$F_2 = \text{No}$	$F_2 = \text{Yes}$
0.5	0.2	0.2	0.1

Table 11.5 No common extension of the distributions in Tables 11.3 and 11.4 exists

	$F_3 = \text{No}$		$F_3 = \text{Yes}$	
	$F_2 = \text{No}$	$F_2 = \text{Yes}$	$F_2 = \text{No}$	$F_2 = \text{Yes}$
$F_1 = \text{No}$	–	???	0.6	???
$F_1 = \text{Yes}$	0.5	???	???	???

Table 11.6 The structure of the cross-classification of Original Grade and Change

		Change		
		Higher	Same	Lower
Original	A	–	x	x
grade	B	x	x	x
	C	x	x	x
	D	x	x	x
	F	x	x	–

In this case, because of the incomplete Cartesian product structure, the algebras of events on the three-dimensional space generated by the (F_1, F_2) and by the (F_2, F_3) marginals are not qualitatively conditionally independent, see Bártfai, Rudas [14],[5] and Arató et al. [8]. And in that case, weak compatibility does not imply strong compatibility.

Further examples of sample spaces that are incomplete Cartesian products include many situations, where change is being measured. For example, a test is graded with letter grades A, B, C, D, F, and then a second test is given, and only the change in the grade is noted. It can be Higher, Same, and Lower, compared to the original grade. One may be interested in how the change in the grade relates to the original grade. Thus, Original Grade, with five categories, may be cross-classified with Change with three categories. Obviously, for those whose grade was A, no higher, and for those whose grade was F, no lower grade may be observed, leading to an incomplete Cartesian product.

Interestingly, the incomplete nature of the data in Table 11.6 is related to structures where effect is not associated with variables that define the Cartesian product, but rather with other subsets of variables. If one cross-classifies original grade with second grade, then one has a complete Cartesian product, but the effects of interest are not all associated with the variables forming the table. Table 11.7

[5] See also in Sect. 2.9.

Table 11.7 Change in the cross-classification of original grade with the second grade

		Second grade				
		A	B	C	D	F
Original	A	S	L	L	L	l
grade	B	H	S	L	L	L
	C	H	H	S	L	L
	D	H	H	H	S	L
	F	H	H	H	H	S

shows the categories of Change in the 5×5 complete Cartesian product, with H = Higher, S = Same, and L = Lower.

Quasi log-linear models (see Goodman [67], Fienberg [55], or Bishop et al. [23]) extend the possibility of applying log-linear models to incomplete Cartesian products. First, assume that the nonexisting cells exist, and thus obtain a complete Cartesian product. Then, consider a log-linear model on the complete Cartesian product, with an extra multiplicative parameter included in every cell that does not exist in reality. Call this model the augmented log-linear model. Fit the augmented log-linear model to the data on the complete Cartesian product, so that the frequencies in the cells that do not exist are assumed to be zero. Then, the estimates for the cells in the incomplete Cartesian product are considered the estimates under the quasi log-linear model.

More precisely, let V be a set of variables, and let $\mathscr{T}$ be the Cartesian product of their ranges, which is the complete sample space. Let v denote a cell of the table and let the cells $v \in \mathscr{U}$ be nonexistent, where $\mathscr{U} \subseteq \mathscr{T}$. Let Γ be a descending class of subsets of V. Then, the augmented log-linear model for Γ assumes that

$$P(v) = d_v \prod_{\gamma \in \Gamma} g_\gamma((v)_\gamma), \tag{11.2}$$

where

$$d_v = 1, \text{ if } v \notin \mathscr{U}.$$

The augmented log-linear model has two important properties. First, the parameter d_v has no effect in the case of existing cells, because in that case it is set to 1. If a cell v does not exist, then the parameter d_v only appears in that cell and its value is to be estimated. Therefore, the model contains no restriction for the cells that do not exist. Second, the log-linear model is equivalent to assuming that certain higher-order odds ratios are equal to 1. If such an odds ratio only contains existing cells, then the quasi log-linear model assumes that this is equal to 1, because no d_v parameter enters it. If, however, such an odds ratio contains any nonexisting cell, then the odds ratio is unrestricted by the quasi log-linear model, because it contains d_v parameters. This is formulated in the next proposition.

Proposition 11.1 *Let V be a set of variables and let $\mathscr{T}$ be the Cartesian product of their ranges. Assume the cells in $\mathscr{U} \subseteq \mathscr{T}$ do not exist. Let P be a distribution on $\mathscr{T}$ in the quasi log-linear model (11.2). Further, let Q be a distribution on $\mathscr{T}$ in the log-linear model defined as in (11.2), but with $\mathscr{U} = \emptyset$. Then, if $v \in \mathscr{U}$, then $P(v)$ is not restricted by the quasi log-linear model.*

Further, if the log-linear model implies that

$$COR_Q(W|V \setminus W = (c)_{V\setminus W}) = 1,$$

then

$$COR_P(W|V \setminus W = (c)_{V\setminus W}) = 1$$

is implied by the quasi log-linear model, if

$$(w, (c)_{V\setminus W}) \notin \mathscr{U}, \text{ for all indices } w \text{ of } W.$$

□

For example, if the sample space is as shown in Table 11.1, then

$$\mathscr{T} = \{(0, 0), (0, 1), (1, 0), (1, 1)\},$$

and

$$\mathscr{U} = \{(0, 0)\}.$$

In this case, for the log-linear model of independence, there is only one odds ratio set to 1 by the model:

$$\frac{Q(0, 0)Q(1, 1)}{Q(0, 1)Q(1, 0)} = 1,$$

and as the cell (0, 0) is involved, the condition of the second part of Proposition 11.1 does not hold, and

$$\frac{P(0, 0)P(1, 1)}{P(0, 1)P(1, 0)}$$

is not restricted by the quasi-independence model. In fact, the quasi-independence model in the case of the sample space in Table 11.1 is not restrictive, at all, and is not a model, therefore.

For the $2 \times 2 \times 2$ sample space with nonexisting (0,0,0) cell, with the structure shown in Table 11.5, and the log-linear model of conditional independence for the completed table

$$F_1 \perp\!\!\!\perp F_3|F_2,$$

the model is that

$$COR_Q(F_1, F_3|F_2 = 0) = 1 \text{ and } COR_Q(F_1, F_3|F_2 = 1) = 1.$$

The quasi conditional independence model is that

$$COR_P(F_1, F_3|F_2 = 1) = 1,$$

because in this conditional odds ratio, the cell $(0, 0, 0)$ does not occur, while it does when conditioning is on $F_2 = 0$.

Heuristically, one may say that the quasi log-linear model maintains the restrictions of those (conditional) odds ratios that are meaningful, when the nonexistence of some of the cells is taken into account.

Assuming independence of F_1 and F_3, when $F_2 = 1$ but not when $F_2 = 0$ is also a simple example of context-specific models, see Colombi and Forcina [30] and Nicolussi and Cazzaro [123].

The models, that is, simple structures to be discussed in this chapter, generalize, among others, quasi log-linear models and context-specific models that apply to sample spaces that are incomplete Cartesian products. But the models also apply to finite sample spaces with no further structure, at all. Important examples of this situation include when one wishes to analyze responses to a single question. Then, the sample space consists of as many categories as the number of options the respondents may choose from, and every observation falls into exactly one category. Thus, there is no cross-classification, at all. An example may be analyzing the countries of origin of international students at a university. In such cases, effects may be associated, in theory, with arbitrary subsets of the sample space. For example, one effect may be associated with the country being European, another with the country having a per capita GDP above $ 30.000. An alternative approach may be to define these and possibly other features as variables and consider the Cartesian product of the ranges. But this table may have empty or nonexisting cells, similarly to the example of Original Grade and Change in grade. The problem of sample spaces that are incomplete Cartesian products is closely related to the coordinate-free approach where effects may be associated with arbitrary subsets of the variables.

Note, however, that even if the sample space is a complete Cartesian product, and a log-linear model is considered, the effects allowed in the model are also associated with subsets of the variables. In 2×2, $A \times B$ table, with the model of independence, the multiplicative parameter β_0^A is present in the cells $(0, 0)$ and $(0, 1)$ or the multiplicative parameter β_0^B is present in the cells $(0, 0)$ and $(1, 0)$. Such sets are called cylinder sets. More generally, a cylinder set is defined as

$$\{c : (c)_W = w\},$$

for a $W \subseteq V$, and joint index w of the variables W.

In the next section, multiplicative models will be considered, where each effect is associated with, that is, is present in a subset of the sample space.

11.3 Relational Models

A fairly general model class, called relational models, was suggested by Klimova et al. [87] that contains models that may be applied in the variety of situations illustrated in Sect. 11.2. These models generalize log-linear models on sample spaces that are complete Cartesian products, and, in general, effects may be associated with arbitrary subsets of cells, not only cylinder sets. Such models, in social mobility research, were considered by Hauser [73], Hout [75], Kovach et al. [89], Zelterman and Youn [174], Klimova and Rudas [80], and in modeling rater agreement, see Tanner and Young [162], and see also Rudas [143].

One of the surprising aspects of relational models is that in this generality, whether the data were generated by a Multinomial or a multivariate Poisson distribution may lead to different MLEs. This is in sharp contrast with the case of unrestricted MLEs (see Theorem *LCDA* 4.8), and with the case of log-linear models or regular exponential families on a sample space that is a complete Cartesian product, see Theorems *LCDA* 6.6, 7.4, 12.2, and with the case of homogeneous models (see Theorem 10.4). Therefore, the Multinomial and the multivariate Poisson sampling procedures will be treated parallel but separately.

Let the sample space be denoted by $\mathscr{I}$. This may be a complete Cartesian product or an incomplete Cartesian product, and it also may be any finite set. The language, nevertheless, will be borrowed from the contingency table case, thus, $\mathscr{I}$ will be called a table, and an $i \in \mathscr{I}$ a cell or an index. The index i will sometimes be treated as if the cells were numbered from 1 to $|\mathscr{I}|$. Following Klimova et al. [87], the parameters δ_i associated with the cells $i \in \mathscr{I}$ are either probabilities:

$$\delta_i = p_i, \quad p_i > 0, \quad \sum_{i \in \mathscr{I}} p_i = 1,$$

which will be used when sampling is Multinomial, or are intensities

$$\delta_i = \lambda_i, \quad \lambda_i > 0,$$

which will be used when sampling is multivariate Poisson.

A relational model assumes that the cell parameters δ_i are products of other parameter values, say α. Each parameter α is associated with a subset of the cells of $\mathscr{I}$ and appears in those cells, which belong to this subset. Thus, a relational model is defined by specifying subsets $S_1, \ldots S_k$ of $\mathscr{I}$. The model assumes that

$$\delta_i = \prod_{j: i \in S_j} \alpha_j, \tag{11.3}$$

for some (strictly) positive parameters $\alpha_1, \ldots, \alpha_k$. Using a design matrix $\mathbf{A}$ of size $|\mathscr{I}| \times k$, where the entries of $\mathbf{A}$ are

$$a_{i,j} = 1 \text{ if } i \in S_j, \text{ and } a_{i,j} = 0 \text{ if } i \notin S_j,$$

because of the assumed positivity of the cell parameters, the model may also be written as

$$\log \delta = \mathbf{A}\phi, \tag{11.4}$$

where the components of ϕ are the logarithms of the α_j values. This model will be referred to as the relational model defined by $\mathbf{A}$.

This definition describes the parameters δ that are in a relational model. In each cell i, the parameter δ_i, whether it is a probability or an intensity, is the product of parameters, such that each is associated with one of the sets S_j that the particular cell i belongs to.

The sense in which relational models may be coordinate-free is that the sets S_j are not restricted to being cylinder sets of a sample space of a product structure. Even if the sample space does have a product structure, the domain of an effect may be entirely unrelated to (groups of) variables. Further, the sample space does not need to have a product structure at all.

Of course, if the design matrix is not of full column rank, a reparameterization of the distributions, more precisely of their logarithms, in the relational model as described in Theorem 8.1 is possible. Such a reparameterization is possible, as was shown in the proof of Theorem 8.1, by retaining the linearly independent columns of the design matrix. Therefore, in the sequel, without loss of generality, it will be assumed that the design matrix $\mathbf{A}$ is a $0-1$ matrix of full column rank.

Further, it will be assumed that every row of $\mathbf{A}$ contains at least one 1; otherwise, the product in (11.3) would be empty for that particular cell.

As a simple example, let the sample space have four cells, denoted as a, b, c, d. Different choices of the subsets S lead to models with fundamentally different properties.

Let first

$$S_1 = \{a, b\}, \quad S_2 = \{c, d\}, \quad S_3 = \{a, b, c, d\}. \tag{11.5}$$

Here the design matrix is

$$\begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

Clearly, the columns of this matrix are not linearly independent, and a straightforward reparameterization would be to define the model with the subsets:

$$S_1 = \{a, b\}, \quad S_2 = \{c, d\} \tag{11.6}$$

with the design matrix

$$\begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}.$$

This leads to the following parameterization of the cell parameters:

$$\delta_a = \alpha_1, \quad \delta_b = \alpha_1, \quad \delta_c = \alpha_2, \quad \delta_d = \alpha_2.$$

The same model could also be obtained by having

$$S_1 = \{a, b\}, \quad S_2 = \{a, b, c, d\}. \tag{11.7}$$

and the design matrix

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}.$$

In the case of this parameterization, one has, using βs for the parameters,

$$\delta_a = \beta_1\beta_2, \quad \delta_b = \beta_1\beta_2, \quad \delta_c = \beta_2 \quad \delta_d = \beta_2.$$

The equivalence of the two parameterizations of the same model is seen by setting

$$\beta_1 = \frac{\alpha_1}{\alpha_2}, \quad \beta_2 = \alpha_2,$$

and

$$\alpha_1 = \frac{\beta_1}{\beta_2}, \quad \alpha_2 = \beta_2.$$

Another model, for distributions on the same sample space, is defined by

$$S_1 = \{a, b\}, \quad S_2 = \{b, c\}, \quad S_3 = \{c, d\} \tag{11.8}$$

and design matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

The distributions in this model may be written as

$$\delta_a = \alpha_1,$$

$$\delta_b = \alpha_1\alpha_2,$$

$$\delta_c = \alpha_2\alpha_3,$$

$$\delta_d = \alpha_3.$$

The last relational model to consider for this sample space is defined by

$$S_1 = \{a, b\}, \quad S_2 = \{c, d\}, \quad S_3 = \{a, c\}, \quad S_4 = \{b, d\}. \tag{11.9}$$

The design matrix of this model is

$$\begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}.$$

This parameterization of the model is not of full rank. Full rank parameterizations of the same model may be based on the subsets

$$S_1 = \{a, b\}, \quad S_2 = \{c, d\}, \quad S_3 = \{a, c\}, \tag{11.10}$$

or

$$S_1 = \{a, b\}, \quad S_2 = \{a, c\}, \quad S_3 = \{a, b, c, d\}. \tag{11.11}$$

If the sample space is a 2×2 contingency table, with

$$a = (1, 1), \quad b = (1, 2), \quad c = (2, 1), \quad d = (2, 2),$$

the relational model defined by (11.9) is the independence of the two one-way marginal distributions. The parameterization in (11.11) allows an overall effect and an effect associated with the first row and one associated with the first column.

As will be seen in the sequel, an important characteristic of a relational model is whether or not it contains an overall effect. The model is said to contain (or have) the overall effect,[6] if a column vector of 1s may be obtained as a linear

[6] Strictly speaking, one has to distinguish between the effect that is present in every cell and a particular value of that effect in the case of a distribution. The former will be referred to as the overall effect, the latter as an overall effect, but in most cases, reference will be made to both.

combination of the columns of the matrix **A** in (11.4). This property is not affected by reparameterization. Clearly, the relational models defined in (11.5), in (11.8), and in (11.9) contain an overall effect. For example, in the case of model (11.8), the sum of the first and the last columns of the design matrix gives a vector of 1s.

The following is a simple relational model with no overall effect. Let

$$S_1 = \{a, b, c\}, \quad S_2 = \{c, d\}. \tag{11.12}$$

Then, the design matrix is

$$\begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Under this model, the cell parameters are

$$\delta_a = \alpha_1, \quad \delta_b = \alpha_1, \quad \delta_c = \alpha_1\alpha_2, \quad \delta_d = \alpha_2.$$

In addition to (11.3) or the equivalent (11.4), relational models have a dual characterization. To illustrate this, consider first the model defined by (11.6). From the form of the cell parameters, one sees that

$$\frac{\delta_a}{\delta_b} = 1, \quad \frac{\delta_c}{\delta_d} = 1. \tag{11.13}$$

This follows from either (11.6) or (11.7). This characterization is based on two odds being equal to 1. This is equivalent to the model, that is, this also implies the parameterization obtained from (11.6) or (11.7). This is immediate, as (11.13) implies that

$$\delta_a = \delta_b, \quad \delta_c = \delta_d,$$

and from this, the parameterization based on (11.6) follows.

A similar representation of the model in (11.8) is

$$\frac{\delta_b}{\delta_a} = \frac{\delta_c}{\delta_d}$$

or

$$\frac{\delta_a\delta_c}{\delta_b\delta_d} = 1. \tag{11.14}$$

For the model (11.9) one has, of course, that

$$\frac{\delta_a \delta_d}{\delta_c \delta_b} = 1, \tag{11.15}$$

see Theorem *LCDA* 10.7.

The characterizations of the relational models given in (11.13), (11.14), and (11.15) are all of the type that an odds ratio[7] is equal to 1.

For the model defined in (11.12), a similar characterization is

$$\frac{\delta_a}{\delta_b} = 1, \quad \frac{\delta_c}{\delta_b \delta_d} = 1. \tag{11.16}$$

These two requirements are clearly implied by the model. To see that these also imply the model, set

$$\alpha_1 = \delta_a, \quad \alpha_2 = \delta_d,$$

and then (11.16) implies that

$$\delta_a = \delta_b = \alpha_1, \delta_c = \alpha_1 \alpha_2.$$

The characterization in (11.16) contains an odds, and the quantity

$$\frac{\delta_c}{\delta_b \delta_d}.$$

This is the ratio of products of cell parameters, but it differs from the odds ratios seen so far. Here, the number of cell parameters in the numerator (1, in this case) is different from the number of cell parameters in the denominator (2, in this case). Such an expression will be called a non-homogeneous odds ratio. All the odds ratios seen in the characterizations of the other examples in (11.13), (11.14), (11.15) are homogeneous. The concept of homogeneity was used in Sect. 10.3.2. The meaning of homogeneity (or the lack of it) formulated in terms of odds ratios is that if all cell parameters are multiplied by the same number, then the value of a homogeneous odds ratio does not change, while the value of a non-homogeneous odds ratio does. Relational models defined by restrictions on homogeneous odds ratios are homogeneous, but if among the ones restricted, there is a non-homogeneous odds ratio, then the model is not homogeneous. Lack of homogeneity of a model leads to surprising properties, some of which will be discussed in the next section.

In general, one has the following.

Theorem 11.1 *There exists a matrix* $\mathbf{D_A}$*, such that the cell parameter distribution* δ_i *is in the relational model defined by* $\mathbf{A}$ *if and only if the first one out of the following two conditions holds in the case of intensities, and both of the following two conditions hold in the case of probabilities:*

[7] As before, an odds is considered a special case of an odds ratio.

$$\mathbf{D}_\mathbf{A} \log \delta = 0$$

and

$$1'\delta = 1.$$

Proof The logarithms of the distributions δ_i in the relational model (11.4) are, in the case of intensities, a linear subspace of $\mathbb{R}^{|\mathscr{I}|}$, and in the case of probabilities a subset of this.[8] In both cases, the dimension of the set of the log cell parameters is equal to the column rank of $\mathbf{A}$, that is, the number of linearly independent columns of $\mathbf{A}$. When $\mathbf{A}$ is of full rank, then its column rank is k. This is straightforward for the log intensities, and is true for the log probabilities, because no linear restriction applies to the log probabilities that would make them part of a lower-dimensional linear subspace. Of course, the log probabilities are a lower-dimensional manifold in this linear subspace, see Sect. 8.2.

Let $\mathbb{D}_\mathbf{A}$ denote the linear subspace of $\mathbb{R}^{|\mathscr{I}|}$ that is orthogonal to the smallest linear subspace containing the logarithms of the cell parameters in the model, that is, the collection of those vectors $d \in \mathbb{R}^{|\mathscr{I}|}$, for which

$$d' \log \delta = 0,$$

for all δ that is in the relational model defined by $\mathbf{A}$. This linear subspace is called the kernel of $\mathbf{A}$, because

$$0 = d' \log \delta = d'(\mathbf{A}\phi) = (d'\mathbf{A})\phi$$

for all ϕ parameters implies that

$$d'\mathbf{A} = 0 \text{ for all } d \in \mathbb{D}_\mathbf{A}. \tag{11.17}$$

The dimension of the linear subspace $\mathbb{D}_\mathbf{A}$ is $|\mathscr{I}| - k$. Consider a basis of $\mathbb{D}_\mathbf{A}$ that is, $|\mathscr{I}| - k$ vectors $d_1, \ldots, d_{|\mathscr{I}|-k}$ in $\mathbb{R}^{|\mathscr{I}|}$ that have the property that any vector in $\mathbb{D}_\mathbf{A}$ is a linear combination of them. Arrange the transposes of these vectors into a matrix of size $(|\mathscr{I}| - k) \times \mathbb{R}^{|\mathscr{I}|}$ to obtain $\mathbf{D}_\mathbf{A}$. Then, (11.17) implies that

$$\mathbf{D}_\mathbf{A}\mathbf{A} = 0,$$

thus, if $\log \delta = \mathbf{A}\phi$ for some ϕ, that is, δ is in the relational model, then $\mathbf{D}_\mathbf{A}\mathbf{A}\phi = \mathbf{D}_\mathbf{A} \log \delta = 0$. Conversely, if $\mathbf{D}_\mathbf{A} \log \delta = 0$, then $\log \delta$ is in the subspace that is the orthogonal complement of the kernel of $\mathbf{A}$ that is in the linear subspace spanned by $\mathbf{A}$. Thus, $\log \delta$ may be written as $\mathbf{A}\phi$ for some ϕ, so $\log \delta$ is in the model. This proves the claim for intensities.

[8] A subset, because the probabilities need to sum to 1.

For probabilities, in addition to the first condition in the theorem, the second one also needs to hold, and also these two, if hold, imply that the vector of cell parameters is in the model for probabilities.

□

To link the matrix $\mathbf{D_A}$ to the odds ratios characterizing the relational model, one needs the following result from Klimova et al. [87]. The theorem is given without proof, as it would require more advanced knowledge in linear algebra than assumed here.

Theorem 11.2 *The matrix* $\mathbf{D_A}$ *in Theorem 11.1 may be selected to have integer entries.*

□

From now on, $\mathbf{D_A}$ will be assumed to have integer entries. Then, for a row d'_j of $\mathbf{D_A}$ and a δ vector of cell parameters in the relational model (11.4), consider the vector $\delta^{d'_j}$, where exponentiation is interpreted coordinate-wise. Theorem 11.1 implies immediately the following.

Corollary 11.1 *A vector* δ *of cell parameters is in the relational model (11.4) if and only if the following first condition holds in the case of intensities and both of the following two conditions hold in the case of probabilities. For all rows* d'_j *of* $\mathbf{D_A}$*,*

$$\delta^{d'_j} = 1$$

and

$$1'\delta = 1.$$

□

The quantities $\delta^{d'_j}$ are called generalized odds ratios. If one defines $[d'_j]_+$ as the vector d'_j with the negative entries replaced by 0, and $[d'_j]_-$ as the vector d'_j with the positive entries replaced by 0 and the signs of the negative entries are changed to positive , then a generalized odds ratio is

$$\frac{\delta^{[d'_j]_+}}{\delta^{[d'_j]_-}}.$$

According to Theorem 11.2, the exponents may be selected to be integers.

For example, in the case of the relational model defined by (11.12), one obtains for the product $\mathbf{D_A A}$ that

$$\begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & -1 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

The rows of the first matrix give the exponents of the odds ratios defining the model as shown in (11.16).

When, for a particular j, the sums of the components of $[d'_j]_+$ and that of $[d'_j]_-$ are equal, the corresponding odds ratio is homogeneous, and when the sums are different, it is non-homogeneous. Of course, the generalized odds ratios defining a relational model are not uniquely determined. Further, every model may be defined with at most one non-homogeneous generalized odds ratio, see Klimova and Rudas [84].

Theorem 11.3 *Every relational model may be defined by assuming that generalized odds ratios are equal to* 1*, so that the number of non-homogeneous odds ratios is at most* 1.

Proof The exponents of the generalized odds ratios defining the relational model based on $\mathbf{A}$ are the rows of the matrix $\mathbf{D_A}$. A generalized odds ratio is homogeneous if and only if the corresponding row sum is zero. As the matrix $\mathbf{D_A}$ is defined by the property of spanning the linear subspace orthogonal to the linear subspace of the cell parameters, linear transformations not using zero coefficients may be applied to any $\mathbf{D_A}$. According to Theorem 11.2, the entries in the matrix $\mathbf{D_A}$ may be assumed to be integers.

If all the rows of $\mathbf{D_A}$ sum to zero, then all defining generalized odds ratios are homogeneous.

In the remainder of the proof we show that if there are 2 rows of $\mathbf{D_A}$ that do not sum to zero, then one of them may be replaced with a row that does sum to zero, without changing the model defined.

Let d'_1 and d'_2 be two rows that do not sum to zero. Let the row sums be $1'd_1$ and $1'd_2$, where 1 is a vector of 1s. Then, replace d_2 by

$$d = (1'd_2)d_1 - (1'd_1)d_2.$$

Then, d' is a linear combination of the rows d'_1 and d'_2 consisting of integer entries, and their sum is

$$1'd = 1'((1'd_2)d_1 - (1'd_1)d_2) = 1'((1'd_2)d_1) - 1'((1'd_1)d_2)$$

$$= (1'd_2)(1'd_1) - (1'd_1)(1'd_2) = 0.$$

Thus, the generalized odds ratio associated with d is homogeneous. The rows of the modified matrix $\mathbf{D_A}$ are linearly independent and orthogonal to $\mathbf{A}$, thus, it spans the same linear subspace as the original one.

□

The homogeneity or non-homogeneity of the generalized odds ratios that may be used to define a relational model is related to the presence or lack of the overall effect. Klimova et al. [87] gave the following result.

Theorem 11.4 *The relational model has an overall effect if and only if all the generalized odds ratios that are assumed to be equal to* 1 *in its definition are homogeneous.*

Proof A relational model has an overall effect, if the column space of the design matrix $\mathbf{A}$ contains the vector of 1. As the rows of $\mathbf{D_A}$ span the linear subspace orthogonal to $\mathbf{A}$, having an overall effect implies that all the rows of the matrix $\mathbf{D_A}$ are orthogonal to 1. Thus, having an overall effect implies that

$$d_j' 1 = 1' d_j = 0, \tag{11.18}$$

for any row d_j' of $\mathbf{D_A}$, that is, the generalized odds ratio corresponding to any row d_j' is homogeneous. On the other hand, if (11.18) holds for all rows of $\mathbf{D_A}$, then the vector 1 is not in the kernel of $\mathbf{A}$, thus, the columns of $\mathbf{A}$ span this vector.

□

Note that the matrix $\mathbf{D_A}$ is, in general, not unique. The previous result applies to any possible choice of $\mathbf{D_A}$.

Log-linear models are special cases of relational models. Log-linear models apply to sample spaces that are Cartesian products, are specified using cylinder sets, have an overall effect, and may be specified by setting homogeneous odds ratios equal to 1.

The following theorem establishes that relational models are smooth manifolds.

Theorem 11.5 *When the design matrix* $\mathbf{A}$ *is of size* $|\mathscr{I}| \times k$ *and is of full rank, then the generated relational model is a smooth manifold in* $\mathbb{R}^{|\mathscr{I}|}$ *of dimension* k *for intensities, and of dimension* $k - 1$ *for probabilities.*

Proof As the cell parameters were assumed to be (strictly) positive, in the case of intensities $\{\log \delta_i : i \in \mathscr{I}\}$ is a general distribution with a full rank linear parameterization (11.4). The domain of ϕ is $\mathbb{R}^k$, which is a k-dimensional open set, and the Jacobian of the transformation is $\mathbf{A}$, which is of full rank. This implies that $\{\log \delta_i : i \in \mathscr{I}\}$ is a smooth k-dimensional manifold in $\mathbb{R}^{|\mathscr{I}|}$. Then, Theorem 8.6 yields the required result for intensities.

In the case of probabilities, the proof is different depending on whether the relational model has or does not have an overall effect.

When $\mathbf{A}$ has an overall effect, then there is a reparameterization of the relational model with a design matrix such that one of the columns is a vector of 1s. Because of the assumed full rank, the other columns are linearly independent from this column and also of each other. Then, the parameter associated with this column is the normalizing factor $a(\theta)$ in (8.13), and the remaining $k - 1$ parameters are unrestricted. From this, an argument similar to that of the case of intensities yields the result.

When $\mathbf{A}$ does not have an overall effect, the probability distributions in the relational model are obtained as the intersection of the relational model for

intensities, using the same design matrix, and of the probability simplex for $|\mathscr{I}|$ categories:

$$\{\delta : \delta = \exp \mathbf{A}\phi\} \cap \Delta_{|\mathscr{I}|}.$$

The relational model for intensities was already seen to be a smooth manifold of dimension k. The probability simplex is a smooth manifold of dimension $k-1$, see Theorem 8.1. Then, using Theorem 8.6, it may be assumed that they have respective smooth parameterizations using a k-dimensional open set, say $\mathscr{U}_1$, and a $k-1$-dimensional open subset of it, say $\mathscr{U}_2$. The intersection of these sets is the $k-1$-dimensional open set $\mathscr{U}_2$. To see that the Jacobian is of full rank on $\mathscr{U}_2 \subseteq \mathscr{U}_1$, note that it is of full rank on $\mathscr{U}_1$, but then it is also of full rank on a subset of $\mathscr{U}_1$.

□

Relational models are obviously exponential families. An exponential family is called regular if its parameter space is an open set (see Section *LCDA* 7.2). As implied by the previous theorem, relational models for intensities are regular exponential families with a k-dimensional minimal representation. Further, relational models for probabilities with an overall effect are also regular exponential families with a $k-1$-dimensional minimal representation. The parameter space of relational models for probabilities without the overall effect is a $k-1$-dimensional smooth manifold in the k-dimensional Euclidean space as their parameter space. Those manifolds are not open sets in the k-dimensional Euclidean space but are, of course, images of a $k-1$-dimensional open set. Such an exponential family is called a curved exponential family.

Klimova and Rudas [84] studied the implications of including the overall effect in an exponential family, if it does not have one, and of removing the overall effect if it does have one. In the context of the relational models considered here, if a model is defined by using homogeneous and one non-homogeneous generalized odds ratio (see Theorem 11.3), then the model obtained by including the overall effect is specified by the homogeneous odds ratios of the original model being equal to 1. The paper also describes relational models in the language of algebraic geometry, and several operations in algebraic geometry are given statistical interpretations. Among others, it is shown that augmenting a sample space, for example, making an incomplete Cartesian product complete, is identical to the homogeneization of an inhomogeneous variety. These results are not discussed here.

An interesting special class of relational models applies to incomplete binary sample spaces of k features (variables), where the cell in which none of the features is present does not exist. Table 11.1 showed such a sample space with two variables and Table 11.5 with three variables. Such a sample space may be thought of as a register, but there are also other problems leading to similar structures. For example, customers may be asked why they chose a particular product or service. If various possible reasons are listed, including "other," then everybody chooses at least one reason. Let the data-generating procedure be multivariate Poisson. In the

incomplete 2×2 table, independence in the usual sense does not apply. A variant of independence for the cell intensities is defined by

$$\lambda_{(1,1)} = \lambda_{(1,0)}\lambda_{(0,1)}, \tag{11.19}$$

or

$$\frac{\lambda_{(1,1)}}{\lambda_{(1,0)}\lambda_{(0,1)}} = 1.$$

The latter is a restriction on a non-homogeneous odds ratio. It may be interpreted by saying that the joint intensity, with which both features occur, is the product of the individual intensities. Models defined by restrictions on such non-homogeneous odds ratios were apparently first mentioned by Aitchison and Silvey [3]. When one has three variables, the structure of the conditional sample spaces of two variables given the third depend on which category of the third variable is being conditioned upon. When conditioning is on the category that the feature is not present, the conditional table is an incomplete Cartesian product, and when conditioning is on the category that the feature is present, the conditional table is complete. In his case, instead of conditional odds ratios, one has, for the first two features, given the third a generalized and a not generalized odds ratio, called Aitchison-Silvey ratios:

$$\frac{\lambda_{(1,1,0)}}{\lambda_{(0,1,0)}\lambda_{(1,0,0)}}, \quad \frac{\lambda_{(1,1,1)}\lambda_{(0,0,1)}}{\lambda_{(0,1,1)}\lambda_{(1,0,1)}}.$$

From this, higher-order conditional Aitchison-Silvey ratios may be defined, and models similar to log-linear models may be developed that are based on, instead of conditional odds ratios, conditional Aitchison-Silvey ratios which are being set to 1, see Klimova and Rudas [86].

The relational model defined for intensities in (11.19) may be used to illustrate how different relational models may be when defined for intensities of for probabilities, even if the same design matrix is used. The same model for probabilities assumes that $P(1, 1) = P(1, 0)P(0, 1)$. This means that the probability of both features being present is less than the probability that either one of the features is present. In the case of intensities, the corresponding inequality is not implied by the model and may or may not be true.

Relational models were defined here for (strictly) positive cell parameters. Klimova and Rudas [83] studied ways of extending relational models to certain cell parameters without the (strict) positivity restriction. They defined an extended relational model as the collection of the pointwise limits[9] of sequences of distributions in the relational model. They showed this extended model to be the closure of the relational model according to the so-called Bregman divergence [25], which is a

[9] That is, limits for every cell $i \in \mathscr{I}$.

generalization of the Kullback-Leibler divergence (see Section *LCDA* 12.2) which also applies to general distributions, where the distribution totals are not equal.

In the next section, maximum likelihood estimation for relational models is studied.

11.4 Estimation and Testing

Maximum likelihood estimates under Multinomial or multivariate Poisson sampling for log-linear models were characterized in Theorem *LCDA* 12.1 as the unique distribution that, on the one hand, is in the model and, on the other hand, its marginal distributions for the marginals that are allowed interactions in the log-linear model are the same as that of the observed distribution.

In the case of relational models, instead of marginal distributions, certain subset sums characterize the MLEs, although the situation is more complex than in the case of log-linear models.

Relational models are defined by a class of subsets of the sample space, $S_1, \ldots, S_k$, and the columns of the design matrix $\mathbf{A}$ are the indicators of the subsets. Let $\{f(i) : i \in \mathscr{I}\}$ be a general distribution on the sample space, without the assumption of (strict) positivity. It may be observed frequencies, but also the cell parameters, whether intensities or probabilities. Then, the subset sums are defined as

$$f(S_j) = \sum_{i \in S_j} f(i), \quad j = 1, \ldots, k.$$

The vector of subset sums may also be written as

$$(f(S_1), f(S_2), \ldots, f(S_k))' = \mathbf{A}' f.$$

In the proof of the next theorem, the following forms of the definition of the relational model will be useful:

$$\delta_i = \prod_{j=1}^{k} \alpha_j^{a_{i,j}}, \quad \log \delta_i = \sum_{j=1}^{k} a_{i,j} \phi_j, \quad i \in \mathscr{I}.$$

Klimova et al. [87] gave the following result.

Theorem 11.6 *Let* $\mathbf{A}$ *be the design matrix of a relational model and* $(f(i),\ i \in \mathscr{I})$ *be the observed frequencies. Assume the MLEs of the intensities,* $\hat{\lambda}$ *and the MLEs of the probabilities* $\hat{p}$ *exist. Then, with*

$$N = \sum_{i \in \mathscr{I}} f(i),$$

one has

$$\hat{\lambda}_i = N\hat{p}_i, \quad i \in \mathscr{I},$$

if and only if $\mathbf{A}$ *contains the overall effect.*

Proof For probabilities, the kernel of the log-likelihood, augmented with a Lagrangian term (see Section *LCDA* 4.1), where $-\kappa$ is the Lagrange multiplier,[10] is

$$\left(\sum_{i\in\mathscr{I}} f(i)\sum_{j=1}^{k} a_{i,j}\phi_j\right) - \kappa\left(\sum_{i\in\mathscr{I}}\prod_{j=1}^{k}\alpha_j^{a_{i,j}} - 1\right)$$

or

$$\left(\sum_{i\in\mathscr{I}} f(i)\sum_{j=1}^{k} a_{i,j}\log\alpha_j\right) - \kappa\left(\sum_{i\in\mathscr{I}}\prod_{j=1}^{k}\alpha_j^{a_{i,j}} - 1\right).$$

As the parameters are assumed to be (strictly) positive, the partial derivative according to $\alpha_{j'}$, for a fixed j', is

$$\left(\sum_{i\in\mathscr{I}} f(i)\frac{a_{i,j'}}{\alpha_{j'}}\right) - \kappa\left(\sum_{i\in\mathscr{I}} a_{i,j'}\alpha_{j'}^{a_{i,j'}-1}\prod_{j\neq j'}\alpha_j^{a_{i,j}}\right).$$

If this is set to zero and then multiplied by $\alpha_{j'}$, one obtains that

$$\left(\sum_{i\in\mathscr{I}} f(i)a_{i,j'}\right) - \kappa\left(\sum_{i\in\mathscr{I}} a_{i,j'}\alpha_{j'}^{a_{i,j'}}\prod_{j\neq j'}\alpha_j^{a_{i,j}}\right)$$

$$= \left(\sum_{i\in\mathscr{I}} f(i)a_{i,j'}\right) - \kappa\left(\sum_{i\in\mathscr{I}} a_{i,j'}\prod_{j=1}^{k}\alpha_j^{a_{i,j}}\right) = 0.$$

Combined for every j', one has

$$\mathbf{A}'f - \kappa\mathbf{A}'\hat{p} = 0. \tag{11.20}$$

The partial derivative according to κ gives:

$$1'\hat{p} = 1.$$

[10] The negative sign is irrelevant and is used for later convenience only.

In the case of the multivariate Poisson distribution (see Section *LCDA* 2.5), the kernel of the log-likelihood is

$$\sum_{i\in\mathscr{I}}\left(\sum_{j=1}^{k} f(i)a_{i,j}\phi_j - \prod_{j=1}^{k}\alpha_j^{a_{i,j}}\right)$$

or

$$\sum_{i\in\mathscr{I}}\left(\sum_{j=1}^{k} f(i)a_{i,j}\log\alpha_j - \prod_{j=1}^{k}\alpha_j^{a_{i,j}}\right).$$

The partial derivative according to $\alpha_{j'}$ for a fixed j' is

$$\sum_{i\in\mathscr{I}}\left(f(i)\frac{a_{i,j'}}{\alpha_{j'}} - a_{i,j'}\alpha_{j'}^{a_{i,j'}-1}\prod_{j\neq j'}\alpha_j^{a_{i,j}}\right).$$

Setting this equal to zero and multiplying both sides by $\alpha_{j'}$ gives:

$$\sum_{i\in\mathscr{I}}\left(f(i)a_{i,j'} - a_{i,j'}\prod_{j=1}^{k}\alpha_j^{a_{i,j}}\right) = 0,$$

or in matrix form for all j'

$$\mathbf{A}'f - \mathbf{A}'\hat{\lambda} = 0. \tag{11.21}$$

Equations (11.20) and (11.21) are properties of the respective MLEs. Further, the MLEs are in the model, meaning that the defining generalized odds ratios have to be equal to 1. When the model has an overall effect, the defining odds ratios are homogeneous (see Theorem 11.4), thus, if the MLE $\hat{\lambda}$ is in the model, so is any p that may be written as

$$\kappa p = \hat{\lambda} \tag{11.22}$$

for some κ. With an appropriate choice of κ, this p may be made such that also (11.20) holds. But then, as the MLE is the unique solution of the likelihood equations in the model, this p is the MLE, and the proportionality of the MLEs under the two sampling schemes follows.

Because the overall effect is present, there exists a vector $d \in \mathbb{R}^k$ such that

$$\mathbf{A}d = 1, \text{ or } d'\mathbf{A}' = 1',$$

and multiplying both sides of (11.20) from the left with d' yields that

$$1'f - \kappa 1'\hat{p} = 0,$$

and thus $\kappa = N$, so the instead of (11.20), one has

$$\mathbf{A}'f - N\mathbf{A}'\hat{p} = 0. \tag{11.23}$$

Comparing this to (11.21) gives:

$$\mathbf{A}'\hat{\lambda} = N\mathbf{A}'\hat{p},$$

and from this one has that

$$\mathbf{A}'(\hat{\lambda} - N\hat{p}) = 0.$$

Because the vector 1 is a linear combination of the columns of $\mathbf{A}$, one has

$$1'(\hat{\lambda} - N\hat{p}) = 0,$$

thus

$$1'\hat{\lambda} = N1'\hat{p} = N,$$

and because of the proportionality (11.22),

$$\hat{p} = \frac{1}{N}\hat{\lambda} \tag{11.24}$$

for the MLEs.

To see the converse, assume that (11.24) holds for the MLEs. Under the relational model

$$\log \hat{\lambda} = \mathbf{A}\phi_{\hat{\lambda}}$$

for some $\phi_{\hat{\lambda}}$, and also, by assumption,

$$\log \hat{\lambda} = \log(N\hat{p}) = \log N 1 + \mathbf{A}\phi_{\hat{p}},$$

for some ϕ_p. Thus (with a vector 1 on the left-hand side and a number 1 in the numerator),

$$1 = \mathbf{A}\frac{1}{\log N}(\phi_{\hat{\lambda}} - \phi_{\hat{p}}),$$

so that the vector 1 is a linear combination of the columns of $\mathbf{A}$.

□

Further, the details of the previous proof, in particular (11.20) and (11.21), imply the following.

Corollary 11.2 *If the maximum likelihood estimates of the cell parameters* $\hat{\lambda}$ *and* $\hat{p}$ *exist for a set of observed frequencies* $(f(i),\ i \in \mathscr{I})$ *under a relational model, then*

$$\hat{\lambda}(S_j) = f(S_j), \quad j = 1, \ldots, k,$$

and

$$\hat{p}(S_j) = \gamma \frac{1}{N} f(S_j), \quad j = 1, \ldots, k.$$

Further, if the relational model has an overall effect, then $\gamma = 1$.

□

The value of γ is called the adjustment factor. It applies to the subset sums of the MLEs of the probabilities under Multinomial sampling. When the model contains no overall effect, the adjustment factor is not equal to 1, thus, the observed subset sums are only reproduced by the MLEs up to a constant[11] multiplier.

The sum of the MLEs of the probabilities is, of course, equal to 1. The sum of the MLEs of intensities is not necessarily equal to the observed total. In fact, one has the following.

Corollary 11.3 *If the relational model contains an overall effect, then the sum of the observed frequencies is reproduced by the sum of the MLEs of the intensities.*

Proof As the sum of the probabilities is 1, Theorem 11.6 implies that if there is an overall effect, then the sum of the observations, N, is equal to the sum of the MLEs of the cell intensities.

□

Section 10.3 used the homogeneity property of marginal log-linear models to prove the proportionality of the MLEs for Multinomial and multivariate Poisson sampling. Relational models, even if formulated for cell intensities, that is, general distributions, are homogeneous in the sense used in Sect. 10.3, if and only if they are defined by homogeneous odds ratios, that is, if and only if they contain the overall effect, see Theorem 11.4. In that case, one also has the proportionality of the MLEs under Multinomial and multivariate Poisson sampling, see Theorem 11.6, with the constant of proportionality N.

To illustrate some of the foregoing results, consider a very small example, with the sample space shown in Table 11.1, and apply the model of Aitchison-Silvey independence of (11.19) and assume multivariate Poisson sampling.

[11] The multiplier is constant in the sense that for a given model and set of data, it is the same for all subsets.

Table 11.8 Observed frequencies on an incomplete 2×2 contingency table

		F_2	
		No	Yes
F_1	No	–	$f(0,1)$
	Yes	$f(1,0)$	$f(1,1)$

If observations f are available, as shown in Table 11.8, then, if the MLEs exist, by Corollary 11.2, one has

$$\hat{\lambda}_{(1,0)}\hat{\lambda}_{(0,1)} + \hat{\lambda}_{1,0} = f(1,1) + f(1,0)$$

and

$$\hat{\lambda}_{(1,0)}\hat{\lambda}_{(0,1)} + \hat{\lambda}_{0,1} = f(1,1) + f(0,1),$$

that is, the sums of the subsets defining the model are reproduced by the MLEs. Note that as $\lambda_{(1,1)}$ is assumed to be the product of the other two intensities, this also applies to their respective MLEs.

If, in addition, the total number of observations was also reproduced, then one would also have

$$\hat{\lambda}_{(1,0)}\hat{\lambda}_{(0,1)} + \hat{\lambda}_{1,0} + \hat{\lambda}_{1,0} = f(1,1) + f(1,0) + f(1,0).$$

It is easy to see that the last three equations may all hold if and only if

$$\hat{\lambda}_{(1,0)}\lambda_{(0,1)} = f(1,1), \quad \hat{\lambda}_{(1,0)} = f(1,0), \quad \hat{\lambda}_{(0,1)} = f(0,1).$$

That is, the sum of the observations is reproduced by the MLEs of the cell intensities, if and only if the observed frequencies, interpreted as intensities, are in the model. This was an example of a model that does not have an overall effect.

The results obtained so far allow a characterization of the MLEs under a relational model for multivariate Poisson sampling.

Theorem 11.7 *Let* $\mathbf{A}$ *be a design matrix for a relational model, and let* $\mathbf{D}_\mathbf{A}$ *be a matrix described in Theorems 11.1 and 11.2. For the observations* $(f(i), \ i \in \mathscr{I})$, $\hat{\lambda}$ *is the maximum likelihood estimate under the relational model generated by* $\mathbf{A}$ *in the case of multivariate Poisson sampling, if and only if*

$$\mathbf{D}_\mathbf{A} \log \hat{\lambda} = 0,$$

and

$$\mathbf{A}'\hat{\lambda} = \mathbf{A}'f.$$

Proof The first condition means that $\hat{\lambda}$ is in the relational model, see Theorem 11.1. The second condition characterizes the solution of the likelihood equation in the case of multivariate Poisson sampling, see (11.21).

Thus, the conditions mean that $\hat{\lambda}$ is in the model and is a solution of the likelihood equation. This clearly holds for the MLE.

That the conditions imply that $\hat{\lambda}$ is the MLE follows from Theorem *LCDA* 7.4. In that result, the derivative of the augmented likelihood function was given in *LCDA* (7.15), in a way identical[12] to the derivative in (11.21). It was proved in Theorem *LCDA* 7.4 that the second partial derivative matrix, at the point where this derivative was zero, was negative definite, thus, the solution gave the maximum of the log-likelihood function.

□

The situation in the case of probabilities is more complex. One immediately has the following.

Corollary 11.4 *Let* $\mathbf{A}$ *be a design matrix for a relational model, and let* $\mathbf{D}_{\mathbf{A}}$ *be a matrix described in Theorems 11.1 and 11.2. For the observations* $(f(i),\ \ i \in \mathscr{I})$*, the maximum likelihood estimate, if it exists under the relational model generated by* $\mathbf{A}$*, in the case of Multinomial sampling, is the unique distribution* $\hat{p}$ *with*

$$\mathbf{D}_{\mathbf{A}} \log \hat{p} = 0,$$

$$1'\hat{p} = 1,$$

and

$$\kappa \mathbf{A}'\hat{p} = \mathbf{A}'f$$

for some positive κ*. Further, if the model has an overall effect, then* $\kappa = N$*, the total sample size.*

Proof The conditions mean that $\hat{p}$ is in the model (see Theorem 11.1) and that it is the solution of the likelihood equations, see (11.20). The unicity follows from the unicity of the MLE. The last claim follows from Corollary 11.2.

□

Klimova and Rudas [87] proved the converse of the result above. It is included here without proof, as the proof requires more knowledge in linear algebra than assumed here.

Theorem 11.8 *Let* $\mathbf{A}$ *be a design matrix for a relational model, and let* $\mathbf{D}_{\mathbf{A}}$ *be a matrix described in Theorems 11.1 and 11.2. For the observations* $(f(i),\ \ i \in \mathscr{I})$*,*

[12] The following notations link the presentation here to *LCDA* (7.15): $x = f$, $\mathbf{T} = \mathbf{A}$, $p = \lambda/N$, $n = N$.

$\hat{p}$ is the maximum likelihood estimate under the relational model generated by $\mathbf{A}$ *and Multinomial sampling, if $\hat{p}$ is such that*

$$\mathbf{D}_{\mathbf{A}} \log \hat{p} = 0,$$

$$1' \hat{p} = 1,$$

and

$$\kappa \mathbf{A}' \hat{p} = \mathbf{A}' f$$

for some positive κ. Further, if the model has an overall effect, then $\kappa = N$, where N is the total sample size.

□

Note that the characterization just given generalizes the characterization of the MLEs under log-linear models in Section *LCDA* 12.2, given after Theorem *LCDA* 12.8. That characterization said that the MLE was the unique distribution in the intersection of the log-linear model (a multiplicative family) and of a linear family, defined by the observed distributions on those marginals that were allowed interactions in the log-linear model. Theorem 11.7 generalizes this result in the case of multivariate Poisson sampling, with the first condition meaning that the distribution $\hat{\lambda}$ is in the model, while the second one means that it is in the linear family defined by the observed subset sums. For Multinomial sampling and probabilities, Corollary 11.4, if the MLE exists, gives the same characterization of it.

For log-linear models, Theorems *LCDA* 12.6 and 12.8 imply that if there exists a (strictly) positive distribution with the same observed marginal distributions as the observed data had for the marginals that are interactions (including the main effects) in the model, then there exists an MLE, whether the sampling procedure was Multinomial or multivariate Poisson. Obviously, if the observed distribution is (strictly) positive, then this condition holds. These results, although were formulated for log-linear models, hold for any regular exponential family. For a general discussion, see Andersen [4] and Barndorff-Nielsen [10].

A necessary and sufficient condition of the existence for the MLEs under a relational model was given in Klimova and Rudas [83], which generalizes the results mentioned above and which is included here with a proof that does not contain all the details. In that paper, another condition that implies the existence of the MLE in the case of extended relational models was also given. First, a preliminary result is needed.

Theorem 11.9 *Consider a relational model on a sample space $\mathscr{I}$ and observations $(f(i) : i \in \mathscr{I})$. If f is (strictly) positive, then MLEs under the relational model exist, whether sampling was Multinomial or multivariate Poisson.*

Proof When the parameters of interest are intensities which is sampling is multivariate Poisson, then the relational model is a regular exponential family, see the comment after Theorem 11.5. Then, the comment before this theorem implies the required result.

In the case of probabilities, let μ be an unspecified, positive constant. For any μ, let $\hat{\lambda}_{\mu f}$ be the MLE for the data μf in the same model for intensities, which exists by the first part of the theorem.

Then, by Theorem 11.7,

$$\mathbf{D_A} \log \hat{\lambda}_{\mu f} = 0,$$

and

$$\mathbf{A}' \hat{\lambda}_{\mu f} = \mathbf{A}' \mu f.$$

This may be seen as a system of equations for the cell parameters $\hat{\lambda}_{\mu f}$. Because all involved functions are continuous, it can be seen[13] that $\hat{\lambda}_{\mu f}$ is a continuous function of μ.

As every row of $\mathbf{A}$ contains at least one 1,

$$0 < 1' \hat{\lambda}_{\mu f} < 1' \mathbf{A}' \hat{\lambda}_{\mu f} = 1' \mathbf{A}' \mu f = 1' \mu \mathbf{A}' f.$$

Therefore, $1' \hat{\lambda}_{\mu f}$ may be made arbitrarily small (positive) by the appropriate choice of μ.

Further, for arbitrary $i \in \mathscr{I}$,

$$1' \hat{\lambda}_{\mu f} > \mu f(i),$$

and because of this, $1' \hat{\lambda}_{\mu f}$ may be made arbitrarily large by choosing an appropriate μ.

These, together with the continuity imply that there exists a μ^*, such that

$$1' \hat{\lambda}_{\mu^* f} = 1.$$

Then, $\hat{\lambda}_{\mu^* f}$ is a (strictly) positive probability distribution on $\mathscr{I}$. For this, the conditions in Theorem 11.8 hold, and thus it is the MLE in the case of Multinomial sampling.

□

After this sufficient condition, a necessary and sufficient condition is given.

Theorem 11.10 *Consider a sample space $\mathscr{I}$ and the relational model on it for the cell parameters δ defined by the design matrix* $\mathbf{A}$*, and observations* $(f(i),\ i \in \mathscr{I})$.

[13] For a formal proof, see Klimova and Rudas [82].

Then, the MLE exists if and only if there exists a (strictly) positive vector $e \in \mathbb{R}^{|\mathscr{I}|}$, *such that*

$$\mathbf{A}'e = \mathbf{A}'f,$$

in the case of intensities, and

$$\mathbf{A}'e = \frac{1}{N}\mathbf{A}'f,$$

in the case of probabilities.

Proof The relational model for intensities is a regular exponential family, see the comments after Theorem 11.5. Therefore, the condition of the existence of the MLE in regular exponential families discussed before Theorem 11.9 applies.

For relational models for probabilities, first suppose that for the given observations f, the MLE $\hat{p}$ exists. This $\hat{p}$ is in the relational model, and thus is (strictly) positive. By Corollary 11.2,

$$\hat{p}(S_j) = \gamma \frac{1}{N} f(S_j),$$

for all j, and thus

$$\mathbf{A}'\hat{p} = \gamma \mathbf{A}' \frac{1}{N} f$$

for some positive γ. This implies that for some vector $d \in \mathbb{R}^{|\mathscr{I}|}$, which is in the orthogonal complement of the linear subspace spanned by the columns of $\mathbf{A}$, that is, for a d with $\mathbf{A}'d = 0$,

$$\hat{p} = \gamma \frac{1}{N} f + d.$$

Let

$$e = \frac{1}{\gamma}\hat{p} = \frac{1}{N} f + \frac{1}{\gamma} d.$$

This is e is (strictly) positive because $\hat{p}$ is (strictly) positive, and as

$$\mathbf{A}' \frac{1}{\gamma} d = \frac{1}{\gamma} \mathbf{A}' d = 0,$$

one has that

$$\mathbf{A}'e = \mathbf{A}'\frac{1}{N}f =' \frac{1}{N}\mathbf{A}'f, \tag{11.25}$$

as required.

To see the converse, assume that a (strictly) positive e exists, for which (11.25) holds. Then, there exists a vector d in the subspace that is the orthogonal complement of the subspace spanned by the columns of $\mathbf{A}$, such that

$$e = \frac{1}{N}f + d.$$

Let

$$d_1 = \frac{1}{1 + 1'd}d.$$

Note that the numerator $1 + 1'd$ is a positive number, as

$$1 + 1'd = 1'(\frac{1}{N}f + d) = 1'e,$$

and e is (strictly) positive.

Then, define the vector $q \in \mathbb{R}^{|\mathscr{I}|}$ as

$$q = (1 - 1'd_1)\frac{1}{N}f + d_1.$$

The vector q sums to 1. Indeed,

$$1'q = 1'((1 - 1'd_1)\frac{1}{N}f + d_1) = (1 - 1'd_1)1'\frac{1}{N}f + 1'd_1 = 1 - 1'd_1 + 1'd_1 = 1.$$

Further, the components of q are (strictly) positive:

$$q = (1 - 1'd_1)\frac{1}{N}f + d_1 = (1 - 1'\frac{1}{1 + 1'd}d)\frac{1}{N}f + \frac{1}{1 + 1'd}d$$

$$= (1 - \frac{1}{1 + 1'd}1'd)\frac{1}{N}f + \frac{1}{1 + 1'd}d = (\frac{1}{1 + 1'd})\frac{1}{N}f + \frac{1}{1 + 1'd}d$$

$$= \frac{1}{1 + 1'd}(\frac{1}{N}f + d) = \frac{1}{1 + 1'd}e,$$

and e is a (strictly) positive vector.

Thus, q is (strictly) positive and sums to 1; consequently it may be taken as a probability distribution on the sample space.

Then, by Theorem 11.9, the MLE of q, $\hat{p}_q$ in the relational model exists. For this distribution, by Corollary 11.4

$$\mathbf{D_A} \log \hat{p}_q = 0,$$

$$1' \hat{p}_q = 1,$$

and

$$\kappa_q \mathbf{A}' \hat{p}_q = \mathbf{A}' q,$$

for some positive κ_q. Note that the first two requirements are not related to the observed distribution. Thus, if the condition above holds for f instead of q, for a possibly different κ, then $\hat{p}_q$ is also the MLE of the probability distribution based on the data f.

Indeed, one has

$$\mathbf{A}' \hat{p}_q = \frac{1}{\kappa_q} \mathbf{A}' q = \frac{1}{\kappa_q} (1 - 1' d_1) \frac{1}{N} \mathbf{A}' f$$

that is, $\hat{p}_q$ is also the MLE of f.

□

Determining the MLE in the case of intensities only requires a slight modification of the IPFP (see Section *LCDA* 12.2). The MLEs reproduce the observed sums of the subsets defining the model, while in the case of log-linear models, marginal distributions, which are the subset sums of cylinder sets, were reproduced.

In the case of probabilities, the algorithm proposed by Klimova and Rudas [82] to find the MLE under a relational model has two components. One that finds a (not necessarily probability) distribution[14] $p^*(\gamma)$, for an arbitrary positive adjustment factor $\gamma = 1/\kappa$ with the properties that

$$\mathbf{D_A} \log p^*(\gamma) = 0$$

and

$$\mathbf{A}' p^*(\gamma) = \gamma \mathbf{A}' \frac{1}{N} f,$$

and then another one, which finds a value γ^*, such that, in addition,

$$1' p^*(\gamma^*) = 1.$$

This is the characterization of the MLE, given in Corollary 11.4 and Theorem 11.8.

The first algorithm is the following.

[14] That is, the sum of the $p_i^*(\gamma)$ values is not necessarily 1.

Algorithm IPF(γ)

Step 0: Set $d = 0$, $\delta_i^d = 1$, for $i \in \mathscr{I}$, $g = f$, if δ is λ and $g = (1/N)f$ if δ is p.
Step 1: Let $d = d + 1$ and $1 \leq j \leq J$ be such that $d = j + tJ$ for some integer t.
Step 2: Compute

$$\delta_i^d = \gamma \left(\frac{\sum_{i \in \mathscr{I}} a_{i,j} g_i}{\sum_{i \in \mathscr{I}} a_{i,j} \delta_i^{d-1}} \right)^{a_{i,j}} \delta_i^{d-1}, \text{ for } i \in \mathscr{I}.$$

Step 3: If $\mathbf{A}'\delta^d = \gamma \mathbf{A}' g$, then $\delta^*(\gamma) = \delta^d$ and finish.
Step 4: Otherwise go to Step 1.

If δ is λ, then $\delta^*(1)$ is the MLE in the relational model. For probabilities, one needs to apply the second algorithm.

Algorithm G-IPF

Step 0: If δ is λ, then compute $\hat{\lambda}$ as $\delta^*(1)$ and finish.
Step 1: if δ is p, then compute $\delta^*(1)$ using IPF(1).
Step 2: If $1'\delta^*(1) = 1$, then $\hat{p} = \delta^*(1)$, and finish.
Step 3: Set

$$\gamma_R = (1'\mathbf{A}'g)^{-1}$$

and

$$\gamma_L = min((g(S_j))^{-1}, \quad j = 1, \ldots, J)$$

Step 4: Set

$$\gamma = \frac{\gamma_R + \gamma_L}{2}$$

and determine $\delta^*(\gamma)$ using IPF(γ).
Step 5: If $1'\delta^*(\gamma) = 1$, then set $\hat{p} = \delta^*(\gamma)$ and finish.
Step 6: If $1'\delta^*(\gamma) < 1$, then

$$\gamma_L = \frac{\gamma_R + \gamma_L}{2}$$

and go to Step 4. If $1'\delta^*(\gamma) > 1$, then

$$\gamma_R = \frac{\gamma_R + \gamma_L}{2}$$

and go to Step 4.

Of course, in any implementation of these algorithms, the equalities in Step 3 of IPF(γ) and in Steps 2 and 5 of G-IPF have to be checked up to a given precision.

The ideas behind these algorithms are quite straightforward. When the adjustment factor γ is known, IPF(γ) applies the IPFP procedure, starting with a uniform distribution. The only difference is that instead of marginal adjustments, that is, adjustments according to cylinder sets, adjustments apply to the subsets of the sample space defining the model and mean multiplying the cell parameters in all cells of the subsets with the ratio of the required and of the actual subset sums. The algorithm G-IPF finds an adjustment factor that makes the sum of the probabilities equal to 1, by applying bisection to an interval that surely contains the true adjustment factor.

Klimova and Rudas [82] gave proofs of convergence of these algorithms. In particular, they showed that the adjustment factor for the subset sums that makes the probabilities sum to 1 is between the bounds given in Step 3 of G-IPF.

These algorithms are implemented in the R package `gIPFrm`, see Klimova and Rudas [81].

Of course, general optimization algorithms may also be applied to find the MLEs under relational models. These include the algorithms described in Evans and Forcina [52]. Further, Forcina [60] gave an algorithm which is, in a sense, dual to the one presented here. Klimova and Rudas [82] and Forcina [60] discuss the relative merits of the two approaches.

For testing relational models, Klimova and Rudas [85] derived the asymptotic normality of the MLEs under Multinomial and multivariate Poisson sampling, with or without the presence of the overall effect. As in the case of multivariate Poisson sampling the observed total is, generally, not reproduced by the MLEs, the likelihood ratio statistic is not related to the minimum value of the Kullback-Leibler divergence between observations and estimates (see Section *LCDA* 12.2), but rather the more general Bregman divergence is relevant. While some of the proofs are more involved than in the case of log-linear models, the result about the asymptotic chi-squared distributions of the Pearson chi-squared and of the likelihood ratio statistics hold. As relational models are smooth manifolds, the number of degrees of freedom associated with these tests is implied by the number of cells of the sample space and of the dimension of the model.

11.5 Summary

This chapter reviewed results concerning a very broad model class that generalizes log-linear models in a way different from the generalization given by marginal log-linear models. Relational models do not assume a specific structure of the sample space. In particular, the sample space may be an incomplete Cartesian product or may have no structure, at all. Relational models for the cell parameters are multiplicative, and each multiplicative parameter pertains to a subset of the cells of the sample space. The models are defined by specifying the subsets of cells that

have a common parameter in their respective multiplicative representations, and the indicator functions of these subsets are the columns of the design matrix.

When the columns of the design matrix give, as a linear combination, the vector of 1s, the model is said to have an overall effect, because in this case it may be reparameterized to have the whole sample space as one of the subsets with a common parameter. Somewhat surprisingly, whether or not the model has an overall effect has important implications for the MLEs under relational models. Two sampling schemes were investigated, Multinomial and multivariate Poisson.

When the overall effect is present, the properties of the MLEs are parallel to those seen in the case of log-linear models. The MLEs for probabilities reproduce the observed relative frequencies of the subsets defining the model, and the MLEs for intensities reproduce the observed subset sums. The MLEs for cell parameters under the two sampling schemes are proportional, with the ratio of the intensities to the probabilities being the total sample size.

When there is no overall effect in the model, the MLEs of probabilities under Multinomial sampling reproduce the subset relative frequencies only up to a constant of proportionality, called the adjustment factor, and the total of the estimates of the intensities is not necessarily equal to the observed sample size.

Relational models may also be defined by generalized odds ratios. These are ratios of products of the cells parameters (probabilities or intensities). These are homogeneous, that is, have the same sum of the exponents of the cell parameters in the numerator as in the denominator, if and only if the model has an overall effect. Otherwise, some of the odds ratios are non-homogeneous, and it was shown that one non-homogeneous among the odds ratios is always sufficient to define the model.

Beyond proving the results above, it was also shown that the MLEs exist if and only if there is a (strictly) positive distribution with the same subset sums as the observed data.

An iterative algorithm was described to find the MLEs, and results about testing were summarized.

11.6 Things to Do

1. In the case when the data consist of a single classification, a popular approach is to define an indicator variable for each category, and analyze these indicators, often called dummy variables. That is, if there are k categories, define k binary variables V_i, $i = 1, \ldots, k$, and pretend their cross-classification was observed. Show that the resulting sample space is an incomplete Cartesian product.
2. Describe the relationships among the variables V_k defined in the previous item. Many sources suggest to use these "dummy" variables as predictors in a linear regression. Explain why this is highly problematic.
3. Prove Proposition 11.1.

4. In Chapters *LCDA* 10 and 11, balanced and corner parameters were used in the context of log-linear parameterizations. Can such parameters be obtained using a parameterization like the one in (11.3)?
5. Prove that if the column space of the matrix **A** in (11.14) contains a vector of 1s, that is, the vector of 1s is a linear combination of its columns, then there is a reparameterization of the relational model such that the design matrix contains a column of 1s.
6. Work out the details of the various reparameterizations of the relational model defined in (11.9).
7. Prove that (11.9) defines independence in a 2×2 table.
8. Define conditional independence in a $2 \times 2 \times 2$ table as a relational model.
9. Prove that (11.8) is equivalent to (11.14).
10. Check that Corollary 11.2 is correct.
11. Prove that the observed subset sums are sufficient statistics when sampling is Multinomial in the case of relational models, that is, the likelihood of a given set of observations may be written as a function of the subset sums.
12. Prove that the observed sample total is also part of the sufficient statistics when sampling is multivariate Poisson, for relational models without an overall effect.
13. What are the sufficient statistics in the case of a log-linear model? Are they different in the case of Multinomial and multivariate Poisson sampling?
14. Consider a log-linear model and find the maximum likelihood estimates of the expected values of the sufficient statistics.
15. Is it true in the case of log-linear models that the expected values of the sufficient statistics under the maximum likelihood estimate are the same as their observed values?
16. Under what condition is the claim in the item above true for relational models?

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